

Transaction Security Manual

BASE24[®]



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What's New

The following tables highlight the major changes that have been made to the ***BASE24 External Message Manual*** since its publication for BASE24 release 6.0 version 10. The first column of each table lists the sections and appendixes in which major changes have been made. The second column of each table describes the major changes for the section or appendix.

September 2012

Section/ Appendix	Major Changes
All	Adds support for the Atalla Network Security Processor (NSP) model A10160 throughout the manual.

ACI Worldwide, Inc.

Preface

BASE24 transaction security includes PIN encryption, PIN verification, message authentication, card verification, chip authentication, and dynamic key management. The ***BASE24 Transaction Security Manual*** describes BASE24 support in each of these areas, along with the database settings required to implement this support.

The manual also describes how hardware security modules are integrated into BASE24 processing and the duties they can assume as a part of that processing.

This manual does not describe the implementation of the Transaction Security Services process for transaction security processing. The Transaction Security Services process must be implemented to support full triple DES processing across the BASE24 system (i.e., in Device Handler processes or modules, Authorization processes or modules, and Interchange Interface processes) or when migrating Atalla security module software to support the Atalla Key Block (AKB) format for key storage. Triple DES translation and message authentication (using double-length keys) can be supported in certain Interchange Interface processes only without using the Transaction Security Services process. This manual describes this partial support of triple DES in Interchange Interface processes without using the Transaction Security Services process. For detailed information on the implementation, configuration, and operation of the Transaction Security Services process in a BASE24 system, refer to the ***BASE24 Transaction Security Services Processing Guide***.

Audience

This manual is intended for the system administrators and operational staff who determine how the BASE24 system is to handle transaction security and who set up the BASE24 database.

Prerequisites

This manual discusses a number of general topics relating to transaction security. These discussions have been included primarily to provide a stronger insight into the way BASE24 handles transaction security. The manual is not intended as a substitute for the industry literature relating to transaction security. It is recommended that individuals making decisions on BASE24 transaction security processing be familiar with the industry literature available for the type of transaction security processing being performed on their BASE24 system. Refer to the following ANSI and ISO standards for additional information:

- The ANSI X3.92 standard, *Data Encryption Algorithm Standard*
- The ANSI X9.8 standard, *American National Standard for Personal Identification Number (PIN) Management and Security*
- The ANSI X9.9 standard, *Financial Institution Message Authentication (Wholesale)*
- The ANSI X9.17 standard, *Financial Institution Key Management (Wholesale)*
- The ANSI X9.24-1 standard, *American National Standard for Symmetric Key Management using Symmetric Techniques*
- The ANSI X9.2 standard, *Interchange Message Specification for Debit and Credit Card Message Exchange Among Financial Institutions*
- The 1987 ISO 8583 standard, *Bank Card Originated Messages—Interchange Message Specifications*
- The ISO 8732 standard, *Banking—Key Management (Wholesale)*
- The ISO 9564 standard, *Banking—Personal Identification Number Management and Security*

Also important is an understanding of the basic functionality of the security modules to be integrated with BASE24. Security modules are all integrated into the BASE24 system in a similar fashion; however, the individual functions of each module and the ways in which these functions are accessed vary by vendor and module. Thus, users of this manual should have a working knowledge of the security modules they plan to use, especially in the area of key generation and management.

Additional Documentation

The BASE24 documentation set is arranged so that each BASE24 manual presents a topic or group of related topics in detail. When one BASE24 manual presents a topic that has already been covered in detail in another BASE24 manual, the topic is summarized and the reader is directed to the other manual for additional information. Information has been arranged in this manner to be more efficient for readers who do not need the additional detail and at the same time provide the source for readers who require the additional information. This manual contains references to the following BASE24 publications:

- The ***BASE24 Base Files Maintenance Manual*** provides data entry screens and field descriptions for base files.
- The ***BASE24 ERACOM Security Control Module (SCM) Reference Manual*** and the ***BASE24 ERACOM Security Module (ESM) Reference Manual*** describe BASE24 support and configuration requirements for SafeNet (Eracom) security devices.
- The ***BASE24-atm Files Maintenance Manual*** provides data entry screens and field descriptions for BASE24-atm files.
- The ***BASE24-pos Files Maintenance Manual*** provides data entry screens and field descriptions for BASE24-pos files.
- The ***BASE24-atm EMV Support Manual*** and the ***BASE24-pos EMV Support Manual*** describe the configuration requirements and processing performed by BASE24 for the EMV card authentication methods.
- The ***BASE24-teller Files Maintenance Manual*** provides data entry screens and field descriptions for BASE24-teller files.
- The ***BASE24 BIC ISO Interface Manual*** provides information regarding the BASE24 Interchange (BIC) message processing.
- The ***BASE24 BIC ISO Standards Manual*** provides information regarding the BASE24 Interchange (BIC) ISO external message structure.
- The ***BASE24 External Message Manual*** describes the BASE24 External Message. Configured using the External Message File (EMF), it is the ISO-based message used to communicate with hosts.
- The ***BASE24 ISO Host Interface Manual*** provides information regarding the BASE24 ISO Host Interface external message structure.
- The ***BASE24 Logical Network Configuration File Manual*** contains additional information regarding BASE24 assigns and params in the Logical Network Configuration File (LCONF).

- The ***BASE24 Text Command Reference Manual*** provides descriptions and syntax for BASE24 send or deliver process and interprocess commands.
- The ***BASE24 Transaction Security Services Processing Guide*** provides detailed information on implementing the Transaction Security Services process into a BASE24 system.
- The ***Remote Application Manager Implementation and Installation Guide*** provides information on configuring and using the Remote Application Manager process (referred to as the Racal Access Module process when used with Thales e-Security devices) with Transmission Control Protocol/Internet Protocol (TCP/IP). This informal document is a TGAL document provided in the BA10RAM.RAMUPDT file on the HP NonStop system and is also available from ACI upon request.
- The ***RAM/UDP Update Guide*** describes the use of the URAM process with User Datagram Protocol/Internet Protocol (UDP/IP) for Thales e-Security devices. This informal document is a Word document provided in the BA10RAM.URAMUDOC file on the HP NonStop system and is also available from ACI upon request.

This manual contains references to the following Atalla publications:

- The ***Atalla Key Block Banking Command Reference Manual*** describes the commands available for the A8100, A8150, A9100, A9150, A101000, A10150, and A10160 NSP models.
- The ***Banking Command Reference Manual*** describes the commands available for the A8000, A8100V, A8150V, A9000, A9100V, A9150V, A10000E, A10100V, A10150V, and A10160V NSP models.
- The ***Installation and Operations Guide for Atalla NSP Series products using Atalla Variants*** describes installation and operations procedures for the A8100V, A8150V, A9100V, A9150V, A10100V, A10150V, and A10160V NSP models.
- The ***Installation and Operations Guide for Atalla NSP Series products with the Atalla Key*** describes installation and operations procedures for the A8100, A8150, A9100, A9150, A10100, A10150, and A10160 NSP models.
- The ***A8000 Installation, Operations, and Command Reference Guide*** describes installation and operations procedures for the A8000 model.
- The ***A8000-AKB Installation and Operations Guide*** describes installation and operations procedures for the A8000-AKB NSP model.
- The ***A9000 Installation and Operations Guide*** describes installation and operations procedures for the A9000 NSP model.

- The *A9000-AKB Installation and Operations Guide* describes installation and operations procedures for the A9000-AKB NSP model.
- The *A10000E Installation and Operations Guide* describes installation and operations procedures for the A10000E NSP model.
- The *A10000E-AKB Installation and Operations Guide* describes installation and operations procedures for the A10000E-AKB NSP model.

This manual also contains references to the following Hewlett-Packard (HP) NonStop publications:

- The HP NonStop *Himalaya S-Series Hardware Installation Guide* provides information on adding a 3861 Ethernet 4 ServerNet Adapter (E4SA) controller.
- The HP NonStop *TLAM Configuration and Management Manual* provides information on adding an Ethernet controller and line.

Software

This manual documents standard processing as of its publication date. Software that is not current and custom software modifications (CSMs) may result in processing that differs from the material presented in this manual. The customer is responsible for identifying and noting these changes.

Documentation of standard transaction processing for the BASE24-telebanking and BASE24-billpay products is addressed in the *BASE24 Integrated Server Transaction Security Manual*.

Manual Summary

The *BASE24 Transaction Security Manual* is organized by security function (e.g., PIN verification, message authentication), then by process (e.g., Device Handler, Authorization, or Interchange Interface process). Sections that discuss Device Handler and Authorization processes are organized by product (i.e., BASE24-atm, BASE24-pos, or BASE24-teller). When configuring a process, it is important that all applicable security functions are included in the setup procedure.

The following is a summary of the contents of this manual.

“Conventions Used in this Manual” follows this preface and describes notation and documentation conventions necessary to understand the information in the manual.

Section 1, “Introduction,” introduces the reader to transaction security, including overviews of key management, PIN encryption, message authentication, card and PIN verification, and their support by BASE24.

Section 2, “BASE24 PIN Encryption Support,” summarizes how BASE24 manages PIN encryption in conjunction with the various external processors to which they are connected. These processors include hosts, interchanges, and EFT terminals. This section also includes a discussion of PIN blocks and their role in PIN encryption.

Section 3, “BASE24 Interface PIN Encryption Support,” describes how BASE24 supports PIN encryption in Interchange Interface and Host Interface processes.

Section 4, “BASE24 Device Handler PIN Encryption Support,” describes how BASE24 supports PIN encryption in Device Handler processes.

Section 5, “BASE24 Message Authentication Support,” summarizes how BASE24 manages message authentication in conjunction with the various external processors to which they are connected. These processors include hosts, interchanges, and EFT terminals.

Section 6, “BASE24 Interface Process Message Authentication Support,” describes how BASE24 supports message authentication in Interchange Interface and Host Interface processes.

Section 7, “BASE24 Device Handler Message Authentication Support,” describes how BASE24 supports message authentication in Device Handler processes.

Section 8, “BASE24 PIN Verification Support,” describes how PIN verification is performed by the BASE24-atm, BASE24-pos, and BASE24-teller Authorization processes. The basic options available, the fields used in the BASE24 internal messages for PIN verification, and the processing performed by the BASE24 Authorization processes are presented in this section.

Section 9, “IBM DES (3624-Format) PIN Verification Support,” describes BASE24 support of IBM DES (3624-format) PIN verification, including how the BASE24 database must be set up to allow BASE24 to verify PINs under this method of verification.

Section 10, “Diebold PIN Verification Support,” describes BASE24 support of Diebold PIN verification, including how the BASE24 database must be set up to allow BASE24 to verify PINs under this method of verification.

Section 11, “Identikey PIN Verification Support,” describes BASE24 support of Identikey PIN verification, including how the BASE24 database must be set up to allow BASE24 to verify PINs under this method of verification.

Section 12, “Visa PVV PIN Verification Support,” describes BASE24 support of Visa PVV PIN verification, including how the BASE24 database must be set up to allow BASE24 to verify PINs under this method of verification.

Section 13, “BASE24 Card Verification Support,” describes how card verification is performed by the BASE24-atm, BASE24-pos, and BASE24-teller Authorization processes. The basic options available, the fields used in the BASE24 internal messages for card verification, and the processing performed by the BASE24 Authorization processes are presented in this section.

Section 14, “BASE24-pos Secure Internet Validation Support,” describes how e-commerce transactions are validated by the BASE24-pos Device Handler/Router/Authorization process. This section presents the validation processing performed by the Router/Authorization module, the fields used in the BASE24-pos Standard Internal Message (PSTM), and the options you can configure in the BASE24 database.

Section 15, “BASE24 Device Key Management,” describes the dynamic key management options available for BASE24-atm and BASE24-pos devices.

Section 16, “BASE24 Interface Key Management,” describes the dynamic key management options provided by the BASE24 Host Interface and BASE24 Interchange (BIC) Interface processes. It also describes the Cryptographic Service Message (CSM) element in network management messages used for dynamic key management.

Section 17, “Security Module Support,” describes the usage of security modules with BASE24 and the functions of each that are supported by BASE24.

Appendix A, “BASE24 Supercrypt Utility,” describes the BASE24 Supercrypt utility, a program that is required for encrypting inbound and outbound keys for entry into the BASE24 database when BASE24 perform external message PIN encryption in software.

Appendix B, “Atalla Security Modules,” describes how Atalla security modules must be configured to work with BASE24. It does not describe the proprietary workings of any of the security modules, but rather the functions of each that are supported by BASE24 and the information a user requires for configuring the modules for use by BASE24.

Appendix C, “Thales e-Security Host Security Modules,” describes how Thales e-Security (Racal) Host security modules must be configured to work with BASE24. It does not describe the proprietary workings of any of the security modules, but rather the functions of each that are supported by BASE24 and the information a user requires for configuring the modules for use by BASE24.

Readers can use the index by field name to locate information about a particular screen field and the index by data name to locate information about a particular field from a file or record structure.

Publication Identification

Three entries appearing at the bottom of each page uniquely identify this BASE24 publication. The publication number (for example, BA-AE000-13 for the ***BASE24 Transaction Security Manual***) appears on every page to assist readers in identifying the manual from which a page of information was printed. The publication date (for example, Sep-2012 for September, 2012) indicates the issue of the manual. The software release information (for example, R6.0v10 for release 6.0, version 10) specifies the software that the manual describes. This information matches the document information on the copyright page of the manual.

Conventions Used in this Manual

This section explains how terminology, syntax notation, and blank characters are documented in this manual.

Definition of Terms

The following definitions are provided to give the reader a clear understanding of specific terms as they are used in this manual.

External Processors

This manual uses the term *external processor* to describe a transaction processing entity that is external to BASE24. External processors can include the following:

- A data processing center (DPC), which is the BASE24 designation for host computers connected online to a BASE24 system.
- An interchange, which is the BASE24 designation for interchange systems connected online to a BASE24 system.
- A co-network, which is another BASE24 system or another transaction processing network connected online to a BASE24 system using the BIC ISO Interface process.

When this manual is referring to processing performed by one of these specific entities, that entity is referred to by the name shown in the list above. If the processing can be performed by any of these entities (DPC, interchange, or co-network), the term *external processor* is used.

BASE24

For purposes of this manual, all generic references to *BASE24* are meant to include only the following specific BASE24 products:

- BASE24-atm
- BASE24-pos
- BASE24-teller

Magnetic Stripe Data

The term *magnetic stripe data* refers to the data contained in the physical magnetic stripe of a card. The data may include Track 1 data, Track 2 data, or both Track 1 and Track 2 data depending on preferences of the card issuer.

SYSGEN Utility and DSC Utility

This manual frequently references the terms *SYSGEN utility* and *DSC utility*. These terms refer to the HP NonStop System Generation (SYSGEN) and the HP NonStop Dynamic System Configuration (DSC) utilities.

Terminal Data Files Terminology

Data entered on BASE24-atm Terminal Data files (ATD) screens is stored in different files, depending on the type of terminal being defined. For Diebold 10XX/478X and NCR 5XXX Device Handler terminals, data is stored in the BASE24-atm Terminal Data Dynamic File—general data (ATDD1) and the BASE24-atm Terminal Data Static File—general data (ATDS1) files. Dynamic data is stored in the ATDD1 file and static data is stored in the ATDS1 file. For all other types of ATM terminals, data is stored in the Terminal Data File (TDF).

Data entered on BASE24-pos Terminal Data files (PTD) screens is actually stored in different files, depending on the type of terminal being defined. For Hypercom and Standard POS Device Handler terminals, dynamic data is stored in the BASE24-pos Terminal Data Dynamic File—general data (PTDD1) and static data is stored in the BASE24-pos Terminal Data Static File—general data (PTDS1). For all other types of POS terminals, all data from the PTD screens is stored in the POS Terminal Data File (PTDF).

For documentation purposes, the terms BASE24-atm Terminal Data files and BASE24-pos Terminal Data files are used throughout this manual with the ATD or PTD acronym when referring to a screen and without an acronym when referring to a file.

Syntax Notation

Syntax descriptions in this manual use the following symbols and conventions to denote how commands must be entered:

Notation	Meaning
UPPERCASE LETTERS	Indicate key words and literals that you must enter exactly as shown.
<i>lowercase italics</i>	Identify variables that you must provide.
Brackets []	Enclose optional syntax items that you can enter in the command.
Braces { }	Enclose a group of items from which you are required to choose one item. The items are separated within the braces by a vertical bar ().
Vertical bar	Separates alternatives in a list of items.
Ellipsis ...	Indicates the immediately preceding syntactical item can occur one or more times.
Punctuation	Must be entered as shown. This includes parentheses, commas, semicolons, and other symbols not previously described.
Spaces	Are required as delimiters wherever it would be otherwise impossible to discriminate between adjacent words or symbols. You can insert additional spaces between any adjacent words or symbols, but not within a word or multiple character symbol.
Indented lines	Indicates a continuation of the preceding line of syntax. If the syntax of a command is too long to fit on a single line, each continuation line is indented.

Blank Characters

Field descriptions for internal and external messages, tokens, files maintenance screens, and reports often include lists of valid values. When the value contains a blank character, this manual uses the symbol *b* to indicate the blank character.

1: Introduction

The transaction security requirements of any organization can vary greatly according to the environment in which the organization is operating. An organization often requires an integrated system of software, industry hardware, and manual procedures to properly implement transaction security.

BASE24 is designed to be flexible in its transaction security support. It allows for a wide range of software and hardware options geared toward the individual needs of institutions. BASE24 also addresses the diverse needs of a large-scale transaction processing system where the originator of a transaction can be operating under an entirely different transaction security system from the authorizer of a transaction.

BASE24 transaction security includes several components that work together:

- PIN encryption ensures that PIN information is maintained in a confidential manner.
- PIN verification ensures that the true cardholder is performing the transaction.
- Message authentication ensures that data remains tamper-free while passing between an EFT device and its authorization destination.
- Card verification ensures that a valid card is being used to initiate the transaction.
- Chip authentication ensures that a credit or debit card with an embedded microprocessor chip is valid.

This section includes overviews of each of these transaction security components, how they are supported by BASE24, and the role transaction security plays in the electronic funds transfer (EFT) industry. It also provides an introduction to the security modules supported by BASE24, the use of check digits with hardware-encrypted keys, and key management tools provided with BASE24.

In reviewing the information presented in this section, note that the term *external processor* can refer to a data processing center (DPC), an interchange, or a co-network as defined in the “Conventions Used in this Manual” section of this manual.

Data Encryption and Decryption

Data encryption and decryption are the means by which sensitive data such as a PIN can be rendered unintelligible for storage in a database or transmission over communications lines and then subsequently returned to its original form when necessary.

The support provided by BASE24 for data encryption and decryption is based on the use of the Data Encryption Standard (DES) algorithm, a data encryption algorithm widely used throughout the financial industry for various cryptographic functions. Currently, there are variations of data encryption and decryption using the DES algorithm—single DES and triple DES. Both of these variations are described below.

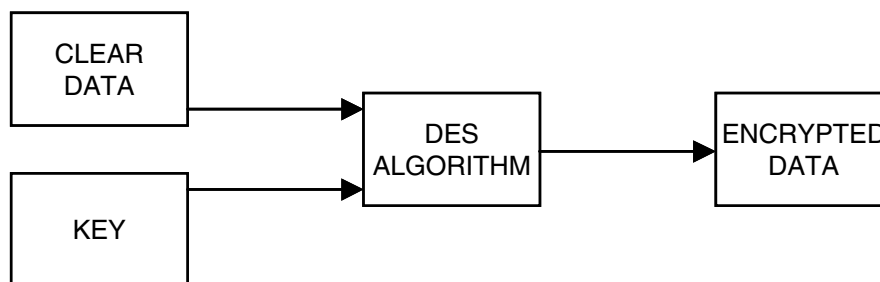
Single DES

Single DES encryption starts with a data item in its original, or clear, form. Then, together with a second data item called a key, that clear data is used as input to the DES algorithm, which yields a new, encrypted value bearing no outward resemblance to the original.

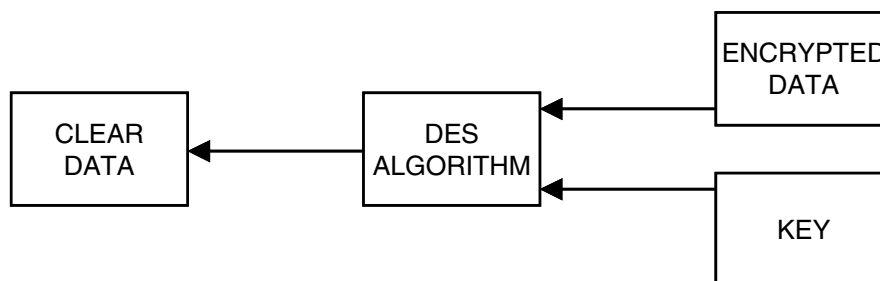
Single DES decryption is the reverse of the encryption operation. It starts with the encrypted data and the same key that was used in the encryption process. In this case, the DES algorithm is applied in reverse to the two data items, bringing the unintelligible data back to its original, or clear, form. All keys used in single DES operations are single-length keys.

As shown below, there are three components of a single DES encryption or decryption operation: the DES algorithm, a key, and the data to be encrypted or decrypted.

The following diagram illustrates the processing involved in single DES data encryption. The clear data to be encrypted is used as input to the DES algorithm, along with a single length encryption key. The output is the encrypted data.



The next diagram illustrates the processing used to decrypt data that has been encrypted using single DES data encryption. The encrypted data is used as input to the DES algorithm, along with the same single length encryption key that was used originally to encrypt the data.



Triple DES

Triple DES is a variation of the existing DES algorithm, where the algorithm is applied three times using a double- or triple-length key instead of just once using a single-length key. The output from the first DES operation is used as the data input to the second DES operation. The output from the second DES operation is then used as the data input to the third DES operation, yielding the final result. Because triple DES uses longer keys, triple DES provides additional security against exhaustive criminal trial and error attacks to determine a DES key and thereby obtain a cardholder's clear PIN. Government bodies, retail banking associations, interchange organizations, and terminal vendors are beginning to mandate the use of triple DES now or in the near future.

Note: This manual describes the use of triple DES in Interchange Interface processes only. To support full triple DES throughout a BASE24 system, you must implement the Transaction Security Services process. For detailed information on implementing the Transaction Security Services process, refer to the ***BASE24 Transaction Security Services Processing Guide***.

The following Interchange Interface processes support the translation of PIN blocks from triple DES to single DES, and vice versa:

Banknet Interchange Interface process
BIC ISO Interchange Interface process
EPS-Net Interchange Interface process
ISO Host Interface process
LINK ISO Interchange Interface process
MDS ISO Interchange Interface process
NBGC Interchange Interface process
VisaNet ISO Interchange Interface process

The BIC ISO Interchange Interface process and the ISO Host Interface process also support triple DES message authentication using double length keys without using the Transaction Security Services process.

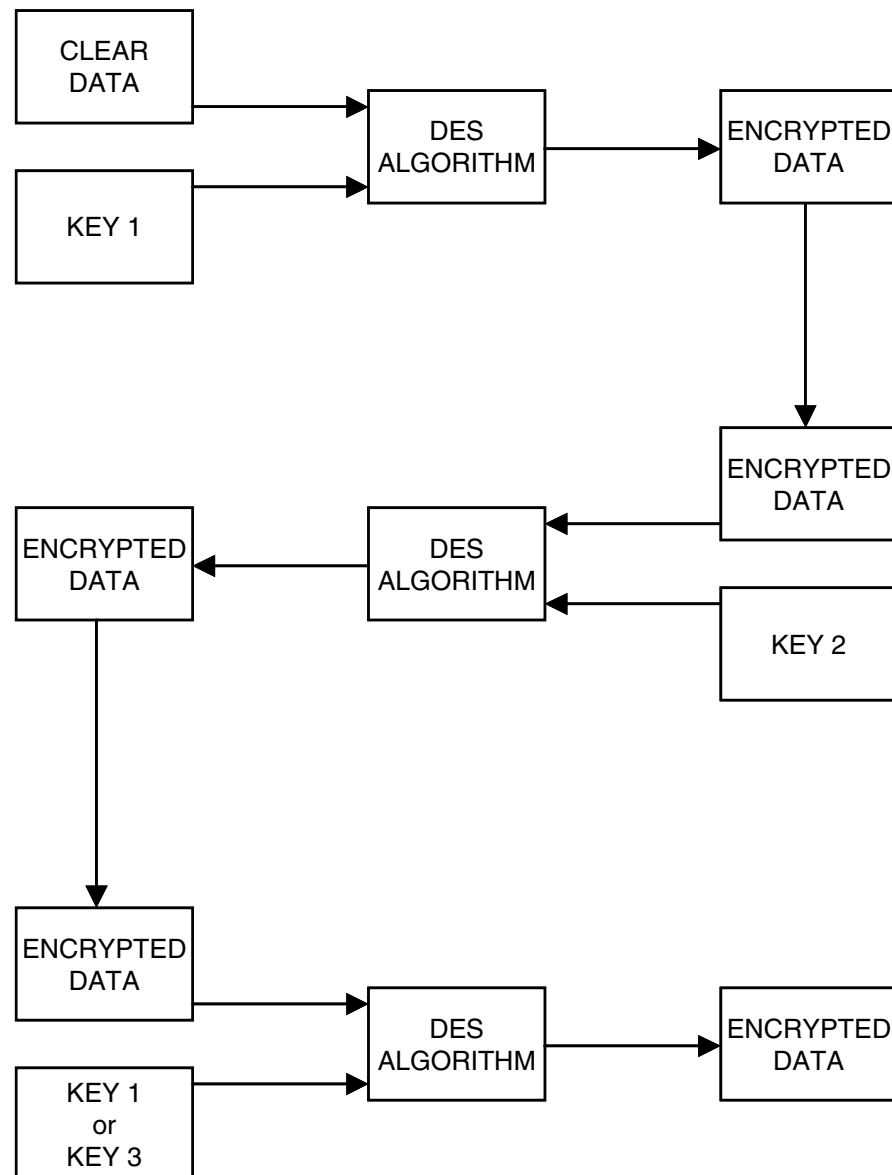
Although triple DES allows for different modes of encryption (i.e., encrypt/encrypt/encrypt and encrypt/decrypt/encrypt) and decryption (i.e., decrypt/decrypt/decrypt and decrypt/encrypt/decrypt), BASE24 implements the encrypt/decrypt/encrypt mode for encryption and the decrypt/encrypt/decrypt mode for decryption. For migration purposes from single DES to triple DES, these modes allow triple DES to yield the same results as single DES encryption and decryption when the same single length key is used for all three DES operations. In triple DES, the same single DES operations are used, but they are applied three times using two or three different keys.

When using double length keys, the first length of the key, referred to as the left half, is used in the first and third DES operations. The second length of the key, referred to as the right half, is used in the second DES operation.

When using triple length keys, the first length, referred to as the left key, is used in the first DES operation, the second length, referred to as the middle key, is used in the second DES operation, and the third length, referred to as the right key, is used in the third operation.

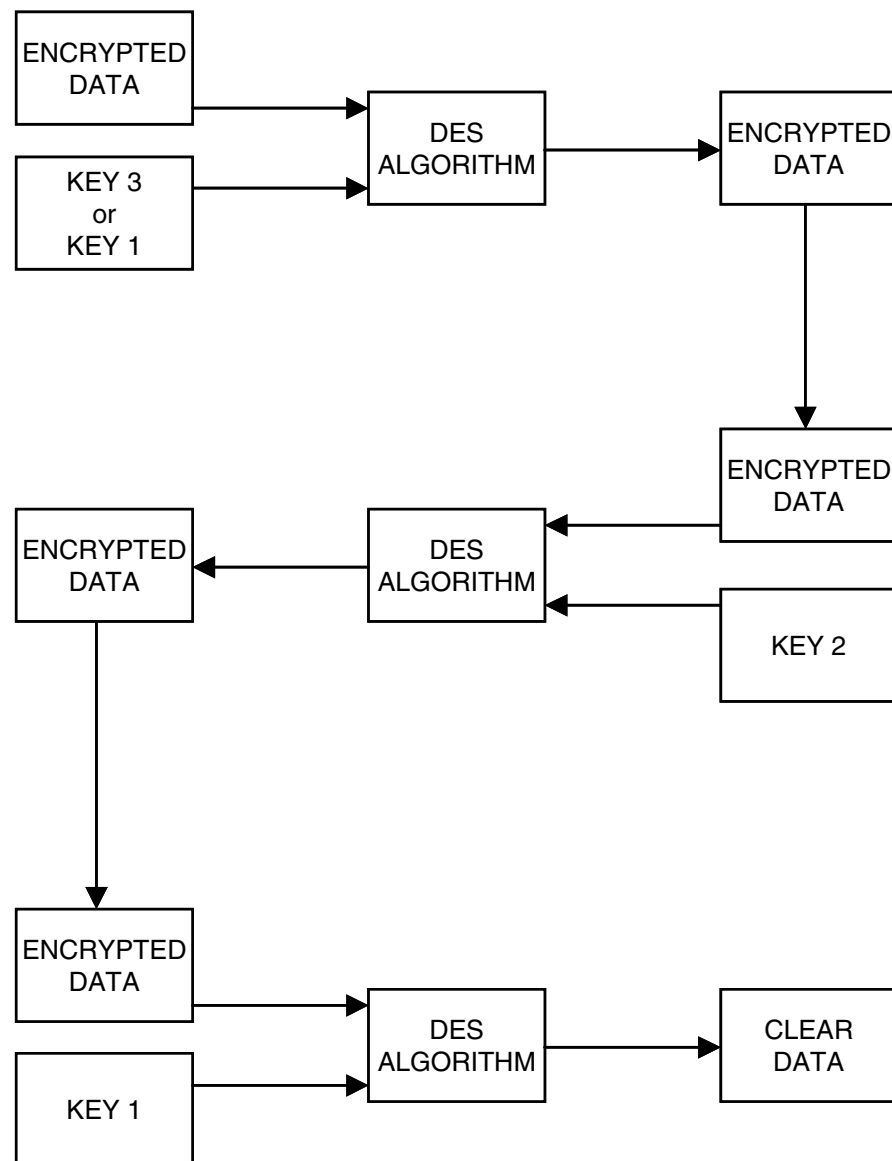
Encryption

The following diagram illustrates the operations involved in triple DES encryption. The clear data to be encrypted is first input to the DES algorithm, along with an encryption key (i.e., the first length of a double or triple length key). The encrypted data output from the first operation is used as input to the DES algorithm, along with a second key (i.e., the second length of a double or triple length key) to obtain encrypted data. Finally, the encrypted data output from the second operation is then used as input to the DES algorithm, along with the same encryption key used in the first operation (for double length keys) or the third length of a triple length key to obtain the final triple DES encrypted data.



Decryption

The following diagram illustrates the operations involved in triple DES decryption. The encrypted data to be decrypted is first used as input to the DES algorithm, along with an encryption key (i.e., the first length of a double length key or the third length of a triple length key). The encrypted data output from the first operation is used as input to the DES algorithm, along with a second key (i.e., the second length of a double or triple length key) to obtain encrypted data. Finally, the encrypted data output from the second operation is then used as input to the DES algorithm, along with the same encryption key used in the first operation (for double length keys) or the first length of a triple length key to obtain the final clear version of the triple DES encrypted data.



DES Algorithm

The DES algorithm is a set of mathematical operations widely used throughout the financial industry that uses a 56-bit encryption key for data encryption and decryption. It is also known as the data encryption algorithm (DEA), which is approved by the International Organization for Standardization (ISO) for use with ISO standards related to PIN security and message authentication. As stated previously, it is the DEA on which BASE24 cryptographic support is based.

The triple DES algorithm is the same as the DES algorithm, except that it uses a single length (56-bit), double length (112-bit) or triple length (168-bit) encryption key for data encryption and decryption and it applies the DES algorithm three times. The triple DES algorithm is also known as the triple data encryption algorithm (T-DEA). When triple DES is implemented with one key, it is referred to as *one-key* triple DES. This method is used only when translating a PIN block from triple DES encryption to single DES encryption. When triple DES is implemented with two keys, it is referred to as *two-key* triple DES. When triple DES implemented with three keys, it is referred to a *three-key* triple DES.

Note: Although a single length key ostensibly consists of 64 bits, only 56 bits are actually used by the DES algorithm. Eight bits are used solely for checking parity, and are thereafter discarded. Hence the effective key length is 56 bits, and it is usually quoted as such. Thus, a single length (56-bit) key is stored as 64 bits, a double length (112-bit) key is stored as 128 bits, and a triple length (168-bit) key is stored as 192 bits.

The DES algorithm is available royalty-free for public use. The algorithm operations are not documented in this manual. For information on the actual workings of the DES algorithm, refer to the ANSI X3.92 standard, ***Data Encryption Algorithm Standard***.

Keys

Both single DES and triple DES encryption require what is called a key. A key is a sequence of numbers used as one of the two variables input into the DES algorithm. Its role in the encryption and decryption process is major.

DES encryption and decryption require two input data blocks. From these two values a third is computed. When encrypting a block of data, the inputs are the clear version of the data to be encrypted and the encryption key. The result of the operation is the encrypted form of the clear data. When decrypting a block of data,

the inputs are the encrypted version of the data and the same encryption key that was used to encrypt the data. The output in this case is the clear form of the encrypted data.

In both cases, the encryption key is one of the two given blocks of data which allows for derivation of the third.

Triple DES encryption and decryption also require two input data blocks to each individual encrypt or decrypt operation. Triple DES encryption is comprised of an encrypt, a decrypt, and a second encrypt operation, while triple DES decryption is comprised of a decrypt, encrypt, and a second decrypt operation. The result of each operation is used as one of the two input blocks to the next operation. Two or three encryption keys can be used.

Encryption keys can consist of different lengths. A single-length key is 56 bits in length, a double-length key is 112 bits in length, and a triple-length key is 168 bits in length. Basically, the longer a key is in length, the harder it is to determine through a brute force trial and error attack. Single DES encryption and decryption uses single-length encryption keys while triple DES encryption and decryption uses double- or triple-length encryption keys. Regardless of key length, encryption keys are often comprised of parts when placed in security modules, with each part known only to one person for security reasons.

Encryption Security

The security of DES encryption and decryption is dependent on keeping the encryption keys confidential.

Since the DES algorithm is widely used, if an encryption key becomes compromised, the data encrypted with it becomes vulnerable to decryption by unauthorized parties.

For encrypted data stored on a database, security is fairly straightforward. The encryption key can be maintained and kept secure by the party in control of the database. When security modules are used for data encryption and decryption, the key itself is often comprised of parts, with each part known only to one person. Under this method, when a key must be placed in the security module, each person enters the part known to them separately, such that no one person has knowledge of the entire key.

File maintenance screens are used to place encryption keys in the BASE24 databases. Each key is entered in its entirety; therefore, the option of maintaining key information in parts is not available when entering keys in the BASE24 databases.

For encrypted data transmitted between two parties or networks, encryption key security becomes slightly more involved because both parties must know the encryption key and, at the same time, maintain its secrecy.

Translation

Another data security concept that is closely related to encryption and decryption is translation. In situations involving more than one data format or key (e.g., an EFT device is using one key and the external processor is using another key, or an interchange has mandated triple DES encrypted PINs using double-length keys and BASE24 authorization still uses single DES encryption and single-length keys), the data must be translated as necessary so that both entities can accurately read the data. This differs from encryption and decryption because the data is in encrypted form before and after translation without appearing in clear form, as shown below.

Process	Input Data	Output Data
Encryption	Clear	Encrypted
Decryption	Encrypted	Clear
Translation	Encrypted	Encrypted

PIN Management

In an EFT environment, transactions are authorized electronically and can be initiated from unattended terminals. Even when transactions are initiated at terminals operated by retail merchants or financial institution personnel, plastic cards are often used as the primary form of customer identification. Thus, a means of ensuring that the card is in the hands of the true cardholder is required. The PIN serves as the basis for authenticating each cardholder.

PIN is an acronym for personal identification number, which is a number assigned to a cardholder to allow that cardholder to initiate EFT transactions.

Because of the major role that PINs play in the authorization of EFT transactions, all parties involved in the processing of EFT transactions are obliged to maintain PIN confidentiality.

If an EFT transaction is accompanied by the cardholder's PIN, the authorizer of the transaction can assume that the transaction is being made by the true cardholder or on the authority of the true cardholder.

However, if an EFT transaction is accompanied by a PIN other than the cardholder's, the authorizer of the transaction must assume that the transaction is being attempted without the consent of the true cardholder and therefore must decline the transaction.

Not only must cardholders keep their PINs secret, but their confidentiality also must be ensured as they travel from the point at which a transaction is originated to the point at which the transaction is authorized—an electronic journey that can span multiple processors and can cross numerous communications links.

Because of the relative ease with which data can be tapped while being transmitted over various commercial communications lines or discerned during processing, PINs and other sensitive data items are often encrypted to guard against unauthorized detection. This encryption, while effective in disguising data, is itself only viable if certain aspects of the encryption are kept confidential.

The problem of preserving PIN confidentiality is further compounded by the fact that the PIN must ultimately be verified in an electronic medium, requiring that an authorizer be able to electronically determine the validity of a PIN entered in a transaction. This not only requires that the authorizer be able to deal with an encrypted version of the PIN, but also requires that keys used to encrypt the PIN be kept confidential to ensure that PINs are not generated in an unauthorized manner.

There are a number of methods of PIN verification used throughout the financial industry. Many approach PIN verification in different ways. However, most are based on the derivation of certain items of control information that can be kept on file for a cardholder, in lieu of the actual PIN, and then mathematically linked to the cardholder's PIN for verification at the time of a transaction.

A number of industry-wide standards and practices have been developed to effectively guard PINs while traveling through various electronic media and during verification. BASE24 allows for integrating many of these standards and practices in day-to-day processing, and provides various options to its users for tailoring PIN security to their own situations and needs. BASE24 PIN support can be divided into two areas, each having its own distinct set of options:

- Encryption, decryption, and translation
- Verification

PIN Encryption, Decryption, and Translation

BASE24 PIN encryption and decryption duties are handled by the BASE24 processes that send transactions to and receive transactions from processing entities outside the system. These processes include the following:

- Host Interface processes
- Interchange Interface processes
- Device Handler processes

When a transaction is received from outside the BASE24 system, the PIN may have to be encrypted, decrypted, or translated from one encrypted form to another, depending on how the BASE24 system and external processor are set up to handle PINs.

BASE24 support of PIN encryption is described in sections 2, 3, and 4 of this manual.

PIN Verification

While PINs can be verified outside of the BASE24 system, PIN verification within the BASE24 system is the responsibility of the BASE24 Authorization processes or modules. When transactions are received by an Authorization process or module, part of the authorization procedure is verifying the PIN.

The mechanics of BASE24 PIN verification is described in section 8.

BASE24 supports four methods of PIN verification:

- IBM DES (3624-format) PIN verification
- Diebold PIN verification
- Identikey PIN verification
- Visa PIN verification value (PVV) verification

Each of these methods has its own unique set of requirements and options.

Information specific to the individual PIN verification methods is contained in sections 9 through 12.

Section	PIN Verification Method
Section 9	IBM DES (3624-format) PIN verification
Section 10	Diebold PIN verification
Section 11	Identikey PIN verification
Section 12	Visa PVV PIN verification

PIN Verification Value Security

The PIN verification value, referred to as the PIN verification number for the Identikay PIN verification method, the PIN offset for the IBM DES and Diebold PIN verification methods, or the PIN Verification Key Indicator (PVKI) and PIN verification value for the Visa PVV PIN verification method, is entered and displayed in the POFST/PVV field on Cardholder Authorization File (CAF) screen 1.

Because there are known cryptographic attacks that can be launched when an adversary is provided with a customer's card number and PIN verification value (PVV), the BASE24 system can display asterisks instead of the actual PVV in this field. If asterisks are displayed instead of the actual PVV in this field, users still have the ability to enter a new PVV in this field, however, once the CAF record is saved, the POFST/PVV field will display asterisks only.

The Logical Network Configuration File (LCONF) param MASK-POFST-PVV specifies whether asterisks or the actual PVV are displayed in the POFST/PVV field on CAF screen 1. Valid values for this param are as follows:

Y or y = Yes, mask the PVV in the POFST/PVV field on CAF screen 1.
N or n = No, do not mask the PVV in the POFST/PVV field on CAF screen 1.
Default.

You can set this param to not mask the PVV for testing or customer service environments where users need to access the actual PVV.

Message Authentication

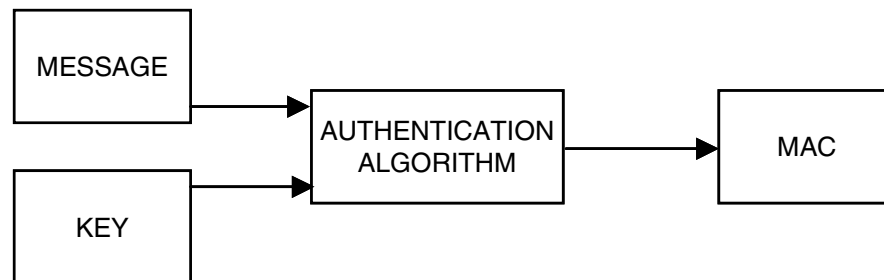
In an EFT environment, data passes between an EFT device and its authorization destination electronically, regardless of whether the information is entered by a customer at an ATM, by a merchant at a point-of-sale (POS) device, or by a teller at a financial institution's teller terminal. The integrity of this data is a key to the success of all of these systems. Message authentication is a method used to monitor the integrity of data flowing between these EFT devices and the transaction authorizer.

The message authentication performed in BASE24 is based on the use of the authentication algorithm defined in the ANSI X9.9 standard, *Financial Institution Message Authentication (Wholesale)* and the ANSI X9.17 standard, *Financial Institution Key Management (Wholesale)*. This algorithm is used to generate a value called a message authentication code (MAC).

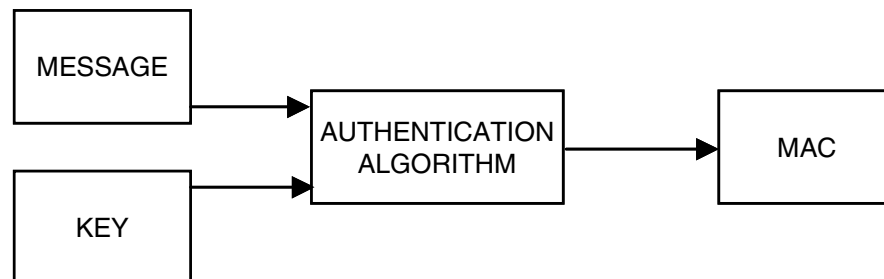
Like PIN encryption, message authentication also uses keys. While PIN encryption uses keys to encrypt and decrypt data, message authentication uses keys to encrypt the same data twice and compare the results. The MAC is part of the message that passes between a transaction originator and a transaction authorizer. The originator uses the key from its database and the information that is already in the message to generate a MAC and then adds the MAC to the message before sending it. The transaction recipient verifies the message by generating a second MAC based on the same message information used by the transaction originator, then comparing the MAC it generates with the MAC contained in the message. If the two MACs agree, the message can be accepted because it has not changed between the originator and the recipient. If the two MACs do not agree, the message should not be accepted because it has changed in some way between the originator and the recipient. A message can change as the result of tampering or because of communications difficulties. The MACs also do not agree if the keys are out of synchronization (i.e., the originator and recipient are not using the same key).

MACs can be generated and verified in messages moving between the BASE24 system and EFT devices attached to it as well as in messages moving between the BASE24 system and external processors. BASE24 support of message authentication is described in sections 5, 6, and 7 of this manual.

The following illustrations demonstrate message authentication. Before a message is sent, the message authentication key and selected fields from the message are used as input to the authentication algorithm to generate a MAC that is carried in the message.



When a message is received, the same message authentication key and fields from the message are used as input to the authentication algorithm to generate a MAC that is then compared to the MAC received in the message.



Card Verification

Card verification provides a means of confirming the validity of the card presented by a customer. BASE24 can perform card verification in software or with a security module.

Note: Dynamic card verification for contactless magnetic stripe cards, or proximity cards, requires the use of the Transaction Security Services process and is described in the ***BASE24 Transaction Security Services Processing Guide*** rather than in this manual.

Magnetic stripe cards or integrated circuit cards (ICCs), also referred to as chip cards or smart cards, can contain a value referred to as the card verification digits (CVD) in the magnetic stripe on the card. Europay, MasterCard, and Visa (EMV) cards used in the BASE24 system are chip cards. All EMV chip cards are currently *hybrid* cards, i.e. they contain both a magnetic stripe and a chip. The magnetic stripe contains a conventional CVD. The chip contains a data element known as the Track-2 Equivalent Data, which, as its name suggests, contains data similar to the Track 2 data on the magnetic stripe. One of the differences between the CVD in the magnetic stripe and the CVD on the chip, is that the CVD on the chip could have been generated on the chip using a service code value that is different from the actual service code from the track data. To distinguish the CVD obtained from the magnetic stripe from the CVD obtained from the chip, the CVD in the Track-2 Equivalent Data on the chip is referred to as the iCVD throughout this manual. The iCVD is designed to prevent someone from using the Track-2 Equivalent Data to create counterfeit Track 2 data on a magnetic stripe. If this is attempted, card verification should fail because the card issuer assumes that the service code from the magnetic stripe was used in the CVD generation, but the CVD was actually generated using a different service code.

The CVD can also be printed on the front of a card or on the signature panel on the back of a card for manual entry of card verification data when a magnetic stripe or chip card reader is unavailable.

The card issuer generates the CVD from two card verification keys plus the primary account number, card expiration date, and service code from the card. To verify a card, the BASE24 system uses the same card verification keys and magnetic stripe data read from the card or manually entered information to compute another CVD and then compares the computed CVD with the CVD read from the card or the CVD that was manually entered by the merchant (e.g, for the CVD printed on the card). The card is considered valid if the two CVDs match because the same card-read or printed card information and card verification keys are required to obtain the same CVD.

Note: When the iCVD is obtained from a chip card, card verification processing is the same as for a magnetic stripe card except that a service code value specified in the CV DATA field on Card Prefix File (CPF) screen 11 is used to compute the iCVD compared with the iCVD from the card.

For BASE24-pos, the card verification data can be manually entered by the card acquirer and/or electronically captured by swiping the card through a magnetic card reader. When this data is manually entered, BASE24 card verification processing is the same, except that a service code value of 000 is used to compute the CVD compared with the manually entered CVD and card verification can be performed once or twice with different date formats for the CVD. For American Express cards, card verification can only be performed once with the correct date format. The manual entry of card verification data allows BASE24-pos to detect and prevent the use of counterfeit cards when a magnetic stripe or chip card reader is unavailable, or as an additional security check even when a magnetic stripe reader is available. Both an electronically and manually acquired CVD can be verified in the same transaction. In this case, the electronically acquired CVD is referred to as CVD1 and the manually acquired CVD is referred to as CVD2.

The CVD serves a different purpose than the PIN. The card user need not be aware of the CVD. Unlike the PIN, which the customer must memorize and physically enter with each transaction, the card issuer places the CVD in either Track 1 or Track 2 of the magnetic stripe on the card and prints the CVD on the card. The issuer then uses the CVD each time the card is presented for a transaction. The CVD provides a method of guarding against a counterfeit or fraudulent card, while the PIN provides a method of identifying the customer using the card.

The following list presents the actual names used by Visa, MasterCard, American Express, and Europay, MasterCard, and Visa (EMV) for the generic CVD and iCVD terms. All terms refer to the same value. The American Express CSC can be a manually entered three- or four-digit value or a five-digit value obtained from the magnetic stripe. The MasterCard CVC2, Visa CVV2, and three- or four-digit American Express CSC are used for manual entry of the CVD and are supported only with BASE24-pos. The Visa iCVV and MasterCard Chip CVC are used for the three-digit value obtained from the chip on an EMV card.

Processor	Name Used	Abbreviation
Visa	Card Verification Value	CVV
Visa	Card Verification Value2	CVV2
MasterCard	Card Validation Code	CVC

Processor	Name Used	Abbreviation
MasterCard	Card Validation Code2	CVC2
American Express	Card Security Code	CSC
Visa EMV	Integrated Circuit Card Verification Value	iCVV
MasterCard EMV	Integrated Circuit Card Validation Code	Chip CVC

For Visa and MasterCard, the three-digit CVV2 or CVC2 number is printed on the signature panel on the back of the card immediately after the card's account number. For American Express, the three-digit CSC number is printed on the signature panel on the back of the card and the four-digit CSC number is printed on the front of the card above the card account number. The four-digit CSC location is currently occupied by the four-digit batch code (4DBC) or Card Identification Number (CID). The four-digit CSC is unembossed (flat) rather than embossed (raised).

BASE24 support for card verification is described in section 13 of this manual.

Secure Internet Validation

Secure Internet validation provides the means to validate Internet and electronic commerce (e-commerce) transactions for credit or debit cards (magnetic stripe and smart cards) when the ACI Commerce Gateway product interfaces with the BASE24-pos system. BASE24-pos supports cardholder payer authentication methods approved by Visa and MasterCard issuers for online purchases.

The Visa-approved authentication method for online purchases is called the Three-Domain Secure (3-D Secure) Electronic Commerce Verification Service (also referred to as Verified by Visa and Visa Payer Authentication). The authentication method for MasterCard is called Secure Payment Application (SPA) – Universal Cardholder Authentication Field (UCAF). BASE24-pos also supports an authentication method called Virtual PAN, which uses a virtual primary account number (PAN) associated with the actual PAN. The Virtual PAN method can be used with or without the SPA UCAF method. All of these services are intended to reduce risk, reduce fraud, decrease chargebacks, and increase consumer Internet purchases.

For the 3-D Secure method and the SPA UCAF method, an Access Control Server (ACS) associated with the card issuer cryptographically generates a value that can be used to authenticate the payer. For Visa, this value is called the Cardholder Authentication Verification Value (CAVV). For MasterCard, this value is called the Accountholder Authentication Value (AVV). The CAVV or AVV is returned to the online merchant and is included in the purchase transaction sent for authorization. For the Virtual PAN method, the ACS generates a virtual PAN the first time a card is used with the service and an expiration date each time the card is used with the service. This information is also returned to the online merchant.

BASE24-pos provides the means to validate the CAVV, AVV, and virtual PAN received in a purchase, preauthorization purchase, or preauthorization purchase completion transaction. The BASE24-pos support for secure Internet validation is provided in section 14.

Chip Authentication

Chip authentication provides a means of confirming the validity of a debit or credit card that contains an embedded microprocessor chip. Such cards are referred to as *integrated circuit cards* or *chip cards*. BASE24 supports chip authentication with a security module for integrated circuit cards adhering to specifications developed by Europay, MasterCard, and Visa (EMV) for interoperability across competing payment systems.

BASE24 provides chip authentication when the EMV add-on product is purchased for BASE24-atm or BASE24-pos. The EMV add-on product provides card authentication method (CAM) utilities to capture and verify security-related data held on the microchip on the credit or debit card. This data is known as the card authentication method (CAM) data.

The security module commands used for chip authentication in Atalla and Thales e-Security security modules are listed in appendix B and appendix C, respectively.

The configuration requirements and processing performed by BASE24 for the EMV card authentication methods are described in detail in the ***BASE24-atm EMV Support Manual*** and the ***BASE24-pos EMV Support Manual***.

Note: EMV card chip authentication should not be confused with chip authentication program (CAP) token validation, which is discussed below for the [“CAP Token Validation”](#) topic.

Offline PIN Management

EMV cards have the capability to hold the cardholder's PIN within the integrated circuit card (ICC). This allows for offline PIN verification (i.e., where the PIN entered by the cardholder is verified by the ICC) to be provided as an alternative cardholder verification method for electronic transactions. Because many payment terminals will not be enhanced to offer offline PIN verification, online PIN verification (i.e., where the PIN entered by the cardholder is verified by the issuer's system) will continue to be widely used as the cardholder verification method for EMV cards.

When both online and offline PIN verification are used as the cardholder verification method for an EMV card, cardholder confusion could result if the offline PIN value (i.e., the reference PIN value held on the ICC) differed from the online PIN value (i.e., the PIN offset or verification value held on the issuer's system). Also, if a cardholder exceeds the allowable bad PIN tries when entering the offline PIN, a mechanism is required to allow the bad PIN try counter to be reset. Therefore, it is critical that card issuers provide the following administrative functions for the management offline PIN verification:

- Support for offline PIN changes (i.e., the cardholder changes his or her PIN to a new value at an ATM). The new PIN must be transmitted to the card so that the PIN value on the card can be synchronized with the new PIN offset value in the issuer's database.
- Support for offline PIN unblocking. In this case, the PIN try counter on the EMV card has reached zero and needs to be reset to the normal value (i.e., the maximum PIN try limit). This is achieved through an issuer script command.

Both of these tasks are technically achieved by delivering a script message to the card during an authorization response message, although it is impractical for this to take place during a routine point-of-sale (POS) transaction. Instead, an ATM is the ideal secure terminal at which to offer this service.

To ensure that card issuers can successfully manage the offline PIN function for EMV cards, (i.e., ensuring that the offline PIN is synchronized with the online PIN known to the issuer), the BASE24-atm EMV add-on product provides an offline PIN management service.

The BASE24-atm EMV add-on product support for PIN management comprises both PIN change and PIN unblock (offline PIN try counter reset) functions. It also ensures that the issuer system is notified if the card cannot be updated with the PIN management information in the script message.

Card management systems may require knowledge of the new PIN when a PIN Change transaction is performed successfully. These systems may have to include the new PIN on the card if/when the card is re-issued, e.g. if it is lost or when it expires. BASE24-atm allows for a new PIN block encryption key on the Transaction Security Services (TSS) database to act as a transport key in the *zone* between the BASE24-atm system and the card management system. A key locator to this PIN block encryption key can be configured on the BASE24 database. Before writing a BASE24-atm Transaction Log File (TLF) record, the PIN block containing the new PIN is translated to encryption under the new key. The card management system also stores the new key on its database. Therefore, when it receives the new PIN block (e.g., in a TLF extract file), the card management system is able to decrypt the new PIN block.

Note: You must use the Transaction Security Services process with the BASE24-atm EMV add-on product to support offline PIN changes (i.e., the generation of PIN CHANGE scripts). The Transaction Security Services process is not required by issuers who want to support offline PIN unblocking only (i.e., the generation of PIN UNBLOCK scripts only), or who support offline PIN management transactions as an acquirer function only (i.e., where the transactions themselves are authorized on a back-end host). For more information on the offline PIN management function support provided by the Transaction Security Services process, refer to the *BASE24 Transaction Security Services Processing Guide*.

Refer to the *BASE24-atm EMV Support Manual* for further information on offline PIN management support provided by the BASE24-atm EMV add-on product.

Session Key Derivation Support

Through the use of the Transaction Security Services process, BASE24 supports the session key derivation function, which complies with EMV 2000 (v4.0) and EMV 2004 (v4.1). Examples of EMV implementations that use this method include Visa version 1.4.0 (VIS 1.4.0), MasterCard M/Chip version 4.0 (M/Chip 4.0), and common core definitions (CCD) processing standards. This method generates a unique session key for each ICC application transaction by generating a *tree* of keys. The Transaction Security Services process supports session key derivation for Atalla NSPs, Bull BNTng hardware security modules (HSMs), SafeNet (Eracom) Mark II security modules, and Thales e-Security host security modules (HSMs). For detailed information on the configuration requirements for session key derivation, refer to the *BASE24 Transaction Security Services Processing Guide*.

CAP Token Validation

Through the use of the Transaction Security Services process, BASE24 supports chip authentication program (CAP) token validation for MasterCards's OneSMART Authentication solution, Visa's Dynamic Passcode Authentication (DPA) solution, and the APACS Minimum Implementation of the MasterCard Chip Authentication Program Functional Architecture. CAP token validation provides two factor authentication (2FA) in card-not-present point-of-sale (POS) environments. Because two factor authentication uses two security items—the chip on the card and a PIN—it is more secure than using just chip or PIN verification. The Transaction Security Services process supports CAP token validation in Thales e-Security host security modules (HSMs). For detailed

information on the configuration requirements for CAP token validation, refer to the ***BASE24-pos EMV Support Guide*** and the ***BASE24 Transaction Security Services Processing Guide***.

Note: Because BASE24-pos uses card verification transactions for both secure internet validation and CAP token validation, only one type of validation can be implemented in a BASE24 system.

Security Modules

A number of vendors offer Tamper Resistant Security Modules (TRSMs) that are designed to perform PIN protection and verification, as well as other security-related functions within a physically secure environment. BASE24 currently supports the following security module models from three vendors—Atalla, Thales e-Security, SafeNet (Eracom), and Bull:

- Atalla Network Security Processor (NSP) models A8000, A8100, A8150, A9000, A9100, A9150, A10000E, A10100, A10150, and A10160

All Axx100, Axx150, and Axx160 models can run using the older variant key format or the Atalla Key Block (AKB) format. These devices can be configured to support the variant format or the AKB format simply by installing the appropriate CD-ROM in the device, rebooting, and re-establishing the security policy using the 105, 108, and 109 commands. All Axx100, Axx150, and Axx160 models come standard with an AKB software CD-ROM. Appending the letter V to the Axx100, Axx150, or Axx160 model number when ordered (e.g., A10160V, A10150V, A10160V, A10100V, A9100V, A9150V, A8100V, and A8150V) ensures that the module is shipped from the factory with the variant software CD-ROM as well. If the letter V is not appended to the model number when ordering, the Axx100, Axx150, or Axx160 model is shipped from the factory only with the standard AKB software CD-ROM. Also note that CD-ROMs for the Axx100 are not compatible with the Axx150/Axx160 or vice versa.

All other Atalla models can also be ordered with variant or AKB version software preloaded. While on-site upgrades from variant to AKB software are possible on the other Atalla models, this process is not as convenient as on the Axx100, Axx150, or Axx160 series.

Note: When using AKB software, the Transaction Security Services process is required. For more information on the Transaction Security Services process and configuration requirements for AKB software, refer to the *BASE24 Transaction Security Services Processing Guide*.

- Thales e-Security Host Security Module models RG7100 and RG7110, RG7300 and RG7310, RG7400, and HSM 8000

Thales e-Security Host Security Modules were formerly named Racal-Guardata Host Security Modules.

- SafeNet (Eracom) ProtectHost White Mark II host security module (HSM) and the ProtectHost White AMB HSM

BASE24 support and configuration requirements for SafeNet (Eracom) security devices are described in the ***BASE24 ERACOM Security Control Module (SCM) Reference Manual*** and the ***BASE24 ERACOM Security Module (ESM) Reference Manual***.

- Bull BNTng hardware security modules (HSMs)

Note: When using Bull BNTng HSMs, the Transaction Security Services process is required. For more information on the Transaction Security Services process and configuration requirements for Bull BNTng HSMs, refer to the ***BASE24 Transaction Security Services Processing Guide***.

The functions within these modules are proprietary. However, most of the principles of PIN encryption, PIN verification, message authentication, and card verification remain the same regardless of whether they are performed in hardware or software. The difference is that instead of performing the transaction security processing in software, BASE24 sends information to the security module and accepts the results.

The advantage of using a security module is that instead of BASE24 having the keys in the clear, which is a requirement for software encryption, the keys themselves can be encrypted under a key known only to the security module. The use of security modules is a requirement for compliance with American National Standards Institute (ANSI) TG-3 PIN protection and key management guidelines, as well as most card association and network security mandates.

Thus, BASE24 is only given access to encrypted versions of the keys; the security module itself is responsible for decrypting the keys it is given prior to processing. The use of security modules is required by most card associations and processing networks.

For general information on the functions supported by BASE24 for each type of security module and how security modules are used in the system, refer to section 17. For information on how devices must be configured for use with BASE24 systems, refer to either appendix B, for information regarding Atalla Security Modules, or appendix C, for information regarding Thales e-Security Host Security Modules.

Check Digits

Throughout the BASE24 database, many of the fields that can contain a hardware-encrypted key are accompanied by an optional four-character check digits field. For example, screen 2 of the Key File (KEYF), or screens 2 and 3 of the Key 6 File (KEY6), display eight fields containing keys used for PIN encryption or translation and message authentication. A check digits field appears next to each of these key fields.

When calculating an encrypted value, also known as a cryptogram, to be used as a key, security modules also assign check digits. These check digits can be used to verify the cryptogram was entered or generated correctly in the BASE24 database.

BASE24 uses check digits in message authentication performed by the Atalla Network Security Processor (NSP). When the MAC received with a message and the MAC generated by the BASE24 system do not match, the BASE24 system compares the check digits in the BASE24 database with the check digits returned by the NSP to determine the response code that the BASE24 system should return.

If the BASE24 and NSP check digits match, the BASE24 system returns an invalid MAC response code indicating the message cannot be authenticated because the data has been modified. If the BASE24 and NSP check digits do not match, the BASE24 system returns a key synchronization error response code indicating the message cannot be authenticated because the MAC key check digits in the BASE24 database do not match the check digits returned by the NSP.

A mismatch of the two check digits is known as a key synchronization error and indicates new keys are necessary. The Thales e-Security Host Security Module (HSM) does not return check digits to the BASE24 system with the MAC, so the HSM always returns an invalid MAC response code when the MAC received with a message and the MAC generated by BASE24 do not match.

Key Management

Keys play an important role in every transaction security process, whether it is data encryption, decryption, or translation; PIN verification; message authentication; card verification; or security device operation. BASE24 provides the tools listed below to assist in managing the keys used in these processes.

- Supercrypt utility
- Dynamic key management

Supercrypt Utility

ACI provides the Supercrypt utility for encrypting the inbound and outbound PIN encryption working keys when the external processor is encrypting PINs and the BASE24 system is carrying them in the clear.

Appendix A provides the step-by-step operating procedures for the Supercrypt utility.

BASE24-atm Device Dynamic Key Management

The BASE24-atm Dynamic Key Management add-on product enables you to automatically change transaction keys in ATM terminal based on a number of configurable thresholds. You can establish thresholds based on time, number of transactions, number of errors, number of consecutive MAC errors, or any combination of these factors. Both of the transaction keys used by ATM terminals (PIN key and MAC key) are automatically changed when any of the defined thresholds are surpassed.

When a threshold is exceeded for an ATM terminal, the BASE24 system generates new PIN and MAC keys for the terminal, downloads the new keys to the terminal, updates the keys in the BASE24 database, and resets all threshold counters for the terminal.

The BASE24-atm Dynamic Key Management add-on product and the procedures for configuring and maintaining it are discussed in section 15 of this manual.

BASE24-pos Device Dynamic Key Management

BASE24-pos dynamic key management enables you to automatically change transaction keys in a POS terminal based on a number of configurable thresholds. Thresholds can be established based on time, the number of transactions, number of errors, number of consecutive errors, or any combination of these factors. One or more working keys can be changed when any of the defined thresholds is surpassed. Thresholds are supported for PIN keys, MAC keys, and data encryption keys.

Currently, dynamic key management is supported for the SPDH Device Handler module only.

When a threshold is exceeded for one of the transaction keys in an SPDH-compliant terminal, the BASE24 system generates a new key for the terminal, downloads the new key to the terminal, updates the key in the BASE24 database, and resets all threshold counters for that type of key for the terminal.

BASE24-pos dynamic key management and the procedures for configuring and maintaining it are discussed in section 15 of this manual.

Note: Dynamic key management of data encryption keys is only supported when using the Transaction Security Services process.

Interface Dynamic Key Management

BASE24 Host Interface and Interchange Interface processes use working keys to protect the personal identification numbers (PINs) and message authentication codes (MACs) moving through the BASE24 network.

Working keys should be changed regularly to maintain proper security. The monitoring and changing of these keys is known as key management, and can be performed manually or automatically. Automated key management is often known as dynamic key management. BASE24 dynamic key management requires the use of security modules, and is not supported in interface processes if security modules are not configured.

Dynamic key management allows an external processor to automatically change working keys based on a number of factors. Thresholds can be established based on time, number of transactions, number of errors, number of consecutive errors, or any combination of these factors. Operator commands are also available for

changing working keys as required. In order for working keys to be changed automatically, another key, known as the key exchange key (KEK), is used to encrypt the working keys for distribution.

BASE24 dynamic key management is based on the ANSI X9.17-1985 standard, ***Financial Institution Key Management (Wholesale)*** and the ISO 8732 standard, ***Banking—Key Management (Wholesale)***.

A number of Interchange Interface processes provided by ACI include varying types of key management. Each interchange determines the type of key management it will use for transactions exchanged between itself and its external processor. ACI develops Interchange Interface processes specifically to meet the individual key management requirements of each interchange.

While interchanges dictate the functionality of the Interchange Interface processes ACI develops to allow BASE24 systems to communicate with them, ACI dictates the functionality of the following interface processes:

- BASE24 Interchange (BIC) ISO Interface process
- BASE24 ISO Host Interface process

BASE24 support of dynamic key management for these interface processes is described in section 16 of this manual.

2: BASE24 PIN Encryption Support

This section provides an introduction to the PIN encryption support BASE24 provides in the Host Interface, Interchange Interface, and Device Handler processes.

BASE24 PIN Encryption

BASE24 products provide PIN encryption support in its Host Interface, Interchange Interface, and Device Handler processes. Encryption refers to the process of rendering PINs unintelligible for transmission, and decryption refers to subsequently returning PINs to their original form. The whole process includes encrypting, decrypting, and translating PINs.

Within BASE24, PINs travel in the clear or encrypted under a secure key (i.e., a key that is in the clear only within a security module).

When a security module is being used by BASE24 for PIN encryption, PINs are never in the clear within BASE24. They are passed from process to process in their encrypted forms.

When a security module is not being used for PIN encryption, BASE24 decrypts PINs upon entry to the system, and the PINs remain in the clear until they are verified or sent to an external system.

Because BASE24 can be linked to a number of different external processing entities—all of which can have their own types of PIN encryption and decryption requirements—BASE24 must provide for a flexible interface between its own PIN handling and the PIN handling of these entities.

BASE24 does this by integrating certain PIN handling responsibilities into the specific processes designed to communicate with each of these entities.

The various external processing entities with which BASE24 interacts can be divided into three categories: hosts, interchanges, and EFT terminals. BASE24 provides a specific interface process for each, as follows:

- Communications with hosts are controlled by Host Interface processes.
- Communications with interchanges are controlled by Interchange Interface processes.
- Communications with EFT terminals are controlled by Device Handler processes.

These processes assume responsibility for the PIN encryption between BASE24 and their respective external entities.

PIN Blocks

During encryption, a PIN is not simply fed into the encryption routines as is. Rather, it is presented to the routines in what is called a PIN block.

A PIN block is a 16-digit series of hexadecimal characters, of which the PIN is a part.

The DES algorithm accepts only 8-byte or 16-hexadecimal-digit blocks of data for input, and since PINs can be from 4 to 12 digits in length, the PIN block type defines what to do with the gap between the PIN length and DES input requirements.

Thus, when encrypting a PIN, the routines are really encrypting a clear PIN block. Likewise, when decrypting a PIN, what is really being decrypted is an encrypted PIN block. This is true, regardless of whether the encryption or decryption is being performed in software by BASE24 or by a hardware module.

BASE24 supports three types of PIN blocks—PIN/PAN PIN blocks, PIN/PAD PIN blocks, and derived unique key per transaction (DUKPT) PIN blocks—all of which are commonly used throughout the industry.

Note: Whereas PIN/PAN PIN blocks and PIN/PAD PIN blocks are supported throughout BASE24 products, DUKPT PIN blocks are supported by the Standard POS Device Handler (SPDH) module and BASE24-pos Hypercom Device Handler (HPDH) module only with or without the Transaction Security Service process. Both the SPDH and HPDH modules translate DUKPT PIN blocks into PIN/PAN PIN blocks before they are sent to the Router/Authorization module.

PIN/PAN PIN Blocks

BASE24 supports three PIN/PAN PIN blocks, including one commonly known as the ANSI PIN block. The ANSI PIN block is the standard PIN block prescribed by the ANSI X9.8 standard, *American National Standard for Personal Identification Number (PIN) Management and Security*. ANSI PIN blocks also are called Format 0 PIN blocks by the ISO 9564 standard, *Banking—Personal Identification Number Management and Security*.

Within BASE24, references to an ANSI PIN block (e.g., ANSI PAN FORMAT or ANSI OFFSET field) actually refer to all three PIN/PAN PIN block formats, not just the ANSI standard. A particular PIN/PAN PIN block format is identified by stating which PAN digits are used to create the PIN block.

Building PIN/PAN PIN blocks is a multistep process. Two strings—a PIN string and a PAN string—must first be created. BASE24 uses the PIN, the PIN length, and the pad character F to create a PIN string. BASE24 then uses a subset of the PAN to create a PAN string. Finally, these strings are combined using an exclusive-OR operation to obtain the final PIN/PAN PIN block.

To build a PIN/PAN PIN block, BASE24 performs the following steps:

1. Builds a 16-character PIN string.
 - a. Places a zero in the first, left-most position.
 - b. Places a hexadecimal representation of the PIN length in the second position. (For example, if a PIN is 10 digits in length, the length is specified by the hexadecimal value A.)
 - c. Places the clear PIN in the string, starting in the third position.
 - d. Pads (fills in) the remaining positions to the right with the hexadecimal value F.

Example: A PIN of 8051 (4 digits in length) would result in a PIN string of 048051FFFFFFFFF.

2. Builds a 16-character PAN string.
 - a. Enters zeros in the first four positions, starting from the left.
 - b. Selects 12 digits from the PAN of the card involved in the transaction being processed.

The ANSI standard uses the right-most 12 digits of the PAN, excluding the check digit. Therefore, BASE24 always uses these 12 digits when forming PIN/PAN PIN blocks to use within the BASE24 system when external processors are not involved.

When building PIN blocks for external processors, BASE24 uses the value in the ANSI PAN FORMAT field on KEYF or KEY6 screen 1 to determine the 12 digits used by the external processor to form the PIN block. Valid values are as follows:

- 0 = Use the 12 right-most digits, excluding the check digit (ANSI standard).
- 1 = Use the 12 right-most digits, including the check digit.
- 2 = Use the 12 left-most digits.

If the PAN does not contain enough significant digits to provide the 12 digits specified by the ANSI standard or the value in the ANSI PAN FORMAT field on KEYF or KEY6 screen 1, BASE24 uses leading zeros to reformat the PAN used in creating the PAN string.

- c. Places the selected 12 digits from the PAN in the right-most positions of the PAN string.

Example: Using the 12 right-most digits, excluding the check digit, a PAN of 1040000533921277526 results in a PAN string of 0000053392127752.

3. Combines the PIN string and PAN string through an exclusive-OR operation.
 - a. Converts each hexadecimal digit in the PIN and PAN strings into its binary representation. As an example, a hexadecimal 2 has a binary representation of 0010 and a hexadecimal A has a binary representation of 1010.
 - b. Combines, using the exclusive-OR operation, the binary representation (four bits) for the first digit in the PIN string with the binary representation (four bits) for the first digit in the PAN string so that the result is still a binary representation. This exclusive-OR operation is also known as modulo 2 addition. For each bit, if the values are the same (both 0 or both 1), the resulting bit is 0, and if the values are different (one is 0 and the other is 1), the resulting bit is 1. As an example, 0010 plus 1010 equals 1000.

BASE24 then repeats this process for each digit of the two strings.

- c. Converts the binary representations obtained in step b back to their hexadecimal equivalents. As an example, the binary representation of 1000 converts to a hexadecimal 8.

The following is an example of the creation of a PIN/PAN PIN block using the 12 right-most PAN digits, excluding the check digit:

INPUTS:

PIN: 8 0 5 1

PAN: 1 0 4 0 0 0 0 5 3 3 9 2 1 2 7 7 5 2 6

STRINGS:

PIN STRING: 0 4 8 0 5 1 F F F F F F F F F F

PAN STRING: 0 0 0 0 0 5 3 3 9 2 1 2 7 7 5 2

RESULTING PIN BLOCK: 0 4 8 0 5 4 C C 6 D E D 8 8 A D

PIN/PAD PIN Blocks

PIN/PAD PIN blocks are another form of PIN block that can be used in PIN encryption.

The PIN and a hexadecimal pad character of A through F are used to build a PIN/PAD PIN block.

To build a PIN/PAD PIN block, BASE24 performs the following steps:

1. Places the PIN in the left-most positions.
2. Pads each of the remaining positions after the PIN with the hexadecimal pad character specified in the PIN PAD CHARACTER field on KEYF or KEY6 screen 1.

Note: The Thales e-Security security module restricts the pad character to hexadecimal F. For other security module pad character requirements, refer to the vendor documentation for the security module being used.

The following is an example of the creation of a PIN/PAD PIN block:

INPUTS:

PIN: **8 0 5 1**

PAD CHARACTER: **F**

RESULTING PIN BLOCK: **8 0 5 1 F F F F F F F F F F F**

DUKPT PIN Blocks

DUKPT PIN blocks are different from PIN/PAN and PIN/PAD PIN blocks because they are derived using a unique key for each transaction from a POS device rather than using the same master/session encryption key for all transactions sent from a POS device.

Using the DUKPT method, each terminal security module (TSM), or PIN pad, derives a unique key for each transaction. However, the TSM never contains any information that could lead to the determination of any past transaction keys used by the device, or of any past or future keys used by any other TSM. The originating TSM derives the current transaction key from an initial key, loaded into the TSM at initialization. When the BASE24 system receives the transaction, a

host security module (HSM) determines the current transaction key by using the host derivation key stored in the Derivation Key File (KEYD), and non-secret information contained in the transaction message.

Note: If you are using the Transaction Security Services process, the actual host derivation key is stored in the TSS Derivation Key File (DRVKD) and the derivation key in the KEYD is set to a key locator value. For more information on the DRVKD, refer to the ***BASE24 Transaction Security Services Processing Guide***.

Initialization

A host derivation key (KHD), also known as the base key, is used to generate a unique initial terminal key for each TSM. An HSM derives the initial terminal key by encrypting a key serial number (KSN). KSNs can vary in length from 12–20 bytes and consist of up to four fields, as described below:

KHD ID — The identifier of the KHD, and also known as the key set identifier. One KHD can be used for the key management of multiple terminals, and the KHD ID is used to select the KHD appropriate to a particular TSM. This field is mandatory.

Field Length: 5–9 hexadecimal characters

KHD SUB ID — The subkey identifier of the KHD. This field is reserved for future use and is currently set to 0. This field is optional.

TSM ID — The unique identifier of the TSM or PIN pad. Because the PIN pad packs the left-most bit of the transaction counter as the right-most bit of the TSM identifier, this field must always be even (i.e., the right-most bit must always be set to 0). This field is mandatory.

Field Length: 2–5 hexadecimal characters

Transaction Counter — A hexadecimal value, set to zero for the initial terminal key generation. Each of the five hexadecimal characters (0–9, and A–F) for the counter are represented by four bits as shown in the following table.

Bit Values	Character Represented
0000	0
0001	1
0010	2
0011	3
0100	4
0101	5
0110	6
0111	7
1000	8
1001	9
1010	A
1011	B
1100	C
1101	D
1110	E
1111	F

The TSM increases this counter by one with every transaction, unless the number of bits turned on in the hexadecimal representation for the next number would exceed 10. If so, the TSM skips that number until it reaches another number where the number of bits turned on in the hexadecimal representation for the number does not exceed 10. The counter continues to be increased in this way for each transaction until it reaches 1,000,000. At this time, the TSM must be reinitialized with a new initial terminal key.

Field Length: 5 hexadecimal characters

For Thales e-Security security modules, an additional field called the key descriptor is required by the HSM to encrypt the KSN. This field is described below. For SPDH modules, the key descriptor is transmitted in the message from the terminal. For HPDH modules, the key descriptor defined in the LCONF param POS-DH-KSN-DESCR is used. SPDH modules use the key descriptor defined in the POS-DH-KSN-DESCR param, only if it is not present in the message.

Key Descriptor — A three-character field that defines the length of the first three fields in the KSN as follows:

- Left character. KHD identifier length.
- Middle character. KHD subkey identifier length. Set to 0 if a subkey identifier is not used.
- Right character. TSM identifier length.

The initial terminal key generated by encrypting the KSN under the KHD is entered into the TSM. The KSN (and the key descriptor, if applicable) is also loaded into the TSM, but the KHD is not. At this time, the TSM generates a table of future (or *derived*) keys, by taking the KSN (with differing values of the transaction counter), and encrypting it under the initial key. Once a table of future keys has been created, the initial key is erased from the TSM.

Note that the initial terminal key is a function of the KHD, the KHD ID, and the TSM ID. Therefore, no two originating TSMs are given the same initial key, provided that no two originating TSMs with the same KHD ID have identical TSM IDs.

All KHDs used to generate initial terminal keys must be stored in the KEYD or DRVKD.

Originating TSM Processing

At the beginning of a new transaction, the TSM increments the transaction counter in the KSN and then uses the new counter value to select a derived key from a key storage table. The selected key becomes the current transaction key, and is erased from the key storage table on completion of the transaction. The current transaction key is used to encrypt the PIN into an ANSI PIN block as described earlier. The encrypted PIN block and the KSN are transmitted in the transaction message sent to the BASE24 system. The key descriptor is also transmitted in the message if you are using the SPDH module and Thales e-Security security modules.

Receiving HSM Processing

When the SPDH or HPDH module receives a transaction for a terminal using DUKPT encryption, it checks the KHD ID, KHD subkey ID, TSM ID and transaction counter received in the KSN in the message, performs optional DUKPT message authentication (for the SPDH module only using the Transaction Security Services process), and translates the DUKPT-encrypted PIN block into an ANSI PIN block.

KSN Validation. When the SPDH or HPDH module receives a transaction, it extracts the KSN and key descriptor (if applicable), and depending on whether this is the first transaction received from the terminal or a subsequent transaction, it performs one of the following:

- On the first transaction received from the terminal, the SPDH or HPDH module stores the KSN from the message into the device-dependent data of the BASE24-pos Terminal Data files (PTD).
- For all subsequent transactions received from the terminal, the SPDH or HPDH module compares the TSM ID in the KSN of the transaction to the TSM ID in the KSN stored in the device-dependent of the PTD. If the two values do not match, the SPDH or HPDH module performs the action indicated by the POS-DH-KSN-APPRV-OPT param in the LCONF and does not check the transaction counter. If the two values do match, the SPDH or HPDH module checks the transaction counter as described below.

The SPDH or HPDH module also ensures that the transaction counter in the KSN of the transaction is within an allowable range as follows:

1. Converts the transaction counters in both the KSN of the transaction and the KSN of the PTD to integers.
2. Checks the POS-DH-KSN-SAVE-OPT param in the LCONF to determine whether the KSN from the previous transaction is restored in the device-dependent area of the BASE24-pos Terminal Data files (PTD) when an out-of-sequence condition occurs (i.e., the module declines a transaction with a response code of 899).
3. Determines whether the transaction counter in the message is less than or equal to the transaction counter in the PTD. If so, the SPDH or HPDH module performs the action indicated by the POS-DH-KSN-CNTR-APPRV-OPT param in the LCONF. Otherwise, it continues with the next step.
4. Subtracts the transaction counter in the PTD from the transaction counter in the message. If the difference between the two counters is greater than the maximum difference allowed by the POS-DH-KSN-MAX-DIFFERENCE

param in the LCONF, the SPDH or HPDH module performs the action indicated by the POS-DH-KSN-CNTR-APPRV-OPT param in the LCONF. Otherwise, the SPDH or HPDH module sets the KSN and KSN descriptor (SPDH module only) in the device-dependent area of the PTD to the values from the KSN in the message and continues processing.

DUKPT Message Authentication. The SPDH module supports optional DUKPT message authentication while the HPDH module does not. The following paragraphs describe SPDH module processing for DUKPT message authentication.

The SPDH module determines whether the MAC TYPE field on PTD screen 7 is set to a value of 7 (DUKPT message authentication/PIN translation) or 8 (DUKPT message authentication/no PIN translation). If this field is set to a value of 8, the SPDH module calls the Transaction Security Services process to perform message authentication immediately. If this field is set to a value of 7, this processing is performed at the same time the PIN is translated. For DUKPT message authentication, both the MAC and the KSN are included in the message authentication calculation along with the derivation key instead of the MAC key. If DUKPT message authentication is to be performed, the SPDH module retrieves the KHD key locator value from the KEYD before calling the Transaction Security Services process. The SPDH module sends the following information to the Transaction Security Services process to verify the DUKPT MAC:

- The KHD key locator value from the KEYD
- The KSN from the request message
- The length of the KSN and the KSN key descriptor
- The data from the message to be included in the MAC calculation and the length of the data
- The MAC value from the request message

The Transaction Security Services process sends a command to the HSM to verify the MAC. If the calculated MAC does not match the MAC received in the message, the transaction is denied.

Before the transaction response is sent back to the terminal, the SPDH module calls the Transaction Security Services process again to generate a DUKPT MAC for the response message. In this case, the SPDH module sends the following information to the Transaction Security Services process to generate the DUKPT MAC:

- The KHD key locator value from the KEYD

- The KSN from the request message
- The length of the KSN and the KSN key descriptor
- The data from the message to be included in the MAC calculation and the length of the data

The Transaction Security Services process sends a command to the HSM to generate the MAC. The SPDH module includes the generated MAC in the response message sent back to the terminal.

DUKPT PIN Translation. If necessary, the SPDH or HPDH module retrieves the KHD or key locator value from the KEYD. If the key descriptor is needed, but it is not present in the message, the SPDH or HPDH module uses the value of the POS-DH-KSN-DESCR LCONF param.

The SPDH or HPDH module determines whether the PIN ENCRYPTION TYPE field on PTD screen 7 is set to a value of 7 (Derived unique key per transaction). If this field is set to a value of 7, the module retrieves the intermediate PIN encryption key or key locator value from the PIN MASTER KEY field on PTD screen 7. (If you are using the Transaction Security Services process, the actual intermediate PIN encryption key is retrieved from the Inbound Key field on the PIN Information tab of the Channel Security Configuration window of the ACI Desktop user interface.) If the PIN ENCRYPTION TYPE field is not set to a value of 7, the KSN is present in the transaction message, and the POS-DH-DUKPT-UPDATE-METHOD param is set to a value of Y, the module automatically generates an intermediate PIN encryption key using a key encryption key value of all zeroes, stores the key value in the PIN MASTER KEY field, and sets the PIN ENCRYPTION TYPE field to a value of 7. (If you are using the Transaction Security Services process, the Transaction Security Services process generates the intermediate PIN encryption key using the PIN key exchange key and stores the key value in the PIN Information Inbound Key field on the Channel Security Configuration window of the ACI Desktop user interface.)

If the MAC TYPE field on PTD screen 7 is set to a value of 7 (DUKPT message authentication/PIN translation), the SPDH module sends a request to the Transaction Security Services process to perform DUKPT message authentication verification and PIN translation. See the [“DUKPT Message Authentication”](#) topic on the previous page.

The SPDH or HPDH module then sends a request to the HSM or the Transaction Security Services process to translate the DUKPT PIN block into an ANSI PIN block.

The hardware security module encrypts the key serial number (with the transaction counter set to zero) using the selected host derivation key, resulting in the initial terminal key loaded into the originating TSM. It then uses the initial key and current transaction counter as inputs to a nonreversible transformation process, resulting in the current transaction key. Finally, it uses the current transaction key and the intermediate PIN encryption key to translate the DUKPT PIN block into an ANSI PIN block. The HSM decrypts the DUKPT PIN block using the current transaction key and then re-encrypts the result using the intermediate PIN encryption key into an ANSI PIN block. The ANSI PIN block and the intermediate PIN encryption key are then placed in the PSTM for standard processing (i.e., PIN verification by the Router/Authorization module or PIN translation and routing to an interchange or host).

Future Key Generation

Once a transaction is completed, the TSM generates some number of future keys (sometimes none, sometimes one or more) by transforming the current transaction key as a function of varying values of the transaction counter. The TSM places these newly generated future keys into key storage and erases the current transaction key.

The nonreversible transformation process performed by the TSM is different but related to the nonreversible transformation process performed by the HSM. Future keys are generated, stored, and selected at the originating TSM in a manner such that the receiving HSM is able to determine the current transaction key.

Changing the TSM in an SPDH-compliant or Hypercom Terminal

When a new terminal security module (TSM), or PIN pad, is installed in an SPDH-compliant or Hypercom terminal using the DUKPT method of PIN encryption, you must update the static KSN data stored in device dependent data of the PTD if the POS-DH-KSN-APPRV-OPT param in the LCONF is set to a value of 1 or 2. You can do this using either of the following procedures, depending on whether batch, shift, and day totals have been cleared from the PTD for the terminal.

If batch, shift, and day totals have not been cleared from the PTD for the terminal, perform the following procedure when changing the TSM.

1. Set the POS-DH-KSN-APPRV-OPT param in the LCONF to a value of 3.

2. Warmboot the Device Handler/Router/Authorization process that contains the SPDH or HPDH module for the terminal.
3. Change the TSM in the SPDH-compliant or Hypercom device and send a transaction from the terminal that contains the Key Serial Number (KSN) and Descriptor field (i.e., FID 6, subFID T) for an SPDH-compliant device, or the DUKPT Key Serial Number field (field 33) for a Hypercom device, in a request to the BASE24 system.

The SPDH or HPDH module compares the TSM ID received in the transaction request to the TSM ID stored in the KSN static data maintained in PTD device dependent data. Because the values no longer match, it performs the action specified by the POS-DH-KSN-APPRV-OPT param, which in this case, is to update the KSN static data in the PTD with the KSN static data from the message. Once the KSN static data is updated in PTD device dependent data, the TSM ID should match that received from the terminal.

4. Change the value of the POS-DH-KSN-APPRV-OPT param in the LCONF back to its previous value of 1 or 2.
5. Warmboot the Device Handler/Router/Authorization process that contains the SPDH or HPDH module for the terminal.

If batch, shift, and day totals have been cleared from the PTD for the terminal, perform the following procedure when changing the TSM.

1. Change the TSM in the SPDH-compliant or Hypercom device.
2. Log on to the BASE24 user access system.
3. On BASE24-pos Terminal data files (PTD) screen 1, read the PTD record for the terminal.
4. Delete the terminal record. The device dependent data for the terminal is deleted from the PTD.
5. Without changing any field values displayed on the screen, immediately add the terminal record back.

When the SPDH or HPDH module receives the next transaction message from the terminal that contains the Key Serial Number (KSN) and Descriptor field (i.e., FID 6, subFID T) for an SPDH-compliant device, or the DUKPT Key Serial Number field (field 33) for a Hypercom device, it stores the KSN from the message into the device-dependent data of the BASE24-pos Terminal Data files (PTD). Because the record was just added, the SPDH or HPDH module regards this transaction as the first transaction received from the terminal.

Thales e-Security Host Security Module Requirements

Special requirements exist for PIN blocks in configurations using the Thales e-Security Host Security Module (HSM). These requirements, explained below, affect translations between two PIN/PAN PIN blocks, the IBM DES (3624-format) PIN verification method, and the Visa PVV PIN verification method.

PIN Translations

PIN translation using a HSM requires that the same PAN digits used to create the input PIN block in a PIN/PAN-to-PIN/PAN translation are also used to create the output PIN block. This requirement does not apply to translations between a PIN/PAN PIN block and a PIN/PAD PIN block.

As an example, BASE24 receives a transaction from an interchange and the transaction has a PIN/PAN PIN block created using a PAN string containing the 12 right-most digits, excluding the check digit. In this case, the HSM can translate the PIN block into a PIN/PAD PIN block or another PIN/PAN PIN block that uses a PAN string containing the 12 right-most digits, excluding the check digit. The HSM cannot translate the PIN block into another PIN/PAN PIN block using a PAN string containing the 12 left-most digits or one using a PAN string containing the 12 right-most digits, including the check digit.

Refer to sections 3 and 4 for the specific BASE24 file maintenance settings to meet this requirement.

IBM DES (3624-Format) PIN Verification

IBM DES (3624-format) PIN verification using a HSM requires that the validation data include the five right-most positions of the PAN, excluding the check digit. Therefore, the PAN VERIFY OFFSET and PAN VERIFY LENGTH fields in the KEYA must be set appropriately to capture this portion of the PAN.

Refer to section 9 for the BASE24 file maintenance settings for this verification method.

Visa PVV PIN Verification

Visa PIN verification value (PVV) method of PIN verification using a HSM requires that external processors use PIN/PAD PIN blocks or ANSI X9.8 standard PIN/PAN PIN blocks in the messages sent to BASE24. The Visa PVV PIN

verification method uses the right-most digits of the PAN excluding the check digit. The ANSI X9.8 standard PIN/PAN PIN block uses the 12 right-most digits of the PAN excluding the check digit. The HSM returns a PIN block error if the external processors use a PIN/PAN PIN block other than the ANSI X9.8 standard.

This requirement does not affect any BASE24 file maintenance settings. The external processor must be instructed to use either PIN/PAD PIN blocks or ANSI X9.8 PIN/PAN PIN blocks.

For more information on Visa PVV PIN verification, refer to section 12.

BASE24 Internal Messages

Transactions move through BASE24 systems in the form of standard internal messages, which are passed among the BASE24 processes. All information relating to PIN encryption, and the key under which the PIN is encrypted, is carried in the internal message.

When a message is received from an external source, BASE24 converts that message from the native format of the external source to a standard internal format recognizable to the BASE24 processes. The standard internal message formats used by BASE24-atm (Standard Internal Message, STM), BASE24-pos (POS Standard Internal Message, PSTM), and BASE24-teller (Teller Standard Internal Message, TSTM) are not identical; however, they do contain many of the same fields.

When BASE24 receives a transaction from an external source, the receiving process creates the internal message, performs whatever actions are required on the PIN (decryption, encryption, translation, or no action), and updates the PIN-related fields in the internal message accordingly. Any process subsequently receiving the message can then determine from the internal message the form in which the PIN is being carried.

When a message is being sent out from BASE24, the process responsible for creating the external message determines the form the PIN is in from the fields in the internal message and performs any necessary actions (decryption, encryption, or translation) based on the current form of the PIN and the needs of the external entity.

Internal messages contain the following information for PIN encryption. Each description includes the name of the STM, PSTM, and TSTM field that contains the information.

ANSI Offset — Indicates the starting position in the PAN of the 12 digits used to create the PIN/PAN PIN block if that type of PIN block was used in encrypting the PIN. This value can be greater than, less than, or equal to 0. (A 0 indicates the first position of the PAN.)

BASE24-atm and BASE24-pos use the 12 right-most digits of the PAN, excluding the check digit to create the PAN string. BASE24-teller uses the value of the ANSI PAN FORMAT field in the Teller Terminal Data File (TTDF) to determine which PAN digits are used to create the PAN string.

Example: A PAN of 1049000451234534567 would result in a PAN string of 045123453456 and an ANSI offset of 6 for BASE24-atm and BASE24-pos. BASE24-teller would obtain the same PAN string and ANSI offset values if the ANSI PAN FORMAT field in the TTDF is set to 0 (12 right-most digits excluding check digit). BASE24-teller would obtain a PAN string of 451234534567 and an ANSI offset of 7 if the ANSI PAN FORMAT field in the TTDF is set to 1 (12 right-most digits including check digit).

If the PAN does not contain enough significant digits to provide the 12 digits necessary (i.e., the PAN is less than 13 digits in length), BASE24 adds leading zeros to the PAN used in creating the PIN block and sets the ANSI offset to a negative value. A negative ANSI offset identifies the number of leading zeros BASE24 has added to the PAN after right-justifying the available significant digits.

Example: A PAN of 1234534567 would result in a PAN string of 000123453456 and an ANSI offset of -3 for BASE24-atm and BASE24-pos. BASE24-teller would obtain the same PAN string and ANSI offset values if the ANSI PAN FORMAT field in the TTDF is set to 0 (12 right-most digits excluding check digit). BASE24-teller would obtain a PAN string of 001234534567 and an ANSI offset of -2 if the ANSI PAN FORMAT field in the TTDF is set to 1 (12 right-most digits including check digit).

STM Field:	RQST.ANSI-OFST
PSTM Field:	ANSI-OFST
TSTM Field:	PIN-TKN.ANSI-OFST

Error Flag — Provides additional information describing processing errors. Each BASE24 Authorization, Device Handler, Host Interface, and Interchange Interface process can be configured to communicate with a security device to perform PIN management. The following errors can occur when BASE24 communicates with a security module to process PINs:

- **Sanity check error or PIN block error.** The PIN block, as decrypted by the security device, does not contain valid data. A sanity check error or PIN block error can result when two parties sharing a PIN encryption key lose key synchronization. For example, if a valid PIN block is encrypted with a DES DEA key, but the security module is given a different key to decrypt the PIN block, a sanity check error or PIN block error results when the security device attempts to extract the clear PIN from the decrypted PIN block. Atalla describes this error as a sanity check error. Thales e-Security describes this error as a PIN block error.
- **Security device failure.** The BASE24 process could not communicate with the security device (e.g., timeout).

- Invalid PIN processing data or parameters. The BASE24 process responsible for communicating with the security device received incomplete or inaccurate information (e.g., configuration error).

Depending on the BASE24 process, a single response code in the internal message can identify a transaction that has been declined for a security device failure, an error in PIN processing data or parameters, or a sanity check error. External processors want to distinguish between a sanity check error, also known as a key synchronization error, and the other errors so they can issue new PIN encryption keys.

STM Field: AT50-TKN.ERR-FLG
PSTM Field: PS50-TKN.ERR-FLG
TSTM Field: TSTMH.RESP-HDR.ERR-FLG

PIN — Contains the PIN in the clear or in encrypted form. If the PIN is carried in encrypted form, it is really the PIN block that is carried in the message. If the PIN is in the clear, it is left-justified and blank filled to the right.

STM Field: RQST.PIN
PSTM Field: PIN
TSTM Field: PIN-TKN.PIN

PIN Format — Indicates whether the PIN is in the clear or encrypted. If the PIN is encrypted, this field also indicates the type of PIN block used in the encryption, PIN/PAN PIN block or PIN/PAD PIN block.

STM Field: RQST.PIN-FRMT
PSTM Field: PIN-FRMT
TSTM Field: PIN-TKN.PIN-FRMT

PIN Key — The encrypted form of the key under which the PIN is currently encrypted, if the PIN is encrypted.

PINs travel through BASE24 in the clear or encrypted under a secure key. This field carries the secure key.

STM Field: RQST.PIN-KEY
PSTM Field: PIN-KEY
TSTM Field: PIN-TKN.PIN-KEY

PIN Pad Character — Indicates the PIN pad character used to create the PIN/PAD PIN block, if that type of PIN block was used in encrypting the PIN.

STM Field: RQST.PINPAD-CHAR
PSTM Field: PINPAD-CHAR
TSTM Field: PIN-TKN.PIN-PAD-CHAR

PIN Size — Contains the size of the PIN, if the PIN is in the clear. If the PIN is being carried in encrypted form, the value in this field is set to 16 (the size of the PIN block).

STM Field: RQST.PIN-SIZE
PSTM Field: PIN-SIZE
TSTM Field: PIN-TKN.PIN-SIZE

Previous Key — The encrypted form of the previous key used by an external source to encrypt PINs. After a key exchange has taken place between BASE24 and an external source, the interface process saves the previous key and places it and the new key in the BASE24 internal message. This allows for situations in which PINs are subsequently sent from the external source encrypted under the previous key.

Some interface processes use the CLEAR OLD KEY TIMER VALUE field on Key File (KEYF) screen 3 or Key 6 File (KEY6) screen 4 to control how long the previous key can be used.

STM Field: RQST.USER-KEY
PSTM Field: USER-KEY
TSTM Field: PIN-TKN.USER-KEY

Originator — When using Thales e-Security HSMs, the security utilities use the value of this field to determine whether terminal or interchange commands for the correct LMK pair are sent to the security device. For example, if this field is set to a value of 1, indicating the transaction originated from a device controlled by BASE24, the security utilities use terminal commands. If this field is set to a value of 7, indicating that the transaction originated from an interchange, the security utilities use interchange commands.

When a transaction originates from a POS device using the derived unique key per transaction (DUKPT) encryption method and Thales e-Security host security modules are used, an additional value is required. The Standard POS Device

Handler (SPDH) or BASE24-pos Hypercom Device Handler (HPDH) module sets this field to a value of 9 when an incoming request contains a DUKPT-encrypted PIN block and the SECURE-DEV-TYP param is set to a value of RSM or VSM. A value of 9 in this field ensures that the security utilities build the correct terminal command in the call to a Thales e-Security HSM device when translating a DUKPT-encrypted PIN block into an ANSI PIN block encrypted under a master/session key. For this value, the security utilities append the single-length input encryption key and the output encryption key with a key scheme tag indicating that the key is double length

PSTM Field: ORIGINATOR

3: BASE24 Interface PIN Encryption Support

This section presents a detailed discussion of the PIN encryption support provided by BASE24 interface processes.

In reviewing the information presented in this section, note that the term *external processor* can refer to a data processing center (DPC), an interchange, or a co-network as defined in the “Conventions Used in this Manual” section of this manual.

BASE24 Interface Processes

The BASE24 ISO Host Interface and Interchange Interface processes handle PIN encryption between BASE24 systems and their respective external processors. In the case of the ISO Host Interface process, the external processor is a data processing center (DPC). In the case of an Interchange Interface process, the external processor is an interchange.

External processors are afforded identical options by BASE24 for PIN encryption, and the BASE24 ISO Host Interface processes and Interchange Interface processes approach PIN encryption in the same way. Therefore, their PIN handling capabilities and options can be described simultaneously. The only difference is that interchanges set the rules to which BASE24 conforms, while host DPCs are more flexible and can make use of the options offered by BASE24.

BASE24 supports a variety of interface processes, with each process having its own specific PIN encryption requirements. The following documentation includes the PIN encryption options provided by BASE24 in general. The BASE24 options selected for each interface process are based on a combination of interface process capabilities and user preferences.

BASE24 interface processes provide external messages that are based on ISO message formats. The BASE24 ISO Host Interface and BASE24 Interchange (BIC) ISO Interface processes use the ISO format, which is based on the 1987 ISO 8583 standard, *Bank Card Originated Messages—Interchange Message Specifications*.

This section applies to BASE24-atm, BASE24-pos, and BASE24-teller. The BASE24 ISO Host Interface process supports all three BASE24 products. However, interchange interface processes do not support BASE24-teller.

Note: The external processor for a BASE24 Interchange (BIC) Interface process is often called a co-network instead of an interchange. In this section, all references to interchanges include BIC co-networks.

External Processor Keys

BASE24 identifies five types of keys for use in handling PINs between BASE24 and an external processor:

- Inbound encryption keys
- Outbound encryption keys
- Intermediate keys
- Previous inbound and outbound keys
- PIN key exchange keys

In most cases, the keys required by BASE24 must be physically placed in the BASE24 database through file maintenance. BASE24 utility programs are provided to assist in the preparation of this information. The key fields in the database can be left with their default values (zeros) if keys are provided by an automatic key change when the interface process is started; however, the PIN key exchange key, if used, must be entered manually.

Each interchange determines the type of key management to be used for transactions exchanged between that interchange and BASE24. ACI develops the BASE24 interface processes for each of these interchanges to meet its key management requirements. Dynamic key management, if allowed, can include inbound encryption keys, outbound encryption keys, or both. While interchanges dictate the functionality of the interface processes ACI develops for them, ACI dictates the functionality of the following interface processes:

- BASE24 Interchange (BIC) ISO Interface process
- BASE24 ISO Host Interface process

Both of these interface processes offer dynamic key management for PIN encryption keys. Refer to section 16 for additional information about dynamic key management for interface processes.

Inbound Encryption Keys

An inbound encryption key is the working key BASE24 uses for dealing with incoming PINs from an external processor. If a PIN is encrypted coming from the external processor, this is the key under which it is encrypted. If a PIN is in the clear coming from the external processor and must be encrypted by BASE24, this is the key under which it is to be encrypted.

When encryption is performed by a security module, the value entered in the BASE24 database is a cryptogram (i.e., encrypted version) of the inbound encryption key. This cryptogram is obtained using the Atalla master file key (MFK) or the Thales e-Security local master key (LMK). If the external processor provides a cryptogram of the key, this cryptogram is encrypted under the PIN key exchange key (KEK). The security module translates the inbound encryption key from encryption under the KEK to encryption under the MFK or LMK. The clear version of the inbound encryption key does not appear in the BASE24 database.

Partial Triple DES

If the interface supports triple DES-encrypted PINs but the BASE24 system does not, the triple DES-encrypted PIN block is translated into a single DES-encrypted PIN block. This special translation is performed only when both the external PIN and the internal PIN are encrypted in hardware.

A process-specific LCONF param, DES-TRIPLE-SINGLE, indicates whether an Interchange Interface process translates PINs from triple DES encryption to single DES encryption when received from an interchange.

If the DES-TRIPLE-SINGLE param is set to a value of Y, the Interchange Interface process uses the Key 6 File (KEY6) instead of the Key File (KEYF). The KEY6 supports double-length PIN encryption keys, which are required for triple DES encryption.

When an Interchange Interface process receives a transaction from the interchange and encryption is performed in hardware, the process checks the value of the DES-TRIPLE-SINGLE param. If the param is set to a value of Y, the process calls the security utilities to translate the PIN. The process passes the double-length KPE and the single-length encrypted intermediate key from the KEY6 along with the triple DES-encrypted PIN block to the HSM. The HSM translates the PIN block from encryption under the double-length KPE to encryption under the single-length intermediate key. If PINs are handled in the clear internally, the security utilities bound into the Interchange Interface process decrypt the PIN block in software using the clear intermediate key from the KEY6. The clear PIN block is then placed in the STM/PSTM sent to the BASE24 Authorization process. If PINs are encrypted internally, the translated PIN and the encrypted intermediate key are placed in the STM/PSTM that is then sent to the BASE24 Authorization process.

Each interchange that supports triple DES and dynamic key management sends a double length PIN encryption key that is stored in the KEY6. This key is sent either in a new field or in an expanded existing key field, as determined by each interchange. The following Interchange Interface processes support partial triple DES:

Banknet Interchange Interface process
BIC ISO Interchange Interface process
EPS-Net Interchange Interface process
ISO Host Interface process
LINK ISO Interchange Interface process
MDS ISO Interchange Interface process
NBGC Interchange Interface process
VisaNet ISO Interchange Interface process

Outbound Encryption Keys

An outbound encryption key is the working key under which a PIN being sent to an external processor must be encrypted, if encryption is required by the external processor.

When encryption is performed by a security module, the value entered in the BASE24 database is a cryptogram (i.e., encrypted version) of the outbound encryption key. This cryptogram is obtained using the Atalla MFK or the Thales e-Security LMK. If the external processor provides a cryptogram of the key, this cryptogram is encrypted under the KEK. The security module translates the outbound encryption key from encryption under the KEK to encryption under the MFK or LMK. The clear version of the outbound encryption key does not appear in the BASE24 database.

Partial Triple DES

If the interface supports triple DES-encrypted PINs but the BASE24 system does not, the single DES-encrypted PIN block is translated into a triple DES-encrypted PIN block. This special translation is performed for the following scenarios:

- Internal clear PIN to external PIN encrypted in hardware
- Internal PIN encrypted in hardware to external PIN encrypted in hardware

A process-specific LCONF param, DES-TRIPLE-SINGLE, indicates whether an Interchange Interface process translates PINs from single DES encryption to triple DES encryption when sent to the interchange.

If the DES-TRIPLE-SINGLE param is set to a value of Y, the Interchange Interface process uses the Key 6 File (KEY6) instead of the Key File (KEYF). The KEY6 supports double-length PIN encryption keys, which are required for triple DES encryption.

When an Interchange Interface process sends a transaction to an interchange, the process checks the value of the DES-TRIPLE-SINGLE param. If the param is set to a value of Y, the process calls the security utilities to translate the PIN. The process passes the double length KPE and the single length encrypted intermediate key from the KEY6 along with the single DES-encrypted PIN block to the HSM. The HSM decrypts the PIN using the single length intermediate key and then re-encrypts it under the double-length KPE. The translated PIN and the double length KPE are then placed in the external message that is then sent to the interchange.

Each interchange that supports triple DES and dynamic key management sends a double length PIN encryption key that is stored in the KEY6. This key is sent either in a new field or in an expanded existing key field, as determined by each interchange. The following Interchange Interface processes support partial triple DES:

- Banknet Interchange Interface process
- BIC ISO Interchange Interface process
- EPS-Net Interchange Interface process
- ISO Host Interface process
- LINK ISO Interchange Interface process
- MDS ISO Interchange Interface process
- NBGC Interchange Interface process
- VisaNet ISO Interchange Interface process

Intermediate Keys

An intermediate key is a single-length key for which both the clear and encrypted versions are available to BASE24 (the encrypted version being encrypted under the key of a security module). BASE24 does not support double- or triple-length intermediate keys.

BASE24 uses intermediate keys to decrypt PINs that are encrypted under keys managed by a security module. BASE24 also uses intermediate keys to take PINs that are received in the clear and encrypt them under a key that is managed by a security module and to translate triple DES-encrypted PINs into single DES-encrypted PINs.

When an encrypted PIN must be brought into the clear for processing or for sending to an external processor, BASE24 can have the security module re-encrypt the PIN under the intermediate key using the encrypted version, allowing BASE24 to decrypt the PIN in software using the clear version of the key.

When a PIN that is received in the clear must be encrypted for processing by a security module, BASE24 can encrypt the PIN in software using the clear version of the key, allowing the security module to translate the PIN using the encrypted version of the key.

Previous Keys

After a key change has taken place between BASE24 and an external processor, PINs can be sent between BASE24 and the external processor encrypted under the new key or the previous key. Because the external processor does not specify the key under which a PIN is encrypted, the interface process saves the previous key and places it in the BASE24 internal message along with the new key.

During BASE24 PIN verification, the BASE24-atm, BASE24-pos, or BASE24-teller Authorization process or module sends both encryption keys to a security module before allowing the transaction to be declined due to a sanity check error (Atalla security modules only) or PIN block error (Thales e-Security security modules only). The Authorization process or module tries the new key first, and if a sanity check error or PIN block error occurs, tries to use the previous key.

When translating PINs, interface processes send both encryption keys to a security module before allowing a transaction to be declined due to a sanity check error or PIN block error. The interface processes try the new key first, and if a sanity check error or PIN block error occurs, try to use the previous key. Interface processes can translate PINs in the following situations:

- Incoming messages, when internal messages carry clear PINs and external messages carry encrypted PINs.
- Outgoing messages, when internal messages carry encrypted PINs and external messages carry clear PINs.

- Outgoing messages, when both internal and external messages carry encrypted PINs.

Some interface processes use the CLEAR OLD KEY TIMER VALUE field on KEYF screen 3 or on KEY6 screen 4 to control how long the previous key can be used.

Note: Only certain external processors support the use of a previous key.

PIN Key Exchange Keys

A PIN key exchange key (KEK) is the key BASE24 uses to encrypt or decrypt inbound and outbound PIN encryption keys when BASE24 and the external processor use dynamic key management to automatically change the inbound and outbound PIN encryption keys.

A KEK can be single length (16 hexadecimal characters), double length (32 hexadecimal characters), or triple length (48 hexadecimal characters), however, it must be of equal or greater length than the PIN encryption key. Using double-length and triple-length KEKs provides additional security by increasing the difficulty of using trial and error to obtain the clear value of the working keys, with triple-length KEKs providing the most security. The KEKs used for encrypting PIN encryption working keys and message authentication working keys both must be the same length, whether single, double, or triple.

Note: Triple-length KEKs are only supported when implementing the Transaction Security Services process. For more information on the Transaction Security Services process, refer to the *BASE24 Transaction Security Services Processing Guide*.

Dynamic Key Management

Dynamic key management allows BASE24 and an external processor to change working keys based on time, number of transactions, number of errors, number of consecutive errors, or any combination of these factors.

The following Interchange Interface processes support dynamic key management:

- Banknet Interchange Interface process
- BASE24 Interchange (BIC) ISO Interface process
- Generic Interchange Interface process
- ISO Host Interface process
- LINK ISO Interchange Interface process

- MDS ISO Interchange Interface process
- PLUS ISO Interchange Interface process
- VisaNet ISO Interchange Interface process

Refer to interchange-specific documentation to determine whether other Interchange Interface processes support dynamic key management.

Dynamic key management requires the use of security modules and is not supported in software. Supported hardware modules include the following:

- Atalla Network Security Processor
- Thales e-Security Host Security Module

For detailed information regarding hardware support, refer to the appropriate security module vendor manual. Refer to section 16 for additional information on BASE24 dynamic key management for interface processes.

PIN Encryption Options

An external processor can use the following three PIN encryption options:

- Clear PIN processing. All PINs carried in the external message passed between BASE24 and the external processor are in the clear.
- Encrypted PINs—security module accessed for PIN processing. All PINs carried in the external message passed between BASE24 and the external processor are encrypted under secure keys using the single DES or triple DES algorithms.
- Encrypted PINs—software DES PIN processing. All PINs carried in the external message passed between BASE24 and the external processor are encrypted using software, using the single DES algorithm. No security module is in use.

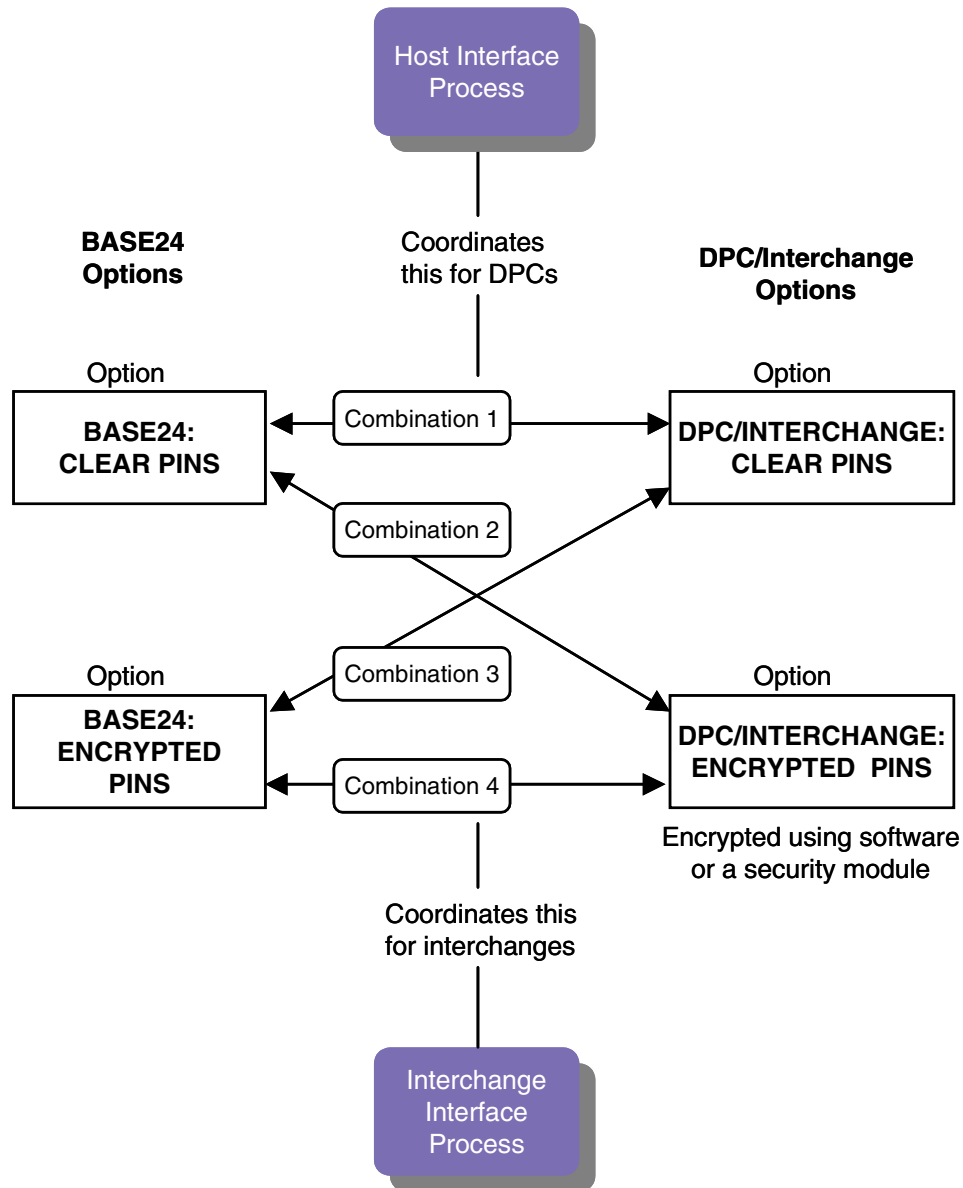
BASE24 can use the following two PIN encryption options internally:

- Clear PIN processing. All PINs carried in the BASE24 internal message are in the clear.
- Encrypted PINs—security module accessed for PIN processing. All PINs carried in the BASE24 internal message are encrypted under secure keys using the single DES algorithm.

Note: The Transaction Security Services process is required to carry triple DES-encrypted PINs in the BASE24 internal message. The Transaction Security Services process is not required if you only need to translate PINs from single DES to triple DES, and vice versa, between the BASE24 system and select interchanges. For more information on the Transaction Security Services process, refer to the ***BASE24 Transaction Security Services Processing Guide***.

The three external processor options and the two BASE24 options offer the following combinations, as illustrated on the following page:

1. Internal and external messages carry clear PINs.
2. Internal messages carry clear PINs and external messages carry PINs encrypted in software or under secure keys.
3. Internal messages carry PINs encrypted under secure keys and external messages carry clear PINs.
4. Internal and external messages carry PINs encrypted under secure keys.



Note: Software security can be used instead of a security module for encrypting PINs in the external messages when internal messages carry PINs encrypted under a secure key. However, ACI recommends hardware security for the following reasons:

- Security modules that are already in place for internal message PIN management can also be used for external message PIN management.

- A higher level of security exists because keys are controlled within security modules and intermediate keys are not required.
- Transaction processing is more efficient because PIN management involves less data manipulation.

BASE24 allow for each of these combinations, and each combination requires its own file settings in the BASE24 database. This section discusses each of these combinations and describes how the BASE24 database must be set up for each.

When a transaction is received from an external processor, BASE24 may be required to move the PIN from one type of encryption to another (e.g., from triple DES to single DES) for use internally by the system. Similarly, when a transaction is sent to an external processor, BASE24 may be required to move the PIN from one type of encryption to another (e.g., from single DES to triple DES) prior to sending it.

BASE24 and External Processor Carry Clear PINs

If PINs are carried in clear form by both BASE24 and an external processor, no special PIN handling is required.

Processing Outgoing Messages

In the case of outgoing messages, the BASE24 interface process sends the clear PIN to the external processor.

Processing Incoming Messages

In the case of incoming messages, the BASE24 interface process accepts the PIN sent by the external processor as the clear PIN for the transaction.

Setting Up the BASE24 Database

Fields on screen 1 of the Key File (KEYF) record for each external processor are used when both the external processor and BASE24 carry PINs in clear form. Fields pertain to inbound and outbound keys. Fields on KEYF screen 3 deal exclusively with dynamic key management. Screens 2 and 3 are not used when all PINs are carried in clear form. Refer to the ***BASE24 Base Files Maintenance Manual*** for additional information about these screens.

The following fields on KEYF screen 1 must be set as indicated when both the external processor and BASE24 carry PINs in clear form.

ENCRYPT TYPE — Specifies the type of PIN encryption used or expected by the external processor. This field must be set to a value of 0 (clear PINs) to indicate that PINs are carried in the clear by the external processor.

BASE24 ENCRYPT TYPE — Specifies the type of PIN management used by BASE24. This field must be set to a value of 0 (clear PINs) to indicate that PINs are carried in the clear by BASE24.

PIN BLOCK FORMAT — Specifies the PIN block format of the PIN in the inbound and outbound external messages. This field must be set to a value of 0 (clear PINs) because PINs are carried in the clear, meaning PIN blocks are not used by the external processor or BASE24.

ANSI PAN FORMAT — Specifies which PAN digits are used in the formation of the PIN/PAN PIN block in the external message. This field must be set to a value of 0 (12 right-most digits, excluding the check digit) even though PIN blocks are not used by the external processor or BASE24.

PIN PAD CHARACTER — Specifies the PAD character used in the formation of the PIN block in the external message. This field must be set to F even though PIN blocks are not used by the external processor or BASE24.

NUMBER OF KEYS — Identifies whether the inbound and outbound keys defined on KEYF screen 2 are the same or different. The value in this field is used when PIN or MAC keys are exchanged automatically.

- If message authentication is not being performed because PIN keys are not required when PINs are carried in the clear both internally and externally, this field must be set to the default value of 1 .
- If message authentication is being performed using the same key for inbound and outbound messages, this field must be set to 1.
- If message authentication is being performed using different keys for inbound and outbound messages, this field must be set to 2.

KEY LENGTH — Identifies whether single- or double-length Key Exchange Keys (KEKs) are used with this external processor. The value in this field is used when PIN and MAC keys are exchanged automatically. Single-length keys contain 16 characters and double-length keys contain 32 characters.

- If message authentication is not being performed because KEKs are not required when PINs are carried in the clear both internally and externally, this field must be set to the default value of 1.
- If single-length KEKs are being used for message authentication key exchanges, this field must be set to a value of 1.
- If double-length KEKs are being used for message authentication key exchanges, this field must be set to a value of 2.

BASE24 Carries Clear PINs; External Processor Carries Encrypted PINs

If PINs are carried in clear form by BASE24, but are carried in encrypted form by the external processor, BASE24 PIN processing is carried out as described below.

PINs are carried in clear form in the BASE24-atm Standard Internal Message (STM), BASE24-pos Standard Internal Message (PSTM), or BASE24-teller Standard Internal Message (TSTM), and either software or hardware security can be used for encrypting PINs in the external messages between BASE24 and the external processor.

Processing Outgoing Messages

In the case of outgoing messages, BASE24 must encrypt PINs prior to sending them to the external processor. The BASE24 interface process performs as follows:

1. Creates a PIN block using the clear PIN.
 - If the PIN BLOCK FORMAT field on KEYF or KEY6 screen 1 is set to a value of 1 (ANSI PIN block), BASE24 creates a PIN/PAN PIN block using the appropriate characters from the cardholder's PAN, as specified in the ANSI PAN FORMAT field on KEYF or KEY6 screen 1.
 - If the PIN BLOCK FORMAT field on KEYF or KEY6 screen 1 is set to a value of 3 (PIN/PAD PIN block), BASE24 creates a PIN/PAD PIN block using the appropriate pad character specified in the PIN PAD CHARACTER field on KEYF or KEY6 screen 1.

Note: If the ENCRYPT TYPE field on KEYF or KEY6 screen 1 is set to a value of 1 (security module PIN management), a security module must be brought into use. In this case, the interface process formats a PIN/PAD PIN block, regardless of the setting in the PIN BLOCK FORMAT field on KEYF or KEY6 screen 1, because the PIN block is an intermediate PIN block that is translated in step 5 to the format required by the external processor.

2. Determines the outbound encryption key to use to encrypt the PIN.
 - If the OUTBOUND KEYS: CURRENT INDEX field on KEYF or KEY6 screen 2 is set to a value of 1 (use PIN key 1), the encryption key value in the OUTBOUND KEYS: PIN KEY1 field on KEYF or KEY6 screen 2 is used.
 - If the OUTBOUND KEYS: CURRENT INDEX field on KEYF or KEY6 screen 2 is set to a value of 2 (use PIN key 2), the encryption key value in the OUTBOUND KEYS: PIN KEY2 field on KEYF or KEY6 screen 2 is used.
3. Determines the encryption type and continues accordingly.
 - If the ENCRYPT TYPE field in the KEYF is set to 1 (security module PIN management), processing continues with step 4.
 - If the ENCRYPT TYPE field on KEYF or KEY6 screen 1 is set to a value of 2 (software DES PIN management), the BASE24 interface process encrypts the PIN block in software using the DES algorithm and the outbound encryption key selected in step 2. The interface process then sends the encrypted PIN block to the external processor.
4. Encrypts the PIN block in software under the clear intermediate key taken from the INTERMEDIATE KEYS: CLEAR field on KEYF or KEY6 screen 1.
5. Requests that the security module translate the PIN block for sending to the external processor.

The interface process provides the security module with the following information:

- The encrypted PIN block.
- The encrypted version of the intermediate key taken from the INTERMEDIATE KEYS: ENCRYPTED field on KEYF or KEY6 screen 1.
- The encrypted version of the outbound encryption key selected in step 2.
- The type of PIN block required by the external processor, specified in the PIN BLOCK FORMAT field on KEYF or KEY6 screen 1.
- The PAN digits specified by the ANSI PAN FORMAT field or the PIN PAD CHARACTER field on KEYF or KEY6 screen 1, depending on the type of PIN block required by the external processor.

The security module performs the following steps:

- a. Decrypts the intermediate key and the outbound encryption key.
- b. Decrypts the PIN block using the decrypted intermediate key.
- c. Extracts the clear PIN from the PIN block.
- d. Creates a PIN block of the type required by the external processor.
- e. Encrypts the PIN block using the decrypted outbound encryption key.
- f. Returns the encrypted PIN block to the interface process.

The interface process then sends the encrypted PIN block to the external processor.

Processing Incoming Messages

In the case of incoming messages, the BASE24 interface process must decrypt PINs upon arrival from the external processor. The BASE24 interface process performs the following steps:

1. Determines the inbound encryption key to use for decrypting the PIN.
 - If the INBOUND KEYS: CURRENT INDEX field on KEYF or KEY6 screen 2 is set to a value of 1 (use PIN key 1), the encryption key value in the INBOUND KEYS: PIN KEY1 field on KEYF or KEY6 screen 2 is used first and the encryption key value in the INBOUND KEYS: PIN KEY2 field on KEYF or KEY6 screen 2, if one exists, is used as the previous key in step 4.
 - If the INBOUND KEYS: CURRENT INDEX field on KEYF or KEY6 screen 2 is set to a value of 2 (use PIN key 2), the encryption key value in the INBOUND KEYS: PIN KEY2 field on KEYF or KEY6 screen 2 is used and the encryption key value in the INBOUND KEYS: PIN KEY1 field on KEYF or KEY6 screen 2, if one exists, is used as the previous key in step 4.
2. Determines the encryption type and continues accordingly.
 - If the ENCRYPT TYPE field on KEYF or KEY6 screen 1 is set to a value of 1 (security module PIN management), the BASE24 interface process must have the PIN translated with an intermediate key prior to decrypting it. Processing continues with step 3.

- If the ENCRYPT TYPE field on KEYF or KEY6 screen 1 is set to a value of 2 (software DES PIN management), the BASE24 interface process decrypts the PIN block in software using the DES algorithm and the inbound encryption key selected in step 1. Processing then continues with step 6.
3. Requests that the security module translate the PIN block for decryption by BASE24.

The interface process provides the security module with the following information:

- The encrypted PIN block taken from the external message.
- The encrypted version of the key under which the PIN is currently encrypted. This is the inbound key selected in step 1.
- The encrypted version of the intermediate key taken from the INTERMEDIATE KEYS: ENCRYPTED field on KEYF or KEY6 screen 1.
- The type of PIN block used by the external processor, specified in the PIN BLOCK FORMAT field on KEYF or KEY6 screen 1.
- The PAN digits specified by the ANSI PAN FORMAT or PIN PAD CHARACTER field on KEYF or KEY6 screen 1, depending on the type of PIN block used by the external processor.

The security module then performs the following steps:

- a. Decrypts the inbound encryption key and the intermediate key.
 - b. Decrypts the PIN block using the decrypted inbound encryption key.
 - c. Extracts the clear PIN from the PIN block.
 - d. Creates a PIN/PAN PIN block using the clear PIN.

A PIN/PAN PIN block with a PAN string containing the 12 right-most digits, excluding the check digit, is always requested by the BASE24 interface process when the PIN is being translated to encryption under the intermediate key.
 - e. Encrypts the PIN/PAN PIN block using the decrypted intermediate key.
 - f. Returns the encrypted PIN block to the interface process.
4. Repeats step 3 with the previous inbound encryption key, if necessary.
- If the security module successfully translated the PIN block in step 3 using the current key identified in step 1, processing continues with step 5.

- If the security module returned a sanity check error or PIN block error in step 3, the interface process repeats step 3 using the previous key identified in step 1. If the security module successfully translates the PIN block, processing continues with step 5.
 - If the security module returns a sanity check error or PIN block error when repeating step 3 with the previous key, the interface process declines the transaction.
 - If no previous key exists (i.e., the previous key field contains zeros or blanks), the interface process declines the transaction.
5. Decrypts the PIN block using the clear version of the intermediate key.
- Since the PIN block is now encrypted under the intermediate key, the BASE24 interface process can use the clear version of the key to decrypt the PIN block in software.
- The clear version of the intermediate key is taken from the INTERMEDIATE KEYS: CLEAR field on KEYF or KEY6 screen 1.
6. Extracts the clear PIN from the PIN block and places it in the PIN field of the internal message.
7. Computes the size of the clear PIN and notes the size in the PIN-SIZE field of the internal message.

Setting Up the BASE24 Database

Fields on screens 1 and 2 of the Key File (KEYF) or KEY 6 File (KEY6) record for each external processor are used when BASE24 carries PINs in clear form and the external processor carries PINs in encrypted form. Fields on KEYF screen 3 and KEY6 screen 4 deal exclusively with dynamic key management and are described in section 16. Refer to the ***BASE24 Base Files Maintenance Manual*** for additional information about these screens.

KEYF or KEY6 Screen 1

The following fields on KEYF or KEY6 screen 1 must be set as indicated when BASE24 carries PINs in clear form and the external processor carries PINs in encrypted form.

ENCRYPT TYPE — Specifies the type of PIN encryption used or expected by the external processor.

- If a security module is being used, this field must be set to a value of 1 (security module PIN management).
- If PINs are encrypted by software for the external processor, this field must be set to a value of 2 (software DES PIN management).

BASE24 ENCRYPT TYPE — Specifies the type of PIN management used by BASE24. This field must be set to a value of 0 (clear PINs) to indicate that PINs are carried in the clear in the BASE24 internal message.

PIN BLOCK FORMAT — Specifies the PIN block format of the PIN in the inbound and outbound external messages.

- If PINs are encrypted for or by the external processor using a PIN/PAN PIN block, this field must be set to a value of 1 (ANSI PIN block).
- If PINs are encrypted for or by the external processor using a PIN/PAD PIN block, this field must be set to a value of 3 (PIN/PAD PIN block).

ANSI PAN FORMAT — Specifies which PAN digits are used in the formation of the PIN/PAN PIN block in the external message.

- If PIN/PAN PIN blocks are used for the external processor (PIN BLOCK FORMAT field contains a value of 1), this field must specify the PAN digits to use for forming a PIN/PAN PIN block in external messages.
- If PIN/PAD PIN blocks are used for the external processor (PIN BLOCK FORMAT field contains a value of 3), this field must be set to 0.
- If a Thales e-Security Host Security Module is used, this field must be set to 0 (12 right-most PAN digits, excluding check digit) to match the PAN string used by BASE24 in internal messages.

PIN PAD CHARACTER — Specifies the PAD character used in the formation of the PIN block in the external message.

- If PIN/PAD PIN blocks are used for the external processor (PIN BLOCK FORMAT field contains a value of 3), this field must specify the pad character to use for forming a PIN/PAD PIN block in external messages.
- If PIN/PAN PIN blocks are used for the external processor (PIN BLOCK FORMAT field contains a value of 1), this field must be set to F.

NUMBER OF KEYS — Identifies whether the inbound and outbound keys defined on KEYF screen 2 or on KEY6 screens 2 and 3 are the same. The value in this field is used when PIN or MAC keys are exchanged automatically. The same number of keys must be used for PINs and MACs.

- If the same key is used for inbound and outbound messages, this field must be set to 1.
- If different keys are used for inbound and outbound messages, this field must be set to 2.

KEY LENGTH — Identifies whether single- or double-length Key Exchange Keys (KEKs) are used with this external processor. The value in this field is used when PIN and MAC keys are exchanged automatically. The PIN KEK and MAC KEK must be the same length. Single-length keys contain 16 characters and double-length keys contain 32 characters.

- If single-length KEKs are being used for key exchanges, this field must be set to a value of 1.
- If double-length KEKs are being used for key exchanges, this field must be set to a value of 2.

INTERMEDIATE KEYS

The following three fields are used whenever a PIN that has been encrypted under a secure key (security module PIN encryption) must be decrypted into the clear or when a PIN is in the clear and needs to be encrypted under a secure key.

CLEAR — Specifies the clear version of the intermediate key. Valid values for each character in this field are 0 through 9 and A through F. This field must contain 16 valid characters.

- If the ENCRYPT TYPE field is set to a value of 1 (security module PIN management), this field must contain the clear version of the intermediate key.
- If the ENCRYPT TYPE field is set to a value of 2 (software DES PIN management), this field must be set to zeros.

CHECK DIGITS — Specifies the check digits associated with the clear version of the intermediate key. Valid values for each character in this field are 0 through 9 and A through F. This field must contain four valid characters. A value of 0000 in this field indicates check digits are not being used with the clear version of the intermediate key.

ENCRYPTED — Specifies the encrypted version of the intermediate key. Valid values for each character in this field are 0 through 9 and A through F. This field must contain 16 valid characters.

- If the ENCRYPT TYPE field is set to a value of 1 (security module PIN management), this field must contain the encrypted version of the intermediate key.
- If the ENCRYPT TYPE field is set to a value of 2 (software DES PIN management), this field must be set to zeros.

EXCHANGE KEYS

The following two fields are used to create keys that BASE24 uses when PIN keys are automatically exchanged between the Interchange Interface process and the interchange.

PIN KEY — Identifies the security module-encrypted form of the Key Exchange Key (KEK) value used when inbound and outbound PIN keys are exchanged using dynamic key management. This two-part field contains a single- or double-length KEK, depending on the value in the KEY LENGTH field. There are two sets of 16 hexadecimal characters. The first set is used for single-length keys and both sets are used for double-length keys. Valid values for each character in this field are 0 through 9 and A through F. Both sets must contain 16 valid characters. If the value in the KEY LENGTH field is 1 (single-length keys), the second set of characters must be zeros. Refer to section 16 for additional information about dynamic key management.

CHECK DIGITS — Specifies the check digits associated with the PIN KEK. Valid values for each character in this field are 0 through 9 and A through F. This field must contain four valid characters. A value of 0000 in this field indicates check digits are not being used with the PIN KEK.

KEYF or KEY6 Screen 2

The following fields on KEYF or KEY6 screen 2 must be set as indicated when BASE24 carries PINs in clear form and the external processor carries PINs in encrypted form.

PIN KEY INFORMATION

The PIN Key Information section of KEYF or KEY6 screen 2 contains two groups of fields, Outbound Keys and Inbound Keys, each containing six fields.

OUTBOUND KEYS

The following six fields pertain to PIN encryption and PIN translation.

PIN KEY1 — Identifies the first key value to be used for encrypting PINs to be sent to the external processor. This field must contain 16 valid characters. Valid values for each character in this field are 0 through 9 and A through F.

Note: On KEY6 screen 2, this field occurs twice. Each field must contain 16 valid characters to define a double-length PIN encryption key.

Two keys can be specified; the one actually used is indicated by the value in the CURRENT INDEX field. When two keys are specified, BASE24 considers the key not currently in use to be the older of the two keys.

- If the ENCRYPT TYPE field on KEYF or KEY6 screen 1 is set to a value of 1 (security module PIN management), these keys must be encrypted by or for a security module, such that the security module can decrypt the key when it is sent by BASE24.
- If the ENCRYPT TYPE field on KEYF or KEY6 screen 1 is set to a value of 2 (software DES PIN management), these keys must be encrypted by the BASE24 Supercrypt utility prior to being entered in the KEYF or KEY6. For information on the Supercrypt utility, refer to appendix A.

CHECK DIGITS1 — Identifies the check digits associated with the key entered in the PIN KEY1 field. Valid values for each character in this field are 0 through 9 and A through F. This field must contain four valid characters. A value of 0000 in this field indicates check digits are not being used with the key.

If an external processor supports automatic outbound key changes, this field can be set to the default value of zero, and the interface process automatically sets this field when it is first started.

CURRENT INDEX — Indicates which outbound PIN encryption key is currently being used by the interface process. Valid values are as follows:

- 1 = Value in the PIN KEY1 field
- 2 = Value in the PIN KEY2 field

If an external processor supports automatic outbound key changes, this field can be set to the default value of 1, and the interface process automatically sets this field when it is first started.

PIN KEY2 — Indicates a second key value to be used for encrypting PINs to be sent to the external processor. This field must contain 16 valid characters. Valid values for each character in this field are 0 through 9 and A through F.

Note: On KEY6 screen 2, this field occurs twice. Each field must contain 16 valid characters to define a double-length PIN encryption key.

Two keys can be specified; the one actually used is indicated by the value in the CURRENT INDEX field. When two keys are specified, BASE24 considers the key not currently in use to be the older of the two keys.

- If the ENCRYPT TYPE field on KEYF or KEY6 screen 1 is set to a value of 1 (security module PIN management), these keys must be encrypted by or for a security module, such that the security module can decrypt the key when it is sent by BASE24.
- If the ENCRYPT TYPE field on KEYF or KEY6 screen 1 is set to a value of 2 (software DES PIN management), these keys must be encrypted by the BASE24 Supercrypt utility prior to being entered in the KEYF or KEY6. For information on the Supercrypt utility, refer to appendix A.

If an external processor supports automatic outbound key changes, this field can be set to the default value of zero, and the interface process automatically sets this field when it is first started.

CHECK DIGITS2 — Identifies the check digits associated with the key entered in the PIN KEY2 field. Valid values for each character in this field are 0 through 9 and A through F. This field must contain four valid characters. A value of 0000 in this field indicates check digits are not being used with the key.

If an external processor supports automatic outbound key changes, this field can be set to the default value of zero, and the interface process automatically sets this field when it is first started.

KEY COUNTER — Identifies the number of times the outbound PIN key has been changed by dynamic key management processing. This counter is increased whenever the outbound PIN key is changed by BASE24 or the external processor. Valid values for each character in this field are 0 through 9 and A through F. This field must contain 6 valid characters.

- If the external processor expects a value other than zero when dynamic key management is established, this field must be set to the value specified by the external processor.
- If dynamic key management is not used or if the external processor does not require a specific initial counter value when dynamic key management is established, this field can default to zero. The Host Interface process or BASE24 Interchange (BIC) Interface process increments the counter automatically and attempts to resynchronize it with the external processor. Refer to the appropriate interface documentation to determine whether a specific Interchange Interface process uses this field.

INBOUND KEYS

The following six fields contain the PIN keys used for inbound transactions (that is transactions sent by the external processor to BASE24).

PIN KEY1 — Identifies the first key value used by the external processor to encrypt PINs in the transactions sent to BASE24. This field must contain 16 valid characters. Valid values for each character in this field are 0 through 9 and A through F.

Note: On KEY6 screen 2, this field occurs twice. Each field must contain 16 valid characters to define a double-length PIN encryption key.

Two keys can be specified; the one actually used is indicated by the value in the CURRENT INDEX field. When two keys are specified, BASE24 considers the key not currently in use to be the older of the two keys.

- If the ENCRYPT TYPE field on KEYF or KEY6 screen 1 is set to a value of 1 (security module PIN management), these keys must be encrypted by or for a security module, such that the security module can decrypt the key when it is sent by BASE24.

- If the ENCRYPT TYPE field on KEYF or KEY6 screen 1 is set to a value of 2 (software DES PIN management), these keys must be encrypted by the BASE24 Supercrypt utility prior to being entered in the KEYF or KEY6. For information on the Supercrypt utility, refer to appendix A.

If an external processor supports automatic inbound key changes, this field can be set to the default value of zero, and the interface process automatically sets this field when it is first started.

CHECK DIGITS 1 — Identifies the check digits associated with the key entered in the PIN KEY1 field. Valid values for each character in this field are 0 through 9 and A through F. This field must contain four valid characters. A value of 0000 in this field indicates check digits are not being used with the key.

If an external processor supports automatic outbound key changes, this field can be set to the default value of zero, and the interface process automatically sets this field when it is first started.

CURRENT INDEX — Indicates which outbound PIN encryption key is currently being used by the interface process. Valid values are as follows:

- 1 = Value in the PIN KEY1 field
- 2 = Value in the PIN KEY2 field

If an external processor supports automatic outbound key changes, this field can be set to the default value of 1, and the interface process automatically sets this field when it is first started.

PIN KEY2 — Identifies a second key value used by the external processor to encrypt PINs in the transactions sent to BASE24. This field must contain 16 valid characters. Valid values for each character in this field are 0 through 9 and A through F.

Note: On KEY6 screen 2, this field occurs twice. Each field must contain 16 valid characters to define a double-length PIN encryption key.

Two keys can be specified; the one actually used is indicated by the value in the CURRENT INDEX field. When two keys are specified, BASE24 considers the key not currently in use to be the older of the two keys.

- If the ENCRYPT TYPE field on KEYF or KEY6 screen 1 is set to a value of 1 (security module PIN management), these keys must be encrypted by or for a security module, such that the security module can decrypt the key when it is sent by BASE24.
- If the ENCRYPT TYPE field on KEYF or KEY6 screen 1 is set to a value of 2 (software DES PIN management), these keys must be encrypted by the BASE24 Supercrypt utility prior to being entered in the KEYF or KEY6. For information on the Supercrypt utility, refer to appendix A.

If an external processor supports automatic outbound key changes, this field can be set to the default value of zero, and the interface process automatically sets this field when it is first started.

CHECK DIGITS2 — Identifies the check digits associated with the key entered in the PIN KEY2 field. Valid values for each character in this field are 0 through 9 and A through F. This field must contain four valid characters. A value of 0000 in this field indicates check digits are not being used with the key.

If an external processor supports automatic inbound key changes, this field can be set to the default value of zero, and the interface process automatically sets this field when it is first started.

KEY COUNTER — Identifies the number of times the inbound PIN key has been changed by dynamic key management processing. This counter is increased whenever the inbound PIN key is changed by BASE24 or the external processor. Valid values for each character in this field are 0 through 9 and A through F. This field must contain 6 valid characters.

- If the external processor expects a value other than zero when dynamic key management is established, this field must be set to the value specified by the external processor.
- If dynamic key management is not used or if the external processor does not require a specific initial counter value when dynamic key management is established, this field can default to zero. The Host Interface process or BASE24 Interchange (BIC) Interface process increments the counter automatically and attempts to resynchronize it with the external processor. Refer to the appropriate interface documentation to determine whether a specific Interchange Interface process uses this field.

BASE24 Carries Encrypted PINs; External Processor Carries Clear PINs

If PINs are carried in encrypted form by BASE24 and are carried in clear form by the external processor, BASE24 PIN processing is carried out as described below.

Security modules must be used to perform internal message PIN management when PINs are carried in encrypted form in the STM, PSTM, or TSTM. PINs in the external messages between BASE24 and the external processor are carried in the clear.

Processing Outgoing Messages

In the case of outgoing messages, BASE24 must decrypt PINs prior to sending them to the external processor. The BASE24 interface process performs the following steps:

1. Requests that the security module translate the PIN block for decryption by BASE24.

The interface process provides the security module with the following information:

- The encrypted PIN block taken from the PIN field in the internal message.
- The type of PIN block used by BASE24, taken from the PIN-FRMT field in the internal message.
- The PAN digits specified by the ANSI offset value in the internal message, or the PIN pad character value from the internal message, depending on the type of PIN block used.
- The encrypted version of the key under which the PIN block is currently encrypted, taken from the PIN-KEY field in the internal message.
- The encrypted version of the intermediate key, taken from the INTERMEDIATE KEYS: ENCRYPTED field on KEYF screen 1.

The security module performs the following steps:

- a. Decrypts the intermediate key and the key under which the PIN block is encrypted.
 - b. Decrypts the PIN block using the decrypted version of the key used to encrypt the PIN block.
 - c. Extracts the clear PIN from the PIN block.
 - d. Creates a PIN/PAN PIN block using the clear PIN.

A PIN/PAN PIN block with a PAN string containing the 12 right-most digits, excluding the check digit, is always requested by BASE24 when the PIN is being translated to encryption under the intermediate key.
 - e. Encrypts the PIN/PAN PIN block using the decrypted intermediate key.
 - f. Returns the encrypted PIN block to the interface process.
2. Repeats step 2 with the previous encryption key, if necessary.
 - If the security module successfully translated the PIN block in step 1 using the key taken from the PIN key field in the internal message, processing continues with step 3.
 - If the security module returned a sanity check error or PIN block error in step 1, the interface process repeats step 1 using the previous encryption key, taken from the previous key field in the internal message. If the security module successfully translates the PIN block, processing continues with step 3.
 - If the security module returns a sanity check error or PIN block error when repeating step 1 with the previous key, the interface process declines the transaction.
 - If no previous key exists (i.e., the previous key field in the internal message contains zeros or blanks), the interface process declines the transaction.
 3. Decrypts the PIN block using the clear version of the intermediate key.

Since the PIN block has been encrypted under the intermediate key, BASE24 can use the clear version of the key to decrypt the PIN block in software.

The clear version of the intermediate key is taken from the INTERMEDIATE KEYS: CLEAR field on KEYF screen 1.
 4. Extracts the PIN from the clear PIN/PAN PIN block.

Processing Incoming Messages

In the case of incoming messages, BASE24 must encrypt PINs upon arrival from the external processor. The BASE24 interface process performs the following steps:

1. Creates a PIN/PAN PIN block using the right-most 12 digits of the PAN, excluding the check digit.
2. Encrypts the PIN block using the DES algorithm and the clear intermediate key taken from the INTERMEDIATE KEYS: CLEAR field on KEYF screen 1.
3. Determines the inbound encryption key to use to encrypt the PIN.
 - If the INBOUND KEYS: CURRENT INDEX field on KEYF screen 2 is set to a value of 1 (use PIN key 1), the encryption key value in the INBOUND KEYS: PIN KEY1 field on KEYF screen 2 is used.
 - If the INBOUND KEYS: CURRENT INDEX field on KEYF screen 2 is set to a value of 2 (use PIN key 2), the encryption key value in the INBOUND KEYS: PIN KEY2 field on KEYF screen 2 is used.
4. Requests that the security module translate the PIN block to encryption under the inbound encryption key.

The interface process provides the security module with the following information:

- The encrypted PIN block.
- The encrypted version of the inbound encryption key selected in step 3.
- The encrypted version of the intermediate key taken from the INTERMEDIATE KEYS: ENCRYPTED field on KEYF screen 1.
- The type of PIN block created by BASE24 in step 1.
- The PAN digits used for the PIN block created in step 1.

The security module then performs the following steps:

- a. Decrypts the inbound encryption key and the intermediate key.
 - b. Decrypts the PIN block using the decrypted intermediate key.
 - c. Encrypts the PIN block using the decrypted inbound encryption key.
 - d. Returns the encrypted PIN block to the interface process.
5. Places the encrypted PIN block in the PIN field of the internal message, and sets the PIN-SIZE field in the internal message to 16.

6. Sets the PIN-FRMT field in the internal message to indicate the PIN is encrypted and that a PIN/PAN PIN block was used.
7. Sets the ANSI offset field in the internal message to identify the starting position of the PAN string used to compute the PIN/PAN PIN block.
8. Sets the PIN key and previous key fields in the internal message.
 - If the INBOUND KEYS: CURRENT INDEX field on KEYF screen 2 is set to a value of 1 (use PIN key 1), the encryption key value in the INBOUND KEYS: PIN KEY1 field is the current key and the encryption key value in the INBOUND KEYS: PIN KEY2 field is the previous key.
 - If the INBOUND KEYS: CURRENT INDEX field on KEYF screen 2 is set to a value of 2 (use PIN key 2), the encryption key value in the INBOUND KEYS: PIN KEY2 field is the current key and the encryption key value in the INBOUND KEYS: PIN KEY1 field is the previous key.

Setting Up the BASE24 Database

Fields on screens 1 and 2 of the Key File (KEYF) record for each external processor are used when BASE24 carries PINs in encrypted form and the external processor carries PINs in clear form. Fields on KEYF screen 3 deal exclusively with dynamic key management and are described in section 16. Refer to the *BASE24 Base Files Maintenance Manual* for additional information about these screens.

KEYF Screen 1

The following fields on KEYF screen 1 must be set as indicated when BASE24 carries PINs in encrypted form and the external processor carries PINs in clear form.

ENCRYPT TYPE — Specifies the type of PIN encryption used or expected by the external processor. This field must be set to a value of 0 (clear PINs) to indicate that PINs are carried in the clear by the external processor.

BASE24 ENCRYPT TYPE — Specifies the type of PIN management used by BASE24. This field must be set to a value of 1 (security module PIN management) to indicate that PINs are carried in encrypted form by BASE24.

PIN BLOCK FORMAT — Specifies the PIN block format of the PIN in the inbound and outbound external messages. This field must be set to a value of 1 (ANSI PIN block) to identify the format of the PIN block returned to the interface process by the security module.

ANSI PAN FORMAT — Specifies which PAN digits are used in the formation of the PIN/PAN PIN block in the external message. This field must be set to a value of 0 (12 right-most digits, excluding check digit) to identify the PAN digits carried in the PIN block returned to the interface process by the security module.

PIN PAD CHARACTER — Specifies the PAD character used in the formation of the PIN block in the external message. This field must be set to F.

NUMBER OF KEYS — Identifies whether the inbound and outbound keys defined on KEYF screen 2 are combined or separate. The value in this field is used when PIN or MAC keys are exchanged automatically. The same number of keys must be used for PINs and MACs.

- If message authentication is not being performed because inbound and outbound PIN keys are not used when PINs are carried in the clear externally, this field must be set to the default value of 1.
- If message authentication is being performed using the same key for inbound and outbound messages, this field must be set to 1.
- If message authentication is being performed using different keys for inbound and outbound messages, this field must be set to 2.

KEY LENGTH — Identifies whether single- or double-length Key Exchange Keys (KEKs) are being used with this external processor. The value in this field is used when PIN and MAC keys are exchanged automatically. Single-length keys contain 16 characters and double-length keys contain 32 characters.

- If message authentication is not being performed because KEKs are not required when PINs are carried in the clear externally, this field must be set to the default value of 1.
- If single-length KEKs are being used for message authentication key exchanges, this field must be set to a value of 1.
- If double-length KEKs are being used for message authentication key exchanges, this field must be set to a value of 2.

INTERMEDIATE KEYS

The following three fields are used when a PIN that has been encrypted under a secure key (security module PIN encryption) must be decrypted into the clear or when a PIN is in the clear and needs to be encrypted under a secure key.

CLEAR — Identifies the clear version of the intermediate key to be used for translating PINs for software decryption (outbound) or software encryption (inbound). Valid values for each character in this field are 0 through 9 and A through F. This field must contain 16 valid characters.

CHECK DIGITS — Specifies the check digits associated with the clear version of the intermediate key. Valid values for each character in this field are 0 through 9 and A through F. This field must contain four valid characters. A value of 0000 in this field indicates check digits are not being used with the clear intermediate key.

ENCRYPTED — Identifies the encrypted version of the intermediate key to be used for translating PINs for software decryption (outbound) or software encryption (inbound). Valid values for each character in this field are 0 through 9 and A through F. This field must contain 16 valid characters.

KEYF Screen 2

The following fields on KEYF screen 2 must be set as indicated when BASE24 carries PINs in encrypted form and the external processor carries PINs in clear form.

PIN KEY Information

The PIN Key Information section of KEYF screen 2 contains two groups of fields, Outbound Keys and Inbound Keys, each containing six fields.

INBOUND KEYS

The following fields contain the PIN keys used for inbound transactions (that is transactions sent by the external processor to BASE24).

PIN KEY1 — Identifies the first key used by the external processor to encrypt PINs in the transactions sent to BASE24. This field must contain 16 valid characters. Valid values for each character in this field are 0 through 9 and A through F.

Two keys can be specified; the one actually used is indicated by the value in the CURRENT INDEX field. When two keys are specified, BASE24 considers the key not currently in use to be the older of the two keys.

- If the ENCRYPT TYPE field on KEYF screen 1 is set to a value of 1 (security module PIN management), these keys must be encrypted by or for a security module, such that the security module can decrypt the key when it is sent by BASE24.
- If the ENCRYPT TYPE field on KEYF screen 1 is set to a value of 2 (software DES PIN management), these keys must be encrypted by the BASE24 Supercrypt utility prior to being entered in the KEYF. For information on the Supercrypt utility, refer to appendix A.

If an external processor supports automatic inbound key changes, this field can be set to the default value of zero, and the interface process automatically sets this field when it is first started.

CHECK DIGITS1 — Identifies the check digits associated with the key entered in the PIN KEY1 field. Valid values for each character in this field are 0 through 9 and A through F. This field must contain four valid characters. A value of 0000 in this field indicates check digits are not being used with the key.

If an external processor supports automatic outbound key changes, this field can be set to the default value of zero, and the interface process automatically sets this field when it is first started.

CURRENT INDEX — Indicates which outbound PIN encryption key is currently being used by the interface process. Valid values are as follows:

- 1 = Value in the PIN KEY1 field
- 2 = Value in the PIN KEY2 field

If an external processor supports automatic outbound key changes, this field can be set to the default value of 1, and the interface process automatically sets this field when it is first started.

PIN KEY2 — Identifies the second key used by the external processor to encrypt PINs in the transactions sent to BASE24. This field must contain 16 valid characters. Valid values for each character in this field are 0 through 9 and A through F.

Two keys can be specified; the one actually used is indicated by the value in the CURRENT INDEX field. When two keys are specified, BASE24 considers the key not currently in use to be the older of the two keys.

- If the ENCRYPT TYPE field on KEYF screen 1 is set to a value of 1 (security module PIN management), these keys must be encrypted by or for a security module, such that the security module can decrypt the key when it is sent by BASE24.
- If the ENCRYPT TYPE field on KEYF screen 1 is set to a value of 2 (software DES PIN management), these keys must be encrypted by the BASE24 Supercrypt utility prior to being entered in the KEYF. For information on the Supercrypt utility, refer to appendix A.

If an external processor supports automatic outbound key changes, this field can be set to the default value of zero, and the interface process automatically sets this field when it is first started.

CHECK DIGITS2 — Identifies the check digits associated with the key entered in the PIN KEY2 field. Valid values for each character in this field are 0 through 9 and A through F. This field must contain four valid characters. A value of 0000 in this field indicates check digits are not being used with the key.

If an external processor supports automatic outbound key changes, this field can be set to the default value of zero, and the interface process automatically sets this field when it is first started.

BASE24 and External Processor Carry Encrypted PINs

If PINs are carried in encrypted form by both BASE24 and the external processor, BASE24 PIN processing is carried out as described below.

Security modules must be used to perform internal message PIN management when PINs are carried in encrypted form in the STM, PSTM, or TSTM. While either software or hardware security can be used for encrypting PINs in the external messages between BASE24 and the external processor, ACI recommends hardware security for the following reasons:

- Security modules that are already in place for internal message PIN management can also be used for external message PIN management.
- A higher level of security exists because keys are controlled within security modules and intermediate keys are not required.
- Transaction processing is more efficient because PIN management involves less data manipulation.

The processing descriptions and database settings that follow are based on security modules being used to perform PIN management in external messages as well as internal messages.

Processing Outgoing Messages

In the case of outgoing messages, BASE24 must translate PINs prior to sending them to the external processor.

Note: In situations where BASE24 has received a message containing an encrypted PIN from an interchange or device and is routing the message to an external processor that expects an encrypted PIN, the PIN block carried in the internal message at this point has not been modified from the way it was received from the device or first interchange.

To translate PINs, the BASE24 interface process performs the following steps:

1. Determines the encryption key to use to encrypt the PIN for sending to the external processor.
 - If the OUTBOUND KEYS: CURRENT INDEX field on KEYF or KEY6 screen 2 is set to a value of 1 (use PIN key 1), the encryption key value in the OUTBOUND KEYS: PIN KEY1 field is used.
 - If the OUTBOUND KEYS: CURRENT INDEX field on KEYF or KEY6 screen 2 is set to a value of 2 (use PIN key 2), the encryption key value in the OUTBOUND KEYS: PIN KEY2 field is used.
2. Requests that a security module translate the PIN block to encryption under the outbound encryption key.

The interface process provides the security module with the following:

- The encrypted PIN block taken from the PIN field in the internal message.
- The encrypted version of the key under which the PIN is currently encrypted, taken from the PIN-KEY field in the internal message.
- The type of PIN block used by BASE24, taken from the PIN-FRMT field in the internal message.
- The PAN digits specified by the ANSI offset value in the internal message, or the PIN pad character value from the internal message, depending on the type of PIN block used.
- The encrypted outbound encryption key selected in step 1.
- The type of PIN block required by the external processor taken from the PIN BLOCK FORMAT field on KEYF or KEY6 screen 1.

Note: Some security modules require that the same PAN digits used to create the input PIN block in a PIN/PAN-to-PIN/PAN translation are also used to create the output PIN block. This requirement does not apply to translations between a PIN/PAN PIN block and a PIN/PAD PIN block. The Thales e-Security Host Security Module has this requirement.

The security module then performs the following steps:

- a. Decrypts the outbound key and the key under which the PIN block is encrypted.
 - b. Decrypts the PIN block using the decrypted version of the key under which the PIN block is encrypted.
 - c. Extracts the clear PIN from the clear PIN block.
 - d. Creates a new PIN block of the type required by the external processor.
 - e. Encrypts the PIN block using the decrypted outbound key.
 - f. Returns the encrypted PIN block to the interface process.
3. Repeats step 1 with the previous encryption key, if necessary.
 - If the security module successfully translated the PIN block in step 2. using the key taken from the PIN-KEY field in the internal message, the PIN block is ready for sending to the external processor.
 - If the security module returned a sanity check error or PIN block error in step 1, the interface process repeats step 1 using the previous encryption key, taken from the previous key field in the internal message. If the security module successfully translates the PIN block, the PIN block is ready for sending to the external processor.
 - If the security module returns a sanity check error or PIN block error when repeating step 1 with the previous key, the interface process declines the transaction.
 - If no previous key exists (i.e., the previous key field in the internal message contains zeros or blanks), the interface process declines the transaction.

Processing Incoming Messages

In the case of incoming messages, no translation is required by BASE24. The BASE24 interface process performs the following steps:

1. Moves the PIN block from the external message into the PIN field of the internal message and sets the PIN-SIZE field in the internal message to 16.

2. Sets the PIN-KEY and previous key fields in the internal message.
 - If the INBOUND KEYS: CURRENT INDEX field on KEYF or KEY6 screen 2 is set to a value of 1 (use PIN key 1), the encryption key value in the INBOUND KEYS: PIN KEY1 field is the PIN key and the encryption key value in the INBOUND KEYS: PIN KEY2 field is the previous key.
 - If the INBOUND KEYS: CURRENT INDEX field on KEYF or KEY6 screen 2 is set to a value of 2 (use PIN key 2), the encryption key value in the INBOUND KEYS: PIN KEY2 field is the PIN key and the encryption key value in the INBOUND KEYS: PIN KEY1 field is the previous key.
3. Sets the PIN block information in the internal message, based upon the value of the PIN BLOCK FORMAT field on KEYF or KEY6 screen 1.
 - If the PIN BLOCK FORMAT field on KEYF or KEY6 screen 1 is set to a value of 1 (ANSI PIN block), the interface process sets the PIN format field in the internal message to indicate a PIN/PAN PIN block was used by the external processor. It also sets the ANSI offset field in the internal message to identify the starting position of the PAN string used to compute the PIN/PAN PIN block.
 - If the PIN BLOCK FORMAT field on KEYF or KEY6 screen 1 is set to a value of 3 (PIN/PAD PIN block), the interface process sets the PIN format field in the internal message to indicate a PIN/PAD PIN block was used by the external processor. It also sets the PIN pad character field in the internal message from the PIN PAD CHARACTER field on KEYF or KEY6 screen 1.

Setting Up the BASE24 Database

Fields on screens 1 and 2 of the Key File (KEYF) or Key 6 File (KEY6) record for each external processor are used when both the external processor and BASE24 carry their PINs in encrypted form. Fields on KEYF screen 3 or KEY6 screen 4 deal exclusively with dynamic key management and are described in section 16. Refer to the ***BASE24 Base Files Maintenance Manual*** for additional information about these screens.

KEYF or KEY6 Screen 1

The following fields on KEYF or KEY6 screen 1 must be set as indicated when BASE24 and the external processor carry PINs in encrypted form.

ENCRYPT TYPE — Specifies the type of PIN encryption used or expected by the external processor. This field must be set to a value of 1 (security module PIN management).

BASE24 ENCRYPT TYPE — Specifies the type of PIN management used by BASE24. This field must be set to a value of 1 (security module PIN management) to indicate that PINs are carried in encrypted form by BASE24.

PIN BLOCK FORMAT — Specifies the PIN block format of the PIN in the inbound and outbound external messages.

- If PINs are encrypted by or for the external processor using a PIN/PAN PIN block, this field must be set to a value of 1 (ANSI PIN block).
- If PINs are encrypted by or for the external processor using a PIN/PAD PIN block, this field must be set to a value of 3 (PIN/PAD PIN block).

ANSI PAN FORMAT — Specifies which PAN digits are used in the formation of the PIN/PAN PIN block in the external message.

- If PIN/PAN PIN blocks are used for the external processor (PIN BLOCK FORMAT field on KEYF or KEY6 screen 1 contains a value of 1), this field must specify the PAN digits to use for forming a PIN/PAN PIN block.
- If PIN/PAD PIN blocks are used for the external processor (PIN BLOCK FORMAT field on KEYF or KEY6 screen 1 contains a value of 3), this field must be set to 0.

The Thales e-Security Host Security Module, if used, requires that the PAN string used to create the input PIN block in a PIN/PAN to PIN/PAN translation is also used to create the output PIN block.

PIN PAD CHARACTER — Specifies the PAD character used in the formation of the PIN block in the external message.

- If PIN/PAD PIN blocks are used for the external processor, this field must specify the pad character to use for forming a PIN/PAD PIN block.
- If PIN/PAN PIN blocks are used for the external processor, this field must be set to F.

NUMBER OF KEYS — Identifies whether the inbound and outbound keys defined on KEYF or KEY6 screen 2 are combined or separate. The value in this field is used when PIN or MAC keys are exchanged automatically. The same number of keys must be used for PINs and MACs.

- If the same key is used for inbound and outbound messages, this field must be set to 1.
- If different keys are used for inbound and outbound messages, this field must be set to 2.

KEY LENGTH — Identifies whether single- or double-length Key Exchange Keys (KEKs) are used with this external processor. The value in this field is used when PIN and MAC keys are exchanged automatically. The PIN KEK and MAC KEK must be the same length. Single-length keys contain 16 characters and double-length keys contain 32 characters.

- If single-length KEKs are being used for key exchanges, this field must be set to a value of 1.
- If double-length KEKs are being used for key exchanges, this field must be set to a value of 2.

EXCHANGE KEYS

The following two fields must be set as indicated to allow PIN keys to be exchanged between the Interchange Interface process and an interchange in encrypted form.

PIN KEY — Identifies the security module-encrypted form of the Key Exchange Key (KEK) used when inbound and outbound PIN keys are exchanged using dynamic key management. This two-part field contains a single- or double-length KEK, depending on the value in the KEY LENGTH field. There are two sets of 16 hexadecimal characters. The first set is used for single-length keys and both sets are used for double-length keys. Valid values for each character in this field are 0 through 9 and A through F. Both sets must contain 16 valid characters. If the value in the KEY LENGTH field is 1 (single-length keys), the second set of characters must be zeros. Refer to section 16 for additional information about dynamic key management.

CHECK DIGITS — Identifies the check digits associated with the PIN KEK. Valid values for each character in this field are 0 through 9 and A through F. This field must contain four valid characters. A value of 0000 in this field indicates check digits are not being used with the PIN KEK.

KEYF or KEY6 Screen 2

The following fields on KEYF or KEY6 screen 2 must be set as indicated when BASE24 and the external processor carries PINs in clear form.

PIN KEY Information

The PIN Key Information section of KEYF or KEY6 screen 2 contains two groups of fields, Outbound Keys and Inbound Keys, each containing six fields.

OUTBOUND KEYS

The following six fields pertain to PIN encryption and PIN translation.

PIN KEY1 — Identifies the first key to be used for encrypting PINs to be sent to the external processor. This field must contain 16 valid characters. Valid values for each character in this field are 0 through 9 and A through F.

Note: On KEY6 screen 2, this field occurs twice. Each field must contain 16 valid characters to define a double-length PIN encryption key.

Two keys can be specified; the one actually used is indicated by the value in the CURRENT INDEX field. When two keys are specified, BASE24 considers the key not currently in use to be the older of the two keys.

- If the ENCRYPT TYPE field on KEYF or KEY6 screen 1 is set to a value of 1 (security module PIN management), these keys must be encrypted by or for a security module, such that the security module can decrypt the key when it is sent by BASE24.
- If the ENCRYPT TYPE field on KEYF or KEY6 screen 1 is set to a value of 2 (software DES PIN management), these keys must be encrypted by the BASE24 Supercrypt utility prior to being entered in the KEYF or KEY6. For information on the Supercrypt utility, refer to appendix A.

CHECK DIGITS1 — Identifies the check digits associated with the key entered in the PIN KEY1 field. Valid values for each character in this field are 0 through 9 and A through F. This field must contain four valid characters. A value of 0000 in this field indicates check digits are not being used with the key.

If an external processor supports automatic outbound key changes, this field can be set to the default value of zero, and the interface process automatically sets this field when it is first started.

CURRENT INDEX — Indicates which outbound PIN encryption key is currently being used by the interface process. Valid values are as follows:

- 1 = Value in the PIN KEY1 field
- 2 = Value in the PIN KEY2 field

If an external processor supports automatic outbound key changes, this field can be set to the default value of 1, and the interface process automatically sets this field when it is first started.

PIN KEY2 — Indicates a second key to be used for encrypting PINs to be sent to the external processor. This field must contain 16 valid characters. Valid values for each character in this field are 0 through 9 and A through F.

Note: On KEY6 screen 2, this field occurs twice. Each field must contain 16 valid characters to define a double-length PIN encryption key.

Two keys can be specified; the one actually used is indicated by the value in the CURRENT INDEX field. When two keys are specified, BASE24 considers the key not currently in use to be the older of the two keys.

- If the ENCRYPT TYPE field on KEYF or KEY6 screen 1 is set to a value of 1 (security module PIN management), these keys must be encrypted by or for a security module, such that the security module can decrypt the key when it is sent by BASE24.
- If the ENCRYPT TYPE field on KEYF or KEY6 screen 1 is set to a value of 2 (software DES PIN management), these keys must be encrypted by the BASE24 Supercrypt utility prior to being entered in the KEYF or KEY6. For information on the Supercrypt utility, refer to appendix A.

If an external processor supports automatic outbound key changes, this field can be set to the default value of zero, and the interface process automatically sets this field when it is first started.

CHECK DIGITS2 — Identifies the check digits associated with the key entered in the PIN KEY2 field. Valid values for each character in this field are 0 through 9 and A through F. This field must contain four valid characters. A value of 0000 in this field indicates check digits are not being used with the key.

If an external processor supports automatic outbound key changes, this field can be set to the default value of zero, and the interface process automatically sets this field when it is first started.

KEY COUNTER — Identifies the number of times the outbound PIN key has been changed by dynamic key management processing. This counter is increased whenever the outbound PIN key is changed by BASE24 or the external processor. Valid values for each character in this field are 0 through 9 and A through F. This field must contain 6 valid characters.

- If the external processor expects a value other than zero when dynamic key management is established, this field must be set to the value specified by the external processor.
- If dynamic key management is not used or if the external processor does not require a specific initial counter value when dynamic key management is established, this field can default to zero. The Host Interface process or BASE24 Interchange (BIC) Interface process increments the counter automatically and attempts to resynchronize it with the external processor. Refer to the appropriate interface documentation to determine whether a specific Interchange Interface process uses this field.

INBOUND KEYS

The following fields contain the PIN keys used for inbound transactions (that is transactions sent by the external processor to BASE24).

PIN KEY1 — Identifies the first key used by the external processor to encrypt PINs in the transactions sent to BASE24. This field must contain 16 valid characters. Valid values for each character in this field are 0 through 9 and A through F.

Note: On KEY6 screen 2, this field occurs twice. Each field must contain 16 valid characters to define a double-length PIN encryption key.

Two keys can be specified; the one actually used is indicated by the value in the CURRENT INDEX field. When two keys are specified, BASE24 considers the key not currently in use to be the older of the two keys.

- If the ENCRYPT TYPE field on KEYF or KEY6 screen 1 is set to a value of 1 (security module PIN management), these keys must be encrypted by or for a security module, such that the security module can decrypt the key when it is sent by BASE24.
- If the ENCRYPT TYPE field on KEYF or KEY6 screen 1 is set to a value of 2 (software DES PIN management), these keys must be encrypted by the BASE24 Supercrypt utility prior to being entered in the KEYF or KEY6. For information on the Supercrypt utility, refer to appendix A.

If an external processor supports automatic inbound key changes, this field can be set to the default value of zero, and the interface process automatically sets this field when it is first started.

CHECK DIGITS1 — Identifies the check digits associated with the key entered in the PIN KEY1 field. Valid values for each character in this field are 0 through 9 and A through F. This field must contain four valid characters. A value of 0000 in this field indicates check digits are not being used with the key.

If an external processor supports automatic outbound key changes, this field can be set to the default value of zero, and the interface process automatically sets this field when it is first started.

CURRENT INDEX — Indicates which outbound PIN encryption key is currently being used by the interface process. Valid values are as follows:

1 = Value in the PIN KEY1 field

2 = Value in the PIN KEY2 field

If an external processor supports automatic outbound key changes, this field can be set to the default value of 1, and the interface process automatically sets this field when it is first started.

PIN KEY2 — Identifies a second key used by the external processor to encrypt PINs in the transactions sent to BASE24. This field must contain 16 valid characters. Valid values for each character in this field are 0 through 9 and A through F.

Note: On KEY6 screen 2, this field occurs twice. Each field must contain 16 valid characters to define a double-length PIN encryption key.

Two keys can be specified; the one actually used is indicated by the value in the CURRENT INDEX field. When two keys are specified, BASE24 considers the key not currently in use to be the older of the two keys.

- If the ENCRYPT TYPE field on KEYF or KEY6 screen 1 is set to a value of 1 (security module PIN management), these keys must be encrypted by or for a security module, such that the security module can decrypt the key when it is sent by BASE24.
- If the ENCRYPT TYPE field on KEYF or KEY6 screen 1 is set to a value of 2 (software DES PIN management), these keys must be encrypted by the BASE24 Supercrypt utility prior to being entered in the KEYF or KEY6. For information on the Supercrypt utility, refer to appendix A.

If an external processor supports automatic outbound key changes, this field can be set to the default value of zero, and the interface process automatically sets this field when it is first started.

CHECK DIGITS2 — Identifies the check digits associated with the key entered in the PIN KEY2 field. Valid values for each character in this field are 0 through 9 and A through F. This field must contain four valid characters. A value of 0000 in this field indicates check digits are not being used with the key.

If an external processor supports automatic outbound key changes, this field can be set to the default value of zero, and the interface process automatically sets this field when it is first started.

KEY COUNTER — Identifies the number of times the outbound PIN key has been changed by dynamic key management processing. This counter is increased whenever the outbound PIN key is changed by BASE24 or the external processor. Valid values for each character in this field are 0 through 9 and A through F. This field must contain 6 valid characters.

- If the external processor expects a value other than zero when dynamic key management is established, this field must be set to the value specified by the external processor.
- If dynamic key management is not used or if the external processor does not require a specific initial counter value when dynamic key management is established, this field can default to zero. The Host Interface process or BASE24 Interchange (BIC) Interface process increments the counter automatically and attempts to resynchronize it with the external processor. Refer to the appropriate interface documentation to determine whether a specific Interchange Interface process uses this field.

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4: BASE24 Device Handler PIN Encryption Support

This section presents a detailed discussion of the PIN encryption support provided by BASE24-atm, BASE24-pos, and BASE24-teller Device Handler processes or modules.

BASE24 Device Handler Processes or Modules

BASE24 Device Handler processes or modules are responsible for the interface between BASE24 and the EFT terminals attached to it. As such, they assume responsibility for PIN encryption between BASE24 and its EFT terminals. They manage any keys required for encryption. They also pass on or decrypt PINs on entry into the system, depending on the requirements of the terminal and system.

The following documentation includes the PIN encryption options provided for BASE24-atm, BASE24-pos, and BASE24-teller devices. While similar, each product group has its own unique characteristics and is discussed separately.

BASE24-atm, BASE24-pos, and BASE24-teller support a variety of Device Handler processes or modules, with each Device Handler process or module supporting its own specific PIN encryption requirements. The following documentation includes the PIN encryption options provided by BASE24 products in general. The BASE24 options selected for each Device Handler process or module are based on a combination of device capabilities and user preferences.

BASE24-atm Device Handler Processes

BASE24-atm Device Handler processes are responsible for the interface between BASE24-atm and the terminals attached to it. The following documentation addresses PIN encryption within BASE24-atm, including the database settings available and resulting internal message values.

Refer to vendor documentation for PIN encryption-related configuration requirements for each device. These are settings that must be established at the device itself, and may or may not affect the BASE24 database.

Device Handler Process Keys

There are two types of keys that are used in PIN encryption between terminals and the BASE24-atm Device Handler processes: transaction keys and master keys.

Transaction Keys

Terminals can send PINs to the Device Handler process in the clear or in encrypted form. If PINs are transmitted in the clear, no special key management is required between the terminal and the Device Handler process. However, if PINs are encrypted by a terminal, they must be encrypted under what is known as a transaction key (also known as a session key).

A transaction key is an encryption key that the Device Handler process distributes electronically to terminals using the same lines over which all other communications travel.

Transaction keys are stored in the BASE24-atm Terminal Data files so that the key is available for decrypting PINs coming from the terminal. However, this transaction key is never displayed or logged anywhere in the system.

Master Keys

Since transaction keys are distributed using the same lines as other communications, a means of sending transaction keys in a secure manner is required. This security is provided by a separate master key known to both the terminal and BASE24-atm. This master key is used to encrypt transaction keys for their movement over the communications lines.

The standard BASE24-atm product does not support the transmission of master keys over communications lines. Each master key must be physically entered at the terminal. The BASE24-atm Automated Key Distribution System add-on product automates the otherwise error-prone and labor-intensive process of generating, protecting, and hand-entering ATM master keys at each ATM in your system. The BASE24-atm Automated Key Distribution System add-on product, which requires the use of the Transaction Security Services process, is described in the ***BASE24-atm NCR 5XXX Device Support Manual***, ***BASE24-atm Diebold 10XX/478X Device Support Manual***, and the ***BASE24 Transaction Security Services Processing Guide***.

ACI hard codes a terminal master key in each of its BASE24-atm Device Handler processes. If transaction keys are to be encrypted in software, the hard-coded master keys must be used as the terminal master keys. This method does not require that a master key be placed in the BASE24 database for each terminal.

If transaction keys are to be encrypted by a security module for the standard BASE24-atm product, an individual master key must be resident in the BASE24-atm Terminal Data files record (or in the Channel Security File (CSECD) record if using the Transaction Security Services process) for each terminal. Under this latter method, clear and encrypted versions of the master key can be determined individually for each terminal. The clear version of the master key must be entered at the terminal and the encrypted version is entered in the terminal's BASE24-atm Terminal Data files or CSECD record.

When using the BASE24-atm Automated Key Distribution System add-on product, you do not need to manually enter ATM master keys. They are automatically generated and downloaded to the terminals and stored in the Transaction Security Service (TSS) database.

Transmitting Transaction Keys to the Device

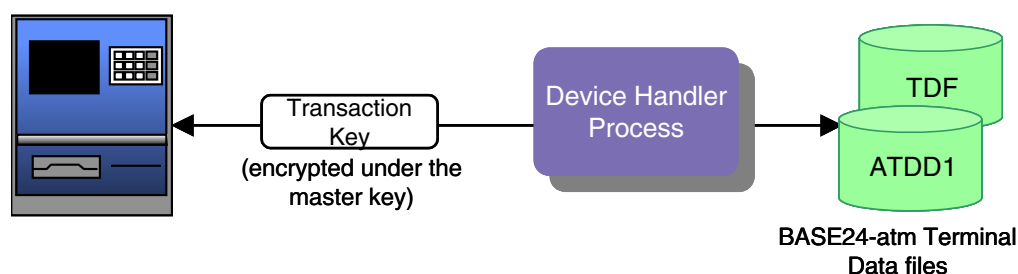
Device Handler processes normally transmit new transaction keys to the terminals during load and open sequences. Some Device Handler processes also periodically transmit new transaction keys once the terminals are up and running. In addition, the transmission of a new transaction key can also result from a command to the Device Handler process from the BASE24 Device Control Terminal (DCT). When a new transaction key must be sent to a terminal, the Device Handler process checks the PIN ENCRYPT TYPE field on the device-specific BASE24-atm Terminal Data files (ATD) screen 9 to determine whether PIN management is performed in software or using a security module.

If a transaction key load is unsuccessful, the Device Handler process prevents the ATM from opening until a successful transaction key load occurs. This keeps transaction keys used by the ATM and the Device Handler process synchronized. BASE24-atm denies transactions for invalid PIN or sanity check errors when transaction keys are not synchronized.

When a new transaction key must be sent to a terminal, the Device Handler process checks the PIN ENCRYPT TYPE field on ATD screen 9 for the terminal record.

Software PIN Management

If the value in the PIN ENCRYPT TYPE field on the device-specific ATD screen 9 is set to a value of 01 or 04, indicating that BASE24 PIN management is performed in software, the Device Handler process generates a new transaction key, encrypts that key under its hard-coded master key, and sends the transaction key to the terminal. It then places the new transaction key in the BASE24-atm Terminal Data files record for its own use. The Device Handler process must store the transaction key in the BASE24-atm Terminal Data files so that the key is available for decrypting PINs coming from the terminal. However, this transaction key is never displayed or logged anywhere in the system. The following graphic illustrates this processing.



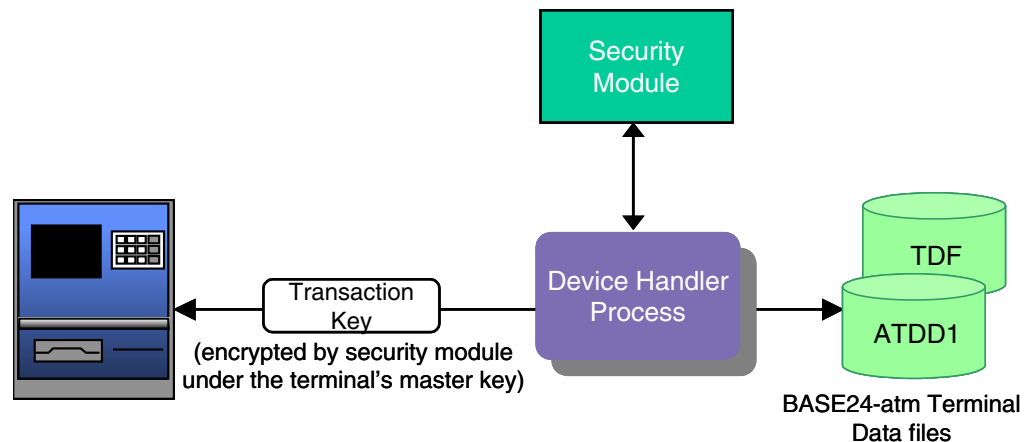
The terminal can decrypt the transaction key using the master key resident at the terminal.

BASE24-atm sends a new transaction key to the terminal encrypted under the terminal encrypted under the terminal master key (hardcoded in the Device Handler process). The Device Handler process then stores a clear version of the transaction key in the BASE24-atm Terminal Data files for its own use.

Security Module PIN Management

If the value in the PIN ENCRYPT TYPE field on the device-specific ATD screen 9 is set to a value of 03 or 05, indicating that BASE24 PIN management is performed using a security module, the Device Handler process requests a new transaction key from the security module used in the system. To do this, the Device Handler process provides the encrypted terminal master key to the security device (taken from the MASTER KEY field on the device-specific ATD screen 9 record).

The security module then randomly generates a transaction key and encrypts it under the terminal master key for sending to the terminal. It also provides the Device Handler process with the new transaction key encrypted under the security module master key for storage in the BASE24-atm Terminal Data files. The Device Handler process must store the encrypted transaction key in the BASE24-atm Terminal Data files so that the key can be included in the internal message for transactions coming from the terminal. Here again, this transaction key is never displayed or logged anywhere in the system. The following graphic illustrates this processing.



The terminal can decrypt the transaction key using the master key resident at the terminal.

BASE24-atm requests a new transaction key from the security module encrypted under the terminal's master key and sends a copy to the terminal. The Device Handler process stores an encrypted version of the transaction key in the BASE24-atm Terminal Data files.

Terminal PIN Block Requirements

The type of PIN blocks required for use by terminals connected directly to BASE24-atm depends on the Device Handler process involved.

The following BASE24-atm Device Handler processes support both PIN/PAN and PIN/PAD PIN blocks:

- Diebold 910/911/912
- Diebold 10XX/478X
- Fujitsu 7000
- NCR 5XXX
- Olivetti 6000

The following BASE24-atm Device Handler processes support PIN/PAN PIN blocks only:

- Diebold CashSource Plus
- NCR Cash
- Tidel AnyCard
- Triton

The following BASE24-atm Device Handler processes support PIN/PAD PIN blocks only:

- Burroughs 650/750
- Docutel 5100
- Fujitsu 910/911
- IBM 3624

All BASE24-atm Device Handler processes are hard coded to require the use of a pad character of a hexadecimal F with PIN/PAD PIN blocks.

Setting Up the BASE24 Database

PIN encryption is managed by the Device Handler process at the terminal level. This means that keys can be established separately for each terminal and the decision to transmit encrypted or clear PINs can be made separately for each terminal.

Device Handler processes derive the information they need for each terminal from the BASE24-atm Terminal Data files record.

The fields on the device-specific BASE24-atm Terminal Data files (ATD) screen 9 must be set up as indicated on the following pages for each terminal with which BASE24-atm exchanges messages. Several ATD screen 9 formats exist because different brands and models of ATMs support different functionality. Because of these differences, the ATD screen 9 for a particular Device Handler process may not have all of the fields listed on the following pages. Refer to the ***BASE24-atm Files Maintenance Manual*** and to device-specific BASE24-atm device support manuals for additional information.

MASTER KEY — Specifies the master key used to encrypt transaction keys sent to this terminal. Transaction keys are encrypted based on the value in the PIN ENCRYPT TYPE field on ATD screen 9.

- If transaction keys are transmitted to this terminal in the clear (the PIN ENCRYPT TYPE field is set to a value of 00), this field is not used. In this case, BASE24-atm does not use a master key.
- If transaction keys are encrypted by BASE24-atm for transmission to the terminal (the PIN ENCRYPT TYPE field is set to a value of 01 or 04), this field is not used. In this case, BASE24-atm uses the terminal master key hard-coded in the Device Handler process.
- If transaction keys are encrypted by a security module for transmission to the terminal (the PIN ENCRYPT TYPE field is set to a value of 03 or 05), this field must contain the terminal master key encrypted by or for the specific security module being used in the system.

Note: The same master key is used for PIN encryption and message authentication transaction keys, so settings in the PIN ENCRYPT TYPE and MAC SUPPORT TYPE fields must agree (i.e., none, hardware, or software).

CHECK DIGITS — Specifies the check digits associated with the master key when it is encrypted by or for the specific security module being used in the system. Valid values for each character in this field are 0 through 9 and A through F. This field must contain four valid characters. A value of 0000 in this field indicates check digits are not being used with the master key.

PIN ENCRYPT TYPE — Specifies the type of encryption performed on PINs transmitted from this terminal. This includes all PINs except the new PIN value for PIN change transactions. The value in this field also determines the type of encryption performed on transaction keys transmitted to this terminal.

Values 04 and 05 are valid only for devices that support PIN/PAN PIN blocks. Refer to the topic “Terminal PIN Block Requirements” earlier in this section for a list of Device Handler processes that support PIN/PAN blocks.

- If PINs are transmitted from the terminal in the clear, this field must be set to a value of 00.
- If PINs are transmitted from the terminal in encrypted form using a PIN/PAD PIN block and decrypted by BASE24-atm, this field must be set to a value of 01.
- If PINs are transmitted from the terminal using a PIN/PAD PIN block and encrypted under keys that are only in the clear within a security module, this field must be set to a value of 03.
- If PINs are transmitted from the terminal in encrypted form using a PIN/PAN PIN block and decrypted by BASE24-atm, this field must be set to a value of 04.
- If PINs are transmitted from the terminal using a PIN/PAN PIN block and encrypted under keys that are only in the clear within a security module, this field must be set to a value of 05.

Note: The ANSI standard PIN/PAN PIN block (12 right-most PAN digits, excluding the check digit) is created when the PIN ENCRYPT TYPE field is set to a value of 04 or 05.

PIN CHANGE ENCRYPT TYPE — Specifies the type of encryption performed on the new PIN value transmitted from this terminal during a PIN change transaction. It does not control the type of encryption performed on the existing PIN during a PIN change transaction or on the new PIN entered during the first transaction performed on an account. Values 04 and 05 are valid only for devices

that support PIN/PAN PIN blocks. Refer to the topic “Terminal PIN Block Requirements” previously in this section for a list of Device Handler processes that support PIN/PAN PIN blocks.

A Device Handler process may or may not support clear new PINs. If new PINs must be encrypted, the type of encryption performed on the new PIN value must match the type of encryption performed on existing PINs. If new PINs can be in the clear or be encrypted, the type of encryption performed on the new PIN value must be 00 (clear) or match the value in the PIN ENCRYPT TYPE field, as shown in the following table.

Value in the PIN ENCRYPT TYPE Field	Value in the PIN CHANGE ENCRYPT TYPE Field
00	00
01	00 or 01
03	00 or 03
04	00 or 04
05	00 or 05

Refer to section 8 for operating considerations and additional BASE24 database settings for PIN change transactions.

Processing BASE24-atm Transactions

Terminals can send PINs to BASE24-atm in the clear or encrypted under a transaction key. The type of encryption used by the terminal for the new PIN entered during a PIN change transaction is specified in the PIN CHANGE ENCRYPT TYPE field on the device-specific ATD screen 9 for the terminal record. The type of encryption used by the terminal for all other PINs entered, including the existing PIN entered during a PIN change transaction, is specified in the PIN ENCRYPT TYPE field on the device-specific ATD screen 9 for the terminal record. Therefore, for transactions other than PIN change transactions, a single terminal cannot support a combination of both clear and encrypted PINs. For PIN change transactions, the new PIN information can be encrypted or in the clear when the existing PIN is encrypted.

Processing Incoming Messages

When a transaction is sent to BASE24-atm by a terminal, the Device Handler process creates a Standard Internal Message (STM) for use by BASE24-atm. It sets the PIN-related fields in the STM as described below based on the value in the PIN ENCRYPT TYPE field on the device-specific ATD screen 9 for the terminal record.

No Encryption

If the value in the PIN ENCRYPT TYPE field on the device-specific ATD screen 9 is set to a value of 00 (no encryption), the Device Handler process sets the PIN-related fields in the STM as follows:

PIN Change token	This token contains new PIN information for PIN change transactions. Refer to the topic “Processing Incoming PIN Change Transaction Messages” later in this section for a list of settings for the PIN Change token.
RQST.ANSI-OFST	This field is set to zeros because the PIN is in the clear.
RQST.PIN	This field contains the clear PIN extracted from the terminal request message. The PIN is right-justified and the field is padded to the left with zeros.
RQST.PIN-FRMT	This field is set to a value of 0 (clear PIN).
RQST.PIN-KEY	This field is set to zeros because the PIN is in the clear.
RQST.PINPAD-CHAR	This field is set to zero because the PIN is in the clear.
RQST.PIN-SIZE	This field is calculated from the clear PIN.

Software PIN Management

If the value in the PIN ENCRYPT TYPE field on the device-specific ATD screen 9 is set to a value of 01 or 04, meaning that BASE24 PIN management is performed in software, the Device Handler process decrypts the PIN block using the DES algorithm and the clear transaction key stored in the BASE24-atm Terminal Data files for the terminal. It then extracts the clear PIN from the PIN block and sets the PIN-related fields in the STM as follows:

PIN Change token	This token contains new PIN information for PIN change transactions. Refer to the topic “Processing Incoming PIN Change Transaction Messages” later in this section for a list of settings for the PIN Change token.
RQST.ANSI-OFST	This field is set to zeros because the PIN is in the clear.
RQST.PIN	This field contains the clear PIN extracted from the terminal request message. The PIN is right-justified and the field is padded to the left with zeros.
RQST.PIN-FRMT	This field is set to a value of 0 (clear PIN).
RQST.PIN-KEY	This field is set to zeros because the PIN is in the clear.
RQST.PINPAD-CHAR	This field is set to zero because the PIN is in the clear.
RQST.PIN-SIZE	This field is calculated during extraction of the clear PIN from the clear PIN block.

Security Module PIN Management—PIN/PAD PIN Block

If the value in the PIN ENCRYPT TYPE field on the device-specific ATD screen 9 is set to a value of 03, meaning that BASE24 PIN management is performed using a PIN/PAD PIN block under the keys of a security module, the Device Handler process sets the PIN-related fields in the STM as follows:

PIN Change token	This token contains new PIN information for PIN change transactions. Refer to the topic “Processing Incoming PIN Change Transaction Messages” later in this section for a list of settings for the PIN Change token.
RQST.ANSI-OFST	This field is set to zeros.
RQST.PIN	This field contains the encrypted PIN block from the terminal request message.
RQST.PIN-FRMT	This field is set to a value of 3 (PIN/PAD PIN block).
RQST.PIN-KEY	This field contains the encrypted transaction key stored in the BASE24-atm Terminal Data files.
RQST.PINPAD-CHAR	This field is set to F.
RQST.PIN-SIZE	This field is set to 16 (the size of the PIN block).

Security Module PIN Management—PIN/PAN PIN Block

If the value in the PIN ENCRYPTION TYPE field on the device-specific ATD screen 9 is set to a value of 05, meaning that BASE24 PIN management is performed using a PIN/PAN PIN block under the keys of a security module, the Device Handler process sets the PIN-related fields in the STM as follows:

PIN Change token	This token contains new PIN information for PIN change transactions. Refer to the topic “Processing Incoming PIN Change Transaction Messages” later in this section for a list of settings for the PIN Change token.
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RQST.ANSI-OFST	This field indicates the starting position in the PAN of the 12 digits used to create the PIN/PAN PIN block used in encrypting the PIN. This value can be greater than, less than, or equal to 0. (A 0 indicates the first position of the PAN.) Refer to the topic “PIN/PAN PIN Blocks” in section 2 for more information on how BASE24 creates the PIN/PAN PIN block.
RQST.PIN	This field contains the encrypted PIN block from the terminal request message.
RQST.PIN-FRMT	This field is set to a value of 1 (ANSI PIN block).
RQST.PIN-KEY	This field contains the encrypted transaction key stored in the BASE24-atm Terminal Data files.
RQST.PINPAD-CHAR	Not applicable.
RQST.PIN-SIZE	This field is set to 16 (the size of the PIN block).

Processing Incoming PIN Change Transaction Messages

When a PIN change transaction is sent to BASE24-atm by one of its terminals, the Device Handler process adds a PIN Change token to the STM, and sets the fields in the PIN Change token as described below based on the values in the PIN ENCRYPT TYPE and PIN CHANGE ENCRYPT TYPE fields on the device-specific ATD screen 9 for the terminal record.

No Encryption

If the value in the PIN CHANGE ENCRYPT TYPE field on the device-specific ATD screen 9 is set to a value of 00 (no encryption), the Device Handler process sets the fields in the PIN Change token as follows:

PINC-TKN.NEW-PIN-FRMT	This field is set to a value of 0 (clear PIN).
PINC-TKN.NEW-PIN-OFST	This field is set to zeros. During PIN verification, the Authorization process generates the PIN verification digits for the new PIN and places the value in this field. The Authorization process also places this value in the POFST/PVV field on screen 1 of the Cardholder Authorization File (CAF) when the PIN verification digits are maintained on the BASE24 database.
PINC-TKN.NEW-PIN-SIZE	This field is calculated from the clear PIN.
PINC-TKN.NEW-PIN-1	This field contains the clear PIN extracted from the terminal request message. The PIN is left-justified and the remainder of the field contains blanks.
PINC-TKN.NEW-PIN-2	This field contains the second clear PIN extracted from the terminal request message. The PIN is left-justified and the remainder of the field contains blanks. This field contains blanks if the device compares the two new PIN entries and finds that they match.
PINC-TKN.PIN-CNT	This field is set to 1 or 2, depending on the number of new PIN entries received from the device.

Software PIN Management

If the PIN CHANGE ENCRYPT TYPE field on the device-specific ATD screen 9 contains a value of 01 or 04 (encryption in software), the Device Handler process sets the fields in the PIN Change token as follows:

PINC-TKN.NEW-PIN-FRMT	This field is set to a value of 0 (clear PIN).
PINC-TKN.NEW-PIN-OFST	This field is set to zeros. During PIN verification, the Authorization process generates the PIN verification digits for the new PIN and places the value in this field. The Authorization process also places this value in the POFST/PVV field on screen 1 of the Cardholder Authorization File (CAF) when the PIN verification digits are maintained on the BASE24 database.
PINC-TKN.NEW-PIN-SIZE	This field is calculated during extraction of the clear PIN from the clear PIN block.
PINC-TKN.NEW-PIN-1	This field contains the first clear PIN obtained from decrypting the PIN block in the terminal request message. The PIN is right-justified and the field is padded to the left with zeros.
PINC-TKN.NEW-PIN-2	This field contains the second clear PIN entry obtained from decrypting the PIN block in the terminal request message. The PIN is right-justified and the field is padded to the left with zeros. This field contains blanks if the device compares the two new PIN entries and finds that they match.
PINC-TKN.PIN-CNT	This field is set to 1 or 2, depending on the number of new PIN entries received from the device.

Security Module PIN Management

If the PIN CHANGE ENCRYPT TYPE field on the device-specific ATD screen 9 contains a value of 03 or 05, meaning that BASE24 PIN management is performed under the keys of a security module, the Device Handler process sets the fields in the PIN Change token as follows:

PINC-TKN.NEW-PIN-FRMT	This field is set to a value of 3 (security module PIN management).
PINC-TKN.NEW-PIN-OFST	This field is set to zeros. During PIN verification, the Authorization process generates the PIN verification digits for the new PIN and places the value in this field. The Authorization process also places this value in the POFST/PVV field on screen 1 of the Cardholder Authorization File (CAF) when the PIN verification digits are maintained on the BASE24 database.
PINC-TKN.NEW-PIN-SIZE	This field is set to 16 (the size of the PIN block).
PINC-TKN.NEW-PIN-1	This field contains the encrypted PIN block from the terminal request message for the first new PIN entry.
PINC-TKN.NEW-PIN-2	This field contains the encrypted PIN block from the terminal request message for the second new PIN entry. This field contains blanks if the device compares the two new PIN entries and finds that they match.
PINC-TKN.PIN-CNT	This field is set to 1 or 2, depending on the number of new PIN entries received from the device.

Processing Outgoing Messages

BASE24-atm sets the PIN fields to zeros when the transaction is authorized. PINs are not carried back to the terminals in response messages.

BASE24-pos Device Handler Modules

BASE24-pos Device Handler modules are responsible for the interface between BASE24-pos and the EFT terminals attached to it. The following documentation addresses PIN encryption within BASE24-pos, including the database settings available and resulting internal message values.

Refer to vendor documentation for PIN encryption-related configuration requirements for each device. These are settings that must be established at the device itself, and may or may not affect the BASE24 database.

Device Handler Module Keys

There are two types of keys that are used in PIN encryption between terminals and their BASE24-pos Device Handler modules: transaction keys and master keys.

Transaction Keys

Terminals can send PINs to the Device Handler module in the clear or in encrypted form. If PINs are transmitted in the clear, no special key management is required between the terminal and the Device Handler module. However, if PINs are encrypted by a terminal, they must be encrypted under what is known as a transaction key (also known as a session key).

A transaction key is an encryption key that the Device Handler module distributes electronically to terminals using the same lines over which all other communications travel.

Master Keys

Since transaction keys are distributed using the same lines as other communications, a means of sending transaction keys in a secure manner is required. This security is provided by a separate master key, known to both the terminal and BASE24-pos. This master key is used to encrypt transaction keys for their movement over the communications lines.

Master keys are by nature long-term, and, once in place, are usually left unchanged unless they are compromised. BASE24-pos does not support the transmission of master keys over communications lines. Master keys are not hard coded into BASE24-pos Device Handler modules. Each master key must be physically

entered at the terminal and in the BASE24-pos Terminal Data files or Merchant Host Configuration File (MHCF) record, and optionally in the Derivation Key File (KEYD) or TSS Derivation Key File (DRVKD) for that terminal. The MHCF is used only by the Merchant Host Interface (MHI) module when it operates in host control mode. The KEYD is used only by the Standard POS Device Handler (SPDH) module or BASE24-pos Hypercom Device Handler (HPDH) module when it supports terminals using the derived unique key per transaction (DUKPT) method of encrypting PINs. When using the Transaction Security Services process, the derivation key in the KEYD is set to a key locator value and the actual derivation key is stored in the DRVKD. The DRVKD is maintained from the Derivation Key Configuration window of the ACI Desktop user interface.

If transaction keys are to be encrypted in software, clear versions of these master keys must be entered in the BASE24-pos Terminal Data files and are then scrambled by the Pathway application prior to being stored on disk.

If transaction keys are to be encrypted by a security module, clear and encrypted versions of the master key are determined individually for each terminal. The clear version of the key is entered at the terminal and the encrypted version is entered in the BASE24-pos Terminal Data files or MHCF record.

Transmitting Transaction Keys to the Terminal

There are several instances in which Device Handler modules transmit new transaction keys to their terminals. Device Handler modules normally transmit new transaction keys to their terminals during load sequences. The Standard POS Device Handler (SPDH) module also can periodically transmit new transaction keys once the terminals are up and running, based on settings in the ACI Standard Configuration File (ACNF).

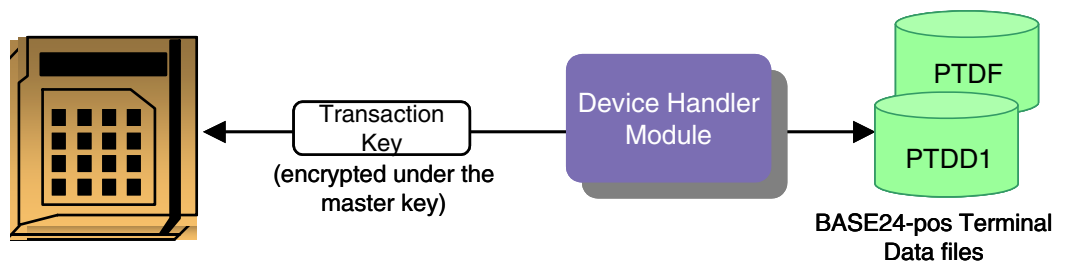
When a new transaction key must be sent to a terminal, the Device Handler module checks the PIN ENCRYPTION TYPE field on BASE24-pos Terminal Data files (PTD) screen 7 for the terminal record to determine whether PIN management is performed in software or using a security module.

When the Merchant Host Interface (MHI) module runs in host control mode, the PIN ENCRYPTION TYPE field on MHCF screen 2 indicates whether the transaction key must be generated in software or using a security module prior to sending the new transaction key. The UPDATE PTDF field on MHCF screen 1 indicates whether the MHI module runs in terminal or host control mode.

Software PIN Management

If the value in the PIN ENCRYPTION TYPE field on PTD screen 7 is set to a value of 01 or 02, indicating that BASE24 PIN management is performed in software, the Device Handler module randomly generates a new transaction key, encrypts that key under the master key in the PIN MASTER KEY field on PTD screen 7, and sends the transaction key to the terminal. It then places the new transaction key in the BASE24-pos Terminal Data files terminal record for its own use. The Device Handler module must store the transaction key in the BASE24-pos Terminal Data files so that the key is available for decrypting PINs coming from the terminal. However, this transaction key is never displayed or logged anywhere in the system. The following graphic illustrates this processing.

When the MHI module runs in host control mode, PIN management processing is similar to that described above except that it uses the PIN ENCRYPTION TYPE and MASTER KEY fields on MHCF screen 2 and stores the generated transaction key in the PTDF. The following graphic illustrates the transmission of transaction keys generated by the Device Handler module.



The terminal can decrypt the transaction key using the master key resident at the terminal.

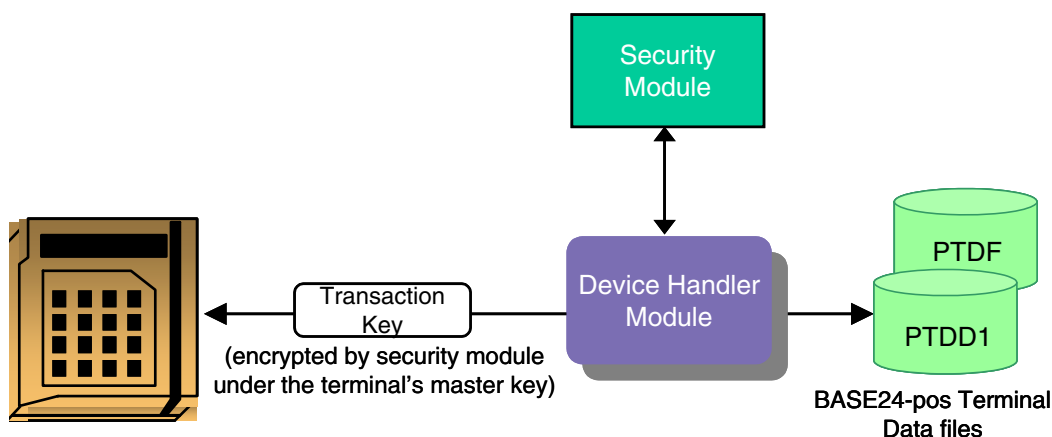
BASE24-pos sends a new transaction key to the terminal encrypted under the terminal master key. The Device Handler module then stores a clear version of the transaction key in the BASE24-pos Terminal Data files for its own use.

Security Module PIN Management

If the value in the PIN ENCRYPTION TYPE field on PTD screen 7 is 03 or 04, indicating that BASE24 PIN management is performed using a security module, the Device Handler module requests a new transaction key from the security module used in the system. To do this, the Device Handler module provides the encrypted terminal master key (taken from the PIN MASTER KEY field on PTD screen 7) to the security device.

The security module then randomly generates a transaction key and encrypts it under the terminal master key for sending to the terminal. It also provides the Device Handler module with the new transaction key encrypted under the security module master key for storage in the BASE24-pos Terminal Data files. The Device Handler module must store the encrypted transaction key in the BASE24-pos Terminal Data files so that the key can be included in the internal message for transactions coming from the terminal. This transaction key is never displayed or logged anywhere in the system. The following graphic illustrates this processing.

When the MHI module runs in host control mode, PIN management processing is similar to that described above except that it uses the PIN ENCRYPTION TYPE and MASTER KEY fields on MHCF screen 2 and stores the generated transaction key in the MHCF. The following graphic illustrates the transmission of transaction keys generated by a security module.



The terminal can decrypt the transaction key using the master key resident at the terminal.

BASE24-pos requests a new transaction key from the security module encrypted under the terminal's master key and sends a copy to the terminal. The Device Handler module stores an encrypted version of the transaction key in the BASE24-pos Terminal Data files.

Setting Up the BASE24 Database

PIN encryption is managed by the Device Handler module at the terminal level. Device Handler modules derive the information they need for each terminal from the BASE24-pos Terminal Data files record or Merchant Host Configuration File (MHCF) record, and optionally from a Derivation Key File (KEYD) or TSS Derivation Key File (DRVKD) record and Logical Network Configuration File (LCONF) records. Fields in the MHCF record are used by the MHI module only. Fields in the KEYD or DRVKD record and assign and param records in the LCONF are used only by Standard POS Device Handler (SPDH) and BASE24-pos Hypercom Device Handler (HPDH) modules for terminals using the derived unique key per transaction (DUKPT) method of PIN encryption.

MHCF Screen 1

MHCF screen 1 contains one field that impacts PIN management processing performed by the MHI module. Refer to the ***BASE24-pos Merchant Host Interface Manual*** for more information on this field.

UPDATE PTDF — Indicates whether the DPC is running in terminal or host control mode.

- If this field is set to a value of 0, the MHI module operates in host control mode and uses the PIN management parameters stored in the MHCF.
- If this field is set to a value of 1, the MHI module operates in terminal mode and uses the PIN management parameters stored in the BASE24-pos Terminal Data files.

MHCF Screen 2

MHCF screen 2 contains PIN encryption data for the MHI module when it operates in host control mode. Refer to the ***BASE24-pos Merchant Host Interface Manual*** for more information on this screen.

PIN ENCRYPTION TYPE — Specifies the type of encryption performed on PINs transmitted from this DPC. The value in this field also determines the type of encryption performed on transaction keys transmitted to this DPC.

- If PINs are not used or are transmitted from the DPC in the clear, this field must be set to a value of 00.

- If PINs are transmitted from the DPC in encrypted form using a PIN/PAD PIN block and decrypted by BASE24-pos, this field must be set to a value of 01.
- If PINs are transmitted from the DPC in encrypted form using a PIN/PAN PIN block and decrypted by BASE24-pos, this field must be set to a value of 02.
- If PINs are transmitted from the DPC using a PIN/PAD PIN block and encrypted under keys that are only in the clear within a security module, this field must be set to a value of 03.
- If PINs are transmitted from the DPC using a PIN/PAN PIN block and encrypted under keys that are only in the clear within a security module, this field must be set to a value of 04.

Note: The ANSI standard PIN/PAN PIN block (12 right-most PAN digits, excluding the check digit) is created when the PIN ENCRYPTION TYPE field is set to a value of 02 or 04.

MASTER KEY — Specifies the master key used to encrypt transaction keys sent to this DPC. Transaction keys are encrypted based on the value in the PIN ENCRYPTION TYPE field.

- If transaction keys for PINs are transmitted to the DPC in the clear (the PIN ENCRYPTION TYPE field is set to a value of 00), this field should be set to zeros.
- If transaction keys for PINs are encrypted by BASE24-pos for transmission to the DPC (the PIN ENCRYPTION TYPE field is set to a value of 01 or 02), this field must contain a clear version of the terminal master key.
- If transaction keys for PINs are encrypted by a security module for transmission to the DPC (the PIN ENCRYPTION TYPE field is set to a value of 03 or 04), this field must contain the terminal master key encrypted by or for the specific security module being used in the system.

PIN PAD CHARACTER — Specifies the pad character used in the formation of the PIN block in the external message if the PIN/PAD PIN block format is used in PIN encryption. This field must contain a value in the range of 0 through 9 or A through F.

PIN KEY VARIANT — Specifies the key variant to be applied to the PIN key exchange key (KEK) when translating PIN keys.

- If the MHI module and the external processor use Atalla hardware security modules, this field must be set to a value of 1 (variant 1).
- If the MHI module uses an Atalla hardware security module and the external processor uses a Thales e-Security hardware security module, this field must be set to a value of 1 (variant 1).
- If the MHI module uses a Thales e-Security hardware security module and the external processor uses an Atalla security module, this field must be set to a value of 1 (variant 1).
- If the MHI module and the external processor use Thales e-Security hardware security modules, this field can be set to a value of 0 (variant 0) or 1 (variant 1); however, the MHI module and the external processor must use the same setting.

PTD Screen 7

Fields on BASE24-pos Terminal Data files (PTD) screen 7 must be set up as indicated below for each terminal with which BASE24-pos exchanges messages. Refer to the ***BASE24-pos Files Maintenance Manual*** and to device-specific BASE24-pos device support manuals for additional information.

PIN PAD CHARACTER — Specifies the pad character used in the formation of the PIN block in the external message if the PIN/PAD PIN block format is used in PIN encryption. This field must contain a value in the range of 0 through 9 or A through F.

PIN ENCRYPTION TYPE — Specifies the type of encryption performed on PINs transmitted from this terminal. The value in this field also determines the type of encryption performed on transaction keys transmitted to this terminal.

- If PINs are not used or are transmitted from the terminal in the clear, this field must be set to a value of 00.
- If PINs are transmitted from the terminal in encrypted form using a PIN/PAD PIN block and decrypted by BASE24-pos, this field must be set to a value of 01.
- If PINs are transmitted from the terminal in encrypted form using a PIN/PAN PIN block and decrypted by BASE24-pos, this field must be set to a value of 02.

- If PINs are transmitted from the terminal using a PIN/PAD PIN block and encrypted under keys that are only in the clear within a security module, this field must be set to a value of 03.
- If PINs are transmitted from the terminal using a PIN/PAN PIN block and encrypted under keys that are only in the clear within a security module, this field must be set to a value of 04.
- If PINs are transmitted from the terminal using a derived unique key per transaction (DUKPT) PIN block and encrypted under keys that are only in the clear within a security module, this field must be set to a value of 07. The DUKPT PIN block is translated into a ANSI standard PIN/PAN PIN block before it is sent to the Router/Authorization module for authorization or routing. This value is used only with the Standard POS Device Handler (SPDH) or BASE24-pos Hypercom Device Handler (HPDH) module.

Note: The ANSI standard PIN/PAN PIN block (12 right-most PAN digits, excluding the check digit) is created when the PIN ENCRYPTION TYPE field is set to a value of 02, 04, or 07.

PIN MASTER KEY — Specifies the master key used to encrypt transaction keys sent to this terminal. Transaction keys are encrypted based on the value in the PIN ENCRYPTION TYPE field on PTD screen 7.

- If transaction keys for PINs are transmitted to the terminal in the clear (the PIN ENCRYPTION TYPE field is set to a value of 00), this field should be set to zeros.
- If transaction keys for PINs are encrypted by BASE24-pos for transmission to the terminal (the PIN ENCRYPTION TYPE field is set to a value of 01 or 02), this field must contain a clear version of the terminal master key.
- If transaction keys for PINs are encrypted by a security module for transmission to the terminal (the PIN ENCRYPTION TYPE field is set to a value of 03 or 04), this field must contain the terminal master key encrypted by or for the specific security module being used in the system.
- If an SPDH-compliant or Hypercom terminal uses the derived unique key per transaction (DUKPT) method of PIN encryption, this field must contain an intermediate PIN block encryption key encrypted by or for the specific security module being used in the system. When the PIN block in an incoming transaction is in DUKPT format, the SPDH or HPDH module translates the block to be encrypted under the intermediate PIN block encryption key.

Note: If you are using the Transaction Security Services process, this field must be set to a key locator value used to retrieve the actual terminal master key or intermediate PIN block encryption key from the Inbound Key field on the PIN Information tab of the Channel Security Configuration window of the ACI Desktop user interface.

CHECK DIGITS — Specifies the check digits associated with the PIN master key when it is encrypted by or for the specific security module being used in the system. Valid values for each character in this field are 0 through 9 and A through F. This field must contain four valid characters. A value of 0000 in this field indicates check digits are not being used with the PIN master key.

VALIDATE PIN — Specifies whether the terminal has the capability of accepting PINs with transactions.

- If the terminal has the capability of accepting PINs with transactions, this field must be set to a value of Y. The Device Handler module sets position 3 of the PT-SRV-ENTRY-MDE field in the PSTM to a value of 1 (PIN entry capability).
- If the terminal does not have the capability of accepting PINs with transactions, this field must be set to a value of N. The Device Handler module sets position 3 of the PT-SRV-ENTRY-MDE field in the PSTM to a value of 2 (no PIN entry capability).

DERIVATION KEY GROUP — Specifies the Derivation Key File (KEYD) group with which this terminal is associated for retrieval of the derivation key. This field is an optional field that allows multiple terminal records to be associated with the same KEYD record. This field is used only for ACI Standard POS Device Handler-compliant and Hypercom terminals using the derived unique key per transaction (DUKPT) method of PIN encryption (i.e., the PIN ENCRYPTION TYPE field is set to a value of 07).

KEYD Screen 1

KEYD screen 1 contains two fields that impact PIN management processing performed by the SPDH or HPDH module. Refer to the *BASE24 Base Files Maintenance Manual* for more information on this screen.

DERIVATION KEY — Specifies the security module encrypted form of the double-length derivation key used to derive a unique PIN encryption key for each transaction. This is the same initial key loaded into the terminal security module

(TSM) of an ACI standard POS device or Hypercom device that is used to derive a unique PIN encryption key for each transaction sent to the BASE24 system. The derivation key must be encrypted under a double-length Master File Key (MFK) variant of 8 for Atalla security devices or a double-length Local Master Key (LMK) pair variant of 28–29 for Thales e-Security security devices before it can be entered in this field.

Note: If the SPDH or HPDH module is using the Transaction Security Services process, this field must be set to a key locator value used to retrieve the actual derivation key from the Derivation Key field on the Derivation Key Configuration window of the ACI Desktop user interface.

CHECK DIGITS — Specifies the check digits corresponding to the values in the DERIVATION KEY fields. Valid values for each position in this field are 0 through 9 and A through F. This field must contain four valid characters.

The check digits can be obtained from the utility used to encrypt the key before it is entered in the KEYD.

LCONF

The LCONF contains seven params that impact PIN management processing performed by the SPDH or HPDH module for derived unique key per transaction (DUKPT) processing. Each of these params are described below.

POS-DH-KSN-DESCR — Specifies the default key serial number (KSN) descriptor required by the Thales e-Security security device to perform a translate command when unique keys are derived per transaction at the terminal to encrypt the PIN block.

POS-DH-KSN-APPRV-OPT — Controls the action taken by the SPDH or HPDH module if the static data received in the Key Serial Number (KSN) and Descriptor field (i.e., FID 6, subFID T) for an SPDH-compliant device, or the DUKPT Key Serial Number field (field 33) for a Hypercom device, of a request message does not match the KSN static data maintained in PTD device dependent data. Valid values are as follows:

- 1 = Deny the transaction
- 2 = Approve the transaction but do not update KSN data in the PTD
- 3 = Approve the transaction and update the KSN data in the PTD (default)

POS-DH-KSN-CNTR-APPRV-OPT — Controls the action taken by the SPDH or HPDH module if the transaction counter has not been incremented or the difference in the transaction counter between two transactions from the same terminal exceeds the limit set in the POS-DH-KSN-MAX-DIFFERENCE param. The previous transaction counter is maintained in the Key Serial Number (KSN) static data stored in PTD device dependent data, and the current transaction counter is carried in the Key Serial Number (KSN) and Descriptor field (i.e., FID 6, subFID T) for an SPDH-compliant device, or the DUKPT Key Serial Number field (field 33) for a Hypercom device, of the request message. Valid values are as follows:

- 1 = Deny the transaction
- 2 = Approve the transaction but do not update the KSN data in the PTD
- 3 = Approve the transaction and update the KSN data in the PTD (default)

POS-DH-KSN-MAX-DIFFERENCE — The largest allowable difference in the transaction counter between two transactions from the same terminal. If the transaction counter received in a message is greater than the transaction counter stored in the device-dependent area of the BASE24-pos Terminal Data files (PTD), the SPDH or HPDH module subtracts the transaction counter in the PTD from the transaction counter in the message. If the difference between the two counters is greater than the value of this param, the SPDH or HPDH module performs the action indicated by the POS-DH-KSN-CNTR-APPRV-OPT param. Otherwise, the SPDH or HPDH module sets the KSN and KSN descriptor (SPDH module only) in the device-dependent area of the PTD to the values from the KSN in the message and continues processing.

For each transaction received from the terminal, the SPDH or HPDH module stores the KSN, including the transaction counter, from the request message in the device-dependent area of the BASE24-pos Terminal Data files (PTD). Valid values are 0–4096. A value of 0 indicates that KSN counter difference checking is not performed.

POS-DH-KSN-SAVE-OPT — Specifies whether the Standard POS Device Handler (SPDH) module restores the KSN when an out-of-sequence condition occurs (i.e., the module declines a transaction with a response code of 899). If this param is set to Y, the module saves and restores the KSN from the previous transaction in the device-dependent area of the BASE24-pos Terminal Data files (PTD) for an out-of-sequence condition. If this param is not configured, is invalid,

or is set to N, the module does not save and restore the KSN from the previous transaction in the device-dependent area of the PTD for an out-of-sequence condition. Valid values are as follows:

Y = Yes, save and restore the KSN from the previous transaction in the device-dependent area of the BASE24-pos Terminal Data files (PTD) for an out-of-sequence condition.

N = No, do not save and restore the KSN from the previous transaction in the device-dependent area of the PTD for an out-of-sequence condition.

Note: The counter in the KSN is supposed to increase by one for each transaction processed as a means of preventing fraud. Setting this param to Y prevents the counter in the KSN from being increased after an out-of-sequence transaction.

POS-DH-KEYD-READ-FROM-DISK — Specifies whether the Standard POS Device Handler (SPDH) or BASE24-pos Hypercom Device Handler (HPDH) module reads the Derivation Key File (KEYD) from disk or from memory. Valid values are as follows:

Y = Yes, read the KEYD from disk.

N = No, do not read the KEYD from disk. The file is read from extended memory. The KEYD is loaded into extended memory during initialization.

POS-DH-DUKPT-UPDATE-METHOD — Specifies whether the Standard POS Device Handler (SPDH) or BASE24-pos Hypercom Device Handler (HPDH) module can automatically update the PIN ENCRYPTION TYPE field on BASE24-pos Terminal Data files (PTD) screen 7 to a value of 07 (DUKPT) when the Key Serial Number (KSN) and Descriptor field (i.e., FID 6, subFID T) for an SPDH-compliant device, or the DUKPT Key Serial Number field (field 33) for a Hypercom device, is received in a message for the first time and a Derivation Key File (KEYD) record exists for the SPDH-compliant or Hypercom terminal. Valid values are as follows:

Y = Yes, automatically update the PIN encryption method in the BASE24-pos Terminal Data files.

N = No, do not automatically update the PIN encryption method in the BASE24-pos Terminal Data files.

Processing BASE24-pos Transactions

Terminals can send PINs to BASE24-pos in the clear or encrypted under a transaction key. The type of encryption used by the terminal is specified in the PIN ENCRYPTION TYPE field on the BASE24-pos Terminal Data files (PTD) screen 7 or on Merchant Host Configuration File (MHCF) screen 2 for the terminal record. Therefore, a single terminal cannot support a combination of both clear and encrypted PINs.

Processing Incoming Messages

When a transaction is sent to BASE24-pos by a terminal, the Device Handler module creates a POS Standard Internal Message (PSTM) for use by BASE24-pos. It sets the PIN-related fields in the PSTM as described below based on the PIN ENCRYPTION TYPE field on PTD screen 7 or on MHCF screen 2.

No Encryption

If the value in the PIN ENCRYPTION TYPE field on PTD screen 7 or on MHCF screen 2 is set to a value of 00 (no encryption), the Device Handler module sets the PIN-related fields in the PSTM as follows:

ANSI-OFST	This field is not applicable.
PIN	This field contains the clear PIN extracted from the terminal request message. The PIN is right-justified and the field is padded to the left with zeros.
PIN-FRMT	This field is set to a value of 0 (clear PIN).
PIN-KEY	This field is not applicable.
PINPAD-CHAR	This field is not applicable.
PIN-SIZE	This field is calculated from the clear PIN.

Software PIN Management

If the value in the PIN ENCRYPTION TYPE field on PTD screen 7 or on MHCF screen 2 is set to a value of 01 or 02, meaning that BASE24 PIN management is performed in software, the Device Handler module decrypts the PIN block using

the DES algorithm and the clear transaction key stored in the BASE24-pos Terminal Data files or MHCF for the terminal. It then extracts the clear PIN from the PIN block and sets the PIN-related fields in the PSTM as follows:

ANSI-OFST	This field is not applicable.
PIN	This field contains the clear PIN extracted from the terminal request message. The PIN is right-justified and the field is padded to the left with zeros.
PIN-FRMT	This field is set to a value of 0 (clear PIN).
PIN-KEY	This field is not applicable.
PINPAD-CHAR	This field is not applicable.
PIN-SIZE	This field is calculated during extraction of the clear PIN from the clear PIN block.

Security Module PIN Management—PIN/PAD PIN Block

If the value in the PIN ENCRYPTION TYPE field on PTD screen 7 or on MHCF screen 2 is set to a value of 03, meaning that BASE24 PIN management is performed using a PIN/PAD PIN block under the keys of a security module, the Device Handler module sets the PIN-related fields in the PSTM as follows:

ANSI-OFST	This field is not applicable.
PIN	This field contains the encrypted PIN block from the terminal request message.
PIN-FRMT	This field is set to a value of 3 (PIN/PAD PIN block).
PIN-KEY	This field contains the encrypted transaction key stored in the BASE24-pos Terminal Data files or in the MHCF.
PINPAD-CHAR	This field is set to the value of the PIN PAD CHARACTER field on PTD screen 7 or on MHCF screen 2.
PIN-SIZE	This field is set to 16 (the size of the PIN block).

Security Module PIN Management—PIN/PAN PIN Block

If the value in the PIN ENCRYPTION TYPE field on PTD screen 7 or on MHCF screen 2 is set to a value of 04, meaning that BASE24 PIN management is performed using a PIN/PAN PIN block under the keys of a security module, the Device Handler module sets the PIN-related fields in the PSTM as follows:

ANSI-OFST	This field indicates the starting position in the PAN of the 12 digits used to create the PIN/PAN PIN block used in encrypting the PIN. This value can be greater than, less than, or equal to 0. (A 0 indicates the first position of the PAN.) Refer to the topic “PIN/PAN PIN Blocks” in section 2 for more information on how BASE24 creates the PIN/PAN PIN block.
PIN	This field contains the encrypted PIN block from the terminal request message.
PIN-FRMT	This field is set to a value of 1 (ANSI PIN block).
PIN-KEY	This field contains the encrypted transaction key stored in the BASE24-pos Terminal Data files or in the MHCF.
PINPAD-CHAR	This field is not applicable.
PIN-SIZE	This field is set to 16 (the size of the PIN block).

Security Module PIN Management—DUKPT PIN Block

If the value in the PIN ENCRYPTION TYPE field on PTD screen 7 is set to a value of 07, meaning that BASE24 PIN management is performed using a DUKPT PIN block under the keys of a security module, the Standard POS Device Handler (SPDH) or BASE24-pos Hypercom Device Handler (HPDH) module translates the DUKPT PIN block into a PIN/PAN PIN block and sets the PIN-related fields in the PSTM as described above for a PIN/PAN PIN block. The PIN/PAN PIN block is encrypted under the intermediate PIN block encryption key from the PIN MASTER KEY field on PTD screen 7 (or from the Inbound Key field on the PIN Information tab of the Channel Security Configuration window if you are using the Transaction Security Services process).

If the value in the PIN ENCRYPTION TYPE field on PTD screen 7 is not set to a value of 07, the SPDH or HPDH module requests a security module or the Transaction Security Services process to randomly generate a PIN encryption key. The module stores this value in the PIN MASTER KEY field on PTD screen 7 (or the Transaction Security Services process stores this value in the Inbound Key field on the PIN Information tab of the Channel Security Configuration window) when it receives a transaction containing a KSN and the POS-DH-DUKPT-UPDATE-METHOD param is set to a value of Y. The SPDH or HPDH module also automatically updates the PIN ENCRYPTION TYPE field on PTD screen 7 to a value of 07 the first time a terminal sends a transaction request containing the Key Serial Number (KSN) and Descriptor field (i.e., FID 6, subFID T) for an SPDH-compliant device, or the DUKPT Key Serial Number field (field 33) for a Hypercom device, and a Derivation Key File (KEYD) record exists for the SPDH-compliant or Hypercom terminal.

If the SPDH or HPDH module does not use the Transaction Security Services process, it also checks the value of the SECURE-DEV-TYP param in the LCONF to determine whether Thales e-Security host security modules are being used. If this param is set to a value of RSM or VSM, the SPDH or HPDH module sets the ORIGINATOR field in the PSTM to a value of 9, which allows the security utilities to build the correct commands when interfacing with the security modules. If the SPDH or HPDH module uses the Transaction Security Services process, the SECURE-DEV-TYP param in the LCONF must be set to a value of TSS. In this case, the Transaction Security Services process determines and builds the correct commands based on the first security device configured on the Security Device Configuration window of the ACI Desktop user interface.

Processing Outgoing Messages

BASE24-pos sets the PIN to zeros when the transaction is authorized. PINs are not carried back to the terminals in response messages.

BASE24-teller Device Handler Processes

BASE24-teller Device Handler processes are responsible for the interface between BASE24-teller and the teller terminals attached to it. The following documentation addresses PIN encryption within BASE24-teller, including the database settings available and resulting internal message values.

Refer to vendor documentation for PIN encryption-related configuration requirements for each device. These are settings that must be established at the device itself, and may or may not affect the BASE24 database.

Note: BASE24-teller Device Handler processes are custom modules designed to meet the capabilities of the device and the requirements of the BASE24-teller system.

Device Handler Process Keys

There are two types of keys that are used in PIN encryption between terminals and their BASE24-teller Device Handler processes: transaction keys and master keys.

Transaction Keys

Terminals can send PINs to the Device Handler process in the clear or in encrypted form. If PINs are transmitted in the clear, no special key management is required between the terminal and the Device Handler process. However, if PINs are encrypted by a terminal, they must be encrypted under what is known as a transaction key (also known as a session key).

A procedure for managing the transaction keys must be included in the Device Handler process design if terminal-encrypted PINs are going to be used. The Teller Terminal Data File (TTDF) record includes provisions for storing the transaction key, if one is used.

Master Keys

Since transaction keys, if used, are distributed to BASE24 terminals using the same lines over which all other communications travel, a means of sending transaction keys in a secure manner is required. This security is provided by a separate master key, known to both the terminal and BASE24. This master key is used to encrypt

transaction keys for their movement over the communications lines. As noted, however, the actual procedure for managing the transaction keys and the exact role of the master key in this procedure is part of the Device Handler process design.

Master keys are by nature long-term and, once in place, are usually left unchanged unless they are compromised. BASE24-teller does not support the transmission of master keys over communications lines. Master keys are not hard coded into BASE24-teller Device Handler processes. Each terminal master key must be physically entered at the terminal and in the TTDF record.

Setting Up the BASE24 Database

PIN encryption is managed by the Device Handler process at the terminal level. Device Handler processes derive the information they need for each terminal from the Teller Terminal Data File (TTDF) record.

Fields on TTDF screen 2 must be set as shown below for each terminal with which BASE24-teller exchanges messages. Refer to the ***BASE24-teller Files Maintenance Manual*** for additional information.

PIN ENCRYPT TYPE — Specifies the type of encryption performed on PINs transmitted from this terminal. The value in this field also determines the type of encryption performed on transaction keys transmitted to this terminal. A method for generating a transaction key and moving it between the Device Handler process and the teller terminal or controller is required when the PIN ENCRYPT TYPE field on TTDF screen 2 is set to a value other than 00.

- If PINs are not used or will be transmitted from the terminal in the clear, this field must be set to a value of 00.
- If PINs will be transmitted from the terminal in encrypted form using a PIN/PAD PIN block and decrypted by BASE24-teller, this field must be set to a value of 01.
- If PINs will be transmitted from the terminal in encrypted form using a PIN/PAN PIN block and decrypted by BASE24-teller, this field must be set to a value of 02.
- If PINs will be transmitted from the terminal using a PIN/PAD PIN block and encrypted under keys that are only in the clear within a security module, this field must be set to a value of 03.

- If PINs will be transmitted from the terminal using a PIN/PAN PIN block and encrypted under keys that are only in the clear within a security module, this field must be set to a value of 04.

MASTER KEY — Specifies the master key used to encrypt transaction keys sent to this terminal. Transaction keys are encrypted based on the value in the PIN ENCRYPT TYPE field on TTDF screen 2.

- If transaction keys will be transmitted to the terminal in the clear (the PIN ENCRYPTION TYPE field is set to a value of 00), this field should be set to zeros.
- If transaction keys will be encrypted by BASE24-teller for transmission to the terminal (the PIN ENCRYPTION TYPE field is set to a value of 01 or 02), this field must contain a clear version of the terminal master key.
- If transaction keys will be encrypted by a security module for transmission to the terminal (the PIN ENCRYPTION TYPE field is set to a value of 03 or 04), this field must contain the terminal master key encrypted by or for the specific security module being used in the system.

CHECK DIGITS — Specifies the check digits associated with the master key when it is encrypted by or for the specific security module being used in the system. Valid values for each character in this field are 0 through 9 and A through F. This field must contain four valid characters. A value of 0000 in this field indicates check digits are not being used with the master key.

PIN ENTRY SUPPORT — Specifies whether the teller terminal has PIN entry capabilities.

- If the teller terminal has PIN entry capabilities, this field must be set to a value of Y.
- If the teller terminal does not have PIN entry capabilities, this field must be set to a value of N.

ANSI PAN FORMAT — Specifies which PAN digits are used to form PIN/PAN PIN blocks at this terminal.

- If the 12 right-most digits excluding the check digit are used to form PIN/PAN PIN blocks (PIN ENCRYPT TYPE field is set to 2 or 4), this field must be set to a value of 0.

- If the 12 right-most digits including the check digit are used to form PIN/PAN PIN blocks (PIN ENCRYPT TYPE field is set to 2 or 4), this field must be set to a value of 1.
- If the 12 left-most digits are used to form PIN/PAN PIN blocks (PIN ENCRYPT TYPE field is set to 2 or 4), this field must be set to a value of 2.
- If clear PINs or PIN/PAD PIN blocks are used (PIN ENCRYPT TYPE field is set to 0, 1, or 3), this field can default to a value of 0.

PIN PAD CHARACTER — Specifies the pad character used in PIN/PAD PIN blocks.

- If PIN/PAD PIN blocks are used (PIN ENCRYPT TYPE field is set to 1 or 3), this field must contain a value from A to F.
- If clear PINs or PIN/PAN PIN blocks are used (PIN ENCRYPT TYPE field is set to 0, 2, or 4), this field can default to F.

Processing BASE24-teller Transactions

Terminals can send PINs to BASE24-teller in the clear or encrypted under a key. The type of encryption used by the terminal is specified in the PIN ENCRYPT TYPE field on TTDF screen 2. Therefore, a single terminal cannot support a combination of both clear and encrypted PINs.

Processing Incoming Messages

When a transaction is sent to BASE24-teller by one of its terminals, the Device Handler process creates a Teller Standard Internal Message (TSTM) for use by BASE24-teller. Each BASE24-teller transaction can be configured in the Teller Transaction File (TTF) to be processed one of four ways:

- Authorized by the BASE24-teller Authorization process.
- Authorized by the host.
- Logged to the Teller Transaction Log File (TTLF) by the BASE24-teller Device Handler process.
- Logged to the TTLF by the BASE24-teller Authorization process.

When a transaction is logged by the Device Handler process, the Device Handler process does not create the PIN token, regardless of whether the transaction contains a PIN. When a transaction that contains a PIN is processed in any of the remaining ways, the Device Handler process adds a PIN token to the TSTM and sets the fields in the token as described on the following pages based on the value in the PIN ENCRYPT TYPE field on TTDF screen 2.

No Encryption

If the value in the PIN ENCRYPT TYPE field on TTDF screen 2 is set to a value of 00 (no encryption), the Device Handler process sets the PIN-related fields in the TSTM as follows:

PIN-TKN.ANSI-OFST	This field is set to zeros because the PIN is in the clear.
PIN-TKN.PIN	This field contains the clear PIN extracted from the terminal request message. The PIN is right-justified and the field is padded to the left with zeros.
PIN-TKN.PIN-FRMT	This field is set to 0 (clear PIN).
PIN-TKN.PIN-KEY	This field is not applicable.
PIN-TKN.PIN-PAD-CHAR	This field is not applicable.
PIN-TKN.PIN-SIZE	This field is calculated from the clear PIN.

Software PIN Management

If the value in the PIN ENCRYPT TYPE field on TTDF screen 2 is set to a value of 01 or 02, meaning that BASE24 PIN management is performed in software, the Device Handler process decrypts the PIN block using the DES algorithm and the clear transaction key stored in the TTDF record for the terminal. It then extracts the clear PIN from the PIN block and sets the PIN-related fields in the TSTM as follows:

PIN-TKN.ANSI-OFST	This field is not applicable.
PIN-TKN.PIN	This field contains the clear PIN obtained from decrypting the PIN block in the terminal request message. The PIN is right-justified and the field is padded to the left with zeros.
PIN-TKN.PIN-FRMT	This field is set to 0 (clear PIN).
PIN-TKN.PIN-KEY	This field is not applicable.
PIN-TKN.PIN-PAD-CHAR	This field is not applicable.
PIN-TKN.PIN-SIZE	This field is calculated during extraction of the clear PIN from the clear PIN block. For a PIN/PAN PIN block, the PIN size is the value in the second digit of the PIN string.

Security Module PIN Management—PIN/PAD PIN Block

If the value in the PIN ENCRYPT TYPE field on TTDF screen 2 is set to a value of 03, meaning that BASE24 PIN management is performed using a PIN/PAD PIN block under the keys of a security module, the Device Handler process sets the PIN-related fields in the TSTM as follows:

PIN-TKN.ANSI-OFST	This field is not applicable.
PIN-TKN.PIN	This field contains the encrypted PIN block from the terminal request message.
PIN-TKN.PIN-FRMT	This field is set to 3 (PIN/PAD PIN blocks).
PIN-TKN.PIN-KEY	This field contains the encrypted transaction key stored in the TTDF.

PIN-TKN.PIN-PAD-CHAR	This field is set to the value of the PIN PAD CHARACTER field in the TTDF.
PIN-TKN.PIN-SIZE	This field is set to 16 (the size of the PIN block).

Security Module PIN Management—PIN/PAN PIN Block

If the value in the PIN ENCRYPT TYPE field on TTDF screen 2 is set to a value of 04, meaning that BASE24 PIN management is performed using a PIN/PAN PIN block under the keys of a security module, the Device Handler process sets the PIN-related fields in the TSTM as follows:

PIN-TKN.ANSI-OFST	This field indicates the starting position in the PAN of the 12 digits used to create the PIN/PAN PIN block used in encrypting the PIN. This value can be greater than, less than, or equal to 0. (A 0 indicates the first position of the PAN.) The Device Handler process uses the value in the ANSI PAN FORMAT field on TTDF screen 2 to identify which PAN digits are used, then determines the starting position. Refer to the topic “PIN/PAN PIN Blocks” in section 2 for more information on how BASE24 creates the PIN/PAN PIN block.
PIN-TKN.PIN	This field contains the encrypted PIN block from the terminal request message.
PIN-TKN.PIN-FRMT	This field is set to 1 (ANSI PIN block).
PIN-TKN.PIN-KEY	This field contains the encrypted transaction key stored in the TTDF.
PIN-TKN.PIN-PAD-CHAR	This field is not applicable.
PIN-TKN.PIN-SIZE	This field is set to 16 (the size of the PIN block).

Processing Outgoing Messages

BASE24-teller sets the PIN to zeros when the transaction is authorized. PINs are not carried back to the terminals in response messages.

5: BASE24 Message Authentication Support

This section provides an introduction to the message authentication support BASE24 provides in certain Host Interface, Interchange Interface, and Device Handler processes.

In reviewing the information presented in this section, note that the term *external processor* can refer to a data processing center (DPC), an interchange, or a co-network as defined in the “Conventions Used in this Manual” section of this manual.

BASE24 Message Authentication

BASE24 provides message authentication support in its ISO Host Interface, BASE24 Interchange (BIC) ISO Interface, certain Device Handler processes, and the following Interchange Interface processes:

- Generic Interchange Interface process
- MDS ISO Interchange Interface process
- PLUS ISO Interchange Interface process
- Visanet ISO Interchange Interface process

For other interchanges, refer to the interchange-specific documentation to determine whether message authentication is supported.

Note: The ISO Host Interface and BASE24 Interchange (BIC) ISO Interface processes support triple DES message authentication using double-length keys without using the Transaction Security Services process.

Message authentication is utilized to ensure transaction messages are received exactly as sent by the legitimate originator. It is designed to protect messages against accidental or deliberate alteration.

The message authentication code (MAC) is generated by the originator of the message, based upon message elements identified in advance by the originator and recipient, and included with the message. To verify a MAC it receives, the recipient computes a MAC based upon the same message elements, then compares the computed MAC and the MAC received with the message. If the two MACs match, the message is valid and has not been altered. However, if the two MACs do not match, the message cannot be validated. This can be because alterations may have occurred in the message or because the originator and recipient are not using the same keys to compute the MAC.

Message authentication can occur between a device and a Device Handler process, between a DPC and an ISO Host Interface process, and between a BIC ISO Interface process and its co-network. The MAC is not carried in any of the BASE24 internal messages. The message authentication performed by each of these links is independent. For example, a transaction originating at a device for authorization at an external processor can go through the authentication procedure four times, as follows:

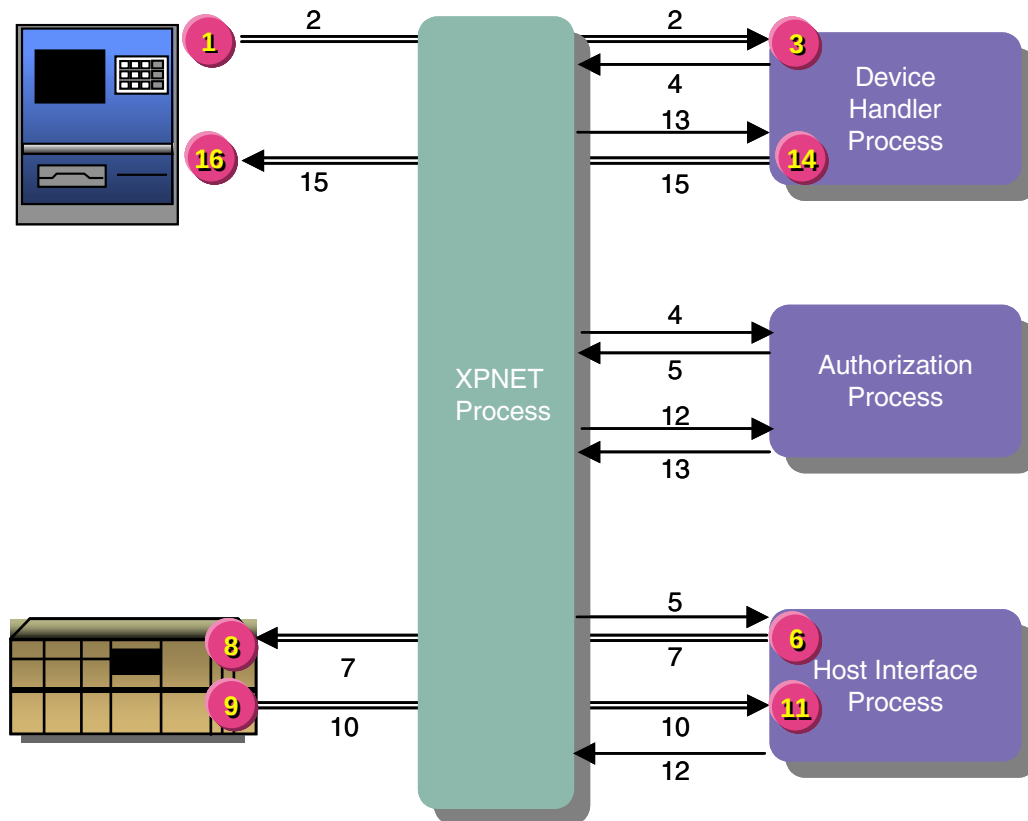
- From the device to the Device Handler process
- From the interface process to the external processor

- From the external processor to the interface process
- From the Device Handler process to the device

Section 6 discusses the message authentication processing of the ISO Host Interface and BIC ISO Interface processes in greater detail, and section 7 discusses the message authentication processing of Device Handler processes in greater detail. For information on the message authentication processing performed by Interchange Interface processes, refer to the interchange-specific documentation.

Transaction Flow

The following transaction flow diagram illustrates message authentication for a transaction originated at a device and authorized by a host. The steps are described following the illustration. The steps in which the transaction message contains a MAC are depicted with double lines in the illustration.



1. The device generates a MAC and includes it in the native-mode message.
2. The device passes the native-mode message to the Device Handler process.
3. The Device Handler process verifies the MAC it receives in the native-mode message.
4. The Device Handler process converts the native-mode message to an internal message and passes the internal message to the Authorization process. The internal message does not contain a MAC.
5. The Authorization process passes the internal message to the Host Interface process because the transaction is being authorized by the host.
6. The Host Interface process generates a MAC and includes it in the external message.

7. The Host Interface process passes the external message to the host.
8. While authorizing the transaction, the host verifies the MAC it receives in the external message.
9. The host generates a new MAC based on information in the response message and includes it in the external message.
10. The host passes the external message to the Host Interface process.
11. The Host Interface process verifies the MAC it receives in the external message.
12. The Host Interface process converts the external message to an internal response message and passes the internal message to the Authorization process. The internal message does not contain a MAC.
13. The Authorization process passes the internal message to the Device Handler process.
14. The Device Handler process converts the internal message to the device's native mode, generates a MAC based on information in the native mode message, and includes the MAC in the native-mode message.
15. The Device Handler process passes the native-mode message to the device.
16. The device verifies the MAC it receives in the native-mode message.

Hardware and Software Authentication

BASE24 performs message authentication in hardware and software. Hardware message authentication requires a security module while software message authentication does not.

The BIC ISO Interface and BASE24 ISO Host Interface processes support message authentication in both hardware and software. For other interchanges, refer to the interchange-specific documentation to determine whether message authentication is supported in hardware and software.

Hardware message authentication can be performed with the Atalla Network Security Processor (NSP) and the Thales e-Security Security Module.

The following BASE24 processes use security modules to perform message authentication:

Process	Product
BASE24 Interchange (BIC) ISO Interface process	BASE24-atm BASE24-pos
BASE24 ISO Host Interface process	BASE24-atm BASE24-pos BASE24-teller
Diebold 910/911 Device Handler process	BASE24-atm
Diebold 912 Device Handler process	BASE24-atm
Diebold 10XX/478X Device Handler process	BASE24-atm
NCR 5XXX Device Handler process	BASE24-atm
Tidel Device Handler process	BASE24-atm
Triton Device Handler process	BASE24-atm
Standard POS Device Handler (SPDH) process	BASE24-pos
Merchant Host Interface (MHI) process	BASE24-pos

A limited number of BASE24 processes also perform message authentication in software:

Process	Product
Standard POS Device Handler (SPDH) process	BASE24-pos
Hypercom POS Device Handler (HPDH) process	BASE24-pos
Merchant Host Interface (MHI) process	BASE24-pos

BASE24-teller Device Handler processes do not support message authentication. The Teller Terminal Data File (TTDF) file maintenance screens contain fields related to message authentication; however, these fields are provided for customization purposes only.

Authentication Inputs

The MAC resulting from generation or verification depends on several variables. All information used to generate a MAC initially must match the information used to verify the MAC when it is received. Parameters for these computations include the following:

- MAC key
- Data to be authenticated
- Location and length of the MAC
- Authentication algorithm

Note: Derived unique key per transaction (DUKPT) message authentication uses a derivation key instead of a MAC key and also includes the key serial number (KSN), length of the KSN, and KSN key descriptor as input for computing the MAC. DUKPT message authentication is only supported in the BASE24-pos Standard POS Device Handler (SPDH) module when using the Transaction Security Services process. For more information on DUKPT message authentication, refer to the ***BASE24 Transaction Security Services Processing Guide***.

MAC Key

The key used to verify the MAC must match the key used to generate the MAC initially. The key used by a device and the key used by the Device Handler process must be the same. Likewise, the key used by an external processor and the key used by the Host Interface or Interchange Interface process must be the same. The key must contain only hexadecimal characters.

Check digits associated with the MAC key are entered with the MAC key in the Key File (KEYF) or Key 6 File (KEY6). New check digits are calculated each time the MAC key is changed. These check digits help verify the accuracy of the key itself and are used each time a MAC is generated or verified to ensure that the security module is using the proper key. BASE24 returns a MAC failure response when a MAC cannot be verified and the check digits returned by the security module match the check digits in the KEYF or KEY6. BASE24 returns a KMAC (MAC key) synchronization error when a MAC cannot be verified and the check digits returned by the security module do not match the check digits in the KEYF or KEY6.

Note: The ISO Host Interface and BASE24 Interchange (BIC) ISO Interface processes must use the KEY6 to support triple DES message authentication using double-length keys without using the Transaction Security Services process.

Data to be Authenticated

The data to be authenticated can be an entire message or selected fields from a message, depending on configuration options provided by the device, the Device Handler process, the external processor, the interface process for the external processor, and the security device. The message originator and recipient must use the same data arranged in the same order when generating and verifying MACs. Data also must be in the same character set (i.e., ASCII or EBCDIC) when generating and verifying MACs.

Note: When using the Atalla NSP, the length of the data being authenticated cannot exceed 4096 bytes. When using the Thales e-Security Security Module, the length of the data being authenticated cannot exceed 400 bytes.

Location and Length of MAC

The MAC generated by the transaction originator is expected to be located in the final 64 bits of the message, whether it is the native-mode message exchanged between a device and a Device Handler process or the ISO-based external message exchanged between an external processor and an interface process.

In an ISO message, the MAC can be in data element P-64 or in data element S-128, depending on whether any secondary data elements are in the message. Data element P-64 contains the MAC when the message contains only primary data elements (P-1 through P-63). Data element S-128 contains the MAC if the message contains any of the other secondary data elements (S-65 through S-127). The MAC can be in data element P-64 or data element S-128, but not both.

According to the ANSI X9.9-1986 standard, *Financial Institution Message Authentication (Wholesale)*, the MAC can be 32, 48, or 64 bits in length. BASE24 supports only the 32-bit, left-justified MAC. The MAC is represented as eight hexadecimal characters.

Authentication Algorithm

The authentication algorithm is the formula used to obtain the MAC from the authentication data and MAC key. BASE24 uses the authentication algorithm defined in the ANSI X9.9-1986 standard, *Financial Institution Message Authentication (Wholesale)* when generating or verifying MACs. The ANSI-standard authentication algorithm is used in all message authentication performed with security modules. It is also used when message authentication is performed in software by the BASE24 Standard POS Device Handler process or BASE24-pos Merchant Host Interface Device Handler process. The same authentication data and MAC key will provide the same MAC, whether authentication is performed in hardware or software.

When you configure double-length MAC keys in the KEY6 for the ISO Host Interface and BASE24 Interchange (BIC) ISO Interface processes, triple DES message authentication is performed without using the Transaction Security Services process.

Note: The BASE24-pos Hypercom Device Handler process uses a proprietary algorithm when performing message authentication in software and uses the ANSI standard algorithm when performing message authentication in hardware. The device configuration and the Device Handler process must agree regarding whether message authentication is performed in hardware or software.

BASE24 Internal Messages

Transactions move through BASE24 systems in the form of standard internal messages, which are passed among the BASE24 processes.

When a message is received from an external entity, BASE24 converts that message from the native-mode format of the external entity to a standard internal format recognizable to the BASE24 processes. The standard internal message formats used by BASE24-atm (STM), BASE24-pos (PSTM), and BASE24-teller (TSTM) are not identical; however, they do contain many of the same fields. When BASE24 completes processing a message, it converts the message from the standard internal format to the native-mode format of the external entity to which BASE24 is sending the message.

Message authentication deals with the messages exchanged between BASE24 and the external entities. A message received from one external entity (e.g., an EFT device) and destined for another external entity (e.g., a DPC or co-network) is authenticated for each external entity independently. As a result, only the following message authentication-related field is carried in the internal messages.

Error Flag — Provides additional information to identify the reason that a system error response code is returned. BASE24 does not return unique response codes for message authentication errors; therefore, BASE24 returns a system error response code and this error flag. The BASE24 utilities responsible for message authentication when it is performed in hardware return reason codes that are used to set the value in this field. Valid values are as follows:

- I = Invalid MAC. This value indicates the MAC received in the message does not match the MAC generated for verification purposes.
- K = KMAC synchronization error. This value indicates the MAC received in the message does not match the MAC generated for verification purposes; however, the two MACs may not match because the sender and receiver are not using the same key.

The Atalla Network Security Processor (NSP) compares the message authentication key check digits returned from the security module with the message authentication key check digits stored in the Key File (KEYF), returning one reason code if the check digits match and another reason code if they do not match. The BASE24 utilities return an error flag value of I if the check digits match and an error flag value of K if the check digits do not match.

The Thales e-Security Host Security Module (HSM) does not compare check digits; therefore, the BASE24 utilities do not return an error flag value of K with the HSM.

M = MAC failure. This value indicates the MAC cannot be generated or verified because of an error involving invalid data or device failure.

STM Field:	AT50-TKN.ERR-FLG
PSTM Field:	PS50-TKN.ERR-FLG
TSTM Field:	TSTMH.RESP-HDR.ERR-FLG

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6: BASE24 Interface Process Message Authentication Support

This section presents a detailed discussion of the message authentication support provided by BASE24 Interface processes.

In reviewing the information presented in this section, note that the term *external processor* can refer to a data processing center (DPC), an interchange, or a co-network as defined in the “Conventions Used in this Manual” section of this manual.

BASE24 Interface Processes

The BASE24 ISO Host Interface and BASE24 Interchange (BIC) ISO Interface processes handle message authentication, or the generation and verification of message authentication codes (MACs), between BASE24 systems and their respective external processors. In the case of a Host Interface process, the external processor is a Data Processing Center (DPC). In the case of a BIC ISO Interface process, the external processor is a co-network.

All external processors are afforded very similar options by BASE24 for message authentication, and the BASE24 Host Interface processes and BIC Interface processes approach message authentication in a similar way. Therefore, their message authentication capabilities and options can be described simultaneously.

BASE24 interface processes provide external messages that are based on ISO message formats. The BASE24 ISO Host Interface and BIC ISO Interface processes use the ISO format, which is based on the 1987 ISO 8583 standard, *Bank Card Originated Messages—Interchange Message Specifications*.

This section applies to BASE24-atm, BASE24-pos, and BASE24-teller. The BASE24 ISO Host Interface process supports all three BASE24 products. The BIC ISO Interface process supports BASE24-atm and BASE24-pos, but does not support BASE24-teller.

For information on Interchange Interface process support of message authentication, refer to the interchange-specific documentation to determine how, and if, message authentication is supported.

Cipher Block Chaining Method Configuration

If you are not using the BASE24 Transaction Security Services process, the LCONF param PSEUDO-3DES-CBC-MAC-TYPE specifies whether to use pseudo 3DES cipher block chaining or 3DES cipher block chaining for message authentication in the ISO Host Interface and BIC ISO Interface processes. Pseudo 3DES cipher block chaining uses a single key while 3DES cipher block chaining requires two keys in the KEY6. If this param is set to Y, pseudo 3DES cipher block chaining is used by the ISO Host Interface and BIC ISO Interface processes. If this param is set to N or is not configured, 3DES cipher block chaining is used by the ISO Host Interface and BIC ISO Interface processes.

This param does not apply when using the BASE24 Transaction Security Services process because you can configure the message authentication method to use in the TSS database.

External Processor Keys

BASE24 identifies four types of keys for use in handling message authentication between BASE24 and an external processor:

- Inbound MAC keys
- Outbound MAC keys
- Previous inbound and outbound keys
- MAC key exchange keys

In most cases, the keys required by BASE24 must be physically placed in the BASE24 database through file maintenance functions. BASE24 utility programs are provided to assist in the preparation of this information. The inbound and outbound key fields in the database can be left with their default values (zeros) if keys are provided by an automatic key change when the interface process is started; however, the MAC key exchange key, if used, must be entered manually.

Inbound MAC Keys

An inbound MAC key is the working key BASE24 uses for verifying MACs in messages coming from an external processor. The external processor maintains this key in its database for generating the MACs in messages being sent to BASE24.

The value entered in the BASE24 database is a cryptogram (i.e., encrypted version) of the inbound MAC key. This cryptogram is obtained using the Atalla master file key (MFK) or the Thales e-Security local master key (LMK). The cryptogram of the key provided by the external processor is encrypted under the MAC key exchange key (KEK). The security module translates the inbound MAC key from encryption under the KEK to encryption under the MFK or LMK. The clear version of the inbound MAC key does not appear in the BASE24 database.

Outbound MAC Keys

An outbound MAC key is the working key BASE24 uses for generating MACs in messages going to an external processor. The external processor maintains this key in its database for verifying the MACs received with messages from BASE24.

The value entered in the BASE24 database is a cryptogram (i.e., encrypted version) of the outbound MAC key. This cryptogram is obtained using the Atalla master file key (MFK) or the Thales e-Security local master key (LMK). The cryptogram of the key provided by the external processor is encrypted under the MAC key exchange key (KEK). The security module translates the outbound MAC key from encryption under the KEK to encryption under the MFK or LMK. The clear version of the outbound MAC key does not appear in the BASE24 database.

Previous Keys

After a key change has taken place between BASE24 and an external processor, BASE24 can receive messages from the external processor containing a MAC generated under the new key or the previous key. Because the external processor does not specify the key under which a message is authenticated, the interface process saves the previous key for use in case the new key does not work.

During BASE24 authentication of an incoming message, the interface process can send both inbound MAC keys to a security module before allowing the transaction to be declined due to a message authentication error. The interface process tries the new key first, and if a message authentication error occurs, tries to use the previous key. A previous key is not used with outbound messages. Only certain external processors support the use of a previous key.

The ISO Host Interface and BIC ISO Interface processes use the CLEAR OLD KEY TIMER VALUE field on Key File (KEYF) screen 3 or Key 6 File (KEY6) screen 4 to control how long the previous key can be used.

MAC Key Exchange Keys

A MAC key exchange key (KEK) is the key BASE24 uses to encrypt or decrypt inbound and outbound MAC keys when BASE24 and the external processor use dynamic key management to change the inbound and outbound MAC keys. This key is an exchange key obtained from the security module. Refer to section 16 for additional information on dynamic key management for interface processes.

A KEK can be single length (16 hexadecimal characters), double length (32 hexadecimal characters), or triple length (48 hexadecimal characters). Using double-length and triple-length KEKs provides additional security by increasing the difficulty of using trial and error to obtain the clear value of the working keys, with triple-length KEKs providing the most security. The KEKs used for encrypting PIN encryption working keys and message authentication working keys both must be the same length, whether single, double, or triple.

Note: Triple-length KEKs are supported only when the Transaction Security Services process is installed. For more information on the Transaction Security Services process, refer to the ***BASE24 Transaction Security Services Processing Guide***.

Message Authentication Options

The message authentication options available on messages between BASE24 and an external processor include the following:

- Messages to be authenticated
- Mandatory authentication data
- Optional authentication data
- Character set

Messages to be Authenticated

The MAC ENCRYPT TYPE field on Key File (KEYF) or Key 6 File (KEY6) screen 1 indicates the type of message authentication supported by BASE24. Most messages exchanged between the BASE24 interface processes and an external processor can be authenticated. BASE24 does not authenticate the following messages:

- Echo test
- Logon
- Logoff
- MAC change key request
- MAC change key response

The MAC is carried in data element P-64 or data element S-128, depending on whether any secondary elements are in the message. Data element P-64 contains the MAC when the message contains only primary data elements (P-1 through P-63). Data element S-128 contains the MAC if the message contains any other secondary data elements (S-65 through S-127). The MAC can be in primary data element P-64 or data element S-128, but not both.

When a BASE24 interface process receives a message, it checks the following message authentication information:

- The value in the MAC ENCRYPT TYPE field on KEYF or KEY6 screen 1 identifies whether message authentication should be performed.

- The External Message File (EMF) record or default settings for the message identify whether data element P-64 or data element S-128 should be in the message. Message authentication is not performed for a message that is not supposed to contain a MAC.
- If message authentication is to be performed, a message authentication code must be present in either data element P-64 or S-128. If a MAC is not present in either data element P-64 or S-128, the message is declined with a 9000-series message.

Mandatory Authentication Data

The data to be authenticated consists of mandatory and optional components. The following components in the BASE24 external message are always included in MAC computations when message authentication is performed:

- Start-of-BASE24-header indicator
- BASE24 header
- Message type identifier
- Primary bit map

Refer to the ***BASE24 External Message Manual*** for additional information about these external message components.

Optional Authentication Data

Any elements in the external message except the ones designated for the MAC (data elements P-64 and S-128) can be included in MAC computations. The BASE24 ISO Host Interface and BIC ISO Interface processes use the External Message File (EMF) and the Key File (KEYF) or Key 6 (KEY6) to determine which elements are included in MAC computations. An EMF record can be defined for each possible combination of interface process, message type (e.g., 0200, 0210, 0420), message direction (i.e., inbound, outbound, or both), and product (i.e., BASE24-atm, BASE24-pos, or BASE24-teller).

Each interface process also has default MAC settings that it can use when an EMF record has not been defined or the EMF is unavailable. Refer to the ***BASE24 External Message Manual*** for default MAC settings for the BASE24 ISO Host Interface process and to the ***BASE24 BIC ISO Standards Manual*** for default MAC settings for the BIC ISO Interface process. An EMF record is not needed if these default settings are appropriate.

The value in the FULL MSG MAC field on EMF screen 1 controls whether selected data elements or the entire message will be considered when computing the MAC. Default settings specify which elements are to be included in a partial message MAC computation; however, default settings do not contain a value for the FULL MSG MAC field on EMF screen 1. When an EMF record has not been defined or the EMF is unavailable, the interface processes use the value in the FULL MESSAGE MAC field on KEYF or KEY6 screen 1 to determine whether to use selected data elements or the entire message to compute the MAC. If an EMF record is available, the value in the FULL MSG MAC field on EMF screen 1 overrides the value in the FULL MESSAGE MAC field on KEYF or KEY6 screen 1.

Note: External messages between the BASE24 interface process and DPCs can contain IMS or CICS transaction code equivalents, which IMS- or CICS-based hosts can establish for their messages. This information is not included in MAC computations.

Character Set

MACs can be generated from data carried in ASCII or EBCDIC character sets; however, the data must be in the same character set when message authentication is performed by the sender and receiver. If the character set is changed between MAC generation and verification, the two MACs will not match. The Host Interface and BIC Interface processes handle the character set the same way when performing message authentication; however, the BIC ISO Interface process performs an extra step related to the character set during logon.

Host Interface Process

When messages exchanged between the BASE24 Host Interface process and a DPC are being authenticated, the data must be in the character set used by the DPC when the MAC is generated or verified. The Host Interface process carries the message internally in the ASCII character set. The MAC DATA TYPE field on KEYF or KEY6 screen 1 identifies the character set being used by the DPC. If the DPC uses the EBCDIC character set, the Host Interface process translates the outgoing message elements being authenticated from ASCII to EBCDIC before passing them to the security module for the MAC to be generated.

BIC ISO Interface Process

When messages exchanged between the BIC ISO Interface process and a co-network are being authenticated, the data must be in the character set carried over data communications lines between the interface process and the co-network. The interface process carries the message internally in the ASCII character set. The MAC DATA TYPE field on KEYF or KEY6 screen 1 identifies the character set being used for data communications. If the EBCDIC character set is used for data communications, the BIC ISO Interface process translates the outgoing message elements being authenticated from ASCII to EBCDIC before passing them to the security module for the MAC to be generated.

When the BIC ISO Interface process receives a logon request from the co-network it performs checks to ensure the following information is the same:

- Data communications character set, identified by the value in the MAC DATA TYPE field on KEYF or KEY6 screen 1. Options are ASCII and EBCDIC.
- MAC encryption type, identified by the value in the MAC ENCRYPT TYPE field on KEYF or KEY6 screen 1. Options are hardware, software, and none.
- Number of keys, identified by the value in the NUMBER OF KEYS field on KEYF or KEY6 screen 1. Options are 1 and 2.

If any of these items do not match between the BIC ISO Interface process and the co-network, the logon request is declined.

Processing Outgoing Messages

If message authentication is supported, a MAC is generated and placed in the outgoing message. If message authentication is not supported, the message continues to be processed normally. The interface process performs the following steps:

1. Determines whether this interface process supports message authentication.
 - If the MAC ENCRYPT TYPE field on KEYF or KEY6 screen 1 contains a 1 (hardware), message authentication is supported and processing continues with step 2.
 - If the MAC ENCRYPT TYPE field on KEYF or KEY6 screen 1 contains a value other than 1, message authentication is not supported and the interface process continues processing the message normally.
2. Determines whether message authentication should be performed for this message by checking the fields on EMF screen 1 for data element P-64 or data element S-128. If the value in the MAC ENCRYPT TYPE field on KEYF or KEY6 screen 1 is set to a value of 1 (hardware) and default values are used instead of the EMF, the interface process continues processing as if data element P-64 and data element S-128 were set to conditional in the EMF.
 - If data element P-64 or data element S-128 is set to conditional, message authentication is performed for this message and processing continues with step 3.
 - If data element P-64 and data element S-128 are set to off, message authentication is not performed for this message and the interface process continues processing the message normally.
3. Determines which fields are to be included in the MAC calculation.
 - If an EMF record is present, the values of the FULL MSG MAC field on EMF screen 1 and the fields on screen 2 of that record control which fields are included.
 - If an EMF record is not present, the value of the FULL MESSAGE MAC field on KEYF or KEY6 screen 1 and the default MAC settings in the interface process control which fields are included.

The mandatory authentication data from the BASE24 external message is always included in the MAC calculation.

4. Builds a data element buffer to contain the portion of the message included in the MAC calculation.

5. Translates the data included in the MAC calculation into the EBCDIC character set, if necessary.
 - If the data is transmitted to the external processor in the EBCDIC character set, the data included in the MAC calculation is translated from ASCII to EBCDIC.
 - If the data is transmitted to the external processor in the ASCII character set, no translation is necessary and processing continues with step 6.
6. Uses a standard set of utilities to perform edit checks on the data to be authenticated. The MAC key must contain only hexadecimal characters.
 - If an Atalla Network Security Processor (NSP) is being used, the length of the data to be authenticated cannot exceed 4096 bytes.
 - If a Thales e-Security Security Module is being used, the length of the data to be authenticated cannot exceed 400 bytes.
7. Requests that a security module generate the MAC.

MAC generation and PIN translation are performed using a single request to the security module when all of the following conditions are met:

- A security module is being used for PIN translation as well as message authentication.
- The outgoing PIN block type is ANSI (PIN/PAN).
- The transaction originates at a BASE24 device or Device Handler process.
- The security module is an NSP. (HSMs do not support MAC generation and PIN translation in a single request.)

If any of these conditions is not met, separate requests are made to the security module for MAC generation and PIN translation. Refer to section 3 for PIN translation procedures.

The interface process provides the security module with the following information:

- The data to be authenticated.
- The length of the data to be authenticated.
- The MAC key taken from the appropriate OUTBOUND MAC KEY field on KEYF screen 2 or KEY6 screen 3.
- The PIN translation information described in section 3, if appropriate.
- The character set of the newly translated PIN.

The security module then performs the following steps:

- a. Translates the PIN, if necessary.
- b. Translates the PIN from ASCII to EBCDIC, if the rest of the message was translated from ASCII to EBCDIC in step 5.
- c. Places the translated PIN in the data element buffer created in step 4, if necessary. This is necessary only if the PIN is translated in this step and is part of the data to be authenticated.
- d. Generates the MAC.
- e. Returns the following to the interface process:
 - The translated PIN block, if PIN translation was performed.
 - The check digits of the PIN key in the internal message and the check digits of the outbound PIN encryption key used by the security module, if PIN translation was performed. (The check digits of the previous encryption key replace the check digits of the PIN key if PIN translation is repeated using the previous encryption key. Refer to step 8.)
 - The MAC.
 - The check digits of the MAC key used by the security module.
8. Repeats step 7 with the previous encryption key, if necessary.

If PIN translation was not performed in step 7, processing continues with step 9.

If the security module successfully translated the PIN block in step 7 using the key taken from the PIN key field in the internal message, processing continues with step 9.

If the security module returned a sanity check error in step 7, the interface process repeats step 7 using the previous PIN encryption key, taken from the previous key field in the internal message. If the security module successfully translates the PIN block, processing continues with step 9.

If the security module returns a sanity check error when repeating step 7 with the previous key or if step 7 cannot be repeated because no previous key exists (i.e., the previous key field in the internal message contains zeros or

blanks), the interface process returns a decline response to the Authorization process with an error code of S (sanity check error) and one of the following response codes:

Product	Response Code
BASE24-atm	70 (system failure)
BASE24-pos	074 (unable to authorize)
BASE24-teller	F0I (system error)

9. Continues processing based on the status code.

The interface process uses a standard set of utilities to communicate with the security module. These utilities return one of the following codes to indicate the outcome of MAC generation processing:

- 0 = No error
- 2 = Device error (recoverable error)
- 3 = Operation failed (unrecoverable error)

If the status code is 0, the interface process places the MAC in data element P-64 or S-128 and places the translated PIN block, if applicable, in data element P-52. The message is ready to be sent to the external processor.

If the status code is 2 and the message is an advice, reversal, representment (POS only), or reconciliation (POS only), the message is placed in the Store-and-Forward File (SAF).

If the status code is 3 and the message is an advice, reversal, representment (POS only), or reconciliation (POS only), the message is dropped.

If the status code is 2 or 3 and the message is not one of those already mentioned, the interface process returns a response to the Authorization process with an error code of M (MAC failure) and one of the following response codes:

Product	Response Code
BASE24-atm	70 (system failure)
BASE24-pos	074 (unable to authorize)
BASE24-teller	F0I (system error)

10. Performs key management, if applicable. The interface process performs the following steps:
 - a. Resets the outgoing consecutive MAC error counter if no error occurred during MAC generation.
 - b. Increments the outgoing total and consecutive PIN error counters if PIN translation was attempted and a sanity check error was returned.
 - c. Increments the outgoing transaction counter for MACs regardless of MAC generation results.
 - d. Increments the outgoing transaction counter for PINs if PIN translation was attempted, regardless of the results.
 - e. Generates change key or new key messages if any counters exceed their thresholds.

Processing Incoming Messages

If message authentication is supported, a MAC received in an incoming message is verified. If message authentication is not supported, the message continues to be processed normally. The interface process performs the following steps:

1. Determines whether this interface process supports message authentication.
 - If the MAC ENCRYPT TYPE field on KEYF or KEY6 screen 1 contains a value of 1 (hardware), message authentication is supported and processing continues with step 2.
 - If the MAC ENCRYPT TYPE field on KEYF or KEY6 screen 1 contains a value other than 1, message authentication is not supported and the BASE24 interface process continues processing the message normally.
2. Determines whether message authentication should be performed for this message. Refer to the topic “Messages to be Authenticated” earlier in this section for the guidelines the interface process uses to make this determination.
 - If this message meets the guidelines, message authentication is performed for this message and processing continues with step 3.
 - If this message does not meet the guidelines, message authentication is not performed for this message and the interface process continues processing the message normally.
3. Determines which fields are to be included in the MAC calculation.
 - If an EMF record is present, the value of the FULL MSG MAC field on EMF screen 1 and the fields in that record control which fields are included.
 - If an EMF record is not present, the value of the FULL MESSAGE MAC field on KEYF or KEY6 screen 1 and the default MAC settings in the interface process control which fields are included.

Note: The mandatory authentication data from the BASE24 external message is always included in the MAC calculation.
4. Builds a data element buffer to contain the portion of the message included in the MAC calculation.

5. Translates the data included in the MAC calculation into the EBCDIC character set, if necessary.
 - If the data is received from the external processor in the EBCDIC character set, the data included in the MAC calculation must be in the EBCDIC character set. If the message has already been translated to the ASCII character set, the data included in the MAC calculation is translated back to the EBCDIC character set.
 - If the data is received from the external processor in the ASCII character set, no translation is necessary and processing continues with step 6.
6. Uses a standard set of utilities to perform edit checks on the data to be authenticated. The MAC key must contain only hexadecimal characters.
 - If an Atalla Network Security Processor (NSP) is being used, the length of the data to be authenticated cannot exceed 4096 bytes.
 - If a Thales e-Security Security Module is being used, the length of the data to be authenticated cannot exceed 400 bytes.
7. Requests that a security module verify the MAC.

The interface process provides the security module with the following information:

- The data to be authenticated.
- The length of data to be authenticated.
- The MAC key taken from the appropriate INBOUND MAC KEY field on KEYF screen 2 or KEY6 screen 3.
- The MAC received from the external processor.

The security module then performs the following steps:

- a. Verifies the MAC.
 - b. Returns the check digits of the MAC key used by the security module to the interface process.
8. Repeats step 7 with the previous MAC key, if necessary.

The interface process uses a standard set of utilities to communicate with the security module. These utilities return one of the following codes to indicate the outcome of MAC verification processing:

- 0 = No error
- 1 = MAC not verified
- 2 = Device error (recoverable error)
- 3 = Operation failed (unrecoverable error)

If the utilities return a status code other than 1 in step 7, processing continues with step 9.

If the utilities return a status code of 1 in step 7 and the MAC key check digits (if available) returned by the security module match the MAC key check digits in the KEYF or KEY6, the interface process repeats step 7 using the MAC key taken from the other INBOUND MAC KEY field on KEYF screen 2 or KEY6 screen 3. Processing continues with step 9.

If step 7 cannot be repeated because no second inbound MAC key exists (i.e., the other INBOUND MAC KEY field on KEYF screen 2 or KEY6 screen 3 contains zeros) or the MAC key check digits do not match, processing continues with step 9.

9. Continues processing based on the status code.

If the utilities return a status code of 0, processing continues with step 10.

If the utilities return a status code of 1, 2, or 3, the interface process returns a reject message to the source of the incoming message and passes a decline response message to the Authorization process. Processing then continues with step 10. Both messages have an error code with one of the following values, based on the status code and MAC check digits:

Status Code	Error Code	Response Code
1	I	Invalid MAC. The two MACs do not match and the MAC check digits returned by the security module match the MAC check digits in the KEYF or KEY6.
1	K	KMAC synchronization error. The two MACs do not match and the MAC check digits returned by the security module do not match the MAC check digits in the KEYF or KEY6. Note: Only NSP security modules can return this error code because other security modules do not return check digits for comparison with information in the KEYF or KEY6.
2 or 3	M	MAC failure. The MACs could not be compared.

Both messages also have one of the following response codes:

Product	Response Code
BASE24-atm	70 (system failure)
BASE24-pos	074 (unable to authorize)
BASE24-teller	FOI (system error)

10. Performs key management, if applicable.
 - a. Resets the incoming consecutive MAC error counter if no error occurred during MAC verification (status code 0 is returned from the utilities using either MAC key).
 - b. Increments the incoming consecutive MAC error counter if the MAC cannot be verified (status code 1 is returned from the utilities). If two MAC keys are available and the MAC cannot be verified using either key, the interface process increments the counter by 1, not 2.
 - c. Increments the incoming transaction error counter for MACs if the MAC cannot be verified (status code 1 is returned from the utilities) and the check digits received in the message match the check digits maintained on the BASE24 database. If two MAC keys are available and the MAC cannot be verified using either key, the interface process increments the counter by 1, not 2.
 - d. Increments the incoming KMAC error counter if the MAC cannot be verified (status code 1 is returned from the utilities) and the check digits received in the message do not match the check digits maintained on the BASE24 database. A second MAC key, if available, is not used when a KMAC error occurs.
 - e. Increments the incoming transaction counters for MACs regardless of MAC verification results. The number of MAC keys used does not affect this counter.
 - f. Generates change key or new key messages if any counters exceed their thresholds.

Setting Up the BASE24 Database

Settings in the Key File (KEYF) or Key 6 File (KEY6), and the External Message File (EMF) must be considered when performing message authentication.

Key File (KEYF) and Key 6 (KEY6) Settings

The following settings are needed in fields on screens 1 and 2 of the Key File (KEYF) or on screens 1 and 3 of the Key 6 File (KEY6) for each external processor performing message authentication. Fields on KEYF screen 3 and KEY6 screen 4 deal exclusively with dynamic key management and are described in section 16. Refer to the *BASE24 Base Files Maintenance Manual* for additional information about these screens.

KEYF or KEY6 Screen 1

KEYF or KEY6 screen 1 contains message authentication parameters, intermediate keys, PIN key exchange keys, and message authentication code (MAC) key exchange keys. Descriptions of the fields required for processing message authentication are as follows:

MAC ENCRYPT TYPE — Specifies how message authentication will be performed for messages between the external processor and the interface process. This field must be set to a value of 1 (security module MAC processing) to indicate message authentication is being performed on messages between BASE24 and this external processor.

MAC DATA TYPE — A code specifying the character set in which messages are being transmitted between the external processor and the interface process.

- If messages are being transmitted in the ASCII character set, this field must be set to a value of 0.
- If messages are being transmitted in the EBCDIC character set, this field must be set to a value of 1.

NUMBER OF KEYS — Identifies whether the inbound and outbound keys defined on KEYF screen 2 or on KEY6 screens 2 and 3 are combined or separate. The value in this field is used when PIN or MAC keys are exchanged automatically. The same number of keys must be used for PINs and MACs.

- If the same key is used for inbound and outbound messages, this field must be set to a value of 1.
- If different keys are used for inbound and outbound messages, this field must be set to a value of 2.

KEY LENGTH — Identifies whether single- or double-length Key Exchange Keys (KEKs) are being used with this external processor. The value in this field is used when PIN and MAC keys are exchanged using dynamic key management. The PIN KEK and MAC KEK must be the same length. Single-length keys contain 16 characters and double-length keys contain 32 characters.

- If single-length KEKs are being used for key exchanges, this field must be set to a value of 1.
- If double-length KEKs are being used for key exchanges, this field must be set to a value of 2.

FULL MESSAGE MAC — Identifies whether selected external message data elements or all external message elements except the ones designated for the MAC (data elements P-64 and S-128) will be considered when computing the MAC. When an EMF record has not been defined or the EMF is unavailable, the interface processes use the value in this field to determine whether to use default groups of data elements or the entire message to compute the MAC. If an EMF record is available, the value in the FULL MSG MAC field on EMF screen 1 overrides the value in this field. Refer to the topic “Optional Authentication Data” earlier in this section for the default settings that are used when no EMF record is available.

- If the default groups of data elements are used to compute the MAC, this field must be set to a value of N.
- If the entire message is used to compute the MAC, this field must be set to a value of Y.

EXCHANGE KEYS

The following two fields are used to create keys that BASE24 uses when MAC keys are automatically exchanged between the Interchange Interface process and the interchange.

MAC KEY — The security module-encrypted form of the Key Exchange Key (KEK) used when inbound and outbound MAC keys are exchanged using dynamic key management. This two-part field contains a single- or double-length KEK, depending on the value in the KEY LENGTH field. Valid values for each character in these fields are 0 through 9 and A through F. Both fields must contain 16 valid characters. If the value in the KEY LENGTH field is 1, the second set of characters must be zeros. Refer to section 16 for additional information about dynamic key management.

CHECK DIGITS — The check digits associated with the MAC KEK. Valid values for each character in this field are 0 through 9 and A through F. This field must contain four valid characters. A value of 0000 in this field indicates check digits are not being used with MAC KEKs.

KEYF Screen 2 or KEY6 Screen 3

KEYF screen 2 or KEY6 screen 3 define the inbound and outbound working keys for MACs. Descriptions of the fields required for processing messages are as follows.

MAC KEY INFORMATION

The MAC Key Information section of KEYF screen 2 or KEY6 screen 3 contains two groups of fields, Outbound Keys and Inbound Keys, each containing six fields.

OUTBOUND KEYS

The following six fields contain the MAC keys used for outbound transactions (that is transactions sent by BASE24 to the external processor).

Note: If an external processor supports automatic key changes, the MAC KEY1, MAC KEY2, CURRENT INDEX, CHECK DIGITS1, and CHECK DIGITS2 fields are automatically set by the interface process when it is first started.

MAC KEY1 — The key used by BASE24 to generate MACs in messages from BASE24 to the external interface. The interface process uses the key in this field when the CURRENT INDEX field contains a value of 1. Valid values for each position in this field are 0 through 9 and A through F. This field must contain 16 valid characters.

Two keys can be specified, although the one to actually use must be indicated using the CURRENT INDEX field. When two keys are specified, BASE24 considers the key not currently in use to be the older of the two keys.

These keys must be encrypted under the BASE24 security module master key and must be known to the external processor.

CHECK DIGITS1 — The check digits corresponding to the value in the MAC KEY1 field. Valid values for each character in this field are 0 through 9 and A through F. This field must contain four valid characters. A value of 0000 in this field indicates check digits are not being used with MAC KEKs.

The check digits can be obtained from the utility used to encrypt the key.

CURRENT INDEX — Indicates which outbound MAC key is currently being used by the interface process. Valid values are as follows:

1 = Value in the MAC KEY1 field

2 = Value in the MAC KEY2 field

MAC KEY2 — The key used by BASE24 to generate MACs in messages from BASE24 to the external interface. The interface process uses the key in this field when the CURRENT INDEX field contains a value of 2. Valid values for each position in this field are 0 through 9 and A through F. This field must contain 16 valid characters.

Two keys can be specified, although the one to actually use must be indicated using the CURRENT INDEX field. When two keys are specified, BASE24 considers the key not currently in use to be the older of the two keys.

These keys must be encrypted under the BASE24 security module master key and must be known to the external processor.

CHECK DIGITS2 — The check digits corresponding to the value in the MAC KEY2 field. Valid values for each character in this field are 0 through 9 and A through F. This field must contain four valid characters. A value of 0000 in this field indicates check digits are not being used with MAC KEKs.

The check digits can be obtained from the utility used to encrypt the key.

MAC KEY COUNTER — Identifies the number of times the outbound MAC key has been changed by dynamic key management processing. This counter is increased whenever the outbound MAC key is changed by BASE24 or the external processor. Valid values for each character in this field are 0 through 9 and A through F. This field must contain 6 valid characters.

- If the external processor expects a value other than zero when dynamic key management is established, this field must be set to the value specified by the external processor.
- If dynamic key management is not used or if the external processor does not require a specific initial counter value when dynamic key management is established, this field can default to zeros. The Host Interface process or BIC ISO Interface process increments the counter automatically and attempts to resynchronize it with the external processor.

INBOUND KEYS

The following six fields contain the MAC keys used for inbound transactions (that is transactions sent by the external processor to BASE24).

Note: If an external processor supports automatic key changes, the MAC KEY1, MAC KEY2, CURRENT INDEX, CHECK DIGITS1, and CHECK DIGITS2 fields will automatically be set by the interface process when it is first started.

MAC KEY1 — The key used by BASE24 to verify MACs generated by the external processor. The interface process uses the key in this field when the CURRENT INDEX field contains a value of 1. Valid values for each position in this field are 0 through 9 and A through F. This field must contain 16 valid characters.

Two keys can be specified, although the one to actually use must be indicated using the CURRENT INDEX field. When two keys are specified, BASE24 considers the key not currently in use to be the older of the two keys.

These keys must be encrypted under the BASE24 security module master key and must be known to the external processor.

CHECK DIGITS1 — The check digits corresponding to the value in the MAC KEY1 field. Valid values for each character in this field are 0 through 9 and A through F. This field must contain four valid characters. A value of 0000 in this field indicates check digits are not being used with MAC KEKs.

The check digits can be obtained from the utility used to encrypt the key.

CURRENT INDEX — Indicates which outbound MAC key is currently being used by the interface process. Valid values are as follows:

- 1 = Value in the MAC KEY1 field
- 2 = Value in the MAC KEY2 field

MAC KEY2 — The key used by BASE24 to verify MACs generated by the external interface. The interface process uses the key in this field when the CURRENT INDEX field contains a value of 2. Valid values for each position in this field are 0 through 9 and A through F. This field must contain 16 valid characters.

Two keys can be specified, although the one to actually use must be indicated using the CURRENT INDEX field. When two keys are specified, BASE24 considers the key not currently in use to be the older of the two keys.

These keys must be encrypted under the BASE24 security module master key and must be known to the external processor.

CHECK DIGITS2 — The check digits corresponding to the value in the MAC KEY2 field. Valid values for each character in this field are 0 through 9 and A through F. This field must contain four valid characters. A value of 0000 in this field indicates check digits are not being used with MAC KEKs.

The check digits can be obtained from the utility used to encrypt the key.

MAC KEY COUNTER — Identifies the number of times the inbound MAC key has been changed by dynamic key management processing. This counter is increased whenever the inbound MAC key is changed by BASE24 or the external processor. Valid values for each character in this field are 0 through 9 and A through F. This field must contain 6 valid characters.

- If the external processor expects a value other than zero when dynamic key management is established, this field must be set to the value specified by the external processor.
- If dynamic key management is not used or if the external processor does not require a specific initial counter value when dynamic key management is established, this field can default to zeros. The Host Interface process or BIC ISO Interface process increments the counter automatically and attempts to resynchronize it with the external processor.

External Message File (EMF) Settings

The following fields in the External Message File (EMF) must be considered for each message being authenticated. Refer to the topic “Optional Authentication Data” earlier in this section for the default settings that are used when no EMF record is available.

EMF screen 1 contains specifications for creating the external messages sent and interpreting the external messages received. EMF screen 2 contains specifications for determining which data elements are to be included when authenticating external messages and is identical in appearance to Screen 1. Refer to the ***BASE24 Base Files Maintenance Manual*** for additional information regarding the use of EMF screens 1 and 2.

EMF Screen 1

The following fields are presented on EMF screen 1.

FULL MSG MAC — Specifies whether all data elements in the message will be included in the MAC computation. This value overrides the value in the FULL MESSAGE MAC field on screen 1 of the KEYF or KEY6.

- If only the data elements specified in the MESSAGE ELEMENTS P-1 through S-128 fields on screen 2 are used to compute the MAC, this field must be set to a value of N.
- If the entire message is used to compute the MAC, this field must be set to a value of Y.

MESSAGE ELEMENTS P-64 and S-128 — The flags specifying whether each data element is required, is conditional, or is not used in the message. A flag exists for each of the 128 elements that a message could contain. Elements P-64 and S-128 should be set to C (conditional) for each message that will be authenticated so they can carry the MAC when necessary.

EMF Screen 2

The following fields are presented on EMF screen 2.

MESSAGE ELEMENTS P-1 through S-128 — The flags specifying whether each data element will be included in the MAC computation. A flag exists for each of the 128 elements that a message could contain.

- If the FULL MSG MAC field on EMF screen 1 contains the value Y, the flags for individual elements do not need to be set.
- If the FULL MSG MAC field on EMF screen 1 contains the value N, the flag for each element to be included in the MAC computation should be set to a value of Y. The flags for elements P-64 and S-128 are not used because the MAC cannot be included in the MAC computation.
- If the FULL MSG MAC field on EMF screen 1 contains the value N, the flag for each element to be omitted from the MAC computation can be set to a value of N or left blank.

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7: BASE24 Device Handler Message Authentication Support

This section describes the BASE24 message authentication support provided by BASE24-atm and BASE24-pos Device Handler processes or modules.

BASE24-teller Device Handler processes do not support message authentication. The file maintenance screens for the Teller Terminal Data File (TTDF) contain fields related to message authentication. However, these fields are provided for customization purposes only.

BASE24 Device Handler Processes or Modules

BASE24-atm and BASE24-pos Device Handler processes or modules handle message authentication, or the generation and verification of message authentication codes (MACs), between a BASE24 system and the terminals attached to it. They also manage any keys required for message authentication.

The documentation in this section discusses the message authentication options provided for BASE24-atm and BASE24-pos devices. While similar, each product has its own unique characteristics and is discussed separately.

BASE24-atm and BASE24-pos support a variety of Device Handler processes or modules, with each Device Handler process or module having its own specific message authentication requirements. The documentation in this section discusses the message authentication options in general. The options selected for each Device Handler process or module are based on a combination of device capabilities and user preference.

BASE24-atm Device Handler Processes

BASE24-atm Device Handler processes are responsible for the interface between BASE24-atm and the terminals attached to it. The following documentation addresses message authentication within BASE24-atm, including the database settings available and incoming and outgoing message processing.

The following BASE24-atm Device Handler processes support message authentication using a security module:

- Diebold 910/911 Device Handler process
- Diebold 912 Device Handler process
- Diebold 10XX/478X Device Handler process
- Fujitsu 7000 Device Handler process
- NCR 5XXX Device Handler process
- Olivetti 6000 Device Handler process
- Tidel Device Handler process
- Triton Device Handler process

The BASE24-atm IBM 4730-series Device Handler process supports message authentication in software instead of using a security module.

Each Device Handler process has its own device support manual. Refer to these manuals for information regarding the messages that can be authenticated and the fields in each message that are used when generating or verifying MACs.

Also refer to vendor documentation for message authentication-related configuration requirements for each device. These are settings that must be established at the device itself, and may or may not affect the BASE24 database.

Device Handler Process Keys

There are two types of keys that are used in message authentication between terminals and their Device Handler processes: transaction keys and master keys.

Transaction Keys

A transaction key, also known as a session key, is distributed electronically by the Device Handler process to its terminals using the same lines over which all other communications travel. There are separate transaction keys for PIN encryption and message authentication.

Master Keys

Since transaction keys are distributed to BASE24-atm terminals using the same lines as other communications, a means of sending transaction keys in a secure manner is required. This security is provided by a separate master key known to both the terminal and the Device Handler process. This master key is used to encrypt transaction keys for their movement over the communication lines. The same master key is used to encrypt the PIN encryption transaction key and message authentication transaction key.

Master keys are by nature long-term, and, once in place, are usually left unchanged unless they are compromised. Unlike transaction keys, master keys cannot be transmitted over communications lines. Each terminal master key must be physically entered at the terminal.

ACI hard codes a terminal master key in each of its Device Handler processes. If transaction keys are to be encrypted by Device Handler processes, the hard-coded master keys must be used as the respective terminal master keys. This method does not require that a master key be placed in the BASE24 database for each terminal.

If transaction keys are to be encrypted by a security module, an individual master key must be resident in each BASE24-atm Terminal Data files record. Under this latter method, clear and encrypted versions of the master key can be determined individually for each terminal. The clear version of the master key must be entered at the terminal and the encrypted version is entered in the BASE24-atm Terminal Data files record.

Note: If you are using the Transaction Security Services process, the master key and transaction keys fields in the BASE24-atm Terminal Data files are set to key locator values used to retrieve the actual encrypted master key or transaction key stored in the Channel Security File (CSECD). In addition, if you are using the BASE24-atm Automated Key Distribution System add-on product with the Transaction Security Services process, you can automatically generate and securely download master keys in ATM terminals online. For more information, refer to the ***BASE24 Transaction Security Services Processing Guide***.

Transmitting Transaction Keys to the Terminal

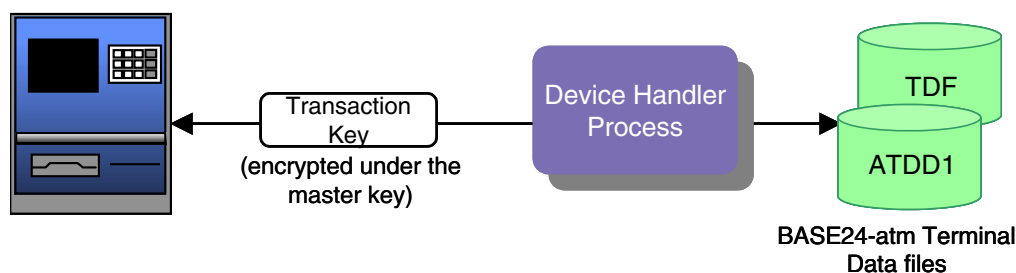
There are several instances in which Device Handler processes transmit new transaction keys to their terminals. Device Handler processes normally transmit new transaction keys to terminals during load and open sequences. Some Device Handler processes also periodically transmit new transaction keys once the terminals are up and running. In addition, the transmission of a new transaction key can also result from a command to the Device Handler process from the BASE24 Device Control Terminal (DCT).

If a transaction key load is unsuccessful, the Device Handler process prevents the ATM from opening until a successful transaction key load occurs. This keeps transaction keys used by the ATM and the Device Handler process synchronized. BASE24-atm denies transactions for invalid PIN or sanity check errors when transaction keys are not synchronized.

When a new transaction key must be sent to a terminal, the Device Handler process checks the MAC SUPPORT TYPE field on the device-specific BASE24-atm Terminal Data files (ATD) screen 9 for the terminal record to determine if a MAC must be generated prior to sending the new transaction key.

Software Message Authentication

If the value in the MAC SUPPORT TYPE field on the device-specific ATD screen 9 is set to a value of 1, indicating that message authentication is performed in software, the Device Handler process generates a new transaction key, encrypts that key under its hard-coded master key, and sends the transaction key to the terminal. It then places the new transaction key in the BASE24-atm Terminal Data files record for its own use. The Device Handler process must store the transaction key in the BASE24-atm Terminal Data files so that the key is available for generating MACs for messages bound for the terminal and verifying MACs in messages received from the terminal. However, this transaction key is never displayed or logged anywhere in the system. The following graphic illustrates this processing.



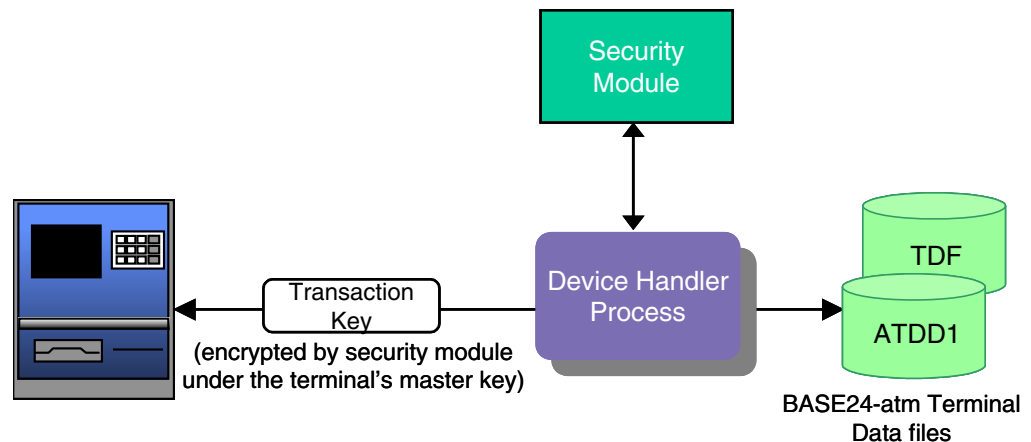
The terminal can decrypt the transaction key using the master key resident at the terminal.

BASE24-atm sends a new transaction key to the terminal encrypted under the terminal encrypted under the terminal master key (hardcoded in the Device Handler process). The Device Handler process then stores a clear version of the transaction key in the BASE24-atm Terminal Data files for its own use.

Security Module Message Authentication

If the value in the MAC SUPPORT TYPE field on the device-specific ATD screen 9 is set to a value of 3, indicating that message authentication is performed using a security module, the Device Handler process requests a new transaction key from the security module used in the system. To do this, the Device Handler process provides the encrypted terminal master key to the security device (taken from the MASTER KEY field on the device-specific ATD screen 9 for the terminal record).

The security module then randomly generates a transaction key and encrypts it under the terminal master key for sending to the terminal. It also provides the Device Handler process with the new transaction key encrypted under the security module master key for storage in the BASE24-atm Terminal Data files. The Device Handler process must store the encrypted transaction key in the BASE24-atm Terminal Data files so that the key can be used to generate MACs for messages bound for the terminal and verify MACs in messages received from the terminal. This transaction key is never displayed or logged anywhere in the system. The following graphic illustrates this processing.



The terminal can decrypt the transaction key using the master key resident at the terminal.

BASE24-atm requests a new transaction key from the security module encrypted under the terminal's master key and sends a copy to the terminal. The Device Handler process stores an encrypted version of the transaction key in the BASE24-atm Terminal Data files.

Time Variant Numbers

Message tampering between Device Handler processes and ATM devices can be accomplished by message modification and message replay. Message modification is changing the information in the message and message replay is copying the transaction being sent from the ATM and sending it again later. Message authentication provides a means of controlling message modification and the use of time variant numbers provides a way to control message replay.

Devices can include time variant numbers in certain messages sent to the Device Handler process when message authentication is performed. The time variant numbers replace the message sequence number in the message and are part of the data used to generate and verify MACs. Devices do not include time variant numbers in messages when message authentication is not performed.

The method used to create a time variant number varies by device. It can be a randomly generated value, it can be based on the time the transaction occurs, or it can be created using another method. Regardless of the method used, the purpose of the time variant number is to uniquely identify a message so that it cannot be duplicated. Some devices automatically include time variant numbers, while other devices can be configured to include or exclude time variant numbers. Refer to documentation from your device vendor for available time variant number options.

When the Device Handler process receives a message containing a time variant number and a MAC, the Device Handler process stores the time variant number with device dependent data in the BASE24-atm Terminal Data files. The Device Handler process then uses the time variant number stored in the BASE24-atm Terminal Data files when creating the response message to the device and generating the MAC included with the response message.

Note: The potential exists for an ATM to process a transaction without generating a customer receipt or journal record. BASE24-atm uses the time variant number it receives with a solicited status message to generate the MAC that it places in the response to the ATM. If the time variant numbers in the solicited status message and response do not match, BASE24-atm uses the original time variant number when generating the MAC in the message it sends to the ATM. The ATM uses the modified time variant number when verifying the MAC in the message it receives from BASE24-atm. The ATM does not generate a customer receipt or journal record when it cannot verify the MAC in the message that it receives.

Setting Up the BASE24 Database

Device Handler processes obtain the message authentication information they need for each terminal from a param in the Logical Network Configuration File (LCONF) and fields on device-specific BASE24-atm Terminal Data files (ATD) screen 9.

LCONF Param

The following LCONF param is required for Device Handler message authentication. For more information on LCONF params refer to the ***BASE24 Logical Network Configuration File Manual***.

ATM-DH-CONS-MAC-ERR-LMT — Specifies how many consecutive message authentication errors can occur before the Device Handler process sends a shutdown message to the terminal.

- If errors are not to be counted, the param is set to 0.
- If the number of consecutive errors is to be specified, the param can be set to a value between 1 and 32,767.

If the param is missing or invalid, the Device Handler process uses a default value of 3.

Note: If you are using the BASE24-atm Dynamic Key Management add-on feature, this param specifies the number of consecutive message authentication errors that can occur before the Device Handler process initiates a download of a new MAC key and a new PIN key to the terminal. For more information on the BASE24-atm Dynamic Key Management add-on feature, refer to the ***BASE24 Transaction Security Services Processing Guide***.

Device-specific ATD Screen 9

Device-specific ATD screen 9 contains device-specific security information, PIN information, and message authentication information. Refer to the ***BASE24-atm Files Maintenance Manual*** for illustrations of device-specific ATD screen 9 that contain some or all of the fields mentioned in the following descriptions.

SECURITY INFORMATION

The following two fields relate to the master key used by the Device Handler process when encrypting the PIN encryption and message authentication transaction keys that it sends to the terminal being configured.

MASTER KEY — Specifies the master key used to encrypt transaction keys used by this terminal.

- If message authentication is not performed (the MAC SUPPORT TYPE field is set to a value of 0), this field is not used.
- If message authentication is performed in software (the MAC SUPPORT TYPE field is set to a value of 1), this field is not used.
- If message authentication is performed in hardware (the MAC SUPPORT TYPE field is set to a value of 3), this field must contain the terminal master key encrypted by or for the specific security module being used in the system.

Note: The same master key is used for PIN encryption and message authentication transaction keys, so settings in the MAC SUPPORT TYPE and PIN ENCRYPT TYPE fields must agree (i.e., none, hardware, or software).

CHECK DIGITS — Specifies the check digits associated with the master key when it is encrypted by or for the specific security module being used in the system. Valid values for each character in this field are 0 through 9 and A through F. This field must contain four valid characters. A value of 0000 in this field indicates check digits are not being used with the master key.

MESSAGE AUTHENTICATION INFORMATION

The following fields relate to message authentication code (MAC) support. MAC support provides a validation method for the contents of transmitted messages thus ensuring the integrity of messages received.

MAC SUPPORT TYPE — Specifies the type of message authentication support.

- If the terminal does not use message authentication, this field must be set to a value of 0.
- If the terminal performs message authentication in software, this field must be set to a value of 1. This value is valid for the IBM 4730-series Device Handler process only.

- If the terminal performs message authentication in hardware, this field must be set to a value of 3.

Note: This field only enables message authentication in the Triton Device Handler process for a device. Once message authentication has been enabled for a Triton device, you must use the MACINGOFF text command to disable message authentication for a Triton device. You can use the MACINGON command as an alternative to setting this field to a value of 3 to enable message authentication in the Triton Device Handler process. Refer to the ***BASE24 Text Command Reference Manual*** for more information on the MACINGOFF and MACINGON commands.

MAC DATA TYPE — A code specifying the character set in which messages are being transmitted between the terminal and the Device Handler process. The device-specific ATD screen 9 contains this field only if it is needed by the Device Handler process to identify the character set.

- If the terminal transmits data in the ASCII character set, this field must be set to a value of 0.
- If the terminal transmits data in the EBCDIC character set, this field must be set to a value of 1.

SOLICITED STATUS MSG — Specifies whether solicited status messages will be authenticated.

- If solicited status messages are not to be authenticated, this field must be set to a value of 0.
- If solicited status messages are to be authenticated, this field must be set to a value of 1.

Only the device-specific ATD screen 9 for the NCR 5XXX Device Handler process contains this field. The Flag 10 setting at the ATM must be consistent with the setting in this field. Refer to vendor documentation for additional device configuration information.

CALCULATION METHOD — Specifies whether part or all of the message is to be authenticated.

- If the entire message is to be authenticated, this field must be set to a value of 0.

- If a portion of the message is to be authenticated, this field must be set to a value of 1. At least one field must be identified in each message that is specified for authentication.

Only the device-specific ATD screen 9 for the NCR 5XXX Device Handler process contains this field. The Flag 9 setting at the ATM must be consistent with the setting in this field. Refer to vendor documentation for additional device configuration information.

The N50NAMS File contains the following seven tables used by the NCR 5XXX Device Handler process to identify the message fields authenticated when the CALCULATION METHOD field is set to a value of 1:

```
ncr_tran_rqst_tbl  
ncr_tran_reply_tbl  
ncr_sol_status_tbl  
ncr_other_msg_tbl  
ncr_track1_tbl  
ncr_track2_tbl  
ncr_track3_tbl
```

Processing Outgoing Messages

If message authentication is supported, a MAC is generated and placed in the outgoing message. If message authentication is not supported, the message continues to be processed normally. The Device Handler process performs the following steps:

1. Determines whether the ATM terminal is configured for support of message authentication.
 - If the MAC SUPPORT TYPE field on the device-specific ATD screen 9 contains a value of 0, message authentication is not supported and the Device Handler process continues processing the message normally.
 - If the MAC SUPPORT TYPE field on the device-specific ATD screen 9 contains a value of 3 (hardware), message authentication is supported and processing continues with step 2.
2. Determines whether message authentication should be performed for this message. Only the NCR 5XXX Device Handler process performs this step.
 - If message authentication is to be performed for this message, processing continues with step 3.

- If message authentication is not performed for this message, message authentication processing is complete and the Device Handler process continues processing the message normally.
3. Obtains the time variant number that was placed in the BASE24-atm Terminal Data files when the incoming message was processed.
 4. Builds a buffer to hold the data to be authenticated.
 5. Determines which fields are to be included in the MAC calculation. Only the NCR 5XXX Device Handler process performs this step.
 - If the Device Handler process includes an option for using the full message or selected fields when generating the MAC, the Device Handler process uses the CALCULATION METHOD field on NCR 5XXX ATD screen 9 to determine whether the full message or selected fields are used to generate the MAC.
 - If the Device Handler process does not include an option for using the full message or selected fields when generating the MAC, the Device Handler process uses the predefined portion of the message to generate the MAC.
 6. Translates the data included in the MAC calculation into the EBCDIC character set, if necessary.
 - If the receiving device uses the EBCDIC character set, the data included in the MAC calculation is translated from ASCII to EBCDIC.
 - If the receiving device uses the ASCII character set, no translation is necessary and processing continues with step 7.
 7. Uses a standard set of utilities to perform edit checks on the data to be authenticated.

The MAC transaction key must contain only hexadecimal characters.

- If software is being used to generate the MAC, the data to be authenticated cannot exceed the length specified in the Device Handler process. Refer to the device-specific device support manual for more information.
 - If an Atalla Network Security Processor (NSP) is being used to generate the MAC, the length of the data to be authenticated cannot exceed 4096 bytes.
 - If a Thales e-Security Security Module is being used to generate the MAC, the length of the data to be authenticated cannot exceed 400 bytes.
8. Performs message authentication in software, if configured to do so.

- If the MAC SUPPORT TYPE on the device-specific ATD screen 9 is set to a value of 1 (software), the MAC is generated in software and processing continues with step 10.
 - If the MAC SUPPORT TYPE on the device-specific ATD screen 9 is set to a value of 3 (hardware), processing continues with step 9.
9. Requests that a security module generate the MAC.

The Device Handler process provides the security module with the following:

- The data to be authenticated.
- The length of the data to be authenticated.
- The MAC transaction key from the BASE24-atm Terminal Data files.

The security module then generates the MAC and returns it to the Device Handler process.

10. Continues processing.

The Device Handler process uses a standard set of utilities to communicate with the security module or generate the MAC in software. These utilities return one of the following codes to indicate the outcome of MAC generation processing in software:

Software	Security Module	Description
Pass	0	MAC generated successfully
Fail	2	Device failed
Fail	3	Operation failed

If the MAC is generated successfully, the Device Handler process places the MAC in the appropriate field of the message being sent to the device. Message authentication processing is complete.

If the MAC cannot be generated and the message is a response message for a financial transaction that has been approved, the Device Handler process sends a reversal message to the Authorization process with a reversal code of 21 (MAC failure).

If the MAC cannot be generated and the message is a downline load, the Device Handler process aborts the load.

When the device cannot verify the MAC in a message it receives from the Device Handler process or the message it receives contains an invalid time variant number, the device sends a completion to the Device Handler process. The Device Handler process performs the following:

1. Increments the consecutive error counter, if one is being maintained.
2. Performs one of the following:
 - If the consecutive error limit is greater than zero (meaning a consecutive error limit is being maintained) and the consecutive error counter is equal to or greater than the limit, the Device Handler process sends a shutdown message to the terminal. The customer cannot attempt another transaction.
 - If the consecutive error limit is greater than zero (meaning a consecutive error limit is being maintained) and the consecutive error counter is less than the limit, the Device Handler process allows the transaction to time out. The customer can attempt another transaction.
 - If the consecutive error limit is zero (meaning a consecutive error limit is not being maintained), the Device Handler process does not perform consecutive error limit processing.
3. Performs one of the following, if applicable:
 - If the message is a response message for a financial transaction that has been approved, the Device Handler process sends a reversal message to the Authorization process with a reversal code of 23 (message replay error) or 24 (invalid MAC), depending on the status returned from the device.
 - If the message is a downline load, the Device Handler process aborts the downline load.

Processing Incoming Messages

If message authentication is supported, a MAC received in an incoming message is verified. If message authentication is not supported, the message continues to be processed normally. The Device Handler process performs the following:

1. Determines whether this ATM terminal is configured for support of message authentication.
 - If the MAC SUPPORT TYPE field on the device-specific ATD screen 9 contains a value of 0, message authentication is not supported and the Device Handler process continues processing the message normally.

- If the MAC SUPPORT TYPE field on the device-specific ATD screen 9 contains a value of 3 (hardware), message authentication is supported and processing continues with step 2.
- 2. Determines whether message authentication should be performed for this message.
 - If the Device Handler process is to perform message authentication for this message, processing continues with step 3.
 - If the Device Handler process is not to perform message authentication for this message, message authentication processing is complete.
- 3. Determines whether the message contains a MAC data field.
 - If the message contains a MAC data field, processing continues with step 4.
 - If the message does not contain a MAC data field, the Device Handler process rejects the message.
- 4. Builds a buffer to hold the data to be authenticated.
- 5. Determines which fields are to be included in the MAC calculation.
 - If the Device Handler process includes an option for using the full message or selected fields when verifying the MAC, the Device Handler process uses the value of the CALCULATION METHOD field on NCR 5XXX ATD screen 9 to determine whether the full message or selected fields are used to verify the MAC.
 - If the Device Handler process does not include an option for using the full message or selected fields when verifying the MAC, the Device Handler process uses the predefined portion of the message to verify the MAC.
- 6. Uses a standard set of utilities to perform edit checks on the data to be authenticated.

The MAC transaction key must contain only hexadecimal characters.

- If software is being used to verify the MAC, the data to be authenticated cannot exceed the length specified in the Device Handler process. Refer to the device-specific device support manual for more information.
- If an Atalla Network Security Processor (NSP) is being used to verify the MAC, the length of the data to be authenticated cannot exceed 4096 bytes.
- If a Thales e-Security Security Module is being used to verify the MAC, the length of the data to be authenticated cannot exceed 400 bytes.

7. Requests that a security module verify the MAC.

The Device Handler process provides the security module with the following:

- The data to be authenticated.
- The length of the data to be authenticated.
- The MAC transaction key from the BASE24-atm Terminal Data files.
- The MAC received from the device.

The security module then verifies the MAC and returns it to the Device Handler process.

8. Continues processing.

The Device Handler process uses a standard set of utilities to communicate with the security module or verify the MAC in software. These utilities return one of the following codes to indicate the outcome of MAC verification processing in software:

Software	Security Module	Description
Pass	0	MAC verified successfully
Fail	1	Invalid MAC
Fail	2	Device failed
Fail	3	Operation failed

If the MAC is verified successfully, the Device Handler process performs the following:

- a. Resets the consecutive error counter.
- b. Obtains the time variant number received with the message and places it in the BASE24-atm Terminal Data files for use when the outgoing message is processed.
- c. Continues normal processing. Message authentication processing is complete.

If the MAC is invalid, the Device Handler process increments the consecutive error counter, if one is being maintained.

If the MAC is invalid or the verification operation fails, the Device Handler process performs one of the following:

- If the consecutive error limit is greater than zero (meaning a consecutive error limit is being maintained) and the consecutive error counter is equal to or greater than the limit, the Device Handler process sends a shutdown message to the terminal. The customer cannot attempt another transaction.
- If the consecutive error limit is greater than zero (meaning a consecutive error limit is being maintained) and the consecutive error counter is less than the limit, the Device Handler process allows the transaction to time out. The customer can attempt another transaction.
- If the consecutive error limit is zero (meaning a consecutive error limit is not being maintained), the Device Handler process does not perform consecutive error limit processing.

BASE24-pos Device Handler Modules

BASE24-pos Device Handler modules are responsible for the interface between BASE24-pos and the terminals attached to it. The following documentation addresses message authentication within BASE24-pos, including the database settings available and incoming and outgoing message processing.

The following BASE24-pos Device Handler modules support message authentication in software or using a security module:

- BASE24 Standard POS Device Handler (SPDH) module

Note: Derived unique key per transaction (DUKPT) message authentication is supported in the BASE24-pos Standard POS Device Handler (SPDH) module when using the Transaction Security Services process. DUKPT message authentication uses a derivation key instead of a MAC key and also includes the key serial number (KSN), length of the KSN, and KSN key descriptor as input for computing the MAC. For more information on DUKPT message authentication for the SPDH module, refer to the ***BASE24 Transaction Security Services Processing Guide***.

- BASE24-pos Merchant Host Interface (MHI) module

Note: The MHI module is included in the Device Handler module category for transaction security purposes. It is bound with the BASE24-pos Router and Authorization modules like a Device Handler module and performs the same functions.

The BASE24-pos Hypercom Device Handler module supports message authentication in software only.

Each Device Handler module has its own device support manual. Refer to these manuals for information regarding the messages that can be authenticated and the fields in each message that are used when generating or verifying MACs.

Also refer to vendor documentation for message authentication-related configuration requirements for each device. These are settings that must be established at the device itself, and may or may not affect the BASE24 database.

Device Handler Module Keys

There are two types of keys that are used in message authentication between terminals and their Device Handler modules: transaction keys and master keys.

Transaction Keys

A transaction key, also known as a session key, is distributed electronically by the Device Handler module to its terminals using the same lines over which all other communications travel. There are separate transaction keys for PIN encryption and message authentication.

Master Keys

Since transaction keys are distributed to BASE24-pos terminals using the same lines as other communications, a means of sending transaction keys in a secure manner is required. This security is provided by a separate master key known to both the terminal and the Device Handler module. This master key is used to encrypt transaction keys for their movement over the communications lines. There are separate master keys for PIN encryption and message authentication.

Master keys are by nature long-term, and, once in place, are usually left unchanged unless they are compromised. Unlike transaction keys, master keys cannot be transmitted over communications lines. Each terminal master key must be physically entered at the terminal.

A PIN encryption master key and a message authentication master key must be resident in each BASE24-pos Terminal Data files or Merchant Host Configuration File (MHCF) record. The clear version of the master keys must be entered at the terminal and the encrypted version of the keys entered in the BASE24-pos Terminal Data files or MHCF record. The master keys in the MHCF are used by the MHI module only when it operates in host control mode.

If you are using the derived unique key per transaction (DUKPT) method of message authentication, a key called a base derivation key is used to generate a unique MAC key for each transaction sent from the terminal that includes a MAC. When verifying the MAC, both the MAC and the KSN are included in the message authentication calculation along with the derivation key. For detailed information on DUKPT message authentication, refer to [section 2](#). The base derivation key must be resident in the Derivation Key File (KEYD).

Note: If you are using the Transaction Security Services process, master key fields in the BASE24-pos Terminal Data files and the derivation key field in the KEYD must be set to key locator values used to retrieve the actual master key or derivation key values from the TSS database.

Transmitting Transaction Keys to the Terminal

There are several instances in which Device Handler modules transmit new transaction keys to their terminals. Device Handler modules normally transmit new transaction keys to their terminals during load sequences. Device Handler modules can also periodically transmit new transaction keys once their terminals are up and running, based on settings in the Device Handler module-specific configuration file. For example, the SPDH module uses the ACI Standard Configuration File (ACNF).

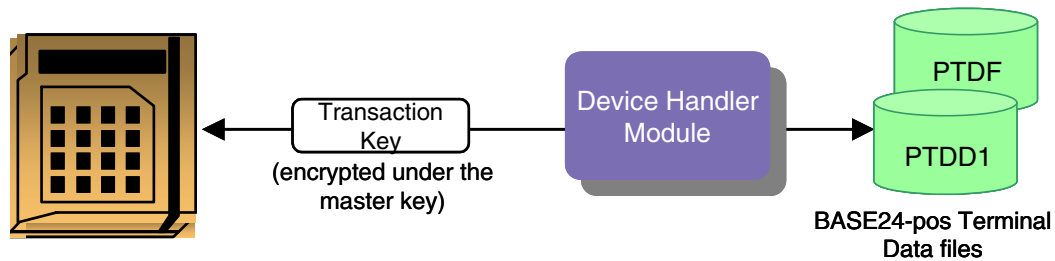
When a new transaction key must be sent to a terminal, the Device Handler module checks the MAC TYPE field on PTD screen 7 to determine if a MAC must be generated prior to sending the new transaction key.

When the Merchant Host Interface (MHI) module runs in host control mode, the value in the MAC ENCRYPTION TYPE field on MHCF screen 2 specifies whether a MAC must be generated prior to sending the new transaction key. The value in the UPDATE PTDF field on MHCF screen 1 indicates whether the MHI module runs in terminal or host control mode. The value in the MAC TYPE field on MHCF screen 1 indicates whether the MHI module uses BASE24 to host MACs or BASE24 to terminal MACs when message authentication is supported. BASE24 to host MACs can be used when the MHI module operates in terminal or host control mode. BASE24 to terminal MACs can only be used when the MHI module runs in terminal mode.

Software Message Authentication

If the value in the MAC TYPE field on PTD screen 7 is set to a value of 01 or 02, indicating that BASE24 message authentication is performed in software, the Device Handler module generates a new transaction key, encrypts that key under the master key in the MAC MASTER KEY field on PTD screen 7, and sends the transaction key to the terminal. It then places the new transaction key in the BASE24-pos Terminal Data files record for its own use. The Device Handler module must store the transaction key in the BASE24-pos Terminal Data files so that the key is available for generating MACs bound for the terminal and verifying

MACs received from the terminal. However, this transaction key is never displayed or logged anywhere in the system. The following graphic illustrates this processing.



The terminal can decrypt the transaction key using the master key resident at the terminal.

BASE24-pos sends a new transaction key to the terminal encrypted under the terminal master key. The Device Handler module then stores a clear version of the transaction key in the BASE24-pos Terminal Data files for its own use.

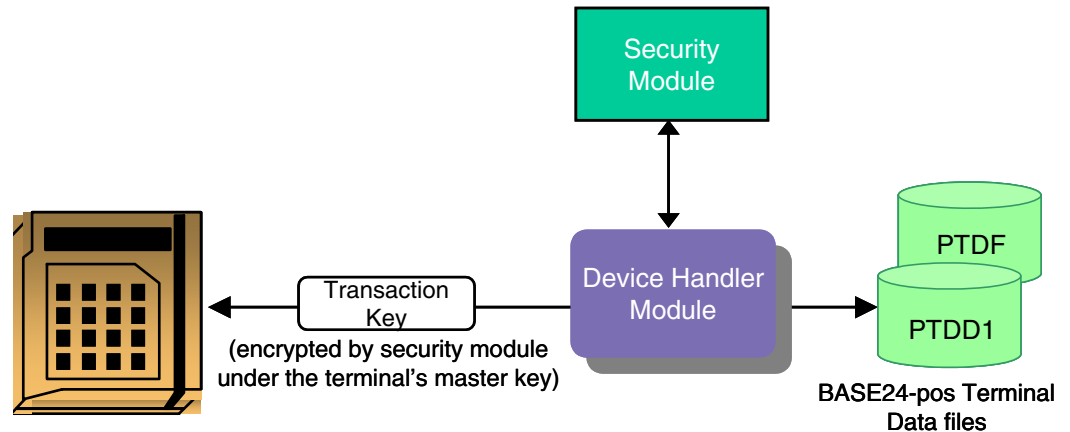
When the MHI module runs in host control mode, message authentication processing is similar to that described above except that it uses the MAC ENCRYPTION TYPE and MASTER KEY fields on MHCF screen 2. The generated transaction key is stored in the PTDF.

Security Module Message Authentication

If the value in the MAC TYPE field on PTD screen 7 is set to a value of 03, indicating that BASE24 message authentication is performed using a security module, the Device Handler module requests a new transaction key from the security module used in the system. To do this, the Device Handler module provides the encrypted terminal master key (taken from the MAC MASTER KEY field on PTD screen 7) to the security device.

The security module then randomly generates a transaction key and encrypts it under the terminal master key for sending to the terminal. It also provides the Device Handler module with the new transaction key encrypted under the security module master key for storage in the BASE24-pos Terminal Data files. The Device Handler module must store the encrypted transaction key in the BASE24-pos Terminal Data files so that the key can be used to generate MACs for

messages bound for the terminal and verify MACs in messages received from the terminal. This transaction key is never displayed or logged anywhere in the system. The following graphic illustrates this processing.



The terminal can decrypt the transaction key using the master key resident at the terminal.

BASE24-pos requests a new transaction key from the security module encrypted under the terminal's master key and sends a copy to the terminal. The Device Handler module stores an encrypted version of the transaction key in the BASE24-pos Terminal Data files.

When the MHI module runs in host control mode, message authentication processing is similar to that described above except that it uses the MAC ENCRYPTION TYPE and MASTER KEY fields on MHCF screen 2 and stores the generated transaction key in the MHCF.

Setting Up the BASE24 Database

Device Handler modules obtain the message authentication information they need for each terminal from a parameter in the Logical Network Configuration File (LCONF) and fields in the BASE24-pos Terminal Data files record or Merchant Host Configuration File (MHCF) record. Fields in the MHCF record are used by the MHI module only.

Note: All messages exchanged between POS devices and BASE24-pos Device Handler modules use the ASCII character set. As a result, the character set is not configured in the BASE24 database or checked during message processing.

LCONF Param

The following LCONF param is required for Device Handler message authentication. For more information on LCONF params refer to the ***BASE24 Logical Network Configuration File Manual***.

POS-DH-CONS-MAC-ERR-LMT — Contains the maximum number of consecutive message authentication code (MAC) errors allowed before the SPDH module takes corrective action. When configured for MAC generation with DUKPT, the SPDH module generates an event message. When configured for MAC generation without DUKPT, the SPDH module generates a new MAC communications key and sends it in the next message sent to the terminal. Valid values are 0 through 32,000 inclusive. A value of 0 indicates that an event message or a new MAC communications key is not generated, regardless of the number of consecutive MAC errors. This param is not required. If this param is not present in the LCONF or is set to an invalid value, the SPDH module logs an event message and uses a default value of 3.

MHCF Screen 1

MHCF screen 1 contains two fields that impact message authentication processing performed by the MHI module. Refer to the ***BASE24-pos Merchant Host Interface Manual*** for an illustration of MHCF screen 1, which contains the fields mentioned in the following descriptions.

MAC TYPE — Indicates the type of message authentication codes (MACs) used by the DPC.

- If message authentication is not performed, this field must be set to a value of 0.
- If BASE24 to host MACs are used, this field must be set to a value of 1. These MACs are used when the MHI module is operating in host or terminal control mode.
- If BASE24 to terminal MACs are used, this field must be set to a value of 2. These MACs are used when the MHI module is operating in terminal control mode.

UPDATE PTDF — Indicates whether the DPC is running in terminal or host control mode.

- If this field is set to a value of 0, the MHI module operates in host control mode and uses the MAC parameters stored in the MHCF.
- If this field is set to a value of 1, the MHI module operates in terminal mode and uses the MAC parameters stored in the PTDF.

MHCF Screen 2

MHCF screen 2 contains MAC encryption data for the MHI module when it operates in host control mode. Refer to the ***BASE24-pos Merchant Host Interface Manual*** for an illustration of MHCF screen 2, which contains the fields mentioned in the following descriptions.

MAC ENCRYPTION TYPE — Specifies the type of authentication performed on messages exchanged with this DPC.

- If message authentication is not performed, this field must be set to a value of 00.
- If message authentication is performed in software using the ANSI standard authentication algorithm, this field must be set to a value of 01.
- If message authentication is performed in software using a proprietary algorithm, this field must be set to a value of 02.
- If message authentication is performed in hardware using the ANSI standard authentication algorithm, this field must be set to a value of 03.

MAC MASTER KEY — Specifies the master key used to encrypt transaction keys used by this DPC for message authentication. Valid values for each character in this field are 0 through 9 and A through F. This field must contain 16 valid characters.

- If message authentication is not performed (the MAC ENCRYPTION TYPE field is set to a value of 00), this field can default to zeros.
- If message authentication is performed in software (the MAC ENCRYPTION TYPE field is set to a value of 01 or 02), this field must contain a clear version of the terminal MAC master key.
- If message authentication is performed in hardware (the MAC ENCRYPTION TYPE field is set to a value of 03), this field must contain the terminal master key encrypted by or for the specific security module being used for messages to and from this DPC.

MAC KEY VARIANT — Specifies the key variant to be applied to the MAC key exchange key (KEK) when translating MAC keys for this DPC. Regardless of the hardware security module used by BASE24 or the external processor this field must be set to a value of 3 (variant 3).

PTD Screen 7

PTD screen 7 contains encryption and decryption data for transactions being sent to and from the terminal as well as transaction security data presented by the terminal being configured. Refer to the ***BASE24-pos Files Maintenance Manual*** for an illustration of PTD screen 7, which contains the fields mentioned in the following descriptions.

MAC TYPE — Specifies the type of authentication performed on messages exchanged with this terminal.

- If message authentication is not performed, this field must be set to a value of 00.
- If message authentication is performed in software using the ANSI standard authentication algorithm, this field must be set to a value of 01.
- If message authentication is performed in software using a proprietary algorithm, this field must be set to a value of 02. This value is valid for Hypercom Device Handler modules only.
- If message authentication is performed in hardware using the ANSI standard authentication algorithm, this field must be set to a value of 03.

- If DUKPT message authentication is performed in hardware at the same time the DUKPT-encrypted PIN block is translated into an ANSI PIN block, this field must be set to a value of 07.
- If DUKPT message authentication is performed in hardware but the DUKPT-encrypted PIN block is not translated into an ANSI PIN block at the same time, this field must be set to a value of 08.

Note: The PIN ENCRYPTION TYPE field on PTD screen 7 must be set to a value of 07 to use the 07 and 08 values in the MAC TYPE field. In addition, you must use the Transaction Security Services process to support values 07 and 08 in the MAC TYPE field.

MASTER KEY — Specifies the master key used to encrypt transaction keys used by this terminal for message authentication. Valid values for each character in this field are 0 through 9 and A through F. This field must contain 16 valid characters.

- If message authentication is not performed (the MAC TYPE field is set to a value of 00), this field can default to zeros.
- If message authentication is performed in software (the MAC TYPE field is set to value of 01 or 02), this field must contain a clear version of the terminal MAC master key.
- If message authentication is performed in hardware (the MAC TYPE field is set to a value of 03), this field must contain the terminal master key encrypted by or for the specific security module being used for messages to and from this device.
- If message authentication is performed in hardware using the Transaction Security Services process (the MAC TYPE field is set to a value of 07 or 08), this field must contain a key locator value.

CHECK DIGITS — The check digits associated with the MAC master key when it is encrypted by or for the specific security module being used in the system. Valid values for each character in this field are 0 through 9 and A through F. This field must contain four valid characters. A value of 0000 in this field indicates check digits are not being used with the MAC master key.

Processing Outgoing Messages

If message authentication is supported, a MAC is generated and placed in the outgoing message. If message authentication is not supported, the message continues to be processed normally.

Note: All messages exchanged between POS devices and BASE24-pos Device Handler modules use the ASCII character set. As a result, the character set is not configured in the BASE24 database or checked during message processing.

When the MHI module operates in host control mode, the following processing is the same except that it uses the MAC ENCRYPTION TYPE field on MHCF screen 2 to determine whether message authentication is supported for the DPC and uses the MAC transaction key to generate the MAC. The MAC transaction key is stored in the PTDF if it was generated in software. It is stored in the MHCF if it was generated by a security module. In addition, the MHI module does not currently support DUKPT or use of the Transaction Security Services process.

The Device Handler module performs the following:

1. Determines whether the POS terminal is configured for support of message authentication.
 - If the MAC TYPE field on PTD screen 7 contains a value of 00, message authentication is not supported, and the Device Handler module continues processing the message normally.
 - If the MAC TYPE field on PTD screen 7 contains a value of 01 (software), message authentication is supported, and processing continues with step 2.
 - If the MAC TYPE field on PTD screen 7 contains a value of 02 (proprietary software), message authentication is supported, and processing continues with step 2. This value is valid for Hypercom Device Handler modules only.
 - If the MAC TYPE field on PTD screen 7 contains a value of 03 (hardware), message authentication is supported, and processing continues with step 2.
 - If the MAC TYPE field on PTD screen 7 contains a value of 07 or 08 (DUKPT message authentication in hardware), message authentication is supported, and processing continues with step 2.
2. Determines whether message authentication should be performed for this message.
 - If the Device Handler module is to perform message authentication for this message, processing continues with step 3.
 - If the Device Handler module is not to perform message authentication for this message, message authentication processing is complete.
3. Builds a buffer to hold the data to be authenticated.

4. Determines which fields are to be included in the MAC calculation. This information is obtained from the Device Handler module-specific configuration file.
5. Uses a standard set of utilities to perform edit checks on the data to be authenticated.

The MAC transaction key must contain only hexadecimal characters.

- If software is being used to generate the MAC, the data to be authenticated cannot exceed the length specified in the Device Handler module. Refer to the device-specific device support manual for more information.
 - If an Atalla Network Security Processor (NSP) is being used to generate the MAC, the length of the data to be authenticated cannot exceed 4096 bytes.
 - If a Thales e-Security Security Module is being used to generate the MAC, the length of the data to be authenticated cannot exceed 400 bytes.
6. Performs message authentication in software, if configured to do so.
 - If the MAC TYPE field on PTD screen 7 is set to a value of 01 or 02 (software), the MAC is generated in software and processing continues with step 8.
 - If the MAC TYPE field on PTD screen 7 is set to a value of 03 (hardware), processing continues with step 7.
 - If the MAC TYPE field on PTD screen 7 is set to a value of 07 or 08 (DUKPT message authentication in hardware), the Device Handler module calls the Transaction Security Services process to generate the MAC and return it to the Device Handler module. Processing continues with step 8.
 7. Requests that a security module generate the MAC.

The Device Handler module provides the security module with the following:

- The data to be authenticated
- The length of the data to be authenticated
- The MAC transaction key from the BASE24-pos Terminal Data files

The security module then generates the MAC and returns it to the Device Handler module.

8. Continues processing.

The Device Handler module uses a standard set of utilities to communicate with the security module or the Transaction Security Services process, or to generate the MAC in software. These utilities return one of the following codes to indicate the outcome of MAC generation processing in software:

Software	Security Module	Description
Pass	0	MAC generated successfully
Fail	2	Device failed
Fail	3	Operation failed

If the MAC is generated successfully, the Device Handler module places the MAC in the appropriate field of the message being sent to the device. Message authentication processing is complete.

If the MAC cannot be generated and the message is a response for a financial transaction that has been approved, the Device Handler module sends a reversal message to the Authorization module with a reversal code of 21 (MAC failure) and sends a bad MAC response to the device.

Processing Incoming Messages

If message authentication is supported, a MAC received in an incoming message is verified. If message authentication is not supported, the message continues to be processed normally.

Note: All messages exchanged between POS devices and BASE24-pos Device Handler modules use the ASCII character set. As a result, the character set is not configured in the BASE24 database or checked during message processing.

When the MHI module operates in host control mode, the following processing is the same except that the process uses the MAC ENCRYPTION TYPE field on MHCF screen 2 to determine whether message authentication is supported for the DPC and uses the MAC transaction key to generate the MAC. The MAC transaction key is stored in the PTD if it was generated in software. It is stored in the MHCF if it was generated by a security module.

The Device Handler module performs the following:

1. Determines whether this POS terminal is configured for support of message authentication.
 - If the MAC TYPE field on PTD screen 7 contains a value of 00, message authentication is not supported and the Device Handler module continues processing the message normally.
 - If the MAC TYPE field on PTD screen 7 contains a value of 01 (software), message authentication is supported and processing continues with step 2.
 - If the MAC TYPE field on PTD screen 7 contains a value of 02 (proprietary software), message authentication is supported and processing continues with step 2. This value is valid for Hypercom Device Handler modules only.
 - If the MAC TYPE field on PTD screen 7 contains a value of 03 (hardware), message authentication is supported and processing continues with step 2.
 - If the MAC TYPE field on PTD screen 7 contains a value of 07 or 08 (DUKPT message authentication in hardware), message authentication is supported, and processing continues with step 2.
2. Determines whether the message contains a MAC data field.
 - If the message contains a MAC data field, processing continues with step 3.

- If the message does not contain a MAC data field, the Device Handler module continues to process the message without performing message authentication unless DUKPT message authentication is configured and the KSN is retrieved. In this case, the SPDH module denies the transaction with a bad MAC error.
3. Builds a buffer to hold the data to be authenticated.
 4. Determines which fields are to be included in the MAC calculation. This information is obtained from the Device Handler module-specific configuration file.
 5. Uses a standard set of utilities to perform edit checks on the data to be authenticated.

The MAC transaction key must contain only hexadecimal characters.

- If software is being used to verify the MAC, the data to be authenticated cannot exceed the length specified in the Device Handler module. Refer to the device-specific device support manual for more information.
 - If an Atalla Network Security Processor (NSP) is being used to verify the MAC, the length of the data to be authenticated cannot exceed 4096 bytes.
 - If a Thales e-Security Security Module is being used to verify the MAC, the length of the data to be authenticated cannot exceed 400 bytes.
6. Performs message authentication in software, if configured to do so.
 - If the MAC TYPE on PTD screen 7 is set to a value of 01 or 02 (software), the MAC is verified in software and processing continues with step 8.
 - If the MAC TYPE on PTD screen 7 is set to a value of 03 (hardware), processing continues with step 7.
 - If the MAC TYPE field on PTD screen 7 is set to a value of 07 or 08 (DUKPT message authentication in hardware), the Device Handler module calls the Transaction Security Services process to verify the MAC and return it to the Device Handler module. Processing continues with step 8.
 7. Requests that a security module verify the MAC.

The Device Handler module provides the security module with the following:

- The data to be authenticated.
- The length of the data to be authenticated.
- The MAC transaction key from the BASE24-pos Terminal Data files.

- The MAC received from the device.

The security module then verifies the MAC and returns it to the Device Handler module.

8. Continues processing.

The Device Handler module uses a standard set of utilities to communicate with the security module or the Transaction Security Services process, or verify the MAC in software. These utilities return one of the following codes to indicate the outcome of MAC verification processing in software:

Software	Security Module	Description
Pass	0	MAC verified successfully
Fail	1	Invalid MAC
Fail	2	Device failed
Fail	3	Operation failed

If the MAC is verified successfully, the Device Handler module resets the consecutive error counter and continues normal processing. Message authentication processing is complete.

If the MAC is invalid, the Device Handler module increments the consecutive error counter, if one is being maintained.

If the MAC is invalid or the verification operation fails, the Device Handler module performs one of the following:

- If the consecutive error limit is greater than zero (meaning a consecutive error limit is being maintained) and the consecutive error counter is equal to or greater than the limit, the Device Handler module returns a bad MAC response to the device. The Device Handler module also resets the consecutive error counter and performs device-specific functionality to generate and load a new MAC transaction key.
- If the consecutive error limit is zero (meaning a consecutive error limit is not being maintained), the Device Handler module does not perform consecutive error limit processing.

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8: BASE24 PIN Verification Support

This section presents an introduction to the PIN verification support provided by BASE24-atm, BASE24-pos, and BASE24-teller. It also presents local PIN verification options provided by BASE24-atm and BASE24-teller, cardholder PIN selection and PIN change options provided by BASE24-atm, and options for identifying preverified PINs in the ISO Host Interface and BIC ISO Interface processes.

Several file maintenance screens are discussed in this section. These screens are illustrated in other BASE24 manuals.

BASE24 PIN Verification

PIN verification, that is, the authentication of a cardholder by a transaction authorizer, is performed by BASE24-atm, BASE24-pos, and BASE24-teller Authorization processes. BASE24-atm and BASE24-teller also support local PIN verification, allowing PINs to be verified at a terminal before the transaction is forwarded to the BASE24 system for authorization. BASE24-atm also supports cardholder PIN selection and PIN change options.

PIN verification confirms that the PIN entered with a transaction is the actual PIN of the cardholder authorized to use the card involved in the transaction. When transactions are received by a BASE24 Authorization process, part of the authorization procedure can include verification of the PIN.

If PINs are to be verified by the Authorization process, they can be verified in software or by a security module. For verifying PINs in software, the PIN and all of the data required for verification must be in the clear so that they can be used by the Authorization process. For verifying PINs using a security module, the PIN is carried in the system encrypted under a key known only to the security module.

The following documentation includes the PIN verification options provided by BASE24-atm, BASE24-pos, and BASE24-teller. Information includes basic configuration options, BASE24 Standard Internal Message field descriptions, and an overview of the procedures performed within the Authorization process. BASE24-atm, BASE24-pos, and BASE24-teller, while similar, have their own unique characteristics and are discussed separately. It also discusses how the ISO Host Interface and BIC ISO Interface processes can identify whether a PIN has already been verified by altering the PIN block.

Pre-verified PIN Identification

If a PIN has already been verified, the ISO Host Interface and BIC ISO Interface processes set the PIN block to all hexadecimal “F” characters or another character defined by the PRE-VERIFIED-PIN-PAD-CHAR LCONF param. Although this parameter defaults to a value of “F”, you may want to set it to a non-hexadecimal character (i.e., an alphanumeric character other than 0-9 and A-F) to ensure that it cannot be confused with a valid PIN block, although the chances of a valid PIN block containing all “F” characters is only 1 in 2^{64} . The host must be able to identify whatever character you use to denote a preverified PIN.

BASE24-pos Note: BASE24-pos transactions may or may not contain a PIN, and authorizing systems must be able to differentiate between transactions with a pre-verified PIN and transactions that do not contain a PIN (i.e., that were not entered with a PIN). Therefore, for incoming and outgoing BASE24-pos transactions, a PIN block containing this LCONF pad character is always used to indicate a PIN has been pre-verified. The absence of the P-52 (Personal Identification Number Data) data element in the external message is used to indicate the transaction did not contain a PIN (for incoming BASE24-pos messages, a space-filled P-52 data element is also acceptable to indicate the transaction did not contain a PIN).

BASE24-atm and BASE24-teller Note: BASE24-atm and BASE24-teller transactions always require a PIN. Therefore, for these product transactions, a pre-verified PIN can be accurately identified by the absence of the P-52 (Personal Identification Number Data) data element in the external message. For outgoing messages, BASE24-atm and BASE24-teller always identify a pre-verified PIN by omitting the P-52 data element. For incoming messages, the host has several options—BASE24-atm and BASE24-teller assumes a PIN has been pre-verified if the P-52 data element is omitted, contains spaces, or is filled by the pad character specified in this param.

BASE24-atm PIN Verification

PIN verification processing specific to BASE24-atm is performed by the BASE24-atm Authorization process. The BASE24-atm Authorization process offers several options related to PIN verification.

Institution and Card Prefix PIN Verification Parameters

BASE24-atm provides each institution the option of establishing a separate set of PIN verification parameters for each of the card prefixes it issues, using the same parameters for all card prefixes, or using a combination of the two options. The following parameters appear on screen 2 of the Institution Definition File (IDF) and on screen 2 of the Card Prefix File (CPF). References to these parameters in the remainder of this section include both files. The appropriate file depends on whether the parameters are being maintained at the institution level or the card prefix level. For more information on these fields, refer to the ***BASE24 Base Files Maintenance Manual***.

PIN VERIFICATION KEYA GROUP — Identifies which Key Authorization File (KEYA) record is used for PIN verification information. The value in this field is matched with the value in the GRP field on the KEYA screens.

This field only appears on CPF screen 2. On IDF screens, the value in the FIID field is used for this purpose.

Note: The GRP field on KEYA screens is just part of the key to KEYA records. A beginning date and an ending date that are compared to a card expiration date and a record type that identifies the PIN verification method complete the information needed to fully identify the correct KEYA record.

PIN CHECK TYPE — Indicates the PIN verification method to be used. The Authorization process determines the type of PIN verification to be performed from the value in this field. Valid values are follows:

- 00 = No PIN verification
- 01 = IBM DES (3624-format) PIN verification
- 02 = Diebold PIN verification
- 03 = Identikey PIN verification
- 04 = Visa PVV PIN verification

Each of these types of PIN verification relies on other field settings in the BASE24 database in order to be performed correctly. For information on how the entire database must be set up for each of these PIN verification methods, refer to sections 9 through 12.

The PIN CHECK TYPE field on CPF screen 2 also indicates whether the card prefix uses its own PIN verification parameters or the institution-level parameters that are defined on IDF screen 2, so there is an additional value (i.e., 99) on CPF screen 2. Valid values are as follows:

- 00 = No PIN verification—Use CPF parameters
- 01 = IBM DES (3624-format) PIN verification—Use CPF parameters
- 02 = Diebold PIN verification—Use CPF parameters
- 03 = Identikay PIN verification—Use CPF parameters
- 04 = Visa PVV PIN verification—Use CPF parameters
- 99 = Use the PIN verification parameters on IDF screen 2

An institution with multiple card prefixes can use institution-level parameters for some prefixes while using card prefix-level parameters for other card prefixes. The only restriction is that all of the parameters used by one prefix be defined on a single screen (i.e., CPF screen 2 or IDF screen 2).

MAX PIN TRIES — Specifies the maximum number of consecutive invalid PIN entries that are allowed.

BAD PIN ACTION — Specifies the action to be taken when the maximum number of invalid PIN entries is exceeded.

CARDHOLDER PIN SELECT — Identifies whether BASE24-atm customers can select their PIN the first time they use their card.

ALGO NUMBER LOC — Specifies the location of the algorithm number used with the Diebold PIN verification method.

CHECK IF HOST ONLINE PIN — Specifies whether a cardholder's PIN is to be checked during transaction screening by BASE24. BASE24-atm can perform this check with authorization levels 1 (online) and 3 (online/offline).

POFST/PVV LOC — Specifies the location of the PIN verification digits, which include PIN offset for IBM DES (3624-format) or Diebold PIN verification methods, PIN Verification Value (PVV) for the Visa PVV PIN verification method, or the PIN Verification Number (PVN) for the Identikay PIN verification method.

PIN TRIES RESET OPTION — Specifies when the bad PIN tries counter is reset.

PIN Verification by Authorization Level

Depending on the authorization level under which a transaction is being processed, PIN verification can be switched on or off for each card prefix. This is controlled by the PIN CHECK TYPE and CHECK IF HOST ONLINE PIN fields on screen 2 of the IDF or CPF.

The PIN CHECK TYPE field identifies the method the BASE24-atm Authorization process is to use in performing PIN verification, and the CHECK IF HOST ONLINE PIN field specifies whether PINs are to be verified during transaction screening. A value of 00 in the PIN CHECK TYPE field means the BASE24-atm Authorization process is not to perform PIN verification at any time, regardless of the CHECK IF HOST ONLINE PIN field setting.

There are three authorization levels available through BASE24: level 1 (online), level 2 (offline), and level 3 (online/offline).

Under authorization level 1 (online), transactions must be sent to a host for authorization. Under this authorization level, BASE24-atm can be configured to perform PIN verification as a part of its transaction screening actions by setting the CHECK IF HOST ONLINE PIN field to a value of Y and the PIN CHECK TYPE field to a value of 01–04. This means that BASE24-atm verifies the cardholder's PIN prior to sending the transaction to the host and declines the transaction if the cardholder's PIN is incorrect.

Under authorization level 2 (offline), transactions are not authorized by a host, but by BASE24-atm exclusively. In this case, the value of the CHECK IF HOST ONLINE PIN field is not used. The PIN CHECK TYPE field determines if and how PINs are verified.

Under authorization level 3 (online/offline), transactions must be sent to a host for authorization but can be subsequently authorized by BASE24-atm if the host is unavailable. Under this authorization level, BASE24-atm can be configured to perform PIN verification as a part of its transaction screening actions by setting the

CHECK IF HOST ONLINE PIN field to a value of Y and the PIN CHECK TYPE field to a value of 01–04. In this case, the BASE24-atm Authorization process verifies the cardholder's PIN prior to sending the transaction to the host and declines the transaction if the cardholder's PIN is incorrect. Subsequently, if the transaction is sent to the host and the host is unavailable, the BASE24-atm Authorization process stands in to authorize the transaction. BASE24-atm then automatically verifies the PIN if it did not verify the PIN prior to sending the transaction to the host.

Maximum PIN Retries

BASE24-atm keeps track of the number of bad PIN tries for a cardholder in the Cardholder Authorization File (CAF) or Usage Accumulation File (UAF), depending on the authorization parameters established by an institution for each of its card prefixes. This allows the number of bad PIN tries to accumulate over a period of time, and institutions can then choose to decline a transaction for a cardholder if that cardholder has an excessive number of incorrect PIN tries.

BASE24-atm offers two methods for resetting a cardholder's bad PIN tries counter. The first is the entry of a correct PIN by the cardholder, and the second is the standard clearance of CAF and UAF accumulation totals at the end of each usage accumulation period.

Because of the way the UAF totals are cleared, the bad PIN tries counter in the UAF is always reset at the end of each usage accumulation period. However, in addition, institutions using the UAF can elect, using the PIN TRIES RESET OPTION field in the IDF or CPF, to have the UAF bad PIN tries counter reset by the entry of a correct PIN.

Unlike the UAF, the bad PIN tries counter in the CAF is not automatically reset at the end of each usage accumulation period. Institutions using the CAF can choose, using the PIN TRIES RESET OPTION field, to have the CAF bad PIN tries counter automatically reset with the rest of the totals at the end of each usage accumulation period. If the CAF bad PIN tries counter is not reset at the end of each usage accumulation period, the entry of a correct PIN is the only means for resetting it other than a file maintenance update or a refresh of the cardholder's record.

The following fields on IDF or CPF screen 2 must be set as indicated to define an institution's bad PIN tries processing:

BAD PIN ACTION — Specifies the action to be taken when an ATM receives a response code that indicates PIN tries were exceeded.

- If the ATM is to return the card to the cardholder when PIN tries are exceeded, this field can be set to a value of 0.
- If the ATM is to retain the card (if the ATM has card retention functionality), this field can be set to a value of 1. Cards cannot be retained by ATMs where the card is swiped instead of taken inside the machine to be read.

MAX PIN TRIES — Specifies the number of times a cardholder can enter an incorrect PIN.

- If the PIN TRIES RESET OPTION field is set to a value of 0, 1, or 3, once this number is reached, subsequent transactions are automatically declined and the action specified in the BAD PIN ACTION field is taken.
- If the PIN TRIES RESET OPTION field is set to a value of 2 or 4, the bad PIN tries counter is incremented with each incorrect PIN, but is not compared to the value in the MAX PIN TRIES field.

PIN TRIES RESET OPTION — A code indicating how the bad PIN tries counters in the Cardholder Authorization File (CAF) and Usage Accumulation File (UAF) are to be reset for an institution's cardholders. Valid values are as follows.

- 0 = Reset the bad PIN tries counter at the end of the usage accumulation period, but not when a correct PIN is entered.
- 1 = Reset the bad PIN tries counter at the end of the usage accumulation period. Also reset the bad PIN tries when a correct PIN is entered and the number of bad PIN tries does not exceed the maximum PIN tries.
- 2 = Reset the bad PIN tries counter at the end of the usage accumulation period. Also reset the bad PIN tries when a correct PIN is entered, regardless of the number of bad PIN tries.
- 3 = Reset the bad PIN tries counter when a correct PIN is entered and the number of bad PIN tries does not exceed the maximum PIN tries.
- 4 = Reset the bad PIN tries counter when a correct PIN is entered, regardless of the number of bad PIN tries.

PIN Selection and PIN Change Options

In certain configurations, BASE24-atm gives institutions the ability to allow their cardholders to select or change their PINs at the ATM. The BASE24-atm PIN select option allows cardholders to select their first PINs at the ATM. The BASE24-atm PIN change option allows cardholders to change their existing PINs at the ATM.

The PIN select option differs from the PIN change option in several ways:

- PIN selection is one step in a BASE24-atm transaction while PIN change is a stand-alone BASE24-atm transaction.
- PIN selection is part of the first customer transaction only, while the PIN change option is a transaction that a customer can perform at any time after the account has been opened.
- PIN selection limits PIN verification to the IBM DES (3624-format) or Diebold methods. The PIN Change transaction can be performed using these methods plus the Visa PVV method in software plus all four methods with security modules.

The PIN select and PIN change options also have several similarities:

- Both options can be used on the same account.
- Both options require the Positive or Positive Balance Authorization methods.
- Both options require the PIN verification digits to be stored in the CAF instead of in magnetic stripe data.
- Local PIN verification (i.e., PIN verification performed at the ATM before the BASE24 system receives the transaction) cannot be performed with either option because local PIN verification requires the PIN verification digits to be carried in the magnetic stripe data instead of being stored in the CAF.
- Both options can be performed in software or hardware.

Note: When using the BASE24-atm EMV add-on product with the Transaction Security Services process, PINs can also be changed offline of the BASE24 system at the ATM itself. Additional options to support this function are discussed in the *BASE24-atm EMV Support Manual* and the *BASE24 Transaction Security Services Processing Guide*.

Cardholder PIN Select Option

The cardholder PIN Select option allows cardholders to select their own PINs at the ATM. When this option is used, BASE24-atm accepts the first PIN received from the cardholder, computes the corresponding PIN verification digits, and places the PIN verification digits in the CAF. From that point on, BASE24-atm will recognize the PIN entered by the cardholder during the first transaction as the cardholder's valid PIN.

The following BASE24-atm configuration requirements apply when cardholder PIN selection is offered:

- The PIN verification method must be IBM DES (3624-format) or Diebold.
- The BASE24-atm authorization method must be Positive or Positive Balance so the cardholder has a CAF record.
- PIN verification digits must be carried in the CAF.
- Cards must be placed on file with a status of open or issued, but not active.
- Local PIN verification cannot be used with the cardholder PIN change option because the PIN verification digits must be in the magnetic stripe data of the card for local PIN verification and must be in the CAF for cardholder PIN selection. The PIN verification digits cannot be carried in both places.

An institution should give consideration to the following items before implementing cardholder PIN selection:

- Customers do not have to identify themselves to verify they are the legitimate cardholders before selecting a PIN, creating the possibility of activating lost, forgotten, or stolen cards. This procedure is not in compliance with the ANSI X9.8 standard or the ISO 9564 standard.
- Customers enter their new PINs only once. The ISO 9564 standard recommends a new PIN be entered twice to minimize the possibility of assigning a PIN not intentionally selected by the cardholder.

The following fields located on the CPF, CAF, IDF, and BASE24-atm Terminal Data files (ATD) screens must contain specific values when cardholder PIN selection is in effect.

CPF Screen 1

POFST/PVV — Contains a value defining the position of the IBM DES (3624-format) PIN offset, Diebold PIN offset, Visa PVV PIN, or the Identikey PVN from the magnetic stripe data on the card. This field must be set to zero to indicate that the POFST/PVV field on CAF screen 1 contains the PIN offset.

CPF or IDF Screen 2

CARDHOLDER PIN SELECT — Contains a code, used by BASE24-atm only, identifying whether cardholders are allowed to select their PIN the first time they use their card. The value in this field is used for the institution's proprietary debit cards only. The CARDHOLDER PIN SELECT field can be set to a value of Y to indicate cardholder PIN selection processing is to be performed only if the PIN CHECK TYPE field is set to a value of 01 for DES (IBM 3624) or 02 (Diebold).

POFST/PVV LOC — A code specifying the location of the IBM DES (3624-format) or Diebold PIN verification method PIN offset, the Visa PVV PIN verification method PIN PVV, or the Identikey PIN verification method PIN PVN. The field must be set to a value of 2 because the PIN offset (for the IBM 3624 or Diebold format) is stored in the CAF.

CAF Screen 1

CARD STATUS — A code identifying the status of the card. This field must be set to a value of 0 (issued, but not active) or 1 (open) because cardholder PIN selection applies to the initial PIN selection only. All other card status values indicate conditions where unsupervised PIN selection is not advised.

POFST/PVV — Contains the PIN offset value, PIN verification number (PVN), or the PIN verification key indicator (PVKI) and the PIN verification value (PVV). This field must be left blank because BASE24-atm interprets any entry other than blanks to mean a PIN has already been selected. This field can be reset to blanks to allow a customer to reselect a PIN at a later time.

ATD Screen 9

LOCAL PIN VERIFY — Specifies whether local PIN verification is allowed for this terminal. This field must be set to a value of 1 because local PIN verification cannot be allowed at the terminal if the cardholder PIN select option is used.

Refer to sections 9 and 10 for additional information on IBM DES (3624-format) and Diebold PIN verification methods.

PIN Change Option

The PIN change option allows cardholders to change their existing PIN at the ATM. When this option is used, BASE24-atm uses the existing PIN to verify the cardholder and then allows the cardholder to enter their new PIN twice, computes the corresponding PIN verification digits, and places the PIN verification digits in the CAF. From that point on, BASE24-atm recognizes the new PIN entered by the cardholder as the cardholder's valid PIN.

The following devices support the BASE24-atm PIN change transaction without any additional hardware requirements:

- IBM 3624-format Version 8
- Diebold 910/911/912
- Diebold 10XX/478X
- Fujitsu 7000
- NCR 5XXX
- Olivetti 6000
- Tidel

The following BASE24-atm configuration requirements apply when the PIN change transaction is offered:

- If PIN verification is performed in software, the PIN verification method must be IBM DES (3624-format), Diebold, or Visa PVV.
- If PIN verification is performed using Atalla Network Security Processor (NSP) security modules, all four PIN verification methods are available.
- If PIN verification is performed using Thales e-Security security modules, all PIN verification methods except Identikey are available.

- The BASE24-atm authorization method must be Positive or Positive Balance so the cardholder has a CAF record.
- PIN verification digits must be carried in the CAF.
- Local PIN verification cannot be used with the cardholder PIN Change option because the PIN verification digits must be in the magnetic stripe data of the card for local PIN verification and must be in the CAF for the cardholder PIN Change option. The PIN verification digits cannot be carried both places.

The following fields located on the various BASE24 files maintenance screens indicated must contain specific values when the PIN change transaction is offered.

APCF Screen 2

The fields on this screen specify whether a transaction is allowed for a specific acquirer transaction profile. The PIN change transaction must be included in the acquirer transaction profile configured for ATMs from which PIN change transactions can be entered. This screen is used to establish acquirer transaction profiles for Diebold 10XX/478X and NCR 5XXX ATMs at the institution level in the IDF or at the terminal level in the BASE24-atm Terminal Data files.

ATD Screen 3

PIN CHANGE — Controls whether the PIN change transaction is included in the set of transactions that another institution's (not-on-us) cardholders can perform at this terminal. This field is used for all device types except for Diebold 10XX/478X and NCR 5XXX. For these two device types, allowed transactions for both on-us and not-on-us cardholders (including the PIN change transaction) are configured using the ACQUIRER TXN PROFILE field on ATD screen 2 and Acquirer Processing Code File (APCF) screen 2.

IDF Screen 9

ACQUIRER TXN PROFILE — Controls whether the PIN change transaction is included in the set of transactions that the terminal-owning institution allows. The value in this field can be overridden at the terminal level in the BASE24-atm Terminal Data files.

IDF Screen 13

HOST PIN CHANGE OPTION — Determines where the new PIN verification digits are computed for a PIN change transaction. BASE24-atm uses the value in this field with an online/offline authorization level when BASE24-atm is prescreening PINs or standing in for the host.

- If BASE24-atm prescreens the existing PIN and computes new PIN verification digits for a PIN change transaction, regardless of whether or not the host is available, this field must be set to a value of 0.
- If BASE24-atm verifies the existing PIN and computes new PIN verification digits for a PIN change transaction only when the host is unavailable, this field must be set to a value of 1.
- If BASE24-atm declines a PIN change transaction when the host is unavailable, this field must be set to a value of 2.

For PIN change transactions performed with authorization level 3, BASE24-atm checks the cardholder's PIN regardless of the setting in the CHECK IF HOST ONLINE PIN field on IDF screen 2 if the setting in the HOST PIN CHANGE OPTION field on IDF screen 13 is set to a value of 0.

IPCF Screen 2

The fields on this screen specify whether a transaction is allowed for a specific issuer transaction profile. The PIN Change transaction must be allowed in the issuer transaction profiles configured for the institution in the IDF, the cardholder's account in the CAF, or the cardholder's account prefix in the CPF.

CPF or IDF Screen 2

POFST/PVV LOC — A code specifying the location of the DES (IBM 3624) or Diebold PIN verification method PIN offset, the Visa PVV, or the Identikey PVN. This field must be set to a value of 2 because the PIN verification digits are stored in the CAF.

CPF Screen 1

POFST/PVV — A value defining the position of the DES (IBM 3624) or Diebold PIN verification method PIN offset, the Visa PVV, or the Identikey PVN. This field must be set to zero to indicate that the POFST/PVV field on CAF screen 1 contains the PIN verification digits.

IDF Screen 9, CPF Screen 4, or CAF Screen 8

ISSUER TXN PROFILE/PRFL — Controls whether the PIN change transaction is included in the set of transactions that the card issuing institution allows for the institution, card prefix, or cardholder account.

CAF Screen 1

POFST/PVV — Contains the PIN verification digits for the current PIN. If this field contains blanks to allow customers to select their initial PINs, the PIN change transaction cannot be performed until after the first PIN has been established.

CAF Screen 8

ISSUER TXN PRFL — Controls whether the PIN change transaction is included in the set of transactions that the card issuing institution allows for the cardholder account.

ATD Screen 9

LOCAL PIN VERIFY — Specifies whether local PIN verification is allowed for this terminal. This field must be set to a value of 1 because local PIN verification cannot be allowed at the terminal if the PIN change option is used.

Offline PIN Management

The BASE24-atm EMV add-on product allows issuers to manage the offline PIN on an EMV card by providing support for the following functions:

- Synchronized changing of the online PIN and the offline PIN
- Resetting of the offline bad PIN try counter

The standard PIN Change transaction is used to support the synchronized change of the online PIN and the offline PIN on an EMV card. The enhanced PIN Change transaction processing is an extension to the processing required for the standard PIN Change transaction. The additional processing results in a PIN CHANGE issuer script command being returned to the card in the transaction response message. A hardware security module is used to encrypt the new PIN data included in the issuer script.

The EMV PIN Unblock transaction is used to support the resetting of the offline bad PIN try counter on an EMV card. The processing performed for the EMV PIN Unblock transaction processing is similar to the processing performed for a standard PIN Change transaction, except that new PIN data is not included in the EMV PIN Unblock transaction. Instead, a PIN CHANGE/UNBLOCK issuer script command is returned to the card in the transaction response message.

The BASE24-atm NCR 5XXX Device Handler process supports offline PIN management transactions when the BASE24-atm EMV add-on product is installed.

The following Interface processes support offline PIN management transactions:

- BIC ISO Interchange Interface process
- EPS-Net Interchange Interface process
- ISO Host Interface process
- LINK ISO Interchange Interface process
- VisaNet ISO Interchange Interface process

The configuration requirements and processing performed by BASE24 for offline PIN management transactions are described in detail in the ***BASE24-atm EMV Support Manual***.

Note: You must use the Transaction Security Services process with the BASE24-atm EMV add-on product to support the generation of PIN CHANGE scripts. The Transaction Security Services process is not required by issuers who

want to support the generation of PIN UNBLOCK scripts only, or who support offline PIN management transactions as an acquirer function only (i.e., where the transactions themselves are authorized on a back-end host). For more information on the offline PIN management function support provided by the Transaction Security Services process, refer to the ***BASE24 Transaction Security Services Processing Guide***.

Local PIN Verification Option

Local PIN verification is the verification of the customer's PIN at an ATM before the transaction is sent to BASE24-atm. Local PIN verification reduces communications and processing costs because only transactions with correct PINs are transmitted to BASE24-atm for authorization. Transactions with invalid PINs do not leave the device.

The following BASE24-atm Device Handler processes support local PIN verification:

- IBM 3624-format Version 8
- Diebold 910/911/912
- Diebold 10XX/478X
- Fujitsu 7000
- NCR 5XXX
- Olivetti 6000

An institution should give consideration to the following items when implementing local PIN verification:

- The customer PIN select and PIN change options cannot be used with locally verified PINs because these options require the PIN verification digits to be carried in the CAF and local PIN verification requires the PIN verification digits to be carried in the magnetic stripe data of the plastic card. The PIN verification digits cannot be carried both places simultaneously.
- ATMs may require additional hardware to verify PINs locally.
- ATMs must be configured to perform local PIN verification. Refer to vendor documentation for additional information.
- Bad PIN tries are not tracked on the BASE24 database when PINs are verified locally.

- Certain external processors may not support locally verified PINs.
- The BASE24 Interchange (BIC) ISO Interface and ISO Host Interface processes support locally verified PINs.

The following fields located on the various BASE24 files maintenance screens indicated must contain specific values when PINs are verified locally.

CPF or IDF Screen 2

CARDHOLDER PIN SELECT — Contains a code, used by BASE24-atm only, identifying whether cardholders are allowed to select their PIN the first time they use their card. This field must be set to a value of N because cardholder PIN select and local PIN verification cannot be performed on the same card.

POFST/PVV LOC — A code specifying the location of the IBM DES (3624-format) or Diebold PIN verification method PIN offset, the Visa PVV PIN verification method PIN PVV, or the Identikey PIN verification method PIN PVN. This field must be set to a value of 1 to indicate that the PIN verification digits are located in the magnetic stripe data of the card.

CAF Screen 1

POFST/PVV — Contains the PIN offset value, PIN verification number (PVN), or the PIN verification key indicator (PVKI) and the PIN verification value (PVV). This field is not used because the PIN verification digits are located in the magnetic stripe data of the card.

CPF Screen 1

POFST/PVV — Specifies the location of the PIN verification digits in the magnetic stripe data of the card.

ATD Screen 9

LOCAL PIN VERIFY — Specifies whether local PIN verification is allowed for this terminal.

- If local PIN verification is allowed for all transactions at the terminal, this field must be set to a value of 0. The ATM device must have PIN verification capability and the ATM configuration file must be configured for local PIN verification. Refer to the device-specific BASE24-atm device support manuals for additional information about the ATM configuration file.
- If local PIN verification is not allowed for any transactions at the terminal, this field must be set to a value of 1.
- If local PIN verification is allowed only for on-us transactions at the terminal, this field must be set to a value of 2. An on-us transaction is one where the same institution issues the card and owns the terminal.

BASE24-atm Internal Messages

When a message is received from an external source, BASE24-atm converts that message from the native format of the external source to a Standard Internal Message (STM) format recognizable to the BASE24-atm processes.

All information relating to the PIN, its method of encryption, and the key under which it is encrypted is carried in the STM.

BASE24-atm internal messages contain the following information for PIN verification. Each description includes the name of the STM field that contains the information.

ANSI Offset — Indicates the starting position in the PAN of the 12 digits used to create the PIN/PAN PIN block if that type of PIN block was used in encrypting the PIN. This value can be greater than, less than, or equal to 0. (A 0 indicates the first position of the PAN.)

BASE24-atm uses the 12 right-most digits of the PAN excluding the check digit to create the PAN string.

Example: A PAN of 1049000451234534567 would result in a PAN string of 045123453456 and an ANSI offset of 6.

If the PAN does not contain enough significant digits to provide the 12 digits necessary (i.e., the PAN is less than 13 digits in length), BASE24-atm adds leading zeros to the PAN used in creating the PIN block and sets the ANSI offset to a negative value. The absolute value of this negative ANSI offset identifies the number of leading zeros BASE24-atm has added to the PAN after right-justifying the available significant digits.

Example: A PAN of 1234534567 would result in a PAN string of 000123453456 and an ANSI offset of -3.

STM Field: RQST.ANSI-OFST

Error Flag — Indicates a sanity check error, fraud possibility, or an error in the new PIN information in a PIN change transaction. BASE24-atm Device Handler, Authorization, Host Interface, and some Interchange Interface processes can set this flag. Valid values pertaining to PIN verification are as follows:

- F = Fraud possibility. The terminal did not send a PIN in the transaction and the terminal is not configured to support local PIN verification.
- L = Invalid PIN length. The PIN sent in the transaction was of an invalid length.
- N = No response received (timeout).
- O = One PIN try remaining.
- P = PIN change error. An error occurred in the new PIN information in a PIN Change transaction.
- R = Sanity check error or PIN block error, previous zone. A sanity check error or PIN block error was received from a security device while processing a transaction involving PIN translation or verification. Dynamic key management error counters, if used, should not be incremented for this error.
- S = Sanity check error or PIN block error, current zone. A sanity check error or PIN block error was received from a security device while processing a transaction involving PIN translation or verification. Dynamic key management error counters, if used, should be incremented for this error.
- T = Token error. The internal message for a PIN change transaction did not contain the PIN Change token.

STM Field: AT50-TKN.ERR-FLG

Local PIN Verify Flag — Indicates whether local PIN verification is allowed at the ATM for all transactions, is allowed for on-us transactions only, or is not allowed.

STM Field: AT50-TKN.LOCL-PIN-VRFY-FLG

New PIN Format — Indicates whether the new PIN in a PIN change transaction is in the clear or encrypted.

STM Field: PINC-TKN.NEW-PIN-FRMT

New PIN Offset — The verification digits of the new PIN in a PIN change transaction.

STM Field: PINC-TKN.NEW-PIN-OFST

New PIN Size — Contains the size (i.e., length in bytes) of the new PIN in a PIN change transaction.

STM Field: PINC-TKN.NEW-PIN-SIZE

New PIN 1 — Contains the first entry, in clear or encrypted form, of the new PIN in a PIN change transaction.

STM Field: PINC-TKN.NEW-PIN-1

New PIN 2 — Contains the second entry, in clear or encrypted form, of the new PIN in a PIN change transaction.

STM Field: PINC-TKN.NEW-PIN-2

PIN — Contains the PIN in the clear or in encrypted form. If the PIN is carried in encrypted form, it is really the PIN block that is carried in the message.

The Authorization process always fills this field with zeros if it approves the transaction or declines it for any reason. The Device Handler process fills this field with zeros if the PIN has been verified locally at the ATM.

STM Field: RQST.PIN

PIN Count — Indicates the number of new PIN entries present in a PIN change transaction.

STM Field: PINC-TKN.PIN-CNT

PIN Format — Indicates whether the PIN is in the clear or encrypted. If the PIN is encrypted, this field also indicates the type of PIN block used in the encryption: PIN/PAN PIN block or PIN/PAD PIN block.

STM Field: RQST.PIN-FRMT

PIN Key — The encrypted form of the key under which the PIN is currently encrypted, if the PIN is encrypted.

PINs travel through the BASE24-atm system in the clear or encrypted under a secure key. This field carries the secure key.

STM Field: RQST.PIN-KEY

PIN Pad Character — Indicates the PIN pad character used to create the PIN/PAD PIN block, if that type of PIN block was used in encrypting the PIN.

STM Field: RQST.PINPAD-CHAR

PIN Size — Contains the size (i.e., length in bytes) of the PIN, if the PIN is in the clear. If the PIN is being carried in encrypted form, the value in this field is set to 16 (the size of the PIN block).

STM Field: RQST.PIN-SIZE

PIN Tries — Indicates the number of times the cardholder has entered the PIN incorrectly during the current session at the terminal. This field is initially set to zero and subsequently incremented each time the PIN is checked and determined to be incorrect.

Once a PIN has been verified, the Authorization process sets the value of this field to Z, indicating additional transactions on the card can be processed during the current session without reverifying the PIN.

With local PIN verification, the Device Handler process sets the value of this field to Z, indicating the PIN has been verified at the ATM and does not need to be verified by the Authorization process.

STM Field: RQST.PIN-TRIES

Previous Key — The encrypted form of the previous key used by an external source to encrypt PINs. After an online key exchange has taken place between BASE24-atm and an external source, the interface process saves the previous key and places it in the BASE24-atm internal message to allow for situations in which PINs are subsequently sent from the external source encrypted under the previous key.

Some interface processes use the CLEAR OLD KEY TIMER VALUE field in the Key File (KEYF) to control how long the previous key can be used. This field is accessed using KEYF screen 3.

STM Field: RQST.USER-KEY

BASE24-atm PIN Verification Processing

The BASE24-atm Authorization process takes the following PIN verification actions during its processing of a transaction.

PIN verification actions constitute only a portion of the overall transaction processing performed by the Authorization process.

The steps summarized here are intended to give a general view of PIN verification. For specific information on how each PIN verification method operates, refer to sections 9 through 12.

The Authorization process performs the following:

1. Determines whether the PIN has been verified locally.
 - If the PIN has been verified by the ATM, the Device Handler process has already set the PIN field in the internal message to zeros, set the PIN tries field in the internal message to Z, and set the local PIN verify flag in the internal message based on the value in the LOCAL PIN VERIFY field on the device-specific ATD screen 9. Processing continues with step 2.
 - If the PIN has not been verified by the ATM, processing continues with step 3.
2. Performs a fraud check on the locally verified PIN, if necessary.
 - If the local PIN verify flag in the internal message contains a value of 0, local PIN verification is allowed. The Authorization process continues processing the transaction without any further PIN processing.
 - If the local PIN verify flag in the internal message contains a value of 2 (local PIN verification is allowed only for on-us transactions) and the terminal-owning institution and card-issuing institution are the same, local PIN verification is allowed. The Authorization process continues processing the transaction without any further PIN processing.

- If the local PIN verify flag in the internal message contains a value of 1 (local PIN verification not allowed) or if the local PIN verify flag in the internal message contains a value of 2 (local PIN verification is allowed only for on-us transactions) and the terminal-owning institution and card-issuing institution are not the same, local PIN verification is not allowed. The Authorization process assigns the transaction a response code of 70 (system error) and an error flag of F (fraud possibility). Processing continues with step 11.
3. Determines whether to use IDF or CPF limits.
 - If the PIN CHECK TYPE field on CPF screen 2 is set to a value of 99 (use IDF parameters) and the PIN CHECK TYPE field on IDF screen 2 is set to a value in the range of 01 through 04 (verify PIN), processing continues with step 4 using the PIN verification limits on IDF screen 2.
 - If the PIN CHECK TYPE field on CPF screen 2 is set to a value in the range of 01 through 04 (verify PIN using CPF parameters), processing continues with step 4 using the PIN verification limits on CPF screen 2.
 4. Checks the number of incorrect PINs the cardholder has entered.

If the institution maintains BASE24-atm authorization files, each cardholder's incorrect PIN tries are accumulated in the BASE24 database. However, if the institution does not maintain BASE24-atm authorization files, meaning that authorizations are performed by a host exclusively, the number of incorrect PIN tries is carried in the PIN tries field of the internal message. In this case, PIN tries can be tracked only during each session at the terminal.

- If the PIN TRIES RESET OPTION field on screen 2 of the IDF or CPF is set to a value of 2 or 4, meaning accept a valid PIN regardless of the number of incorrect PIN tries, processing continues with step 5.
- If the PIN TRIES RESET OPTION field on screen 2 of the IDF or CPF is set to a value of 0, 1, or 3, meaning decline the transaction if the PIN tries limit has been exceeded, the Authorization process checks the number of incorrect PINs entered by the cardholder against the institution's maximum number specified in the IDF or CPF, depending on whether PIN verification parameters are set at the institution level or the card prefix level.
 - If the number of incorrect PIN tries is less than the maximum PIN tries allowed, processing continues with step 5.
 - If the number of incorrect PIN tries is greater than or equal to the maximum PIN tries allowed, the Authorization process declines the transaction with a response code of 62 (PIN tries exceeded) and continues processing with step 11.

5. Determines whether the PIN requires verification.
 - If the transaction is being processed under authorization level 2 (offline), processing continues with step 6 because PIN verification is always performed by BASE24.
 - If the transaction is being processed under authorization level 1 (online) or 3 (online/offline), the Authorization process checks the IDF or CPF to determine whether the PIN is to be verified by BASE24.
 - If the PIN is to be verified, processing continues with step 6.
 - If the PIN is not to be verified, the Authorization process continues processing the transaction without any further PIN processing.

6. Allows for PIN selection, if necessary.

If the institution's cardholders are allowed to select their own PINs and there are no PIN verification digits on file for the cardholder, the Authorization process computes PIN verification digits for the entered PIN and places them in the BASE24 database. Processing continues with step 9.

7. Verifies the PIN in software, if possible.

The Authorization process checks the PIN format in the internal message. If the PIN format in the internal message is 1 or 3, indicating that the PIN is in encrypted form, processing continues with step 8. If the PIN format in the internal message is 0, indicating that the PIN is in the clear, the Authorization process performs the following:

- a. Extracts the clear PIN from the PIN field in the internal message using the PIN size in the internal message.
- b. Verifies the clear PIN in software according to the PIN verification method selected for the card prefix. For an explanation of how the different PIN verification methods function, refer to sections 9 through 12.
 - If the PIN is valid, processing continues with step 9.
 - If the PIN is invalid, the Authorization process declines the transaction with a response code of 53 (invalid PIN) and increments the incorrect PIN tries counter in the CAF or UAF and STM. Processing continues with step 11.

8. Calls a security device to verify the PIN.

In step 7, the Authorization process checked the PIN format in the internal message to determine whether the PIN was in clear or encrypted form. If the PIN is in encrypted form, the Authorization process performs the following:

- a. Extracts the following information from the internal message and sends it to the security module for verifying the PIN:
 - The encrypted PIN block from the PIN field.
 - The encrypted PIN key from the PIN key field.
 - The PIN pad character, if the PIN format is 3, indicating that the PIN block is a PIN/PAD PIN block.
 - The PAN digits specified by the ANSI offset field in the internal message, if the PIN format is 1, indicating that the PIN block is a PIN/PAN PIN block.

Other information is forwarded to the security module, based on the type of verification to be performed. The information required for each type of PIN verification is described in sections 9 through 12.

The Authorization process uses a standard set of PIN utilities to communicate with the security module. These PIN utilities return one of the following codes to indicate the outcome of the PIN verification processing:

- 0 = PIN valid
- 1 = PIN invalid
- 2 = Security device failure (e.g., timeout)
- 3 = Invalid data
- 4 = Invalid parameter
- 5 = Sanity check or PIN block error (security module did not find PIN block as expected)

- b. Verifies the code returned from the PIN utilities.

If the PIN utilities return a status of 0 (PIN validated), processing continues with step 9.

If the PIN utilities return a status of 5, the Authorization process resends the PIN verification information to the security module using the previous key from the internal message rather than the current key. This is performed on the contingency that a different key was provided by an interface process.

- c. Determines the response code and error flag to assign the transaction if the PIN was not validated.

- If the PIN utilities return a status of 2, the Authorization process assigns the transaction a response code of 70 (system error) and an error flag of blank (none). Processing continues with step 11.
 - If the PIN utilities return a status of 5 with the PIN key from the internal message and the internal message does not contain a previous key, the Authorization process assigns the transaction a response code of 70 (system error) and an error flag of S (sanity check). Processing continues with step 11.
 - If the PIN utilities return a status of 5 with the previous key from the internal message, the Authorization process assigns the transaction a response code of 70 (system error) and an error flag of S (sanity check). Processing continues with step 11.
 - If the PIN utilities return a status of 1, 3, or 4, the Authorization process assigns the transaction a response code of 53 (invalid PIN) and an error flag of blank (none). The Authorization process also increments the bad PIN tries counter in the CAF or UAF and STM. Processing continues with step 11.
9. Sets the PIN tries field in the internal message to Z and the PIN field in the internal message to zeros.
10. Resets the bad PIN tries counter in the CAF or UAF, if appropriate.
- If the PIN TRIES RESET OPTION field on screen 2 of the CPF or IDF is set to 2 or 4 and the bad PIN tries counter in the CAF or UAF is greater than 0, the Authorization process resets the bad PIN tries counter to 0.
 - If the PIN TRIES RESET OPTION field on screen 2 of the CPF or IDF is set to 1 or 3 and the bad PIN tries counter in the CAF or UAF is greater than 0 but less than the value in the MAX PIN TRIES field, the Authorization process resets the bad PIN tries counter to 0.
11. Continues with its processing.
- If the transaction was declined, the Authorization process places the appropriate response code in the internal message and returns the transaction to the originator.
 - If the transaction was not declined and is a transaction other than a PIN change, the Authorization process continues processing the transaction.

BASE24-atm PIN Change Processing

A PIN change transaction can be compared to a financial transaction such as a deposit or withdrawal with one exception. A PIN change transaction changes the PIN that is assigned to a customer's card while a deposit or withdrawal transaction affects the balance of a customer's account.

A customer starts a PIN change, deposit, or withdrawal transaction the same way. By entering the current PIN, the customer establishes that he or she can use the card. Then, instead of specifying a transaction account and deposit or withdrawal amount, the customer enters the new PIN information. The customer must then use the new PIN to initiate future transactions with the card.

After verifying the PIN, the Authorization process performs the following:

1. Verifies that new PIN information is entered correctly.
 - If the PIN Change token is missing, the Authorization process places response code 74 (message edit error) and error flag T (token error) in the internal message and returns the transaction to the originator.
 - If the PIN Change token contains a PIN count of two and the new PIN entries are unequal, the Authorization process places response code 53 (invalid PIN) and error flag P (PIN change error) in the internal message and returns the transaction to the originator. Device Handler processes receiving this response give the customer another opportunity to enter the new PIN twice. Some Device Handler processes compare the new PIN entries and require them to be equal before placing them in a message that is passed to the Authorization process.
 - If the PIN Change token contains a PIN count of two and two new PIN entries that are equal, the Authorization process changes the PIN count to one and changes the second new PIN entry to zeros. Processing continues with step 2.
 - If the PIN Change token contains a PIN count of one and one new PIN entry, the transaction originator has already verified that the two new PIN entries are equal. Processing continues with step 2.

Note: The Authorization process does not compare the old and new PIN entries. The new PIN entries can be the same as the old PIN.

2. Generates new PIN verification digits for the CAF.

The Authorization process generates new PIN verification digits in software or using a security device, depending on values in the new PIN format and existing PIN format fields in the internal message. This processing is similar

to the processing performed in step 7 and step 8 of the PIN verification processing described earlier in this section. This time, however, the calculated PIN verification digits replace the existing PIN verification digits in the CAF. In PIN verification, the calculated PIN verification digits are compared to the existing PIN verification digits for accuracy.

Refer to the topic “PIN Change Option” earlier in this section for configuration requirements, including available PIN verification methods, for the PIN change transaction.

BASE24-pos PIN Verification

PIN verification processing specific to BASE24-pos is performed by the BASE24-pos Authorization process. The BASE24-pos Authorization process offers several options related to PIN verification.

Institution and Card Prefix PIN Verification Parameters

BASE24-pos provides each institution the option of establishing a separate set of PIN verification parameters for each of the card prefixes it issues, using the same parameters for all card prefixes, or a combination of the two options. The following parameters appear on screen 2 of the Institution Definition File (IDF) and on screen 2 of the Card Prefix File (CPF). References to these parameters in the remainder of this section include both files. The appropriate file depends on whether the parameters are being maintained at the institution level or the card prefix level. For more information on these fields, refer to the ***BASE24 Base Files Maintenance Manual***.

PIN VERIFICATION KEYA GROUP — Identifies which Key Authorization File (KEYA) record is used for PIN verification information. The value in this field is matched with the value in the GRP field on the KEYA screens.

This field only appears on CPF screen 2. On IDF screens, the value in the FIID field is used for this purpose.

Note: The GRP field on KEYA screens is just part of the key to KEYA records. A beginning date and an ending date that are compared to a card expiration date and a record type that identifies the PIN verification method complete the information needed to fully identify the correct KEYA record.

PIN CHECK TYPE — Indicates the PIN verification method used. The Authorization process determines the type of PIN verification to be performed from the value in this field. Valid values are as follows:

- 00 = No PIN verification
- 01 = IBM DES (3624-format) PIN verification
- 02 = Diebold PIN verification
- 03 = Identikey PIN verification
- 04 = Visa PVV PIN verification

Each of these types of PIN verification relies on other field settings in the BASE24 database in order to be performed correctly. For information on how the entire database must be set up for each of these PIN verification methods, refer to sections 9 through 12.

The PIN CHECK TYPE field on CPF screen 2 also indicates whether the card prefix uses its own PIN verification parameters or the institution-level parameters that are defined on IDF screen 2, so there is an additional value (i.e., 99) on CPF screen 2. Valid values are as follows:

- 00 = No PIN verification—Use CPF parameters
- 01 = IBM DES (3624-format) PIN verification—Use CPF parameters
- 02 = Diebold PIN verification—Use CPF parameters
- 03 = Identikey PIN verification—Use CPF parameters
- 04 = Visa PVV PIN verification—Use CPF parameters
- 99 = Use the PIN verification parameters on IDF screen 2

An institution with multiple card prefixes can use institution-level parameters for some prefixes while using card prefix-level parameters for other card prefixes. The only restriction is that all of the parameters used by one prefix be defined on a single screen (i.e., CPF screen 2 or IDF screen 2).

MAX PIN TRIES — Specifies the maximum number of consecutive invalid PIN entries that are allowed.

BAD PIN ACTION — Specifies the action to be taken when the maximum number of invalid PIN entries is exceeded.

ALGO NUMBER LOC — Specifies the location of the algorithm number used with the Diebold PIN verification method.

CHECK IF HOST ONLINE PIN — Specifies whether a cardholder's PIN is to be checked during transaction screening by BASE24. BASE24-pos can perform this check with authorization levels 1 (online) and 3 (online/offline).

POFST/PVV LOC — Specifies the location of the PIN verification digits, which include PIN offset for IBM DES (3624-format) or Diebold PIN verification methods, PIN Verification Value (PVV) for the Visa PVV PIN verification method, or the PIN Verification Number (PVN) for the Identikey PIN verification method.

PIN TRIES RESET OPTION — Specifies when the bad PIN tries counter is reset.

Optional PINs

A PIN is not always required to complete a BASE24-pos transaction. Many POS transactions are performed by merchants, providing the opportunity to properly identify the customer with a driver's license or other form of identification instead of a PIN.

For contactless transactions that originate at unattended terminals, a PIN is not required. However, you can optionally require PINs for contactless transactions that originate at attended terminals with PIN capabilities.

For each prefix, the card issuer can specify, using the PIN PROCESSING FLAG on screen 7 of the Card Prefix File (CPF), whether a PIN is required. Options include the following:

- 0 = PIN entry is not required to complete a transaction.
- 1 = PIN entry is required for every transaction. Transactions from terminals without PIN capabilities are declined.
- 2 = PIN entry is required with transactions from terminals with PIN capabilities; however, transactions from attended terminals without PIN capabilities are processed without the PIN. Transactions from unattended terminals without PIN capabilities are declined.
- 3 = PIN entry is required with transactions from terminals with PIN capabilities; however, transactions from terminals without PIN capabilities are processed without the PIN.
- 4 = PIN entry is required with transactions from terminals with PIN capabilities, except for contactless transactions; however, transactions from terminals without PIN capabilities are processed without the PIN.

The Authorization process uses the value carried in the first two bytes of the PT-SRV-ENTRY-MDE field in the internal message to determine whether the transaction is contactless. Contactless EMV transactions have a PT-SRV-ENTRY-MDE value of 07 and contactless magnetic stripe transactions have a PT-SRV-ENTRY-MDE value of 91.

The Authorization process uses the value carried in the third byte of the PT-SRV-ENTRY-MDE field in the internal message to determine whether the terminal originating the transaction has PIN entry capabilities.

When a transaction is received from a terminal that does not have PIN entry capabilities and the PIN PROCESSING FLAG on screen 7 of the Card Prefix File (CPF) is set to a value of 2, the Authorization process uses the value carried in the PT-SRV-COND-CDE field in the internal message to determine whether the terminal is attended or unattended. A value of 02 or 27 in this field indicates the terminal is unattended. Since the card owner cannot be verified at an unattended terminal, the transaction is declined.

Note: In situations where the PIN has been defined as being optional, BASE24-pos always attempts to verify any PIN it receives and respond accordingly.

PIN Verification by Authorization Level

Depending on the authorization level under which a transaction is being processed, PIN verification can be switched on or off for each card prefix. This is controlled by the PIN CHECK TYPE and CHECK IF HOST ONLINE PIN fields on screen 2 of the IDF or CPF.

The PIN CHECK TYPE field identifies the method the BASE24-pos Authorization process is to use in performing PIN verification, and the CHECK IF HOST ONLINE PIN field specifies whether PINs are to be verified during transaction screening. A value of 00 in the PIN CHECK TYPE field means the BASE24-pos Authorization process is not to perform PIN verification at any time, regardless of the CHECK IF HOST ONLINE PIN field setting.

There are three authorization levels available through BASE24: level 1 (online), level 2 (offline), level 3 (online/offline).

Under authorization level 1 (online), transactions must be sent to a host for authorization. Under this authorization level, BASE24-pos can be configured to perform PIN verification as a part of its transaction screening actions by setting the CHECK IF HOST ONLINE PIN field to a value of Y and the PIN CHECK TYPE field to a value in the range of 01 through 04. This means that BASE24-pos verifies the cardholder's PIN prior to sending the transaction to the host and declines the transaction if the cardholder's PIN is incorrect.

Under authorization level 2 (offline), transactions are not authorized by a host, but rather by BASE24-pos exclusively. In this case, the value of the CHECK IF HOST ONLINE PIN field is not used. The PIN CHECK TYPE field determines if and how PINs are verified.

Under authorization level 3 (online/offline), transactions must be sent to a host for authorization, but can be subsequently authorized by BASE24-pos if the host is unavailable. Under this authorization level, BASE24-pos can be configured to perform PIN verification as a part of its transaction screening actions by setting the CHECK IF HOST ONLINE PIN field to a value of Y and the PIN CHECK TYPE field to a value in the range of 01 through 04. In this case, the BASE24-pos Authorization process verifies the cardholder's PIN prior to sending the transaction to the host and declines the transaction if the cardholder's PIN is incorrect. Subsequently, if the transaction is sent to the host and the host is unavailable, the BASE24-pos Authorization process stands in to authorize the transaction. BASE24-pos automatically verifies the PIN at this time, if it did not verify the PIN before sending the transaction to the host.

Maximum PIN Retries

BASE24-pos keeps track of the number of bad PIN tries for a cardholder in the Cardholder Authorization File (CAF), Usage Accumulation File (UAF), or Administrative Card File (ADMN), depending on the authorization parameters established by an institution for each card prefix. This allows bad PIN tries to accumulate over a period of time, and institutions can then choose to decline a transaction for a cardholder if that cardholder has an excessive number of incorrect PIN tries.

BASE24-pos offers two methods for resetting a cardholder's bad PIN tries counter. The first is the entry of a correct PIN by the cardholder, and the second is the standard clearance of CAF, UAF, and ADMN accumulation totals at the end of each usage accumulation period.

Because of the way the UAF totals are cleared, the bad PIN tries counter in the UAF is always reset at the end of each usage accumulation period. However, in addition, institutions using the UAF can elect—using the PIN TRIES RESET OPTION field—to have the UAF bad PIN tries counter reset by the entry of a correct PIN.

Unlike the UAF, the bad PIN tries counters in the CAF and ADMN are not automatically reset at the end of each usage accumulation period. Institutions using the CAF or ADMN can choose—using the PIN TRIES RESET OPTION field—to have their CAF and ADMN bad PIN tries counters automatically reset with the rest of their totals at the end of each usage accumulation period. If the CAF and ADMN bad PIN tries counters are not reset at the end of each usage accumulation period, the entry of a correct PIN is the only means for resetting them other than a file maintenance update or a refresh of the cardholder's record.

The following fields on IDF or CPF screen 2 must be set as indicated to define an institution's bad PIN tries processing:

BAD PIN ACTION — Specifies the action to be taken by the device when it receives a response code that indicates PIN tries were exceeded.

- If the device is to return the card to the cardholder when PIN tries are exceeded, this field must be set to a value of 0.
- If the device is to retain the card when PIN tries are exceeded, this field must be set to a value of 1.

MAX PIN TRIES — Specifies the number of times a cardholder can enter an incorrect PIN.

- If the PIN TRIES RESET OPTION field is set to a value of 0, 1, or 3, once this number is reached, subsequent transactions are automatically declined and the action specified in the BAD PIN ACTION field is taken.
- If the PIN TRIES RESET OPTION field is set to a value of 2 or 4, the number of bad PIN tries is incremented with each incorrect PIN, but is not compared to the value in the MAX PIN TRIES field.

PIN TRIES RESET OPTION — A code indicating how the bad PIN tries counters in the Cardholder Authorization File (CAF), Usage Accumulation File (UAF), and Administrative Card File (ADMN) are to be reset for an institution's cardholders. (The ADMN is used by BASE24-pos to keep track of bad PIN tries for administrative cards.) This field must be set to one of the following values to specify when the bad PIN tries counters are reset.

- 0 = Reset the bad PIN tries counter at the end of the usage accumulation period, but not when a correct PIN is entered.
- 1 = Reset the bad PIN tries counter at the end of the usage accumulation period. Also reset the bad PIN tries when a correct PIN is entered and the number of bad PIN tries does not exceed the maximum PIN tries.
- 2 = Reset the bad PIN tries counter at the end of the usage accumulation period. Also reset the bad PIN tries when a correct PIN is entered, regardless of the number of bad PIN tries.
- 3 = Reset the bad PIN tries counter when a correct PIN is entered and the number of bad PIN tries does not exceed the maximum PIN tries.
- 4 = Reset the bad PIN tries counter when a correct PIN is entered, regardless of the number of bad PIN tries.

BASE24-pos Internal Messages

When a message is received from an external source, BASE24-pos converts that message from the native format of the external source to the POS Standard Internal Message (PSTM) format recognizable to the BASE24-pos processes.

All information relating to the PIN, its method of encryption, and the key under which it is encrypted is carried in the PSTM.

BASE24-pos internal messages contain the following information for PIN verification. Each description includes the name of the PSTM field that contains the information.

ANSI Offset — Indicates the starting position in the PAN of the 12 digits used to create the PIN/PAN PIN block if that type of PIN block was used in encrypting the PIN. This value can be greater than, less than, or equal to 0. (A 0 indicates the first position of the PAN.)

BASE24-pos uses the 12 right-most digits of the PAN excluding the check digit to create the PAN string.

Example: A PAN of 1049000451234534567 would result in a PAN string of 045123453456 and an ANSI offset of 6.

If the PAN does not contain enough significant digits to provide the 12 digits necessary (i.e., the PAN is less than 13 digits in length), BASE24-pos adds leading zeros to the PAN used in creating the PIN block and sets the ANSI offset to a negative value. The absolute value of this negative ANSI offset identifies the number of leading zeros BASE24-pos has added to the PAN after right-justifying the available significant digits.

Example: A PAN of 1234534567 would result in a PAN string of 000123453456 and an ANSI offset of -3.

PSTM Field: ANSI-OFST

Condition Code — Identifies whether a POS device is attended or unattended. If the POS device is not capable of accepting PINs, this value is checked to determine whether an attendant is present to verify the cardholder's identification.

PSTM Field: PT-SRV-COND-CDE

Entry Mode — Identifies whether a POS device is capable of accepting PINs.

PSTM Field: PT-SRV-ENTRY-MDE

Error Flag — Indicates whether a sanity check error or PIN block error was received from a security device while processing a BASE24-pos transaction. BASE24-pos Authorization, Host Interface, and some Interchange Interface processes can set this flag. Valid values pertaining to PIN verification are as follows:

- L = Invalid PIN length. The PIN received in the transaction was of an invalid length.
- R = Sanity check error or PIN block error, previous zone. A sanity check error or PIN block error was received from a security device while processing a transaction involving PIN translation or verification. Dynamic key management error counters, if used, should not be incremented for this error.
- S = Sanity check error or PIN block error, current zone. A sanity check error or PIN block error was received from a security device while processing a transaction involving PIN translation or verification. Dynamic key management error counters, if used, should be incremented for this error.

PSTM Field: PS50-TKN.ERR-FLG

PIN — Contains the PIN in the clear or in encrypted form. If the PIN is carried in encrypted form, it is really the PIN block that is carried in the message.

The Authorization process always fills this field with zeros if it approves the transaction or declines it for any reason.

PSTM Field: PIN

PIN Format — Indicates whether the PIN is in the clear or encrypted. If the PIN is encrypted, this field also indicates the type of PIN block used in the encryption: PIN/PAN PIN block or PIN/PAD PIN block.

PSTM Field: PIN-FRMT

PIN Key — The encrypted form of the key under which the PIN is currently encrypted if the PIN is encrypted.

PINs travel through the BASE24-pos system in the clear or encrypted under a secure key. This field carries the secure key.

PSTM Field: PIN-KEY

PIN Pad Character — Indicates the PIN pad character used to create the PIN/PAD PIN block if that type of PIN block was used in encrypting the PIN.

PSTM Field: PINPAD-CHAR

PIN Size — Contains the size of the PIN, if the PIN is in the clear. If the PIN is being carried in encrypted form, the value in this field is set to 16 (the size of the PIN block).

PSTM Field: PIN-SIZE

PIN Tries — Indicates the number of times the cardholder has entered the PIN incorrectly during the current session at the terminal. This field is initially set to zero and subsequently incremented each time the PIN is checked and determined to be incorrect.

Once a PIN has been verified, the Authorization process sets the value in this field to Z, indicating additional transactions on the card can be processed during the current session without reverifying the PIN.

PSTM Field: PIN-TRIES

Previous Key — The encrypted form of the previous key used by an interchange to encrypt PINs. After an online key exchange has taken place between the BASE24 Interchange Interface process and an interchange, the Interchange Interface process saves the previous key and places it in the BASE24 internal message to allow for situations in which PINs are subsequently sent from the interchange encrypted under the previous key.

PSTM Field: USER-KEY

Originator — This field applies only to BASE24-pos when a transaction originates from a device using the derived unique key per transaction (DUKPT) encryption method and Thales e-Security host security modules are used, but the Transaction Security Services process is not used. The SPDH or HPDH module sets this field to a value of 9 when it receives a transaction containing a DUKPT-encrypted PIN block and the SECURE-DEV-TYP param is set to a value of VSM or RSM. This allows the security utilities to use the correct calls when interfacing with the security modules.

PSTM Field: ORIGINATOR

BASE24-pos PIN Verification Processing

The BASE24-pos Authorization process takes the following PIN verification actions during its processing of a transaction.

PIN verification actions constitute only a portion of the overall transaction processing performed by the Authorization process.

The steps summarized here are intended to give a general view of PIN verification. For specific information on how each PIN verification method operates, refer to sections 9 through 12.

The Authorization process performs the following:

1. Determines whether a PIN is present.
 - If the internal message contains a PIN, processing continues with step 3.
 - If the internal message does not contain a PIN, processing continues with step 3.
2. Determines whether the transaction is allowed without a PIN.

If the PIN PROCESSING FLAG field on CPF screen 7 is set to a value of 0 (PIN never required), the Authorization process continues processing the transaction without any further PIN processing.

If the PIN PROCESSING FLAG field on CPF screen 7 is set to a value of 2 (PIN required if terminal has PIN capabilities), the entry mode field in the internal message indicates the terminal does not have PIN capabilities, and the condition code field in the internal message indicates an operator is present to identify the customer, the Authorization process sets the response code to 003 (approve with identification) and continues processing with step 10.

The Authorization process declines the transaction with a response code of 096 (PIN required) and continues processing with step 10 in any of the following situations:

- The PIN PROCESSING FLAG field on CPF screen 7 is set to a value of 1 (PIN always required).
- The PIN PROCESSING FLAG field on CPF screen 7 is set to a value of 2 (PIN required if terminal has PIN capabilities) and the entry mode field in the internal message indicates the terminal has PIN capabilities.

- The PIN PROCESSING FLAG field on CPF screen 7 is set to a value of 2 (PIN required if terminal has PIN capabilities) and the entry mode field in the internal message indicates the terminal does not have PIN capabilities, but the condition code field in the internal message indicates the terminal is unattended.
3. Determines whether to use IDF or CPF limits.
 - If the PIN CHECK TYPE field on CPF screen 2 is set to a value of 99 (use IDF parameters) and the PIN CHECK TYPE field on IDF screen 2 is set to a value in the range of 01 through 04 (verify PIN), processing continues with step 4 using the PIN verification limits on IDF screen 2.
 - If the PIN CHECK TYPE field on CPF screen 2 is set to a value in the range of 01 through 04 (verify PIN using CPF parameters), processing continues with step 4 using the PIN verification limits on CPF screen 2.
 - If the PIN CHECK TYPE field on CPF screen 2 is set to a value of 99 (use IDF parameters) and the PIN CHECK TYPE field on IDF screen 2 is set to a value of 0 (do not verify PIN) or if the PIN CHECK TYPE field on CPF screen 2 is set to a value of 0 (do not verify PIN), the Authorization process continues processing the transaction without any further PIN processing.

4. Checks the number of incorrect PINs the cardholder has entered.

If the institution maintains BASE24-pos authorization files, each cardholder's incorrect PIN tries are accumulated in the BASE24 database. However, if the institution does not maintain BASE24-pos authorization files, meaning that authorizations are performed by a host exclusively, the number of incorrect PIN tries is carried in the PIN tries field of the internal message. In this case, PIN tries can only be tracked during each session at the terminal.

- If the PIN TRIES RESET OPTION field on screen 2 of the IDF or CPF is set to a value of 2 or 4, meaning accept a valid PIN regardless of the number of incorrect PIN tries, processing continues with step 5.
- If the PIN TRIES RESET OPTION field on screen 2 of the IDF or CPF is set to a value of 0, 1, or 3, meaning decline the transaction if the PIN tries limit has been exceeded, the Authorization process checks the number of incorrect PINs entered by the cardholder against the institution's maximum number specified in the IDF or CPF, depending on whether PIN verification parameters are set at the institution level or the card prefix level.
 - If the number of incorrect PIN tries is less than the maximum PIN tries allowed, processing continues with step 5.

- If the number of incorrect PIN tries is greater than or equal to the maximum PIN tries allowed, the transaction is declined with a response code of 052 (PIN tries exceeded—return card) or 900 (PIN tries exceeded—retain card) and processing continues with step 10.
5. Determines whether the PIN requires verification by BASE24.
 - If the transaction is being processed under authorization level 2 (offline), processing continues with step 6 because PIN verification is always performed by BASE24.
 - If the transaction is being processed under authorization level 1 (online) or 3 (online/offline), the Authorization process checks the CHECK IF HOST ONLINE PIN field on screen 2 of the IDF or CPF to determine whether the PIN is to be verified. If the PIN is not to be verified, the Authorization process continues without any further PIN processing.

6. Verifies the PIN in software, if possible.

The Authorization process checks the PIN format in the internal message. If the PIN format in the internal message is 1 or 3, indicating that the PIN is in encrypted form, processing continues with step 7. If the PIN format in the internal message is 0, indicating that the PIN is in the clear, the Authorization process performs the following:

- a. Extracts the clear PIN from the PIN field in the internal message using the PIN size value in the internal message.
 - b. Verifies the clear PIN in software according to the PIN verification method selected for the card prefix. For an explanation of how the different PIN verification methods function, refer to sections 9 through 12.
 - If the PIN is valid, processing continues with step 8.
 - If the PIN is invalid, the Authorization process declines the transaction with a response code of 201 (incorrect PIN) and increments the incorrect PIN tries counter in the CAF or UAF and PSTM. Processing continues with step 10.
7. Calls a security device to verify the PIN.

In step 6, the Authorization process checked the PIN format in the internal message to determine whether the PIN was in clear or encrypted form. If the PIN is in encrypted form, the Authorization process performs the following:

- a. Extracts the following information from the internal message and sends it to the security module for verifying the PIN:
 - The encrypted PIN block from the PIN field.

- The encrypted PIN key from the PIN key field.
- The PIN pad character, if the PIN format is 3, indicating that the PIN block is a PIN/PAD PIN block.
- The PAN digits specified by the ANSI offset field in the internal message, if the PIN format is 1, indicating that the PIN block is a PIN/PAN PIN block.

Other information is forwarded to the security module, based on the type of verification to be performed. The information required for each type of PIN verification is described in sections 9 through 12.

The Authorization process uses a standard set of PIN utilities to communicate with the security module. These PIN utilities return one of the following codes to indicate the outcome of the PIN verification processing.

- 0 = PIN valid
- 1 = PIN invalid
- 2 = Security device failure (e.g., timeout)
- 3 = Invalid data
- 4 = Invalid parameter
- 5 = Sanity check or PIN block error (security module did not find PIN block as expected)

- b. Verifies the code returned from the PIN utilities.

If the PIN utilities return a status of 0 (PIN validated), processing continues with step 8.

If the PIN utilities return a status of 5, the Authorization process resends the PIN verification information to the security module using the previous key from the internal message rather than the current key. This is performed on the contingency that a different key was provided by an interface process.

- c. Determines the response code and error flag to assign the transaction if the PIN was not validated.
- If the PIN utilities return a status of 2, the Authorization process assigns the transaction a response code of 054 (no security device) and an error flag of blank (none). Processing continues with step 10.
 - If the PIN utilities return a status of 5 with the PIN key from the internal message and the internal message does not contain a previous key, the Authorization process assigns the transaction a response code of 074 (unable to authorize) and an error flag of S (sanity check). Processing continues with step 10.

- If the PIN utilities return a status of 5 with the previous key from the internal message, the Authorization process assigns the transaction a response code of 074 (unable to authorize) and an error flag of S (sanity check). Processing continues with step 10.
 - If the PIN utilities return a status of 1, 3, or 4, the Authorization process assigns the transaction a response code of 201 (incorrect PIN) and an error flag of blank (none). The Authorization process also increments the bad PIN tries counter in the CAF or UAF and PSTM. Processing continues with step 10.
8. Sets the PIN tries field in the internal message to Z and the PIN field in the internal message to zeros.
 9. Resets the bad PIN tries counter in the CAF or UAF, if appropriate.
 - If the value in the PIN TRIES RESET OPTION field is 2 or 4 and the bad PIN tries counter in the CAF or UAF is greater than 0, the Authorization process resets the bad PIN tries counter to 0.
 - If the value in the PIN TRIES RESET OPTION field is 1 or 3 and the bad PIN tries counter in the CAF or UAF is greater than 0 but less than the value in the MAX PIN TRIES field, the Authorization process resets the bad PIN tries counter to 0.
 10. Continues with its processing.
 - If the transaction was declined, the Authorization process places the appropriate response code in the internal message and returns the transaction to the originator.
 - If the transaction was not declined, the Authorization process continues processing the transaction.

BASE24-teller PIN Verification

PIN verification processing specific to BASE24-teller is performed by the BASE24-teller Authorization process. The BASE24-teller Authorization process offers several options related to PIN verification.

Logical Network Configuration File (LCONF) Parameter

At the logical network level, BASE24-teller provides the option of specifying whether Authorization processes can use the Card Prefix File (CPF) or the Key Authorization File (KEYA) during transaction processing. The BASE24-teller Authorization processes use the following parameter in the Logical Network Configuration File (LCONF) during initialization and warmbooting to determine whether to place these files in extended memory.

TLR-PROCESS-CRD-PIN — Indicates whether BASE24-teller Authorization processes use the Card Prefix File (CPF) or Key Authorization File (KEYA) during transaction processing.

- If card processing can be performed, but no security processing can be performed, this field must be set to a value of C (card processing). The CPF is required.
- If card and security processing can be performed, this field must be set to a value of B (both). The CPF and KEYA are required.
- If no card or security processing can be performed, this field must be set to a value of N (neither).

Security processing cannot be performed without cards.

Institution and Card Prefix PIN Verification Parameters

BASE24-teller provides each institution the option of establishing a separate set of PIN verification parameters for each of the card prefixes it issues, using the same parameters for all card prefixes, or a combination of the two options. The following parameters appear on screen 2 of the Institution Definition File (IDF) and on screen 2 of the Card Prefix File (CPF). References to these parameters in the remainder of this section include both files. The appropriate file depends on whether the parameters are being maintained at the institution level or the card prefix level. For more information on these fields, refer to the ***BASE24 Base Files Maintenance Manual***.

PIN VERIFICATION KEYA GROUP — Identifies which Key Authorization File (KEYA) record is used for PIN verification information. The value in this field is matched with the value in the GRP field on the KEYA screens.

This field only appears on CPF screen 2. On IDF screens, the value in the FIID field is used for this purpose.

Note: The GRP field on KEYA screens is just part of the key to KEYA records. A beginning date and an ending date that are compared to a card expiration date and a record type that identifies the PIN verification method complete the information needed to fully identify the correct KEYA record.

PIN CHECK TYPE — Indicates the PIN verification method used. The Authorization process determines the type of PIN verification to be performed from the value in this field. Valid values are as follows:

- 00 = No PIN verification
- 01 = IBM DES (3624-format) PIN verification
- 02 = Diebold PIN verification
- 03 = Identikey PIN verification
- 04 = Visa PVV PIN verification

Each of these types of PIN verification relies on other field settings in the BASE24 database to be performed correctly. For information on how the entire database must be set up for each of these PIN verification methods, refer to sections 9 through 12.

The PIN CHECK TYPE field on CPF screen 2 also indicates whether the card prefix uses its own PIN verification parameters or the institution-level parameters that are defined on IDF screen 2, so there is an additional value (i.e., 99) on CPF screen 2. Valid values are as follows:

- 00 = No PIN verification—Use CPF parameters
- 01 = IBM DES (3624-format) PIN verification—Use CPF parameters
- 02 = Diebold PIN verification—Use CPF parameters
- 03 = Identikey PIN verification—Use CPF parameters
- 04 = Visa PVV PIN verification—Use CPF parameters
- 99 = Use the PIN verification parameters on IDF screen 2

An institution with multiple card prefixes can use institution-level parameters for some prefixes while using card prefix-level parameters for other card prefixes. The only restriction is that all of the parameters used by one prefix be defined on a single screen (i.e., CPF screen 2 or IDF screen 2).

MAX PIN TRIES — Specifies the maximum number of consecutive invalid PIN entries that are allowed.

BAD PIN ACTION — Specifies the action to be taken when the maximum number of invalid PIN entries is exceeded.

ALGO NUMBER LOC — Specifies the location of the algorithm number used with the Diebold PIN verification method.

CHECK IF HOST ONLINE PIN — Specifies whether a cardholder's PIN is to be checked during transaction screening by BASE24. BASE24-teller can perform this check with host authorization.

POFST/PVV LOC — Specifies the location of the PIN verification digits, which include PIN offset for IBM DES (3624-format) or Diebold PIN verification methods, PIN Verification Value (PVV) for the Visa PVV PIN verification method, or the PIN Verification Number (PVN) for the Identikye PIN verification method.

PIN TRIES RESET OPTION — Specifies when the bad PIN tries counter is reset.

PIN Verification by Authorization Level

Depending on the authorization level under which a transaction is being processed, PIN verification can be switched on or off for each card prefix. This is controlled by the PIN CHECK TYPE and CHECK IF HOST ONLINE PIN fields on screen 2 of the IDF or CPF.

The PIN CHECK TYPE field identifies the method the BASE24-teller Authorization process is to use in performing PIN verification, and the CHECK IF HOST ONLINE PIN field specifies whether PINs are to be verified during transaction screening. A value of 00 in the PIN CHECK TYPE field means the BASE24-teller Authorization process is not to perform PIN verification at any time, regardless of the CHECK IF HOST ONLINE PIN field setting.

There are two authorization levels available through BASE24-teller: host (online) and BASE24 (offline).

With host authorization (online), transactions must be sent to a host for authorization. Under this authorization level, BASE24-teller can be configured to perform PIN verification as a part of its transaction screening actions by setting the CHECK IF HOST ONLINE PIN field to a value of Y and the PIN CHECK TYPE field to a value in the range of 01 through 04. This means that BASE24-teller verifies the cardholder's PIN prior to sending the transaction to the host and declines the transaction if the cardholder's PIN is incorrect.

With BASE24-teller authorization (offline), transactions are not authorized by a host, but by BASE24-teller exclusively. In this case, the value of the CHECK IF HOST ONLINE PIN field is not used. The PIN CHECK TYPE field determines if and how PINs are verified.

Maximum PIN Retries

BASE24-teller keeps track of the number of bad PIN tries for a cardholder in the Cardholder Authorization File (CAF). This allows the number of bad PIN tries to accumulate over a period of time, and institutions can then choose to decline a transaction for a cardholder if that cardholder has an excessive number of incorrect PIN tries.

BASE24-teller does not reset the bad PIN tries counter in the CAF at the end of each usage accumulation period. This is because BASE24 uses customer processing dates in the product-specific segments of the IDF to track usage accumulation periods and only the BASE24-atm and BASE24-pos segments of the IDF have these dates. Therefore, for institutions using BASE24-teller without BASE24-atm or BASE24-pos, the entry of a correct PIN is the only means for resetting the bad PIN tries counter other than a file maintenance update or a refresh of the cardholder's record. Refer to the topics "BASE24-atm PIN Verification Maximum PIN Retries" and "BASE24-pos PIN Verification Maximum PIN Retries" subsections instead of this subsection if an institution is using either of those products with BASE24-teller.

The following fields on IDF or CPF screen 2 must be set as indicated to define bad PIN tries processing for an institution using BASE24-teller without BASE24-atm or BASE24-pos:

BAD PIN ACTION — Specifies the action to be taken by the device when it receives a PIN tries exceeded response code.

- If the device is to display a return card message to the teller when it receives a PIN tries exceeded response code, this field must be set to a value of 0.

- If the device is to display a retain card message when it receives a PIN tries exceeded response code, this field must be set to a value of 1.

MAX PIN TRIES — Specifies the number of times a cardholder can enter an incorrect PIN.

- If the PIN TRIES RESET OPTION field is set to a value of 1 or 3, once this number is reached, subsequent transactions are automatically declined and the action specified in the BAD PIN ACTION field is taken.
- If the PIN TRIES RESET OPTION field is set to a value of 2 or 4, the number of bad PIN tries is incremented with each incorrect PIN, but is not compared to the value in the MAX PIN TRIES field.

PIN TRIES RESET OPTION — A code indicating how the bad PIN tries counter in the Cardholder Authorization File (CAF) is to be reset for an institution's cardholders. This field must be set to one of the following values to specify when the bad PIN tries counter is reset.

- 1 or 3 = Reset the bad PIN tries counter when a correct PIN is entered and the number of bad PIN tries does not exceed the maximum PIN tries.
- 2 or 4 = Reset the bad PIN tries counter when a correct PIN is entered, regardless of the number of bad PIN tries.

Local PIN Verification Option

Local PIN verification is the verification of the customer's PIN at a teller device or controller before the transaction is sent to BASE24-teller. Local PIN verification reduces communications and processing costs because only transactions with correct PINs are transmitted to BASE24-teller for authorization. Transactions with invalid PINs do not leave the device or controller.

An institution should give consideration to the following items when implementing local PIN verification:

- Additional hardware may be required in order to verify PINs locally.
- Teller terminals must be configured to perform local PIN verification. Refer to vendor documentation for additional information.
- Bad PIN tries are not tracked on the BASE24 database when PINs are verified locally.

The following fields located in the various BASE24 files indicated must contain specific values when PINs are verified locally.

CPF or IDF Screen 2

POFST/PVV LOC — A code specifying the location of the DES (IBM 3624) or Diebold PIN verification method PIN offset, the Visa PVV, or the Identkey PVN. This field must be set to a value of 1 to indicate that the PIN verification digits are located in the magnetic stripe data of the card.

CAF Screen 1

POFST/PVV — Contains the PIN offset value, PIN verification number (PVN), or the PIN verification key indicator (PVKI) and the PIN verification value (PVV). This field is not used because the PIN verification digits are located in the magnetic stripe data of the card.

CPF Screen 1

POFST/PVV — Specifies the location of the PIN verification digits in the magnetic stripe data of the card.

TTDF Screen 2

CONTROLLER VALIDATE PIN — Specifies whether local PIN verification is allowed for this terminal.

- If local PIN verification is allowed for transactions at this terminal, this field must be set to a value of Y.
- If local PIN verification is not allowed for transactions at this terminal, this field must be set to a value of N.

BASE24-teller Internal Messages

When a message is received from an external source, BASE24-teller converts that message from the native format of the external source to the BASE24-teller Standard Internal Message (TSTM) format recognizable to the BASE24-teller processes.

All information relating to the PIN, its method of encryption, and the key under which it is encrypted is carried in the TSTM.

BASE24-teller internal messages contain the following information for PIN verification. Each description includes the name of the TSTM field that contains the information.

ANSI Offset — Indicates the starting position in the PAN of the 12 digits used to create the PIN/PAN PIN block if that type of PIN block was used in encrypting the PIN. This value can be greater than, less than, or equal to 0. (A 0 indicates the first position of the PAN.)

BASE24-teller uses the value in the ANSI PAN FORMAT field in the Teller Terminal Data File (TTDF) to determine which PAN digits are used to create the PAN string.

Example: A PAN of 1049000451234534567 and a value of 0 (12 right-most digits excluding check digit) in the ANSI PAN FORMAT field in the TTDF would result in a PAN string of 045123453456 and an ANSI offset of 6. The same PAN and a value of 1 (12 right-most digits including check digit) in the ANSI PAN FORMAT field in the TTDF would result in a PAN string of 451234534567 and an ANSI offset of 7.

If the PAN does not contain enough significant digits to provide the 12 digits necessary (i.e., the PAN is less than 13 digits in length), BASE24-teller adds leading zeros to the PAN used in creating the PIN block and sets the ANSI offset to a negative value. A negative ANSI offset identifies the number of leading zeros BASE24-teller has added to the PAN.

Example: A PAN of 1234534567 and a value of 0 (12 right-most digits excluding check digit) in the ANSI PAN FORMAT field in the TTDF would result in a PAN string of 000123453456 and an ANSI offset of -3. The same PAN and a value of 1 (12 right-most digits including check digit) in the ANSI PAN FORMAT field in the TTDF would result in a PAN string of 001234534567 and an ANSI offset of -2.

TSTM Field: PIN-TKN.ANSI-OFST

Card Present Flag — Indicates whether the transaction was initiated by a card at the teller terminal.

TSTM Field: TSTMH.RQST.CRD-PRESENT

Error Flag — Indicates whether a sanity check error or PIN block error was received from a security device while processing a BASE24-teller transaction. BASE24-teller Device Handler, Authorization, and Host Interface processes can set this flag. Valid values pertaining to PIN verification are as follows:

- R = Sanity check error or PIN block error, previous zone. A sanity check error or PIN block error was received from a security device while processing a transaction involving PIN translation or verification. Dynamic key management error counters, if used, should not be incremented for this error.
- S = Sanity check error or PIN block error, current zone. A sanity check error or PIN block error was received from a security device while processing a transaction involving PIN translation or verification. Dynamic key management error counters, if used, should be incremented for this error.

TSTM Field: TSTMH.RESP-HDR.ERR-FLG

PIN — Contains the PIN in the clear or in encrypted form. If the PIN is carried in encrypted form, it is really the PIN block that is carried in the message.

The Authorization process always fills this field with zeros if it approves the transaction or declines it for any reason. The Device Handler process fills this field with zeros if the PIN has been verified at the teller device or controller.

TSTM Field: PIN-TKN.PIN

PIN Format — Indicates whether the PIN is in the clear or encrypted. If the PIN is encrypted, this field also indicates the type of PIN block used in the encryption: PIN/PAN PIN block or PIN/PAD PIN block.

TSTM Field: PIN-TKN.PIN-FRMT

PIN Key — The encrypted form of the key under which the PIN is currently encrypted, if the PIN is encrypted.

PINs travel through the BASE24-teller system in the clear or encrypted under a secure key. This field carries the secure key.

TSTM Field: PIN-TKN.PIN-KEY

PIN Pad Character — Indicates the PIN pad character used to create the PIN/PAD PIN block, if that type of PIN block was used in encrypting the PIN.

TSTM Field: PIN-TKN.PIN-PAD-CHAR

PIN Size — Contains the size of the PIN, if the PIN is in the clear. If the PIN is being carried in encrypted form, the value in this field is set to 16 (the size of the PIN block).

TSTM Field: PIN-TKN.PIN-SIZE

PIN Tries — Indicates the number of times the cardholder has entered the PIN incorrectly during the current session at the terminal. This field is initially set to zero and subsequently incremented each time the PIN is checked and determined to be incorrect.

Once a PIN has been verified, the Authorization process sets the value in this field to Z, indicating additional transactions on the card can be processed during the current session without reverifying the PIN.

With local PIN verification, the Device Handler process sets the value in this field to Z, indicating the PIN has been verified at the teller device or controller and does not need to be verified by the Authorization process.

TSTM Field: PIN-TKN.PIN-TRIES

BASE24-teller PIN Verification Processing

The BASE24-teller Authorization process takes the following PIN verification actions during its processing of a transaction.

PIN verification actions constitute only a portion of the overall transaction processing performed by the Authorization process.

The steps summarized here are intended to give a general view of PIN verification. For specific information on how each PIN verification method operates, refer to sections 9 through 12.

BASE24-teller transactions can be initiated with or without a card; however, PIN verification can be performed only on card-initiated transactions. BASE24-teller offers a card-initiated transaction with the sole purpose of verifying the PIN, as well as offering PIN verification as a step in the processing performed for other card-initiated transactions.

Unlike BASE24-atm and BASE24-pos, BASE24-teller returns many response codes that can be overridden by the teller. The first character of a response code can be A, O, or F, meaning approved, override required, or fatal (i.e., cannot be overridden). Transaction processing does not stop when a response code requiring an override is assigned. Instead, BASE24-teller collects all of the override required response codes that may be generated, then returns all of them to the teller in a single response. The teller has the opportunity to override all of the response codes, permitting the transaction to be processed. Many of the PIN-related response codes can be overridden, permitting transactions to be processed with PINs that are missing or invalid.

The Authorization process performs the following:

1. Determines whether the transaction is card-initiated.
 - If the card-present flag in the internal message indicates that a card is not being used to initiate this transaction, the Authorization process continues processing the transaction without any further PIN processing.
 - If the card-present flag in the internal message indicates that a card is being used to initiate this transaction, processing continues with step 2.

2. Determines whether the PIN has been verified locally.
 - If the PIN has been verified by the teller device or controller, the Device Handler process has already set the PIN field in the internal message to zeros and set the PIN tries field in the internal message to Z. The Authorization process continues processing the transaction without any further PIN processing.
 - If the PIN has not been verified by the teller device or controller, processing continues with step 3.
3. Determines whether to use IDF or CPF limits.
 - If the PIN CHECK TYPE field on CPF screen 2 is set to a value of 99 (use IDF parameters) and the PIN CHECK TYPE field on IDF screen 2 is set to a value in the range of 01 through 04 (verify PIN), processing continues with step 4 using the PIN verification limits on IDF screen 2.
 - If the PIN CHECK TYPE field on CPF screen 2 is set to a value in the range of 01 through 04 (verify PIN using CPF parameters), processing continues with step 4 using the PIN verification limits on CPF screen 2.
 - If the PIN CHECK TYPE field on CPF screen 2 is set to a value of 99 (use IDF parameters) and the PIN CHECK TYPE field on IDF screen 2 is set to a value of 0 (do not verify PIN) or if the PIN CHECK TYPE field on CPF screen 2 is set to a value of 0 (do not verify PIN), the Authorization process continues processing the transaction without any further PIN processing.
4. Checks the number of incorrect PINs the cardholder has entered.

If the institution maintains BASE24-teller authorization files, each cardholder's incorrect PIN tries are accumulated in the BASE24 database. However, if the institution does not maintain BASE24-teller authorization files, meaning that authorizations are performed by a host exclusively, the number of incorrect PIN tries is carried in the PIN tries field of the internal message. In this case, PIN tries can only be tracked during each session at the terminal.

- If the PIN TRIES RESET OPTION field on screen 2 of the IDF or CPF is set to a value of 2 or 4, meaning accept a valid PIN regardless of the number of incorrect PIN tries, processing continues with step 5.
- If the PIN TRIES RESET OPTION field on screen 2 of the IDF or CPF is set to a value of 0, 1, or 3, meaning decline the transaction if the PIN tries limit has been exceeded, the Authorization process checks the number of incorrect PINs entered by the cardholder against the

institution's maximum number specified in the IDF or CPF, depending on whether PIN verification parameters are set at the institution level or the card prefix level.

- ❑ If the number of incorrect PIN tries is less than the maximum PIN tries allowed, processing continues with step 5.
- ❑ If the number of incorrect PIN tries is greater than or equal to the maximum PIN tries allowed, the Authorization process adds response code O32 (PIN tries exceeded) to the list of response codes that can be overridden and continues processing with step 11.

5. Determines whether the PIN requires verification.

- If the transaction is being authorized by BASE24-teller and the internal message contains a PIN, processing continues with step 6.
- If the transaction is being authorized by BASE24-teller and the internal message does not contain a PIN, the Authorization process adds response code O34 (PIN method—no PIN present) to the list of response codes that can be overridden and continues processing with step 11.
- If the transaction is being authorized by a host, the Authorization process checks the IDF or CPF to determine whether the PIN is to be verified by the BASE24 system before the transaction is passed to the host.
 - ❑ If the PIN is to be verified and the internal message contains a PIN, processing continues with step 6.
 - ❑ If the PIN is to be verified and the PIN verification digits are located on the CAF, the Authorization process adds response code O33 (unable to verify PIN) to the list of response codes that can be overridden and continues processing with step 11.
 - ❑ If the PIN is not to be verified, the Authorization process continues processing the transaction without any further PIN processing.

6. Determines whether magnetic stripe data must be obtained electronically.

The magnetic stripe data must be obtained electronically when the PIN verification digits are located on the card, which is indicated by a value of 1 in the POFST/PVV LOC field on screen 2 of the IDF or CPF. A code in the magnetic stripe data in the internal message indicates whether the data has been obtained manually or electronically.

- If the PIN verification digits are located on the card and the magnetic stripe data is obtained electronically, processing continues with step 7.
- If the PIN verification digits are not located on the card and the magnetic stripe data is obtained manually, processing continues with step 7.

- If the PIN verification digits are located on the card and the magnetic stripe data is obtained manually, the Authorization process assigns the transaction a response code of F35 (card must be swiped). Processing continues with step 11.

7. Verifies the PIN in software, if possible.

The Authorization process checks the PIN format in the internal message. If the PIN format in the internal message is 1 or 3, indicating that the PIN is in encrypted form, processing continues with step 8. If the PIN format in the internal message is 0, indicating that the PIN is in the clear, the Authorization process performs the following:

- a. Extracts the clear PIN from the PIN field in the internal message using the PIN size value in the internal message.
- b. Verifies the clear PIN in software according to the PIN verification method selected for the card prefix. For an explanation of how the different PIN verification methods function, refer to sections 9 through 12.
 - If the PIN is valid, processing continues with step 9.
 - If the PIN is invalid, the Authorization process adds response code O31 (invalid PIN) to the list of response codes that can be overridden and increments the incorrect PIN tries counter in the CAF and TSTM. Processing continues with step 11.

8. Has the PIN verified by a security device.

In step 7, the Authorization process checked the PIN format in the internal message to determine whether the PIN was in clear or encrypted form. If the PIN is in encrypted form, the Authorization process performs the following:

- a. Extracts the following information from the internal message and sends it to the security module for verifying the PIN:
 - The encrypted PIN block from the PIN field.
 - The encrypted PIN key from the PIN key field.
 - The PIN pad character, if the PIN format is 3, indicating that the PIN block is a PIN/PAD PIN block.
 - The PAN digits specified by the ANSI offset field in the internal message, if the PIN format is 1, indicating that the PIN block is a PIN/PAN PIN block.

Other information is forwarded to the security module, based on the type of verification to be performed. The information required for each type of PIN verification is described in sections 9 through 12.

The Authorization process uses a standard set of PIN utilities to communicate with the security module. These PIN utilities return one of the following codes to indicate the outcome of the PIN verification processing:

- 0 = PIN valid
- 1 = PIN invalid
- 2 = Security device failure (e.g., timeout)
- 3 = Invalid data
- 4 = Invalid parameter
- 5 = Sanity check or PIN block error (security module did not find PIN block as expected)

- b. Verifies the code returned from the PIN utilities.

If the PIN utilities return a status of 0 (PIN validated), processing continues with step 9.

If the PIN utilities return a status of 5, the Authorization process resends the PIN verification information to the security module using the previous key from the internal message rather than the PIN key. This is performed on the contingency that a different key was provided by an interface process.

- c. Determines the response code and error flag to assign the transaction if the PIN was not validated.

- If the PIN utilities return a status of 1, the Authorization process adds an error flag of blank (none) and adds response code O31 (invalid PIN) to the list of response codes that can be overridden. Processing continues with step 11.
- If the PIN utilities return a status of 2, the Authorization process assigns the transaction a fatal response code of FOI (system error) and an error flag of blank (none). Processing continues with step 11.
- If the PIN utilities return a status of 3 or 4, the Authorization process adds an error flag of blank (none) and adds response code O31 (invalid PIN) to the list of response codes that can be overridden. The Authorization process also increments the bad PIN tries counter in the CAF and TSTM. Processing continues with step 11.
- If the PIN utilities return a status of 5 with the PIN key from the internal message and the internal message does not contain a previous key, the Authorization process assigns the transaction a response code of FOI (system error) and an error flag of S (sanity check). Processing continues with step 11.

- If the PIN utilities return a status of 5 with the previous key from the internal message, the Authorization process assigns the transaction a response code of F0I (system error) and an error flag of S (sanity check). Processing continues with step 11.
9. Sets the PIN tries field in the internal message to Z and the PIN field in the internal message to zeros.
 10. Resets the bad PIN tries counter in the CAF, if appropriate.
 - If the value in the PIN TRIES RESET OPTION field on screen 2 of the CPF or IDF is 2 or 4 and the bad PIN tries counter in the CAF is greater than 0, the Authorization process resets the bad PIN tries counter to 0.
 - If the value in the PIN TRIES RESET OPTION field on screen 2 of the CPF or IDF is 1 or 3 and the bad PIN tries counter in the CAF is greater than 0 but less than the value in the MAX PIN TRIES field in the IDF or CPF, the Authorization process resets the bad PIN tries counter to 0.
 11. Continues with its processing.
 - If the transaction is declined with a fatal response code, the Authorization process places the appropriate response code in the internal message and returns the transaction to the originator.
 - If the transaction is assigned one or more response codes that require an override to be completed, the Authorization process places the response codes in the internal message and continues processing the transaction.
 - If the transaction has not been assigned any response codes that are fatal or that require an override to be completed, the Authorization process continues processing the transaction.

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9: IBM DES (3624-Format) PIN Verification Support

This section describes BASE24 support of the IBM DES (3624-format) PIN verification method, including how the BASE24 database must be set up to allow BASE24 to verify PINs under this method of verification.

BASE24 allows for implementation of this PIN verification method in software or in a security module.

IBM DES (3624-Format) PIN Verification

IBM DES (3624-format) PIN verification is based on the use of a PIN verification key, a decimalization table, and a PIN offset. IBM DES (3624-format) PIN verification can be performed in software or using a hardware security module linked to the processor on which the BASE24 system is running.

At card-issuing time, the card issuer generates a unique PIN and PIN offset combination for each card, using validation data taken from the cardholder's account number. If the PIN is issuer-assigned or chosen by the cardholder, the PIN offset is generated to correspond to the PIN; if the PIN offset is a set value, the PIN is generated to correspond to the PIN offset. The PIN offset is then placed in the magnetic stripe data of the card or in the BASE24 database.

The following is a summary of the steps taken to verify a PIN using the IBM DES (3624-format) PIN verification method in software or hardware. If a security module is being used, however, it must be given additional information to decrypt the PIN block and extract a clear PIN for verification and also to decrypt the PIN verification key.

The verifying entity performs the following steps:

1. Creates a validation data item using the cardholder account number.
2. Encrypts the validation data using the PIN verification key.
3. Decimalizes the encrypted validation data using the decimalization table.
4. Creates an intermediate PIN by computing the length of the cardholder's PIN offset and taking that same number of characters from the left of the decimalized validation data.
5. Adds the cardholder's PIN offset (using the modulo 10 technique) to the intermediate PIN to derive a PIN check number.
6. Compares the PIN check number to the PIN entered by the cardholder. If they match, the cardholder entered the PIN correctly. If they do not match, the cardholder entered the PIN incorrectly.

PIN Verification Inputs

IBM DES (3624-format) PIN verification requires the following information as input. Some of these items vary from cardholder to cardholder, while others are constant for an institution or card prefix.

- Validation data
- PIN verification key
- Decimalization table
- PIN offset
- PIN

These input items are described separately within this section. They are discussed in the approximate order that they come into play in the PIN verification routine.

Validation Data

Validation data, as applied to IBM DES (3624-format) PIN verification, is a 16-digit string of data unique to the cardholder. This data is used in the IBM DES (3624-format) PIN verification routines for generating a PIN.

The basis for creating validation data is the primary account number (PAN) associated with the card. The PAN equates to the card number stored in the magnetic stripe data of the card.

To derive validation data for use with IBM DES (3624-format) PIN verification, three values are required in addition to the PAN: the PAN offset, the PAN length and a PAN pad character.

The PAN offset value and length are used to extract a string of characters from the account number. The PAN pad character is then used to pad that string of characters out to the 16 digits required for the validation data.

To illustrate, if the PAN verification offset is 3 and the PAN verification length is 10, BASE24 would skip to position 4 of the account number and, starting with that position, pick up 10 consecutive characters of the account number. Then, BASE24 would add six of the PAN pad characters to the right of the account digits to expand the number out to 16 positions. The PAN must be at least as long as the PAN verification offset plus the PAN verification length.

An example of this follows, using a PAN pad character of F.

ACCOUNT NUMBER:	5	0	1	2	7	2	1	3	6	7	2	5	3	3	1	1
OFFSET:	1	2	3													
ACCOUNT NUMBER STRING:				2	7	2	1	3	6	7	2	5	3			
VALIDATION DATA:				2	7	2	1	3	6	7	2	5	3	F	F	F

To perform the above operation, BASE24 requires that the PAN verification offset, PAN verification length, and PAN pad character be specified in the Key Authorization File (KEYA). These values are specified in the PAN VERIFY OFFSET, PAN VERIFY LENGTH, and PAN PAD CHARACTER fields on KEYA screen 2, respectively.

Note: If the value in the PAN VERIFY LENGTH field is set to 16, no pad characters are used. The 16 digits are taken from the account number and used as the validation data.

PIN Verification Key

IBM DES (3624-format) PIN verification requires that the validation data be encrypted for processing. The PIN verification key is the key used to encrypt that validation data. In this case, encryption is used to generate a different sequence of digits and to tie that string of digits to a secret value, the key.

There are two versions of the PIN verification key that can be required, depending on whether the verification is being done exclusively in software by BASE24 or by a hardware module.

In the case of verification performed by BASE24 software, a clear key is required for BASE24 to encrypt the validation data.

In the case of verification performed by a hardware module, an encrypted key (encrypted by or for the hardware module) is required that can be passed to the module for encrypting the validation data.

If some verification is done in software and some in hardware, both a clear PIN verification key and an encrypted PIN verification key are required.

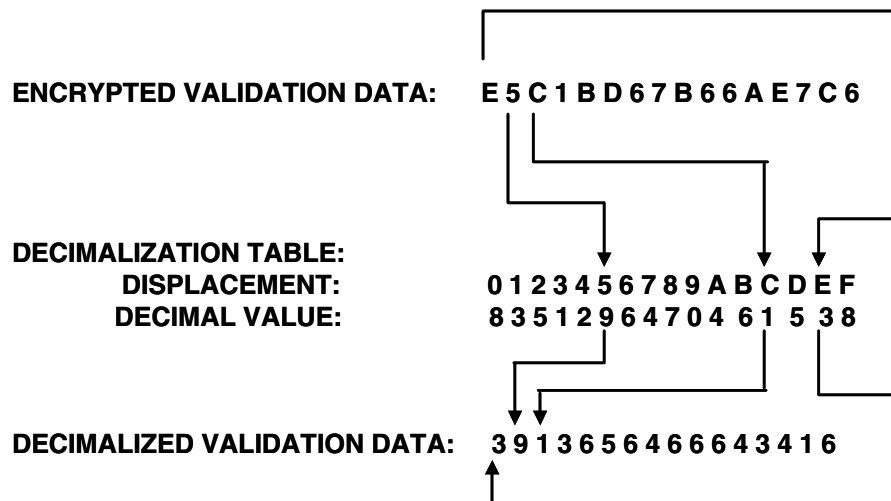
The clear PIN verification key must be specified in the CLEAR KEY field on KEYA screen 2. The encrypted PIN verification key must be specified in the ENCRYPTED KEY field on KEYA screen 2.

Decimalization Table

Because encryption of the validation data used in IBM DES (3624-format) PIN verification yields hexadecimal characters, the encrypted version of the validation data must be decimalized prior to further processing. When data is decimalized, all of the hexadecimal characters are converted to decimal characters.

The technique used to decimalize the encrypted validation data requires a decimalization table, which is a random series of 16 digits selected by the BASE24 system administrator. These digits must be in the range of 0 through 9.

To decimalize encrypted validation data, each character in the data is mapped to its equivalent position in the table, and the character in that position of the table is then substituted for the one in the validation data. The 16 table positions are numbered 0 through 15. An example is shown below:



In the example shown, the 5 in the validation data is replaced with a 9 because the sixth position of the decimalization table contains a 9. Likewise, each of the Cs in the validation data (the hexadecimal equivalent for the decimal number 12) is replaced with a 1 because the thirteenth position of the decimalization table contains a 1. Each position in the validation data is converted to a decimal value in this manner.

The decimalization table to be used by BASE24 must be placed in the DECIMALIZATION TABLE field on KEYA screen 2. There are no requirements for the decimalization table except that all 16 characters be numeric (0 through 9).

PIN Offset

Once the validation data has been decimalized, the left-most digits are taken to form an intermediate PIN. The number of digits to take is determined by the length of the cardholder's PIN offset. The PIN offset can be located in the Cardholder Authorization File (CAF) or in the magnetic stripe data of the cardholder's card. An example is shown below:

DECIMALIZED VALIDATION DATA:	3 9 1 3 6 5 6 4 6 6 6 4 3 4 1 6
PIN OFFSET:	8 7 6 5 (Length of 4)
INTERMEDIATE PIN:	3 9 1 3

The PIN offset, then, is the value which, when added (using the modulo 10 technique) to the intermediate PIN, yields the PIN check number to be matched against the PIN entered by the cardholder. Modulo 10 addition is a technique for adding numbers without carryover. For example, under this technique, 8 plus 4 would be 2, rather than 12, because the carried 1 is disregarded. Similarly, 8888 plus 4444 would be 2222 rather than 13332 because all of the carries are disregarded. An example of this is shown below:

INTERMEDIATE PIN:	3 9 1 3
PIN OFFSET (added Mod-10):	8 7 6 5
PIN CHECK NUMBER:	1 6 7 8

BASE24 allows institutions to specify how PIN offsets are to be obtained for verification using the PIN OFFSET LOC field on Institution Definition File (IDF) screen 2 or on Card Prefix File (CPF) screen 2, depending on whether PIN verification is controlled at the institution level or the card prefix level. The following options are available:

- A standard PIN offset value of 0000 can be used for all verification actions.
- The PIN offset can be taken from the Cardholder Authorization File (CAF). This option requires that an institution use an authorization method that uses the CAF. In this case, the PIN offset can be up to 12 digits in length.
- The PIN offset can be taken from the magnetic stripe data of the card used in the transaction. If PIN offsets are to be placed in the magnetic stripe data, institutions must specify where in the magnetic stripe data they can be found. This is done using the CPF. When PIN offsets are placed in the magnetic stripe data, BASE24 assumes that they are four digits in length and picks up this number of digits from the magnetic stripe data.

PIN

In IBM DES (3624-format) PIN verification, the PIN entered by the cardholder is compared to the PIN check number computed by the verification routines.

If PINs are verified by software, they must be in the clear when it comes time to verify them.

If PINs are verified by a hardware module, they must be passed to the module in encrypted form, along with the key used to encrypt them. The module then decrypts the PIN for comparing to the PIN check number it computes.

BASE24-atm allows cardholders to select their own PINs. The requirements for this are that PIN offsets be maintained in the CAF and that they be initially left blank. When using the Transaction Security Services process, cardholder-selected PINs can be verified in software or in a hardware security module. If you are not using the Transaction Security Services process, cardholder-selected PINs can only be verified in software. For more information on the Transaction Security Services process, refer to the ***BASE24 Transaction Security Services Processing Guide***. Upon receiving the first transaction from a cardholder, BASE24-atm computes an appropriate PIN offset based on the PIN entered by the cardholder and places that offset in the CAF. From that point on, that PIN entered by the cardholder is the only one allowed, unless the cardholder subsequently changes his or her PIN. BASE24-pos and BASE24-teller do not allow cardholder PIN selection.

BASE24-atm also allows cardholders to change their existing PINs. The PIN offset must already exist in the CAF before this transaction is performed. The PIN change transaction can be performed with IBM DES (3624-format) PIN verification in hardware or software. Upon receiving a PIN change transaction, BASE24-atm or the security module computes an appropriate PIN offset based on the PIN entered by the cardholder and uses it to replace the existing offset in the CAF. From that point on, the new PIN is the only one allowed, unless the cardholder changes his or her PIN again. BASE24-pos and BASE24-teller do not allow PIN change transactions.

PINs can be from four to 12 digits in length. However, the length of the PIN offset is the minimum required length of the PIN. For example, if the PIN offset is four digits, the PIN must be at least four digits as well. The PIN can be longer than the PIN offset; however, if the PIN is longer than the PIN offset, only the right-most digits of the PIN are used in the verification.

Setting Up IBM DES (3624-Format) PIN Verification

To support IBM DES (3624-format) PIN verification, BASE24 requires certain settings in its database files. These file settings, in turn, require various decisions by an institution on how it plans to issue its cards, how it wishes verification to be handled, and whether or not it plans to use a hardware module for performing verification.

The remainder of this section discusses BASE24-related requirements for IBM DES (3624-format) PIN verification. The discussion is divided into three parts:

- The BASE24 requirements placed on the cards issued by the institution
- The hardware-specific requirements associated with this verification method
- The BASE24 database fields that must be set up to perform this type of PIN verification

Card Requirements

If PIN offsets are to be maintained on the cards, the position of the PIN offset in the magnetic stripe data must be made available to BASE24. This position is specified in the CPF. PIN offset locations can vary by card prefix; however, the location must be consistent for all cards carrying the same prefix.

If PIN offsets are maintained on the cards, BASE24 assumes that they are four digits in length.

Note: PIN offsets cannot be carried on the cards if cardholders are to be allowed to select their own PINs, if cardholders are to be allowed to use PIN change transactions to change their PINs, or if the PIN offsets to be used are longer than four digits.

Hardware Module Requirements

IBM DES (3624-format) PIN verification can be performed by the following hardware modules supported by BASE24:

- Atalla Network Security Processor (NSP)
- Thales e-Security Host Security Module (HSM)

For instructions on configuring these devices, refer to section 17, appendix B and appendix C.

Note: IBM DES (3624-format) PIN verification using a HSM requires that the validation data include the five right-most positions of the PAN, excluding the check digit. Therefore, the PAN VERIFY OFFSET and PAN VERIFY LENGTH fields in the KEYA must be set appropriately to capture this portion of the PAN.

Institution Definition File (IDF) Settings

The following fields must be configured in the Institution Definition File (IDF) for each institution using the IBM DES (3624-format) PIN verification method. Refer to the *BASE24 Base Files Maintenance Manual* for illustrations of IDF screens 2, 9, and 13.

IDF Screen 2

IDF screen 2 enables an institution to set parameters for its cardholders or consumers. The following fields must be completed when using the IBM DES (3624-format) PIN verification method.

Note: All PIN verification fields appearing on IDF screen 2 also appear on CPF screen 2, allowing this information to be defined at the institution level or at the card prefix level.

CHECK IF HOST ONLINE PIN — Controls whether BASE24 verifies the PIN before sending a card-initiated transaction to the host.

- If BASE24 is to verify the PIN, this field must be set to a value of Y.
- If BASE24 is not to verify the PIN, this field must be set to a value of N.

When performing a PIN change transaction with authorization level 3 (online/offline), BASE24-atm verifies the existing PIN without checking the value in this field if the HOST PIN CHANGE OPTION field on IDF screen 13 is set to a value of 0 (do not send to host).

MAX PIN TRIES — Specifies the number of times an incorrect PIN can be entered before the action specified in the BAD PIN ACTION field is taken.

BAD PIN ACTION — Specifies the action to be taken by the device when it receives a PIN tries exceeded response code.

- If the card is to be returned to the cardholder, this field can be set to a value of 0. An ATM returns the card and a POS or teller device displays a message instructing the operator to return the card to the customer.
- If the card is to be retained, this field can be set to a value of 1. A POS or teller device displays a message instructing the operator to keep the card instead of returning it to the customer. An ATM retains the card, if the ATM has card retention functionality. This option is not valid for ATMs where the card is swiped instead of taken inside the machine to be read.

POFST/PVV LOC — Specifies whether a PIN offset other than 0000 is to be used and, if so, where the PIN offset is to be found on the card or in the CAF. Valid values are as follows:

- 0 = Use a PIN offset of 0000.
- 1 = Use the PIN offset in the magnetic stripe data of the card. With this option, BASE24-teller declines the transaction if the magnetic stripe data is not read electronically.
- 2 = Use the PIN offset in the CAF.

PIN CHECK TYPE — Specifies a code indicating the PIN verification method used by the institution. This field must be set to 01 to specify IBM DES (3624-format) PIN verification.

ALGO NUMBER LOC — Specifies the code indicating the location of the algorithm number. Currently, the algorithm number is required only for the Diebold PIN verification method; therefore, this field must be set to zero because this type of PIN verification does not use algorithm numbers.

CARDHOLDER PIN SELECT — Specifies a code used by only BASE24-atm that identifies whether cardholders are allowed to select their PINs the first time they use their cards. Valid values are as follows:

- Y = Yes, cardholders are allowed to select their own PINs at ATMs.
- N = No, PINs are assigned to the cardholders by the institution.

Cardholder PIN selection is not the same as the PIN change transaction. Cardholder PIN selection permits cardholders to establish a PIN when they first use their cards. A PIN change transaction permits a cardholder to change an existing PIN.

To use this option, PIN offsets must be stored in the CAF (i.e., the POFST/PVV LOC field must be set to a value of 2), PIN verification must be performed by BASE24 in software, and the PIN offset in the CAF must be left blank.

PIN TRIES RESET OPTION — A code specifying when the PIN tries counter is reset and whether the bad PIN action is taken when the bad PIN tries limit is exceeded. Refer to the topic “Maximum PIN Retries” within the BASE24-atm, BASE24-pos, or BASE24-teller PIN verification discussions in section 8 for valid values.

IDF Screen 9

IDF screen 9 enables institutions to specify acquirer and issuer transaction profiles.

ACQUIRER TXN PROFILE — Controls whether the PIN change transaction is included in the set of transactions that the terminal-owning institution allows. The value in this field can be overridden at the terminal level in the BASE24-atm Terminal Data files.

ISSUER TXN PROFILE — Controls whether the PIN change transaction is included in the set of transactions that the card issuing institution allows for its cardholders. This profile can be overridden at the card prefix level in the Card Prefix File (CPF) or at the cardholder level in the Cardholder Authorization File (CAF).

IDF Screen 13

IDF screen 13 enables institutions to set their BASE24-atm processing control parameters. The following field must be completed when using the IBM DES (3624-format) PIN verification method.

HOST PIN CHANGE OPTION — Determines where the new PIN offset is computed for a PIN change transaction with an online/offline authorization level when BASE24-atm is prescreening PINs or standing in for the host.

- If BASE24-atm prescreens the existing PIN and computes a new PIN offset for a PIN change transaction, this field must be set to a value of 0.

- If BASE24-atm verifies the existing PIN and computes a new PIN offset for a PIN change transaction only when the host is unavailable, this field must be set to a value of 1.
- If BASE24-atm declines a PIN change transaction when the host is unavailable, this field must be set to a value of 2.

Key Authorization File (KEYA) Settings

The following fields must be configured in the Key Authorization File (KEYA) for each FIID or CPF group using the IBM DES (3624-format) PIN verification method. Refer to the ***BASE24 Base Files Maintenance Manual*** for illustrations of all KEYA screens.

KEYA Screen 1

KEYA screen 1 allows users to identify the KEYA record to be accessed. The following fields must be completed when using the IBM DES (3624-format) PIN verification method.

GRP — The code used in the IDF or CPF to identify this record.

- If the PIN CHECK TYPE field on IDF screen 2 contains a value of 01, 02, 03, or 04 (verify PINs) and the PIN CHECK TYPE field on CPF screen 2 for one or more of the institution's card prefixes contains a value of 99 (use the IDF for PIN verification), a KEYA record must exist with the institution's FIID in this field.
- If the PIN CHECK TYPE field on CPF screen 2 contains a value of 01, 02, 03, or 04 (verify PINs), a KEYA record must exist with an entry in this field that matches the entry in the PIN VERIFICATION KEYA GROUP field on CPF screen 2.

BEGIN DATE — The first date in the range of card expiration dates to which this KEYA record applies. The date in this field is expressed in YYYYMM format and is compared to the expiration date of the card being used to initiate the transaction. Valid values are as follows:

- 000000 = Data in this KEYA record is valid for cards with expiration dates that are less than or equal to the date specified in the END DATE field.
- YYYYMM = Data in this KEYA record is valid for cards with expiration dates that are equal to or greater than this date but less than or equal to the date specified in the END DATE field.
- ***** = Data in this KEYA record is valid regardless of the card expiration date. The END DATE field also must contain asterisks.

The BEGIN DATE and END DATE fields permit a financial institution to maintain different sets of verification parameters based on the card expiration date. For example, an institution using the IBM DES (3624-format) PIN verification method plans to change the decimalization table it uses to generate PINs or PIN offsets. All outstanding cards use the existing decimalization table. The first cards to use the new decimalization table will have an expiration date of June, 2002, which appears in YYYYMM format as 200206. The existing decimalization table would be effective for cards issued with expiration dates through May, 2002, which appears in YYYYMM format as 200205.

Two KEYA records are necessary. The first KEYA record contains the old decimalization table and has a value in the END DATE field of 200205. The BEGIN DATE field for the first KEYA record can be 000000 because all outstanding cards use the verification parameters it contains. The second KEYA record contains the new decimalization table and has a value in the BEGIN DATE field of 200206. The END DATE field for the second KEYA record must contain a future date.

If more than one set of verification parameters had already been in use, there would have been additional KEYA records and the BEGIN DATE field for the first KEYA record would contain the date that its verification parameters became effective.

If beginning and ending dates are unnecessary, the BEGIN DATE and END DATE fields should be set to asterisks (*).

END DATE — The last date in the range of card expiration dates to which this KEYA record applies. The date in this field is expressed in YYYYMM format and is compared to the expiration date of the card being used to initiate the transaction. Refer to the BEGIN DATE field description for an example of how this field is used. Valid values are as follows:

YYYYMM = Data in this KEYA record is valid for cards with expiration dates that are less than or equal to this date but equal to or greater than the date in the BEGIN DATE field.

***** = Data in this KEYA record is valid regardless of the card expiration date. The BEGIN DATE field also must contain asterisks.

Note: 000000 is not a valid value in this field.

RECORD TYPE — Specifies the type of verification information contained in this record. A separate KEYA screen contains the information for each verification type. For the IBM DES (3624-format) PIN verification method, this field must be set to a value of 01.

KEYA Screen 2

KEYA screen 2 contain data for the IBM DES (3624-format) PIN verification information. They are used only when the value in the Record Type field on KEYA screen 1 is set to a value of 01.

DECIMALIZATION TABLE — The decimalization table to be used by BASE24. This table must be 16 characters in length. Valid values for each character in this field are 0 through 9. For protection in the database, this table is scrambled for storage in the file by the KEYA Server process.

PAN VERIFY OFFSET — The offset to the starting position in the PAN for extracting validation data. Position 00 identifies the first digit in the PAN.

CLEAR KEY — The clear PIN verification key used when PIN verification is performed without a security module.

- If verification is, at least some of the time, to be performed without a security module, this field must contain a clear PIN verification key. This key must be 16 hexadecimal characters. Valid values for each character in this field are 0 through 9 and A through F. For protection in the database, this key is scrambled for storage by the KEYA Server process.
- If PIN verification is to be performed by a hardware module exclusively, this field is not used .

PAN VERIFY LENGTH — The length of the data (number of digits) to be extracted from the PAN for the formation of validation data. Valid values are 04 through 16.

ENCRYPTED KEY — The encrypted PIN verification key used when PIN verification is performed by a hardware security module.

- If verification is, at least some of the time, to be performed by a hardware module, this field must contain an encrypted PIN verification key. The PIN verification key must be encrypted by or for the specific security module being used, such that the module is able to decrypt the key when it is provided by BASE24. The encrypted key must be 16 characters in length. Valid values for each character in this field are 0 through 9 and A through F.
- If PIN verification is to be performed by BASE24 software exclusively, this field is not used.

PAN PAD CHARACTER — The character to use for padding the validation data out to 16 digits. This can be 0 through 9 or A through F.

CHECK DIGITS — The check digits associated with the encrypted PIN verification key used when PIN verification is performed by a hardware security module. Valid values for each character in this field are 0 through 9 and A through F. A value of 0000 in this field indicates check digits are not being used with PIN verification keys.

Card Prefix File (CPF) Settings

The following fields must be configured in the Card Prefix File (CPF) for each prefix processed for the institution using the IBM DES (3624-format) PIN verification method. Refer to the ***BASE24 Base Files Maintenance Manual*** for illustrations of CPF screens 1, 2, 4, and 7.

CPF Screen 1

CPF screen 1 contains general information about card prefixes. The following fields are required when using the IBM DES (3624-format) PIN verification method.

POFST/PVV — Identifies the relative position of the PIN offset in the magnetic stripe data.

- If PIN offsets are to be stored in the magnetic stripe data of the card (i.e., the POFST/PVV LOC field on IDF or CPF screen 2 is set to a value of 1), this value identifies the relative position of the PIN offset, and must be specified for each CPF record.
- If PIN offsets are not to be stored in the magnetic stripe data of the card, this field can be allowed to default to zero.

ALGO #/PVKI — The value defining the position of the algorithm number in the magnetic stripe data. This field must be left to default to zeros because the IBM DES (3624-format) PIN verification method does not use algorithm numbers or PIN verification key indicators.

CPF Screen 2

CPF screen 2 contains transaction security information for the card prefix. The following fields must be completed when using the IBM DES (3624-format) PIN verification method.

Note: All but one of the PIN verification fields appearing on CPF screen 2 also appear on IDF screen 2, allowing this information to be defined at the institution level or at the card prefix level. The FIID field on IDF screen 2 serves the same purpose as the PIN VERIFICATION KEYA GROUP field on CPF screen 2.

PIN VERIFICATION KEYA GROUP — The value used to select the proper Key Authorization File (KEYA) record when verifying PINs for cards with this prefix. The value in this field is matched with the value in the GRP field on screen 1 of the KEYA record being used with this card prefix.

PIN CHECK TYPE — Specifies the code indicating the cardholder PIN verification method to be used. This field must be set to 01 to specify IBM DES (3624-format) PIN verification.

MAX PIN TRIES — The number of times an incorrect PIN can be entered before the action specified in the BAD PIN ACTION field is taken.

BAD PIN ACTION — Specifies the action to be taken by the device when it receives a PIN tries exceeded response code.

- If the card is to be returned to the cardholder, this field can be set to a value of 0. An ATM returns the card and a POS or teller device displays a message instructing the operator to return the card to the customer.
- If the card is to be retained, this field can be set to a value of 1. A POS or teller device displays a message instructing the operator to keep the card instead of returning it to the customer. An ATM retains the card, if the ATM has card retention functionality. This option is not valid for ATMs where the card is swiped instead of taken inside the machine to be read.

CARDHOLDER PIN SELECT — Specifies a code used by only BASE24-atm that identifies whether cardholders are allowed to select their PINs the first time they use their cards. Valid values are as follows:

Y = Cardholders are to be allowed to select their own PINs at ATMs.

N = PINs are to be assigned to the cardholders by the institution.

Cardholder PIN selection is not the same as the PIN change transaction. Cardholder PIN selection permits cardholders to establish a PIN when they first use their card. A PIN change transaction permits a cardholder to change an existing PIN.

To use this option, PIN offsets must be stored in the CAF (i.e., the POFST/PVV LOC field must be set to a value of 2), PIN verification must be performed by BASE24 in software, and the PIN offset in the CAF must be left blank.

ALGO NUMBER LOC — The code indicating the location of the algorithm number. Currently, the algorithm number is required only for the Diebold PIN verification method; therefore, this field must be set to zero because this type of PIN verification does not use algorithm numbers.

CHECK IF HOST ONLINE PIN — Controls whether BASE24 verifies the PIN before sending a card-initiated transaction to the host.

- If BASE24 is to verify the PIN, this field must be set to a value of Y.
- If BASE24 is not to verify the PIN, this field must be set to a value of N.

When performing a PIN change transaction with authorization level 3 (online/offline), BASE24-atm verifies the existing PIN without checking the value in this field if the HOST PIN CHANGE OPTION field on IDF screen 13 is set to a value of 0 (do not send to host).

POFST/PVV LOC — Specifies whether a PIN offset other than 0000 is to be used and, if so, where the PIN offset is to be found, on the card or in the CAF. Valid values are as follows:

- 0 = Use a PIN offset of 0000.
- 1 = Use the PIN offset in the magnetic stripe data of the card. With this option, BASE24-teller declines the transaction if the magnetic stripe data is not read electronically.
- 2 = Use the PIN offset in the CAF.

PIN TRIES RESET OPTION — A code specifying when the PIN tries counter is reset and whether the bad PIN action is taken when the bad PIN tries limit is exceeded. Refer to the topic “Maximum PIN Retries” within the BASE24-atm, BASE24-pos, or BASE24-teller PIN verification discussions in section 8 for valid values.

CPF Screen 4

CPF screen 4 enables institutions to specify the issuer transaction profile, which determines the transactions allowed for an institution’s cardholders at the card prefix level. This profile can be overridden at the cardholder level in the Cardholder Authorization File (CAF).

ISSUER TXN PRFL — Controls whether the PIN change transaction is included in the set of transactions that the card issuing institution allows for this card prefix.

CPF Screen 7

CPF screen 7 displays flags indicating how BASE24-pos handles preauthorization holds, chargebacks, representments, PIN processing, and address verification, as well as default values for combination card and account types and the maximum number of refunds allowed by BASE24-pos. The following field must be completed when using the IBM DES (3624-format) PIN verification method.

PIN PROCESSING FLAG — Identifies whether PINs are required with transactions from POS devices.

- If a PIN is not required to complete a transaction, this field must be set to a value of 0.
- If a PIN is required with every transaction and transactions from terminals without PIN capabilities are declined, this field must be set to a value of 1.
- If PINs are required with transactions from terminals with PIN capabilities and transactions from attended terminals without PIN capabilities can be processed without PINs, this field must be set to a value of 2.

Regardless of this setting, BASE24-pos attempts to verify any PIN it receives from a POS device.

Cardholder Authorization File (CAF) Settings

The following fields must be configured on CAF screen 1 for each cardholder processed for the institution using the IBM DES (3624-format) PIN verification method. Refer to the *BASE24 Base Files Maintenance Manual* for an illustration of CAF screens 1 and 8.

CARD STATUS — Specifies the code identifying the status of the card. This field must be set to a value of 0 (issued, but not active) or a value of 1 (open) if cardholders are permitted to select their PINs on the first BASE24-atm transaction. All other card status values indicate conditions where unsupervised PIN selection is not advised.

POFST/PVV — The PIN offset. Each institution specifies the settings for PIN verification in the IDF or the CPF. The settings are as follows:

- If PIN offsets are to be stored in the CAF (i.e., the POFST/PVV LOC field on IDF or CPF screen 2 is set to a value of 2), a PIN offset must be placed in this field for each CAF record. If one zero is entered in this field, at least four zeros must be entered. Although spaces are allowed in this field and spaces can trail an entry, spaces cannot be entered before a numeric entry or be embedded in an entry.
- If PIN offsets are to be stored in the CAF but cardholders are allowed to select their own PINs, this field must be left blank in any CAF record for which a cardholder has yet to select the PIN. BASE24 will then fill in the PIN offset when the cardholder first uses the card.

- If PIN offsets are not to be stored in the CAF, this field can be allowed to default to blanks.

CAF Screen 8

CAF screen 4 enables institutions to specify the issuer transaction profile, which determines the transactions allowed for an institution's cardholders at the cardholder account level. This profile overrides the issuer transaction profile specified at the terminal level in the IDF or at the card prefix level in the Card Prefix File (CPF).

ISSUER TXN PROFILE — Controls whether the PIN change transaction is included in the set of transactions that the card issuing institution allows for this cardholder account.

BASE24-atm Terminal Data Files Settings

For installations that include BASE24-atm, the following fields must be configured on Acquirer Processing Code File (APCF) screen 2 and BASE24-atm Terminal Data files (ATD) files maintenance screens for each ATM owned by the institution. Refer to the ***BASE24 Base Files Maintenance Manual*** for an illustration of APCF screen 2 and the ***BASE24-atm Files Maintenance Manual*** for illustrations of ATD screens 3 and 9.

ATD Screen 3

ATD screen 3 determines the types of special transactions allowed for cardholders at a particular terminal. The transaction classes listed can vary depending on parameters set up at configuration, although they should include all special transaction classes allowed at the terminal. The following field must be completed when using the IBM DES (3624-format) PIN verification method with all device types except for the Diebold 10XX/478X or the NCR 5XXX. For these two device types, allowed transactions for both on-us and not-on-us cardholders (including the PIN change transaction) are configured using the ACQUIRER TXN PROFILE field on ATD screen 2 and Acquirer Processing Code File (APCF) screen 2.

PIN CHANGE — Controls whether the PIN change transaction is included in the set of transactions that another institution's cardholders can perform at this terminal. Valid values are as follows:

- 0 = Not allowed
- 1 = Allowed within the country
- 2 = Allowed within the state
- 3 = Allowed nationally
- 4 = Allowed internationally

ATD Screen 9

ATD screen 9 displays device-specific security, PIN, and message authentication information. The device information displayed is determined by the value in the DEVICE TYPE field on ATD screen 1. The following field is required when using the IBM 3624-format PIN verification method.

LOCAL PIN VERIFY — Specifies whether local PIN verification is allowed for this terminal.

- If local PIN verification is allowed for all transactions at the terminal, this field must be set to a value of 0.
- If local PIN verification is not allowed at the terminal, this field must be set to a value of 1.
- If local PIN verification is allowed only for on-us transactions at the terminal, this field must be set to a value of 2. An on-us transaction is one where the same institution issues the card and owns the terminal.

Acquirer Processing Code File (APCF) Settings

The fields on APCF screen 2 specify whether a transaction is allowed for a specific acquirer transaction profile. The PIN Change transaction must be allowed in the acquirer transaction profile configured for ATMs from which it can be entered. This screen is used to establish acquirer transaction profiles for Diebold 10XX/478X and NCR 5XXX ATMs.

Issuer Processing Code File (IPCF) Settings

The fields on IPCF screen 2 specify whether a transaction is allowed for a specific issuer transaction profile. The PIN Change transaction must be allowed in the issuer transaction profiles configured for the institution in the IDF, the cardholder's account in the CAF, or the cardholder's account prefix in the CPF.

BASE24-pos Terminal Data Files Settings

No fields need to be configured on the BASE24-pos Terminal Data files (PTD) files maintenance screens. POS devices do not perform local PIN verification or support PIN change transactions.

Teller Terminal Data File (TTDF) Settings

For installations that include BASE24-teller, the following field on TTDF screen 2 must be configured in the Teller Terminal Data File (TTDF) for each teller terminal owned by the institution. Refer to the ***BASE24-teller Files Maintenance Manual*** for an illustration of TTDF screen 2.

CONTROLLER VALIDATE PIN — Specifies whether local PIN verification is allowed for this terminal.

- If PINs can be verified at the controller before being sent to BASE24-teller, this field must be set to a value of Y.
- If PINs cannot be verified at the controller before being sent to BASE24-teller, this field must be set to a value of N.

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10: Diebold PIN Verification Support

This section describes BASE24 support of the Diebold PIN verification method, including how the BASE24 database must be set up to allow BASE24 to verify PINs under this method of verification.

BASE24 allows for implementation of this PIN verification method in software or a security module.

Diebold PIN Verification

Diebold PIN verification is based on the use of a Diebold number table, an algorithm number, and a PIN offset to verify a cardholder's PIN. Diebold PIN verification can be performed in software or using a hardware module linked to a processor on which the BASE24 system is running; however, an institution cannot perform this verification method in both software and hardware. Each institution using this verification method must determine whether to use software or hardware.

At card-issuing time, the card issuer generates a PIN offset for each card, using the Diebold number table, an algorithm number, and other card-specific data. This offset is then placed in the magnetic stripe data of the card or in the BASE24 database.

The following is a summary of the steps taken to verify a PIN using Diebold PIN verification in software or hardware.

The verifying entity performs the following steps:

1. Processes each digit of the cardholder's PAN against the Diebold number table, using the selected algorithm number as an index into the table.
2. Processes the right-most four digits against the Diebold number table a second time to derive an intermediate PIN.
3. Subtracts the cardholder's PIN offset from the intermediate PIN (using the modulo 10 technique).
4. Compares the result from step 3 with the PIN entered by the cardholder. If they match, the cardholder entered the PIN correctly. If they do not match, the cardholder entered the PIN incorrectly.

PIN Verification Inputs

Diebold PIN verification requires the following information as input. Some of these items vary from cardholder to cardholder, while others are constant for the institution or card prefix.

- Diebold number table (DNT)
- Primary account number (PAN)
- Algorithm number
- PIN offset
- PIN

These input items are described separately within this section. They are discussed in the approximate order that they come into play in the PIN verification routine.

Diebold Number Table

A Diebold number table (DNT) is a random set of 256 nonrepeating values used in Diebold PIN verification.

BASE24 requires access to the card issuer's Diebold number table in order to perform PIN verification in software. The table must be placed in the Key Authorization File (KEYA) using KEYA screen 3.

Each value in the Diebold number table consists of a pair of hexadecimal characters (e.g., AA, A1, 9F, etc.). There are 256 possible combinations of hexadecimal pairs, and these combinations can be scrambled to derive a unique set of table entries, subject to the following rules:

- There must be 256 values in the table.
- No hexadecimal pair can be repeated.
- The first character of the first hexadecimal pair of the table must be a C. Thus, the first hexadecimal pair can be any one of the 16 combinations from C0 to CF.

Note: This table functions in much the same way as a DES key; therefore, it must be kept confidential.

Primary Account Number (PAN)

A primary account number (PAN) is the card number used to identify each card. It must be carried in the magnetic stripe data of the card and is used as input to Diebold PIN verification.

Under Diebold PIN verification, each digit of the PAN is processed individually against the Diebold number table (regardless of the length of the PAN), starting from the right and moving left.

Each digit is processed against the table seven times. The right-most four digits of the PAN are then processed against the table again to derive an intermediate PIN.

Algorithm Number

An algorithm number is the two-digit number used as a starting index in the Diebold number table for calculations on the PAN. An algorithm number can be any value from 00 to 99. Like the Diebold number table, the algorithm number can be chosen randomly and it must be kept confidential. The confidentiality of the DNT and algorithm number enforce the security of the processing.

Institutions must specify how algorithm numbers are to be handled by BASE24 using the ALGO NUMBER LOC field on Institution Definition File (IDF) screen 2 or Card Prefix File (CPF) screen 2, depending on whether PIN verification is being controlled at the institution level or the card prefix level. There are two options available for Diebold PIN verification:

- The algorithm number can be taken from the KEYA. In this case, a single algorithm number is used for all verification actions.
- The algorithm number can be taken from the magnetic stripe data of the card used in the transaction. In this case, algorithm numbers can vary from card to card.

Note: If algorithm numbers are to be placed in the magnetic stripe data, institutions must specify where in the magnetic stripe data they can be found. This is done using the CPF.

PIN Offset

A PIN offset is subtracted (using the modulo 10 technique) from the intermediate PIN to determine a computed PIN. The computed PIN is then compared to the PIN entered by the cardholder. Modulo 10 subtraction is a technique for subtracting numbers without borrowing. For example, under this technique, 2 minus 8 would be 4: 8 cannot be subtracted from 2; therefore, the 2 would automatically be assumed to have a value of 12. Similarly, 3333 minus 8888 would be 5555, because each number is treated separately. An example of this is shown below:

Intermediate PIN:	2 9 9 2
PIN Offset (subtracted Mod-10):	8 7 6 5
Comparison Number:	4 2 3 7

BASE24 allows institutions to specify how PIN offsets are to be derived for verification using the POFST/PVV LOC field on IDF screen 2 or CPF screen 2. Under Diebold PIN verification, PIN offsets must be four digits in length. The following options are available:

- A standard PIN offset value of 0000 can be used for all verification actions.
- The PIN offset can be taken from the Cardholder Authorization File (CAF). This option requires that an institution use an authorization method that uses the CAF.
- The PIN offset can be taken from the magnetic stripe data of the card used in the transaction.

Note: If PIN offsets are to be placed in the magnetic stripe data, institutions must specify where in the magnetic stripe data they can be found. This is done in the CPF. When the PIN offset is placed in the magnetic stripe data, BASE24-teller declines transactions if the magnetic stripe data is not read electronically.

PIN

In Diebold PIN verification, the PIN entered by the cardholder is compared to the PIN computed by the verification routines.

If PINs are verified by software, they must be in the clear when it comes time to verify them.

If PINs are verified by a hardware module, they must be passed to the module in encrypted form, along with the key used to encrypt them. The module then decrypts the PIN for comparing to the PIN it computes.

BASE24-atm allows cardholders to select their own PINs. This requires PIN offsets to be maintained in the CAF, that they are initially left blank, and that PINs are verified in software. Upon receiving the first transaction from a cardholder, BASE24-atm computes an appropriate PIN offset based on the PIN entered by the cardholder and places that offset in the CAF. From that point on, that PIN entered by the cardholder is the only one allowed, unless the cardholder subsequently changes his or her PIN. BASE24-pos and BASE24-teller do not allow cardholder PIN selection.

BASE24-atm also allows cardholders to change their existing PINs. The PIN offset must already exist in the CAF before this transaction is performed. The PIN change transaction can be performed with Diebold PIN verification in hardware or software. Upon receiving a PIN change transaction, BASE24-atm or the hardware security module computes an appropriate PIN offset based on the PIN entered by the cardholder and uses it to replace the existing offset in the CAF. From that point on, the new PIN is the only one allowed, unless the cardholder changes his or her PIN again. BASE24-pos and BASE24-teller do not allow PIN change transactions.

Under this form of PIN verification, PINs must be four digits in length.

Setting Up Diebold PIN Verification

To support Diebold PIN verification, BASE24 requires certain settings in its database files. These file settings, in turn, require various decisions by an institution on how it plans to issue its cards, how it wishes verification to be handled, and whether it plans to use a hardware module for performing verification.

The remainder of this section discusses BASE24-related requirements for Diebold PIN verification. The discussion is divided into three parts:

- The BASE24 requirements placed on the cards issued by the institution
- The hardware-specific requirements associated with this verification method
- The BASE24 database fields that must be set up to perform this type of PIN verification

Card Requirements

If PIN offsets and algorithm numbers are maintained on the cards, the positions of this data on the magnetic stripe must be made available to BASE24. These positions are specified in the CPF. PIN offset and algorithm number locations can vary by card prefix; however, the location must be consistent for all cards carrying the same prefix.

Note: PIN offsets cannot be carried on the cards if cardholders are allowed to select their own PINs or use PIN change transactions to change their PINs.

Hardware Module Requirements

Diebold PIN verification can be performed by the following hardware modules supported by BASE24:

- Atalla Network Security Processor (NSP)
- Thales e-Security Host Security Module (HSM)

For instructions on configuring these devices, refer to section 17, appendix B, and appendix C.

An institution cannot perform Diebold PIN verification in both software and hardware.

NSP support of Diebold PIN verification requires that a clear Diebold number table (DNT) be entered in the DIEBOLD TABLE field on KEYA screen 3. It also requires that a key be provided, in clear and encrypted form, to encrypt the DNT for sending to the NSP. The clear version of the key must be placed in the CLEAR KEY field on KEYA screen 3. The encrypted version of the key must be placed in the ENCRYPTED KEY field on KEYA screen 3. During initialization, the Authorization process uses the clear key for encrypting the DNT and sends the encrypted version of the key to the NSP along with the encrypted DNT to allow the NSP to decrypt the DNT.

Thales e-Security HSM support of Diebold PIN verification requires only that the DNT be accessible to the HSM. The DIEBOLD TABLE field on KEYA screen 3 can contain all zeros or an encrypted version of the DNT returned from the HSM. When Diebold PIN verification is specified, the Authorization process, during initialization, calls a utility and passes the DNT from the KEYA to the HSM. If the DNT from the KEYA contains all zeros, the utility asks the HSM to verify that the HSM already has a valid DNT. If the DNT in the KEYA contains values other than zero, the utility loads the KEYA DNT to the HSM.

In NSP and Thales e-Security HSM support, a relative location for the DNT in the security module must be specified in the NUMBER TABLE RELATIVE LOCATION field on KEYA screen 3. This allows for keeping the DNTs, which could be different for each institution or card prefix, from overlaying each other in the security module.

Diebold number tables are loaded or verified during process initialization, process warmboot, or when a security administrator enters either an INITDNT command for a BASE24-atm Authorization process or a 9531DNT command for a BASE24-pos Device Handler/Router/Authorization process.

Note: If you are using an Atalla NSP model A8000 or A8100 in your BASE24 system, only one BASE24-atm Authorization process or BASE24-pos Device Handler/Router/Authorization process can download Diebold number tables (DNTs) from the Key Authorization File (KEYA) to the device during initialization or process warmboots. The A8000 and A8100 security devices fail when more than one Authorization process attempts to load DNTs during initialization or warmboots. The ATM-A4000-MASTER or POS-A4000-MASTER param must be configured in the LCONF to specify the symbolic name of the process responsible for downloading the DNT to an A8000 or A8100 device. For more information on these params, refer to appendixes B and C.

Institution Definition File (IDF) Settings

The following fields must be configured in the Institution Definition File (IDF) for each institution using the Diebold PIN verification method. Refer to the ***BASE24 Base Files Maintenance Manual*** for illustrations of IDF screens 2, 9, and 13.

IDF Screen 2

IDF screen 2 enables an institution to set up parameters for its cardholders or consumers. The following fields must be completed when using the Diebold PIN verification method.

Note: All PIN verification fields appearing on IDF screen 2 also appear on CPF screen 2, allowing this information to be defined at the institution level or at the card prefix level.

CHECK IF HOST ONLINE PIN — Controls whether BASE24 verifies the PIN before sending a card-initiated transaction to the host.

- If BASE24 is to verify the PIN, this field must be set to a value of Y.
- If BASE24 is not to verify the PIN, this field must be set to a value of N.

When performing a PIN change transaction with authorization level 3 (online/offline), BASE24-atm verifies the existing PIN without checking the value in this field if the HOST PIN CHANGE OPTION field on IDF screen 13 is set to a value of 0 (do not send to host).

MAX PIN TRIES — Specifies the number of times an incorrect PIN can be entered before the action specified in the BAD PIN ACTION field is taken.

BAD PIN ACTION — Specifies the action to be taken by the device when it receives a PIN tries exceeded response code.

- If the card is to be returned to the cardholder, this field can be set to a value of 0. An ATM returns the card and a POS or teller device displays a message instructing the operator to return the card to the customer.

- If the card is to be retained, this field can be set to a value of 1. A POS or teller device displays a message instructing the operator to keep the card instead of returning it to the customer. An ATM retains the card, if the ATM has card retention functionality. This option is not valid for ATMs where the card is swiped instead of taken inside the machine to be read.

POFST/PVV LOC — Specifies whether a PIN offset other than 0000 is to be used and, if so, where the PIN offset is to be found, on the card or in the CAF. Valid values are as follows:

- 0 = Use a PIN offset of 0000.
- 1 = Use the PIN offset in the magnetic stripe data of the card. With this option, BASE24-teller declines the transaction if the magnetic stripe is not read electronically.
- 2 = Use the PIN offset in the CAF.

PIN CHECK TYPE — Specifies a code indicating the PIN verification method used by the institution. This field must be set to a value of 02 to specify Diebold PIN verification.

ALGO NUMBER LOC — Specifies where the algorithm number is to be found—on the card or in the KEYA. Valid values are as follows:

- 1 = Use the algorithm number in the KEYA.
- 2 = Use the algorithm number in the magnetic stripe data of the card.

CARDHOLDER PIN SELECT — Specifies a code used by only BASE24-atm that identifies whether cardholders are allowed to select their PINs the first time they use their cards. Valid values are as follows:

- Y = Yes, cardholders are to be allowed to select their own PINs at ATMs.
- N = No, PINs are to be assigned to the cardholders by the institution.

Cardholder PIN selection is not the same as the PIN change transaction. Cardholder PIN selection permits cardholders to establish PINs when they first use their cards. A PIN change transaction permits a cardholder to change an existing PIN.

To use this option, PIN offsets must be stored in the CAF (i.e., the POFST/PVV LOC field must be set to a value of 2), PIN verification must be performed by BASE24 in software, and the PIN offset in the CAF must be left blank.

PIN TRIES RESET OPTION — A code specifying when the PIN tries counter is reset and whether the action specified in the BAD PIN ACTION field is taken when the bad PIN tries limit is exceeded. Refer to the topic “Maximum PIN Retries” within the BASE24-atm, BASE24-pos, or BASE24-teller PIN verification discussions in section 8 for valid values.

IDF Screen 9

IDF screen 9 enables institutions to specify acquirer and issuer transaction profiles.

ACQUIRER TXN PROFILE — Controls whether the PIN change transaction is included in the set of transactions that the terminal-owning institution allows. The value in this field can be overridden at the terminal level in the BASE24-atm Terminal Data files.

ISSUER TXN PROFILE — Controls whether the PIN change transaction is included in the set of transactions that the card issuing institution allows for its cardholders. This profile can be overridden at the card prefix level in the Card Prefix File (CPF) or at the cardholder level in the Cardholder Authorization File (CAF).

IDF Screen 13

IDF screen 13 enables institutions to set their BASE24-atm processing control parameters. The following field must be completed when using the Diebold PIN verification method.

HOST PIN CHANGE OPTION — Determines where the new PIN offset is computed for a PIN change transaction with an online/offline authorization level when BASE24-atm is prescreening PINs or standing in for the host.

- If BASE24-atm prescreens the existing PIN and computes a new PIN offset for a PIN change transaction, this field must be set to a value of 0.
- If BASE24-atm verifies the existing PIN and computes a new PIN offset for a PIN change transaction only when the host is unavailable, this field must be set to a value of 1.
- If BASE24-atm declines a PIN change transaction when the host is unavailable, this field must be set to a value of 2.

Key Authorization File (KEYA) Settings

The following fields must be configured in the Key Authorization File (KEYA) for each FIID or CPF group using the Diebold PIN verification method. Refer to the *BASE24 Base Files Maintenance Manual* for illustrations of all KEYA screens.

KEYA Screen 1

KEYA screen 1 allows users to identify the KEYA record to be accessed. The following fields must be completed when using the Diebold PIN verification method.

GRP — The code that is used in the IDF or CPF to identify this record.

- If the PIN CHECK TYPE field on IDF screen 2 contains a value of 01, 02, 03, or 04 (verify PINs) and the PIN CHECK TYPE field on CPF screen 2 for one or more of the institution's card prefixes contains a value of 99 (use the IDF for PIN verification), a KEYA record must exist with the institution's FIID in this field.
- If the PIN CHECK TYPE field on CPF screen 2 contains a value of 01, 02, 03, or 04 (verify PINs), a KEYA record must exist with an entry in this field that matches the entry in the PIN VERIFICATION KEYA GROUP field on CPF screen 2.

BEGIN DATE — The first date in the range of card expiration dates to which this KEYA record applies. The date in this field is expressed in YYYYMM format and is compared to the expiration date of the card being used to initiate the transaction. Valid values are as follows:

- | | |
|--------|---|
| 000000 | = Data in this KEYA record is valid for cards with expiration dates that are less than or equal to the date specified in the END DATE field. |
| YYYYMM | = Data in this KEYA record is valid for cards with expiration dates that are equal to or greater than this date but less than or equal to the date specified in the END DATE field. |
| ***** | = Data in this KEYA record is valid regardless of the card expiration date. The END DATE field also must contain asterisks. |

The BEGIN DATE and END DATE fields permit a financial institution to maintain different sets of verification parameters based on the card expiration date. For example, an institution using the Diebold PIN verification method plans to change the Diebold number table (DNT) it uses to generate PINs or PIN offsets. All outstanding cards use the existing DNT. The first cards to use the new DNT will have an expiration date of June, 2002, which appears in YYYYMM format as 200206. The existing DNT would be effective for cards issued with expiration dates through May, 2002, which appears in YYYYMM format as 200205.

Two KEYA records are necessary. The first KEYA record contains the old DNT and has a value in the END DATE field of 200205. The BEGIN DATE field for the first KEYA record can be 000000 because all outstanding cards use the verification parameters it contains. The second KEYA record contains the new DNT and has a value in the BEGIN DATE field of 200206. The END DATE field for the second KEYA record must contain a future date.

If more than one set of verification parameters had already been in use, there would have been additional KEYA records and the BEGIN DATE field for the first KEYA record would contain the date that its verification parameters became effective.

If beginning and ending dates are unnecessary, the BEGIN DATE and END DATE fields should be set to asterisks (*).

END DATE — The last date in the range of card expiration dates to which this KEYA record applies. The date in this field is expressed in YYYYMM format and is compared to the expiration date of the card being used to initiate the transaction. Refer to the BEGIN DATE field description for an example of how this field is used. Valid values are as follows:

- YYYYMM = Data in this KEYA record is valid for cards with expiration dates that are less than or equal to this date but equal to or greater than the date in the BEGIN DATE field.
- ***** = Data in this KEYA record is valid regardless of the card expiration date. The BEGIN DATE field also must contain asterisks.

Note: 000000 is not a valid value in this field.

RECORD TYPE — Specifies the type of verification information contained in this record. A separate KEYA screen contains the information for each verification type. Because this KEYA record is for the Diebold PIN verification method, this field must be set to a value of 02.

KEYA Screen 3

KEYA screen 3 contains Diebold PIN verification information. The following field must be completed when using the Diebold PIN verification method.

NUMBER TABLE RELATIVE LOCATION — Specifies the relative location of the Diebold number table (DNT) in the security device. If PIN verification is to be performed by a hardware module, this field must contain the relative location of the table within the module.

When the security device contains one DNT, the value entered in this field is 1. When the security device contains multiple DNTs, the relative location of the DNT used with this KEYA record is entered in this field.

ALGO NUMBER — The algorithm number used for card verification.

- If a single algorithm number is to be used for verification on all of the cards for this institution or card prefix (i.e., the ALGO NUMBER LOC field on IDF screen 2 or CPF screen 2 is set to 1), that single algorithm number must be specified here. Valid values are hexadecimal characters 00 through 99. (The first character allows values of 0 through 9; the second character allows values of 0 through 9 or A through F, up to the allowed limit of 99.)
- If algorithm numbers are to be stored in the magnetic stripe data of the card, this field can be allowed to default to zero.

CLEAR KEY — The clear key used for encrypting the Diebold number table (DNT).

- If Diebold PIN verification is to be performed by an Atalla Network Security Processor (NSP), the DNT must be passed to it in encrypted form. This field must contain the clear key used by BASE24-atm to encrypt the DNT. This key must be 16 hexadecimal characters. Valid values for each character in this field are 0 through 9 and A through F. For protection in the database, this key is scrambled for storage by the KEYA file server process.
- If the Thales e-Security Host Security Modules (HSMs) are being used, this field is not required and must be set to zeros. The DNT is encrypted by the security module before it is uploaded to the BASE24 system. It is stored in encrypted form and sent back to the HSM in encrypted form as well.

ENCRYPTED KEY — The encrypted version of the key used for encrypting the Diebold number table (DNT).

- If Diebold PIN verification is to be performed by an NSP, the DNT must be passed to it in encrypted form. This field must contain the encrypted version of the key in the CLEAR KEY field.

This key must be encrypted by or for the specific security module being used, such that the module is able to decrypt the key when it is provided by BASE24. Once the key is decrypted, the module uses the clear version for decrypting the DNT. This value must be 16 hexadecimal characters.

- If the Thales e-Security HSM is used because the DNT, if present, is already encrypted by a local master key (LMK) pair, this field is not required.

DIEBOLD TABLE — The DNT to be used by BASE24. The table must consist of 256 hexadecimal number pairs. For protection in the database, this table is scrambled for storage by the KEYA file server process.

The DNT should be in the clear for Atalla security modules, encrypted for Thales e-Security security modules, or contain zeros. Refer to the topic “Hardware Module Requirements” earlier in this section for additional information.

Card Prefix File (CPF) Settings

The following fields must be configured in the Card Prefix File (CPF) for each prefix processed for the institution using the Diebold PIN verification method. Refer to the ***BASE24 Base Files Maintenance Manual*** for illustrations of CPF screens 1, 2, and 7.

CPF Screen 1

CPF screen 1 contains general information about card prefixes. The following fields are required when using the Diebold PIN verification method.

POFST/PVV — Identifies the relative position of the PIN offset in the magnetic stripe data.

- If PIN offsets are to be stored in the magnetic stripe data of the card (i.e., the POFST/PVV LOC field on IDF or CPF screen 2 is set to a value of 1), this value identifies the relative position of the PIN offset in the magnetic stripe data and must be specified for each CPF record.
- If PIN offsets are not to be stored in the magnetic stripe data of the card, this field can be allowed to default to zero.

ALGO #/PVKI — Identifies the relative position of the algorithm number in the magnetic stripe data.

- If algorithm numbers are to be stored in the magnetic stripe data of the card (i.e., the ALGO NUMBER LOC field on IDF or CPF screen 2 is set to a value of 2), this value identifies the relative position of the algorithm number in the magnetic stripe data and must be specified for each CPF record.
- If algorithm numbers are not to be stored in the magnetic stripe data of the card, this field can be allowed to default to zero.

CPF Screen 2

CPF screen 2 contains transaction security information for the card prefix. The following fields must be completed when using the Diebold PIN verification method.

Note: All but one of the PIN verification fields appearing on CPF screen 2 also appear on IDF screen 2, allowing this information to be defined at the institution level or at the card prefix level. The FIID field on IDF screen 2 serves the same purpose as the PIN VERIFICATION KEYA GROUP field on CPF screen 2.

PIN VERIFICATION KEYA GROUP — The value used to select the proper Key Authorization File (KEYA) record when verifying PINs for cards with this prefix. The value in this field is matched with the value in the GRP field on screen 1 of the KEYA record being used with this card prefix.

PIN CHECK TYPE — Specifies the code indicating the cardholder PIN verification method to be used. This field must be set to a value of 02 to specify Diebold PIN verification.

MAX PIN TRIES — The number of times an incorrect PIN can be entered before the action specified in the BAD PIN ACTION field is taken.

BAD PIN ACTION — Specifies the action to be taken by the device when it receives a PIN tries exceeded response code.

- If the card is to be returned to the cardholder, this field can be set to a value of 0. An ATM returns the card and a POS or teller device displays a message instructing the operator to return the card to the customer.

- If the card is to be retained, this field can be set to a value of 1. A POS or teller device displays a message instructing the operator to keep the card instead of returning it to the customer. An ATM retains the card, if the ATM has card retention functionality. This option is not valid for ATMs where the card is swiped instead of taken inside the machine to be read.

CARDHOLDER PIN SELECT — Specifies a code used only by BASE24-atm that identifies whether cardholders are allowed to select their PINs the first time they use their cards. Valid values are as follows:

Y = Yes, cardholders are allowed to select their own PINs at ATMs.

N = No, PINs are assigned to the cardholders by the institution.

Cardholder PIN selection is not the same as the PIN change transaction. Cardholder PIN selection permits cardholders to establish a PIN when they first use their cards. A PIN change transaction permits a cardholder to change an existing PIN.

To use this option, PIN offsets must be stored in the CAF (i.e., the POFST/PVV LOC field must be set to a value of 2), PIN verification must be performed by BASE24 in software, and the PIN offset in the CAF must be left blank.

ALGO NUMBER LOC — Specifies where the algorithm number is to be found, on the card or in the KEYA. Valid values are as follows:

1 = Use the algorithm number in the KEYA.

2 = Use the algorithm number in the magnetic stripe data of the card.

CHECK IF HOST ONLINE PIN — Controls whether BASE24 verifies the PIN before sending a card-initiated transaction to the host.

- If BASE24 is to verify the PIN, this field must be set to a value of Y.
- If BASE24 is not to verify the PIN, this field must be set to a value of N.

When performing a PIN change transaction with authorization level 3 (online/offline), BASE24-atm verifies the existing PIN without checking the value in this field if the HOST PIN CHANGE OPTION field on IDF screen 13 is set to a value of 0 (do not send to host).

POFST/PVV LOC — Specifies whether a PIN offset other than 0000 is to be used and, if so, where the PIN offset is to be found, on the card or in the CAF. Valid values are as follows:

- 0 = Use a PIN offset of 0000.
- 1 = Use the PIN offset in the magnetic stripe data of the card. With this option, BASE24-teller declines the transaction if the magnetic stripe is not read electronically.
- 2 = Use the PIN offset in the CAF.

PIN TRIES RESET OPTION — A code specifying when the PIN tries counter is reset and whether the bad PIN action is taken when the bad PIN tries limit is exceeded. Refer to the topic “Maximum PIN Retries” within the BASE24-atm, BASE24-pos, or BASE24-teller PIN verification discussions in section 8 for valid values.

CPF Screen 7

CPF screen 7 displays flags indicating how BASE24-pos handles preauthorization holds, chargebacks, representments, PIN processing, and address verification, as well as default values for combination card and account types and the maximum number of refunds allowed by BASE24-pos. The following field must be completed when using the Diebold PIN verification method.

PIN PROCESSING FLAG — Identifies whether PINs are required with transactions from POS devices.

- If a PIN is not required to complete a transaction, this field must be set to a value of 0.
- If a PIN is required with every transaction and transactions from terminals without PIN capabilities are declined, this field must be set to a value of 1.
- If PINs are required with transactions from terminals with PIN capabilities and transactions from attended terminals without PIN capabilities can be processed without PINs, this field must be set to a value of 2.

Regardless of this setting, BASE24-pos attempts to verify any PIN it receives from a POS device.

Cardholder Authorization File (CAF) Settings

The following fields must be configured on CAF screen 1 for each cardholder processed for the institution using the Diebold PIN verification method. Refer to the *BASE24 Base Files Maintenance Manual* for an illustration of CAF screen 1.

CARD STATUS — Specifies the code identifying the status of the card. This field must be set to a value of 0 (issued, but not active) or 1 (open) if cardholders are permitted to select their PINs on the first BASE24-atm transaction. All other card status values indicate conditions where unsupervised PIN selection is not advised.

POFST/PVV — The PIN offset. Each institution specifies the settings for PIN verification in the IDF or the CPF. The settings are as follows:

- If PIN offsets are to be stored in the CAF (i.e., the POFST/PVV LOC field on IDF or CPF screen 2 is set to 2), a PIN offset must be placed in this field for each CAF record. If one zero is entered in this field, at least four zeros must be entered. Although spaces are allowed in this field and spaces can trail an entry, spaces cannot be entered before a numeric entry or be embedded in an entry.
- If cardholders are allowed to select their own PINs, this field must be left blank in any CAF record for which a cardholder has yet to select a PIN. BASE24 then fills in the PIN offset when each cardholder first uses the card.
- If PIN offsets are not to be stored in the CAF, this field can be allowed to default to blanks.

BASE24-atm Terminal Data Files Settings

For installations that include BASE24-atm, the following fields must be configured on BASE24-atm Terminal Data files (ATD) files maintenance screens for each ATM owned by the institution. Refer to the *BASE24-atm Files Maintenance Manual* for illustrations of ATD screens 3 and 9.

ATD Screen 3

ATD screen 3 determines the types of special transactions allowed for cardholders at a particular terminal. The transaction classes listed can vary depending on parameters set up at configuration, although they should include all special transaction classes allowed at the terminal. The following field must be completed

when using the Diebold PIN verification method with all device types except for the Diebold 10XX/478X or the NCR 5XXX. For these two device types, allowed transactions for both on-us and not-on-us cardholders (including the PIN change transaction) are configured using the ACQUIRER TXN PROFILE field on ATD screen 2 and Acquirer Processing Code File (APCF) screen 2.

PIN CHANGE — Controls whether the PIN change transaction is included in the set of transactions that another institution's cardholders can perform at this terminal. Valid values are as follows:

- 0 = Not allowed
- 1 = Allowed within the country
- 2 = Allowed within the state
- 3 = Allowed nationally
- 4 = Allowed internationally

ATD Screen 9

ATD screen 9 displays device-specific security, PIN, and message authentication information. The device information displayed is determined by the value in the DEVICE TYPE field on ATD screen 1. The following field is required when using the Diebold PIN verification method.

LOCAL PIN VERIFY — Specifies whether local PIN verification is allowed for this terminal.

- If local PIN verification is allowed for all transactions at the terminal, this field must be set to a value of 0.
- If local PIN verification is not allowed at the terminal, this field must be set to a value of 1.
- If local PIN verification is allowed only for on-us transactions at the terminal, this field must be set to a value of 2. An on-us transaction is one where the same institution issues the card and owns the terminal.

Acquirer Processing Code File (APCF) Settings

The fields on APCF screen 2 specify whether a transaction is allowed for a specific acquirer transaction profile. The PIN Change transaction must be allowed in the acquirer transaction profile configured for ATMs from which it can be entered. This screen is used to establish acquirer transaction profiles for Diebold 10XX/478X and NCR 5XXX ATMs.

Issuer Processing Code File (IPCF) Settings

The fields on IPCF screen 2 specify whether a transaction is allowed for a specific issuer transaction profile. The PIN Change transaction must be allowed in the issuer transaction profiles configured for the institution in the IDF, the cardholder's account in the CAF, or the cardholder's account prefix in the CPF.

BASE24-pos Terminal Data Files Settings

No fields need to be configured in the BASE24-pos Terminal Data files. POS devices do not perform local PIN verification or support PIN change transactions.

Teller Terminal Data File (TTDF) Settings

For installations that include BASE24-teller, the following field must be configured in the Teller Terminal Data File (TTDF) for each teller terminal owned by the institution. Refer to the ***BASE24-teller Files Maintenance Manual*** for an illustration of TTDF screen 2.

CONTROLLER VALIDATE PIN — Specifies whether local PIN verification is allowed for this terminal.

- If PINs can be verified at the controller before being sent to BASE24-teller, this field must be set to a value of Y.
- If PINs cannot be verified at the controller before being sent to BASE24-teller, this field must be set to a value of N.

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11: Identikey PIN Verification Support

This section describes BASE24 support of the Identikey PIN verification method, including how the BASE24 database must be set up to allow BASE24 to verify PINs under this method of verification.

BASE24 only allows for implementation of this type of PIN verification using Atalla hardware modules.

Identikey PIN Verification

Identikey PIN verification is based on the use of the Atalla Identikey algorithm, which is proprietary to Atalla. Therefore, BASE24 does not support this type of PIN verification in software.

Under this verification type, at card-issuing time, the card issuer generates a PIN verification number for each card according to the specifications provided by Atalla. This PIN verification number is then placed on the magnetic stripe of the card or in the BASE24 database.

To verify a cardholder's PIN in a transaction, BASE24 passes this PIN verification number to an Atalla module, along with the PIN entered by the cardholder and other parameters set by the institution. The Atalla module then uses this information to verify the PIN and notify BASE24 of the outcome.

PIN Verification Inputs

Identikey PIN verification requires the following information as input. Note that some of these items change from cardholder to cardholder, while others are constant for the institution or card prefix.

- Primary account number (PAN)
- PIN
- Bank ID
- Compare indicator
- PIN verification number (PVN)

These input items are described separately in this section. They are discussed in the approximate order that they come into play in the PIN verification routine.

Primary Account Number (PAN)

A primary account number (PAN) is the card number used to identify each card. It must be carried on the magnetic stripe of the card, and a portion of it is used as input for Identikey PIN verification.

BASE24 must derive the portion of the PAN to be sent as input to the Atalla module. To do this, it requires two other values: an offset and a length.

The offset denotes the starting point in the PAN for extracting data, and the length indicates the number of characters to take from the PAN. For example, if the PAN offset is 3 and the length is 10, BASE24 skips to position 4 of the PAN and, starting with that position, picks up 10 consecutive characters of the PAN.

The offset and length must be specified in the PARTIAL PAN OFFSET and PARTIAL PAN LENGTH fields respectively on Key Authorization File (KEYA) screen 4.

PIN

In Identikey PIN verification, the customer-entered PIN must be sent to the Atalla module along with the other information for PIN verification.

If PINs are to be verified by an Atalla Network Security Processor (NSP), they can be in the clear or encrypted. If they are presented to the module in encrypted form, the key used to encrypt them also must be presented.

BASE24-atm allows for cardholders to change their existing PINs. The PIN verification number (PVN) must already exist in the CAF before this transaction is performed. Only Atalla Network Security Processors (NSPs) support this transaction. Upon receiving a PIN change transaction, the NSP computes an appropriate PVN based on the PIN entered by the cardholder and uses it to replace the existing PVN in the CAF. From that point on, the new PIN is the only one allowed, until the cardholder changes his or her PIN again. BASE24-pos and BASE24-teller do not allow PIN change transactions.

Cardholders cannot select their own PINs as part of the first transaction with Identikey PIN verification.

Bank ID

A Bank ID is a two-, six-, or eight-digit value associated with the card issuer for use by the Identikey algorithm. It can be a two-digit backward index, a six-digit ISO number, or an eight-digit routing and transit number.

The use of this value is prescribed by the Identikey algorithm. BASE24 passes it to the hardware module. In order for BASE24 to pass it to the module, it must be specified in the KEYA. It is specified in the INSTITUTION IDENTIFIER field on KEYA screen 4.

Compare Indicator

The compare indicator is a value indicating which portion of the Identikkey number derived by the hardware module should be compared to the cardholder's PIN verification number.

Like the Bank ID, the function of the compare indicator is prescribed by the Identikkey algorithm. BASE24 gets the compare indicator value from the KEYA and passes it to the module. Therefore, the value must be specified in the COMPARE INDICATOR field on KEYA screen 4.

PIN Verification Number (PVN)

A PIN verification number (PVN) is a computed value, unique to each card, created using a portion of the PAN, the cardholder's PIN, and a set of other values prescribed by the Identikkey algorithm.

BASE24 passes this value to the hardware module.

PVNs can be stored either in the magnetic stripe data of the card or in the Cardholder Authorization File (CAF) on the BASE24 system. This is specified using the POFST/PVV LOC field on Institution Definition File (IDF) screen 2 or Card Prefix File (CPF) screen 2, depending on whether PIN verification is being controlled at the institution level or the card prefix level.

If the PVN is to be carried in the magnetic stripe data of the card, the location of the value on the magnetic stripe must be specified using the POFST/PVV field on CPF screen 1. If the PVN is to be carried in the CAF, it must be placed in the POFST/PVV field on CAF screen 1.

PVNs can be four, six, or eight digits in length. If they are stored on the cards, BASE24 assumes that they are four digits in length and picks up this number of digits from the magnetic stripe data. However, if PVNs are stored in the CAF, BASE24 uses all of the digits placed in the CAF.

Note: When the PVN is carried in the magnetic stripe data of the card, BASE24-teller declines transactions if the magnetic stripe is not read electronically.

Setting Up Identikey PIN Verification

To support Identikey PIN verification, BASE24 requires certain settings in its database files. These file settings, in turn, require various decisions by an institution on how it plans to issue its cards and how it wishes verification to be handled.

The remainder of this section discusses BASE24-related requirements for Identikey PIN verification. The discussion is divided into three parts:

- The BASE24 requirements placed on the cards issued by the institution
- The hardware-specific requirements associated with this verification method
- The BASE24 database fields that must be set up to perform this type of PIN verification

Card Requirements

If PVNs are to be maintained on the cards, the position of the PVN on the magnetic stripe must be made available to BASE24. This position is specified in the Card Prefix File. PVN locations can vary by card prefix; however, the location must be consistent for all cards carrying the same prefix.

Also, if PVNs are to be maintained on the card, BASE24 assumes that they are four digits in length.

Note: PVNs cannot be carried on the cards if cardholders are to be allowed to use PIN change transactions to change their PINs.

Hardware Module Requirements

Identikey PIN verification can be performed by Atalla Network Security Processors supported by BASE24.

For instructions on configuring these devices, refer to section 17 and appendix B.

Institution Definition File (IDF) Settings

The following fields must be configured in the Institution Definition File (IDF) for each institution using the Identikey PIN verification method. Refer to the ***BASE24 Base Files Maintenance Manual*** for illustrations of IDF screens 2, 9, and 13.

IDF Screen 2

IDF screen 2 enables an institution to set up parameters for its cardholders or consumers. The following fields must be completed when using the Identikey PIN verification method.

Note: All PIN verification fields appearing on IDF screen 2 also appear on CPF screen 2, allowing this information to be defined at the institution level or at the card prefix level.

CHECK IF HOST ONLINE PIN — Controls whether BASE24 verifies the PIN before sending a card-initiated transaction to the host.

- If BASE24 is to verify the PIN, this field must be set to a value of Y.
- If BASE24 is not to verify the PIN, this field must be set to a value of N.

When performing a PIN change transaction with authorization level 3 (online/offline), BASE24-atm verifies the existing PIN without checking the value in this field if the HOST PIN CHANGE OPTION field on IDF screen 13 is set to a value of 0 (do not send to host).

MAX PIN TRIES — Specifies the number of times an incorrect PIN can be entered before the action specified in the BAD PIN ACTION field is taken.

BAD PIN ACTION — Specifies the action to be taken by the device when it receives a PIN tries exceeded response code.

- If the card is to be returned to the cardholder, this field can be set to a value of 0. An ATM returns the card and a POS or teller device displays a message instructing the operator to return the card to the customer.

- If the card is to be retained, this field can be set to a value of 1. A POS or teller device displays a message instructing the operator to keep the card instead of returning it to the customer. An ATM retains the card, if the ATM has card retention functionality. This option is not valid for ATMs where the card is swiped instead of taken inside the machine to be read.

POFST/PVV LOC — Specifies where the PVN is to be found, on the card or in the CAF. Valid values are as follows:

- 1 = Use the PVN in the magnetic stripe data of the card. With this option, BASE24-teller declines the transaction if the magnetic stripe is not read electronically.
- 2 = Use the PVN in the CAF.

PIN CHECK TYPE — Specifies a code indicating the PIN verification method used by the institution. This field must be set to a value of 03 to specify Identikey PIN verification.

ALGO NUMBER LOC — Specifies the code indicating the location of the algorithm number. Currently, the algorithm number is required only for the Diebold PIN verification method; therefore, this field must be set to zero because this type of PIN verification does not use algorithm numbers.

CARDHOLDER PIN SELECT — Specifies a code used by only BASE24-atm that identifies whether cardholders are allowed to select their PIN the first time they use their cards. Valid values are as follows:

- Y = Yes, cardholders are allowed to select their own PINs at ATMs.
- N = No, PINs are assigned to the cardholders by the institution.

Cardholder PIN selection is not the same as the PIN change transaction. Cardholder PIN selection permits cardholders to establish a PIN when they first use their cards. A PIN change transaction permits a cardholder to change an existing PIN.

PIN TRIES RESET OPTION — A code specifying when the PIN tries counter is reset and whether the bad PIN action is taken when the bad PIN tries limit is exceeded. Refer to the topic “Maximum PIN Retries” within the BASE24-atm, BASE24-pos, or BASE24-teller PIN verification discussions in section 8 for valid values.

IDF Screen 9

IDF screen 9 enables institutions to specify acquirer and issuer transaction profiles.

ACQUIRER TXN PROFILE — Controls whether the PIN change transaction is included in the set of transactions that the terminal-owning institution allows. The value in this field can be overridden at the terminal level in the BASE24-atm Terminal Data files.

ISSUER TXN PROFILE — Controls whether the PIN change transaction is included in the set of transactions that the card issuing institution allows for its cardholders. This profile can be overridden at the card prefix level in the Card Prefix File (CPF) or at the cardholder level in the Cardholder Authorization File (CAF).

IDF Screen 13

IDF screen 13 enables institutions to set their BASE24-atm processing control parameters. The following field must be completed when using the Identikey PIN verification method.

HOST PIN CHANGE OPTION — Determines where the new PVN is computed for a PIN change transaction with an online/offline authorization level when BASE24-atm is prescreening PINs or standing in for the host.

- If BASE24-atm prescreens the existing PIN and computes a new PVN for a PIN change transaction, this field must be set to a value of 0.
- If BASE24-atm verifies the existing PIN and computes a new PVN for a PIN change transaction only when the host is unavailable, this field must be set to a value of 1.
- If BASE24-atm declines a PIN change transaction when the host is unavailable, this field must be set to a value of 2.

Key Authorization File (KEYA) Settings

The following fields must be configured in the Key Authorization File (KEYA) for each FIID or CPF group using the Identikey PIN verification method. Refer to the *BASE24 Base Files Maintenance Manual* for illustrations of all KEYA screens.

KEYA Screen 1

KEYA screen 1 allows users to identify the KEYA record to be accessed. The following fields must be completed when using the Identikey PIN verification method.

GRP — The code that is used in the IDF or CPF to identify this record.

- If the PIN CHECK TYPE field on IDF screen 2 contains a value of 01, 02, 03, or 04 (verify PINs) and the PIN CHECK TYPE field on CPF screen 2 for one or more of the institution's card prefixes contains a value of 99 (use the IDF for PIN verification), a KEYA record must exist with the institution's FIID in this field.
- If the PIN CHECK TYPE field on CPF screen 2 contains a value of 01, 02, 03, or 04 (verify PINs), a KEYA record must exist with an entry in this field that matches the entry in the PIN VERIFICATION KEYA GROUP field on CPF screen 2.

BEGIN DATE — The first date in the range of card expiration dates to which this KEYA record applies. The date in this field is expressed in YYYYMM format and is compared to the expiration date of the card used to initiate the transaction. Valid values are as follows:

- | | |
|--------|---|
| 000000 | = Data in this KEYA record is valid for cards with expiration dates that are less than or equal to the date specified in the END DATE field. |
| YYYYMM | = Data in this KEYA record is valid for cards with expiration dates that are equal to or greater than this date but less than or equal to the date specified in the END DATE field. |
| ***** | = Data in this KEYA record is valid regardless of the card expiration date. The END DATE field also must contain asterisks. |

The BEGIN DATE and END DATE fields permit a financial institution to maintain different sets of verification parameters based on the card expiration date. For example, an institution using the Identikey PIN verification method plans to change the PAN offset it uses to generate PINs or PIN offsets. All outstanding cards use the existing PAN offset. The first cards to use the new PAN offset will have an expiration date of June, 2002, which appears in YYYYMM format as 200206. The existing PAN offset would be effective for cards issued with expiration dates through May, 2002, which appears in YYYYMM format as 200205.

Two KEYA records are necessary. The first KEYA record contains the old PAN offset and has a value in the END DATE field of 200205. The BEGIN DATE field for the first KEYA record can be 000000 because all outstanding cards use the verification parameters it contains. The second KEYA record contains the new PAN offset and has a value in the BEGIN DATE field of 200206. The END DATE field for the second KEYA record must contain a future date.

If more than one set of verification parameters had already been in use, there would have been additional KEYA records and the BEGIN DATE field for the first KEYA record would contain the date that its verification parameters became effective.

If beginning and ending dates are unnecessary, the BEGIN DATE and END DATE fields should be set to asterisks (*).

END DATE — The last date in the range of card expiration dates to which this KEYA record applies. The date in this field is expressed in YYYYMM format and is compared to the expiration date of the card used to initiate the transaction. Refer to the BEGIN DATE field description for an example of how this field is used. Valid values are as follows:

YYYYMM = Data in this KEYA record is valid for cards with expiration dates that are less than or equal to this date but equal to or greater than the date in the BEGIN DATE field.

***** = Data in this KEYA record is valid regardless of the card expiration date. The BEGIN DATE field also must contain asterisks.

Note: 000000 is not a valid value in this field.

RECORD TYPE — Specifies the type of verification information contained in this record. A separate KEYA screen contains the information for each verification type. Because this KEYA record is for the Identikey PIN verification method, this field must be set to a value of 03.

KEYA Screen 4

KEYA screen 4 contains Identikey PIN verification information. The following fields must be completed when using the Identikey PIN verification method.

INSTITUTION IDENTIFIER — Contains the two-, six-, or eight-character bank ID used by the Identikey algorithm to identify the institution. If the bank ID is less than eight characters in length, it must be left-justified with no embedded spaces.

ID LENGTH — Specifies the length of the bank ID. BASE24 computes this value automatically based on the length of the data entered in the INSTITUTION IDENTIFIER field.

PARTIAL PAN OFFSET — The offset to the starting position in the PAN for extracting the partial PAN. Position 00 identifies the first digit in the PAN.

PARTIAL PAN LENGTH — The length of the data (i.e., number of digits) to be extracted from the PAN for the formation of the partial PAN. Valid values are 4 through 19.

COMPARE INDICATOR — The portion of the Identikey number to be compared to the cardholder's PVN. Valid values are as follows:

L = Left-most four digits of the Identikey number.

M = Middle four digits of the Identikey number.

R = Right-most four digits of the Identikey number.

The Identikey number is created by the hardware module during PIN verification. Cardholder PVNs can be four, six, or eight digits in length; however, this value is only used if PVNs are four digits in length.

Card Prefix File (CPF) Settings

The following fields must be configured in the Card Prefix File (CPF) for each prefix processed for the institution using the Identikey PIN verification method. Refer to the *BASE24 Base Files Maintenance Manual* for illustrations of CPF screens 1, 2, and 7.

CPF Screen 1

CPF screen 1 contains general information about card prefixes. The following fields are required when using the Identikey PIN verification method.

POFST/PVV — Identifies the relative position of the PVN on the magnetic stripe.

- If PVNs are to be stored on the magnetic stripe of the card (i.e., the POFST/PVV LOC field on IDF or CPF screen 2 is set to 1), this value identifies the relative position of the PVN on the magnetic stripe and must be specified for each CPF record.
- If PVNs are not to be stored on the magnetic stripe of the card, this field can be allowed to default to zero.

ALGO #/PVKI — The value defining the position of the PIN verification method algorithm number on the magnetic stripe. This field must be left to default to zeros because the Identikey PIN verification method does not use algorithm numbers or PIN verification key indicators.

CPF Screen 2

CPF screen 2 contains transaction security information for the card prefix. The following fields must be completed when using the Identikey PIN verification method.

Note: All but one of the PIN verification fields appearing on CPF screen 2 also appear on IDF screen 2, allowing this information to be defined at the institution level or at the card prefix level. The FIID field on IDF screen 2 serves the same purpose as the PIN VERIFICATION KEYA GROUP field on CPF screen 2.

PIN VERIFICATION KEYA GROUP — The value used to select the proper Key Authorization File (KEYA) record when verifying PINs for cards with this prefix. The value in this field is matched with the value in the GRP field on screen 1 of the KEYA record being used with this card prefix.

PIN CHECK TYPE — Specifies the code indicating the cardholder PIN verification method to be used. This field must be set to 03 to specify Identikey PIN verification.

MAX PIN TRIES — The number of times an incorrect PIN can be entered before the action specified in the BAD PIN ACTION field is taken.

BAD PIN ACTION — Specifies the action to be taken by the device when it receives a PIN tries exceeded response code.

- If the card is to be returned to the cardholder, this field can be set to a value of 0. An ATM returns the card and a POS or teller device displays a message instructing the operator to return the card to the customer.
- If the card is to be retained, this field can be set to a value of 1. A POS or teller device displays a message instructing the operator to keep the card instead of returning it to the customer. An ATM retains the card, if the ATM has card retention functionality. This option is not valid for ATMs where the card is swiped instead of taken inside the machine to be read.

CARDHOLDER PIN SELECT — Specifies a code used by only BASE24-atm that identifies whether cardholders are allowed to select their PINs the first time they use their cards. Valid values are as follows:

Y = Yes, cardholders are allowed to select their own PINs at ATMs.

N = No, PINs are assigned to the cardholders by the institution.

Cardholder PIN selection is not the same as the PIN change transaction. Cardholder PIN selection permits cardholders to establish a PIN when they first use their cards. A PIN change transaction permits a cardholder to change an existing PIN.

ALGO NUMBER LOC — Specifies the code indicating the location of the algorithm number. Currently, the algorithm number is required only for the Diebold PIN verification method; therefore, this field must be set to zero because this type of PIN verification does not use algorithm numbers.

CHECK IF HOST ONLINE PIN — Controls whether BASE24 verifies the PIN before sending a card-initiated transaction to the host.

- If BASE24 is to verify the PIN, this field must be set to a value of Y.
- If BASE24 is not to verify the PIN, this field must be set to a value of N.

When performing a PIN change transaction with authorization level 3 (online/offline), BASE24-atm verifies the existing PIN without checking the value in this field if the HOST PIN CHANGE OPTION field on IDF screen 13 is set to a value of 0 (do not send to host).

POFST/PVV LOC — Specifies where the PVN is to be found, on the card or in the CAF. Valid values are as follows:

- 1 = Use the PVN on the magnetic stripe of the card. With this option, BASE24-teller declines the transaction if the magnetic stripe is not read electronically.
- 2 = Use the PVN in the CAF.

PIN TRIES RESET OPTION — A code specifying when the PIN tries counter is reset and whether the bad PIN action is taken when the bad PIN tries limit is exceeded. Refer to the topic “Maximum PIN Retries” within the BASE24-atm, BASE24-pos, or BASE24-teller PIN verification discussions in section 8 for valid values.

CPF Screen 7

CPF screen 7 displays flags indicating how BASE24-pos handles preauthorization holds, chargebacks, representments, PIN processing, and address verification, as well as default values for combination card and account types and the maximum number of refunds allowed by BASE24-pos. The following field must be completed when using the Identikey PIN verification method.

PIN PROCESSING FLAG — Identifies whether PINs are required with transactions from POS devices.

- If a PIN is not required to complete a transaction, this field must be set to a value of 0.
- If a PIN is required with every transaction and transactions from terminals without PIN capabilities are declined, this field must be set to a value of 1.
- If PINs are required with transactions from terminals with PIN capabilities and transactions from attended terminals without PIN capabilities can be processed without PINs, this field must be set to a value of 2.

Regardless of this setting, BASE24-pos attempts to verify any PIN it receives from a POS device.

Cardholder Authorization File (CAF) Settings

The following field must be configured on CAF screen 1 for each cardholder processed for the institution using the Identikey PIN verification method. Refer to the *BASE24 Base Files Maintenance Manual* for an illustration of CAF screen 1.

POFST/PVV — Specifies the PIN verification number (PVN).

- If PVNs are to be stored in the CAF (i.e., the POFST/PVV LOC field on IDF or CPF screen 2 is set to a value of 2), a PVN must be placed in this field for each CAF record. If one zero is entered in this field, at least four zeros must be entered. Although spaces are allowed in this field and spaces can trail an entry, spaces cannot be entered before a numeric entry or be embedded in an entry.
- If PVNs are not to be stored in the CAF, this field can be allowed to default to blanks.

BASE24-atm Terminal Data Files Settings

For installations that include BASE24-atm, the following fields must be configured on BASE24-atm Terminal Data files (ATD) files maintenance screens for each ATM owned by the institution. Refer to the *BASE24-atm Files Maintenance Manual* for illustrations of ATD screens 3 and 9.

ATD Screen 3

ATD screen 3 determines the types of special transactions allowed for cardholders at a particular terminal. The transaction classes listed can vary depending on parameters set up at configuration, although they should include all special transaction classes allowed at the terminal. The following field must be completed when using the Atalla Identikey PIN verification method with all device types except for the Diebold 10XX/478X or the NCR 5XXX. For these two device types, allowed transactions for both on-us and not-on-us cardholders (including the PIN change transaction) are configured using the ACQUIRER TXN PROFILE field on ATD screen 2 and Acquirer Processing Code File (APCF) screen 2.

PIN CHANGE — Controls whether the PIN change transaction is included in the set of transactions that another institution's cardholders can perform at this terminal. Valid values are as follows:

- 0 = Not allowed
- 1 = Allowed within the country
- 2 = Allowed within the state
- 3 = Allowed nationally
- 4 = Allowed internationally

ATD Screen 9

ATD screen 9 displays device-specific security, PIN, and message authentication information. The device information displayed is determined by the value in the DEVICE TYPE field on ATD screen 1. The following field is required when using the Identikey PIN verification method.

LOCAL PIN VERIFY — Specifies whether local PIN verification is allowed for this terminal.

- If local PIN verification is allowed for all transactions at the terminal, this field must be set to a value of 0.
- If local PIN verification is not allowed at the terminal, this field must be set to a value of 1.
- If local PIN verification is allowed only for on-us transactions at the terminal, this field must be set to a value of 2. An on-us transaction is one where the same institution issues the card and owns the terminal.

Acquirer Processing Code File (APCF) Settings

The fields on this screen specify whether a transaction is allowed for a specific acquirer transaction profile. The PIN Change transaction must be allowed in the acquirer transaction profile configured for ATMs from which it can be entered. This screen is used to establish acquirer transaction profiles for Diebold 10XX/478X and NCR 5XXX ATMs.

Issuer Processing Code File (IPCF) Settings

The fields on this screen specify whether a transaction is allowed for a specific issuer transaction profile. The PIN Change transaction must be allowed in the issuer transaction profiles configured for the institution in the IDF, the cardholder's account in the CAF, or the cardholder's account prefix in the CPF.

BASE24-pos Terminal Data Files Settings

No fields need to be configured in the BASE24-pos Terminal Data files. POS devices do not perform local PIN verification or support PIN change transactions.

Teller Terminal Data File (TTDF) Settings

For installations that include BASE24-teller, the following field must be configured in the Teller Terminal Data File (TTDF) for each teller terminal owned by the institution. Refer to the *BASE24-teller Files Maintenance Manual* for an illustration of TTDF screen 2.

CONTROLLER VALIDATE PIN — Specifies whether local PIN verification is allowed for this terminal.

- If PINs can be verified at the controller before being sent to BASE24-teller, this field must be set to a value of Y.
- If PINs cannot be verified at the controller before being sent to BASE24-teller, this field must be set to a value of N.

12: Visa PVV PIN Verification Support

This section describes BASE24 support of the Visa PVV PIN verification method, including how the BASE24 database must be set up to allow BASE24 to verify PINs under this method of verification.

BASE24 allows for implementation of this PIN verification method in software or a security module.

Visa PVV PIN Verification

The Visa PVV PIN verification method is based on using a PIN verification value (PVV) for verifying a cardholder's PIN. Visa PVV PIN verification can be performed in software or using a hardware module linked to a processor on which the BASE24 system is running.

At card-issuing time, the card issuer generates a control PVV for each card using the standard DES algorithm, a pair of PIN verification keys, the PIN, and other card-specific data. This PVV is then placed on the magnetic stripe of the card or in the BASE24 database.

The following is a summary of the steps taken to verify a PIN using Visa PVV PIN verification in software or hardware.

The verification entity performs the following steps:

1. Creates a 16-digit value using 11 digits from the PAN, the 1-digit PIN verification key indicator, and 4 digits from the PIN.
2. Encrypts the 16-digit value using the DES algorithm and the first key of the PIN verification key (PVK) pair.
3. Performs a decrypt function on the result from step 2 using the DES algorithm and the second key of the PVK pair.
4. Performs an encrypt function on the result from step 3 using the DES algorithm and the first key of the PVK pair.
5. Extracts a 4-digit transaction PVV from the result from step 4.

To do this, the result is scanned—left to right—and the first 4 digits found with a value equal to or less than 9 are taken. The result from step 4 is in hexadecimal representation.

If less than 4 digits are found with a value equal to or less than 9, any additional values needed to complete the 4 digits are derived by rescanning the result from step 4, taking only those digits greater than 9 and subtracting 10 from each until the 4 digits have been obtained.

6. Compares the transaction PVV with the control PVV. If they match, the cardholder entered the PIN correctly. If they do not match, the cardholder entered the PIN incorrectly.

PIN Verification Inputs

Visa PVV PIN verification requires the following information as input. Note that some of these items change from cardholder to cardholder, while others are constant for the institution or card prefix.

- Primary account number (PAN)
- PIN
- PIN verification key (PVK) pair
- PIN verification key indicator (PVKI)
- PIN verification value (PVV)

These input items are described separately in this section. They are discussed in the approximate order that they come into play in the PIN verification routine.

Primary Account Number (PAN)

A primary account number (PAN) is the card number used to identify each card. It must be carried on the magnetic stripe of the card and is used as input to Visa PVV PIN verification.

Visa PVV PIN verification uses the right-most 11 digits of the cardholder's PAN, not including the check digit (last digit), as input to create the PIN verification value. An example is shown below:

PAN:	1 2 3 4	5 6 7 8	9 9 8 7	6 5 4 7
PAN DIGITS USED:		5 6 7 8	9 9 8 7	6 5 4

PIN

In Visa PVV PIN verification, the PIN entered by the cardholder is used in the computation of a transaction PVV.

If the PIN is four digits in length, the entire PIN is used in the computations. If the PIN is longer than four digits, only the left-most four digits are used in the computations.

For PIN verification performed by BASE24 in software, the PIN must be in the clear.

For PIN verification performed by a hardware module, the PIN must be passed to the module in encrypted form, along with the key used to encrypt it. The hardware module then decrypts the PIN for use in generating a transaction PVV.

BASE24-atm allows cardholders to change their existing PINs. The PIN verification value (PVV) must already exist in the CAF before this transaction is performed. Visa PVV PIN verification can be performed in hardware or software. Upon receiving a PIN change transaction, BASE24-atm or the hardware security module computes an appropriate PVV based on the PIN entered by the cardholder and uses it to replace the existing PVV in the CAF. From that point on, the new PIN is the only one allowed, until the cardholder changes his or her PIN again. BASE24-pos and BASE24-teller do not allow PIN change transactions.

Cardholders cannot select their own PINs as part of the first transaction with Visa PVV PIN verification.

PIN Verification Key (PVK) Pair

Two PIN verification keys are required in Visa PVV PIN verification. These keys, called the PVK pair, are used as input into the DES algorithm to alternately encrypt and decrypt transaction data in the computation of a PIN verification value. For information on encryption and decryption and the DES algorithm, refer to section 2.

An issuer can use the same PVK pair to generate PIN verification values for all of its cards; however, BASE24 allows for use of up to six different pairs. As with any DES-based operation, the security of the operation is dependent on keeping the keys confidential. Therefore, by using multiple sets of keys, the risk can be divided in case a pair of keys is compromised. In this situation, only those cards for which PIN verification values were created with the compromised keys need to be reissued.

The PVK pairs used to create the institution's PIN verification values must be specified in the Key Authorization File (KEYA).

If BASE24 performs PIN verification in software, clear versions of the keys must be entered in the CLEAR fields on KEYA screen 5.

If a hardware module performs PIN verification, encrypted versions of the keys must be entered in the ENCRYPTED fields on KEYA screen 5. In this case, these keys must be encrypted such that they can be decrypted by the hardware module.

PIN Verification Key Indicator (PVKI)

Because multiple pairs of PIN verification keys can be used, an indicator is required to specify the set of keys used to create the PIN verification value for each card.

This indicator, called the PIN verification key indicator (PVKI), is a one-digit number associated with a card identifying the pair of keys used to create the control PVV for that card. Since there are up to six different sets of keys possible, PVKI values can be from 1 to 6.

The PVKI can either be carried on the magnetic stripe of the card or in the Cardholder Authorization File (CAF). This is controlled by the POFST/PVV LOC field on Institution Definition File (IDF) screen 2 or Card Prefix File (CPF) screen 2, depending on whether PIN verification is being controlled at the institution level or the card prefix level.

If the PVKI is to be carried on the magnetic stripe of the card, the location of the value on the magnetic stripe must be specified using the ALGO #/PVKI field on Card Prefix File (CPF) screen 1.

If the PVKI is to be carried in the CAF, it must be placed in the left-most position of the POFST/PVV field on CAF screen 1.

Besides noting the PVK pair to be used, the PVKI is also used as part of the actual input to the DES algorithm. This input must be 16 characters in length and is composed of 11 digits from the PAN, the one-digit PVKI, and four digits from the entered PIN. An example is shown below:

PAN DIGITS:	5 6 7 8 9 9 8 7 6 5 4
PVKI:	2
PIN:	4 5 5 0
DES ALGORITHM INPUT:	5 6 7 8 9 9 8 7 6 5 4 2 4 5 5 0

Note: When the PVKI or PVV is carried on the magnetic stripe of the card, BASE24-teller declines transactions if the magnetic stripe is not read electronically.

PIN Verification Value (PVV)

A PIN verification value (PVV) is a computed four-digit value created using the PAN, the cardholder's PIN, and a specific set of encryption and decryption keys as input to the standard DES algorithm.

The basis for Visa PVV PIN verification is comparing a transaction PVV against a control PVV. The control PVV is created when the card is issued; the transaction PVV is created as the PIN is being verified. Thus, it is the control PVV that is required as input to Visa PVV PIN verification.

The control PVV can be stored either on the magnetic stripe of the card or in the CAF. This is specified using the POFST/PVV LOC field on IDF or CPF screen 2.

If the control PVV is to be carried on the magnetic stripe of the card, the location of the value on the magnetic stripe must be specified using the POFST/PVV field on CPF screen 1. If the control PVV is to be carried in the CAF, it must be placed in the POFST/PVV field on CAF screen 1.

Note: When the PVKI or PVV is carried on the magnetic stripe of the card, BASE24-teller declines transactions if the magnetic stripe is not read electronically.

Setting Up Visa PVV PIN Verification

To support Visa PVV PIN verification, BASE24 requires certain settings in its database files. These file settings, in turn, require various decisions by an institution on how it plans to issue its cards, how it wishes verification to be handled, and whether or not it plans to use a hardware module for performing verification.

The remainder of this section discusses BASE24-related requirements for Visa PVV PIN verification. The discussion is divided into three parts:

- The BASE24 requirements placed on the cards issued by the institution
- The hardware-specific requirements associated with this verification method
- The BASE24 database fields that must be set up to perform this type of PIN verification

Card Requirements

If PVVs and PVKIs are to be maintained on the cards, their positions on the magnetic stripe must be made available to BASE24. These positions are specified in the CPF. PVV and PVKI locations can vary by card prefix; however, the locations must be consistent for all cards carrying the same prefix. The PVV and PVKI for a prefix must both be on the card or both be in the CAF.

Note: Control PVVs cannot be carried on the cards if cardholders are allowed to use PIN change transactions to change their PINs.

Hardware Module Requirements

Visa PVV PIN verification can be performed by the following hardware modules supported by BASE24:

- Atalla Network Security Processor (NSP)
- Thales e-Security Host Security Module (HSM)

For instructions on configuring these devices, refer to section 17, appendix B, and appendix C.

Note: Visa PVV PIN verification using a Thales e-Security HSM requires that external processors use PIN/PAD PIN blocks or the ANSI X9.8 standard PIN/PAN PIN blocks in the messages sent to the BASE24 system. The Visa PVV PIN verification method uses the right-most digits of the PAN excluding the check digit (last digit). The ANSI X9.8 standard PIN/PAN PIN block uses the 12 right-most digits of the PAN excluding the check digit. The HSM returns a PIN block error if the external processors use a PIN/PAN PIN block other than the ANSI X9.8 standard.

Institution Definition File (IDF) Settings

The following fields must be configured in the Institution Definition File (IDF) for each institution using the Visa PVV PIN verification method. Refer to the ***BASE24 Base Files Maintenance Manual*** for illustrations of IDF screens 2, 9, and 13.

IDF Screen 2

IDF screen 2 enables an institution to set up parameters for its cardholders or consumers. The following fields must be completed when using the Visa PVV PIN verification method.

Note: All PIN verification fields appearing on IDF screen 2 also appear on CPF screen 2, allowing this information to be defined at the institution level or at the card prefix level.

CHECK IF HOST ONLINE PIN — Controls whether BASE24 verifies the PIN before sending a card-initiated transaction to the host.

- If BASE24 is to verify the PIN, this field must be set to a value of Y.
- If BASE24 is not to verify the PIN, this field must be set to a value of N.

When performing a PIN change transaction with authorization level 3 (online/offline), BASE24-atm verifies the existing PIN without checking the value in this field if the HOST PIN CHANGE OPTION field on IDF screen 13 is set to a value of 0 (do not send to host).

MAX PIN TRIES — Specifies the number of times an incorrect PIN can be entered before the action specified in the BAD PIN ACTION field is taken.

BAD PIN ACTION — Specifies the action to be taken by the device when it receives a PIN tries exceeded response code.

- If the card is to be returned to the cardholder, this field can be set to a value of 0. An ATM returns the card and a POS or teller device displays a message instructing the operator to return the card to the customer.
- If the card is to be retained, this field can be set to a value of 1. A POS or teller device displays a message instructing the operator to keep the card instead of returning it to the customer. An ATM retains the card, if the ATM has card retention functionality. This option is not valid for ATMs where the card is swiped instead of taken inside the machine to be read.

POFST/PVV LOC — Specifies where the control PVV and the PVKI are to be found, on the card or in the CAF. Valid values are as follows:

- 1 = Use the PVV and PVKI on the magnetic stripe of the card. With this option, BASE24-teller declines the transaction if the magnetic stripe is not read electronically.
- 2 = Use the PVV and PVKI in the CAF.

PIN CHECK TYPE — Specifies a code indicating the PIN verification method used by the institution. This field must be set to a value of 04 to specify Visa PVV PIN verification.

ALGO NUMBER LOC — Specifies the code indicating the location of the algorithm number. Currently, the algorithm number is required only for the Diebold PIN verification method; therefore, this field must be set to zero because this type of PIN verification does not use algorithm numbers.

CARDHOLDER PIN SELECT — Specifies a code used only by BASE24-atm that identifies whether cardholders are allowed to select their PINs the first time they use their cards. Valid values are as follows:

- Y = Yes, cardholders are allowed to select their own PINs at ATMs.
- N = No, PINs are assigned to the cardholders by the institution.

This field must be set to a value of N because BASE24-atm does not allow for cardholder-selected PINs under this method of PIN verification.

Cardholder PIN selection is not the same as the PIN change transaction. Cardholder PIN selection permits cardholders to establish a PIN when they first use their cards. A PIN change transaction permits a cardholder to change an existing PIN.

PIN TRIES RESET OPTION — A code specifying when the PIN tries counter is reset and whether the bad PIN action is taken when the bad PIN tries limit is exceeded. Refer to the topic “Maximum PIN Retries” within the BASE24-atm, BASE24-pos, or BASE24-teller PIN verification discussions in section 8 for valid values.

IDF Screen 9

IDF screen 9 enables institutions to specify acquirer and issuer transaction profiles.

ACQUIRER TXN PROFILE — Controls whether the PIN change transaction is included in the set of transactions that the terminal-owning institution allows. The value in this field can be overridden at the terminal level in the BASE24-atm Terminal Data files.

ISSUER TXN PROFILE — Controls whether the PIN change transaction is included in the set of transactions that the card issuing institution allows for its cardholders. This profile can be overridden at the card prefix level in the Card Prefix File (CPF) or at the cardholder level in the Cardholder Authorization File (CAF).

IDF Screen 13

IDF screen 13 enables institutions to set their BASE24-atm processing control parameters. The following field must be completed when using the Visa PVV PIN verification method.

HOST PIN CHANGE OPTION — Determines where the new control PVV is computed for a PIN change transaction with an online/offline authorization level when BASE24-atm is prescreening PINs or standing in for the host.

- If BASE24-atm prescreens the existing PIN and computes a new control PVV for a PIN change transaction, this field must be set to a value of 0.

- If BASE24-atm verifies the existing PIN and computes a new control PVV for a PIN change transaction only when the host is unavailable, this field must be set to a value of 1.
- If BASE24-atm declines a PIN change transaction when the host is unavailable, this field must be set to a value of 2.

Key Authorization File (KEYA) Settings

The following fields must be configured in the Key Authorization File (KEYA) for each FIID or CPF group using the Visa PVV PIN verification method. Refer to the *BASE24 Base Files Maintenance Manual* for illustrations of all KEYA screens.

KEYA Screen 1

KEYA screen 1 allows users to identify the KEYA record to be accessed. The following fields must be completed when using the Visa PVV PIN verification method.

GRP — The code that is used in the IDF or CPF to identify this record.

- If the PIN CHECK TYPE field on IDF screen 2 contains a value of 01, 02, 03, or 04 (verify PINs) and the PIN CHECK TYPE field on CPF screen 2 for one or more of the institution's card prefixes contains a value of 99 (use the IDF for PIN verification), a KEYA record must exist with the institution's FIID in this field.
- If the PIN CHECK TYPE field on CPF screen 2 contains a value of 01, 02, 03, or 04 (verify PINs), a KEYA record must exist with an entry in this field that matches the entry in the PIN VERIFICATION KEYA GROUP field on CPF screen 2.

BEGIN DATE — The first date in the range of card expiration dates to which this KEYA record applies. The date in this field is expressed in YYYYMM format and is compared to the expiration date of the card used to initiate the transaction. Valid values are as follows:

- 000000 = Data in this KEYA record is valid for cards with expiration dates that are less than or equal to the date specified in the END DATE field.
- YYYYMM = Data in this KEYA record is valid for cards with expiration dates that are equal to or greater than this date but less than or equal to the date specified in the END DATE field.
- ***** = Data in this KEYA record is valid regardless of the card expiration date. The END DATE field also must contain asterisks.

The BEGIN DATE and END DATE fields permit a financial institution to maintain different sets of verification parameters based on the card expiration date. For example, an institution using the Visa PVV PIN verification method plans to change the PIN verification key pairs (PVK pairs) it uses to generate PINs or PIN offsets. All outstanding cards use the existing PVK pairs. The first cards to use the new PVK pairs will have an expiration date of June, 2002, which appears in YYYYMM format as 200206. The existing PVK pairs would be effective for cards issued with expiration dates through May, 2002, which appears in YYYYMM format as 200205.

Two KEYA records are necessary. The first KEYA record contains the old PVK pairs and has a value in the END DATE field of 200205. The BEGIN DATE field for the first KEYA record can be 000000 because all outstanding cards use the verification parameters it contains. The second KEYA record contains the new PVK pairs and has a value in the BEGIN DATE field of 200206. The END DATE field for the second KEYA record must contain a future date.

If more than one set of verification parameters had already been in use, there would have been additional KEYA records and the BEGIN DATE field for the first KEYA record would contain the date that its verification parameters became effective.

If beginning and ending dates are unnecessary, the BEGIN DATE and END DATE fields should be set to asterisks (*).

END DATE — The last date in the range of card expiration dates to which this KEYA record applies. The date in this field is expressed in YYYYMM format and is compared to the expiration date of the card being used to initiate the transaction. Refer to the BEGIN DATE field description for an example of how this field is used. Valid values are as follows:

YYYYMM = Data in this KEYA record is valid for cards with expiration dates that are less than or equal to this date but equal to or greater than the date in the BEGIN DATE field.

***** = Data in this KEYA record is valid regardless of the card expiration date. The BEGIN DATE field also must contain asterisks.

Note: 000000 is not a valid value in this field.

RECORD TYPE — Specifies the type of verification information contained in this record. A separate KEYA screen contains the information for each verification type. This field must be set to a value of 04 because this KEYA record is for the Visa PVV PIN verification method.

KEYA Screen 5

KEYA screen 5 contains Visa PVV PIN verification information. The following fields must be completed when using the Visa PVV PIN verification method.

Visa PVV Keys

The following fields contain data for the Visa PVV PIN verification method. They are used only when the value in the Record Type field on KEYA screen 1 is set to a value of 04.

CLEAR — The clear PIN verification keys. At least one clear or encrypted PVK pair must contain a value other than 16 zeros. The fields are labeled 1 through 6. The PVKI contains a value in the range of 1 through 6 to identify which PVK pair should be used.

- If PIN verification is to be performed without a security module, these fields must contain at least one pair of clear PIN verification keys. Up to six pairs can be specified. Each key must be 16 hexadecimal characters. Valid values for each character in the fields making up the PVK pairs are 0 through 9 and A through F. For protection in the database, all of these keys are scrambled for storage by the KEYA file server process.

- If PIN verification is to be performed by a hardware module exclusively, these fields are not used.

ENCRYPT — The encrypted PIN verification keys. At least one clear or encrypted PVK pair must contain a value other than 16 zeros. The fields are labeled 1 through 6. The PVKI contains a value in the range of 1 through 6 to identify which PVK pair should be used.

- If PIN verification is to be performed by a hardware module, these fields must contain at least one pair of encrypted PIN verification keys. Up to six pairs can be specified. Each key must be encrypted by or for the specific security module being used, such that the module is able to decrypt the key when it is provided by BASE24. The encrypted keys entered on the screen must be 16 hexadecimal characters. Valid values for each character in the fields making up the PVK pairs are 0 through 9 and A through F.
- If PIN verification is to be performed by BASE24 software exclusively, these fields are not used.

CHECK DIGITS — The check digits corresponding to the value in the ENCRYPT field located directly above this field on the screen. Valid values for each character in the check digits fields are 0 through 9 and A through F. A value of 0000 in this field indicates check digits are not being used with the key in the ENCRYPT field.

Card Prefix File (CPF) Settings

The following fields must be configured in the Card Prefix File (CPF) for each prefix processed for the institution using the Visa PVV PIN verification method. Refer to the *BASE24 Base Files Maintenance Manual* for illustrations of CPF screens 1, 2, and 7.

CPF Screen 1

CPF screen 1 contains general information about card prefixes. The following fields are required when using the Visa PVV PIN verification method.

POFST/PVV — Identifies the relative position of the PVV on the magnetic stripe.

- If PVVs are to be stored on the magnetic stripe of the card (i.e., the POFST/PVV LOC field on IDF or CPF screen 2 is set to 1), this value identifies the relative position of the PVV on the magnetic stripe and must be specified for each CPF record.
- If PVVs are not to be stored on the magnetic stripe of the card, this field can be allowed to default to zero.

Note: When PVVs are carried on the magnetic stripe of the card, BASE24-teller declines transactions if the magnetic stripe is not read electronically.

ALGO #/PVKI — Identifies the relative position of the PVKI on the magnetic stripe.

- If PVKIs are to be stored on the magnetic stripe of the card (i.e., the POFST/PVV LOC field on IDF or CPF screen 2 is set to a value of 1), this value identifies the relative position of the PVKI on the magnetic stripe and must be specified for each CPF record.
- If PVKIs are not to be stored on the magnetic stripe of the card, this field can be allowed to default to zero.

Note: When PVKIs are carried on the magnetic stripe of the card, BASE24-teller declines transactions if the magnetic stripe is not read electronically.

CPF Screen 2

CPF screen 2 contains transaction security information for the card prefix. The following fields must be completed when using the Visa PVV PIN verification method.

Note: All but one of the PIN verification fields appearing on CPF screen 2 also appear on IDF screen 2, allowing this information to be defined at the institution level or at the card prefix level. The FIID field on IDF screen 2 serves the same purpose as the PIN VERIFICATION KEYA GROUP field on CPF screen 2.

PIN VERIFICATION KEYA GROUP — The value used to select the proper Key Authorization File (KEYA) record when verifying PINs for cards with this prefix. The value in this field is matched with the value in the GRP field on screen 1 of the KEYA record being used with this card prefix.

PIN CHECK TYPE — Specifies the code indicating the cardholder PIN verification method to be used. This field must be set to a value of 04 to specify Visa PVV PIN verification.

MAX PIN TRIES — The number of times an incorrect PIN can be entered before the action specified in the BAD PIN ACTION field is taken.

BAD PIN ACTION — Specifies the action to be taken by the device when it receives a PIN tries exceeded response code.

- If the card is to be returned to the cardholder, this field can be set to a value of 0. An ATM returns the card and a POS or teller device displays a message instructing the operator to return the card to the customer.
- If the card is to be retained, this field can be set to a value of 1. A POS or teller device displays a message instructing the operator to keep the card instead of returning it to the customer. An ATM retains the card, if the ATM has card retention functionality. This option is not valid for ATMs where the card is swiped instead of taken inside the machine to be read.

CARDHOLDER PIN SELECT — Specifies a code used only by BASE24-atm that identifies whether cardholders are allowed to select their PINs the first time they use their cards. Valid values are as follows:

Y = Yes, cardholders are allowed to select their own PINs at ATMs.

N = No, PINs are assigned to the cardholders by the institution.

Cardholder PIN selection is not the same as the PIN change transaction. Cardholder PIN selection permits cardholders to establish a PIN when they first use their cards. A PIN change transaction permits a cardholder to change an existing PIN.

ALGO NUMBER LOC — Specifies the code indicating the location of the algorithm number. Currently, the algorithm number is required only for the Diebold PIN verification method; therefore, this field must be set to zero because this type of PIN verification does not use algorithm numbers.

CHECK IF HOST ONLINE PIN — Controls whether BASE24 verifies the PIN before sending a card-initiated transaction to the host.

- If BASE24 is to verify the PIN, this field must be set to a value of Y.
- If BASE24 is not to verify the PIN, this field must be set to a value of N.

When performing a PIN change transaction with authorization level 3 (online/offline), BASE24-atm verifies the existing PIN without checking the value in this field if the HOST PIN CHANGE OPTION field on IDF screen 13 is set to a value of 0 (do not send to host).

POFST/PVV LOC — Specifies where the control PVV and PVKI are to be found, on the card or in the CAF. Valid values are as follows:

- 1 = Use the PVV and PVKI in the magnetic stripe data of the card. With this option, BASE24-teller declines the transaction if the magnetic stripe data is not read electronically.
- 2 = Use the PVV and PVKI in the CAF.

PIN TRIES RESET OPTION — A code specifying when the PIN tries counter is reset and whether the bad PIN action is taken when the bad PIN tries limit is exceeded. Refer to the topic “Maximum PIN Retries” within the BASE24-atm, BASE24-pos, or BASE24-teller PIN verification discussions in section 8 for valid values.

CPF Screen 7

CPF screen 7 displays flags indicating how BASE24-pos handles preauthorization holds, chargebacks, representments, PIN processing, and address verification, as well as default values for combination card and account types and the maximum number of refunds allowed by BASE24-pos. The following field must be completed when using the Visa PVV PIN verification method.

PIN PROCESSING FLAG — Identifies whether PINs are required with transactions from POS devices.

- If a PIN is not required to complete a transaction, this field must be set to a value of 0.
- If a PIN is required with every transaction and transactions from terminals without PIN capabilities are declined, this field must be set to a value of 1.
- If PINs are required with transactions from terminals with PIN capabilities and transactions from attended terminals without PIN capabilities can be processed without PINs, this field must be set to a value of 2.

Regardless of this setting, BASE24-pos attempts to verify any PIN it receives from a POS device.

Cardholder Authorization File (CAF) Settings

The following field must be configured on CAF screen 1 for each cardholder processed for the institution using the Visa PVV PIN verification method. Refer to the *BASE24 Base Files Maintenance Manual* for an illustration of CAF screen 1.

POFST/PVV — Contains the PVV and PVKI.

- If PVVs are to be stored in the CAF (i.e., the POFST/PVV LOC field on IDF or CPF screen 2 is set to a value of 2), a PVV and PVKI must be placed in this field for each CAF record. The PVKI must be the left-most digit of the field, followed by the four-digit PVV. Although spaces are allowed in this field and spaces can trail an entry, spaces cannot be entered before a numeric entry or be embedded in an entry.
- If PVVs are not to be stored in the CAF, this field can be allowed to default to blanks.

BASE24-atm Terminal Data Files Settings

For installations that include BASE24-atm, the following fields must be configured on the BASE24-atm Terminal Data files (ATD) files maintenance screens for each ATM owned by the institution. Refer to the *BASE24-atm Files Maintenance Manual* for illustrations of ATD screens 3 and 9.

ATD Screen 3

ATD screen 3 determines the types of special transactions allowed for cardholders at a particular terminal. The transaction classes listed can vary depending on parameters set up at configuration, although they should include all special transaction classes allowed at the terminal. The following field must be completed when using the Atalla Identikey PIN verification method with all device types except for the Diebold 10XX/478X or the NCR 5XXX. For these two device types, allowed transactions for both on-us and not-on-us cardholders (including the PIN change transaction) are configured using the ACQUIRER TXN PROFILE field on ATD screen 2 and Acquirer Processing Code File (APCF) screen 2.

PIN CHANGE — Controls whether the PIN change transaction is included in the set of transactions that another institution's cardholders can perform at this terminal. Valid values are as follows:

- 0 = Not allowed
- 1 = Allowed within the country
- 2 = Allowed within the state
- 3 = Allowed nationally
- 4 = Allowed internationally

ATD Screen 9

ATD screen 9 displays device-specific security, PIN, and message authentication information. The device information displayed is determined by the value in the DEVICE TYPE field on ATD screen 1. The following field is required when using the Visa PVV PIN verification method.

LOCAL PIN VERIFY — Specifies whether local PIN verification is allowed for this terminal.

- If local PIN verification is allowed for all transactions at the terminal, this field must be set to a value of 0.
- If local PIN verification is not allowed at the terminal, this field must be set to a value of 1.
- If local PIN verification is allowed only for on-us transactions at the terminal. An on-us transaction is one where the same institution issues the card and owns the terminal, this field must be set to a value of 2.

Acquirer Processing Code File (APCF) Settings

The fields on APCF screen 2 specify whether a transaction is allowed for a specific acquirer transaction profile. The PIN Change transaction must be allowed in the acquirer transaction profile configured for ATMs from which it can be entered. This screen is used to establish acquirer transaction profiles for Diebold 10XX/478X and NCR 5XXX ATMs.

Issuer Processing Code File (IPCF) Settings

The fields on IPCF screen 2 specify whether a transaction is allowed for a specific issuer transaction profile. The PIN Change transaction must be allowed in the issuer transaction profiles configured for the institution in the IDF, the cardholder's account in the CAF, or the cardholder's account prefix in the CPF.

BASE24-pos Terminal Data Files Settings

No fields need to be configured in the BASE24-pos Terminal Data files. POS devices do not perform local PIN verification or support PIN change transactions.

Teller Terminal Data File (TTDF) Settings

For installations that include BASE24-teller, the following field must be configured in the Teller Terminal Data File (TTDF) for each teller terminal owned by the institution. Refer to the *BASE24-teller Files Maintenance Manual* for an illustration of TTDF screen 2.

CONTROLLER VALIDATE PIN — Specifies whether local PIN verification is allowed for this terminal.

- If PINs can be verified at the controller before being sent to BASE24-teller, this field must be set to a value of Y.
- If PINs cannot be verified at the controller before being sent to BASE24-teller, this field must be set to a value of N.

13: BASE24 Card Verification Support

This section describes the card verification support provided by BASE24-atm, BASE24-pos, and BASE24-teller.

Card verification processing performed by BASE24-atm, BASE24-pos, and BASE24-teller is very similar. As a result, the information for the three products is presented together, with the discrepancies noted when necessary.

Note: The support of dynamic card verification for contactless magnetic stripe cards, or proximity cards, requires the use of the Transaction Security Services process and is described in the *BASE24 Transaction Security Services Processing Guide*.

BASE24 Card Verification

Card verification provides a means of confirming the validity of the card presented by a customer. The magnetic stripe data on a card can contain a value BASE24 calls the card verification digits (CVD). For MasterCard and Visa cards, the card issuer generates the CVD from two card verification keys plus the entire primary account number, card expiration date, and service code from the card. For American Express cards, the card issuer generates the CVD from a single double-length card verification key plus part of the primary account number and the card expiration date from the card. Thus, the card issuer only uses part of the primary account number and does not use the service code from the card to generate the CVD for American Express cards.

Verifying a card, whether in software or in a security module, involves using the same card verification keys or key and the magnetic stripe information to compute another CVD, then comparing the computed CVD with the CVD contained in the magnetic stripe data. The card is considered valid if the two CVDs match because the same magnetic stripe information and card verification keys are required to obtain the same CVD.

Note: When the iCVD is obtained from a chip card, card verification processing is the same as for a magnetic stripe card except that you can specify whether the service code from the card or a service code specified in the CV DATA field on CPF screen 11 is used to compute the iCVD compared with the iCVD from the card. Additional fields on CPF screen 11 specify the conditions for which the iCVD is verified for EMV transactions. A value of 05 (Integrated Circuit Card) in the PT-SRV-ENTRY-MDE field of the BASE24-pos Standard Internal Message (PSTM) indicates that a chip card was used to initiate the transaction.

For BASE24-pos, the card verification information can also be entered manually by the acquirer instead of, or in addition to, being read electronically by swiping the card through a magnetic stripe reader. In this case, the CVD, primary account number, and expiration date printed on the card are manually entered by the merchant; BASE24-pos automatically assumes a service code value of 000 for manually acquired MasterCard and Visa cards. The manual entry of card verification data allows BASE24-pos to detect and prevent the use of counterfeit cards when a magnetic stripe reader is unavailable, or as an additional security check even when a magnetic stripe reader is available. Both an electronically and manually acquired CVD can be verified in the same transaction.

Note: For electronically acquired (i.e., card-read) transactions, BASE24 automatically declines the transaction if the service code is set to 000. No configuration is required. This processing is performed to prevent a fraud scheme where phishers use a stolen PAN, expiration date, and CVD2 to encode a card with

track data, using the CVD2 (printed on the card) in place of the CVD1 (encoded on the track), and a service code of zeroes in place of the actual service code. This scheme *tricks* the payment system into thinking the CVD1 is valid when it is actually the CVD2 that is validated.

Card verification processing is the same, whether or not the card verification data is entered manually or electronically, with the exception that manual card verification can be performed once or twice using different date formats for the CVD. For American Express cards, however, card verification can only be performed once using the YYMM date format.

Refer to the “[Card Verification](#)” discussion in section 1 for a list of all terms that refer to the CVD. All terms refer to the same value.

Card verification is configured at the card prefix level and can be performed in software or in hardware by all security modules supported by BASE24.

Note: Card verification of the CSC for an American Express card is only supported in hardware using the Transaction Security Services process. In addition, card verification of the CSC is only supported in Atalla NSPs or Thales e-Security HSMs. It is not currently supported in SafeNet (Eracom) security modules.

Card Verification Keys

For MasterCard and Visa cards, BASE24 uses a pair of single-length keys in the card verification algorithm. For American Express cards, BASE24 uses a single double-length key in the card verification algorithm. Like PIN management and message authentication keys, the security of card verification keys is critical to their effectiveness. The card issuer controls the card verification keys because the keys used to generate the CVD encoded on the magnetic stripe of a card or printed on the card when it is created also must be used to verify the card when it is presented by the customer. For MasterCard and Visa cards, card verification key pairs can be maintained in the BASE24 database in clear or encrypted form, depending on whether card verification is performed in software or using a security module. For American Express cards, the single double-length card verification key must be maintained in the TSS database in encrypted form. Card verification is always performed in a security module for American Express cards.

Card Verification Requirements

BASE24-atm, BASE24-pos, and BASE24-teller can perform card verification when all of the following requirements are met:

- The transaction is card-initiated.
- The magnetic stripe of the card contains a CVD and the magnetic stripe data is obtained by passing the card through a magnetic stripe reader. For BASE24-pos, the CVD and cardholder information can also be manually entered.
- Card verification information is properly defined in the BASE24 database. See the [“Setting Up the BASE24 Database”](#) topic provided later in this section for detailed information.
- The CV CHECK TYPE field on Card Prefix File (CPF) screen 2 indicates that electronic card verification should be performed and the MANUAL CV CHECK TYPE field on CPF screen 2 indicates that manual card verification should be performed. The CVD PROCESSING FLAG and DENY MANUAL CARD ENTRY fields on CPF screen 8 indicate whether the CVD2 is expected in the transaction and what actions the BASE24 system should take if it is not present.
- When obtaining the CVD electronically, the CV KEYA GROUP field on CPF screen 2 must contain a value. If this field is blank, card verification is not performed.
- When entering the CVD manually, the MANUAL CV KEYA GROUP field on CPF screen 2 must contain a value. If this field is blank, card verification is not performed.
- If both the electronically-entered and manually-entered CVDs are to be verified in the same transaction, the value of the CV KEYA GROUP field must match the value of the MANUAL CV KEYA GROUP field on CPF screen 2 for card types other than American Express. For all card types, the value in the CV KEYA GROUP field on CPF screen 2 is used.
- The card expiration date must meet or exceed the value specified in the CV DATE field or MANUAL CV DATE field on CPF screen 2.

If the card expiration date is greater than or equal to the value of the CV DATE field and the CVD is obtained electronically, a card verification value is expected on the card track and electronic card verification processing is performed. If the card expiration date is less than the value of the CV DATE field and the CVD is obtained electronically, the Card Verify Flag in the internal message is set to a value of P and electronic card verification is not performed.

If the card expiration date is greater than or equal to the value of the MANUAL CV DATE field and the CVD is obtained manually, a manual card verification value is expected on the card and manual card verification processing is performed. If a card has an expiration date less than this effective date, the Card Verify Flag in the internal message is set to a value of P and manual card verification processing is not performed.

Note: If both a manually-entered CVD and electronically acquired CVD are verified in the same transaction, the card expiration date must meet or exceed the value specified in both the CV DATE field and MANUAL CV DATE field on CPF screen 2 or card verification is not performed.

- Card verification has not already been performed. For BASE24-pos, if card verification has already been performed for a transaction originating from an interchange, card verification is performed again if the Card Verify Flag in the POS Standard Internal Message (PSTM) does not indicate a track length error or an old card.
- The transaction is authorized by BASE24 or card verification is included in screening checks performed by BASE24 when the transaction is authorized by a host.
- The message type is 0200 (request). BASE24-teller can also perform card verification for message type 0300 (inquiry or update request).
- For EMV transactions, the transaction meets the conditions specified for card verification of the iCVD as configured on Card Prefix File (CPF) screen 11.
- For American Express cards, BASE24-atm accepts only the five-digit CSC value embedded within the magnetic stripe. The Diebold 10XX/478X and NCR 5XXX Device Handler processes support the five-digit CSC.

For BASE24-pos, the rules for CSC values differ according to whether the transaction was obtained by swiping the card through a card reader or manually entered by the merchant. For card swiped transactions, the five-digit CSC is mandatory. The three- and four-digit CSC values (both manually entered) are optional and mutually exclusive. For manually-entered transactions, the five-digit CSC value cannot be submitted and the three- and four-digit CSC values are optional. Thus, manually-entered transactions could be submitted without a CSC value.

BASE24-pos regards the three- and four-digit CSC values as mutually exclusive. Therefore, it only allows one of these values to be submitted within a single BASE24-pos transaction. Note that this is a requirement of BASE24-pos and is not a stipulation of American Express. If the three-digit CSC is submitted with a transaction, only the leftmost three bytes of the CSC are used to verify the card.

Currently, the BASE24-pos Standard POS Device Handler (SPDH) is the only BASE24-pos Device Handler module that supports American Express card verification.

- For BASE24-pos, the electronically acquired CVD takes precedence over a manually-entered CVD, even though both values can be verified at the same time. If the electronically acquired CVD is invalid, the BAD CV ACTION - TRACK DATA COMPLETE or BAD CV ACTION - TRACK DATA UNCERTAIN field on Card Prefix File (CPF) screen 2 is used for the transaction. If the electronically acquired CVD is valid, then the result of verifying the manually-entered CVD is checked. If the manually-entered CVD is valid, then the card is successfully verified and transaction processing continues. If the manually-entered CVD is invalid, then the BAD CV ACTION - MANUAL ENTRY field on CPF screen 2 is used for the transaction.

Track 1 and Track 2 Layouts

BASE24 uses the following information when performing card verification electronically:

- Primary account number (PAN)
- Card expiration date
- Service code
- Card verification digits (CVD) generated by the card issuer

When the above information is obtained from the magnetic stripe of the card, the required data can be retrieved from either Track 1 or Track 2 of the card depending on institution preference. A track preference flag is set in the Card Prefix File (CPF) to determine which track is the primary track.

The starting position for each field, known as the offset, is computed as the number of positions to the right of the start sentinel. The start sentinel is in position 1 and has an offset of zero. BASE24 expects the PAN to begin in position 2 in Track 2 data. The PAN has an offset of 1 because it starts in position 2, which is the first position to the right of the start sentinel. The offsets for the card expiration date, service code, and CVD must be specified in the CPF for each card prefix.

The track data formats used by MasterCard and Visa are similar. The track data formats used by American Express are slightly different. All formats specify the PAN, card expiration date, and service code locations. The MasterCard CVC or Visa CVV are located within the discretionary data field. The CVC and CVV are

three digits in length. The CVC or CVV can start anywhere in the discretionary data as long as there is room for all digits. The five-digit American Express CSC is either located at the beginning of the discretionary data (for Track 1) or in its own defined field (for Track 2).

Card verification can be performed on any card that contains the appropriate card verification information on the magnetic stripe, which contains Track 1 or Track 2. This information can be carried in positions other than the ones shown in these tables. The BASE24 database contains offsets to identify the location of each field that is necessary for card verification. The database fields for card verification are described later in this section.

MasterCard and Visa Track 1 Layout

The following table illustrates the ISO-defined fields on Track 1 for MasterCard and Visa cards. The maximum length of Track 1 data is 76 bytes, excluding the start and end sentinels and the optional longitudinal redundancy check (LRC) character.

Contents	Byte Length
Start sentinel	1
Format code	1
PAN	Up to 19
Separator	1
Name	26
Separator	1
Expiration date	4
Service code	3
Discretionary data	Balance of remaining digits
End sentinel	1
Longitudinal redundancy check	1

MasterCard and Visa Track 2 Layout

The following table illustrates the ISO-defined fields on Track 2 for MasterCard and Visa cards. The maximum length of Track 2 data is 37 bytes, excluding the start and end sentinels and the optional longitudinal redundancy check (LRC) character.

Contents	Byte Length
Start sentinel	1
PAN	Up to 19
Separator	1
Expiration date	4
Service code	3
Discretionary data	Balance of remaining digits
End sentinel	1
Longitudinal redundancy check	1

American Express Track 1 Layout

The following table illustrates the ISO-defined fields on Track 1 for American Express cards. The maximum length of Track 1 data is 76 bytes, excluding the start and end sentinels and the optional longitudinal redundancy check (LRC) character.

Contents	Byte Length
Start sentinel	1
Format code	1
PAN	Up to 19
Separator	1
Name	26

Contents	Byte Length
Separator	1
Expiration date	4
Service code	3
Effective date	4
Card Security Code	5
Remaining discretionary data	Balance of remaining digits
End sentinel	1
Longitudinal redundancy check	1

American Express Track 2 Layout

The following table illustrates the ISO-defined fields on Track 2 for American Express cards. The maximum length of Track 2 data is 37 bytes, excluding the start and end sentinels and the optional longitudinal redundancy check (LRC) character.

Contents	Byte Length
Start sentinel	1
PAN	15
Separator	1
Expiration date	4
Service code	3
Effective date	4
Card security code	5
Reserved field	3
Language code	2

Contents	Byte Length
End sentinel	1
Longitudinal redundancy check	1

BASE24 Internal Messages

When a message is received from an external source, BASE24 converts that message from the native format of the external source to a standard internal format recognizable to the BASE24 processes. The standard internal message formats used by BASE24-atm (Standard Internal Message (STM)), BASE24-pos (POS Standard Internal Message (PSTM)), and BASE24-teller (Teller Standard Internal Message (TSTM)) are not identical; however, they do contain many of the same fields.

Internal messages contain the following information for card verification. Each description includes the name of the STM, PSTM, and TSTM field that contains the information.

Card Present Flag — Indicates a BASE24-teller transaction is card-initiated. BASE24-atm and BASE24-pos do not use this field. Since cards are optional for BASE24-teller, the Device Handler process must set the value in this field to Y (yes) when a card is used to initiate the transaction.

STM Field:	Not applicable
PSTM Field:	Not applicable
TSTM Field:	TSTMH.RQST.CRD-PRESENT

Card Verification Digits — Contains manually-entered CVD values such as the MasterCard Card Validation Codes (CVC2), Visa Card Verification Value 2 (CVV2), or three- or four-digit American Express Card Security Code (CSC) when a transaction also contains electronically-entered CVD values. Only the leftmost three bytes are used in CVC2 and CVV2 verification processing. When a transaction contains only a manually-entered CVD value, the value is carried in the TRAN.TRACK2 field of the BASE24-pos internal message.

This field applies to BASE24-pos only.

STM Field:	Not applicable
PSTM Field:	PS51-TKN.CVD-FLD
TSTM Field:	Not applicable

Card Verify Flag — Indicates the results of previous attempts, if any, to verify this card for electronically-entered CVDs. For card-read transactions, this field specifies the results of verifying the CVV, CVC, or five-digit CSC. Valid values are as follows:

- 0 = Card verification was not performed because the transaction was denied before card verification started.
- C = Card verification was performed and the card verification digits (CVD) were invalid. The situation was noted, and transaction processing continued.
- D = Card verification was performed and the CVD was invalid. The transaction was denied and the Error Flag field in the internal message was set to a value of C.
- J = CVV checking was not performed. The track length was in error. The BAD TRK LEN field on CPF screen 1 indicated that the transaction should be denied. This value is not valid for BASE24-teller.
- K = CVV checking was not performed. The track length was in error. The BAD TRK LEN field on CPF screen 1 indicated that the transaction should be referred. This value is valid for BASE24-pos only.
- L = CVV checking was not performed. The track length was in error. The BAD TRK LEN field on CPF screen 1 indicated that processing should continue. This value is not valid for BASE24-teller.
- N or *␣* = Authorizing entity has not yet attempted card verification or could not verify the CVD due to a security device error. (*␣* indicates a blank character.)
- O = CVD checking was not performed. Either the merchant ignored the CVD on purpose or the cardholder falsely indicated that the CVD was not on the card.

P = CVD checking was not performed because the CVD value was not on the card. Not all cards have an encoded or printed CVD value. This value is not valid for BASE24-teller.

For electronic card verification, the card expiration date must be equal to or greater than the date defined in the CV DATE field on CPF screen 2 to ensure that the CVD field has been encoded. If the card expiration date is equal to or greater than the CPF date, the CVD checks are performed.

For manual card verification, the card expiration date must be equal to or greater than the date defined in the MANUAL CV DATE field on CPF screen 2 to ensure that the manual CVD value is included on the card. If the card expiration date is equal to or greater than the CPF date, card verification is performed.

R = Card verification was performed and the CVD was invalid. The situation was noted, and the transaction should be referred.

U = The issuer has not certified or has not provided the encryption keys to the interchange.

Y = Card verification was performed and the CVD was valid.

Note: The magnetic stripe data is also assumed complete if a BASE24-atm Release 5.0 token for BASE24-atm or a BASE24-pos Release 5.0 token for BASE24-pos is not present.

STM Field: AT50-TKN.CRD-VRFY-FLG

PSTM Field: PS50-TKN.CRD-VRFY-FLG

TSTM Field: TSTMH.RESP-HDR.CRD-VRFY-FLG

Card Verify Flag2 — Indicates the results of previous attempts, if any, to verify this card using manually-entered CVDs. This field specifies the results of verifying the manually-entered CVV2, CVC2, or three- or four-digit CSC. Valid values are as follows:

0 = Card verification was not performed because the transaction was denied before card verification processing started.

C = Card verification was performed and the card verification digits (CVD) were invalid. The situation was noted and the transaction processing continued.

- D = Card verification was performed and the CVD was invalid. The transaction was denied and the ERR-FLG field was set to C.
- N or *b* = Card verification was not attempted or a security device error occurred (where *b* indicates a blank space).
- O = Card verification was not performed. Either the merchant ignored the CVD on purpose or the user falsely indicated no CVD was on the card.
- P = Card verification was not performed, CVD was not on the card. Not all cards have a CVD value encoded. The card expiration date must be equal to or greater than an expiration date defined on the CPF to insure that the CVD field has been encoded. If the card expiration date is equal to or greater than the CPF date, the CVD checks are performed.
- R = Card verification was performed and the CVD was invalid. The situation was noted and the transaction should be referred.
- S = CVV2 should be on the card but the merchant indicates that it is not.
- U = Issuer has not certified or has not provided the encryption keys to the interchange.
- Y = Card verification was performed and the CVD was valid.

PSTM Field: POS-DATA1-TKN.CRD-VRFY-FLG2

Complete Track 2 Data Flag — Indicates whether the transaction acquirer can capture and transmit complete Track 1 or Track 2 data for a card swipe transaction. Valid values are as follows:

- N = The transaction acquirer cannot capture and transmit complete track data.
- b*, 0, or Y = The transaction acquirer captures and transmits complete track data. (*b* indicates a blank character.)

Note: The magnetic stripe data is also assumed complete if a BASE24-atm Release 5.0 token for BASE24-atm or a BASE24-pos Release 5.0 token for BASE24-pos is not present.

STM Field: AT50-TKN.COMPLETE-TRACK2-DATA
PSTM Field: PS50-TKN.COMPLETE-TRACK2-DATA
TSTM Field: TSTMH.RQST.COMPLETE-TRACK2-DATA

CVD Present Flag — A code indicating whether the manual Card Verification Value is printed on the card. Valid values are as follows:

- 0 = CVV2, CVC2, or CSC value is deliberately bypassed or not provided by the merchant
- 1 = CVV2, CVC2, or CSC value is present
- 2 = CVV2, CVC2, or CSC value is on the card, but is illegible
- 9 = Cardholder states that the card has no CVV2, CVC2, or CSC imprint

STM Field: Not applicable
PSTM Field: PS51-TKN.CVD-FLD-PRESENT
TSTM Field: Not applicable

Error Flag — Indicates whether an error occurred while performing card verification. A Card Verify Flag value of D, which indicates an invalid CVD, results in a value of C (card verification error) in this field.

STM Field: AT50-TKN.ERR-FLG
PSTM Field: PS50-TKN.ERR-FLG
TSTM Field: TSTMH.RESP-HDR.ERR-FLG

Message Type — Indicates the type of message being processed. Valid values for card verification are as follows:

- 0200 = Request
- 0300 = Inquiry or update request (BASE24-teller only)

STM Field: TYP
PSTM Field: TYP
TSTM Field: TSTMH.SYS.MSG-TYP

Track Data — Magnetic stripe data obtained by passing the card through a magnetic stripe reader (electronically entered) or entered manually by the terminal operator. The first byte of the magnetic stripe data contains an M to identify manually-entered card data or a start sentinel (any character other than an M) to identify that the card was passed through a magnetic stripe reader. Magnetic stripe data also must include the CVD, service code, card expiration date, and PAN. manually-entered data also must include the CVD, card expiration date, and PAN. If both the electronically-entered CVD and manually-entered CVD are provided in a transaction, the manually-entered CVD is carried in the CVD-FLD of the BASE24-pos Release 5.1 token.

STM Field:	RQST.TRACK2
PSTM Field:	TRAN.TRACK2
TSTM Field:	TSTMH.RQST.TRACK2

Transaction Processing

This subsection describes the transaction processing performed for electronic and manual card verification. Electronic card verification can be performed by BASE24-atm, BASE24-pos, and BASE24-teller. Manual card verification can be performed by BASE24-pos only.

Electronic Card Verification

The BASE24-atm, BASE24-pos, and BASE24-teller Authorization processes take the following card verification actions during their processing of an electronically-entered transaction.

Card verification is performed before PIN verification. Card verification actions constitute only a portion of the overall transaction processing performed by the Authorization processes.

The Authorization process performs the following steps:

1. Determines whether electronic card verification should be performed.

The Authorization process ensures that all electronic card verification requirements and manual card verification requirements (if the manually-entered CVD is also included in the transaction), as listed earlier in this section under the [“Card Verification Requirements”](#) topic, have been met.

If one or more of the requirements is not met, the Authorization process continues processing the transaction without any further card verification processing.

If the first character in the Track Data field of the internal message contains an M, manual card verification is performed instead of electronic card verification.

If all of the requirements are met, processing continues with step 2.

2. Determines whether the internal message contains complete magnetic stripe data.
 - If the COMPLETE-TRACK2-DATA field in the internal message indicates magnetic stripe data is complete, processing continues with step 5.

- If the COMPLETE-TRACK2-DATA field in the internal message indicates magnetic stripe data is incomplete, processing continues with step 3.
3. Determines whether sufficient magnetic stripe data exists to verify the card.

The Authorization process determines the length of the magnetic stripe data. It then uses the values of the TRACK1 CVD OFST or TRACK2 CVD OFST and TRACK1 SRVC CODE OFST or TRACK2 SRVC CODE OFST fields on CPF screen 2 to determine whether the portion of magnetic stripe data containing the CVD and service code is available. Refer to the topic “Track 1 and Track 2 Layouts” earlier in this section for more information.

- If sufficient magnetic stripe data is not available, the Authorization process continues processing the transaction without any further card verification processing.
- If sufficient magnetic stripe data is available, processing continues with step 4.
- If the service code is set to 000, the transaction is declined.

Phishers can use a stolen PAN, expiration date and CVD2 to encode a card with track data, using the CVD2 (printed on the card) in place of the CVD1 (encoded on the track), and a service code of zeroes in place of the actual service code. This check ensures that such fraudulent transactions are declined.

4. Determines whether card verification should be attempted with incomplete magnetic stripe data.
 - If the CV CHECK TYPE field on CPF screen 2 is set to a value of 1 (complete only), the Authorization process continues processing the transaction without any further card verification processing.
 - If the CV CHECK TYPE field on CPF screen 2 is set to a value of 2 (all), processing continues with step 5.
5. Performs the following track length checks depending on the track being used:

Track	Track Length Check Performed
Track 1	If Track 1 is used, the Track 1 length is checked to ensure that it is between the values in the LENGTH MIN/MAX fields on CPF screen 1. If the maximum field of the LENGTH MIN/MAX fields on CPF screen 1 is set to zero, no maximum length is checked.

Track	Track Length Check Performed
Track 2	If Track 2 is used, the Track 2 length is checked to ensure that it is between the values in the LENGTH MIN/MAX fields on CPF screen 1. If the maximum field of the LENGTH MIN/MAX fields on CPF screen 1 is set to zero, no maximum length is checked.

If the magnetic stripe data is not within the range, the BAD TRK LEN field on CPF screen 1 is checked to see how the Card Verify Flag is set in the internal message.

- If the BAD TRK LEN field on CPF screen 1 is set to a value of 0, then the Card Verify Flag is set to a value of L (length error) and the Authorization process continues processing the transaction; however, no additional card verification is performed.
 - If the BAD TRK LEN field on CPF screen 1 is set to a value of 1, the Card Verify Flag is set to a value of J and the transaction is denied.
 - If the BAD TRK LEN field on CPF screen 1 is set to a value of 2, the Card Verify Flag is set to a value of J, the transaction is denied, and the card is retained.
 - If the BAD TRK LEN field on CPF screen 1 is set to a value of 3, the Card Verify Flag is set to a value of K and the transaction is referred. This setting is used only for BASE2-pos.
6. Retrieves the KEYA record based on the value of the CV KEYA GROUP field on CPF screen 2 and the card expiration date if the Transaction Security Services process is not used.
- If the appropriate KEYA record is located, the Authorization process compares the card expiration date to the values of the BEGIN DATE and END DATE fields on KEYA screen 1.
 - If the card expiration date falls within the range of dates specified by these fields, or if these fields use wildcard characters, the Authorization process performs card verification.
 - If the card expiration date does not fall within the range of dates specified by these fields, and the fields do not contain wildcard characters, the Authorization process denies the transaction.
 - If the appropriate KEYA record is not found, the Authorization process denies the transaction.
7. Determines whether the electronically-entered CVD for the card has already been verified.
- If the CRD-VRFY-FLG in the BASE24-atm or BASE24-pos Release 5.0 token is set to a value of Y, the electronically-entered CVD for the card has already been verified and was found to be valid. Electronic card verification processing is complete. Manual card verification or normal transaction processing continues.

- If the CRD-VRFY-FLG in the BASE24-atm or BASE24-pos Release 5.0 token is set to a value of C, the electronically-entered CVD for the card has already been verified and was found to be invalid. Card verification processing is complete. Continue processing based on the setting of the BAD CV ACTION - TRACK DATA COMPLETE or BAD CV ACTION - TRACK DATA UNCERTAIN field on CPF screen 2 and the COMPLETE-TRACK2-DATA field in the internal message.
 - If the COMPLETE-TRACK2-DATA field in the internal message indicates magnetic stripe data is complete, transaction authorization continues using the setting of the BAD CV ACTION - TRACK - TRACK DATA COMPLETE field on CPF screen 2.
 - If the COMPLETE-TRACK2-DATA field in the internal message indicates magnetic stripe data is not complete, transaction authorization continues using the setting of the BAD CV ACTION - TRACK DATA UNCERTAIN field on CPF screen 2.
 - If the CRD-VRFY-FLG is set to a value other than Y or C, continue card verification processing with the next step.
8. Determines whether card verification is performed in software or hardware.
- If the ENCRYPT TYPE field on KEYA screen 6 is set to a value of 00 (clear), card verification is to be performed in software by the Authorization process using the clear card verification keys from the KEYA record.
 - If the ENCRYPT TYPE field on KEYA screen 6 is set to a value of 01 (encrypted), card verification is to be performed in hardware. In this case, the Authorization process will pass the message to the security device for card verification using the encrypted card verification keys from the KEYA record.
 - If no KEYA record is found, the Authorization process passes the message to the Transaction Security Services process for verification in a hardware security device using the encrypted keys in the TSS database.
9. Determines whether the card for a BASE24-pos transaction is an American Express card, whether both the electronically-entered CVD and manually-entered CVD are present in the transaction for a MasterCard or Visa card, and whether the manually-entered CVD should be verified if present in the transaction. This step is skipped for a BASE24-atm transaction.

The MANUAL CV CHECK TYPE field on CPF screen 2 must be set to a value of 2 (Validate the CVD2 when it is present in any transaction) for the manually-entered CVD to be verified when the electronically-entered CVD is also present in the transaction.

- If the transaction meets the above requirement, the Authorization process reads the CVD-FLD field in the BASE24-pos Release 5.1 token and performs as follows:
 - If the CVD-FLD field is blank and the CVD-FLD-PRESENT field is set to a value of 0 or 9, the process sets the CRD-VRFY-FLG2 field in the POS Data token to a value of O and continues electronic card verification processing with the next step.
 - If the CVD-FLD field is blank and the CVD-FLD-PRESENT field is set to a value of 2, the process sets the CRD-VRFY-FLG2 field in the POS Data token to a value of N and continues electronic card verification processing with the next step.
 - If the CVD-FLD field contains data, the process saves the value for manual card verification processing. If the value is a four-digit CSC value, only the leftmost three digits are saved. If the card expiration date in the track data is less than the date in the MANUAL CV DATE field on CPF screen 2, the process sets the CRD-VRFY-FLG2 field in the POS Data token to a value of P and continues electronic card verification processing with the next step using the Transaction Security Services process. If the Transaction Security Services process is not configured, the process generates an event message and continues authorization processing using the setting of the BAD CV ACTION - TRACK - TRACK DATA COMPLETE or BAD CV ACTION - TRACK DATA UNCERTAIN field on CPF screen 2.
 - If the BASE24-pos Release 5.1 token is not available, processing abends.
 - If the card is not an American Express card or the manually-entered CVD is not to be verified, continue card verification processing with the next step.
10. Verifies the card in hardware or software.

Whether performed in hardware or software, the following data is used for card verification:

- Both card verification keys from KEYA screen 6 or from the TSS database.
- The primary account number (PAN) from the magnetic stripe data.
- The card expiration date from the magnetic stripe data.
- The service code from the magnetic stripe data (for MasterCard and Visa cards only).

Note: For verification of the iCVD obtained from a chip card, the CHECK METHOD field on CPF screen 11 indicates whether the service code is obtained from the magnetic stripe data or from the CV DATA field on CPF screen 11.

- Card verification digits (CVD) from the magnetic stripe data.

The card verification utilities used by the Authorization process return one of the following codes to indicate the outcome of the card verification:

- 0 = Verification performed, CVD is valid.
- 1 = Verification performed, CVD is not valid.
- 2 = Verification not performed, security device failure.
- 3 = Verification not performed, invalid data.
- 4 = Verification not performed, invalid parameter.

In addition for BASE24-pos transactions, for American Express cards, or MasterCard or Visa cards that contain both electronically acquired and manually-entered CVD values, the card verification utilities return additional fields to indicate which of the CVD values (electronically acquired or manually entered) passed or failed verification. In cases where one value passes and another value fails, the [precedence rules](#) described earlier are used.

11. Determines which of two BAD CV ACTION fields on CPF screen 2 will control processing.
 - If the COMPLETE-TRACK2-DATA field in the internal message indicates magnetic stripe data is complete, transaction authorization continues with step 12 using the setting of the BAD CV ACTION - TRACK - TRACK DATA COMPLETE field on CPF screen 2.
 - If the COMPLETE-TRACK2-DATA field in the internal message indicates magnetic stripe data is not complete, transaction authorization continues with step 12 using the setting of the BAD CV ACTION - TRACK DATA UNCERTAIN field on CPF screen 2.
12. Continues transaction authorization based on the status returned from the card verification utilities and the setting of the appropriate BAD CV ACTION field on CPF screen 2.
 - If the utilities return a status of 0 (verification performed, CVD is valid), regardless of the value in either BAD CV ACTION field on CPF screen 2, the Authorization process sets the Card Verify Flag (and the Card Verify2 Flag, if applicable) in the internal message to a value of Y (CVD is valid). For BASE24-pos, processing continues with step 13. For BASE24-atm and BASE24-teller, card verification processing is complete and normal transaction processing continues.

- If the utilities return a status of 1 (verification performed, CVD is not valid), 2 (verification not performed, security device failure), 3 (verification not performed, invalid data), or 4 (verification not performed, invalid parameter) and the appropriate BAD CV ACTION field on CPF screen 2 contains a value of 0 (denote and continue), the Authorization process sets the Card Verify Flag (and the Card Verify2 Flag, if applicable) in the internal message to a value of C (verification performed, CVD invalid). Card verification processing is complete. Normal transaction processing continues.
- If the utilities return a status of 1 (verification performed, CVD is not valid), 2 (verification not performed, security device failure), 3 (verification not performed, invalid data), or 4 (verification not performed, invalid parameter) and the appropriate BAD CV ACTION field on CPF screen 2 contains a value of 1 (decline and return) or 3 (decline and return, BASE24-atm only), the Authorization process sets the Card Verify Flag (and the Card Verify2 Flag, if applicable) in the internal message to a value of D (verification performed, CVD invalid, transaction declined) and sets the Error Flag in the internal message to C (card verification error). The transaction is declined with a response code indicating the card should be returned to the customer.
- If the utilities return a status of 1 (verification performed, CVD is not valid), 2 (verification not performed, security device failure), 3 (verification not performed, invalid data), or 4 (verification not performed, invalid parameter) and the appropriate BAD CV ACTION field on CPF screen 2 contains a value of 2 (decline and retain), the Authorization process sets the Card Verify Flag (and the Card Verify2 Flag, if applicable) in the internal message to a value of D (verification performed, CVD invalid, transaction declined) and sets the Error Flag in the internal message to a value of C (card verification error). The transaction is declined with a response code indicating the card should not be returned to the customer.
- If the utilities return a status of 1 (verification performed, CVD is not valid), 2 (verification not performed, security device failure), 3 (verification not performed, invalid data), or 4 (verification not performed, invalid parameter) and the appropriate BAD CV ACTION field on CPF screen 2 contains a value of 3 (refer the transaction), the Authorization process sets the Card Verify Flag (and the Card Verify2 Flag, if applicable) in the internal message to a value of R (verification performed, CVD invalid, transaction referred) and sets the Error Flag in the internal message to a value of C (card verification error). The transaction is declined with a response code indicating that the merchant should issue a call.

Note: A value of 3 (refer the transaction) is only valid for BASE24-pos.

13. Determines whether the track data exceeds the maximum track length specified in the appropriate LENGTH MIN/MAX field on CPF screen 1 if the CVD is determined to be valid in the previous step. If the track length exceeds this maximum value, the Authorization process takes the action specified in the BAD TRK LEN field on CPF screen 1. Card verification processing is complete. Normal transaction processing continues.

Manual Card Verification

The BASE24-pos Authorization process takes the following card verification actions during the processing of a transaction with only manually-entered CVDs present in a transaction. See the [“Electronic Card Verification”](#) topic for processing of a transaction with both electronically acquired and manually-entered CVDs.

Card verification is performed before PIN verification. Card verification actions constitute only a portion of the overall transaction processing performed by the Authorization process.

The Authorization process performs the following steps:

1. Determines whether card verification should be performed.

The Authorization process ensures that all manual card verification requirements, as listed earlier in this section under the topic [“Card Verification Requirements”](#) have been met.

If one or more of the requirements is not met, the Authorization process continues processing the transaction without any further card verification processing.

If the first character of the Track Data field in the internal message is not an M, electronic card verification is performed instead of manual card verification.

If all of the requirements are met, processing continues with step 2.

2. Determines whether card verification should be attempted for manually-entered card verification information.
 - If the MANUAL CV CHECK TYPE field on CPF screen 2 is set to a value of 1 (validate the CVD2 when it is present only in manually-entered transactions) or 2 (validate the CVD2 when it is present in any

transaction), the Authorization process checks the first character of the track data in the internal message. If the first character is not an M, the transaction is denied. Otherwise, processing continues with step 3.

- If the MANUAL CV KEYA GROUP field on CPF screen 2 is blank, no further card verification processing is performed. Otherwise, processing continues with step 3.
3. Determines whether the card expiration date in the track data is greater than or equal to the date in the MANUAL CV DATE field on CPF screen 2.
 - If the expiration date is less than the value in the CPF, the Card Verify Flag2 field in the internal message is set to a value of P and no further card verification processing is performed.
 - If the expiration date is greater than or equal to the value in the CPF, processing continues with step 4.
 4. Determines whether the transaction is for a recurring payment. If so, the checks of the DENY MANUAL CARD ENTRY and CVD PROCESSING FLAG on CPF screen 8 are not performed in the next step.
 5. Reads the BASE24-pos Release 5.1 token and performs the following:
 - If the Card Verification Digits field is blank and the CVD Present Flag is set to a value of 0, 2 or 9, the process continues based on the setting of the DENY MANUAL CARD ENTRY field on CPF screen 8:
 - If the DENY MANUAL CARD ENTRY field is set to 0, the process continues with step 6.
 - If the DENY MANUAL CARD ENTRY field is set to 1 and the CVD Present Flag is set to 9, the process sets the Card Verify Flag2 field to a value of O, and declines the transaction. No further card verification processing is performed. Otherwise, processing continues with step 6.
 - If the DENY MANUAL CARD ENTRY field is set to 2 and the CVD Present Flag is set to 0, 2, or 9, the sets the Card Verify Flag2 field to a value of O, and declines the transaction. No further card verification processing is performed. Otherwise, processing continues with step 6.
 - If the DENY MANUAL CARD ENTRY field is set to 3, the process sets the Card Verify Flag2 field to a value of O, and declines the transaction. No further card verification processing is performed. Otherwise, processing continues with step 6.

- If the Card Verification Digits field is blank or the release 5.1 token is not present, the process continues based on the setting of the CVD PROCESSING FLAG field on CPF screen 8:
 - ❑ If the CVD PROCESSING FLAG is set to 0, processing continues with step 6.
 - ❑ If the CVD PROCESSING FLAG is set to 1, the action specified in the BAD CV ACTION - MANUAL ENTRY field on CPF screen 2 is taken and no further card verification processing is performed.
 - ❑ If the CVD PROCESSING FLAG is set to 2 and the cardholder was not present, the action specified in the BAD CV ACTION - MANUAL ENTRY field on CPF screen 2 is taken and no further card verification processing is performed. If the CVD PROCESSING FLAG is set to 2 and the cardholder was present, processing continues with step 6.
 - If the Card Verification Digits field contains data, the process moves the CVD data to the Track Data field and continues with step 6.
 - If the transaction does not contain a CVD and is still being processed after these checks, no further card verification processing is performed.
6. Reads the BASE24-pos Release 5.1 token and performs the following:
- If the Card Verify Flag2 field is set to a value of L or O, no further card verification processing is performed.
 - If the Card Verify Flag2 field is set to a value of Y, the card has already been verified and was found to be valid. No further card verification processing is performed.
 - If the Card Verify Flag2 field is set to a value of C, the card has already been verified and was found to be invalid. Card verification processing is complete. Continue processing based on the setting of the BAD CV ACTION - TRACK DATA COMPLETE or BAD CV ACTION - TRACK DATA UNCERTAIN field on CPF screen 2 and the COMPLETE-TRACK2-DATA field in the internal message.
 - ❑ If the COMPLETE-TRACK2-DATA field in the internal message indicates magnetic stripe data is complete, transaction authorization continues using the setting of the BAD CV ACTION - TRACK - TRACK DATA COMPLETE field on CPF screen 2.
 - ❑ If the COMPLETE-TRACK2-DATA field in the internal message indicates magnetic stripe data is not complete, transaction authorization continues using the setting of the BAD CV ACTION - TRACK DATA UNCERTAIN field on CPF screen 2.

- If the Card Verify Flag2 field is set to any other value, the process continues with step 6.
7. Retrieves the KEYA record based on the value of the MANUAL CV KEYA GROUP field on CPF screen 2 and the card expiration date if the Transaction Security Services process is not used.
 - If the appropriate KEYA record is located, the Authorization process compares the card expiration date to the values of the BEGIN DATE and END DATE fields on KEYA screen 1.
 - If the card expiration date falls within the range of dates specified by these fields, or if these fields use wildcard characters, the Authorization process performs card verification.
 - If the card expiration date does not fall within the range of dates specified by these fields, and the fields do not contain wildcard characters, the Authorization process denies the transaction.
 - If the appropriate KEYA record is not found, the Authorization process denies the transaction.
 8. Determines the format of the expiration date to use for card verification.
 - If the DATE CHECK TYPE field on CPF screen 2 is set to a value of 1 or 3, the expiration date is placed in MMY Y format.
 - If the DATE CHECK TYPE field on CPF screen 2 is set to a value of 0 or 2, the expiration date is placed in YYMM format.
 9. Performs card verification.
 - If the ENCRYPT TYPE field on KEYA screen 6 is set to 00 (clear), the Authorization process verifies the card in software using the clear card verification keys from the KEYA record.
 - If the ENCRYPT TYPE field on KEYA screen 6 is set to 01 (encrypted), the Authorization process passes the message to the security device for card verification using the encrypted card verification keys from the KEYA record.
 - If no KEYA record is found, the Authorization process passes the message to the Transaction Security Services process for verification on a hardware security device using the encrypted keys in the TSS database.

Whether performed in hardware or software, the following data is used for card verification:

- Both card verification keys from KEYA screen 6 or from the TSS database.
- The manually-entered primary account number (PAN).

- The manually-entered card expiration date.
- A service code value of 000 (for MasterCard and Visa cards only).
- The manually-entered card verification digits (CVD).

The card verification utilities used by the Authorization process return one of the following codes to indicate the outcome of the card verification:

- 0 = Verification performed, CVD is valid.
- 1 = Verification performed, CVD is not valid.
- 2 = Verification not performed, security device failure.
- 3 = Verification not performed, invalid data.
- 4 = Verification not performed, invalid parameter.

10. Continues transaction authorization based on the status returned from the card verification utilities and the setting of the BAD CV ACTION - MANUAL ENTRY field on CPF screen 2.

- If the utilities return a status of 0 (verification performed, CVD is valid), regardless of the value in the BAD CV ACTION - MANUAL ENTRY field on CPF screen 2, the Authorization process sets the Card Verify Flag2 field in the internal message to a value of Y (CVD is valid).
- If the utilities return a status of 1 (verification performed, CVD is not valid), the process checks the value of the DATE CHECK TYPE field on CPF screen 2.
 - If the DATE CHECK TYPE field is set to a value of 2 or 3 and card verification has been performed once, the process switches the year and month digits in the expiration date and repeats step 8.
 - If the DATE CHECK TYPE field is set to a value of 0 or 1, or if card verification has been performed twice, the process performs according to the value of the BAD CV ACTION - MANUAL ENTRY field on CPF screen 2 as follows.

Value	Action
0	The Authorization process sets the Card Verify Flag2 field in the internal message to a value of C (verification performed, CVD invalid). Card verification processing is complete. Normal transaction processing continues.

Value	Action
1	The Authorization process sets the Card Verify Flag2 field in the internal message to a value of D (verification performed, CVD invalid, transaction declined) and sets the Error Flag in the internal message to C (card verification error). The transaction is declined with a response code indicating the card should be returned to the customer.
2	The Authorization process sets the Card Verify Flag2 field in the internal message to a value of D (verification performed, CVD invalid, transaction declined) and sets the Error Flag in the internal message to a value of C (card verification error). The transaction is declined with a response code indicating the card should not be returned to the customer.
3	The Authorization process sets the Card Verify Flag2 field in the internal message to a value of R (verification performed, CVD invalid, transaction referred) and sets the Error Flag in the internal message to a value of C (card verification error). The transaction is referred with a response code indicating that the merchant should issue a call.

- If the utilities return a status of 2 (verification not performed, security device failure), 3 (verification not performed, invalid data), or 4 (verification not performed, invalid parameter) and the BAD CV ACTION - MANUAL ENTRY field on CPF screen 2 contains a value of 0 (denote and continue), the Authorization process sets the Card Verify Flag2 field in the internal message to a value of C (verification performed, CVD invalid). Card verification processing is complete. Normal transaction processing continues.
- If the utilities return a status of 2 (verification not performed, security device failure), 3 (verification not performed, invalid data), or 4 (verification not performed, invalid parameter) and the appropriate BAD CV ACTION field on CPF screen 2 contains a value of 1 (decline and return), the Authorization process sets the Card Verify Flag2 field in the internal message to a value of D (verification performed, CVD invalid, transaction declined) and sets the Error Flag in the internal message to C (card verification error). The transaction is declined with a response code indicating the card should be returned to the customer.

- If the utilities return a status of 2 (verification not performed, security device failure), 3 (verification not performed, invalid data), or 4 (verification not performed, invalid parameter) and the appropriate BAD CV ACTION field on CPF screen 2 contains a value of 2 (decline and retain), the Authorization process sets the Card Verify Flag2 field in the internal message to a value of D (verification performed, CVD invalid, transaction declined) and sets the Error Flag in the internal message to a value of C (card verification error). The transaction is declined with a response code indicating the card should not be returned to the customer.
- If the utilities return a status of 2 (verification not performed, security device failure), 3 (verification not performed, invalid data), or 4 (verification not performed, invalid parameter) and the appropriate BAD CV ACTION field on CPF screen 2 contains a value of 3 (refer the transaction), the Authorization process sets the Card Verify Flag2 field in the internal message to a value of R (verification performed, CVD invalid, transaction referred) and sets the Error Flag in the internal message to a value of C (card verification error). The transaction is referred with a response code indicating that the merchant should issue a call.

Setting Up the BASE24 Database

Settings in the Card Prefix File (CPF), Host Configuration File (HCF), Key Authorization File (KEYA), BASE24-pos Terminal Data files, and Teller Terminal Data File (TTDF) must be considered when performing card verification.

Card Prefix File (CPF) Settings

The following settings are needed in the Card Prefix File (CPF) for each card prefix using card verification.

Fields on CPF screens 1 and 2 specify card verification settings that apply to verification of all types of cards, including chip cards. Fields on CPF screen 11 specify card verification settings for chip cards only. Refer to the ***BASE24 Base Files Maintenance Manual*** for illustrations of CPF screens 1 and 2. Refer to the ***BASE24-atm EMV Support Manual*** or the ***BASE24-pos EMV Support Manual*** for illustrations of CPF screen 11.

CPF Screen 1

CPF screen 1 contains general information about card prefixes. The following fields are required when using the BASE24 electronic card verification method.

CARD TRACK INFORMATION

The following field defines the track preference for cards with this prefix. This field must be completed when using the BASE24 electronic card verification method.

TRACK PREFERENCE — Indicates which track (Track 1 or Track 2) is to be used to obtain magnetic stripe data. Valid values are as follows:

- 0 = Track 2
- 1 = Track 1

TRACK1 SETTINGS

The following fields are used to locate certain pieces of information on Track 1 of cards with this prefix. These fields must be completed when using the BASE24 card verification method.

EXP DATE — The starting position, expressed as an offset, of the card expiration date on Track 1 of the card. For example, a value of 18 indicates the card expiration date begins in the nineteenth position on the magnetic stripe. (A value of zero indicates the first position.)

LENGTH MIN/MAX — The minimum and maximum lengths of Track 1 data. Valid values are 0-31. A value of zero in the first part of this field (the minimum length) indicates that no minimum check is performed on the length. A value of zero in the second part of this field (the maximum length) indicates that no maximum check is performed on the length. If the maximum length is not zero, it must be greater than or equal to the minimum length. This field is required only when Track 1 is the preferred track.

TRACK 2 SETTINGS

The following fields are used to locate certain pieces of information on Track 2 of cards with this prefix. These fields must be completed when using the BASE24 card verification method.

EXP DATE — The starting position, expressed as an offset, of the card expiration date on Track 2 of the card. For example, a value of 18 indicates the card expiration date begins in the nineteenth position on the magnetic stripe. (A value of zero indicates the first position.)

LENGTH MIN/MAX — The minimum and maximum lengths of Track 2 data. Valid values are 0-40. A value of zero in the first part of this field (the minimum length) indicates that no minimum check is performed on the length. A value of zero in the second part of this field (the maximum length) indicates that no maximum check is performed on the length. If the maximum length is not zero, it must be greater than or equal to the minimum length. This field is required only when Track 2 is the preferred track.

BAD TRK LEN — The action BASE24 is to take when the card length does not match the expected length. Valid values are as follows:

- 0 = Continue processing
- 1 = Deny the transaction and return the card
- 2 = Deny the transaction and retain the card
- 3 = Refer the transaction (POS only)

CPF Screen 2

CPF screen 2 contains transaction security information for the card prefix.

CARD VERIFICATION INFORMATION

The following fields contain the parameters used by BASE24 to perform card verification. The following fields must be completed when using the BASE24 electronic or manual card verification method.

CV KEYA GROUP — The value used to select the proper Key Authorization File (KEYA) record when verifying PINs or when performing electronic card verification for cards with this prefix. The value in this field is matched with the value in the GRP field on screen 1 of the KEYA record being used with this card prefix if not using the Transaction Security Services process. If using the Transaction Security Services process, the value in this field is matched with the value in the Profile field on the Authentication Profile Configuration window.

CV CHECK TYPE — A code indicating the type of card verification to be performed for cards with this prefix. This field must be set to one of the following values:

- 0 = Do not perform card verification.

- 1 = Perform card verification only when complete magnetic stripe data is available. A field in the internal message identifies whether complete magnetic stripe data is available.

If complete magnetic stripe data is not available or the available data is uncertain, authorization continues without verifying the card.

If complete magnetic stripe data is available and the card verification digits (CVD) on the magnetic stripe match the CVD calculated by BASE24 or a security module, authorization continues after verifying the card.

If complete magnetic stripe data is available and the CVD on the magnetic stripe does not match the CVD calculated by BASE24 or a security module, the value in the BAD CV ACTION - TRACK DATA COMPLETE field specifies the action taken.
- 2 = Perform card verification for all transactions that contain the magnetic stripe information necessary for card verification, regardless of whether complete magnetic stripe data is available. A field in the internal message identifies whether complete magnetic stripe data is available.

If the magnetic stripe data necessary to perform card verification is not available or the data is uncertain, the value in the BAD TRK LEN field on CPF screen 1 specifies the action taken.

If the magnetic stripe data necessary to perform card verification is available and the CVD on the magnetic stripe match the CVD calculated by BASE24 or a security module, authorization continues after verifying the card.

If complete magnetic stripe data is available and the CVD on the magnetic stripe does not match the CVD calculated by BASE24 or a security module, the value in the BAD CV ACTION - TRACK DATA COMPLETE field specifies the action taken.

If the magnetic stripe data is incomplete or uncertain but the data necessary to perform card verification is available, and the CVD on the magnetic stripe of the card does not match the CVD calculated by BASE24 or a security module, the value in the BAD CV ACTION - TRACK DATA UNCERTAIN field specifies the action taken.

MANUAL CV KEYA GROUP — The value used to select the proper Key Authorization File (KEYA) record when verifying the manually-entered card verification digits for cards with this prefix. The value in this field can be the same or different from the value specified in the CARD VERIFICATION KEYA GROUP field. The value in this field is matched with the value in the GRP field on screen 1 of the KEYA record being used with this card prefix if not using the

Transaction Security Services process. If using the Transaction Security Services process, the value in this field is matched with the value in the Profile field on the Authentication Profile Configuration window.

MANUAL CV CHECK TYPE — A code indicating when the validation of manually-entered card verification digits (CVD2) is to be performed for cards with this prefix. This field must be set to one of the following values:

- 0 = CVD2 verification is disabled. Do not perform card verification of the CVDs.
- 1 = Validate the CVD2 when it is present only in manually-entered transactions.
- 2 = Validate the CVD2 when it is present in any transaction (electronic or manually-entered).

CHECK IF HOST ONLINE CV — Indicates whether BASE24 should perform card verification during transaction screening before sending a transaction to the host. For BASE24-atm and BASE24-pos, the value in this field is only used with authorization level 1 (online) and authorization level 3 (online/offline) since transaction screening is only done with these authorization levels. For BASE24-teller, the value in this field is only used with the online authorization level because the online/offline authorization level is not supported.

This field should be set as follows:

- If card verification is to be performed during transaction screening, this field must be set to a value of Y. Requests with invalid CVDs are declined and not sent to the host.
- If card verification is not to be performed during transaction screening, this field must be set to a value of N. Card verification is still performed if the transaction is authorized on BASE24.

DATE CHECK TYPE — A flag that indicates the date format to be used for the expiration date when verifying the card verification digits during manual card verification processing. For values 2 and 3, manual card verification processing can be performed twice using different date formats. For American Express cards, or if you want to validate the CVD1 and CVD2 in the same transaction for

MasterCard or Visa cards, this field must be set to a value of 0 because the message sent to the Transaction Security Services process requires an expiration date format of YYMM. This field defaults to a value of 1. Valid values are as follows:

- 0 = YYMM
- 1 = MMY
- 2 = YYMM first, then MMY
- 3 = MMY first, then YYMM

TRACK1 SRVC CODE OFST — The starting position, expressed as an offset, of the service code data on Track 1 of the card. Valid values when card verification is to be performed are 0 through 99; however, 0 is valid only when the value in the CV CHECK TYPE field is 0 (do not perform card verification) or Track 1 is not used.

TRACK1 CVD OFST — The starting position, expressed as an offset, of the card verification digits (CVD) on Track 1 of the card. Valid values when card verification is to be performed are 0 through 99; however, 0 is valid only when the value in the CV CHECK TYPE field is 0 (do not perform card verification) or Track 1 is not used.

CV DATE — The expiration date (YYMM) that acts as the effective date for electronic card verification processing. If a card has an expiration date greater than or equal to this effective date, a card verification value is expected on the card and card verification processing will occur. If a card has an expiration date less than this effective date, the card may not have card verification data and card verification processing will not occur. In this case, the Authorization process checks the length of the track data against the maximum value of the appropriate LENGTH MIN/MAX field on CPF screen 1. If the track length exceeds this maximum, the Authorization process takes the action specified in the BAD TRK LEN field on CPF screen 1.

TRACK2 CVD OFST — The starting position, expressed as an offset, of the card verification digits (CVD) on Track 2 of the card. Valid values when card verification is to be performed are 0 through 99; however, 0 is valid only when the value in the CV CHECK TYPE field is 0 (do not perform card verification) or Track 2 is not used.

TRACK2 SRVC CODE OFST — The starting position, expressed as an offset, of the service code data on Track 2 of the card. Valid values when card verification is to be performed are 0 through 99; however, 0 is valid only when the value in the CV CHECK TYPE field is 0 (do not perform card verification) or Track 2 is not used.

MANUAL CV DATE — The expiration date (YYMM) that acts as the effective date for manual card verification processing. If a card has an expiration date greater than or equal to this effective date, a manual card verification value is expected on the card and manual card verification processing will occur. If a card has an expiration date less than this effective date, the card may not have manual card verification data and manual card verification processing will not occur. This field defaults to a value of 9901.

BAD CV ACTION - MANUAL ENTRY — A code indicating the action to be taken by BASE24 when the manually-entered CVD does not match the CVD calculated by BASE24 or a security module. This field defaults to a value of 1. Valid values are as follows:

- 0 = Set the Card Verify Flag2 in the internal message to C and continue authorizing the transaction without any further card verification. A Card Verify Flag2 value of C means manual card verification was performed and the CVD was invalid; therefore, manual card verification should not be attempted again.
- 1 = Decline the transaction and return the card.
- 2 = Decline the transaction and retain the card.
- 3 = Refer the transaction (BASE24-pos only).

BAD CV ACTION - TRACK DATA COMPLETE — A code indicating the action to be taken by BASE24 when complete magnetic stripe data is present and the CVD on the magnetic stripe of the card does not match the CVD calculated by BASE24 or a security module. Valid values are as follows:

- 0 = Set the Card Verify Flag in the internal message to C and continue authorizing the transaction without any further card verification. A Card Verify Flag value of C means electronic card verification was performed and the electronically acquired CVD was invalid; therefore, card verification should not be attempted again.
- 1 = Decline the transaction and return the card.
- 2 = Decline the transaction and retain the card.
- 3 = Refer the transaction (BASE24-pos only).
Decline the transaction and return the card (BASE24-atm only).

BAD CV ACTION - TRACK DATA UNCERTAIN — A code indicating the action to be taken by BASE24 when the condition of magnetic stripe data is uncertain and the CVD on the magnetic stripe of the card does not match the CVD calculated by BASE24 or a security module. Valid values are as follows:

- 0 = Set the Card Verify Flag in the internal message to C and continue authorizing the transaction without any further card verification. A Card Verify Flag value of C means electronic card verification was performed and the electronically acquired CVD was invalid; therefore, card verification should not be attempted again.
- 1 = Decline the transaction and return the card.
- 2 = Decline the transaction and retain the card.
- 3 = Refer the transaction (BASE24-pos only).
Decline the transaction and return the card (BASE24-atm only).

CPF Screen 8

CPF screen 8 contains two fields for processing manually-entered BASE24-pos transactions.

Note: These fields do not apply when a manually-entered CVD value is present in a transaction that also contains an electronically acquired CVD value.

CVD PROCESSING FLAG — A code specifying whether manually-entered card verification digits are required for all manually-entered BASE24-pos transactions, except for recurring payment transactions. For recurring payment transactions, acquirers are typically not allowed to retain card verification data. This field defaults to a value of 0. Valid values are as follows:

- 0 = No, a CVD2 is not required.
- 1 = Yes, a CVD2 is required for all manually-entered transactions.
- 2 = Yes, a CVD2 is required only when the cardholder is not present.

DENY MANUAL CARD ENTRY — A code specifying whether manually-entered BASE24-pos transactions are denied when the manually-entered card verification digits are not provided in the transaction, except for recurring payment

transactions. For recurring payment transactions, acquirers are typically not allowed to retain card verification data. This field defaults to a value of 0. Valid values are as follows:

- 0 = No, do not deny manually-entered transactions.
- 1 = Yes, deny manually-entered transactions without a CVD2 if the cardholder does not provide the CVD2 value (i.e., the CVD-FLD-PRESENT field in the BASE24-pos Release 5.1 (C0) token is set to 9).
- 2 = Yes, deny manually-entered transactions without a CVD2 if the cardholder does not provide the CVD2 value (i.e., the CVD-FLD-PRESENT field in the BASE24-pos Release 5.1 (C0) token is set to 9), the CVD2 is illegible (i.e., the CVD-FLD-PRESENT field in the C0 token is set to 2), or the CVD2 is not provided by the merchant (i.e., the CVD-FLD-PRESENT field in the C0 token is set to 0).
- 3 = Yes, deny manually-entered transactions.

CPF Screen 11

CPF screen 11 contains card verification settings for verifying the iCVD on chip cards. The following fields are required for card verification of the iCVD.

CARD VERIFY - CHECK TYPE — A code specifying the conditions under which card verification (i.e., verifying the iCVD) should be performed for EMV transactions. This field defaults to a value of 0. Valid values are as follows:

- 0 = Do not perform card verification.
- 1 = Perform card verification only when an Authorization Request Cryptogram (ARQC) is not present.
- 2 = Perform card verification when an ARQC is not present or when the ARQC has not been successfully verified.
- 3 = Always perform card verification.

EMV CV DATE — The expiration date (YYMM) that acts as the effective date for electronic card verification of the iCVD for EMV transactions. If an EMV card has an expiration date greater than or equal to this effective date, an iCVD is expected on the card and card verification processing can be performed. If an EMV card has an expiration date less than this effective date, the card may not have an iCVD and card verification processing is not performed. This field defaults to a value of 0501 (January, 2005).

CHECK METHOD — A code specifying the algorithm and data to be used when card verification is performed on EMV transactions. Valid values are as follows:

- 0 = Use non-EMV card verification for cards expiring on or after the expiration date configured in the EMV CV DATE field; also use non-EMV card verification for cards expiring before the expiration date configured in the EMV CV DATE field.
- 1 = Replace the service code with the data configured in the CV DATA field for cards expiring on or after the expiration date configured in the EMV CV DATE field; use non-EMV card verification for cards expiring before the expiration date configured in the EMV CV DATE field.
- 2 = Use non-EMV card verification for cards expiring on or after the expiration date configured in the EMV CV DATE field; do not perform card verification on EMV transactions for cards expiring before the expiration date configured in the EMV CV DATE field.
- 3 = Replace the service code with the data configured in the CV DATA field for cards expiring on or after the expiration date configured in the EMV CV DATE field; do not perform card verification on EMV transactions for cards expiring before the expiration date configured in the EMV CV DATE field.

CV DATA — Additional data required when verifying the iCVD for an EMV card. Currently, this field is used to specify the service code used when calculating the iCVD compared with the iCVD on the card when the CHECK METHOD field is set to a value of 1 or 3. This field accepts numeric data only and defaults to a value of 000.

Key Authorization File (KEYA) Settings

The following settings are needed in the Key Authorization File (KEYA) for each CPF group using card verification. Refer to the *BASE24 Base Files Maintenance Manual* for illustrations of KEYA screens 1 and 6.

Note: Records in the KEYA are only needed if you are not using the Transaction Security Services process.

KEYA Screen 1

KEYA screen 1 allows users to identify the KEYA record to be accessed. The following fields must be completed when using the BASE24 card verification method.

GRP — The code that matches the entry in the CARD VERIFICATION KEYA GROUP field or MANUAL CV KEYA GROUP field on CPF screen 2.

BEGIN DATE — The first date in the range of card expiration dates to which this KEYA record applies. The date in this field, expressed in YYYYMM format, is compared to the expiration date of the card used to initiate the transaction. Valid values are as follows:

- 000000 = Data in this KEYA record is valid for cards with expiration dates that are less than or equal to the date specified in the END DATE field.
- YYYYMM = Data in this KEYA record is valid for cards with expiration dates that are equal to or greater than this date but less than or equal to the date specified in the END DATE field.
- ***** = Data in this KEYA record is valid regardless of the card expiration date. The END DATE field also must contain asterisks.

The BEGIN DATE and END DATE fields permit a financial institution to maintain different sets of verification parameters based on the card expiration date. For example, an institution using card verification based on Track 2 data on the magnetic stripe plans to change to new cards using card verification based on Track 1 data on the magnetic stripe. The first cards to use the Track 1 based magnetic stripe data will have an expiration date of June, 2002, which appears in YYYYMM format as 200206. The existing cards, which use Track 2 data, would have expiration dates through May, 2002, which appears in YYYYMM format as 200205.

Two KEYA records are necessary. The first KEYA record, for the old Track 2 data based cards, has a value in the END DATE field of 200205. The BEGIN DATE field for the first KEYA record can be 000000 because all outstanding cards use the verification parameters it contains. The second KEYA record, for the new Track 1 data based cards, has a value in the BEGIN DATE field of 200206. The END DATE field for the second KEYA record must contain a future date.

If more than one set of verification parameters had already been in use, there would have been additional KEYA records and the BEGIN DATE field for the first KEYA record would contain the date that its verification parameters became effective.

If beginning and ending dates are unnecessary, the BEGIN DATE and END DATE fields should be set to asterisks (*).

END DATE — The last date in the range of card expiration dates to which this KEYA record applies. The date in this field, expressed in YYYYMM format, is compared to the expiration date of the card used to initiate the transaction. Valid values are as follows:

- | | | |
|--------|---|---|
| YYYYMM | = | Data in this KEYA record is valid for cards with expiration dates that are less than or equal to this date but equal to or greater than the date in the BEGIN DATE field. |
| ***** | = | Data in this KEYA record is valid regardless of the card expiration date. The BEGIN DATE field also must contain asterisks. |

Note: 000000 is not a valid value in this field.

RECORD TYPE — Specifies the type of verification information contained in this record. A separate KEYA screen contains the information for each verification type. This field must be set to a value of 11 because this KEYA record is for card verification.

KEYA Screen 6

KEYA screen 6 contains fields required for performing card verification information. They are used only when the Record Type field on KEYA screen 1 is set to the value 11.

ENCRYPT TYPE — Indicates the type of keys used when performing card verification.

- If card verification is performed in software using clear keys, this field must be set to a value of 00 (clear keys).
- If card verification is performed with security modules using encrypted keys, this field must be set to a value of 01 (encrypted keys).

CLEAR KEY — Contains the clear card verification key pair.

- If the ENCRYPT TYPE field is set to a value of 00 (software), these fields must contain a pair of clear card verification keys. Each key must be 16 hexadecimal characters. Valid values for each character in these fields are 0 through 9 and A through F. For protection in the database, these keys are scrambled for storage by the KEYA Server process.
- If the ENCRYPT TYPE field is set to a value of 01 (hardware module), these fields are not used.

ENCRYPTED KEY — Contains the encrypted PIN verification keys.

- If the ENCRYPT TYPE field is set to a value of 01 (hardware module), these fields must contain a pair of encrypted card verification keys. Each key must be encrypted by or for the specific security module being used, such that the module is able to decrypt the key when it is provided by BASE24. The encrypted keys entered on the screen must be 16 hexadecimal characters in length. Valid values for each character in these fields are 0 through 9 and A through F.
- If the ENCRYPT TYPE field is set to a value of 00 (software), these fields are not used.

CHECK DIGITS — The check digits associated with the encrypted card verification key used when card verification is performed by a hardware security module. Valid values for each character in this field are 0 through 9 and A through F. A value of 0000 in this field indicates check digits are not being used with card verification keys.

BASE24-pos Terminal Data Files Settings

For installations that include BASE24-pos, the following setting is required on BASE24-pos Terminal Data files (PTD) screen 2 for each POS terminal. Refer to the *BASE24-pos Files Maintenance Manual* for an illustration of PTD screen 2.

COMPLETE TRACK DATA — Indicates whether the terminal has the capability to capture and transmit complete magnetic stripe data on card swipe transactions.

- If the terminal can capture and transmit complete magnetic stripe data, this field must be set, or allowed to default, to a value of Y.
- If the terminal cannot capture and transmit complete magnetic stripe data, this field must be set to a value of N.

Teller Terminal Data File (TTDF) Settings

For installations that include BASE24-teller, the following setting is required on Teller Terminal Data File (TTDF) screen 2 for each teller terminal. Refer to the *BASE24-teller Files Maintenance Manual* for an illustration of TTDF screen 2.

COMPLETE TRACK DATA — Indicates whether the terminal has the capability to capture and transmit complete magnetic stripe data on card swipe transactions.

- If the terminal can capture and transmit complete magnetic stripe data, this field must be set, or allowed to default, to a value of Y.
- If the terminal cannot capture and transmit complete magnetic stripe data, this field must be set to a value of N.

ACI Worldwide, Inc.

14: BASE24-pos Secure Internet Validation Support

This section describes the secure Internet validation support provided by the BASE24-pos product.

Secure Internet Validation

Secure Internet validation provides the means to validate Internet and electronic commerce (e-commerce) transactions for credit or debit cards (magnetic stripe and smart cards) when the ACI Commerce Gateway product interfaces with the BASE24-pos system. The BASE24-pos product supports cardholder payer authentication methods approved by Visa and MasterCard credit or debit card issuers for Internet and e-commerce purchases.

The Visa-approved authentication method for Internet or e-commerce purchases is called the Three-Domain Secure (3-D Secure) Electronic Commerce Verification Service (also referred to as Verified by Visa and Visa Payer Authentication). The authentication method for MasterCard is called Secure Payment Application (SPA) – Universal Cardholder Authentication Field (UCAF). BASE24-pos also supports an authentication method called Virtual PAN, which uses a virtual primary account number (PAN) associated with the actual PAN. The Virtual PAN method can be used with or without both the 3-D Secure and SPA UCAF methods. All of these services are intended to reduce risk, reduce fraud, decrease chargebacks, and increase consumer Internet purchases.

ACI Commerce Gateway

The ACI Commerce Gateway product provides support for both Internet and electronic commerce (e-commerce) card issuers and merchants.

For Internet and e-commerce card issuers, the ACI Commerce Gateway product supports the capability for cardholders to register for a secure Internet validation service and authenticate cardholders during an Internet or e-commerce purchase transaction. ACI Commerce Gateway interfaces with an Access Control Server (ACS) compliant with Visa or MasterCard rules for the card issuer to perform these functions. Once a cardholder is registered, the ACS is used for payer authentication when the cardholder initiates an Internet or e-commerce purchase transaction. For an ACS compliant with MasterCard rules, the ACS must also interface to the BASE24-pos system to perform a card verification transaction to complete payer authentication. An interface to the BASE24-pos system from the ACS is not needed for Visa payer authentication.

Once a cardholder is authenticated (payer authentication), the Internet or e-commerce merchant sends a purchase, preauthorization purchase, preauthorization completion, or mail/telephone order transaction to the card issuer for authorization. The ACI Commerce Gateway product allows Internet or

e-commerce merchants to interface with the BASE24-pos system for authorizing these transactions or sending them through the Banknet, EPS-Net, MDS, or Visanet interchange interfaces for authorization.

For both Internet and e-commerce card issuers and acquirers (merchants), the ACI Commerce Gateway product communicates with the BASE24-pos system using an ISO host interface.

Payer Authentication

Payer authentication is the means by which the card issuer determines that the person using a credit or debit card number to make an Internet or e-commerce purchase is legitimate. This typically involves the validation of a password or other data entered by the payer at the time of purchase as well as cryptographic verification of the card used for the transaction.

The following paragraphs describe the processing performed for each of the supported payer authentication methods. Besides password validation, the SPA UCAF and Virtual PAN methods for payer authentication also require the BASE24-pos product to authorize a card verification transaction prior to authorizing the Internet or e-commerce purchase. The 3-D Secure method does not require authorization of this additional transaction.

3-D Secure

When using a Visa card with the Verified by Visa service, cardholders are prompted to enter their primary account number (PAN), address, and a password when initiating an Internet or e-commerce purchase. This information is passed in a payer authentication transaction to the card issuer's Access Control Server (ACS). If the Internet or e-commerce merchant uses the ACI Commerce Gateway product, the transaction is passed to the e-commerce gateway to communicate with Visa. The transaction is routed through the Visa interchange to the card issuer. The card issuer's ACS uses the password in the transaction to authenticate the payer and creates a Cardholder Authentication Verification Value (CAVV) that is sent back to the Internet or e-commerce merchant. The ACS uses the Visa CVV validation method or an alternative method to cryptographically calculate the CAVV.

Optionally, the issuer's ACS can send a card verification transaction to the BASE24 system using an ISO Host Interface connection where the CAVV is passed in the Trans Stain XID token. The Router/Authorization module authorizes the card verification transaction request, writes the transaction to the E-commerce

Authentication File (EAF) if approved, and returns the transaction response to the issuer ACS using the ISO Host Interface process. This processing is required if you are validating CAVVs using the EAF.

The CAVV created by the issuer ACS is included in the purchase, preauthorization purchase, or preauthorization purchase completion transaction sent to the issuer for authorization. You can configure whether the BASE24 system verifies the CAVV using the Visa CVV validation method when a transaction contains a CAVV or whether it verifies the CAVV against the CAVV stored in the EAF.

When using the CVV validation method, the Router/Authorization module performs card verification of the CAVV included in the purchase, preauthorization purchase, or preauthorization purchase completion transaction. The Visa CVV validation method is essentially the same method used by the BASE24 system to perform card verification of a Visa card verification value. The CAVV is used as the CVV in the card verification processing. For detailed information on card verification processing, refer to section 13 of this manual.

When using EAF validation, the CAVV included in the purchase, preauthorization purchase, or preauthorization purchase completion transaction subsequently sent to the issuer for authorization, is validated against the CAVV stored in the EAF.

SPA UCAF

When a cardholder uses a MasterCard card with the SPA UCAF authentication method to initiate an Internet or e-commerce purchase, the data required for payer authentication can be entered by the cardholder, populated with a wallet, or filled in from data stored by the merchant for the cardholder. The cardholder's SPA-enabled Wallet application (loaded on their PC) acquires the data, and sends a payer authentication request to the card issuer's ACS. The ACS compliant with MasterCard rules is called an SPA Wallet Server. The SPA Wallet Server authenticates the cardholder using a user ID and password, smart card, or other cardholder verification method (CVM), and cryptographically or randomly generates an Accountholder Authentication Value (AAV). If the card issuer resides on the BASE24-pos system, the SPA Wallet Server must also send a card verification transaction with the AAV to BASE24 for authorization. The transaction is sent over an ISO host interface connection. The AAV is carried in the Authentication Data token. When the Router/Authorization module receives a card verification request with this token, it creates a new record in the E-commerce Authentication File (EAF) that contains the cardholder's PAN and the AAV generated for the transaction, before responding to the SPA Wallet Server. The AAV is included in the purchase, or preauthorization purchase, or preauthorization purchase completion transaction subsequently sent to the issuer for authorization.

You can configure whether the BASE24 system verifies the AAV using the MasterCard CVC validation method when a transaction contains an AAV or whether it verifies the AAV against the AAV stored in the EAF.

When using the CVC validation method, the Router/Authorization module performs card verification of the AAV included in the purchase, preauthorization purchase, or preauthorization purchase completion transaction. The CVC validation method is essentially the same method used by the BASE24 system to perform card verification of a MasterCard card validation code. The AAV is used as the CVC in the card verification processing. For detailed information on card verification processing, refer to section 13 of this manual.

When using EAF validation, the AAV included in the purchase, preauthorization purchase, or preauthorization purchase completion transaction subsequently sent to the issuer for authorization, is validated against the AAV stored in the EAF

Virtual PAN

The Virtual PAN method of payer authentication can be used with or without either the 3-D Secure or SPA UCAF method. When a cardholder uses a card with the Virtual PAN authentication method to initiate an Internet or e-commerce purchase, the ACS creates a virtual PAN. The ACS authenticates the cardholder using a user ID and password, smart card, or other cardholder verification method (CVM), generates a virtual PAN and expiration date, and optionally generates a CAVV or AAV. The ACS generates a virtual PAN only for the first transaction submitted for a card. The virtual PAN remains unchanged for all subsequent transactions, while the expiration date is changed on each transaction. If the card issuer resides on the BASE24-pos system, the ACS must also send a card verification transaction with the virtual PAN, expiration date, and optional CAVV or AAV to BASE24 for authorization. The transaction is sent over an ISO Host Interface connection. The virtual PAN and expiration date are carried in the Pseudo Card Number token. The optional CAVV is carried in the Trans Stain XID token while an optional AAV is carried in the Authentication Data token. When the Router/Authorization module receives a card verification request with these tokens, it creates a new record in the E-commerce Authentication File (EAF) that contains the cardholder's PAN, the virtual PAN, and the CAVV or AAV (if provided). If the CAVV or AAV is provided, it is included in the purchase, preauthorization purchase, preauthorization purchase completion, or mail/telephone order transaction sent to the issuer for authorization.

Purchase and Preauthorization Purchase Transactions

Once the payer has been authenticated, the Internet or e-commerce merchant sends a purchase, preauthorization purchase, preauthorization purchase completion, or mail/telephone order transaction to the card issuer for authorization. If the card issuer resides on the BASE24-pos system, the BASE24 system authorizes the transaction as described below according to the card type and payer authentication method used.

Visa Cards

If the customer is using a Visa card, the merchant will have captured all of the data necessary to complete the transaction during the payer authentication and created a CAVV. The CAVV is included in the purchase, preauthorization purchase, or preauthorization purchase completion transaction sent to the Visa interchange for authorization. If the issuer institution resides on the BASE24 system, the Visanet Interchange Interface process receives the transaction and sends it to the Router/Authorization module. The Router/Authorization module determines whether the transaction is an e-commerce transaction, whether the CAVV is present, and whether it needs to be validated. If the CAVV is to be validated using the EAF, BASE24 validates the CAVV in the transaction against the CAVV stored in the EAF. If the CAVV is to be validated using the CVV method and the CAVV-AAV-RSLT-CDE field in the BASE24-pos Release 5.1 token indicates that it has not yet been validated, the CAVV is validated using the CVV method.

If a customer makes a single purchase with multiple authorizations such as a late fulfillment (i.e., the transaction date or shipment exceeds the time limit for qualification), the acquirer must reverse the original transaction and send in another authorization using the same authentication data. If the online merchant splits the fulfillment of an order, each purchase transaction can be submitted with the authentication data of the original purchase.

MasterCard

If the customer is using a MasterCard payment instrument, the merchant will have captured all of the data necessary to complete the transaction during the payer authentication and created an AAV. The AAV is included in the purchase, preauthorization purchase, or preauthorization purchase completion transaction sent to the Banknet or MDS interchange for authorization. If the issuer institution resides on the BASE24 system, the Banknet or MDS Interchange Interface process receives the transaction and sends it to the Router/Authorization module. The Router/Authorization module determines whether the transaction is an

e-commerce transaction, whether the AAV is present, and whether it needs to be validated. If so, the module attempts to locate the transaction in the E-commerce Authentication File (EAF) and validates the AAV in the transaction against the AAV stored in the EAF. If the AAV is successfully validated, the module sets the CAVV-AAV-RSLT-CDE field in the BASE24-pos Release 5.1 token accordingly and marks the transaction record from the EAF as having been used, provided it is not a split payment transaction. If the transaction is a split payment, the module marks the transaction in the EAF to allow further authorizations to complete with the same AAV. If the AAV cannot be validated, the module determines whether to deny the transaction or set the CAVV-AAV-RSLT-CDE field and continues processing based on the value of the BAD SIV ACTION field on CPF screen 6.

If a customer makes a single purchase with multiple authorizations, such as split payments, each purchase transaction can be submitted with the authentication data of the original purchase. However, the control byte of the AAV must be set to a hexadecimal “02” for subsequent authorizations and to hexadecimal “82” for the initial authorization or preauthorization request. The issuer may decline initial authorization requests with AAVs older than 30 calendar days; no time limit exists on the submission of subsequent authorizations.

MasterCard recurring payments should only include the AAV data for the initial authorization. Any subsequent recurring payments are not considered SPA eligible.

Virtual PANs

If the transaction contains a virtual PAN, it is processed just like a transaction using a MasterCard or Visa payment instrument, with the following exceptions. The module reads the EAF using a partial key to determine the actual card number. The module places the real card number and expiration date from the EAF in the Track2 fields of the PSTM and places the virtual PAN and virtual PAN expiration date in the Pseudo Card Number token. The module uses the real card number and expiration date for processing the transaction and logging to the PTLF. However, prior to responding to the acquiring interface (i.e., HISO, Banknet, EPS-Net, MDS, or Visanet), the module replaces TRACK2 fields in the PSTM with the virtual PAN and virtual PAN expiration date so the acquiring interface can find the transaction in suspense. The Pseudo Card Number token should be logged to the PTLF in case it is needed when during settlement with MasterCard or Visa. CAVV or AAV generation and validation is optional when using virtual PANs.

Virtual PANs (VPANs) can be used in mail/telephone order transactions in addition to purchase, preauthorization purchase, or preauthorization purchase completion transactions. If a VPAN is used with an original transaction, the VPAN must be used with any chargebacks and all settlement with an interchange.

E-commerce Authentication File (EAF)

The E-commerce Authentication File (EAF) is central to securing e-commerce transactions using the Virtual PAN method of payer authentication. Use of the EAF is optional for securing e-commerce transactions for both the SPA UCAF (MasterCard) and 3-D Secure (Visa) payer authentication methods. For both methods, the EAF is used only if AVVs or CAVVs are validated using the EAF validation method, or if the Virtual PAN method is used in conjunction with the SPA UCAF (MasterCard) and 3-D Secure (Visa) methods. When the Virtual PAN method is used, the transaction is secured using the EAF validation method, regardless of the SIV check type method configured in the CPF.

The EAF contains one record for each primary account number (PAN) involved in an outstanding e-commerce transaction. If a virtual PAN (VPAN), is used for an e-commerce transaction, the relationship between the PAN and the VPAN is maintained for the life of the VPAN. EAF records are created from the card verification transaction processed during payer authentication. Up to 46 outstanding transactions are supported in the EAF for each PAN.

Because member numbers are not supported for e-commerce transactions, multiple cardholders on the same PAN will have one EAF record unless they use VPANs. VPANs and PANs are used as the primary key to the EAF to allow multiple VPANs to be matched to a single PAN.

Each record stores the CAVV, AAV, or VPAN associated with a PAN, the expiration date of the CAVV, AAV, or VPAN, the full amount available for future transactions, the type of card used, a flag indicating whether the transaction is a split transaction, and the date and time the card verification transaction was logged. For split transactions, the amount is decreased by the amount of each individual transaction processed.

When the Router/Authorization module receives an e-commerce purchase, preauthorization purchase, preauthorization purchase completion, or mail/telephone order transaction, it reads the EAF for a corresponding record. The module validates the CAVV, AAV, or VPAN provided in the transaction against the value stored in the EAF record. Once the purchase is validated, the module marks the record in the EAF as having been used (by setting the expiration date in the EAF record to spaces), unless the transaction is a split payment. In this case, the transaction amount is adjusted in the EAF to allow further authorizations to complete with the same authentication information. Once the transaction amount reaches zero, the expiration date in the EAF record is set to spaces, which marks the record as having been used.

After an EAF record is marked as used, it is still retained for possible reversal processing. A reversal of a card verification transaction causes the transaction entry in the EAF record to be marked as unused.

BASE24 maintains a current and previous EAF. The POS-SETL-EAF-RETENTION-PERIOD param indicates how long the current EAF is retained before it is renamed to the previous EAF, a new current EAF is created, and the old previous EAF is purged. It is recommended that the EAF be maintained for as long as it is possible for your system to receive subsequent requests (e.g., split payments, back orders) related to the original authorization request.

Internal Message

The following paragraphs discuss the BASE24-pos Standard Internal Message (PSTM) fields and tokens required for secure Internet validation processing. For detailed descriptions of tokens, refer to the ***BASE24 Tokens Manual***. For a detailed description of the BASE24-pos Standard Internal Message (PSTM), refer to the ***BASE24-pos Transaction Processing Manual***.

Authentication Data Token

The Accountholder Authentication Value (AAV) for the SPA UCAF or Virtual PAN payer authentication methods is carried by the Authentication Data token. When the Router/Authorization module receives a card verification request with this token, it creates a new record in the E-commerce Authentication File (EAF) that contains the virtual PAN, expiration date, and the AAV generated for the transaction.

BASE24-pos Release 5.1 Token

The BASE24-pos Release 5.1 token carries the electronic commerce flag for Visa transactions, the authentication collection indicator for MasterCard transactions, and the result of CAVV or AAV validation.

For Visa cards, the electronic commerce flag (E-COM-FLG) indicates whether a transaction originated as an e-commerce transaction and whether secure Internet validation can be performed.

For MasterCard transactions, the authentication collection indicator (AUTHN-COLL-IND) field indicates whether the SPA UCAF authentication method is supported by the Internet or e-commerce merchant and whether the data required for the method was collected.

The result of CAVV or AAV validation is stored in the CAVV-AAV-RSLT-CDE field.

Pseudo Card Number Token

The Pseudo Card Number token carries the virtual PAN and expiration date for transactions using the SPA UCAF or Virtual PAN payer authentication methods. You should configure this token to be logged to the BASE24-pos Transaction Log File (PTLF).

Trans Stain XID Token

The Cardholder Authentication Verification Value (CAVV) for the 3-D Secure payer authentication method is carried in the Trans Stain XID token. When the Router/Authorization module receives a card verification request with this token, it determines whether the CAVV needs to be verified, and if so, sends a card verification request to the security module to verify the CAVV or validates the CAVV against the EAF. The SIV CHECK TYPE field on Card Prefix File (CPF) screen 6 specifies the method used to verify the CAVV.

Switch Token

For the Visanet interface, the Switch token carries the result of CAVV validation.

BASE24-pos Standard Internal Message (PSTM)

The TRAN.TRACK2 fields in the BASE24-pos Standard Internal Message (PSTM) normally contain the card number and expiration date for a transaction. However, when the Router/Authorization module receives a transaction containing a virtual PAN, it removes the virtual PAN and expiration date from the TRACK2 fields in the PSTM and places them in the Pseudo Card Number token. The module then reads the E-Commerce Authentication File (EAF) to determine the actual card number with which the virtual PAN is associated. It places the actual card number and expiration date in the TRACK2 fields of the PSTM.

If, at any time, the transaction is sent back to the originator (as denied or approved), the TRACK2 fields in the PSTM are replaced with the virtual PAN and expiration date from the Pseudo Card Number token prior to responding to the originator. However, if the transaction is written to the PTLF, the actual card number and expiration date must exist in the TRACK2 fields in the PSTM (if a corresponding record was found in the EAF). In this case, the virtual PAN and expiration date are stored in the Pseudo Card Number token.

Setting Up the BASE24 Database

Settings in the following files must be considered when configuring secure Internet validation:

- BASE24 Logical Network Configuration File (LCONF)
- Card Prefix File (CPF)
- Key Authorization File (KEYA) or Card Verification File (CRDVD)
- External Message File (EMF)
- Host Configuration File (HCF)

BASE24 Logical Network Configuration File (LCONF) Settings

The following LCONF assigns and params must be configured to support secure Internet validation. The BASE24-pos Main and Secondary Settlement processes use these assigns and params to create the current and previous E-commerce Authentication File (EAF), cut over the EAF, and notify BASE24-pos Device Handler/Router/Authorization processes when the EAF is cut over.

POS-EAF assign — The fully qualified file name of the E-commerce Authentication File (EAF). The Router/Authorization module uses this assign to access the current EAF. The BASE24-pos Main Settlement process uses the current EAF template file to create the current EAF during initialization and at EAF cutover.

POS-EAF-PREV assign — The fully qualified file name of the previous E-commerce Authentication File (EAF). The Router/Authorization module uses this assign to access the previous EAF. The BASE24-pos Main Settlement process uses the previous EAF template file to create the previous EAF during initialization and at EAF cutover to purge the existing previous EAF and rename the current EAF to the previous EAF.

POS-SETL-EAF-RETENTION-PERIOD param — The number of days the current EAF should be retained before it is renamed to the previous EAF. At that time, a new current EAF is created and the old previous EAF is purged. The BASE24-pos Main Settlement process uses this param during logical network cutover to determine whether the current EAF should be cut over. This is determined by the age of the current EAF (file creation date) and the value of the POS-SETL-EAF-RETENTION-PERIOD param. If the EAF file creation date

plus the value of the retention period (e.g., 150 days) is greater than the current date, the EAF is cut over. If this is the case, the process notifies BASE24-pos Device Handler/Router/Authorization processes that the EAF is being cut over, waits ten minutes, and then purges the old previous EAF.

For AAV validation of split payments, the EAF must be retained for a minimum of 150 days because a split payment can be received up to 150 days after the initial payment. The CAVV does not need to be validated for split payments. If you are not validating the AAV for split payments, you should retain the EAF for a minimum of 30 days when the transaction becomes stale for chargebacks.

POS-SETL-EAF-NOTIFY*identifier* param — The symbolic name of a BASE24-pos Device Handler/Router/Authorization process to be notified after EAF cutover is performed. A maximum of 500 processes can be notified. The BASE24-pos Main Settlement process uses this param.

Each POS-SETL-EAF- NOTIFY param contains the name of one process. When multiple processes must be notified, this param can include an optional identifier to specify params used for different processes in the same logical network. The optional identifier is usually a number but can be any unique set of characters that has not already been declared by another POS-SETL-EAF-NOTIFY*identifier* param. For example, POS-SETL-EAF-NOTIFY1 could contain the P1A^RTAU1 process and POS-SETL-EAF-NOTIFY2 could contain the P1A^RTAU2 process.

POS-EAF-NOTIFY-INTERVAL param — The number of writes made to E-commerce Authentication File (EAF) before calculating the percentage of file full. Each time a write is made to the EAF, a counter is increased by one. When the counter is equal to the value specified in this param, the BASE24-pos Device Handler/Router/Authorization process calculates how full the EAF is. The counter is then reset to zero.

When the file reaches the percent full specified in the POS-EAF-NOTIFY-PERCENTAGE param, the BASE24-pos Device Handler/Router/Authorization process generates an event message indicating that the file is becoming full. Until more space is made available, BASE24-pos Device Handler/Router/Authorization process continues to calculate the percentage of file full at the interval specified in this param. If the percentage has increased by one percent or more since the preceding calculation, the BASE24-pos Device Handler/Router/Authorization process generates an event message indicating that the file is becoming full. Valid values are 5 through 32766. The default value is 5.

POS-EAF-NOTIFY-PERCENTAGE param — The percentage of file full at which the BASE24-pos Device Handler/Router/Authorization process generates an event message indicating that the E-commerce Authentication File (EAF) is becoming full. The BASE24-pos Device Handler/Router/Authorization process calculates the percentage of file full and continues to calculate the percentage at the interval indicated in the POS-EAF-NOTIFY-INTERVAL param as described in the following paragraphs until more file space is made available. If the percentage has increased by one percent or more since the preceding calculation, the BASE24-pos Device Handler/Router/Authorization process generates an event message indicating that the file is becoming full.

Each time the BASE24-pos Device Handler/Router/Authorization process writes the number of EAF records specified in the POS-EAF-NOTIFY-INTERVAL param, it determines how full the file is. The calculation to determine how full the file is assumes that the record length is fixed, rather than variable. Because the records can be variable in length and may be shorter than the maximum length assumed by the calculation, the result of this check may be a percentage of file used that is greater than or equal to the actual percentage of file space used. Therefore, this calculation and the resulting event messages may not accurately reflect the EAF approaching its capacity. If the calculation results in the percentage specified in the POS-EAF-NOTIFY-PERCENTAGE param, or more, of its capacity, the BASE24-pos Device Handler/Router/Authorization process generates an event message indicating the condition. Thereafter, the process generates an event message each time it writes the number of records specified in the POS-EAF-NOTIFY-INTERVAL param and as long as the transaction log file calculation results in the percentage specified in the POS-EAF-NOTIFY-PERCENTAGE param, or above, of its capacity.

Valid values are 10 through 99. The default value is 70.

POS-SETL-NOTIFY*identifier* — The symbolic name of an ISO Host Interface process that the Main Settlement process must notify at the time specified in the POS-LN-CUTOVER param. The Main Settlement process uses this param. A maximum of 500 processes can be notified.

Each POS-SETL-NOTIFY param contains the name of one process or service. When multiple processes or services must be notified, this param can include an optional *identifier* to specify params used for different processes or services in the same logical network. The optional *identifier* is usually a number but can be any unique set of characters that has not already been declared by another POS-SETL-NOTIFY*identifier* param.

When this param is configured to notify the ISO Host Interface process, the Main Settlement process sends a 9502 cutover command to the specified process. When the ISO Host Interface process receives this command, it updates a global posting date value that it uses when the posting date is not included in a transaction. Because the ACI Commerce Gateway product does not send the posting date in messages sent to the ISO Host Interface process, this is necessary to properly cut over the posting date.

Card Prefix File (CPF) Settings

The following settings are required in the Card Prefix File (CPF) for each card prefix using secure Internet validation. Refer to the *BASE24 Base Files Maintenance Manual* for an illustration of CPF screen 6.

CPF Screen 6

CPF screen 6 contains BASE24-pos authorization information for card prefixes. The following fields are required when implementing secure Internet validation.

SIV CHECK TYPE — A code indicating the type of secure Internet validation to be performed for cards with this prefix. Valid values are as follows:

- 0 = SIV disabled; do not perform secure Internet validation (default).
- 1 = SIV enabled; validate the CAVV or AAV using the EAF.
- 2 = SIV enabled; verify the CAVV using the CVV method, validate the AAV using the EAF.
- 3 = SIV enabled; verify the CAVV using the CVV method.
- 4 = SIV enabled; verify the CAVV or AAV using the CVV method.
- 5 = SIV enabled; verify the AVV using the CVV method.

SIV KEYA GROUP — The value used to select the proper Key Authorization File (KEYA) record or Card Verification File (CRDVD) record for secure Internet validation of cards with this prefix. If you are using the Transaction Security Services process, this value is used to access the CRDVD record by matching a value entered in the Profile field on the Authentication Profile Configuration window of the ACI Desktop user interface. If you are not using the Transaction Security Services process, this value is used to access the KEYA record by matching a value entered in the GRP field on screen 1 of the KEYA.

The value in this field can be the same or different from the value specified in the CARD VERIFICATION KEYA GROUP or MANUAL CARD VERIF KEYA GROUP fields on CPF screen 2.

CHECK IF ONLINE SIV — Indicates whether BASE24 should perform secure Internet validation during transaction screening before sending a transaction to the host. For BASE24-pos, the value in this field is only used with authorization level 1 (online) and authorization level 3 (online/offline) since transaction screening is only done with these authorization levels. This field defaults to a value of N.

This field should be set as follows:

- If secure Internet validation is to be performed during transaction screening, this field must be set to a value of Y. Requests with an invalid CAVV, AAV, or virtual PAN are declined and not sent to the host.
- If secure Internet validation is not to be performed during transaction screening, this field must be set to a value of N. Secure Internet validation is still performed if the transaction is authorized on the BASE24 system.

BAD SIV ACTION — A code indicating the action to be taken by BASE24 when the CAVV, AAV, or virtual PAN associated with a transaction request is considered to be invalid. This field defaults to a value of 1.

Valid values are as follows:

- 0 = Continue processing.
- 1 = Decline the transaction with a response code of 074 (unable to authorize) and return the card.

SIV NOT PRESENT ACTION — A code indicating the action to perform if one of the following inconsistencies is detected with the data for the transaction:

- For a MasterCard transaction, the data is considered to be inconsistent when the AUTHN-COLL-IND field in the BASE24-pos Release 5.1 token is set to a value of 2 (UCAF data collection is supported by the merchant, and UCAF data was populated), but the AUTHN-IND-DATA field in the Authentication Data token is blank.
- For a Visa transaction, the data is considered to be inconsistent when the E-COM-FLG field in the BASE24-pos Release 5.1 token is set to a value of 5 (Secure electronic transaction with cardholder certificate), 6 (Secure

electronic transaction without cardholder certificate), or 7 (Channel-encrypted electronic commerce transaction), but the TRANS-STAIN field in the Trans Stain XID token is blank.

When a data inconsistency is detected as described above, this field indicates the action to perform. This field defaults to a value of 1. Valid values are as follows:

- 0 = Set the CAVV-AAV-RSLT-CDE field in the BASE24-pos Release 5.1 token to a value of 0 (Not validated due to erroneous data) and continue processing.
- 1 = Set the CAVV-AAV-RSLT-CDE field in the BASE24-pos Release 5.1 token to a value of 0 (Not validated due to erroneous data) and decline the transaction with response code of 074 (unable to authorize).

KEYA or CRDVD Settings

If you are using the Transaction Security Services process, the card verification keys must be stored in the Card Verification File (CRDVD). If you are not using the Transaction Security Services process, the card verification keys used for CAVV or AAV validation must be stored in the Key Authorization File (KEYA). CRDVD and KEYA configuration settings are described below.

CRDVD Settings

If you are using the Transaction Security Services process, the Profile field on the Authentication Profile Configuration window of the ACI desktop must have a matching value in the SIV KEYA GROUP field on CPF screen 6. The card verification key pair to be used for CAVV and AAV validation must be configured on the Card Verification tab of the Authentication Profile Configuration window. Card verification data entered on the Authentication Profile Configuration window is stored in the CRDVD.

KEYA Settings

If you are not using the Transaction Security Services process, the GRP field on KEYA screen 1 must have a matching value in the SIV KEYA GROUP field on CPF screen 6. The card verification key pair to be used for CAVV and AAV validation must be configured on KEYA screen 6.

External Message File (EMF) Settings

The Local Transaction Time (P-12) and Local Transaction Date (P-13) data elements must be included in 0430 response messages sent to the ACI Commerce Gateway product. Therefore, you must update or add the External Message File (EMF) record used for outbound BASE24-pos 0430 messages to include data elements P-12 and P-13.

Host Configuration File (HCF) Settings

When the ACI Commerce Gateway product sends a reversal to the BASE24 system for a BASE24-pos card verification transaction, a response is required. Therefore, you must set the ACK TO DPC field on HCF screen 1 to a value of Y for the ISO Host Interface process to communicate with the ACI Commerce Gateway product.

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15: BASE24 Device Key Management

This section describes the dynamic key management options available for BASE24-atm and BASE24-pos devices.

BASE24-atm Dynamic Key Management

The following paragraphs describe the dynamic key management options available for the BASE24-atm Dynamic Key Management add-on product. This add-on product can be purchased for the following BASE24-atm Device Handler processes:

- Diebold 10XX/478X (running in leased line or dial-up mode)
- NCR 5XXX (running in leased line or dial-up mode)
- Tidel
- Triton

Note: The BASE24-atm Dynamic Key Management add-on product can be used with or without the Transaction Security Services process for the Device Handler processes listed above. When using the Transaction Security Services process, the SECURE-DEV1 and SECURE-DEV2 assigns or the SECURE-DEV-xx assigns in the BASE24 Logical Network Configuration File (LCONF) must be set to the PPD name of a Transaction Security Services process for these Device Handler processes. In addition, the SECURE-DEV-TYP LCONF param for these processes must also be set to a value of TSS.

The SECURE-DEV-ACCESS-TYP param indicates whether a BASE24 application process communicates with a primary and secondary Transaction Security Services process identified by the SECURE-DEV1 and SECURE-DEV-2 assigns or with up to 20 Transaction Security Services processes identified by SECURE-DEV-xx assigns, where *xx* is any two characters. The SECURE-DEV1 assign specifies the primary Transaction Security Services process, while the SECURE-DEV2 assign specifies the secondary Transaction Security Services process. The SECURE-DEV-xx assigns specify up to 20 Transaction Security Services processes accessed on a *round-robin* (i.e., rotating) basis.

ATM Dynamic Key Management

BASE24-atm Device Handler processes use transaction keys to protect the transaction data moving between ATM terminals and the BASE24 system. The installation and changing of these keys is known as key management, and can be performed manually or automatically. Automated key management is often known as dynamic key management.

Dynamic key management enables you to automatically change transaction keys based on a number of factors. Thresholds can be established based on time, number of transactions, number of errors, number of consecutive MAC errors, or any combination of these factors. Both of the transaction keys used by ATM terminals can be changed when any of the defined thresholds is surpassed. Transaction keys can also be changed using a network control command.

Dynamic key management requires the use of security modules to perform PIN encryption, PIN verification, and message authentication, and is not supported when these functions are performed in software.

Transaction Keys

BASE24-atm Device Handler processes use two transaction keys (also known as a session keys), which are described in the paragraphs that follow.

PIN Key — If a PIN is encrypted for transmission from the terminal, this is the key under which it is encrypted. If a PIN is in the clear when transmitted from the terminal and must be encrypted by BASE24, this is the key under which it is to be encrypted.

Message Authentication Key — If inbound messages from the terminal contain a message authentication code (MAC), this is the key the BASE24 interface process uses to verify the MAC. If outbound messages for the terminal contain a MAC, this is the key the BASE24-atm Device Handler process uses to generate the MAC.

Master Keys

Since transaction keys are distributed using the same lines as other communications, a means of sending transaction keys in a secure manner is required. This security is provided by a separate master key known to both the terminal and BASE24-atm. This master key is used to encrypt PIN and MAC transaction keys for their movement over the communications lines.

Dynamic Key Management Thresholds

The BASE24-atm Dynamic Key Management add-on product enables you to automate the changing of terminal PIN and MAC keys when predetermined limits are exceeded, or a key change interval has expired, since the last PIN or MAC key change. These limits and interval are set in BASE24 Logical Network Configuration File (LCONF) params configured for BASE24-atm Device Handler processes. You can configure PIN and MAC keys to be changed in an ATM for the following thresholds:

- The number of transactions to or from an ATM for which the MAC is generated or authenticated exceeds a set limit
- The number of transactions from an ATM for which the PIN is verified exceeds a set limit
- The number of transactions from an ATM on which a MAC error occurred exceeds a set limit
- The number of transactions from an ATM on which a PIN error occurred exceeds a set limit
- The number of consecutive MAC failures for an ATM exceeds a set limit
- The interval (number of hours) since the last PIN or MAC key change

MAC Authentication and PIN Verification Transactions Limits

The ATM-DH-DKM-MAC-TXN-LMT and ATM-DH-DKM-PIN-TXN-LMT LCONF params indicate the maximum number of transactions for an ATM for which the MAC is authenticated or generated, or the PIN is verified, respectively, before the MAC key and PIN key must be changed in the ATM. Valid values for these params are 0–32767. The default value for these params is 0, which indicates that the limit is not used.

The ATM-DH-DKM-MAC-TXN-LMT param specifies the maximum number of transactions to or from an ATM for which the MAC is generated or authenticated, before both the MAC key and PIN key in the ATM are changed. When the number of transactions for a terminal exceeds this limit, the BASE24-atm Device Handler process initiates a download of a new MAC key and PIN key to the terminal, resets all dynamic key management counters for the terminal to 0, and updates the key change timestamp for the terminal.

The ATM-DH-DKM-PIN-TXN-LMT param specifies the maximum number of transactions from an ATM for which the PIN is verified, before both the PIN key and MAC key in the ATM are changed. When the number of transactions for a terminal exceeds this limit, the BASE24-atm Device Handler process initiates a download of a new PIN key and a new MAC key to the terminal, resets all dynamic key management counters for the terminal to 0, and updates the key change timestamp for the terminal.

MAC or PIN Failures Limits

The ATM-DH-DKM-MAC-ERR-LMT and ATM-DH-DKM-PIN-ERR-LMT LCONF params indicate the maximum number of transactions from an ATM on which MAC or PIN errors occurred, respectively, before the MAC key and PIN key must be changed in the ATM. Valid values for these params are 0–32767. The default value for these params is 0, which indicates that the limit is not used.

The ATM-DH-DKM-MAC-ERR-LMT param specifies the maximum number of transactions from an ATM on which MAC errors occurred, before both the MAC key and PIN key in the ATM are changed. When the number of transactions from an ATM on which MAC errors occurred exceeds this limit, the BASE24-atm Device Handler process initiates a download of a new MAC key and a new PIN key to the terminal, resets all dynamic key management counters for the terminal to 0, and updates the key change timestamp for the terminal.

Note: The Triton Device Handler module does not use the ATM-DH-DKM-MAC-ERR-LMT LCONF param. The Triton Device Handler module loads a new MAC key whenever a MAC verification fails.

The ATM-DH-DKM-PIN-ERR-LMT param specifies the maximum number of transactions from an ATM on which PIN errors occurred, before both the PIN key and MAC key in the ATM are changed. When the number of transactions from an ATM on which PIN errors occurred exceeds this limit, the BASE24-atm Device Handler process initiates a download of a new PIN key and MAC key to the terminal, resets all dynamic key management counters for the terminal to 0, and updates the key change timestamp for the terminal.

Consecutive MAC Failures Limit

The ATM-DH-CONS-MAC-ERR-LMT LCONF param indicates the maximum number of consecutive MAC errors for an ATM, before the MAC key and PIN key must be changed in the ATM. Valid values for this param are 0–32767. The default value is 3. A value of 0 indicates that the limit is not used.

When the number of consecutive MAC errors for a terminal exceeds this limit, the BASE24-atm Device Handler process initiates a download of a new MAC key and a new PIN key to the terminal, resets all dynamic key management counters for the terminal to 0, and updates the key change timestamp for the terminal.

Note: The Triton Device Handler module does not use the ATM-DH-CONS-MAC-ERR-LMT LCONF param. The Triton Device Handler module loads a new MAC key whenever a MAC verification fails.

Key Change Interval

The ATM-DH-DKM-KEY-CHG-INTRVL LCONF param specifies the maximum time allowed (in hours) between key changes for a terminal. The difference between the current time and the DKM key change timestamp for a terminal can not exceed the value specified by this param without a key change being performed. When a key change is performed, due to a dynamic key change limit or this interval being exceeded, the key change timestamp for the terminal is updated. Valid values for this param are 0–32767. The default value is 0, which indicates that a key change interval is not used.

Dynamic Key Management Counters and Timer

When the BASE24-atm Dynamic Key Management add-on product is installed, transaction counters and a key change interval timer for each applicable ATM terminal are maintained in the BASE24-atm Terminal Data files. All the counters for a terminal are reset to 0, and the key change timestamp is updated to the current time, whenever the MAC key or PIN key is changed for the terminal. This occurs regardless of whether the MAC key or PIN key was changed automatically because a threshold was exceeded or because the key was changed manually by a network control facility command.

For Diebold 10XX/478X and NCR 5XXX terminals, the transaction counters and timer are maintained in the BASE24-atm Terminal Data Dynamic File—general data (ATDD1). For Tidel and Triton terminals, the transaction counters and timer are maintained in the the Terminal Data File (TDF).

Dynamic key management counters cannot be viewed from BASE24-atm Terminal Data file screens.

Setting Up the BASE24 and TSS Databases

Settings on the Logical Network Configuration File (LNCf) screens and the BASE24-atm Terminal Data files (ATD) screens must be considered when using the BASE24-atm Dynamic Key Management add-on product. Refer to the *BASE24 Logical Network Configuration File Manual* and the *BASE24-atm Files Maintenance Manual* for illustrations of LNCf and ATD screens, respectively.

If you are using the Transaction Security Services process, you must also consider settings on the Channel Security Configuration window of the ACI Desktop user interface. Refer to the *BASE24 Transaction Security Services Processing Guide* for detailed information on this window.

LNCf Screens

Use the LNCf Assign and LNCf Param screens to configure the LCONF assigns and params discussed earlier in this appendix to the applicable BASE24-atm Device Handler processes in your system.

ATD Screens

The PIN ENCRYPT TYPE field on device-specific ATD screen 9 must be set to a value of 00 (No encryption), 03 (ANSI standard PIN/PAD, with a security device), or 05 (ANSI standard PIN/PAN, with a security device) for each applicable ATM terminal. The MAC SUPPORT TYPE field on device-specific ATD screen 9 must be set to a value of 0 (No MAC support) or 3 (Hardware MAC support) for each applicable terminal. The MASTER KEY field on device-specific ATD screen 9 must be set to the ATM master key of each applicable ATM terminal.

Note: If you are using the Transaction Security Services process, the MASTER KEY field must be set to the terminal ID. This value is used as the key locator to records in the Channel Security Configuration File (CSECD).

Key Change Commands

With or without the BASE24-atm Dynamic Key Management add-on product installed, you can manually change the MAC key or PIN key in an ATM terminal using a network control command entered from a network control facility. The

network control commands to generate and download new PIN or MAC keys to ATM terminals are documented in the ***BASE24 Text Command Reference Manual***.

For Diebold 10XX/478X and NCR 5XXX terminals, these commands can also be entered from Device Control Terminal (DCT) screens as well as from a network control facility. For more information on the DCT screens used to download keys to Diebold 10XX/478X and NCR 5XXX terminals, refer to the ***BASE24 Device Control Manual***.

Any time you manually change an ATM terminal PIN or MAC key using a network control command, all the dynamic key management counters and the key change timestamp for the terminal are reset.

Operations Considerations

Whenever new MAC and PIN keys are downloaded to a Diebold 10XX/478X or NCR 5XXX terminal, the BASE24-atm Device Handler process sends a shutdown message to the ATM prior to downloading the new keys. Shutting down an ATM closes the terminal, taking it out of service. After the new keys are successfully downloaded, the Device Handler process sends a startup message to the terminal to reopen the terminal, placing it back in service. If a message to load a new PIN or MAC key fails, the device will remain closed until a load is successfully sent to the terminal.

The Tidel and Triton Device Handler processes do not issue shutdown commands. New keys are sent in a transaction response rather than a download message.

BASE24-pos Dynamic Key Management

The following paragraphs describe the dynamic key management options for the BASE24-pos product. Currently, dynamic key management is supported for the SPDH Device Handler module only.

Note: BASE24-pos dynamic key management requires the Transaction Security Services process for the SPDH Device Handler module. The SPDH Device Handler module cannot back out a failed key exchange without using the Transaction Security Services process. When using the Transaction Security Services process, the SECURE-DEV1 and SECURE-DEV2 assigns or the SECURE-DEV-xx assigns in the BASE24 Logical Network Configuration File must be set to the process pair directory (PPD) name of a Transaction Security Services process. In addition, the SECURE-DEV-TYP LCONF param for these processes must also be set to a value of TSS.

The SECURE-DEV-ACCESS-TYP param indicates whether a BASE24 application process communicates with a primary and secondary Transaction Security Services process identified by the SECURE-DEV1 and SECURE-DEV-2 assigns or with up to 20 Transaction Security Services processes identified by SECURE-DEV-xx assigns, where *xx* is any two characters. The SECURE-DEV1 assign specifies the primary Transaction Security Services process, while the SECURE-DEV2 assign specifies the secondary Transaction Security Services process. The SECURE-DEV-xx assigns specify up to 20 Transaction Security Services processes accessed on a *round-robin* (i.e., rotating) basis.

Dynamic Key Management

BASE24-pos Device Handler modules use working keys to protect the transaction data moving through the BASE24 network. Currently, dynamic key management is supported for the SPDH Device Handler module only. The installation and changing of these keys is known as key management, and can be performed manually or automatically. Automated key management is often known as dynamic key management.

Dynamic key management enables you to automatically change working keys based on a number of factors. Thresholds can be established based on time, the number of transactions, number of errors, number of consecutive errors, or any combination of these factors. One or more working keys can be changed when any of the defined thresholds is surpassed. Working keys can also be changed using a network control command.

Dynamic key management requires the use of security modules to perform PIN or data encryption, PIN verification, and message authentication, and is not supported when these functions are performed in software.

Working Keys

SPDH Device Handler modules can use three working keys, which are described in the paragraphs that follow.

PIN Key — If a PIN is encrypted coming from an SPDH-compliant device, this is the key under which it is encrypted. If a PIN is in the clear coming from the device and must be encrypted by BASE24, this is the key under which it is to be encrypted.

MAC Key — If inbound messages from the terminal contain a message authentication code (MAC), this is the key the SPDH Device Handler module uses to verify the MAC. If outbound messages for the terminal contain a MAC, this is the key the module uses to generate the MAC.

Data Encryption Key — If a data field is encrypted in a message sent to the terminal, this is the key the SPDH Device Handler module uses to encrypt the data field.

Note: Data encryption keys are only supported when using the Transaction Security Services process.

Key Exchange Keys

Dynamic key management requires key exchange keys (KEKs) in addition to the working keys. Working keys must be encrypted before being exchanged between BASE24 and the terminal. Three KEKs are used to encrypt the working keys: one for the PIN key, one for MAC key, and one for the data encryption key. These KEKs must be manually exchanged and entered in the TSS database before dynamic key management can occur.

Atalla and SafeNet (Eracom) use the term *key exchange key* (KEK) to describe the keys used for exchanging PIN, MAC, or data keys. Thales e-Security uses the term *zone master key* (ZMK) to describe the key used for exchanging PIN keys and MAC keys. References to the key exchange key (KEK) in BASE24 documentation include both the Atalla, SafeNet (Eracom), and Thales e-Security keys.

Device Thresholds

Currently, dynamic key management is supported for SPDH-compliant terminals only. These thresholds are configured on the Key Thresholds screen of the ACI Standard Device Configuration File (ACNF). In addition, Secondary Settlement processes ensure that the PIN, MAC, and data encryption keys for an SPDH-compliant terminal are changed at least once during each terminal's business day. Secondary Settlement processes perform this check during terminal cutover processing.

BASE24-pos dynamic key management supports two types of thresholds for changing PIN keys and data encryption keys, and three types of thresholds for changing message authentication keys. One or more of the thresholds can be established for each type of key; however, dynamic key management only occurs when at least one threshold is established for each type of key.

The daily cutover processing check performed by Secondary Settlement processes is performed for each type of key if that type of encryption is supported in a security device for the terminal, regardless of whether any dynamic key thresholds are established.

Counters are maintained for each threshold for each SPDH-compliant terminal. When the first threshold is exceeded for a key type, the appropriate key (i.e., PIN key, MAC key, or data encryption key) is generated and the counter is reset. These thresholds are set on the Key Thresholds screen of the ACI Standard Device Configuration File (ACNF).

BASE24-pos dynamic key management thresholds are based on the following information:

- Length of time (i.e., the daily cutover processing check used with all three key types)
- Number of transactions (used with all three key types)
- Number of errors (used with all three key types)
- Number of consecutive errors (used with MAC keys only)

The length-of-time threshold is based on the terminal business day. If a key type (PIN, MAC, or data) is used in a security device, it must be changed at least once per cutover period. The remaining thresholds use counters to track the number of transactions or errors that have occurred since the last key change. The

circumstances that cause the length-of-time threshold or counter to be increased or reset depend on the threshold being monitored. These thresholds are described in the following subsections.

Daily Cutover Threshold

The length-of-time threshold is based solely on whether a key has been changed since the terminal was last cut over to a new business day. The time is not increased based on transaction activity and this threshold is not checked during transaction processing. When the daily cutover timer expires for a terminal, a Secondary Settlement process determines whether the PIN key, MAC key, and data encryption key have been changed since the last terminal cutover. This check is only performed if the key is used in transaction processing within a security device. For example, if PIN encryption is not used for a terminal or if it is used in software only, this check is not performed for the PIN key.

For each key that is checked and has not been changed since the last terminal cutover, the process sets a flag in the terminal's BASE24-pos Terminal Data files (PTD) record to indicate that a new key is required. These flags are checked in the next transaction response message sent to the terminal, and the appropriate working keys are sent to the terminal in the response. This ensures that the PIN key, MAC key, and data encryption key are changed at least once during each terminal's business day. Once a new key has been downloaded to the terminal, the flag for that key is updated in the terminal's PTD record to indicate that a key change is not pending for the indicated key. A new key can be downloaded due to the setting of the key pending flag in the PTD or because another threshold has been exceeded.

Transaction Thresholds

Each of the transaction thresholds is based on the number of transactions that have occurred since its corresponding transaction counter was set. Each time a counter is increased, it is compared to its threshold. When the counter exceeds its threshold, the SPDH Device Handler module sets the applicable key change pending flag in the terminal's PTD record. This flag is checked in the next transaction response message sent to the terminal, and the corresponding working key is sent to the terminal in the response. For each threshold, a counter can be maintained for each of the three working keys.

Number of Transactions

The message content determines which transaction counter is increased by the SPDH Device Handler module:

- The PIN transaction counter is increased each time a message is received from the terminal containing an encrypted PIN.
- The message authentication transaction counter is increased each time the SPDH Device Handler module processes a message for the terminal that requires MAC verification or MAC generation.
- The data key transaction counter is increased each time a message is processed for the terminal that requires data encryption or decryption.

A transaction counter is reset in the following situations:

- When the SPDH Device Handler module is initialized.
- When a counter has reached its threshold and the new key is downloaded to the terminal.
- When terminal cutover is performed and the new key is downloaded to the terminal.

Number of Errors

The message content determines which error counter is increased by the SPDH Device Handler module:

- The PIN error counter is increased each time a key synchronization error occurs while attempting to validate a PIN. When the SPDH Device Handler module receives an authorization response indicating that the transaction was denied due to a PIN synchronization error, it increases the PIN error counter.
- The message authentication error counter is increased each time an external message for the terminal requires MAC verification or MAC generation and the verification or generation attempt fails due to a synchronization error.
- The data key error counter is increased each time an error occurs while attempting to encrypt or decrypt a data field in a message for the terminal.

An error counter is reset in the following situations:

- When the SPDH Device Handler module is initialized.
- When a counter has reached its threshold and the new key is downloaded to the terminal.
- When terminal cutover is performed and the new key is downloaded to the terminal.

Number of Consecutive MAC Errors

The message authentication consecutive error counter is increased each time the SPDH Device Handler module processes a message for the terminal that requires MAC verification or MAC generation and the verification or generation attempt fails. The consecutive MAC error counter is increased without regard to whether the check digits received in the response from the security module match the check digits maintained in the TSS database.

A consecutive MAC error counter is reset in the following situations:

- When the SPDH Device Handler module is initialized.
- When a counter has reached its threshold and the new key is downloaded to the terminal.
- When terminal cutover is performed and the new key is downloaded to the terminal.
- When the current MAC key it is associated with is used successfully.

Device Threshold Configuration

Settings in the BASE24-pos Terminal Data files (PTD) and the ACI Standard Device Configuration File (ACNF) must be considered when using dynamic key management. Refer to the ***BASE24-pos Files Maintenance Manual*** and the ***BASE24-pos Standard POS Device Configuration Manual*** for detailed information on the screens and fields presented in this section.

If you are using the Transaction Security Services process, you must also consider settings on the Channel Security Configuration window of the ACI Desktop user interface. Refer to the ***BASE24 Transaction Security Services Processing Guide*** for detailed information on this window.

PTD Settings

For dynamic key management thresholds to be used for PIN master, MAC master, and data encryption keys, the corresponding encryption type field in the PTD must be set to perform the associated security operation in a security device as follows:

- For dynamic key management of the PIN key, the PIN ENCRYPTION TYPE field on PTD screen 7 must be set to a value of 03 (DES algorithm (security device, PIN/PAD PIN block) or 04 (DES algorithm (security device, ANSI standard PIN/PAN PIN block, 12 rightmost digits of PAN excluding the check digit).
- For dynamic key management of the MAC key, the MAC TYPE field on PTD screen 7 must be set to a value of 03 (Security module message authentication).
- For dynamic key management of the data encryption key, the DATA ENCRYPTION TYPE field on PTD screen 7 must be set to a value of 01 (ASCII Encryption) or 02 (Binary Encryption).

In addition, the exchange keys used to encrypt working keys for transmission to a terminal, as well as the working keys themselves must be stored in the BASE24-pos Terminal Data files or in the Channel Security Configuration File (CSECD). The CSECD is maintained from the Channel Security Configuration window on the ACI Desktop user interface. If you are using the Transaction Security Services process, the keys are stored in the CSECD. In this case, the PIN MASTER KEY and MAC MASTER KEY fields on PTD screen 7 must be set to a key locator value (i.e., the terminal ID) for the terminal.

ACNF Settings

The dynamic key management transaction thresholds for an SPDH terminal must be set on the Key Thresholds screen. You must set a transaction threshold field to a nonzero value for that threshold to be used. Refer to the *BASE24-pos Standard POS Device Configuration Manual* for a detailed description of this screen and its available fields.

16: BASE24 Interface Key Management

This section describes the dynamic key management options provided by the following interfaces:

- BASE24 Interchange (BIC) ISO Interface process
- BASE24 ISO Host Interface process

The information presented includes the database requirements for each option and describes the network management messages for dynamic key management, including the information contained in the Cryptographic Service Message (CSM) element of the ISO external messages.

Note: Other Interchange Interface processes provided by ACI support varying types of key management depending on the requirements of the interchange. Each interchange typically dictates the type of key management that its processors must use. For information on Interchange Interface process support of message authentication, refer to the interchange-specific documentation to determine how, and if, message authentication is supported.

In reviewing the information presented in this section, note that the term *external processor* can refer to a data processing center (DPC) or a co-network as defined in the “Conventions Used in this Manual” section of this manual.

Dynamic Key Management

BASE24 Host Interface and BIC Interface processes use working keys to protect the transaction data moving through the BASE24 network. The installation and changing of these keys is known as key management, and can be performed manually or automatically. Automated key management is often known as dynamic key management.

Dynamic key management allows an external processor to automatically change working keys based on a number of factors. Thresholds can be established based on time, number of transactions, number of errors, number of consecutive errors, or any combination of these factors. One or more working keys can be changed when any of the defined thresholds is surpassed. Working keys can also be changed using a network control command.

BASE24 dynamic key management is based on the ANSI X9.17 standard, *Financial Institution Key Management (Wholesale)* and the ISO 8732 standard, *Banking—Key Management (Wholesale)*.

Dynamic key management requires the use of security modules to perform PIN encryption, PIN verification, and message authentication, and is not supported when these functions are performed in software.

ISO Host Interface Thales e-Security HSM Encryption Method

If you are using Thales e-Security HSMs without the Transaction Security Services process, the LCONF param RACAL-ANSI-METHOD specifies whether to use the ANSI X9.17 encryption method for key exchanges instead of the default variant encryption method used by the ISO Host Interface and BIC ISO Interface processes. If this param is set to Y, these interface processes operate using the ANSI X9.17 encryption method. If this param is set to N or is not present, these interface processes use the variant encryption method.

This param does not apply when using the Transaction Security Services process because you can configure the encryption method used in the TSS database.

Working Keys

BASE24 Host Interface and BIC Interface processes use four working keys, which are described in the paragraphs that follow.

Inbound PIN Encryption Keys — If a PIN is encrypted coming from the external processor, this is the key under which it is encrypted. If a PIN is in the clear coming from the external processor and must be encrypted by BASE24, this is the key under which it is to be encrypted.

Outbound PIN Encryption Keys — If PIN encryption is required by the external processor, this is the key under which the PIN is to be encrypted by BASE24.

Inbound Message Authentication Keys — If inbound messages from the external processor contain a message authentication code (MAC), this is the key the BASE24 interface process uses to verify the MAC.

Outbound Message Authentication Keys — If outbound messages for the external processor contain a MAC, this is the key the BASE24 interface process uses to generate the MAC.

Key Exchange Keys

Dynamic key management requires key exchange keys (KEKs) in addition to the working keys. Working keys must be encrypted before being exchanged between BASE24 and the external processor. Two KEKs are used to encrypt the working keys: one for PIN encryption working keys and one for message authentication working keys. These KEKs must be manually exchanged and entered in the BASE24 database before dynamic key management can occur.

Atalla uses the term *key exchange key* (KEK) to describe the keys used for exchanging PIN keys and MAC keys. Thales e-Security uses the term *zone master key* (ZMK) to describe the key used for exchanging PIN keys and MAC keys. References to the key exchange key (KEK) in BASE24 documentation include both the Atalla and Thales e-Security keys.

Key Management Processors

Key management involves two parties: the BASE24 interface process and the external processor. A co-network is the external processor for a BIC Interface process and a DPC is the external processor for a BASE24 Host Interface process. Each party has specific key management responsibilities.

The processor type assigned to each party identifies its responsibilities when performing key management, whether key management is performed manually or automatically. There are three possible processor types, each of which is described below:

Main Key Processor — The main key processor controls all key generation. When one key management party is the main processor, the other party must be the secondary processor. The main processor can generate a key itself or in response to a request from the secondary processor.

Secondary Key Processor — The secondary key processor has no responsibility for key generation. When one key management party is the secondary processor, the other party must be the main processor. The secondary processor cannot generate a key, but can initiate a key change by requesting the main processor to generate a new key. The secondary processor can, however, accept or reject a key generated by the main processor.

Co-Network Key Processor — A co-network key processor shares key management responsibilities with another co-network key processor. When one key management party is a co-network, the other party also must be a co-network.

The distinction between inbound and outbound keys is important in this configuration. Each party is responsible for generating the key used as its outbound key. The key that serves as the inbound key for one co-network serves as the outbound key for the other co-network, as shown below.

In this example, co-network 1 (BASE24) is responsible for generating key A because co-network 1 uses it for outbound messages. Co-network 2 (external processor) is responsible for generating key B because co-network 2 uses it for outbound messages.

Key	Co-network 1 (BASE24)	Co-network 2 (external processor)
A	Outbound key	Inbound key
B	Inbound key	Outbound key

Note: The term co-network typically applies to the BIC Interface external processor. When defining key processing responsibilities, however, the term co-network applies to BASE24 and the DPC as well.

Key Management Messages

Key management messages make up a subset of network management (0800) messages. There are four key management messages, each of which is described below.

Note: The change key and new key messages can be generated automatically as the result of thresholds being surpassed. Thresholds are described in the topic “Dynamic Key Management Thresholds” presented later in this section. In addition, all four of these messages can be initiated by an operator issuing a network control command. For more information on using network control commands to generate key management messages, and for complete syntax of the commands involved, refer to the *BASE24 Text Command Reference Manual*.

Change Key Message — A change key message requests that a processor generate a new PIN encryption or message authentication key. The key-generating processor sends a response to the change key message and then, in a separate action, generates the new key and sends it to the requesting processor in a new key message.

The requesting processor sends this type of message to a key-generating processor when the requesting processor has determined that the key needs to be changed. The requesting processor can make that determination based on a threshold being passed or in response to a KEY CHANGE command (the KEY CHANGE command has the same effect as passing a threshold).

New Key Message — A new key message carries a new PIN encryption or message authentication key between processors. The key-using processor verifies and begins using the new key and returns a response to the key-generating processor indicating that the new key has been implemented.

A key-generating processor sends this type of message to a key-using processor when the key-generating processor has either determined that the key needs to be changed or been requested by the key-using processor to change or repeat the key. This can occur as a result of a threshold being passed, notification from the key-using processor (using a change key or repeat key message), or in response to a KEY CHANGE command (the KEY CHANGE command has the same effect as passing a threshold).

Repeat Key Message — A repeat key message requests that a key-generating processor resend a key that was generated earlier. Upon receipt of a repeat key message, the key-generating processor obtains the requested key from its database and sends it to the key-using processor in a new key message. The key itself is not changed, nor are any threshold counters affected.

A BASE24 interface process that is the key-using processor generates the repeat key message only as the result of receiving a KEY REPEAT command. A BASE24 interface process that is the key-generating processor responds to a repeat key message from the key-using processor with a new key message carrying the requested key.

Verify Key Message — A verify key message requests that a key-generating processor verify the key by comparing check digits. The verify key message contains check digits for the key in question, and the processor receiving the message compares the check digits in the message with the check digits it has on file for the particular key. The key-generating processor then responds to the verify key message with the results of the check digit comparison. (If the check digits do not match, the key-using processor that sent the verify key message should use a change key message to request a new key.)

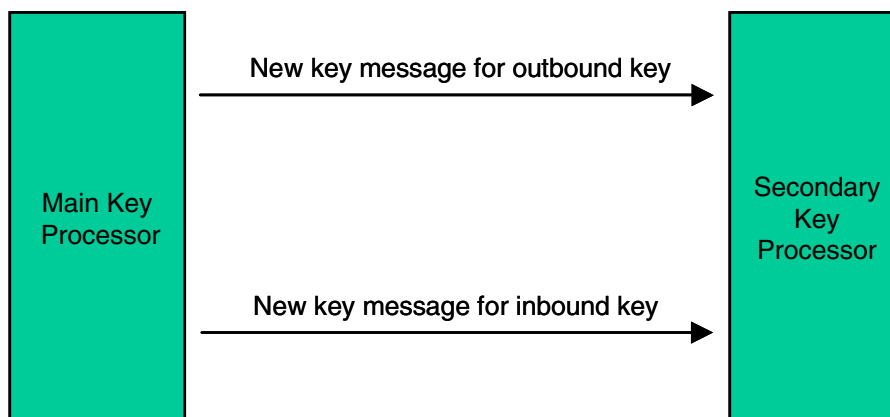
A BASE24 interface process that is the key-using processor generates the verify key message only as the result of receiving a KEY VERIFY command. A BASE24 interface process that is the key-generating processor responds to a verify key message from the key-using processor by comparing check digits for the specified key and returning the results.

Processing Key Management Messages

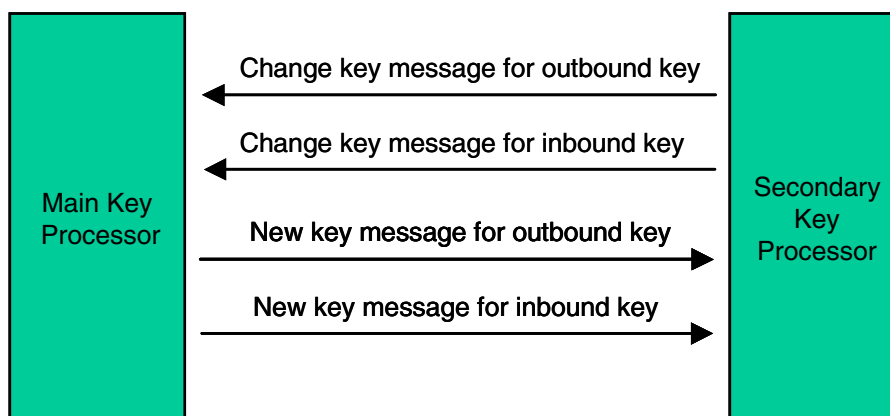
All four key management messages can be requested for the inbound key only, the outbound key only, or both keys. However, the key management messages generated by the interface processes contain only one key, so a request affecting both keys results in two key management messages being generated if the inbound and outbound keys are separate and unique. If the keys are the same, as set up in the NUMBER OF KEYS field on KEYF or KEY6 screen 1 and the KEY PROCESSING TYPE field on KEYF screen 3 or KEY6 screen 4, both the outbound and inbound keys are replaced with one key. If the external processor is a co-network, the inbound and outbound keys must be separate and unique. The NUMBER OF KEYS and KEY PROCESSING TYPE fields in the KEYF or KEY6 must be set in a consistent manner.

This processing is demonstrated in the following example. In this case, an interface process receives a KEY CHANGE command for both (inbound and outbound) keys.

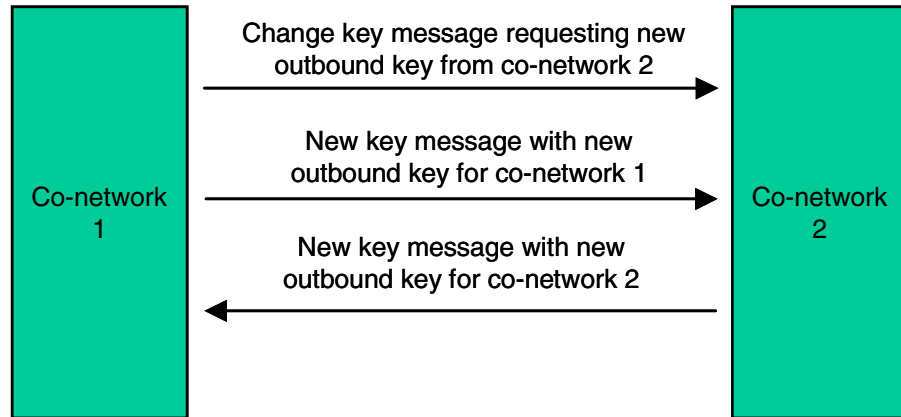
When the interface process is the main key processor and is configured for two keys, it sends two new key messages to the secondary key processor as shown below: one for the key used by the main key processor as its outbound key and one for the key used by the main key processor as its inbound key.



When the interface process is the secondary key processor and is configured for two keys, it sends two change key messages to the main key processor as shown below: one for the key used by the secondary key processor as its outbound key and one for the key used by the secondary key processor as its inbound key. The main key processor responds by sending a new key message for each key.



When the interface process is a co-network key processor and is configured for two keys, it sends a change key message and a new key message to the second co-network key processor as shown below. The change key message is a request to change the key used as an outbound key by the second co-network key processor. The new key message contains the new key to be used as an outbound key by the first co-network key processor. The second co-network receives the change key message and responds by sending a new key message that contains the new outbound key used by the second co-network key processor.



Note: When an acquiring Host or Interchange Interface process generates a new key using the Transaction Security Services process, the new key is not activated (i.e., the current index for the key is changed to point to the new key) until after the Host or Interchange Interface process receives a response from the secondary key processor or co-network indicating that the new key was successfully processed. This processing was included in a critical fix to TSS release 02.4 and applies to any acquirer Host or Interchange Interface process that provides this functionality.

Network Management Messages

Key management messages are network management messages. Each key management message is a request or a response. For ISO messages, the request is an 0800 message and the response is an 0810 message. The table on the following page describes the ISO message elements used to carry the data required for key management messages.

For more information on ISO messages, refer to the following manuals:

- ***BASE24 ISO Host Interface Manual***
- ***BASE24 BIC ISO Interface Manual***

- *BASE24 BIC ISO Standards Manual*
- *BASE24 External Message Manual*

Data Element	Description
Network Management Information Code (S-70)	Identifies the type of key management messages being sent. Valid values are as follows: 161 = Change key 162 = New key 163 = Repeat key 164 = Verify key
Response Code (P-39)	Identifies whether the key management process is successful. Valid values are as follows: 00 = Approved 05 = Declined 12 = Declined—invalid check digits 91 = DPC down

Data Element	Description
Network Management Message Additional Data (P-48)	Includes the following six fields: Number of Keys — Indicates the number of keys supported by the interface process. Valid values are as follows: 1 = Combined keys (inbound and outbound keys are equal) $\varnothing$, 0, or 2 = Separate keys ($\varnothing$ means blank) Key Length — Indicates the type of key management supported by the interface process. Valid values are as follows: 0 or 1 = Single-length key exchange keys (KEKs) 2 = Double-length key exchange keys (KEKs) Key Processor Type — Indicates if the interface process is the main, secondary, or co-network key processor. Valid values are as follows: 0 or N = None or not applicable C = Co-network M = Main S = Secondary MAC Type — Indicates the level of MACs supported by the interface process. Valid values are as follows: $\varnothing$ or 0 = No MAC support ($\varnothing$ means blank) 1 = Hardware MAC Support 2 = Software MAC support MAC Data Type — Indicates the character set in which the data will be authenticated. Valid values are as follows: $\varnothing$ or 0 = ASCII ($\varnothing$ means blank) 1 = EBCDIC
Network Management Message Additional Data (P-48) <i>continued</i>	MAC Key Length — Indicates the type of MAC keys supported by the interface process. Valid values are as follows: 0 or 1 = Single-length MAC keys 2 = Double-length MAC keys

Data Element	Description
Security-Related Control Information (P-53)	Includes the following two fields: Key Type — Identifies whether the key being exchanged is used for message authentication or PIN encryption. Valid values are as follows: 00 = PIN key 01 = MAC key Key Direction — Identifies whether the key being exchanged is used for inbound messages, outbound messages, or both. Valid values are as follows: 01 = Outbound messages only 02 = Inbound messages only 03 = Both inbound and outbound messages
BASE24 Key Management (S-120)	Contains the check digits for the key being exchanged.
Cryptographic Service Message (S-123)	Contains the Cryptographic Service Message (CSM), which is used for moving key management data between processors. This message is discussed later in this section.
Primary Message Authentication Code (P-64)	Contains the MAC if message authentication is performed and secondary data elements are not included in the message.
Secondary Message Authentication Code (S-128)	Contains the MAC if message authentication is performed and secondary data elements are included in the message.

Cryptographic Service Message

The ANSI X9.17 standard, *Financial Institution Key Management (Wholesale)*, establishes standards for key management. This standard also defines the Cryptographic Service Message (CSM) used for moving key management data between processors when the keys are distributed automatically. CSM information is contained in the Cryptographic Service Message data element (S-123) of the ISO external message.

Key Distribution Environment

BASE24 uses the point-to-point key distribution environment. In this environment, two parties share a key encryption key so that further data keys can be exchanged and one or both of the parties has the capability to generate or otherwise acquire keys.

Message Classes

BASE24 supports the following classes of CSMs used in point-to-point environments:

- Request Service Initiation Message (RSI)
- Key Service Message (KSM)
- Response Service Message (RSM)
- Error Service Message (ESM)

Message Fields and Subfields

Each CSM class contains several fields, with many of the fields containing subfields. Some fields and subfields are mandatory while others are optional. Because of this flexibility, each field begins with a unique 2- or 3-character identifier and ends with a blank. The CSM itself always begins with the literal CSM, and the message contents are always carried between a pair of parentheses.

The fields shown in the following table are the only ones that appear in the CSMs created by BASE24. There are five CSM formats because the ESM class can have two formats, depending on the type of error being reported. Each of the CSM formats is described in greater detail, following the table.

CSM Class	RSI	KSM	RSM	ESM	
				With Counts	Without Counts
CSM Fields	MCL RCV ORG SVR	MCL RCV ORG KD CTP	MCL RCV ORG	MCL RCV ORG CTP CTR ERF	MCL RCV ORG ERF

When the BASE24 interface process receives CSMs generated by an external processor, it searches only for the fields shown in the preceding table. It is recommended, but not mandatory, that the fields in the message be kept in the order shown in the table.

Message Formats

BASE24 dynamic key management processing uses the CSM to move key information between the BASE24 and the external processor in three of the four key management messages described earlier in this section: Change Key messages, New Key messages, and Repeat Key messages. The verify key message does not use the CSM because it only moves check digits and does not need the additional information contained in the CSM.

The following describes and presents examples of the BASE24 supported CSM classes:

Request Service Initiation Message (RSI) — Used when a key change request is being passed between processors. An example of a CSM containing the RSI format is shown below:

CSM (MCL/RSI/RCV/1234567890123456/ORG/1234567890123456/SVR)

Each field in the message is preceded by one of the following identifiers. The blanks identified by the character *b* in the example are required in the message because they act as field terminators. The lengths given in the following field descriptions are for the data in the field, excluding field identifiers and trailing blanks.

MCL/ = Message class (RSI)
Field Length: 3 alphabetical characters

RCV/ = Receiver of the message
Field Length: 4–16 alphanumeric characters

ORG/ = Originator of the message
Field Length: 4–16 alphanumeric characters

SVR/ = Service request
Field Length: 0

Note: The absence of a value following this identifier implies a request for one data key.

Key Service Message (KSM) — Used when a new key is passed between processors. An example of a CSM containing the KSM format for a double-length key is shown below:

**CSM (MCL/KSMbRCV/1234567890123456bORG/1234567890123456b
KD/12345678901234561234567890123456bCTP/12345678901234b)**

The information in this example is actually a continuous line. It has been broken here due to space constraints. Each field in the message is preceded by one of the following identifiers. The blanks identified by the character *b* in the example are required in the message because they act as field terminators. The lengths given in the following field descriptions are for the data in the field, excluding field identifiers and trailing blanks.

MCL/ = Message class (KSM)
Field Length: 3 alphabetical characters

RCV/ = Receiver of the message
Field Length: 4–16 alphanumeric characters

ORG/ = Originator of the message
Field Length: 4–16 alphanumeric characters

KD/ = New key value, generated by the security module for a new key request and taken from the database for a repeat key message
Field Length: 16, 32, or 48 hexadecimal characters

CTP/ = Hexadecimal key counter
Field Length: 1–14 hexadecimal characters

Note: For the BASE24 Interchange (BIC) ISO Interface, this field must contain at least six bytes populated with hexadecimal digits. If the field contains more than six bytes of data, the interface uses only the last six bytes.

Response Service Message (RSM) — Used to respond to a KSM that is processed successfully. An example of a CSM containing the RSM format is shown below:

CSM (MCL/RSM/RCV/1234567890123456bORG/1234567890123456b)

Each field in the message is preceded by one of the following identifiers. The blanks identified by the character *b* in the example are required in the message because they act as field terminators. The lengths given in the following field descriptions are for the data in the field, excluding field identifiers and trailing blanks.

MCL/ = Message class (RSM)
Field Length: 3 alphabetical characters

RCV/ = Receiver of the message
Field Length: 4–16 alphanumeric characters

ORG/ = Originator of the message
Field Length: 4–16 alphanumeric characters

Error Service Message (ESM) — Used to respond to a Request Service Initiation Message (RSI) or a Key Service Message (KSM) that cannot be processed successfully. The ESM can have two formats, depending on the type of error being reported. The two counter fields are only included in the message when the expected key count (CTP field) and the key count received from the message originator (CTR field) do not match. The error code (ERF field) in the examples below can contain only alphabetical characters. Numbers are used to demonstrate field length.

An example of a CSM containing the ESM format with the key counters is shown below:

```
CSM(MCL/ESMbRCV/1234567890123456bORG/1234567890123456b  
CTP/12345678901234bCTR/12345678901234bERF/1234567890123456b)
```

An example of a CSM containing the ESM format without the key counters is shown below:

```
CSM(MCL/ESMbRCV/1234567890123456bORG/1234567890123456b  
ERF/1234567890123456b)
```

The information in each of these examples is actually a continuous line. It has been broken here due to space constraints. Each field in the messages is preceded by one of the following identifiers. The blanks identified by the character *b* in the example are required in the message because they act as field terminators. The lengths given in the following field descriptions are for the data in the field, excluding field identifiers and trailing blanks.

- MCL/ = Message class (ESM)
Field Length: 3 alphabetical characters
- RCV/ = Receiver of the message
Field Length: 4–16 alphanumeric characters
- ORG/ = Originator of the message
Field Length: 4–16 alphanumeric characters
- CTP/ = Hexadecimal key count expected by the receiver of the KSM.
This identifier and field are only included in the message when error code P is returned.
Field Length: 1–14 hexadecimal characters

Note: For the BASE24 Interchange (BIC) ISO Interface, this field must contain at least six bytes populated with hexadecimal digits. If the field contains more than six bytes of data, the interface uses only the last six bytes.

CTR/ = Hexadecimal key count originally sent in the KSM by the message originator. This identifier and field are only included in the message when error code P is returned.

Field Length: 1–14 hexadecimal characters

Note: For the BASE24 Interchange (BIC) ISO Interface, this field must contain at least six bytes populated with hexadecimal digits. If the field contains more than six bytes of data, the interface uses only the last six bytes.

ERF/ = The following codes identify the errors that were detected during processing.

C = Cannot process (general error)

F = Format error

H = Invalid receiver ID or originator ID

P = The value in the CTP field of the KSM does not match the expected count

Field Length: 1-16 alphabetical characters

Message Length

The use of multiple CSM formats makes the CSM a variable-length element in the ISO external message. The CSM can also be a fixed-length element in the ISO external message.

The BIC ISO Interface process uses the value in the FIXED LENGTH FORMAT field on screen 12 of the Interchange Configuration File (ICF) or Enhanced Interchange Configuration File (ICFE) to determine whether the ISO message is fixed- or variable-length.

The BASE24 ISO Host Interface process uses the values in the MESSAGE FORMAT fields on all applicable screens of the Host Configuration File (HCF). If any of these fields indicates that fixed-length messages are used, the fixed-length format is used for the CSM. If all of these fields indicate that variable-length messages are used, the variable-length format is used for the CSM.

Fixed-Length Messages

When the CSM is used as a fixed-length element, the data is left-justified within each field of the CSM format being used and each field is blank filled, if necessary, to obtain its maximum length. The CSM format is then left-justified in the external message data element and the data element is blank filled to reach the specified length.

In the fixed-length ISO message, the CSM (S-123) has a length of 150 bytes, excluding the 3-position field length indicator. Although the total length of the CSM is 150 bytes, the longest CSM is currently 114 bytes. Therefore, positions 115 through 150 of the CSM always contain blanks.

The ESM can have two lengths, depending on whether it includes the two key count fields (CTP and CTR). The key count fields are only included when the error code field (ERF) contains a P (counts do not match). Neither key count field is included when the error code is a value other than P.

CSM Format	RSI	KSM	RSM	ESM	
				With Counts	Without Counts
CSM Data Length	60	110	55	114	76

Variable-Length Messages

When the CSM is used as a variable-length element, the length of each field in the CSM is determined by the data it carries. The message format field descriptions presented earlier in this section include the range of possible lengths for each field.

All fields are required except for the key count fields (CTP and CTR) in the ESM. The key count fields are conditional in the ESM, depending on whether the error code field (ERF) contains a P (counts do not match). Neither key count field is included when the error code is a value other than P. Data element S-123 can have the following lengths (excluding the 3-position field length indicator), depending on the CSM format it contains.

CSM Format	RSI	KSM	RSM	ESM	
				With Counts	Without Counts

Minimum Length	36	57	31	49	37
Maximum Length	60	110	55	114	76

Examples

The following examples show an RSM when the receiver and originator are less than the maximum length of 16 alphanumeric characters. The first example illustrates a variable-length format and the second example illustrates a fixed-length format. The variable-length RSM shown has a length of 36 characters while the fixed-length RSM shown maintains its length of 55 characters because each field is padded with blanks. Blanks are identified by the character *b* in the examples.

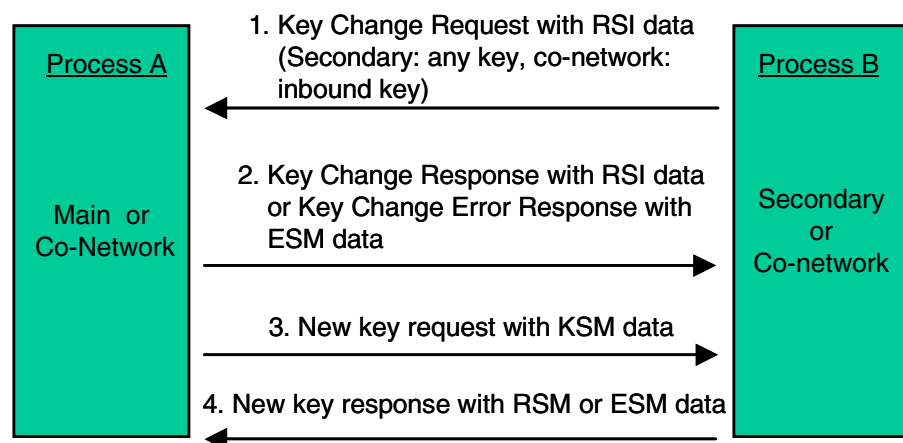
```
CSM(MCL/ESMbRCV/ABC123bORG/DEFG456b)
CSM(MCL/RSMbRCV/ABC123bbbbbbbbbbbbORG/DEFG456bbbbbbbbbb)
```

Message Flow Diagrams

The following subsections present message flow diagrams for key change, new key, and repeat key transactions.

Key Change

The following diagram illustrates how messages containing CSM formats are used in a key change request.

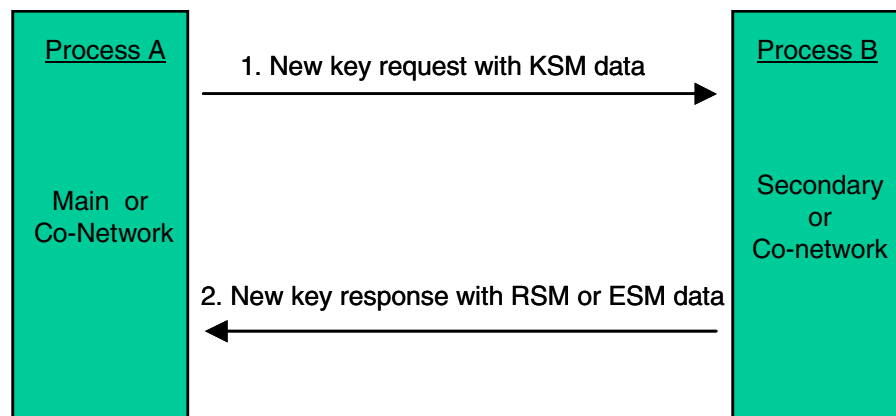


1. The process requesting a key change (process B) sends a request message containing an RSI to the process responsible for generating the key (process A).

2. Process A responds to process B. This response message notifies process B that the key change request has been received, but does not return the new key. The CSM in the response contains the RSI echoed from the original key change request when no errors are detected in the request and the new key can be generated. The CSM in the response contains an ESM when errors are found in the request and the new key cannot be generated.
3. Process A sends a request message containing a KSM with the new key information to process B.
4. Process B responds to process A. This response message contains an RSM when no errors are detected in the KSM or an ESM when errors are found in the KSM.

New Key

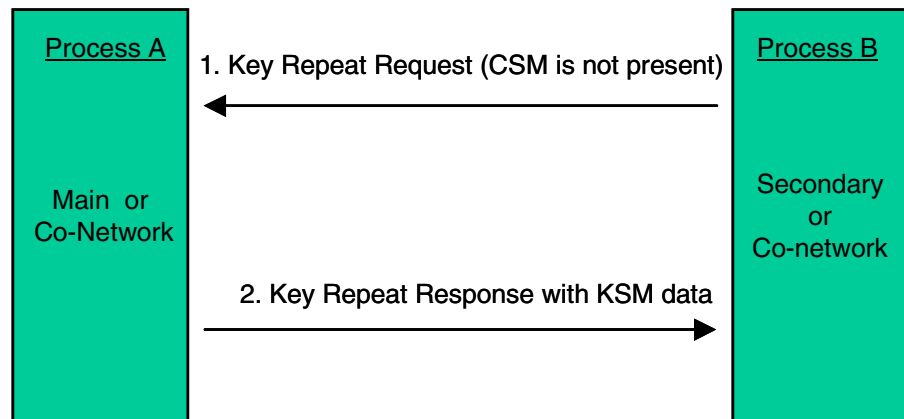
The following diagram illustrates how messages containing CSM formats are used when a new key is passed between processors.



1. The process generating the key (process A) sends a request message containing a KSM with the new key information to the receiving process (process B) without receiving a request from process B.
2. Process B responds to process A. This response message contains an RSM when no errors are detected in the KSM or an ESM when errors are found in the KSM.

Repeat Key

The following diagram illustrates how messages containing CSM formats are used when a key is repeated.



1. The process that originally received the key (process B) sends a key repeat request message to the process that originally generated the key (process A). The CSM is not present in this message unless this element is configured in the External Message File (EMF) as a mandatory element.
2. Process A sends a response message containing a KSM with the existing key information to process B. Process A does not change the key or any key-related counters before sending the response. Since the KSM is in a response, process B cannot return an RSM or ESM.

Note: This command may not work on systems using Thales e-Security security modules (HSMs). The repeat key command uses the GC transaction code. In some HSM firmware or software versions, the GC transaction code may be available only when the HSM is in an authorized state. The HSM should not be in an authorized state during normal online processing. Check with your Thales e-Security customer support representative for more information regarding the capabilities of your HSM and its firmware or software.

Dynamic Key Management Thresholds

BASE24 dynamic key management supports four types of thresholds for changing PIN encryption keys and five types of thresholds for changing message authentication keys. One or more of the thresholds can be established for each type of key; however, dynamic key management only occurs when at least one threshold is established for each type of key. One set of thresholds is used for both inbound and outbound keys and counters are maintained for each. Therefore, if the keys are separate (i.e., the NUMBER OF KEYS field on KEYF or KEY6 screen 1 is set to a value of 2), two sets of counters are maintained. When the first threshold is exceeded, the appropriate key (i.e., PIN key or MAC key) is generated. The first threshold can be exceeded by the secondary process or by the co-network and its key will be generated. These thresholds are set on KEYF screen 3 or KEY6 screen 4.

BASE24 dynamic key management thresholds are based on the following information:

- Length of time (used with both key types)
- Number of transactions (used with both key types)
- Number of errors (used with both key types)
- Number of consecutive errors (used with both key types)
- Number of synchronization errors (used with message authentication keys only)

The length-of-time thresholds use timers to track the amount of time that has elapsed since the last key exchange. The remaining thresholds use counters to track the number of transactions or errors that have occurred since the last key exchange. The circumstances that cause a timer or counter to be increased or reset depend on the threshold being monitored. These thresholds are described in the following subsections.

Note: When new PIN encryption and message authentication working keys are both being generated, the message authentication working keys must be generated first so they can be available for generating the MAC in the new PIN key messages.

Time Threshold

The length-of-time threshold is based solely on the amount of time that has elapsed since the timer was set. The time is not incremented based on transaction activity and this threshold is not checked during transaction processing. When this timer expires, the interface process sends a key change message or a new key message for the key in question to the external processor. A timer can be maintained for each of the four working keys (i.e., inbound PIN encryption, outbound PIN encryption, inbound message authentication, and outbound message authentication) or, if the inbound and outbound keys are the same, one timer can be maintained for PIN encryption keys and one for message authentication keys.

A timer is reset in the following situations:

- When the interface process is initialized or receives a warmboot command. A check is performed at logon and logoff to determine if the time limit has changed and if the timer should be deleted, added, or remain as is (not actually reset).
- When the interface process receives a new key request message for the key from the external processor and successfully processes the message.
- When the interface process receives an approved new key response message from the external processor for a key the interface process sent.

Transaction Thresholds

Each of the transaction thresholds is based on the number of transactions that have occurred since its corresponding transaction counter was set. Each time a counter is incremented, it is compared to its threshold. When the counter exceeds its threshold, the interface process sends a key change message or a new key message for the key in question to the external processor. For each threshold, a counter can be maintained for each of the four working keys or, if the inbound and outbound keys are the same, one counter is maintained for PIN encryption keys and one for message authentication keys.

Number of Transactions

The message content and direction determine which transaction counter is incremented by the interface process:

- The inbound PIN transaction counter is incremented each time an external message is received from an external processor containing an internal or external encrypted PIN.
- The outbound PIN transaction counter is incremented each time an internal message is received from the interface process containing an internal or external encrypted PIN.
- The inbound message authentication transaction counter is incremented each time the external message received from an external processor requires MAC verification.
- The outbound message authentication transaction counter is incremented each time a MAC must be generated before the external message is sent to an external processor.

A transaction counter is reset in the following situations:

- When the interface process is initialized.
- When the interface process receives a new key request message for the key from the external processor and successfully processes the message.
- When the interface process receives an approved new key response message from the external processor for a key the interface process sent.
- When a counter has reached its threshold and a key change message is sent.

Number of Errors

The message content and direction determine which total error counter is incremented by the interface process:

- The inbound PIN total error counter is incremented each time an error occurs while attempting to process an encrypted PIN contained in the external message received from an external processor. This occurs when the external PIN is encrypted and the internal PIN is in the clear.
- The outbound PIN total error counter is incremented each time an error occurs while attempting to generate an encrypted PIN from the internal message to the external message to send to the external processor.

- The inbound message authentication total error counter is incremented each time the external message received from an external processor requires MAC verification and the verification attempt fails, and the check digits received in the response from the security module match the check digits maintained in the BASE24 database.

This counter is also increased when the NUMBER OF KEYS field on KEYF or KEY6 screen 1 is set to a value of 1 (the same key is used for inbound and outbound messages), the message is rejected with a reject code of 197, and the message is not a MAC key change message.

- The outbound message authentication total error counter is increased when the NUMBER OF KEYS field on KEYF or KEY6 screen 1 is set to a value of 1 (the same key is used for inbound and outbound messages), the message is rejected with a reject code of 197, and the message is not a MAC key change message.

A total error counter is reset in the following situations:

- When the interface process is initialized.
- When the interface process receives a new key request message for the key from the external processor and successfully processes the message.
- When the interface process receives an approved new key response message from the external processor for a key the interface process sent.
- When a counter has reached its threshold and a key change message is sent.

Number of Consecutive Errors

The message content and direction determine which consecutive error counter is incremented by the interface process:

- The inbound PIN consecutive error counter is incremented each time an error occurs while attempting to process an encrypted PIN contained in the external message received from an external processor. This occurs when the external PIN is encrypted and the internal PIN is in the clear.
- The outbound PIN consecutive error counter is incremented each time an error occurs while attempting to process an encrypted PIN contained in the external message received from an external processor.
- The inbound message authentication consecutive error counter is incremented each time the external message received from an external processor requires MAC verification and the verification attempt fails. The consecutive error

counter is incremented without regard to whether the check digits received in the response from the security module match the check digits maintained in the BASE24 database.

This counter is also increased when the NUMBER OF KEYS field on KEYF or KEY6 screen 1 is set to a value of 1 (the same key is used for inbound and outbound messages), the message is rejected with a reject code of 197, and the message is not a MAC key change message.

- The outbound message authentication consecutive error counter is increased when the NUMBER OF KEYS field on KEYF or KEY6 screen 1 is set to a value of 1 (the same key is used for inbound and outbound messages), the message is rejected with a reject code of 196 or 197, and the message is not a MAC key change message.

A consecutive error counter is reset in the following situations:

- When the interface process is initialized.
- When the interface process receives a new key request message for the key from the external processor and successfully processes the message.
- When the interface process receives an approved new key response message from the external processor for a key the interface process sent.
- When a counter has reached its threshold and a key change message is sent.
- When the key it is associated with is used successfully. For outbound MAC messages, this is when a response is received.
- When an approved response is received from a request that contained a hardware-encrypted PIN.

Number of Synchronization Errors

The message content and direction determine which message authentication key (KMAC) error counter is incremented by the interface process:

- The inbound KMAC error counter is incremented each time the external message received from an external processor requires MAC verification, the verification attempt fails, and the check digits received in the response from the security module do not match the check digits maintained in the BASE24 database.

This counter is also increased when the NUMBER OF KEYS field on KEYF or KEY6 screen 1 is set to a value of 1 (the same key is used for inbound and outbound messages), the message is rejected with a reject code of 196, and the message is not a MAC key change message.

- The outbound KMAC error counter is increased when the NUMBER OF KEYS field on KEYF or KEY6 screen 1 is set to a value of 1 (the same key is used for inbound and outbound messages), the message is rejected with a reject code of 196, and the message is not a MAC key change message.

A KMAC error counter is reset in the following situations:

- When the interface process is initialized.
- When the interface process receives a new key request message for the key from the external processor and processes it successfully.
- When the interface process receives an approved new key response message from the external processor for a key the interface process sent.
- When a counter has reached its threshold and a key exchange message is sent.

Setting Up the BASE24 Database

Settings in the Key File (KEYF), External Message File (EMF), Host Configuration File (HCF) and Interchange Configuration File (ICF) or Enhanced Interchange Configuration File (ICFE) must be considered when using dynamic key management. Refer to the *BASE24 Base Files Maintenance Manual* for illustrations of all the screens presented in this section.

Key File (KEYF) or Key 6 File (KEY6) Settings

The following settings are needed in the Key File (KEYF) or Key 6 File (KEY6) for each external processor using dynamic key management.

KEYF or KEY6 Screen 1

KEYF or KEY6 screen 1 contains PIN encryption, PIN translation, and message authentication parameters, intermediate keys, PIN key exchange keys, and message authentication code (MAC) key exchange keys. The following fields are required when using key management.

ENCRYPT TYPE — Specifies the type of PIN encryption used or expected by the external processor.

- If any of the PIN key timer fields on KEYF screen 3 or KEY6 screen 4 are set to a nonzero value, this field must be set to a value of 1 (security module PIN management).
- If the CLEAR OLD KEY TIMER VALUE field on KEYF screen 3 or KEY6 screen 4 contains a nonzero value or the KEY PROCESSING TYPE field on KEYF screen 3 or KEY6 screen 4 contains a value other than N (N indicates no dynamic key management is to be performed), this field or the MAC ENCRYPT TYPE field on KEYF or KEY6 screen 1 must be set to a value of 1.

PIN BLOCK FORMAT — Specifies the PIN block format of the PIN in the inbound and outbound external messages.

- If dynamic key management is used, which requires the use of security modules for PIN encryption, this field must be set to a value of 1 (ANSI PIN block) or 3 (PIN/PAD PIN block).

- If dynamic key management is not used for PIN keys, this field must be set to a value of 0 (PINs are in the clear).

MAC ENCRYPT TYPE — Specifies how message authentication will be performed for messages between the external processor and the interface process.

- If any of the MAC key timer fields on KEYF screen 3 or KEY6 screen 4 are set to a nonzero value, this field must be set to a value of 1 (security module MAC processing).
- If the CLEAR OLD KEY TIMER VALUE field on KEYF screen 3 or KEY6 screen 4 contains a nonzero value or the KEY PROCESSING TYPE field on KEYF screen 3 or KEY6 screen 4 contains a value other than N (N indicates no dynamic key management is to be performed), this field or the ENCRYPT TYPE field on KEYF or KEY6 screen 1 must be set to a value of 1.

NUMBER OF KEYS — Identifies whether the inbound and outbound keys defined on KEYF screen 2 or on KEY6 screen 2 and 3 are combined or separate. The value in this field is used when PIN or MAC keys are exchanged automatically. The same number of keys must be used for PINs and MACs.

- If the same key is used for inbound and outbound messages, this field must be set to a value of 1.
- If different keys are used for inbound and outbound messages or the KEY PROCESSING TYPE field on KEYF screen 3 or KEY6 screen 4 is set to a value of C (co-network), this field must be set to a value of 2.

KEY LENGTH — Identifies whether single- or double-length Key Exchange Keys (KEKs) are used with this external processor. The value in this field is used when PIN and MAC keys are exchanged using dynamic key management. The PIN KEK and MAC KEK must be the same length. Single-length keys contain 16 characters and double-length keys contain 32 characters.

- If single-length KEKs are used for key exchanges, this field must be set to a value of 1.
- If double-length KEKs are used for key exchanges, this field must be set to a value of 2.

EXCHANGE KEYS

The following fields must be set as indicated to create keys that BASE24 uses when MAC keys are automatically exchanged between BASE24 and the external processor.

PIN KEY — The security module-encrypted form of the Key Exchange Key (KEK) used when inbound and outbound PIN keys are exchanged using dynamic key management. This two-part field contains a single- or double-length KEK, depending on the value in the KEY LENGTH field. Valid values for each character in these fields are 0 through 9 and A through F. Both fields must contain 16 valid characters. If the value in the KEY LENGTH field is 1, the second set of characters must be zeros.

PIN KEY CHECK DIGITS — The check digits associated with the PIN KEK. Valid values for each character in this field are 0 through 9 and A through F. This field must contain four valid characters. A value of 0000 in this field indicates check digits are not being used with PIN KEKs.

MAC KEY — The security module-encrypted form of the Key Exchange Key (KEK) used when inbound and outbound MAC keys are exchanged using dynamic key management. This two-part field contains a single- or double-length KEK, depending on the value in the KEY LENGTH field. Valid values for each character in these fields are 0 through 9 and A through F. Both fields must contain 16 valid characters. If the value in the KEY LENGTH field is 1, the second set of characters must be zeros.

MAC KEY CHECK DIGITS — The check digits associated with the MAC KEK. Valid values for each character in this field are 0 through 9 and A through F. This field must contain four valid characters. A value of 0000 in this field indicates check digits are not being used with MAC KEKs.

KEYF Screen 2 or KEY6 Screens 2 and 3

KEYF screen 2 or KEY6 screens 2 and 3 consists of inbound and outbound working keys for PINs and MACs. Descriptions of the fields required are as follows.

PIN KEY INFORMATION

The fields in the PIN KEY INFORMATION section on KEYF or KEY6 screen 2 contain working keys for PINs.

OUTBOUND KEYS

The following fields contain the PIN keys used for outbound transactions (that is, transactions sent by the interface from BASE24 to the external processor).

PIN KEY1 — The key used for the encryption or translation of PINs in transactions sent to the external processor. The interface process uses the key in this field when the CURRENT INDEX field contains a value of 1. Valid values for each position in this field are 0 through 9 and A through F. This field must contain 16 valid characters.

Note: On KEY6 screen 2, this field occurs twice. Each field must contain 16 valid characters to define a double-length PIN encryption key.

CHECK DIGITS1 — The check digits corresponding to the value in the PIN KEY1 field. Valid values for each position in this field are 0 through 9 and A through F. This field must contain four valid characters.

The check digits can be obtained from the utility used to encrypt the key.

CURRENT INDEX — Indicates which outbound PIN encryption key is currently being used by the interface process. Valid values are as follows:

- 1 = Value in the PIN KEY1 field.
- 2 = Value in the PIN KEY2 field.

PIN KEY2 — The key used for the encryption or translation of PINs in transactions sent to the external processor. The interface process uses the key in this field when the CURRENT INDEX field contains a value of 2. Valid values for each position in this field are 0 through 9 and A through F. This field must contain 16 valid characters.

Note: On KEY6 screen 2, this field occurs twice. Each field must contain 16 valid characters to define a double-length PIN encryption key.

CHECK DIGITS2 — The check digits corresponding to the value in the PIN KEY2 field. Valid values for each position in this field are 0 through 9 and A through F. This field must contain four valid characters.

The check digits can be obtained from the utility used to encrypt the key.

KEY COUNTER — The number of times the outbound PIN key has been changed by dynamic key management processing.

INBOUND KEYS

The following fields contain the PIN keys used for inbound transactions (that is, transactions sent by the external processor to BASE24).

PIN KEY1 — The key used by the external processor to encrypt PINs in the transaction sent to BASE24. The interface process uses the key in this field when the CURRENT INDEX field contains a value of 1. Valid values for each position in this field are 0 through 9 and A through F. This field must contain 16 valid characters.

Note: On KEY6 screen 2, this field occurs twice. Each field must contain 16 valid characters to define a double-length PIN encryption key.

CHECK DIGITS1 — The check digits corresponding to the value in the PIN KEY1 field. Valid values for each position in this field are 0 through 9 and A through F. This field must contain four valid characters.

The check digits can be obtained from the utility used to encrypt the key.

CURRENT INDEX — Indicates which inbound PIN encryption key is currently being used by the interface process. Valid values are as follows:

- 1 = Value in the PIN KEY1 field.
- 2 = Value in the PIN KEY2 field.

PIN KEY2 — The key used by the external processor to encrypt PINs in the transaction sent to BASE24. The interface process uses the key in this field when the CURRENT INDEX field contains a value of 2. Valid values for each position in this field are 0 through 9 and A through F. This field must contain 16 valid characters.

Note: On KEY6 screen 2, this field occurs twice. Each field must contain 16 valid characters to define a double-length PIN encryption key.

CHECK DIGITS2 — The check digits corresponding to the value in the PIN KEY2 field. Valid values for each position in this field are 0 through 9 and A through F. This field must contain four valid characters.

The check digits can be obtained from the utility used to encrypt the key.

KEY COUNTER — The number of times the inbound PIN key has been changed by dynamic key management processing.

MAC KEY INFORMATION

The fields in the MAC KEY INFORMATION section on KEYF screen 2 or KEY6 screen 3 contain working keys for message authentication codes (MACs).

OUTBOUND KEYS

The following fields contain working keys for message authentication codes (MACs).

Note: If an external processor supports automatic key changes, the MAC KEY1, MAC KEY2, CURRENT INDEX, CHECK DIGITS1, and CHECK DIGITS2 fields will automatically be set by the interface process when it is first started.

MAC KEY1 — The key used by the BASE24 interface process to generate MACs in messages from BASE24 to the external processor. The interface process uses the key in this field when the CURRENT INDEX field contains a value of 1. Valid values for each position in this field are 0 through 9 and A through F. This field must contain 16 valid characters.

Note: On KEY6 screen 3, this field occurs twice. Each field must contain 16 valid characters to define a double-length MAC key.

CHECK DIGITS1 — The check digits corresponding to the value in the MAC KEY1 field. Valid values for each position in this field are 0 through 9 and A through F. This field must contain four valid characters.

The check digits can be obtained from the utility used to encrypt the key.

CURRENT INDEX — Indicates which outbound MAC key is currently being used by the interface process. Valid values are as follows:

- 1 = Value in the MAC KEY1 field.
- 2 = Value in the MAC KEY2 field.

MAC KEY2 — The key used by the BASE24 interface process to generate MACs in messages from BASE24 to the external processor. The interface process uses the key in this field when the CURRENT INDEX field contains a value of 2. Valid values for each position in this field are 0 through 9 and A through F. This field must contain 16 valid characters.

Note: On KEY6 screen 3, this field occurs twice. Each field must contain 16 valid characters to define a double-length MAC key.

CHECK DIGITS2 — The check digits corresponding to the value in the MAC KEY2 field. Valid values for each position in this field are 0 through 9 and A through F. This field must contain four valid characters.

The check digits can be obtained from the utility used to encrypt the key.

KEY COUNTER — The number of times the outbound MAC key has been changed by dynamic key management processing.

INBOUND KEYS

The following fields contain the MAC keys used in inbound transactions (that is, transactions sent by an external processor to BASE24).

Note: If an external processor supports automatic key changes, the MAC KEY1, MAC KEY2, CURRENT INDEX, CHECK DIGITS1, and CHECK DIGITS2 fields will automatically be set by the interface process when it is first started.

MAC KEY1 — The key used by the BASE24 interface process to verify MACs generated by the external processor. The interface process uses the key in this field when the CURRENT INDEX field contains a value of 1. Valid values for each position in this field are 0 through 9 and A through F. This field must contain 16 valid characters.

Note: On KEY6 screen 3, this field occurs twice. Each field must contain 16 valid characters to define a double-length MAC key.

CHECK DIGITS1 — The check digits corresponding to the value in the MAC KEY1 field. Valid values for each position in this field are 0 through 9 and A through F. This field must contain four valid characters.

The check digits can be obtained from the utility used to encrypt the key.

CURRENT INDEX — Indicates which inbound MAC key is currently being used by the interface process. Valid values are as follows:

- 1 = Value in the MAC KEY1 field.
- 2 = Value in the MAC KEY2 field.

MAC KEY2 — The key used by the BASE24 interface process to verify MACs generated by the external processor. The interface process uses the key in this field when the CURRENT INDEX field contains a value of 2. Valid values for each position in this field are 0 through 9 and A through F. This field must contain 16 valid characters.

Note: On KEY6 screen 3, this field occurs twice. Each field must contain 16 valid characters to define a double-length MAC key.

CHECK DIGITS2 — The check digits corresponding to the value in the MAC KEY2 field. Valid values for each position in this field are 0 through 9 and A through F. This field must contain four valid characters.

The check digits can be obtained from the utility used to encrypt the key.

KEY COUNTER — The number of times the inbound MAC key has been changed by dynamic key management processing.

KEYF Screen 3 or KEY6 Screen 4

KEYF screen 3 or KEY6 screen 4 contains the fields used to set the thresholds for dynamic key management. Descriptions of the fields required are as follows. If all limits on KEYF screen 3 or KEY6 screen 4 are set to zero, the interface process using the KEYF record does not initiate key changes automatically based on timers or counters, so the external processor must be responsible for initiating any key changes if dynamic key management is to occur. At least one PIN key limit on KEYF screen 3 or KEY6 screen 4 must be set to a nonzero value for the BASE24 interface process to initiate dynamic key management for PINs and at least one

MAC key limit on KEYF screen 3 or KEY6 screen 4 must be set to a nonzero value for the BASE24 interface process to initiate dynamic key management for MACs.

PIN KEY VARIANT — Specifies the key variant to be applied to the PIN key exchange key (KEK) when translating PIN keys.

- If the BASE24 interface process and the external processor use Atalla hardware security modules, this field must be set to a value of 1 (variant 1).
- If the BASE24 interface process uses an Atalla hardware security module and the external processor uses a Thales e-Security hardware security module, this field must be set to a value of 1 (variant 1).
- If the BASE24 interface process uses a Thales e-Security hardware security module and the external processor uses an Atalla security module, this field must be set to a value of 1 (variant 1).
- If the BASE24 interface process and the external processor use Thales e-Security hardware security modules, this field can be set to a value of 0 (variant 0) or 1 (variant 1); however, the BASE24 interface process and the external processor must use the same setting.

Note: When using the Transaction Security Services process, this field is hardcoded to a value of 1, regardless of the hardware security module used by BASE24 or the external processor.

MAC KEY VARIANT — Specifies the key variant to be applied to the MAC key exchange key (KEK) when translating MAC keys. Regardless of the hardware security module used by BASE24 or the external processor this field must be set to a value of 3 (variant 3).

Note: When using the Transaction Security Services process, this field is hardcoded to a value of 3.

LIMITS

The following fields are used to set the parameters for dynamic key management for this interface. The parameters specify when the key used for PIN encryption and translation should be changed and when the key used for message authentication should be changed. Both types of keys can be changed based on time intervals, transaction activity, the number of errors since the last key change, or the number of consecutive errors. The MAC key can also be changed based on the number of key synchronization errors since the last key change.

PIN KEY TIMER VALUE — Specifies the maximum length of time that a PIN working key should be used, if a time limit is being used to determine how often the key is changed.

The time period can be expressed in minutes, hours, or days, depending on the value in the PIN KEY TIMER INTERVAL field. The shortest length of time is five minutes. Valid values are as follows:

0, 5–1500	If the value in the PIN KEY TIMER INTERVAL field is M (minutes)
0–1000	If the value in the PIN KEY TIMER INTERVAL field is H (hours)
0–180	If the value in the PIN KEY TIMER INTERVAL field is D (days)

BASE24 checks the values in the PIN KEY TIMER VALUE and PIN KEY TIMER INTERVAL fields when the KEYF or KEY6 record is added or updated. If the time is evenly divisible by a longer interval, BASE24 changes the value in the PIN KEY TIMER INTERVAL field to the longer interval and redisplay the value in the PIN KEY TIMER VALUE field expressed in the longer interval. For example, an entry of 60 minutes is automatically changed to an entry of 1 hour and an entry of 48 hours is automatically changed to an entry of 2 days. However, an entry of 61 minutes, even though it is greater than one hour, is not changed because it is not evenly divisible by 60.

The interface process does not use this parameter if the value in this field is set to a value of 0. If the ENCRYPT TYPE field on KEYF or KEY6 screen 1 is set to any value other than 1 (hardware PIN encryption), this field must be set to a value of 0.

MAC KEY TIMER VALUE — Specifies the maximum length of time that a MAC working key should be used, if a time limit is being used to determine how often the key is changed.

The time period can be expressed in minutes, hours, or days, depending on the value in the MAC KEY TIMER INTERVAL field. The shortest length of time is five minutes. Valid values are as follows:

0, 5–1500	If the value in the MAC KEY TIMER INTERVAL field is M (minutes)
0–1000	If the value in the MAC KEY TIMER INTERVAL field is H (hours)
0–180	If the value in the MAC KEY TIMER INTERVAL field is D (days)

BASE24 checks the values in the MAC KEY TIMER VALUE and MAC KEY TIMER INTERVAL fields when the KEYF or KEY6 record is added or updated. If the time is evenly divisible by a longer interval, BASE24 changes the value in the MAC KEY TIMER INTERVAL field to the longer interval and redisplay the value in the MAC KEY TIMER VALUE field expressed in the longer interval. For example, an entry of 60 minutes is automatically changed to an entry of 1 hour and

an entry of 48 hours is automatically changed to an entry of 2 days. However, an entry of 61 minutes, even though it is greater than one hour, is not changed because it is not evenly divisible by 60.

The interface process does not use this parameter if the value in this field is set to 0. If the MAC ENCRYPT TYPE field on KEYF or KEY6 screen 1 is set to any value other than 1 (hardware message authentication), this field must be set to 0.

PIN KEY TIMER INTERVAL — A code specifying the unit of time used with the value in the PIN KEY TIMER VALUE field to set the maximum length of time that a PIN working key should be used. BASE24 changes the value in the PIN KEY TIMER INTERVAL field to the largest possible unit of time when the KEYF or KEY6 record is added or updated. Refer to the PIN KEY TIMER VALUE field for additional information. Valid values are as follows:

D = Days
H = Hours
M = Minutes

MAC KEY TIMER INTERVAL — A code specifying the unit of time used with the value in the MAC KEY TIMER VALUE field to set the maximum length of time that a MAC working key should be used. BASE24 changes the value in the MAC KEY TIMER INTERVAL field to the largest possible unit of time when the KEYF record is added or updated. Refer to the MAC KEY TIMER VALUE field for additional information. Valid values are as follows:

D = Days
H = Hours
M = Minutes

PIN KEY TRAN — The maximum number of transactions that can be performed before changing the current PIN working key.

- If a total transaction limit is being used, this field must contain a value in the range of 50–100,000.
- If a total transaction limit is not being used, this field must contain a value of 0. If the ENCRYPT TYPE field on KEYF or KEY6 screen 1 is set to any value other than 1 (hardware PIN encryption), this field must be set to a value of 0.

MAC KEY TRAN — The maximum number of transactions that can be performed before changing the current MAC working key.

- If a total transaction limit is being used, this field must contain a value in the range of 50–100,000.
- If a total transaction limit is not being used, this field must contain a value of 0. If the MAC ENCRYPT TYPE field on KEYF or KEY6 screen 1 is set to any value other than 1 (hardware message authentication), this field must be set to a value of 0.

PIN KEY ERROR — The maximum number of PIN key-related errors that can occur with the present PIN working key before changing the key.

- If a PIN key error limit is being used, this field must contain a value in the range of 5–9999.
- If a PIN key error limit is not being used, this field must contain a value of 0. If the ENCRYPT TYPE field on KEYF or KEY6 screen 1 is set to any value other than 1 (hardware PIN encryption), this field must be set to a value of 0.

MAC KEY ERROR — The maximum number of MAC generation and verification errors that can occur with the present MAC working key before changing the key.

- If a MAC generation and verification error limit is being used, this field must contain a value in the range of 5–9999.
- If a MAC generation and verification error limit is not being used, this field must contain a value of 0. If the MAC ENCRYPT TYPE field on KEYF or KEY6 screen 1 is set to any value other than 1 (hardware message authentication), this field must be set to a value of 0.

CONSECUTIVE PIN KEY ERROR — The maximum number of consecutive PIN key-related errors that can occur with the present PIN working key before changing the key.

- If a consecutive PIN key error limit is being used, this field must contain a value in the range of 5–9999.
- If a consecutive PIN key error limit is not being used, this field must contain a value of 0. If the ENCRYPT TYPE field on KEYF or KEY6 screen 1 is set to any value other than 1 (hardware PIN encryption), this field must be set to a value of 0.

CONSECUTIVE MAC KEY ERROR — The maximum number of consecutive MAC generation and verification errors that can occur with the present MAC working key before changing the key.

- If a consecutive MAC generation and verification error limit is being used, this field must contain a value in the range of 5–9999.
- If a consecutive MAC generation and verification error limit is not being used, this field must contain a value of 0. If the MAC ENCRYPT TYPE field on KEYF or KEY6 screen 1 is set to any value other than 1 (hardware message authentication), this field must be set to a value of 0.

KMAC SYNCHRONIZATION ERROR — The maximum number of MAC working key synchronization errors that can occur with the present MAC working key before changing the key. Key synchronization is the process of comparing the check digits returned from a security module for the present MAC working key with the check digits stored in the KEYF or KEY6 record for the present MAC working key. These KEYF or KEY6 MAC working key check digit values are set in the CHECK DIGITS1 and CHECK DIGITS2 fields on KEYF screen 2 or KEY6 screen 3.

- If a MAC working key synchronization error limit is being used, this field must contain a value in the range of 3–9999.
- If a MAC working key synchronization error limit is not being used, this field must contain a value of 0. If the MAC ENCRYPT TYPE field on KEYF or KEY6 screen 1 is set to any value other than 1 (hardware message authentication), this field must be set to a value of 0.

CLEAR OLD KEY TIMER VALUE — The maximum length of time, in seconds, that an old key can be used before it is automatically cleared by the interface process.

The INBOUND PIN KEY, OUTBOUND PIN KEY, INBOUND MAC KEY, and OUTBOUND MAC KEY fields on KEYF screen 2 or on KEY screens 2 and 3 can contain information for two keys, identified as KEY1 and KEY2. The value in the CURRENT INDEX field for each type of key identifies which key (1 or 2) is currently in use. At the time a new key is placed in service, the other key becomes the old key. For example, if inbound PIN key number 1 is currently in use when the BASE24 interface process receives a new key, the new key is stored as inbound PIN key number 2. Inbound PIN key number 2 becomes the current key and inbound PIN key number 1 becomes the old key. (If inbound PIN key number 2 is the current key, it becomes the old key and the new key is stored as inbound PIN key number 1.)

When processing a transaction, BASE24 uses the new key for PIN processing or message authentication. If the security device used by BASE24 for PIN processing or message authentication detects an error because the external processor is still using the old key, it can repeat the PIN processing or message authentication using the old key before BASE24 declines the transaction. The value in this field controls how long the old key is available. If the old key is not available, a second PIN processing or message authentication attempt is not made.

- If the old key is to be cleared, this field must contain a value in the range of 1–9999 to specify the time, in seconds, that an old key can be used before being cleared.
- If the old key is not to be cleared, this field must contain a value of 0. If the ENCRYPT TYPE field on KEYF or KEY6 screen 1 or the MAC ENCRYPT TYPE field on KEYF or KEY6 screen 1 is set to any value other than 1, this field must be set to a value of 0.

ORIGINATING ID — An identification code placed in the originator field of the Cryptographic Service Message (CSM) in a message originated by BASE24. The values in the ORIGINATING ID and RECEIVING ID fields must be agreed on by the sender and receiver. The sender's originating ID is the receiver's receiving ID and the sender's receiving ID is the receiver's originating ID. These values do not need to match any specific values in the BASE24 database.

RECEIVING ID — An identification code placed in the receiver field of the Cryptographic Service Message (CSM) in a message originated by BASE24. The values in the ORIGINATING ID and RECEIVING ID fields must be agreed on by the sender and receiver. The sender's originating ID is the receiver's receiving ID and the sender's receiving ID is the receiver's originating ID. These values do not need to match any specific values in the BASE24 database.

KEY PROCESSING TYPE — A code identifying the type of dynamic key management processing this interface process can perform.

- If each interface process is responsible for generating its outbound working keys and can request its inbound working keys from the other interface process in the co-network, this field must be set to a value of C (co-network).
- If this interface process is responsible for all key generation, this field must be set to a value of M (main).
- If this interface process does not generate or receive keys, this field must be set to a value of N (none).

- If this interface process can request the main processor to generate keys but does not generate keys itself, this field must be set to a value of S (secondary).

Dynamic key management requires hardware security devices for PIN management and message authentication. If the ENCRYPT TYPE and MAC ENCRYPT TYPE fields on KEYF or KEY6 screen 1 both are set to values other than 1, meaning security devices are not being used, the value in this field must be set to N (none).

Dynamic key management in a co-network configuration requires the use of separate keys for inbound and outbound messages. This is because each co-network process must generate the key used for its outbound messages and two processes cannot share the responsibility for generating one key. If the NUMBER OF KEYS field on KEYF or KEY6 screen 1 is set to a value of 1, the value in this field cannot be set to a value of C (co-network).

NOTARIZATION SUPPORTED — A code identifying whether this interface process supports key notarization. This field must be set to a value of 0 (not supported) because BASE24 does not currently support key notarization.

External Message File (EMF) Settings

The following field on EMF screen 1 must be considered for network management messages when dynamic key management is used.

MESSAGE ELEMENTS — Identifies the elements contained in ISO messages.

- If default values are being used, an EMF record is not needed for network management (0800 and 0810) messages.
- If default values are not being used, the following elements must be defined as conditional in the EMF record for network management (0800 and 0810) messages:
 - P-39 = Response code
 - P-53 = PIN/MAC flag and inbound/outbound/both flag
 - S-70 = Key management message type
 - S-120 = Check digits for working key
 - S-123 = Cryptographic service message (CSM)

Host Configuration File (HCF) Settings

The following fields on HCF screen 1 must be considered for network management messages when dynamic key management is used.

These fields must be set regardless of the setting of the NMM ENABLED PROCESSING FLAG field on HCF screen 1. Dynamic key management messages are considered a type of network management message, but are not affected by the value in the NMM ENABLED PROCESSING FLAG field because the working keys they are used to manage are necessary for financial transactions.

TIMER LIMITS

The following fields are used to configure timer limits that are not product-specific.

NETWORK MANAGEMENT — Identifies the number of seconds that the BASE24 Host Interface process waits for a response after transmitting a network management message to an external processor station. Network management messages include echo tests, logons, logoffs, and dynamic key management messages.

If the time indicated in this field expires before a response message is received, the Host Interface process marks the station as down. Valid values are 0 through 9999.

EXTENDED NETWORK — Identifies the number of seconds the Host Interface process is to wait, once the initial message has timed out, before resending dynamic key management messages (i.e., new key and change key).

Interchange Configuration File (ICF) or Enhanced Interchange Configuration File (ICFE) Settings

The following fields on ICF or ICFE screen 3 must be considered for network management messages when dynamic key management is used.

These fields must be set regardless of the setting of the NMM ENABLED PROCESSING FLAG field on ICF or ICFE screen 3. Dynamic key management messages are considered a type of network management message, but are not affected by the value in the NMM ENABLED PROCESSING FLAG field because the working keys they are used to manage are necessary for financial transactions.

TIMER LIMITS

The following fields are used to configure timer limits that are not product-specific.

NETWORK MANAGEMENT — Identifies the number of seconds that the BIC Interface process waits for a response after transmitting a network management message to a co-network station. Network management messages include echo tests, logons, logoffs, and dynamic key management messages.

If the time indicated in this field expires before a response message is received, the BIC Interface process marks the station as down. Valid values are 0 through 9999.

EXTENDED NETWORK — Identifies the number of seconds the BIC Interface process is to wait, once the initial message has timed out, before resending dynamic key management messages (i.e., new key and change key).

17: Security Module Support

This section describes the usage of security modules with BASE24 and the functions of each that are supported by BASE24. The information a user requires for configuring specific modules for use with BASE24 is found in appendixes B and C of this manual.

BASE24 Security Module Usage

BASE24 allows for integrating different security modules into its routine transaction processing. It can utilize security modules to perform the following functions:

PIN Verification — Verifying PINs with a security module involves sending prescribed data from a transaction and the BASE24 database to the security module and allowing the module to verify the PIN. BASE24 Authorization processes can be configured to send the appropriate data required for PIN verification to the security module and accept the results returned by the module.

PIN Translation — Translating PINs with a security module involves the translation of PINs from encryption under one key to encryption under another key by the security module. BASE24 Host Interface processes and Interchange Interface processes can be set up to utilize security modules for PIN translation in cases where BASE24 handles PINs under different keys than the external processor with which it communicates.

PIN Changes — Changing PINs with a security module involves receiving old and new PINs entered by the customer at an ATM, verifying the old PIN, and generating PIN verification digits for the new PIN.

PIN verification digits include the following:

- The PIN offset for the IBM DES (3624-format) and Diebold PIN verification methods
- The PIN Verification Value (PVV) for the Visa PVV PIN verification method
- The PIN Verification Number (PVN) for the Identikey PIN verification method

The BASE24 database is updated by replacing the PIN verification digits for the old PIN with the PIN verification digits for the new PIN. A specific transaction, the PIN change transaction, is supported by some BASE24-atm Device Handler processes for this feature. Similar to PIN verification, BASE24-atm Authorization processes can be set up to utilize security modules to verify the existing PIN and generate PIN verification digits for the new PIN.

Message Authentication — Performing message authentication with a security module involves a two-step process. The first step is generating a message authentication code (MAC) based on the contents of a message. This MAC is

added to the message before it is sent. The second step is verifying the MAC when the message is received. The MAC calculation is repeated and the result is compared to the value received in the message. Device Handler processes, ISO Host Interface processes, and BASE24 Interchange (BIC) ISO Interface processes can use security modules to generate the MACs placed in outgoing messages and verify the MACs received in incoming messages.

For customers using the BASE24-atm or BASE24-pos EMV add-on products, issuer scripts carried within transaction messages are also secured with MACs generated by security modules.

Card Verification — Verifying cards with a security module involves comparing the card validation digits (CVD) carried in the magnetic stripe data of a plastic card with a value computed from other data carried in the magnetic stripe data. For BASE24-pos, the CVD can also be manually entered and compared with a value computed from other manually entered data. The CVD is called a card verification code (CVC) by MasterCard, a card verification value (CVV) by Visa, an integrated circuit card (ICC) card verification value (iCVD) by Europay, MasterCard, and Visa (EMV), and a card security code (CSC) by American Express. When manually entered, the CVD is referred to as the CVC2 for MasterCard and the CVV2 for Visa. When both an electronically entered CVD and manually entered CVD are present in a transaction, the electronically entered CVD is referred to as CVD1 and the manually entered CVD is referred to as CVD2. BASE24 Authorization processes can be set up to pass the magnetic stripe information, the manually entered information, or both to a security module and accept the card verification results returned by the security module.

Authorization Request Cryptogram Verification — When a transaction is sent online for a Europay, MasterCard, and Visa (EMV) integrated circuit card (ICC), the chip application on the card calculates an authorization request cryptogram, also known as an ARQC, using card and terminal data. The request cryptogram and the component cryptogram data used to calculate the cryptogram are sent to the issuer with the authorization request.

The issuer checks the request cryptogram, using the cryptogram data and a secure DES key, and calculates an authorization response cryptogram, also known as an ARPC, using response data. This response cryptogram is returned to the card by the acquirer. The chip application on the card can then confirm that the issuer is genuine. During BASE24 processing, one call is made to the security module to both verify the request cryptogram and generate the response cryptogram, although the ARPC must be regenerated if the transaction is subsequently declined. The security module can use EMV 1996 processing or EMV 2000

processing, depending on the key derivation method required. Standard CVV/CVC checking can also be applied as described above, depending on the setting of the CARD VERIFY - CHECK TYPE field on CPF screen 11.

Key Management — BASE24 requires key management by a security module in three situations.

- Generating encrypted transaction keys for a terminal. This is required in cases where PINs and keys are not carried in the clear within the BASE24 system and, thus, cannot be generated by BASE24 Device Handler processes.
- Generating new working keys for an external processor. Working keys can be provided by the BASE24 system or an external processor. A security module can be used to generate the working keys used by the BASE24 system and the external processor when the BASE24 system is responsible for providing them. The inbound and outbound keys carried in the Key File (KEYF) or Key 6 File (KEY6) are working keys.
- Translating working keys. When the BASE24 system and an external processor support online key changes, the BASE24 interface process must take a working key encrypted under the key exchange key (also known as a zone master key (ZMK)) for the external processor and obtain the working key encrypted under the security module master key. Key exchange keys for the external processor are carried in the PIN KEY EXCHANGE KEY and MAC KEY EXCHANGE KEY fields on KEYF or KEY6 screen 1.

BASE24 dynamic key management support automates the generation and translation of new working keys based on parameters such as length of time, number of transactions, number of errors, and number of consecutive errors. Dynamic key management is supported for BASE24 Interchange (BIC) Interface and BASE24 Host Interface processes. BASE24 ISO interface processes offer dynamic key management for PINs and MACs.

Supported Security Modules

BASE24 supports several models of the following security modules in its processing:

- Atalla Network Security Processors (NSPs)
- SafeNet (Eracom) security modules (ESMs)
- Thales e-Security Host Security Module (HSMs)

Each security module supports a subset of BASE24 transaction security processing. Specific support information has been outlined in appendixes to this manual for Atalla NSPs and Thales e-Security HSMs. Support information for SafeNet (Eracom) security modules (ESMs) is provided in the ***ERACOM Security Control Module (SCM) Reference Manual*** and the ***ERACOM Security Module (ESM) Reference Manual***.

While any of the above security modules can be used with BASE24, any one BASE24 process or transaction path can only use one type of security module. For example, an NSP and a Thales e-Security HSM cannot both be used by the same Authorization process. Also, a Device Handler process cannot use a Thales e-Security HSM when the Authorization process that receives transactions from the Device Handler process uses an NSP.

In a configuration with more than one BASE24 product (e.g., BASE24-atm and BASE24-pos), one product can use Thales e-Security HSMs while the other product uses NSPs or ESMs; however, ACI recommends that, whenever possible, one type of security module be used throughout the BASE24 configuration.

Security Module Access

If a security module is to be used by BASE24 for any transaction security processing, ACI recommends that at least two physical devices be configured for the system. This is particularly true in a production system where the malfunction of a security module could halt transaction processing if no backup is available.

Two different schemes are available for accessing Transaction Security Services processes in a BASE24 system, while one scheme is available when accessing security modules without the Transaction Security Services process in a BASE24 system.

When using the Transaction Security Services process, you can configure BASE24 processes to access a primary and secondary Transaction Security Services process, or you can configure BASE24 processes to access up to 20 Transaction Security Services processes on a *round-robin* (i.e., rotating) basis. If you are not using the Transaction Security Services process, you can only configure BASE24 processes to access a primary and secondary security module.

The SECURE-DEV-ACCESS-TYP param in the LCONF dictates the access scheme used for a process. If this param is set to a value of 1 for a BASE24 process, the process accesses up to 20 Transaction Security Services processes identified in SECURE-DEV-xx assigns on a rotating basis, where xx is any two characters. If this param is set to a value of 0 or is not configured for a BASE24

process, the process accesses a primary and secondary security module (or Transaction Security Services process) designated by the SECURE-DEV1 and SECURE-DEV2 assigns, respectively. Each of these security module access schemes is described in more detail below.

With both access methods, you can configure the maximum time a BASE24 process waits for a response from a security module (or Transaction Security Services process) before the request is considered timed out. After three timeouts, BASE24 considers the security module (or Transaction Security Services process) inoperative. The timeout limit is specified in the LCONF param SECURE-DEV-TIM-LMT.

Round-Robin Access

When using the round-robin method to access Transaction Security Services processes (i.e., the SECURE-DEV-ACCESS-TYP param is set to a value of 1), BASE24 processes access up to 20 Transaction Security Services processes on a rotating basis. The security utilities in each BASE24 process maintains a security device control block (SDCB), which indicates the availability of each Transaction Security Services process designated by a SECURE-DEV-xx assign and which Transaction Security Services process is to be used for the next transaction that requires a security module. If a Transaction Security Services process is marked as unavailable in the SDCB, the BASE24 process skips it and accesses the next Transaction Security Services process in the list. Thus, the BASE24 process cycles through all available Transaction Security Services processes as transaction processing progresses, so that all available Transaction Security Services processes are used about equally. When the BASE24 process has accessed the last Transaction Security Services process in the list, it accesses the first available Transaction Security Services process starting at the beginning of the list for the next transaction. With the round-robin access method, all BASE24 processes in a network can share the same list of Transaction Security Services processes and have the usage automatically distributed among them.

When a Transaction Security Services process is marked down, the security utilities in the BASE24 process use the LCONF param SECURE-DEV-REINSTATE-TIME to determine the number of minutes the Transaction Security Services process is excluded from the rotation when it is marked down before the security utilities attempt to reopen the Transaction Security Services process. A value of 0 for this param indicates that the process does not attempt to reopen a Transaction Security Services process based on the time interval. The maximum value for this param is 32000.

When a Transaction Security Services process is marked down, the BASE24 process uses the LCONF param SECURE-DEV-FAIL-NOTIFY to generate a critical event message at regular intervals to notify network operators of the Transaction Security Services process failure. The BASE24 process also attempts to reopen the failed Transaction Security Services process at the interval specified by this param. This param indicates the number of times the Transaction Security Services process comes up in the round-robin list before the BASE24 process attempts to reopen it as well as the interval at which critical event messages are generated.

Primary and Secondary Security Module Access

When the SECURE-DEV-ACCESS-TYP param is not configured or is set to a value of 0 for a BASE24 process, the BASE24 process accesses a primary and secondary security module (or Transaction Security Services process) specified by the SECURE-DEV1 and SECURE-DEV2 assigns, respectively.

For this access method, each BASE24 process that must access a security module (or Transaction Security Services process) must have a primary and secondary module (or Transaction Security Services process) defined for its use. ACI recommends that the two security modules used by a process be the same type (i.e., both be ESMs, both be NSPs, or both be Thales e-Security HSMs). While it is possible to intermix different security module types in primary and backup situations, it should not be done because each type of security module has unique features and processing requirements.

While each process requires a primary and secondary module (or Transaction Security Services process) for this access method, processes do not require exclusive access to a pair of security modules (or Transaction Security Services processes). For example, if an institution has three security modules (or Transaction Security Services processes) and four processes that require a security module (or Transaction Security Services process) for transaction processing, the configuration could be set up as shown below:

Process	Primary	Secondary
Authorization process	Module A or TSS process 1	Module B or TSS process 2
Device Handler process	Module C or TSS process 3	Module A or TSS process 1

Process	Primary	Secondary
Host Interface process	Module B or TSS process 2	Module C or TSS process 3
Interchange Interface process	Module C or TSS process 3	Module B or TSS process 2

When a transaction is received by a process that requires the use of a security module (or Transaction Security Services process), the process first attempts to communicate with its primary security module (or Transaction Security Services process). If the request to the primary security module (or Transaction Security Services process) times out or the security module (or Transaction Security Services process) is considered inoperative, the process attempts to communicate with its secondary security module (or Transaction Security Services process) to process the transaction.

Should both security modules (or Transaction Security Services processes) become inoperative, BASE24 generates critical event messages at regular intervals specified by the LCONF param SECURE-DEV-FAIL-NOTIFY to notify network operators of the failure. If you are using the Transaction Security Services process, the BASE24 process also attempts to reopen the failed Transaction Security Services processes at the interval specified by this param.

The BASE24 system also uses the LCONF param SECURE-DEV-MONITOR-THRESHOLD to provide the following two options when using the primary and secondary security module (or Transaction Security Services process) access scheme:

- Checking the secondary security module (or Transaction Security Services process) at specified intervals
- Balancing transactions between the primary and secondary security modules (or Transaction Security Services processes)

Checking the Secondary Security Module

To minimize the possibility of both security modules (or Transaction Security Services processes) becoming inoperative, the BASE24 system can be configured to occasionally perform one transaction using the secondary security module (or Transaction Security Services process) instead of the primary security module (or Transaction Security Services process). The transaction is performed using the

secondary security module (or Transaction Security Services process) at intervals based on the number of transactions performed by the primary security module (or Transaction Security Services process).

Each process using primary and secondary security devices (or Transaction Security Services processes) maintains its own transaction counter. The value specified in the SECURE-DEV-MONITOR-THRESHOLD param is the number of transactions processed by individual processes using the security device (or Transaction Security Services process), not the total number of transactions processed by the security device (or Transaction Security Services process). For example, if five BASE24 processes use a security device (or Transaction Security Services process) and the param is set to 1000, up to 5000 transactions (1000 by each of the five processes) could be processed by the primary security device (or Transaction Security Services process) before the secondary security device (or Transaction Security Services process) is used.

Occasionally using the secondary security module (or Transaction Security Services process) permits the identification and resolution of any problems with the secondary security module (or Transaction Security Services process) while the primary security module (or Transaction Security Services process) is functioning properly.

Balancing Transactions Between Security Modules

You can also use the SECURE-DEV-MONITOR-THRESHOLD param to balance the number of transactions processed by the primary and secondary security modules (or Transaction Security Services processes). For example, by setting this param to 1, each BASE24 process alternates between the primary and secondary security modules (or Transaction Security Services processes). Each BASE24 process uses the primary security module (or Transaction Security Services process) for one transaction, uses the secondary security module (or Transaction Security Services process) for one transaction, and then repeats the cycle.

Configuring Security Modules

Although the processing within each security module is proprietary to the vendor manufacturing the device, most such devices perform similar functions and are integrated into BASE24 processing in the same way.

All security modules used by BASE24 must be defined to the HP NonStop system configuration and the BASE24 logical network. Depending on the version level of the HP NonStop operating system, modules are defined to the HP NonStop system configuration using the HP NonStop SYSGEN configuration programs or the HP NonStop Dynamic System Configuration (DSC) utility. Modules are defined to each BASE24 logical network using the Logical Network Configuration File (LCONF).

Note: BASE24 application processes interface with security devices directly. Therefore, security modules need not be specified as devices or terminals in the Network Environment File (NEF).

For detailed information regarding configuration of specific security devices, refer to the following sources:

- The HP NonStop configuration manual specific to the method desired for the appropriate operating system used.
- The configuration manual provided by the device vendor.
- Appendix B, “Atalla Security Modules.”
- Appendix C, “Thales e-Security Host Security Modules.”
- The *BASE24 ERACOM Security Control Module (SCM) Reference Manual* and the *BASE24 ERACOM Security Module (ESM) Reference Manual*.

A: BASE24 Supercrypt Utility

BASE24 supports a Supercrypt utility that must be used to encrypt certain keys for storage in the BASE24 database. This utility accepts clear keys as input and provides encrypted key values for entry in the BASE24 database.

The keys encrypted in this manner are the INBOUND PIN KEYS and OUTBOUND PIN KEYS on KEYF or KEY6 screen 2; however, they only need to be encrypted in this manner when BASE24 requires clear PINs for its own processing and the external processor is using encrypted PINs.

Using the Supercrypt Utility

The remainder of this appendix describes how to use the BASE24 Supercrypt utility.

Starting the Supercrypt Utility

The BASE24 Supercrypt utility is a line mode utility that generates prompts and displays lines. This utility is run from a TACL prompt on a HP NonStop computer. The syntax for running the Supercrypt utility is described below.

Syntax

[RUN] [*\$volume.subvolume.*] SKCRYPT

volume — The name of the HP NonStop volume on which the Supercrypt utility is located. An operator must either move to the volume and subvolume on which the utility can be found, prior to entering the command, or fully qualify the name of the utility.

subvolume — The name of the HP NonStop subvolume on which the Supercrypt utility is located. An operator must either move to the volume and subvolume on which the utility can be found, prior to entering the command, or fully qualify the name of the utility.

Exiting the Supercrypt Utility

An operator can exit the utility by pressing the **Return** key or the **Control-Y** key in answer to any of its prompts.

Accessing the Supercrypt Utility

No menu is necessary to operate the Supercrypt utility because it is a single-function utility. The Supercrypt utility requests data for encryption immediately following startup, and repeats the request until the operator exits the utility.

Using the Supercrypt utility involves the following steps:

1. Enter the 16-digit number to be encrypted (e.g., 1234567890123456) at the “Enter data to be ENCRYPTED” prompt. Then press the **Return** key.

The utility replies with the following message, where the variable `xxxxxxxxxxxxxxxx` indicates the encrypted value (e.g., F1843DF1FBEC273A)

Your super-key ENCRYPTED input is `xxxxxxxxxxxxxxxx`

The utility then displays another “Enter data to be ENCRYPTED” prompt.

2. Enter another 16-digit number to be encrypted or press the **Return** key or the **Control-Y** keys to exit the utility.

Error Messages

The Supercrypt utility displays the following possible error messages:

- If alphabetic characters other than A through F are entered, the following message is displayed:

Error - must be valid HEX digits.

The “Enter data to be ENCRYPTED” prompt is then displayed again.

- If fewer than 16 digits are entered before the **Return** key is pressed, the following message is displayed:

Must be 16 digits, try again

Enter the data to be encrypted again.

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B: Atalla Security Modules

This appendix describes how Atalla security modules must be configured to work with BASE24. It does not describe the proprietary workings of any of the security modules, but rather the functions of each that are supported by BASE24 and the information a user will require for configuring the modules for use by BASE24.

BASE24 supports the following Atalla security modules in processing:

- Atalla Network Security Processor (NSP) model A8000
- Atalla Network Security Processor (NSP) model A8100
- Atalla Network Security Processor (NSP) model A8150
- Atalla Network Security Processor (NSP) model A9000
- Atalla Network Security Processor (NSP) model A9100
- Atalla Network Security Processor (NSP) model A9150
- Atalla Network Security Processor (NSP) model A10000E
- Atalla Network Security Processor (NSP) model A10100
- Atalla Network Security Processor (NSP) model A10150
- Atalla Network Security Processor (NSP) model A10160

The Axx100, Axx150, and Axx160 models are the next-generation NSPs offered by Atalla, which include a new hardware design. The A8xx0 series NSPs are marketed as entry-level devices, while the A9xx0 devices provide mid-level performance and the A10xx0 devices provide high-level performance.

All Axx100, Axx150, and Axx160 models can run using the older variant key format or the Atalla Key Block (AKB) format. These devices can be configured to support the variant format or the AKB format simply by installing the appropriate CD-ROM in the device and rebooting. All Axx100, Axx150, and Axx160 models come standard with an AKB software CD-ROM. Appending the letter V to the Axx100, Axx150, or Axx160 model number when ordered (i.e., A10100V, A10150V, A10160V, A9100V, A9150V, A8100V, and A8150V) ensures that the module is shipped from the factory with the variant software CD-ROM as well. If

the letter V is not appended to the model number when ordering, the Axx100, Axx150, or Axx160 model is shipped from the factory only with the standard AKB software CD-ROM.

All other Atalla models can also be ordered with variant or AKB version software preloaded. Appending the letters AKB to the Axx000 model number when ordered (i.e., A10000E-AKB, A9000-AKB, and A8000-AKB) ensures that the module is preloaded with AKB software. Otherwise, these models are preloaded with variant software. While on-site upgrades from variant to AKB software are possible on the other Atalla models, this process is not as convenient as on the Axx100, Axx150, or Axx160 series.

Note: When using AKB software, the Transaction Security Services process is required. For more information on the Transaction Security Services process and configuration requirements for AKB software, refer to the ***BASE24 Transaction Security Services Processing Guide***.

Older models may be compatible with BASE24 under certain circumstances; however, they are not supported or tested for compatibility by ACI.

A command reference manual and an installation and operations guide is provided with each Atalla NSP model ordered as follows:

- The ***Atalla Key Block Banking Command Reference Manual*** describes the commands available for the A8100, A8150, A9100, A9150, A10100, A10150, and A10160 models.
- The ***Banking Command Reference Manual*** describes the commands available for the A8000, A8100V, A8150V, A9000, A9100V, A9150V, A10000E, A10100V, A10150V, and A10160V models.
- The ***Installation and Operations Guide for Atalla NSP Series products using Atalla Variants*** describes installation and operations procedures for the A8100V, A8150V, A9100V, A9150V, A10100V, A10150V, and A10160V models.
- The ***Installation and Operations Guide for Atalla NSP Series products with the Atalla Key*** describes installation and operations procedures for the A8100, A8150, A9100, A9150, A10100, A10150, and A10160 models.
- The ***A8000 Installation, Operations, and Command Reference Guide*** describes installation and operations procedures for the A8000 model.
- The ***A8000-AKB Installation and Operations Guide*** describes installation and operations procedures for the A8000-AKB model.

- The ***A9000 Installation and Operations Guide*** describes installation and operations procedures for the A9000 model.
- The ***A9000-AKB Installation and Operations Guide*** describes installation and operations procedures for the A9000-AKB model.
- The ***A10000E Installation and Operations Guide*** describes installation and operations procedures for the A10000E model.
- The ***A10000E-AKB Installation and Operations Guide*** describes installation and operations procedures for the A10000E-AKB model.

In reviewing the information in this appendix, note that the terms *SYSGEN utility* and *DSC utility* refer to the HP NonStop System Generation (SYSGEN) and the HP NonStop Dynamic System Configuration (DSC) utilities.

Security Processing Support

Each security module supports a subset of BASE24 transaction security processing. The processing limitations may be the result of the security module features or BASE24 product options.

In the following table, a check mark (✓) indicates processing that is supported by each security module. In addition to the various security modules, the table also indicates transaction security processing that can be performed in software (SW).

Function	SW	NSP		
		A8000 A8100 A8150	A9000 A9100 A9150	A10000E A10100 A10150 A10160
PIN encryption	✓	✓	✓	✓
PIN verification—IBM 3624-format	✓	✓	✓	✓
PIN verification—Diebold	✓	✓	✓	✓
PIN verification—Identikey		✓	✓	✓
PIN verification—Visa PVV	✓	✓	✓	✓
Cardholder PIN select—IBM 3624-format	✓			
Cardholder PIN select—Diebold	✓			
Cardholder PIN select—Identikey		Not supported		
Cardholder PIN select—Visa PVV		Not supported		
PIN change—IBM 3624-format	✓	✓	✓	✓
PIN change—Diebold	✓	✓	✓	✓
PIN change—Identikey		✓	✓	✓
PIN change—Visa PVV	✓	✓	✓	✓
Card verification	✓	✓	✓	✓

Function	SW	NSP		
		A8000 A8100 A8150	A9000 A9100 A9150	A10000E A10100 A10150 A10160
Message authentication	✓	✓	✓	✓
Dynamic key management		✓	✓	✓

Configuring Atalla Security Devices

This section presents configuration information for each of the BASE24 supported Atalla security devices. The LCONF assigns and params used in their configuration with BASE24 are also discussed.

Atalla Network Security Processors (NSPs) can be configured using the SYSGEN utility or the HP NonStop DSC utility. The DSC utility supports online reconfiguration of controllers, devices, and some system operational parameters. The DSC changes made remain in effect through all system loads or reloads and remain in effect until the next time you run the SYSGEN utility. To make DSC changes permanent, they can be integrated into the CONFTEXT file for the next time the SYSGEN utility is run.

Asynchronous Configuration

BASE24 can communicate online with the Network Security Processor (NSP) models A8000, A8100, or A8150 using an asynchronous protocol and ASCII data transmission. The A8100 model also can be configured to communicate with BASE24 using the Transmission Control Protocol/Internet Protocol (TCP/IP) over an Ethernet connection. Refer to the “TCP/IP Configuration” topic for more information on this option.

K-Series Servers

Atalla Security Modules must be configured using the SYSGEN utility or Dynamic System Configuration (DSC) utility as 6401 point-to-point asynchronous terminals. The SYSGEN utility or DSC utility definition can use the 6401 default modifiers, but the three modifiers shown in the following example must be changed from the defaults.

The following is an example of a SYSGEN utility or DSC utility definition for a security module named \$NSP1 connected to the controller ASYNCF on port number 25:

<i>device-name</i>	<i>controller.port</i>	<i>product-number</i>	<i>modifier</i>	<i>modifier</i>	<i>modifier</i>
\$NSP1	ASYNCF.25	6401	BAUD19200,	NOECHO,	EIA

Both the A8000 and A8100 models support a baud rate of 9600 or 19200. Using the higher baud rate increases performance.

The device names configured using the SYSGEN utility or DSC utility must be specified in the SECURE-DEV1 and SECURE-DEV2 assigns in the LCONF. The SECURE-DEV1 and SECURE-DEV2 assigns are described later in this appendix.

S-Series Servers

The A8000 and A8100 models can also connect to S-series HP NonStop servers using a ServerNet wide area network (SWAN) concentrator or an asynchronous wide area network (AWAN) server that is attached to an Ethernet local area network (LAN).

The following is an example of the HP NonStop Subsystem Control Facility (SCF) commands used to configure an asynchronous device to an S-series HP NonStop server:

```
ADD PROFILE $ZZWAN.#ATPTerm, FILE $SYSTEM.SYS01.PATPTerm
```

```
ADD DEVICE $ZZWAN.#S0583, PROFILE ATPTerm, IOBJECT &
$SYSTEM.SYS01.OATPCSS, CPU 00, ALTCPU 01, TYPE (53,00), CLIP 1, &
LINE 1, RECSIZE 80, ADAPTER S05, PATH B, NAME #SEC1
```

The device names configured using SCF must be specified in the SECURE-DEV1 and SECURE-DEV2 assigns in the LCONF. The SECURE-DEV1 and SECURE-DEV2 assigns are described later in this appendix.

TCP/IP Ethernet Configuration

The Atalla A8100, A8150, A9000, A9100, A9150, A10000E, A10100, A10150, and A10160 Network Security Processors communicate with the HP NonStop system using Transmission Control Protocol/Internet Protocol (TCP/IP) over an Ethernet connection. Thus, the A8100, A8150, A9000, A9100, A9150, A10000E, A10100, A10150, and A10160 NSP require the configuration and startup of an Ethernet controller, Ethernet port, and the TCP/IP process.

The A8100, A8150, A9000, A9100, A9150, A10000E, A10100, A10150, and A10160 devices use a NonStop system application interface called Boxcar to communicate with BASE24. For the A8100, A8150, A9000, A9100, A9150, A10000E, A10100, A10150, and A10160 devices, it is the Boxcar process name (i.e., \$BOX) that is specified in the SECURE-DEV1 and SECURE-DEV2 assigns in the LCONF. Configuration requirements for the Boxcar interface module are provided later in this appendix.

Note: The A8100 and A8150 NSPs can also be configured as asynchronous devices. For more information on this option, refer to the “Asynchronous Configuration” topic provided earlier in this appendix.

Hardware Configuration for K-Series Servers

Because the A8100, A8150, A9000, A9100, A9150, A10000E, A10100, A10150, and A10160 are TCP/IP devices, each Ethernet controller (e.g., 3613, 3615) must be configured using the SYSGEN utility or DSC utility. For detailed information on configuring an Ethernet controller and line, refer to HP NonStop ***TLAM Configuration and Management Manual***. The following is an example of a SYSGEN utility definition example for a 3615 controller named A10000x:

```
controller-name product-number processor-numbers channel addresses
A10000X          3615          0,1          %030;
```

The following is an example of the SYSGEN utility definition that shows the minimum parameters required to configure a 3615 controller (line):

<i>device-name</i>	<i>controller-name</i>	<i>macro</i>	<i>modifiers</i>
\$LAM1	LAM1.0,LAM1.1	MLAM	TYPE 56, SUBTYPE 0, RSIZE 32000, READBUFFERSIZE 8192, INTERRUPT IOP^INTERRUPT^HANDLER#

After the controller is installed, the TCP/IP process must be started against the Ethernet port. An IP address must be assigned to both the security device and the HP NonStop server. Refer to the “Software Configuration” discussion on the following pages.

Hardware Configuration for S-Series Servers

Because the A8100, A8150, A9000, A9100, A9150, A10000E, A10100, A10150, and A10160 are TCP/IP devices, they must communicate with the S-series server through an Ethernet port on either a 3861 Ethernet 4 ServerNet Adapter (E4SA) controller or through a Processor MultiFunction Unit (PMF). The configuration of a E4SA controller on an S-series server requires the use of the HP NonStop Subsystem Control Facility (SCF) to configure and start objects associated with the E4SA controller. These objects include the adapter, the ServerNet addressable controllers (SACs), the logical interfaces (LIFs), and the physical interfaces (PIFs). For additional information on configuring the 3681 (E4SA) controller, refer to the HP NonStop *Himalaya S-Series Hardware Installation Guide*.

After the controller is installed, the TCP/IP process must be started against the Ethernet port. An IP address must be assigned to both the security device and the HP NonStop server. Refer to the “Software Configuration” discussion on the following pages.

Software Configuration

The software configuration is very similar for the K-series and S-series servers. Examples for both types of servers are provided below. In general, the TCP/IP process must be started against a previously configured port and the Boxcar interface module must be started. Some of the values in these examples, most importantly the IP addresses, are site-specific and must be modified accordingly.

K-Series Server. The following is an example TACL macro for starting the TCP/IP process on a K-series server.

```
?tacl macro
#frame
delete define=tcPIP^host^file
add define=tcPIP^host^file, file $system.ztcPIP.hosts
$system.ztcPIP.tcPIP.lib $system.zqiolib.qiolib,name $ZTCO,&
    nowait, pri 180, cpu 0/1
#push scfin
#append scfin allow all errors
#append scfin assume process $ZTCO
```

```
#append scfin alter subnet, #loop0, ipaddress 127.1
#append scfin add subnet #sn0, type ethernet, devicename $LAN1,&
    ipaddress 130.130.63.16, forceqio on
#append scfin add route #r80, destination 0.0.0.0,&
    gateway 130.130.5.1
#append scfin start route *
#append scfin start subnet *
scfin/inv scfin/
#unframe
```

S-Series Server. The following are examples for starting the TCP/IP process on an S-series server.

The first example starts the TCP/IP process using an SCF OBEY file.

```
=====
= Configure & Start $ZB110 TCPIP process for ADAPTER E1151,LIFL100 ==
=====
ASSUME PROCESS $ZB110
ADD SUBNET #SN1, TYPE ETHERNET, IPADDRESS 192.231.036.99&
DEVICENAME L110
ADD ROUTE #GW, DESTINATION 0, GATEWAY 192.231.036.017,&
DESTTYPE BROADCAST
ALTER SUBNET #LOOP0, IPADDRESS 127.1
START SUBNET *
START ROUTE *
```

The next example starts the TCP/IP process using a TACL macro.

```
TCP/IP /name $ZB110,NOWAIT, CPU 0, PRI 160, TERM $ZHOME/1
```

Starting the Boxcar Process. After the TCP/IP process is started, the Boxcar process must be started using the Atalla-supplied A10KWARM OBEY file. The following is an example of the A10KWARM OBEY file. All A8100, A8150, A9000, A9100, A9150, A10000E, A10100, A10150, and A10160 devices use these OBEY files, which must be edited at installation to match your own host system.

Note: The BOXCAR-DEVICE-NAME param must be set to the IP address of the A8100, A8150, A9000, A9100, A9150, A10000E, A10100, A10150, or A10160 device.

```
CLEAR ALL PARAM
PARAM BOXCAR-LOGFILE $subvol.boxlog.BOXLOG1
PARAM BOXCAR-BACKUPCPU 0
PARAM BOXCAR-DEVICE-NAME 2.2.54.100
PARAM BOXCAR-DEBUG-FILE
PARAM BOXCAR-SOCKET-TIMEOUT 1000
PARAM BOXCAR-TCPIP-PROCESS-NAME $ZB02C

PURGE $subvol.boxlog.BOXLOG1
FUP CREATE $subvol.boxlog.BOXLOG1,TYPE E, REC 200,BLOCK 4096, &
maxextents 200, ext 100

run $system.r0409aac.boxcaro /name $box1,cpu 1,pri 160,nowait, &
term $zhome,highpin on /
```

The PARAM BOXCAR-MAX-SOCKETS should be set to at least 5 for efficiency. The default value is 14. You should configure the NSP to have the same number of sockets as designated by this param. The Axx100 models support up to 64 sockets.

The PARAM BOXCAR-DEBUG-FILE should only be set to a value when debugging in a test environment. Use of this param greatly decreases performance.

Boxcar Interface Module

Each A8100, A8150, A9000, A9100, A9150, A10000E, A10100, A10150, and A10160 device uses a special interface to communicate with the BASE24 system. The Boxcar interface module is a NonStop process provided on the HP NonStop System Utility Tape (SUT) that runs in the HP NonStop Guardian environment. The Boxcar interface module serves as a TCP/IP driver and provides the communications interface between application processes and an Atalla NSP. Refer to the appropriate Atalla device manual for configuration and startup information for the Boxcar process.

The process name assigned to the Boxcar interface module at startup is used as the device name in the SECURE-DEV1 and SECURE-DEV2 assigns in the LCONF.

Note: The Boxcar interface module must be running when BASE24 processes that use an Atalla NSP are initialized or warmbooted. If the Boxcar interface module is stopped, BASE24 processes that use an Atalla NSP must be warmbooted after the interface module is restarted.

Option 023 and the Boxcar Interface Module

When an NSP command response is greater than 1024 bytes, the Atalla Boxcar interface module uses option 023 to determine the end of the command response. The Boxcar interface module retrieves the setting of option 023 for an NSP when the Boxcar interface module initializes. If you change the setting of option 023 after the Boxcar interface module has been initialized, it may not be able to recognize the end of a response message from the NSP that is greater than 1024 bytes and may wait indefinitely.

To avoid this problem, enable option 023 using a command file specified by the `COMMAND_FILE` keyword in the NSP's `CONFIG.PRM` file as shown in the following example. After you have configured the NSP with the desired commands and options, but before starting production, restart the Boxcar interface module.

Example Command File. The following is an example command file named `enable23.txt`.

```
; Start of enable23.txt
S <101#023E#>
R <201#Y#>
; end of enable23.txt
```

To enable this command file in the `CONFIG.PRM` file, add the following two lines to the end of the `CONFIG.PRM` file.

```
[STARTUP]
COMMAND_FILE=enable23.txt
```

Device Options and Banking Commands

Some device options and banking commands required by BASE24 must be configured or enabled in A8000, A8100, A8150, A9000, A9100, A9150, A10000E, A10100, A10150, and A10160 devices before they can be used. These options and commands are discussed below.

Device Options

BASE24 cannot process the message it receives from an Atalla NSP if the message contains carriage return and line feed characters. The `SECURE-DEV-CONF-TXT` param in the `LCONF` causes BASE24 processes that use an Atalla A10000E NSP

to send a banking command that configures the NSP to omit carriage return and line feed characters from the message it returns to BASE24. The SECURE-DEV-CONF-TXT param can also cause BASE24 processes to send a banking command that configures the Atalla A10000E NSP to check for a minimum PIN length and specifies the response code the NSP returns for a PIN length error. More information is provided on the SECURE-DEV-CONF-TXT param later in this appendix.

For the A8000, A8100, A8150, A9000, A9100, A9150, A10100, A10150, and A10160 devices, the option for omitting carriage return and line feed characters from the messages they return to BASE24 must be configured in a 101 (Configure Security Processor Options) command outside of BASE24 (i.e., using the Secure Configuration Terminal). The options these devices use for setting the minimum PIN length and specifying the response code the NSP returns for a PIN length error must be configured using security policy commands (108 and 109) entered using the NSPDIAG tool with the (A0) and (A1) options, respectively. For more information on the 101 command and defining a security policy, refer to Atalla documentation.

The following table lists premium options, which must be purchased from Atalla in a command 105 or command 100 (A10000E only) before they can be used in an Atalla NSP. The first column of the table lists each premium option, while the second column identifies the BASE24 function for which the premium option is required. You only need to purchase the options for the BASE24 functions implemented in your system.

Premium Option	BASE24 Function
62 (Allow command 31DUKPT)	Required for the derived unique key per transaction (DUKPT) encryption method.
66 (Do not validate old PIN for command 37)	Required for PIN Change transaction processing. BASE24 processes require the 37 command to be performed without the old PIN, regardless of the PIN verification method used.

* These options are not premium options when using AKB version software.

Device Banking Commands

Some of the Atalla NSP banking commands used in BASE24 processing may need to be enabled before transaction processing begins. The SECURE-DEV-CONF-CMD100 param in the LCONF defines the text that BASE24 sends to an Atalla A10000E NSP in a 100 (Configure Security Processor Commands) command during initialization or warmboot. A 100 command enables commands that an Atalla A10000E NSP contains in disabled form. The text contained in the 100 command must be obtained directly from Atalla. The banking commands to be enabled may already exist in the Atalla NSP or may have to be obtained from Atalla and installed. More information is provided on the SECURE-DEV-CONF-CMD100 param later in this appendix.

The A8100, A8150, A9000, A9100, A9150, A10100, A10150, and A10160 NSPs support the 105 (Enable Premium Value Command and Options), 108 (Define Security Policy) and 109 (Confirm Security Policy) commands instead of the 100 command to enable Atalla NSP banking commands. The A8000 NSP does not support the 100 or 105 commands because it does not have any premium value commands or options. The NSP banking commands to be enabled in an A8100, A8150, A9000, A9100, A9150, A10100, A10150, or A10160 device must be specified in the security policy established using the 108 and 109 commands.

Some of the banking commands that BASE24 supports are disabled when an A8000, A8100, A8150, A9000, A9100, A9150, A10100, A10150, or A10160 NSP is delivered from the factory because of the security exposure involved with the command. These commands can be enabled using only the 108 and 109 security policy commands. For example, the D0 (Verify Clear PIN) command is disabled in the A8000, A8100, A8150, A9000, A9100, A9150, A10100, A10150, and A10160 due to security exposure. Other banking commands that BASE24 supports are identified as *premium* commands for the A8100, A8150, A9000, A9100, A9150, A10100, A10150, and A10160 NSPs. These commands are disabled when an A8100, A8150, A9000, A9100, A9150, A10100, A10150, or A10160 device is delivered from the factory and can only be enabled in the security policy commands by purchasing a 105 command from Atalla. The 105 command is based on the A8100, A8150, A9000, A9100, A9150, A10100, A10150, or A10160 device serial number and is unique to each device.

The following table lists premium commands, which must be purchased from Atalla in a command 105 or command 100 (A10000E only) before they can be used in an Atalla NSP. The first column of the table lists each premium command,

while the second column identifies the BASE24 function for which the premium command is required. You only need to purchase the commands for the BASE24 functions implemented in your system.

Premium Command	BASE24 Function
30 (Encrypt PIN)	Required for clear PIN encryption.
37 (PIN Change)	Required for PIN Change transaction processing. BASE24 processes require the 37 command to be performed without the old PIN, regardless of the PIN verification method used.
351 (EMV PIN Change)	Required for validation of offline PIN changes to EMV cards.
11D (Generate ATM MAC or Data Encryption Key)*	This command is needed if message authentication is performed and the NSP is used to generate a new MAC transaction key for a terminal. This command is not needed to generate a new MAC key for an interface process.

* The 11D command is replaced by the 10 and 1A commands in AKB version software. The 10 and 1A commands are not premium commands.

The 105, 108 and 109 commands must be entered outside of BASE24. The 105 command to enable the 30, 37, and 11D premium commands must be obtained from Atalla and entered into the A8100, A8150, A9000, A9100, A9150, A10100, A10150, or A10160 device using the NSPDIAG tool. Once this is done, the 108 command can be composed and entered using the NSPDIAG tool. The 208 command response returned for the 108 command then must be entered at the NSP using the Secure Configuration Terminal (SCT). The NSP encrypts this result using variant 30. The result from this is then entered as a 109 command using the NSPDIAG tool. For detailed information on the 105, 108, and 109 commands, NSPDIAG tool, and SCT, refer to Atalla documentation.

To enable banking commands for A8000 devices, use the 108 and 109 commands entered from the SCT. For more information, refer to Atalla documentation.

If you are translating PIN/PAD PIN blocks from single DES to triple DES, and vice versa, as an interim step in migrating interchanges to triple DES, the 335 (PIN Translate) command and options 6A (allow the outgoing double-length PIN encryption key to be a replicated single-length key) and 6C (allow one key-triple DES (single-length) working keys) are required. Option 6A is required if the destination uses single-length keys. Option 6C is required if the source uses single-length keys. This banking command is available in all Atalla models (except the A8000) with AKB 2.1.3 or later software and in the following models for software version 2.9 or later software: A8100V, A8150V, A9000, A9100V, A9150V, A10000E, A10100V, A10150V, and A10160V. Note that the 2.90 equivalent software on Axx100V models is version 1.10. Note that you must use the Transaction Security Services process for Atalla NSPs running AKB software.

If you need options 6A or 6C for the 335 command in your BASE24 system, you must enable them in the NSP's security policy using the 108 and 109 commands. These options are disabled in the NSP's default factory security policy. For more information on the 335 command and its options, refer to Atalla documentation.

Note: For NSPs running software prior to release 2.40 or 1.4, you must obtain option 6A in a 105 command, free of charge from Atalla, before you can enable it.

Logical Network Configuration File (LCONF)

The following section presents the Logical Network Configuration File (LCONF) information required for configuration of the Atalla security devices.

Each BASE24 process uses the value configured in the User Field on the LCONF Assign screen for the SECURE-DEV1 and SECURE-DEV2 assigns to identify the functions available with the security module. The valid User Field values for each Atalla security module model are shown in the following table. For models with multiple valid User Field values, the functional differences for each value are explained in notes provided in the assign descriptions. The SECURE-DEV-CONF-CMD100 and SECURE-DEV-CONF-TXT params are used only when the User Field for these assigns is set to a value of NSP.

Atalla Security Module Model	Valid USER FIELD Values
A8000	A4 or NSP
A8100V,* A8150V,* A9000,* A9100V,* A9150V,* A10100V,* A10150V,* A10160V,* A10000E	NSP

* When using one of these models with a USER FIELD value of NSP for the SECURE-DEV1 and SECURE-DEV2 assigns, the SECURE-DEV-CONF-CMD100 param must be removed from the LCONF.

Note: The following LCONF configuration requirements do not apply when using an Atalla NSP with AKB software. When using AKB software, the Transaction Security Services process must be used, which requires LCONF assigns and params to be configured differently. For detailed information on these LCONF configuration requirements, refer to the *BASE24 Transaction Security Services Processing Guide*.

LCONF Assigns

The following assigns are required in the LCONF to configure a security module.

SECURE-DEV1 — Specifies the name of the primary security device or the name of the interface module used with the primary security device. For the Atalla A8000 security devices and A8100V security devices configured as asynchronous devices, this assign specifies the SYSGEN utility, DSC, or SCF name of the primary security device. For Atalla A9000, A9100V, A9150V, A10000E,

A10100V, A10150V, and A10160V security devices, and A8100V or A8150V security devices configured as TCP/IP devices, this assign specifies the name given to the A8100V, A8150V, A9000, A9100V, A9150V, A10000E, A10100V, A10150V, or A10160V interface module used with the primary security device. The device name or interface module name used must be compatible with the HP NonStop Expand product.

There can be multiple SECURE-DEV1 assigns in the LCONF, depending on whether a primary device is to be specified individually for each process requiring the use of a security device (in which case, each process has its own SECURE-DEV1 assign in the LCONF).

To associate a primary security device with a specific BASE24 process, the symbolic name of the BASE24 process must be placed in the READ BY field on the LCONF Assign screen. If a specific primary security device need not be associated with a specific BASE24 process, the READ BY field can be filled in with asterisks.

When determining its primary security device, a BASE24 process first reads through the LCONF for a SECURE-DEV1 assign with its own name in the READ BY field. If it finds a match, it uses that corresponding assign. However, if it does not find a match, it reads through the LCONF again searching for a SECURE-DEV1 assign with asterisks in the READ BY field.

Note: Each BASE24 process uses the value in the USER FIELD on the LCONF Assign screen to identify the Atalla security device being used. For the A8100V, A8150V, A9000, A9100V, A9150V, A10100V, A1050V, and A10000E models, the USER FIELD must be set to a value of NSP. For the NSP model A8000, the USER FIELD can be set to a value of A4 or NSP. When this field is set to a value of A4 for the A8000, the module functions like the retired A1 module and you can download DNTs to the device. When the User Field is set to a value of NSP for the A8000, the full NSP features of the device are available, except that you cannot download DNTs to the device.

SECURE-DEV2 — Specifies the name of the secondary security device or the name of the interface module used with the secondary security device. For the Atalla A8000 security devices and A8100V security devices configured as asynchronous devices, this assign specifies the SYSGEN utility, DSC, or SCF name of the secondary security device. For Atalla A9000, A9100V, A9150V, A10000E, A10100V, A10150V, and A10160V security devices, and A8100V or A8150 security devices configured as TCP/IP devices, this assign specifies the name given to the A8100V, A8150V, A9000, A9100V, A9150V, A10000E,

A10100V, A10150V, or A10160V interface module used with the secondary security device. The device name or interface module name used must be compatible with the HP NonStop Expand product.

There can be multiple SECURE-DEV2 assigns in the LCONF, depending on whether a secondary device is to be specified individually for each process requiring the use of a security device (in which case, each process has its own SECURE-DEV2 assign in the LCONF).

To associate a secondary security device with a specific BASE24 process, the symbolic name of the BASE24 process must be placed in the READ BY field on the LCONF Assign screen. If a specific secondary security device need not be associated with a specific BASE24 process, the READ BY field can be filled in with asterisks.

When determining its secondary security device, a BASE24 process first reads through the LCONF for a SECURE-DEV2 assign with its own name in the READ BY field. If it finds a match, it uses that corresponding assign. However, if it does not find a match, it reads through the LCONF again searching for a SECURE-DEV2 assign with asterisks in the READ BY field.

Note: Each BASE24 process uses the value in the USER FIELD on the LCONF Assign screen to identify the Atalla security device being used. For the A8100V, A8150V, A9000, A9100V, A9150V, A10100V, A10150V, A10160V, and A10000E models, the USER FIELD must be set to a value of NSP. For the NSP model A8000, the USER FIELD can be set to a value of A4 or NSP. When this field is set to a value of A4 for the A8000, the module functions like the retired A1 module and you can download DNTs to the device. When the User Field is set to a value of NSP for the A8000, the full NSP features of the device are available, except that you cannot download DNTs to the device.

LCONF Params

The following params are configured in the LCONF to configure a security module.

ATM-A4000-MASTER — The symbolic name of the BASE24-atm Authorization process (for example, P1A^AUTH01) responsible for downloading Diebold number tables (DNTs) from the Key Authorization File (KEYA) to an Atalla Network Security Processor (NSP) model A8000 during process initialization or process warmboots. A8000 security devices fail when more than

one Authorization process attempts to load DNTs during initialization or warmboots. This param is required only when using a model A8000 security device with the BASE24-atm product.

POS-A4000-MASTER — The symbolic name of the BASE24-pos Device Handler/Router/Authorization process (for example, P1A^RTAU01) responsible for downloading Diebold number tables (DNTs) from the Key Authorization File (KEYA) to an Atalla Network Security Processor (NSP) model A8000 during process initialization and process warmboots. A8000 security devices fail when more than one Authorization module attempts to load DNTs during initialization or warmboots. This param is required only when using a model A8000 security device with the BASE24-pos product.

SECURE-DEV-CONF-CMD100 — Defines the text that a BASE24 process sends to an Atalla Network Security Processor (NSP) model A10000E in a 100 command during initialization or warmboot. A 100 command enables commands that an NSP contains in disabled form. The text contained in the 100 command must be obtained directly from Atalla. The banking commands to be enabled may already exist in the NSP. The text to be contained in the 100 command must be entered in the Text field of the Param screen in the following format:

100#Text-obtained-from-Atalla

The following NSP banking commands must be enabled with a 100 command before they can be used:

- 30 = Encrypt Clear PIN. This command is needed if a clear PIN is passed to the NSP for encryption.
- 37 = PIN Change Transaction. This command is needed if the BASE24-atm PIN change transaction is offered and the NSP is used to generate the new PIN verification digits. Option 066 (verify old PIN in a separate call to the device) also must be enabled with NSP command 37.
- 11D = Generate MAC Key. This command is needed if message authentication is performed and the NSP is used to generate a new MAC transaction key for a terminal. This command is not needed to generate a new MAC key for an interface process.
- D0 = Verify Clear PIN. This command is needed if a clear PIN is passed to the NSP for verification.

This param is needed only if any of the banking commands listed above are used. A single SECURE-DEV-CONF-CMD100 param can be defined in the LCONF with the READ BY field on the LCONF Param screen set to asterisks (default value). The BASE24 security utilities do not send this param to non-NSP security devices.

Each 100 command received by an NSP can override the settings established by any prior 100 commands if the commands do not contain the same text. If multiple processes (BASE24 or non-BASE24) issue 100 commands to an NSP, care must be taken to ensure that all 100 commands contain the same text.

Note: Because the NSP models A8000, A8100V, A8150V, A9000, A9100V, A9150V, A10100V, A10150V, and A10160V do not support the 100 command, this param is not valid when these models are used. When using an A8100V, A8150V, A9000, A9100V, A9150V, A10100V, A10150V, or A10160V module, banking command and option configuration must be initialized manually outside of BASE24 using security policy commands (108 and 109) entered using the NSPDIAG tool. For more information on defining a security policy, refer to Atalla documentation.

SECURE-DEV-CONF-TXT — Defines the configuration text sent to an Atalla Network Security Processor (NSP) model A10000E in a 101 command during initialization or warmboot. A 101 command can modify three of the default configuration settings used by the A10000E NSP. The text to be contained in the 101 command must be entered in the Text field of the LNCf Param screen and can consist of one or more four-character option words.

Each option word contains a three-digit option ID followed by a one-character action flag. If more than one option word is defined, the option words must be separated by commas. Each of the available option IDs can only be used once in each SECURE-DEV-CONF-TXT param. The valid option IDs are as follows:

023 = Specifies whether the NSP should remove carriage return and line feed characters from all response messages returned to the BASE24. This option ID has the following valid action flag value:

E = Enable. Remove carriage return and line feed characters from all response messages returned to BASE24 by the NSP.

- 024 = Specifies whether the NSP should identify PIN length errors by returning a unique code in response messages to the BASE24. This option ID has the following action flag values:
- D = Disable. Return an S error response for PIN length and sanity check errors. BASE24 processes an S error response as a sanity check error.
 - E = Enable. Return an L error response for a PIN length error and return an S error response for a sanity check error. BASE24 processes an L error response as an invalid PIN and processes an S error response as a sanity check error.
- 040 = Sets the minimum valid PIN length used by the NSP when performing PIN length checks. This option ID has the following valid action flag values:
- 0 = Perform the PIN length check using a minimum PIN length of zero. This value has the effect of disabling the PIN length check because any PIN entered passes.
 - 1–9, A–C = Perform the PIN length check using this value as the minimum PIN length. A, B, and C represent lengths of 10, 11, and 12, respectively. BASE24 products require a minimum PIN length of 4 digits.

As an example, to remove carriage return and line feed characters from all response messages returned to BASE24, return an L response code for a PIN length error, and set a minimum PIN length of 4 characters, the following information would be entered in the Text field on the Param screen:

023E, 024E, 0404

This param is needed only if any of these option IDs (023, 024, or 040) are used. Once a configuration value is changed using one of these option IDs, the new value remains in effect until the NSP is reset or the configuration value is changed by receiving the option ID in another 101 command. To avoid possible inconsistencies caused by one process changing a configuration value and another process expecting the default configuration value, ACI recommends the use of this param with all three option words for all BASE24 processes using NSP security modules. A single SECURE-DEV-CONF-TXT param can be defined in the LCONF with the READ BY field on the LCONF Param screen set to asterisks (i.e., default value). The BASE24 security utilities do not send this param to non-NSP security devices.

Each 101 command received by an NSP can override the settings established by any prior 101 commands if the commands do not contain the same text. If multiple processes (whether they are BASE24 processes or other application processes) issue 101 commands to an NSP, care must be taken to ensure that all 101 commands contain the same text.

Note: Because the NSP models A8100V, A8150V, A9000, A9100V, A9150V, A10100V, A10150V, and A10160V do not support options 024 and 040 in the 101 command, this param is not valid when these models are used. When using an A8100V, A8150V, A9000, A9100V, A9150V, A10100V, A10150V, or A10160V model, these options must be initialized manually outside of BASE24 using security policy commands (108 and 109) entered using the NSPDIAG tool and the (A0) and (A1) options, respectively. The 023 option must be enabled using the 101 command entered outside of BASE24 as well. For more information on the 101 command and defining a security policy, refer to Atalla documentation.

SECURE-DEV-FAIL-NOTIFY — The interval at which critical event messages are generated due to the failure of a primary or secondary security device for a process. Valid values are 10 to 1000; the default value is 20.

If the primary and secondary security devices cannot be initialized during process startup and this param is set to the default of 20, every twentieth transaction that a process must decline because the security devices are inoperative causes a message to be generated indicating the security devices have failed. (Note that a critical event message is always generated immediately upon failure of both devices.)

If a primary security device is marked down and this param is set to the default of 20, every twentieth transaction that is directed to the primary security device causes a message to be generated even though the transactions are being handled properly by the secondary security device. Transactions routed directly to the secondary security device do not affect the primary security device transaction counter. The SECURE-DEV-MONITOR-THRESHOLD param controls when a transaction is directed to the secondary security module instead of the primary security device.

If the primary and secondary security devices fail at some time after initialization, a separate failed transaction counter is maintained for each security device. A transaction that is sent to one security device and then the other increments both counters. An event message is generated each time either counter reaches the value established in this param.

SECURE-DEV-MONITOR-THRESHOLD — The interval, expressed in the number of transactions processed by the security device defined in the SECURE-DEV1 assign, for checking the security device defined by the SECURE-DEV2 assign. Valid values are 0–32000. The default value of 0 indicates no check is performed.

This param can be used to monitor the secondary security device or to balance the load between the primary and secondary security devices. Each process using a security device maintains its own transaction counter. This means the interval specified in the SECURE-DEV-MONITOR-THRESHOLD param needs to be set for the number of transactions processed by individual processes using the security device, not the total number of transactions processed by the security device. For example, if five BASE24 processes use a security device and the param is set to 1000, up to 5000 transactions (1000 by each of the five processes) could be processed by the primary security device before the secondary security device is used. Setting this param to 1 causes each of the BASE24 processes to alternate between the two security devices.

If all processes use the same threshold for monitoring the secondary security device, a single SECURE-DEV-MONITOR-THRESHOLD param can be defined in the LCONF. In this case, the READ BY field on the LCONF Param screen must be set to asterisks (default value).

If different secondary security device monitoring thresholds are used, SECURE-DEV-MONITOR-THRESHOLD params must be defined in the LCONF for each threshold and the READ BY field can be set to identify the processes using that threshold. If a majority of processes use a particular threshold, an alternative to identifying all of its processes can be making that threshold the default by placing asterisks in the READ BY field. Then, only processes using a different threshold need to be defined specifically.

SECURE-DEV-TIM-LMT — The maximum time, in hundredths of a second, to wait for a response from the security device before the request is considered to be timed out. Valid values are 1 to 6000; the default value is 500 (i.e., 5 seconds). The same time limit applies to all processes so there is only one SECURE-DEV-TIM-LMT param in the LCONF, and the READ BY field must be set to asterisks.

SECURE-DEV-TYP — Specifies whether the BASE24 security utilities in a BASE24 process communicate directly with an Atalla security device or do not use a security device. Valid values are as follows:

ASM = Atalla Network Security Processor (NSP)
N/A = No security module required

In some BASE24 logical networks it may be desirable to use more than one security device type. This is possible, since the SECURE-DEV-TYP param is assigned by process. Care must be exercised, however, in assigning different security device types to different processes, especially where the different processes communicate with each other.

For example, if a Device Handler or Interchange Interface process receives a PIN encrypted under a key managed by an Atalla security device, but passes the PIN and the Atalla-encrypted PIN key to an Authorization process that is using a Thales e-Security security device for PIN verification, the PIN cannot be verified unless the Thales e-Security device is configured specifically to correctly decrypt the Atalla-encrypted PIN key.

The same care must be exercised regarding MAC keys. For example, if a Device Handler or interface process receives a MAC generated under a key managed by an Atalla security device, but attempts to verify the MAC using a Thales e-Security security device, the MAC cannot be verified unless the Thales e-Security device is configured specifically to correctly decrypt the Atalla-encrypted MAC key.

Also, the Authorization process sends information from Key Authorization File (KEYA) records to its security device so care must be taken to ensure that any encrypted data passed by the Authorization process from the KEYA to the security device has previously been encrypted by that same security device type with the same master key.

If all processes use the same type of security device, a single SECURE-DEV-TYP param can be defined in the LCONF. In this case, the READ BY field on the LCONF Param screen must be set to asterisks (default value).

If different security device types are used, SECURE-DEV-TYP params must be defined in the LCONF for each type and the READ BY field on the LCONF Param screen for each type can be set to identify the processes using that type. If a majority of processes use a particular type of security device, an alternative to identifying all of its processes can be making that device type the default by placing asterisks in the READ BY field of its SECURE-DEV-TYP param. Then, only processes using a different type of security device need to be defined specifically.

If no security devices are used, a SECURE-DEV-TYP param must still be defined in the LCONF. The READ BY field on the LCONF Param screen must be set to asterisks (default value) and the field on the LCONF Param screen must be set to N/A (no security module required).

Atalla Security Device Support

Summaries of Atalla Network Security Processor (NSP) functions supported by BASE24 products presented in this section include the following:

- The NSP specifications required for generating, encrypting, and translating keys for storage in the BASE24 database.
- The NSP online functions supported by BASE24 products.
- The online functions supported by each model of the NSP.

This material is not meant to be a tutorial on the NSP; it presumes a familiarity with the capabilities and nomenclature as discussed in Atalla documentation.

Preparing Data for the BASE24 Files

A secure key loading device such as the Secure Configuration Terminal (SCT), is used to perform offline key management and the programming of the Master File Key (MFK) directly into the NSP.

A secure key loading device, such as the SCT, can be used to encrypt or translate the keys under the proper MFK variant for storage in the BASE24 files, as shown in the following table. The table identifies the BASE24 field name, BASE24 file maintenance screen, MFK variant, and key type.

Note: The following BASE24 file information does not apply when using an Atalla NSP with AKB software. When using AKB software, the Transaction Security Services process must be used, which requires its own database for hardware key information. For detailed information on the database loading requirements for the Transaction Security Services process, refer to the ***BASE24 Transaction Security Services Processing Guide***.

Field Name in BASE24 File	Screen	MFK Variant	Key Type
MASTER KEY	ATD 9	5	ATM A key (AATM)
		1	PIN encryption key*
PIN MASTER KEY	PTD 7	5	ATM A key (AATM)
		1	PIN encryption key*
MAC MASTER KEY	PTD 7	0	Key exchange key
MASTER KEY	TTDF 2	5	ATM A key (AATM)
MAC MASTER KEY	TTDF 2	0	Key exchange key
CRYPTOGRAPHIC KEYS	KEYI 1	9	EMV ACC master keys
SECURE MESSAGING KEYS: FOR INTEGRITY	KEYI 1	13	EMV MAC master key
PIN KEY EXCHANGE KEY	KEYF 1 KEY6 1	0	Key exchange key
MAC KEY EXCHANGE KEY	KEYF 1 KEY6 1	0	Key exchange key
INTERMEDIATE KEY ENCRYPTED	KEYF 1 KEY6 1	1	PIN encryption key
INBOUND PIN KEY 1	KEYF 2 KEY6 2	1	PIN encryption key
INBOUND PIN KEY 2	KEYF 2 KEY6 2	1	PIN encryption key
OUTBOUND PIN KEY 1	KEYF 2 KEY6 2	1	PIN encryption key
OUTBOUND PIN KEY 2	KEYF 2 KEY6 2	1	PIN encryption key
INBOUND MAC KEY 1	KEYF 2 KEY6 3	3	Message authentication key

Field Name in BASE24 File	Screen	MFK Variant	Key Type
INBOUND MAC KEY 2	KEYF 2 KEY6 3	3	Message authentication key
OUTBOUND MAC KEY 1	KEYF 2 KEY6 3	3	Message authentication key
OUTBOUND MAC KEY 2	KEYF 2 KEY6 3	3	Message authentication key
DES (IBM 3624) ENCRYPTED KEY	KEYA 2	4	PIN verification key
DIEBOLD ENCRYPTED KEY	KEYA 3	0	Key exchange key
DIEBOLD TABLE [†]	KEYA 3	—	
VISA PVV KEYS ENCRYPTED	KEYA 5	4	PIN verification key
CARD VERIFICATION ENCRYPTED KEY 1 [‡]	KEYA 6	3	Card verification key A
CARD VERIFICATION ENCRYPTED KEY 2 [‡]	KEYA 6	3	Card verification key B
DERIVATION KEY	KEYD 1	8	Derivation key

* This key type is used for this field only when it needs to contain an intermediate key.

For ATM or POS devices configured to send PINs in the clear to any interchanges on the BASE24 system, the intermediate key must be stored in the BASE24-atm or BASE24-pos Terminal Data files.

[†] The Diebold number table must be placed in the KEYA in the clear. A BASE24 process will encrypt it for sending to the NSP.

[‡] For verification of MasterCard and Visa cards without using the Transaction Security Services process, these fields must contain a pair of single-length keys encrypted under an MFK variant of 3. For verification of American Express cards, you must use the Transaction Security Services process. In this case, a single encrypted double-length key is stored in the Card Verification data source (CRDVD) instead of in the KEYA.

NSP Online Functions Supported by BASE24

The table below summarizes the NSP message IDs and functions used by the BASE24 Device Handler, Authorization, Interchange Interface, and Host Interface processes.

Message ID	Function	Device Handler Process	Authorization Process	Interchange Interface Process	Host Interface Process
10	Generate Working Key Type J	Y	N	Y	Y
11	Translate Working Key for Distribution	N	N	Y	Y
12	Translate Working Key for Local Storage*	N	N	Y	Y
13	Translate Working Key for Local Storage (switch-to-switch)	N	N	Y	Y
15	Change Transaction Key	Y	N	N	N
17	Translate Working Key from Master File Key encryption to APP-resident Key Exchange Key encryption*	N	N	Y	Y
30	Clear PIN Encryption	N	N	Y	Y
31#1	Translate PIN—ANSI	Y	N	Y	Y
31#3	Translate PIN—PIN/Pad	Y	N	Y	Y
31#7	Translate PIN – VISA DUKPT	Y	N	N	N
32#1	Verify PIN—Identikey	N	Y	N	N
32#2	Verify PIN—IBM 3624-format	N	Y	N	N
32#3	Verify PIN—Visa PVV	N	Y	N	N
32#5	Verify PIN—Diebold	N	Y	N	N

Message ID	Function	Device Handler Process	Authorization Process	Interchange Interface Process	Host Interface Process
33#11	Translate PIN (differing PAN digits)—ANSI to/from PLUS	N	N	Y	Y
33#13	Translate PIN (differing PAN digits)—ANSI to PIN/Pad	N	N	Y	Y
33#33	Translate PIN (differing PAN digits)—PIN/Pad to PIN/Pad	N	N	Y	Y
37#1	Change PIN—Identik [†]	N	Y	N	N
37#2	Change PIN—IBM 3624-format [†]	N	Y	N	N
37#3	Change PIN—Visa PVV [†]	N	Y	N	N
37#5	Change PIN—Diebold [†]	N	Y	N	N
5E#3	Verify CVV/CVC	N	Y	N	N
56#0	Generate MAC [*]	Y	N	Y	Y
5A#0	Verify MAC [*]	Y	N	Y	Y
73	Clear PIN Key Table [‡]	N	Y	N	N
74	Load Diebold Number Table [‡]	N	Y	N	N
98	Generate MAC	Y	N	Y	Y
99	Verify MAC	Y	N	Y	Y
9E	Translate Working Key from Current MFK to Pending MFK	N	N	N	N
BD#1	Translate PIN and Generate MAC—ANSI [§]	N	N	Y	Y
BD#3	Translate PIN and Generate MAC—PIN/Pad [§]	N	N	Y	Y

Message ID	Function	Device Handler Process	Authorization Process	Interchange Interface Process	Host Interface Process
D0#1	Verify PIN—Clear PIN Identkey	N	Y	N	N
100	Security Processor Configuration Command ^{§**}	Y	Y	Y	Y
101	Configure Security Processor Options ^{§**}	Y	Y	Y	Y
11D	Generate MAC Key ^{*§}	Y	N	N	N
335	PIN Translate—Double Length KPE [§]	N	N	Y	Y
350	Verify EMV ARQC [§]	N	Y	N	N
351	EMV PIN Change ^{§††}	N	Y	N	N
352	Generate EMV MAC [§]	N	Y	N	N

* These functions are not supported in any Atalla model using AKB software.

† The 37 message ID can be configured by Atalla to be performed with or without the old PIN, depending on the value used in command 100 or in security policy commands (108 and 109) when this command is activated. For a BASE24 process, it must be configured to be performed without the old PIN. This configuration is called option 66.

‡ The Authorization process or module sends Diebold number tables to the NSP using a 74 message ID during initialization or warmboot, or when a security administrator enters a load DNT command. If the security device being used is an Atalla Network Security Processor (NSP), model A8000 or A8100, the Authorization process uses the 73 message ID to clear the PIN key table in the device prior to loading the Diebold number table.

§ The BD,11D, 335, 350, 351, and 352 functions are not supported by the A8000 model. The A8100V model supports all of these commands; however, you cannot send binary data in the BD command when sending the command over an asynchronous port. The A8100 model does not support the 11D command when it is configured with AKB version software.

****** These functions are not supported by BASE24 when using the NSP model A8000, A8100, A8150, A9000, A9100, A9150, A10100, A10150, or A10160.

- The 100 command is not supported by the A8000, A8100, A8150, A9000, A9100, A9150, A10100, A10150, and A10160 models. Instead, these devices are configured using the 108 and 109 commands.
- The A8000, A8100, A8150, A9000, A9100, A9150, A10100, A10150, and A10160 models do not support options 024 and 040 in the 101 command. These are replaced by options (A0) and (A1), respectively, which are set using the 108 and 109 commands.

†† This function is only supported by BASE24 when using the Transaction Security Services process.

Setting the Security Policy

Before implementing an Atalla NSP into a BASE24 system, you must ensure that the security policy established for each NSP enables the commands and options required by BASE24 as previously discussed in this appendix. You can compare the list of commands and options used by BASE24 processes to the summary of commands and options provided for each NSP in the Atalla vendor documentation. If a command or option required by BASE24 is disabled in the factory default security policy, it must be enabled before you can use the NSP in the BASE24 system. For more information on setting a security policy, refer to the Atalla vendor documentation.

Note: The message IDs and functions supported by BASE24 for the A8000 and A8100V NSPs depend on how the devices are configured to BASE24 as follows:

- When the User Field for the SECURE-DEV1 and SECURE-DEV2, or SECURE-DEV-xx, assigns is set to a value of A4, the A8000 and A8100V models support the same functions as the retired A1 model, as well as the ability to clear the key table in the device and download DNTs to the device.
- When the User Field for the SECURE-DEV1 and SECURE-DEV2, or SECURE-DEV-xx, assigns is set to a value of NSP, the A8000 and A8100V models support the same functions as an NSP, except for the ability to download DNTs to the device. The A8000 model does not support the following NSP functions: BD, 11D, 335, 350, and 352.

C: Thales e-Security Host Security Modules

This appendix describes how Thales e-Security Host Security Modules must be configured to work with BASE24. It does not describe the proprietary workings of any of the security modules, but rather the functions of each that are supported by BASE24 and the information a user will require for configuring the modules for use by BASE24.

BASE24 supports the following Thales e-Security Host Security Module (HSM) models:

- RG7100 and RG7110
- RG7300 and RG7310
- RG7400
- HSM 8000

Note: The RG7110 and RG7310 models are high performance versions of the RG7100 and RG7300 models, respectively. The configuration requirements for these high performance models are the same as that of their lower performance counterparts.

The HSM 8000, available in both standard medium- (220 tps) and high-speed (800 tps) models is designed to complement the RG7000 range by providing higher levels of performance at a reduced price per transaction. The HSM 8000 is fully backward compatible with the RG7000 range of HSM's. Note that security functions within the HSM 8000 are based on software rather than firmware as with the RG7000 models. This functionality is delivered as licensed software modules on a CD-ROM with an associated activation licence. The configuration requirements for HSM 8000 models are the same as that of the RG7100 models. The standard medium-speed model is called the HSM8-SM. The standard high-speed model is called the HSM8-SH.

In reviewing the information in this appendix, note that the terms *SYSGEN utility* and *DSC utility* refer to the HP NonStop System Generation (SYSGEN) and the HP NonStop Dynamic System Configuration (DSC) utilities.

Security Processing Support

Thales e-Security HSMs support a subset of BASE24 transaction security processing. The processing limitations may be the result of the security module features or BASE24 options. In the following table, a check mark (✓) indicates processing that is supported by the security modules. All Thales e-Security HSMs support the same functions with BASE24.

Function	HSM
PIN encryption	✓
PIN verification—IBM 3624-format	✓
PIN verification—Diebold	✓
PIN verification—Identikey	
PIN verification—Visa PVV	✓
Cardholder PIN select—IBM 3624-format	
Cardholder PIN select—Diebold	
Cardholder PIN select—Identikey	
Cardholder PIN select—Visa PVV	
PIN change—IBM 3624-format	✓
PIN change—Diebold	✓
PIN change—Identikey	
PIN change—Visa PVV	✓
Card verification	✓
Message authentication	✓
Dynamic key management	✓

Configuring Thales e-Security Security Devices

This section presents configuration information for the BASE24 supported Thales e-Security security devices.

Thales e-Security Host Security Modules (HSMs) can be configured using the SYSGEN utility or the HP NonStop DSC utility. The DSC utility supports online reconfiguration of controllers, devices, and some system operational parameters. The DSC changes made remain in effect through all system loads or reloads and remain in effect until the next time you run the SYSGEN utility. To make DSC changes permanent, they can be integrated into the CONFTEXT file for the next time the SYSGEN utility is run.

Data Communications Options

BASE24 supports different data communications modes for each of the HSM models as shown in the following table. The first column of the table lists each of the supported HSM models. The second through fourth columns of the table indicate the data communications modes available for each model. A check mark (✓) in columns two through four indicates that the data communications mode is available for the model. Each of the three available data communications modes and their configuration requirements are explained below and on the following pages.

HSM Model	ASYNCH	SDLC	TCP/IP and UDP/IP
RG7100	✓		✓
RG7300	✓	✓	
RG7400	✓		
HSM 8000	✓		✓

Asynchronous Emulation (ASYNCH)

Asynchronous emulation mode is full duplex, allowing the HP NonStop server to send commands to the HSM independently of reading the associated responses from the HSM. With this mode, there is no inherent flow control; the HSM returns

a response as soon as it has finished processing a command. The following paragraphs discuss the configuration requirements for BASE24-supported Thales e-Security security devices using asynchronous emulation mode.

Using asynchronous emulation, BASE24 communicates online with HSMs using an asynchronous protocol and ASCII data transmission.

With this communications mode, HSMs must be configured using the SYSGEN utility or DSC utility as 6603 point-to-point asynchronous terminals. The SYSGEN utility or DSC utility definition can use the 6603 default modifiers, but four modifiers must be changed from the defaults.

The following is an example of a SYSGEN utility definition for a security module named \$HSM1 connected to the controller ASYNCF on port number 25:

<i>device-name</i>	<i>controller.port</i>	<i>product-number</i>	<i>modifier</i>	<i>modifier</i>	<i>modifier</i>	<i>modifier</i>
\$HSM1	ASYNCF.25	6603	BAUD9600,	NOECHO,	PAGE,	PAGETERM %3

Note: Using asynchronous emulation, the Thales e-Security Host Security Module (HSM) must be configured to operate in nontransparent mode. The HSM will not function properly with BASE24 when configured in transparent mode. Refer to the appropriate Thales e-Security documentation for HSM configuration procedures.

When using asynchronous emulation mode, the device name (e.g., \$HSM1) configured using the SYSGEN utility or DSC utility is specified in the SECURE-DEV1 and SECURE-DEV2 assigns in the LCONF.

SDLC Emulation

Synchronous data-link control (SDLC) emulation mode allows the HP NonStop server to view the HSM as data communications equipment (DCE) operating as a non-switched, point-to-point, half duplex device. The electrical interface conforms to the RS-449 standard without the secondary channel.

BASE24 communicates online with RG7300 Thales e-Security Host Security Modules using a 3604 bit-synchronous communications board, or any controller capable of supporting the HP NonStop Envoy ACP/XF interface, that is installed in the I/O bay of a HP NonStop computer system.

The RG7300 device also uses a NonStop system application interface process called the HSM Interface Process (HIP) to communicate with BASE24. For the RG7300 device, it is the HIP process name (i.e., \$HIP) configured using the SYSGEN utility or the DSC utility that is specified in the SECURE-DEV1 and SECURE-DEV2 assigns in the LCONF. Refer to the appropriate Thales e-Security documentation for HIP installation and user procedures.

The HIP requires the use of the HP NonStop Envoy ACP/XF product using bit synchronous SDLC protocol. One HIP is required for each RG7300 device connected to the HP NonStop system. Each HIP accepts requests from one or more BASE24 processes. BASE24 processes are responsible for all queue and error handling since the HIP can be considered to be a single-threaded SDLC device driver.

Ethernet (TCP/IP and UDP/IP)

The HSM acts as a TCP/UDP server supporting connections of up to eight TCP ports or sockets and one UDP port or socket. Eight ports are available when using TCP/IP, while seven are available when using UDP/IP. Applications establish connections to the HSM ports by first connecting to the Well-Known-Port at the IP address. The port number and IP address are defined for the HSM during configuration.

Since the HSM communicates with the HP NonStop server using the Transmission Control Protocol/Internet Protocol (TCP/IP) or User Datagram Protocol/Internet Protocol (UDP/IP) over an Ethernet connection, the configuration and startup of an appropriate Ethernet controller, Ethernet port, and TCP/IP process are required.

The HSM uses an ACI-supplied interface process called the Racal Access Module (RAM) process to communicate with the BASE24 application over a TCP/IP connection.

The HSM uses an ACI-supplied interface process called the Racal Access Module for UDP/IP (URAM) process to communicate with the BASE24 application over a UDP/IP connection.

For the HSM, it is the RAM or URAM PPD name and application service group (i.e., \$RAM.#SECURE1, \$URAM#SECURE1) that is specified for the SECURE-DEV1 and SECURE-DEV2 security device assigns in the LCONF.

Configuration requirements for the RAM interface process are provided in the informal ***Remote Application Manager Implementation and Installation Guide***. This guide is a TGAL document located in the RAMUPDT file on the BA10RAM subvolume.

Configuration requirements for the URAM interface process are provided in the ***RAM/UDP Update Guide***. This guide is a Microsoft Word document. The binary format of this document is located in the URAMUDOC file on the BA10RAM subvolume. To view this document, perform a binary transfer of the URAMUDOC file to your PC and add a .DOC extension to the file name. You can then open the document using the Word application.

URAM-MSG-MDFY Parameter

The URAM-MSG-MDFY parameter in the URAMCNFG file indicates what type of message modification processing the URAM process performs. When using the Transaction Security Services process, you must set this parameter to a value of 0 (no message modification is performed). If you are not using the Transaction Security Services process, you must set this parameter to a value of 1 (STX/ETX characters are stripped and the four byte header is altered between the URAM process and the HSM). For more detailed information on the URAM-MSG-MDFY parameter and its possible values, refer to the global definition comments in the BA10RAM.URAMCNFG file.

Hardware Configuration for K-Series Servers

Because the HSM is a TCP/IP device, each Ethernet controller (e.g., 3613, 3615) must be configured using the SYSGEN utility or DSC utility. For detailed information on configuring an Ethernet controller and line, refer to the HP NonStop ***TLAM Configuration and Management Manual***. The following is an example of a SYSGEN utility definition example for a 3615 controller named RG7100x:

```
controller-name product-number processor-numbers channel addresses
RG7100X          3615          0,1          %030;
```

The following is an example of the SYSGEN utility definition that shows the minimum parameters required to configure a 3615 controller (line):

<i>device-name</i>	<i>controller-name</i>	<i>macro</i>	<i>modifiers</i>
\$LAM1	LAM1.0,LAM1.1	MLAM	TYPE 56, SUBTYPE 0, RSIZE 32000, READBUFFERSIZE 8192, INTERRUPT IOP^INTERRUPT^HANDLER#

After the controller is installed, the TCP/IP process must be started against the Ethernet port. An IP address must be assigned to both the RG7100 or HSM 8000 device and the HP NonStop server. Refer to the “Software Configuration” discussion below.

Hardware Configuration for S-Series Servers

Because the HSM is a TCP/IP device, it must communicate with the S-series server through an Ethernet port on either a 3861 Ethernet 4 ServerNet Adapter (E4SA) controller or through a Processor MultiFunction Unit (PMF). The configuration of a E4SA controller on an S-series server requires the use of the HP NonStop Subsystem Control Facility (SCF) to configure and start objects associated with the E4SA controller. These objects include the adapter, the ServerNet addressable controllers (SACs), the logical interfaces (LIFs), and the physical interfaces (PIFs). For additional information on configuring the 3681 (E4SA) controller, refer to the HP NonStop *Himalaya S-Series Hardware Installation Guide*.

After the controller is installed, the TCP/IP process must be started against the Ethernet port. An IP address must be assigned to both the RG7100 or HSM 8000 device and the HP NonStop server. Refer to the “Software Configuration” discussion below.

Software Configuration

The software configuration is very similar for the K-series and S-series servers. Examples for both types of servers are provided below. In general, the TCP/IP process must be started against a previously configured port and the RAM or URAM interface process must be started. Some of the values in these examples, most importantly the IP addresses, are site-specific and must be modified accordingly.

K-Series Server. The following is an example TACL macro for starting the TCP/IP process on a K-series server.

```
?tacl macro
#frame
delete define=tcPIP^host^file
add define=tcPIP^host^file, file $system.ztcPIP.hosts
$system.ztcPIP.tcPIP.lib $system.zqiolib.qiolib,name $ZTCO,&
    nowait, pri 180, cpu 0/1

#push scfin
#append scfin allow all errors
#append scfin assume process $ZTCO
#append scfin alter subnet, #loop0, ipaddress 127.1
#append scfin add subnet #sn0, type ethernet, devicename $LAN1,&
    ipaddress 130.130.63.16, forceqio on
#append scfin add route #r80, destination 0.0.0.0,&
    gateway 130.130.5.1
#append scfin start route *
#append scfin start subnet *
scfin/inv scfin/
#unframe
```

S-Series Server. The following are examples for starting the TCP/IP process on an S-series server.

The first example starts the TCP/IP process using an SCF OBEY file.

```
=====
= Configure & Start $ZB110 TCPIP process for ADAPTER E1151,LIFL100 ==
=====
ASSUME PROCESS $ZB110
ADD SUBNET #SN1, TYPE ETHERNET, IPADDRESS 192.231.036.99&
DEVICENAME L110
ADD ROUTE #GW, DESTINATION 0, GATEWAY 192.231.036.017,&
DESTTYPE BROADCAST
ALTER SUBNET #LOOP0, IPADDRESS 127.1
START SUBNET *
START ROUTE *
```

The next example starts the TCP/IP process using a TACL macro.

```
TCP/IP /name $ZB110,NOWAIT, CPU 0, PRI 160, TERM $ZHOME/1
```

Racal Access Module (RAM or URAM) Process. The RAM or URAM process runs under control of the XPNET process, and, therefore, must be configured to the XPNET system. For detailed information on configuring and starting the

RAM or URAM process, refer to the following informal documents available from ACI: ***Remote Application Manager Implementation and Installation Guide*** and ***RAM/UDP Update Guide***, respectively.

Logical Network Configuration File (LCONF)

The following section presents the Logical Network Configuration File (LCONF) information required for configuration of the Atalla security devices.

LCONF Assigns

The following assigns are required in the LCONF to configure a security module.

SECURE-DEV1 — Specifies the name of the primary security device or the name of the interface module used with the primary security device. For security devices other than the RG7100 or HSM 8000 security device using Transmission Control Protocol/Internet Protocol (TCP/IP) or User Datagram Protocol/Internet Protocol (UDP/IP) or the RG7300 security device using synchronous data-link control (SDLC) emulation, this assign specifies the SYSGEN utility or DSC name of the primary security device. For an RG7100 or HSM 8000 device using TCP/IP or UDP/IP, this assign specifies the process PPD name and application service group defined for the Racal Access Module process (for example, \$RAM.#SECURE1 or \$URAM.#SECURE1). The Racal Access Module process PPD name is defined in the Network Environment File (NEF) and the application service group is defined in the Racal Access Module Configuration File. For an RG7300 device using SDLC, this assign specifies the process name given to the HSM Interface process (for example, \$HIP).

When a Thales e-Security security device is used with the BASE24-card product, the USER FIELD of this LCONF assign record must contain a value of RG6000 or RG7000 to indicate whether a Version 6000 or Version 7000 security device is being used. (The Version 7000 device accepts a PIN-Length parameter to generate a variable-length PIN. The Version 6000 device only creates four-digit PINs.) If the User Field is blank, the Plastic Card Generation Server (SVPCGEN) process assumes a Version 7000 device is being used.

If your Thales e-Security security device is connected using a synchronous protocol such as TCP/IP or UDP/IP and you are generating or verifying MACs, the USER FIELD of this assign must contain a value of BINARY. This prevents the security utilities from performing binary to hexadecimal conversion of data when generating or verifying a MAC.

There can be multiple SECURE-DEV1 assigns in the LCONF, depending on whether a primary device is to be specified individually for each process requiring the use of a security device (in which case each process has its own SECURE-DEV1 assign in the LCONF).

To associate a primary security device with a specific process, the symbolic name of the process must be placed in the READ BY field on the LCONF Assign screen. If a specific primary security device need not be associated with a specific process, the READ BY field can be filled in with asterisks.

When determining its primary security device, a process first reads through the LCONF for a SECURE-DEV1 assign with its own name in the READ BY field. If it finds a match, it uses that corresponding assign. However, if it does not find a match, it reads through the LCONF again searching for a SECURE-DEV1 assign with asterisks in the READ BY field.

SECURE-DEV2 — Specifies the name of the secondary security device or the name of the interface module used with the secondary security device. For security devices other than the RG7100 or HSM 8000 security device using Transmission Control Protocol/Internet Protocol (TCP/IP) or User Datagram Protocol/Internet Protocol (UDP/IP) or the RG7300 security device using synchronous data-link control (SDLC) emulation, this assign specifies the SYSGEN utility or DSC name of the primary security device. For an RG7100 or HSM 8000 device using TCP/IP or UDP/IP, this assign specifies the process name and application service group defined for the Racal Access Module process (for example, \$RAM.#SECURE2 or \$URAM.#SECURE2). The Racal Access Module process PPD name is defined in the Network Environment File (NEF) and the application service group is defined in the Racal Access Module Configuration File. For an RG7300 device using SDLC, this assign specifies the process name given to the HSM Interface process (for example, \$HIP).

When a Thales e-Security security device is used with the BASE24-card product, the USER FIELD of this LCONF assign record must contain a value of RG6000 or RG7000 to indicate whether a Version 6000 or Version 7000 security device is being used. (The Version 7000 device accepts a PIN-Length parameter to generate a variable-length PIN. The Version 6000 device only creates four-digit PINs.) If the User Field is blank, the Plastic Card Generation Server (SVPCGEN) process assumes a Version 7000 device is being used.

If your Thales e-Security security device is connected using a synchronous protocol such as TCP/IP or UDP/IP and you are generating or verifying MACs, the USER FIELD of this assign must contain a value of BINARY. This prevents the security utilities from performing binary to hexadecimal conversion of data when generating or verifying a MAC.

There can be multiple SECURE-DEV2 assigns in the LCONF, depending on whether a secondary device is to be specified individually for each process requiring the use of a security device (in which case each process has its own SECURE-DEV2 assign in the LCONF).

To associate a secondary security device with a specific process, the symbolic name of the process must be placed in the READ BY field on the LCONF Assign screen. If a specific secondary security device need not be associated with a specific process, the READ BY field can be filled in with asterisks.

When determining its secondary security device, a process first reads through the LCONF for a SECURE-DEV2 assign with its own name in the READ BY field. If it finds a match, it uses that corresponding assign. However, if it does not find a match, it reads through the LCONF again searching for a SECURE-DEV2 assign with asterisks in the READ BY field.

LCONF Params

The following params are required in the LCONF to configure a security module.

RSM-TERM-MASTER-KEY — The encrypted terminal master key used with Thales e-Security security devices. This key is required when the BASE24 ISO Host Interface process or the BASE24 Interchange (BIC) ISO Interface process use Thales e-Security security devices to generate message authentication code (MAC) encryption keys. This param is not required when only BASE24 Device Handler processes use Thales e-Security security devices to generate MAC encryption keys or when Thales e-Security security devices are not used to generate MAC encryption keys.

The BASE24 ISO Host Interface process and the BIC ISO Interface process use this key in an intermediate step that is performed when BASE24 is responsible for generating and distributing MAC encryption keys. The key in this param does not coincide with any other keys being used by the external processor. For ease of maintenance, ACI recommends using one RSM-TERM-MASTER-KEY param for a BASE24 network.

When this param is required, the default value of zeros must be changed. The Thales e-Security console command K can be used to obtain the encrypted key for this param. Refer to the appropriate Thales e-Security manual for information regarding the console command.

SECURE-DEV-FAIL-NOTIFY — The interval at which critical event messages are generated due to the failure of a primary or secondary security device for a process. Valid values are 10 to 1000; the default value is 20.

If the primary and secondary security devices cannot be initialized during process startup and this param is set to the default of 20, every twentieth transaction that a process must decline because the security devices are inoperative causes a message to be generated indicating the security devices have failed. (Note that a critical event message is always generated immediately upon failure of both devices.)

If a primary security device is marked down and this param is set to the default of 20, every twentieth transaction that is directed to the primary security device causes a message to be generated even though the transactions are being handled properly by the secondary security device. Transactions routed directly to the secondary security device do not affect the primary security device transaction counter. The SECURE-DEV-MONITOR-THRESHOLD param controls when a transaction is directed to the secondary security module instead of the primary security device.

If the primary and secondary security devices fail at some time after initialization, a separate failed transaction counter is maintained for each security device. A transaction that is sent to one security device and then the other increments both counters. An event message is generated each time either counter reaches the value established in this param.

SECURE-DEV-MONITOR-THRESHOLD — The interval, expressed in the number of transactions processed by the security device defined in the SECURE-DEV1 assign, for checking the security device defined by the SECURE-DEV2 assign. Valid values are 0–32000. The default value of 0 indicates no check is performed.

This param can be used to monitor the secondary security device or to balance the load between the primary and secondary security devices. Each process using a security device maintains its own transaction counter. This means the interval specified in the SECURE-DEV-MONITOR-THRESHOLD param needs to be set for the number of transactions processed by individual processes using the security device, not the total number of transactions processed by the security device. For example, if five BASE24 processes use a security device and the param is set to 1000, up to 5000 transactions (1000 by each of the five processes) could be processed by the primary security device before the secondary security device is used. Setting this param to 1 causes each of the BASE24 processes to alternate between the two security devices.

If all processes use the same threshold for monitoring the secondary security device, a single SECURE-DEV-MONITOR-THRESHOLD param can be defined in the LCONF. In this case, the READ BY field on the LCONF Param screen must be set to asterisks (default value).

If different secondary security device monitoring thresholds are used, **SECURE-DEV-MONITOR-THRESHOLD** params must be defined in the LCONF for each threshold and the **READ BY** field can be set to identify the processes using that threshold. If a majority of processes use a particular threshold, an alternative to identifying all of its processes can be making that threshold the default by placing asterisks in the **READ BY** field. Then, only processes using a different threshold need to be defined specifically.

SECURE-DEV-TIM-LMT — The maximum time, in hundredths of a second, to wait for a response from the security device before the request is considered to be timed out. Valid values are 1 to 6000; the default value is 500 (i.e., 5 seconds). The same time limit applies to all processes so there is only one **SECURE-DEV-TIM-LMT** param in the LCONF, and the **READ BY** field must be set to asterisks.

SECURE-DEV-TYP — Specifies whether BASE24 security utilities in a BASE24 process communicate directly with a Thales e-Security security device or do not use a security device. Valid values are as follows:

N/A = No security module required
RSM = Thales e-Security Host Security Module (same as VSM)
VSM = Thales e-Security Host Security Module (same as RSM)

In some BASE24 logical networks it may be desirable to use more than one security device type. This is possible, since the **SECURE-DEV-TYP** param is assigned by process. Care must be exercised, however, in assigning different security device types to different processes, especially where the different processes communicate with each other.

For example, if a Device Handler or interface process receives a PIN encrypted under a key managed by an Atalla security device, but passes the PIN and the Atalla-encrypted PIN key to an Authorization process that is using a Thales e-Security security device for PIN verification, the PIN cannot be verified unless the Thales e-Security device is configured specifically to correctly decrypt the Atalla-encrypted PIN key.

The same care must be exercised regarding MAC keys. For example, if a Device Handler or interface process receives a MAC generated under a key managed by an Atalla security device, but attempts to verify the MAC using a Thales e-Security security device, the MAC cannot be verified unless the Thales e-Security device is configured specifically to correctly decrypt the Atalla-encrypted MAC key.

Also, the Authorization process sends information from Key Authorization File (KEYA) records to its security device so care must be taken to ensure that any encrypted data passed by the Authorization process from the KEYA to the security device has previously been encrypted by that same security device type with the same master key.

If all processes utilize the same type of security device, a single SECURE-DEV-TYP param can be defined in the LCONF. In this case, the READ BY field on the LCONF Param screen must be set to asterisks (default value).

If different security device types are utilized, SECURE-DEV-TYP params must be defined in the LCONF for each type and the READ BY field on the LCONF Param screen for each type can be set to identify the processes using that type. If a majority of processes use a particular type of security device, an alternative to identifying all of its processes can be making that device type the default by placing asterisks in the READ BY field of its SECURE-DEV-TYP param. Then, only processes using a different type of security device need to be defined specifically.

If no security devices are used, a SECURE-DEV-TYP param must still be defined in the LCONF. The READ BY field on the LCONF Param screen must be set to a value of asterisks (default value) and the Text field on the LCONF Param screen must be set to a value of N/A (no security module required).

Thales e-Security Host Security Module Support

Summaries of Thales e-Security Host Security Module (HSM) functions supported by BASE24 presented in this section include the following:

- The HSM commands required for generating, encrypting, and translating data for storage in the BASE24 database.
- The HSM online functions supported by BASE24.

This material is not meant to be a tutorial on the HSM; it presumes a familiarity with the HSM's capabilities and nomenclature as discussed in the Thales e-Security documentation. Specifically, it does not address the creation and maintenance of the HSM local master keys (LMKs), which also are discussed in the Thales e-Security documentation.

Note: The Thales e-Security HSM was formerly known as the Visa Security Module (VSM). DDLs and other BASE24 documentation may still reference the VSM. Such references should be considered to apply to the HSM instead. The Visa PVV PIN verification method and the Visa Security Module are not related. The verification method name remains unchanged.

Preparing Data for the BASE24 Files

Offline HSM Master Key loading and key management are accomplished using a user-supplied asynchronous ASCII terminal connected to the HSM asynchronous port. This asynchronous ASCII terminal is known as the HSM console.

The table on the following pages summarizes the HSM console commands used to create, encrypt, and translate keys for storage in the BASE24 database and to load and encrypt the Diebold number table. Note that the HSM Master Keys must have already been loaded and the HSM placed in the authorized state before the commands can be performed.

Field Name in BASE24 File	Screen	Master Key Pair	HSM Console Command
MASTER KEY	ATD 9	14–15	K = Encrypt a terminal master key
		06–07	B = Generate and encrypt a working key*
PIN MASTER KEY	PTD 7	14–15	K = Encrypt a terminal master key
		06–07	B = Generate and encrypt a working key*
MAC MASTER KEY	PTD 7	14–15	K = Encrypt a terminal master key
MASTER KEY	TTDF 2		
MAC MASTER KEY	TTDF 2		
CRYPTOGRAPHIC KEYS	KEYI 1	Variant 1 of 28–29	FK = Format key from components
SECURE MESSAGING KEYS: FOR INTEGRITY	KEYI 1	Variant 2 of 28–29	FK = Format key from components
INTERMEDIATE KEY ENCRYPTED	KEYF 1 KEY6 1	06–07	B = Generate and encrypt a working key*
PIN KEY EXCHANGE KEY	KEYF 1 KEY6 1	04–05	EC = Encrypt clear component FK = Format key from components IK = Import key [†]
MAC KEY EXCHANGE KEY	KEYF 1 KEY6 1	04–05	EC = Encrypt clear component FK = Format key from components IK = Import key [†]
INBOUND PIN KEY 1	KEYF 2 KEY6 2	06–07	FK = Format key from components [‡]
INBOUND PIN KEY 2			
OUTBOUND PIN KEY 1	KEYF 2 KEY6 2	06–07	FK = Format key from components [‡]
OUTBOUND PIN KEY 2			

Field Name in BASE24 File	Screen	Master Key Pair	HSM Console Command
INBOUND MAC KEY 1	KEYF 2 KEY6 3	26–27	FK = Format key from components [§]
INBOUND MAC KEY 2			
OUTBOUND MAC KEY 1	KEYF 2 KEY6 3	26–27	FK = Format key from components [§]
OUTBOUND MAC KEY 2			
DES (IBM 3624) ENCRYPTED KEY	KEYA 2	14–15	K = Encrypt a PIN verification key
DIEBOLD TABLE	KEYA 3	10–11	R = Load and encrypt a clear Diebold table ^{**}
VISA PVV KEYS ENCRYPTED	KEYA 5	14–15	K = Encrypt a PIN verification key
CARD VERIFICATION ENCRYPTED KEY 1	KEYA 6	Variant of 14–15 ^{††}	KA = Generate a CVK pair
CARD VERIFICATION ENCRYPTED KEY 2			
DERIVATION KEY	KEYD 1	28–29	DE = Form a ZMK from clear components DG = Generate a base derivation key

* The intermediate key encrypted under the HSM Master Key pair 06–07 must be stored in the KEYF or KEY6 record for the external processor if a mixed clear and encrypted PIN interface environment is required.

For ATM or POS devices configured to send PINs in the clear to any interchanges on the BASE24 system, the intermediate key encrypted under the HSM Master Key pair 06–07 must be stored in the BASE24-atm or BASE24-pos Terminal Data files.

† A key exchange key is the key under which working keys are encrypted for exchange between BASE24 and an external processor. It equates to Thales e-Security's Zone Master Key (ZMK).

An encrypted ZMK need not be placed in the KEY EXCHANGE KEY fields of the KEYF or KEY6 unless automatic key exchange is supported. (BASE24 only supports automatic key exchange with DPCs and certain interchanges.)

To generate a ZMK, the console command EC must be used first to obtain three encrypted ZMK components. Console command FK is then used to format the key from the three encrypted ZMK components. Finally, the IK command is used to translate the key from encryption under the ZMK to encryption under the LMK for storage in the KEY EXCHANGE KEY fields.

- ‡ If the BASE24 operations staff is responsible for generating the inbound and outbound keys for PIN encryption, console command B or FK can be used to generate and encrypt these keys under the HSM Master Key pair 06–07. These keys can then be placed in the KEYF or KEY6, and the ZMK-encrypted forms of these keys can be conveyed to the external processor.

If the inbound and outbound working keys are generated by an external processor, the keys can be distributed automatically or manually. If these keys are distributed automatically when the interface process is first brought up, inbound and outbound entries in the KEYF or KEY6 must be left as zeros. The ZMK must be provided in the KEY EXCHANGE KEY fields in the KEYF or KEY6 (refer to the second footnote for more information about the ZMK).

Otherwise, the BASE24 operations staff receives ZMK-encrypted versions or clear versions of these keys. The ZMK-encrypted versions of the keys must be used to obtain versions encrypted under the HSM Master Key pair 06–07 for storage in the KEYF or KEY6. Since the inbound and outbound keys must be encrypted or decrypted using the ZMK, depending on whether the keys are being sent to or received from an external processor, the ZMK must be provided in the HSM console command B or FK.

If the operations staff receives clear versions of these keys, they must be encrypted using the HSM console command BK to generate the encrypted zone PIN keys (ZPKs). The key type must be set to a value of 2 for this command. If only one component is used, it should be entered three times. Otherwise, at least two different components should be entered. The resulting cryptogram can then be placed in the KEYF or KEY6, and the ZMK-encrypted forms of these keys can be conveyed to the external processor.

- § If zone authentication keys are not distributed automatically, they must be generated manually and encrypted using the HSM console FK command.

** Initially, the clear Diebold number table (DNT) must be manually entered using the HSM console command R. This command causes the clear DNT to be encrypted and loaded into the HSM. It also returns an encrypted version of the DNT to the BASE24 system administrator.

The BASE24 system administrator can then copy the encrypted form of the DNT and place it in the KEYA manually.

†† The HSM encrypts card verification keys under a variant of key pair 14-15. This variant, unlike the variants used with Atalla security devices, is an unknown value. HSM card verification keys must be encrypted using HSM console command KA. This command incorporates the variant that is required to obtain the proper results.

Double-Length Key Encryption Scheme

BASE24 requires that most double-length keys used by Thales e-Security HSMs to be encrypted using Thales' variant (1A+32H) key encryption scheme instead of the old (ANSI X9.17 method) non-variant (32H) key encryption scheme. For more information on these encryption schemes, refer to the Thales *Host Security Module, RG7000 Programmer's Manual*.

If a variant of the LMK is required for the key type (see the Master Key Pair column in the table for the [“Preparing Data for the BASE24 Files”](#) topic on the previous pages), this variant and is always applied regardless of the key encryption scheme used (ANSI X9.17 or variant) and is applied to the left half of the LMK.

For double-length keys encrypted using the variant key encryption, another two variants are applied to the right half of the LMK. Refer to the “Local Master Key Triple DES Variant Scheme” topic in the Thales *Host Security Module, RG7000 Programmer's Manual* for a description of how these variants are applied.

Note: Although Thales e-Security HSMs treat key pairs like double-length keys, the Transaction Security Services process views key pairs as two single-length keys. Because of this, the Transaction Security Services process does not support the configuration of the key scheme for commands using these keys. For consistency, the BASE24 security utilities have not been enhanced to use the double-length key encryption scheme in commands that use key pairs (i.e., Visa PVV PIN verification keys and card verification keys) stored in the Key Authorization File (KEYA).

If you are currently using double-length keys encrypted under the (ANSI X9.17 method) non-variant (32H) key encryption scheme, you must convert these keys to encryption under the variant (1A+32H) key encryption scheme. ACI provides conversion programs for double-length keys in the following files:

- Derivation Key File (KEYD)
- ICC Key File (KEYI)

The following table lists the keys on the files maintenance screens for the above files that are converted by these programs and how the keys are used in processing.

Screen	Key Field Name	Key Usage
KEYD 1	DERIVATION KEY	Translating a PIN from BDK encryption to interchange key encryption.
KEYI 1	CRYPTOGRAPHIC KEYS	ARQC verification or ARPC generation.
	SECURE MESSAGING KEYS FOR INTEGRITY	Generating a secure message with integrity.

For detailed information on running the above file conversion programs, refer to the ***BASE24 Release 6.0 Version 4 Implementation Guide***.

Double-length keys must be encrypted under the variant method for the following Thales HSM online functions supported by BASE24:

Function	Transaction Code
Translate PIN (encryption under LMK pair 14–15 to encryption under LMK pair 06–07)	CA
Translate PIN (encryption under LMK pair 06–07 to encryption under LMK pair 06–07)	CC
Translate PIN (BDK encryption under LMK pair 28–29 to ZPK encryption under LMK pair 06–07)	CI
ARQC (or TC/ACC) Verification or ARPC Generation	KQ

Function	Transaction Code
Generate secure message with integrity and optional confidentiality and PIN change	KU

HSM Online Functions Supported by BASE24

The following table summarizes the HSM transaction codes and functions used by the BASE24 Device Handler, Authorization, Interchange Interface and Host Interface processes.

Transaction Code	Function	Device Handler Process	Authorization Process	Interchange Interface Process	Host Interface Process
A0	Generate a single length ZPK under LMK pair 06–07	Y	N	N	N
CA	Translate PIN (encryption under LMK pair 14–15 to encryption under LMK pair 06–07)	N	N	Y	Y
CC	Translate PIN (encryption under LMK pair 06–07 to encryption under LMK pair 06–07)	N	N	Y	Y
CE	Generate Diebold PIN Offset	N	Y	N	N
CG	Verify PIN from terminal—Diebold	N	Y	N	N
CI	Translate PIN (BDK encryption under LMK pair 28–29 to ZPK encryption under LMK pair 06–07)	Y	N	N	N
CY	Verify a Visa CVV	N	Y	N	N
DA	Verify PIN from terminal—IBM 3624-format	N	Y	N	N
DC	Verify PIN from terminal—Visa PVV	N	Y	N	N
DE	Generate IBM 3624 PIN Offset	N	Y	N	N
DG	Generate Visa PVV	N	Y	N	N
EA	Interchange PIN Verify—IBM 3624-format	N	Y	N	N
EC	Interchange PIN Verify—Visa PVV	N	Y	N	N
EG	Interchange PIN Verify—Diebold	N	Y	N	N

Transaction Code	Function	Device Handler Process	Authorization Process	Interchange Interface Process	Host Interface Process
FA	Interchange Key Translate (encryption under ZMK to encryption under LMK pair 06–07)	N	N	Y	N
GC	Interchange Key Translate (encryption under LMK pair 06–07 to encryption under ZMK)*	N	N	Y	N
HA	Generate TAK under TMK, TPK, or LMK	Y	N	Y	Y
HC	Generate a New Terminal Master Key or PIN key	Y	N	N	N
IA	Interchange Key Generate (encryption under ZMK and LMK pair 06–07)	N	N	Y	N
IE	Encrypt Docutel Test Pattern under a Terminal Master Key	Y	N	N	N
JC	Translate PIN from TMK or TPK to LMK	N	Y	N	N
JE	Translate PIN from ZWK to LMK	N	Y	N	N
KA	Generate a key check value	N	N	Y	N
KQ	ARQC (or TC/ACC) Verification or ARPC Generation	N	Y	N	N
KU	Generate secure message with integrity and optional confidentiality and PIN change	N	Y	N	N
LA	Load Encrypted Diebold number table to HSM [†]	N	Y	N	N
LC	Verify Diebold number table held in HSM [†]	N	Y	N	N
MG	Translate TAK from LMK to ZMK	N	N	Y	Y
MI	Translate TAK from ZMK to LMK	N	N	Y	Y
MK	Generate MAC [‡]	Y	N	Y	Y

Transaction Code	Function	Device Handler Process	Authorization Process	Interchange Interface Process	Host Interface Process
MM	Verify MAC [‡]	Y	N	Y	Y

* In some HSM firmware or software versions, the GC transaction code may be available only when the HSM is in the authorized state. As a result, the Repeat Key command for dynamic key management may not work. Check with your Thales e-Security customer support representative.

† If Diebold PIN verification is required, the Authorization process restores or verifies the encrypted DNT in the HSM. If the DNT in the KEYA contains any values other than zeros, the Authorization process uses the LA message to restore the DNT in the HSM from the KEYA. If the DNT in the KEYA is zeros (i.e., it was never loaded into the KEYA), the Authorization process uses the LC message to verify the DNT in the HSM.

‡ The MK and MM transaction codes are not supported in the standard Thales e-Security product. They must be obtained from Thales e-Security

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